Virology

Ppt no. 01

VIRUSES

- Definition and meaning

MEANING

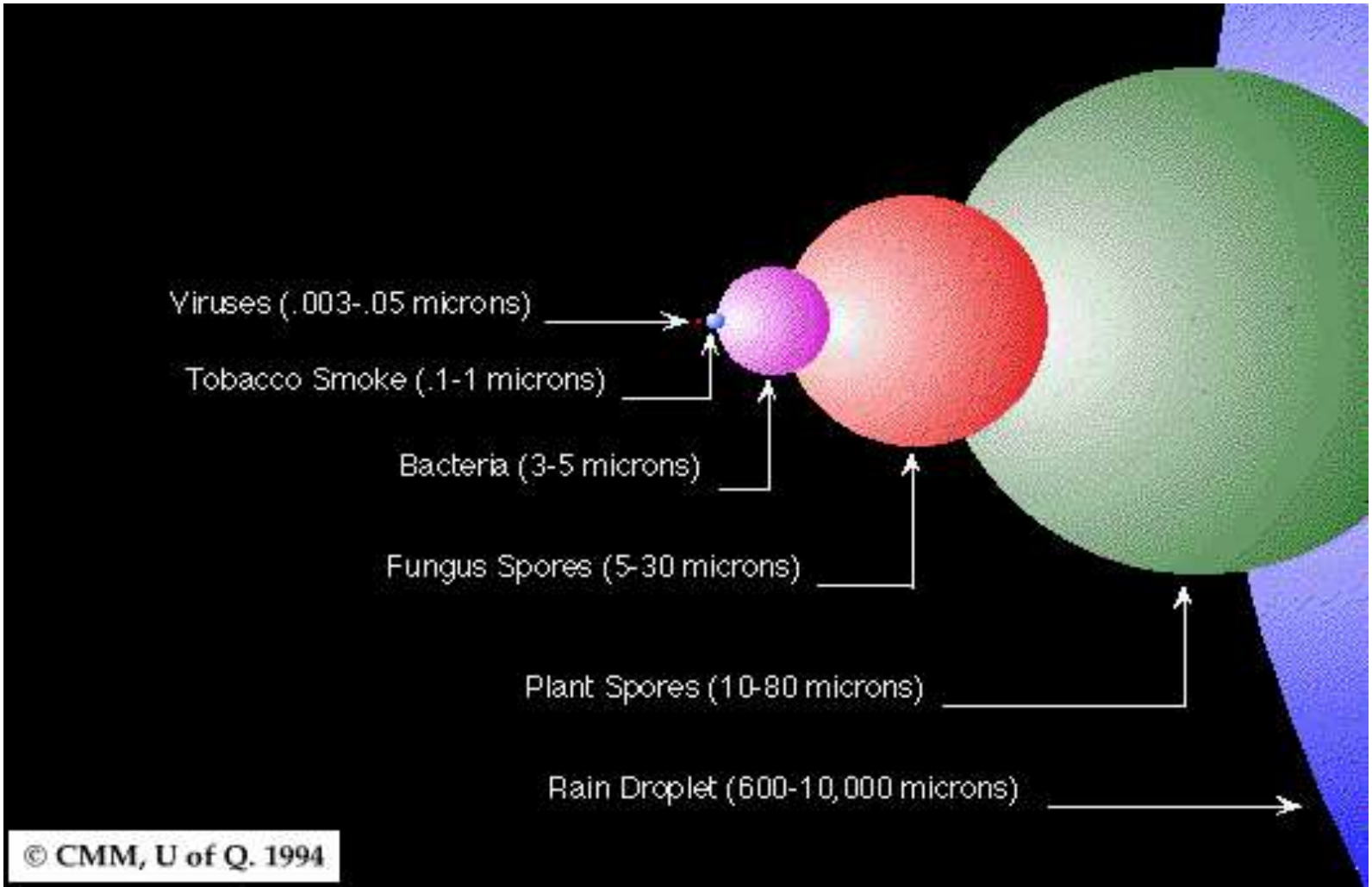
- Latin word
- Meaning **poison** or **venom**
- Given by Beijerinck (1897)

DEFINITION

- Viruses are non-cellular particles made up of genetic material and proteins that can invade living cells
- Obligate intracellular parasites

INTERESTING FACT

- Submicroscopic
  - Smaller than bacteria
  - Smaller than the smallest cell



TAKE HOME QUESTION!

Can you find the size range of viruses?

Ppt no. 02

Discovery of viruses

Ancient Times

First written record

From Egypt around 3700BC

Signs of poliomyelitis in a priest

Smallpox

Edward Jenner (1796)

Smallpox vaccine using milder cowpox virus

No knowledge of viral particles

Iwanowski

- In 1892, Dmitri Iwanowski found out reason for disease in plants
- He thought the bacteria causing disease is small enough to pass the smallest filters
- He did not know the significance of his results

The name "VIRUS"

- In 1898
- By Beijernick
- Studied filtered diseased plant juices and found the caused disease in healthy plants

Tobacco Mosaic Virus3

- Wendell Stanley (1935) discovered that:

– Viruses are composed of protein and nucleic acid

- Studied Diseased tobacco plants
- Discovered isolates of virus
- Won Nobel award

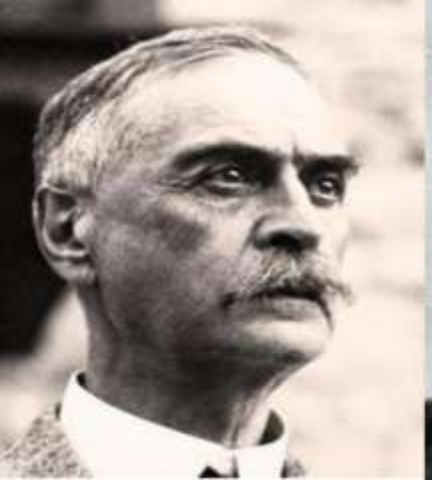
Foot and Mouth disease

- In 1898
- Recognized by

– Fredrich Loeffler

– Pal Froscch

- Disease in cattle



Poliomyelitis

- 1909
- Karl Landsteiner and Erwin Popper
- First human virus to be found

Bacteriophages

- Fredrick Twort (1915) and Felix D'Herelle (1917)
- Recognized viruses that infect bacteria
- Bacteriophages (eaters of bacteria)

END slide

Ppt no. 03

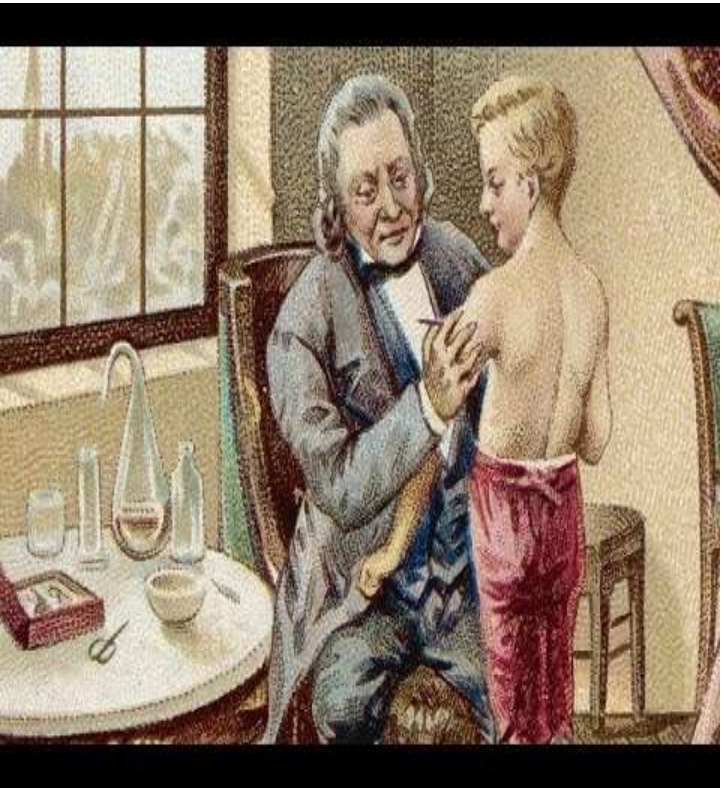
History Of Virology

First written record

- Hieroglyph from Memphis (capital of ancient Egypt)
- 3700 BC
- Temple priest showing signs of paralytic poliomylietis

1196 BC

- Pharoah Ramses V
- Dies in 1196 BC
- Well preserved mummified body kept in Cairo museum
- Believed to have small pox
- Pustular lesions on face



Endemic small pox

- Small pox endemic in China by 1000 BC
- Survivors did not suffer from re-infection
- Variolation was developed

May 1796

- Edward Jenner introduced the concept of vaccination
- Cow pox against small pox
- Adopted worldwide in 19th century

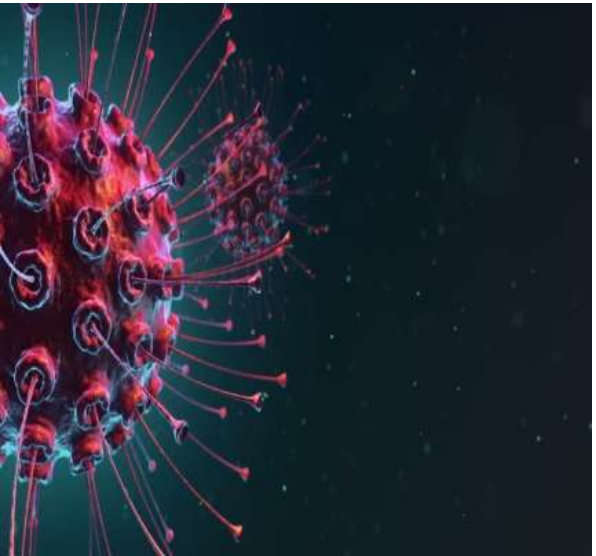
Martinus Beijerinick

- Tobacco Mosaic Virus
- First to develop modern idea of virus
- Referred as *contagium vivum fluidium* (soluble living germ)

Ppt no. 04

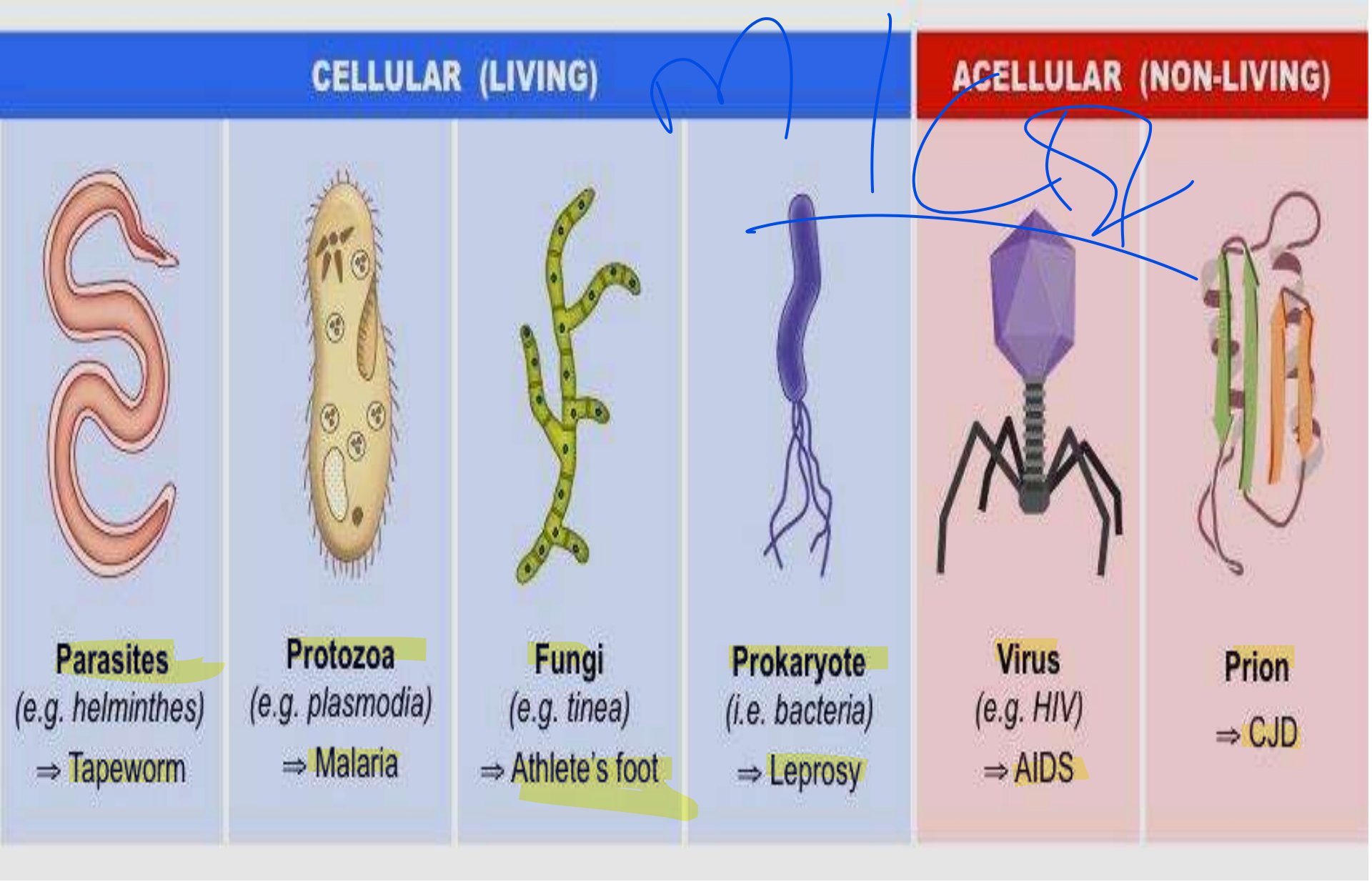
Importance of Virology

UNIQUE PROPERTIES

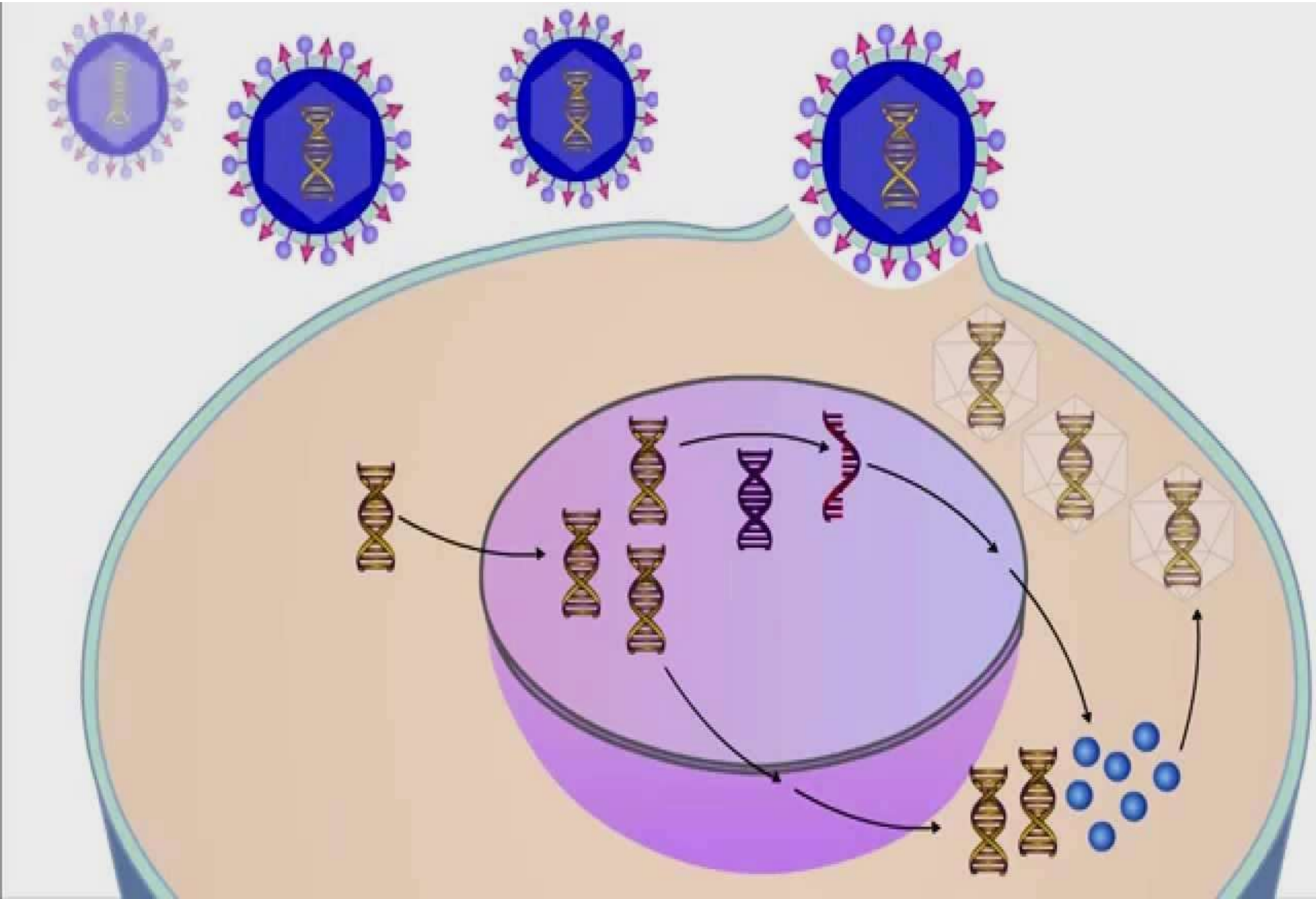


- Living as well as non-living characteristics
- Unable to cultivate in laboratory easily
- Simplest structure
- Complex interaction with host
- Replication within host

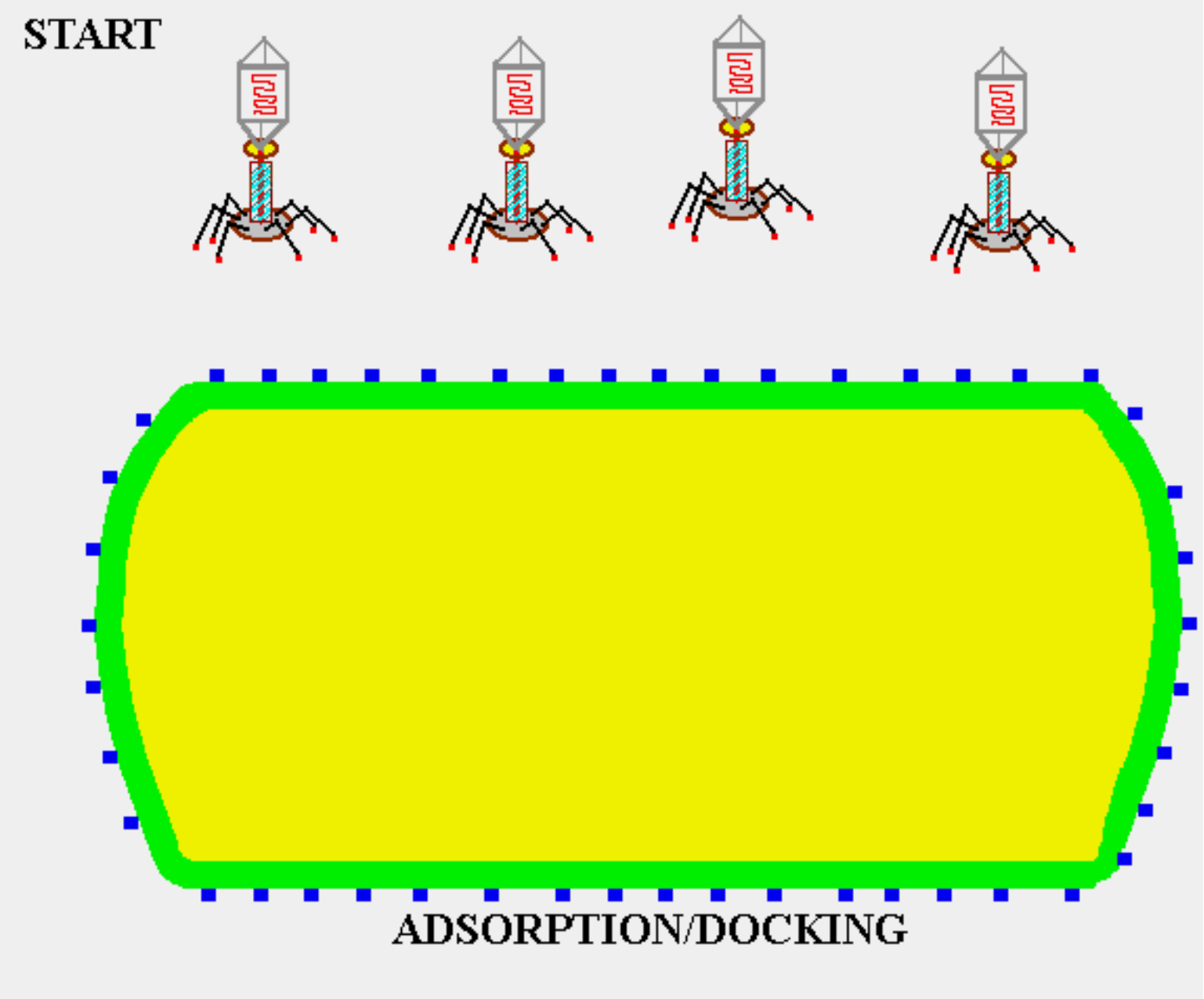
1. To study an important class of pathogens



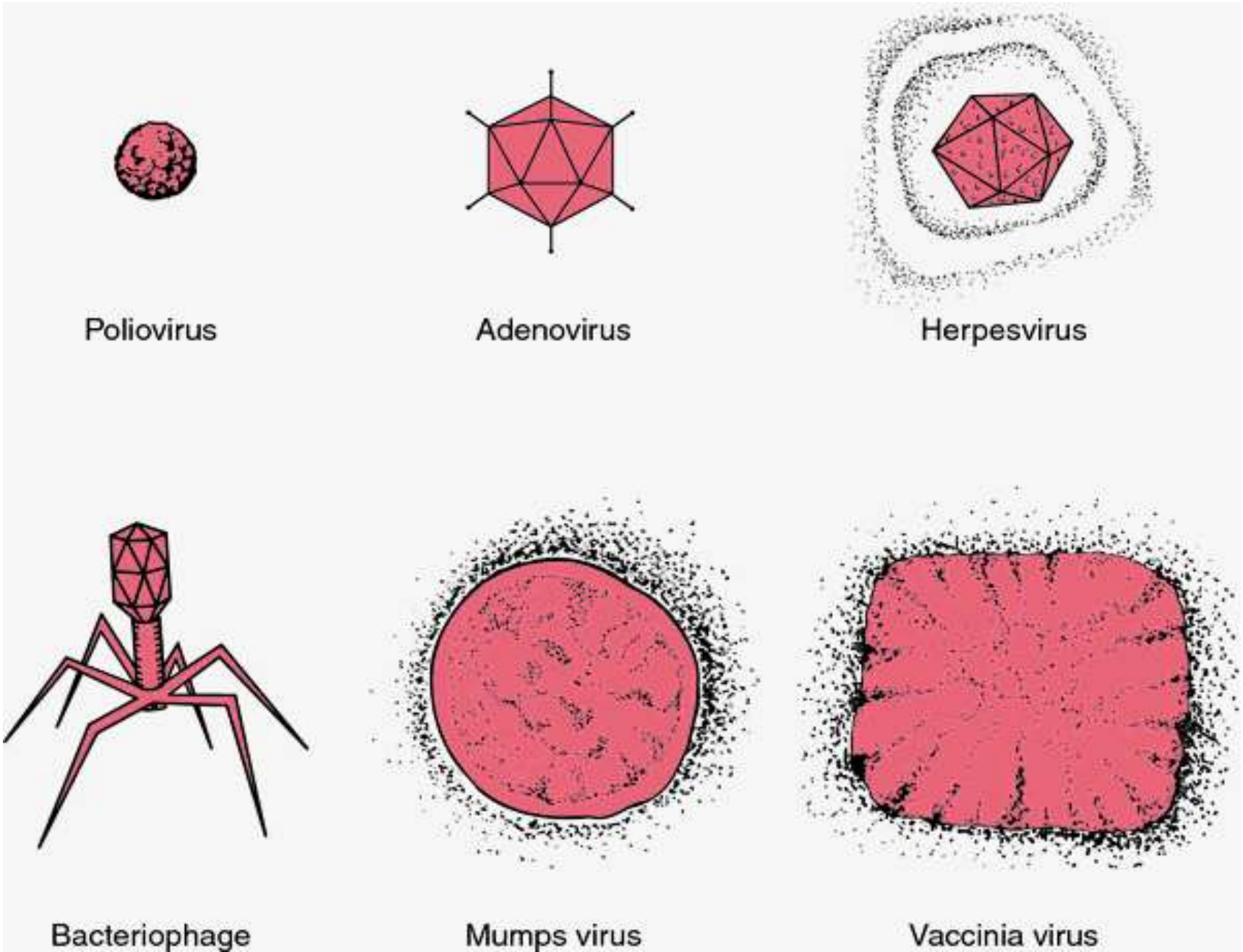
2. To study pathogenesis of viruses



3. To study lifecycle of viruses



4. To study structure of viruses



5. To study viral cultivation in lab

a. Tissue culture
b. Embryonated egg
c. Live animal

Ppt no. 06

VIRUSES

- Living or non-living

Are virus living entities?

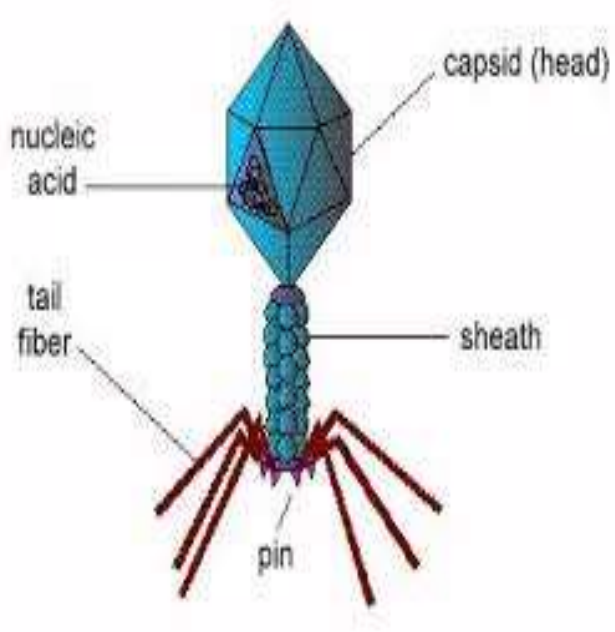
Ambiguous question

What is life?

- Definition of life: A complex set of processes resulting from the actions of proteins specified by nucleic acids
- Nucleic acids of living things are in action all the time

What is Virus ?

- Viruses are complicated assemblies of molecules, including proteins, nucleic acids, lipids, and carbohydrates
- Viruses can only replicate themselves by infecting a host cell and therefore cannot reproduce on their own.



CONCLUSION

- Viruses are:

"exceptionally complex aggregation of non-living chemicals"

OR

" exceptionally simple living organisms"

Ppt no. 07

ORIGIN OF VIRUSES

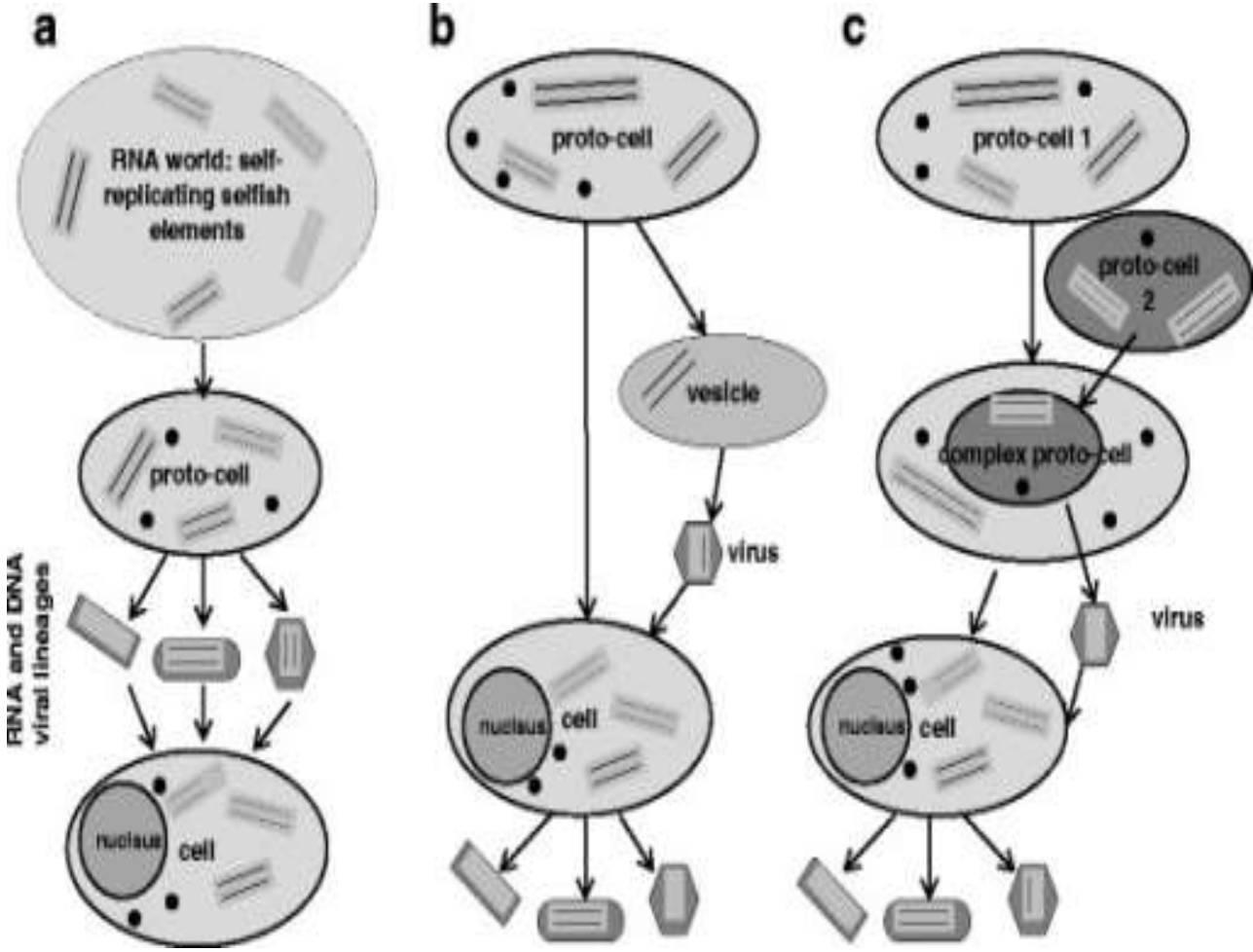
THEORIES

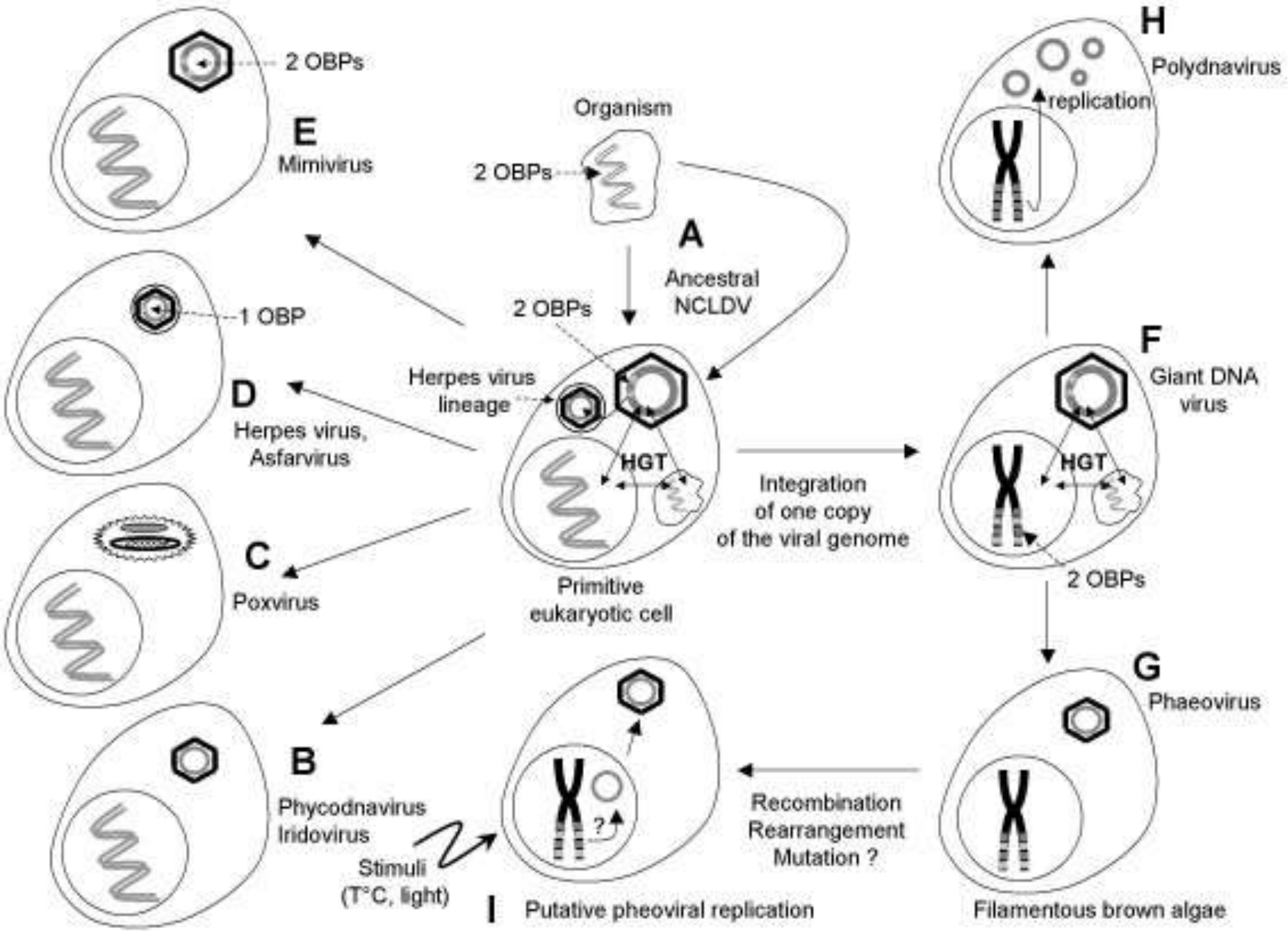
Evolved to be: Economic form of microbial life. Most efficient

Three theories for evolution

FIRST THEORY

They may have originated and evolved with the other life forms




2 OBPs
E
Mimivirus
Organism
2 OBPs
A
Ancestral
NCLDV
H
Polydnavirus
replication
1 OBP
2 OBPs
Herpes virus
lineage
D
Herpes virus,
Asfarvirus
HGT
F
Giant DNA
virus
HGT
Integration
of one copy
of the viral genome
C
Poxvirus
Primitive
eukaryotic cell
2 OBPs
G
Phaeovirus
B
Phycodnavirus
Iridovirus
Stimuli
(T°C, light)
?
Recombination
Rearrangement
Mutation ?
I
Putative pheoviral replication
Filamentous brown algae

SECOND THEORY

They may have arisen from segments of the cellular nucleic acids which acquired the ability to replicate at the expense of host cell

THIRD THEORY

- Regressive theory
- Viruses arose from free living organisms which gradually lost genetic information. Until they became totally dependent on the biosynthetic pathways of their host cells

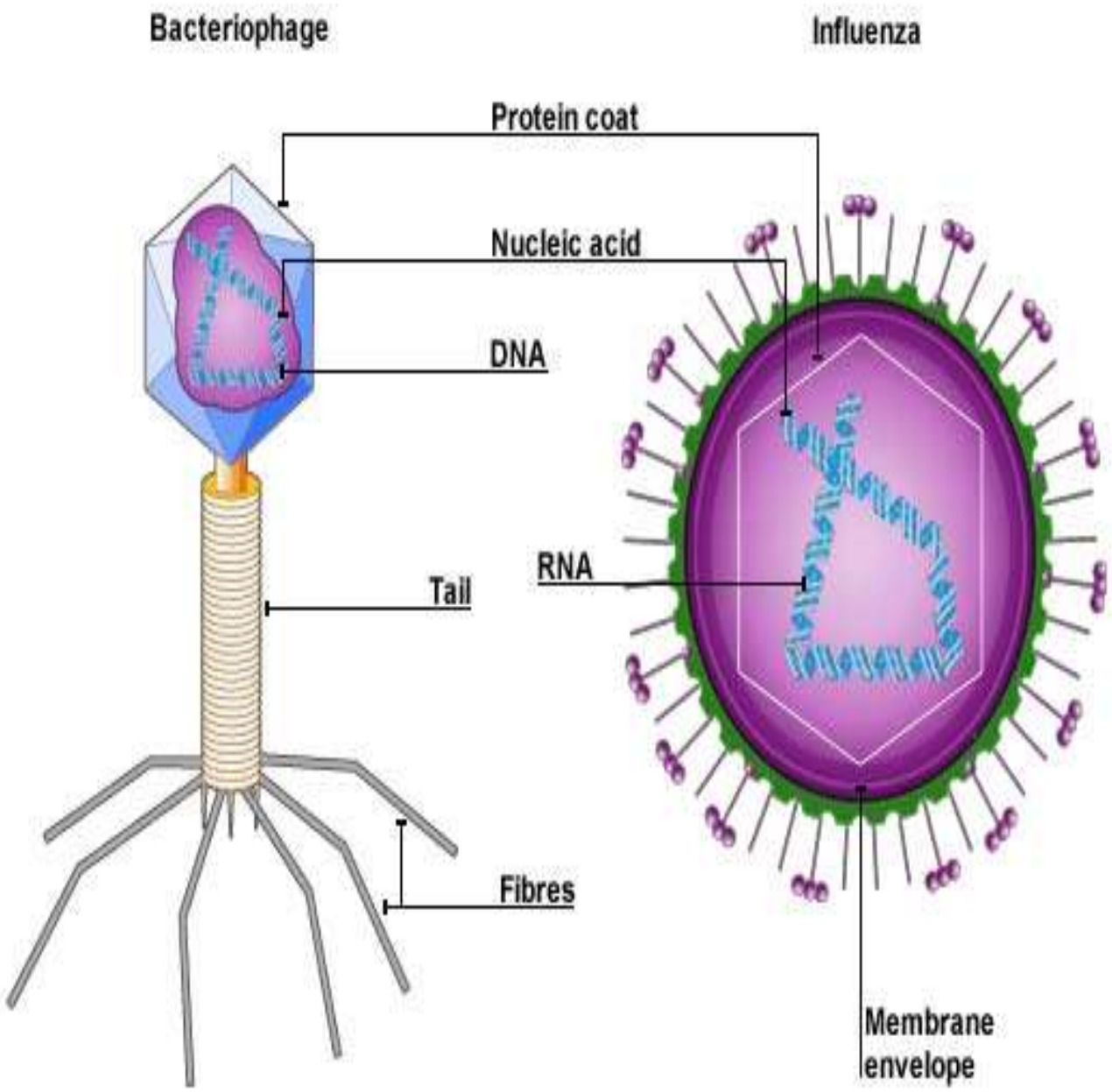
Ppt no. 08

GENERAL PROPERTIES OF VIRUSES

VIRIONS

- Mature viruses
- Variable in size

VIRUS	SIZE
Foot and Mouth Disease virus	27nm
Pox virus	300nm



STRUCTURE

1. Protein
2. Nucleic acid

Deoxyribonucleic Acid (DNA)

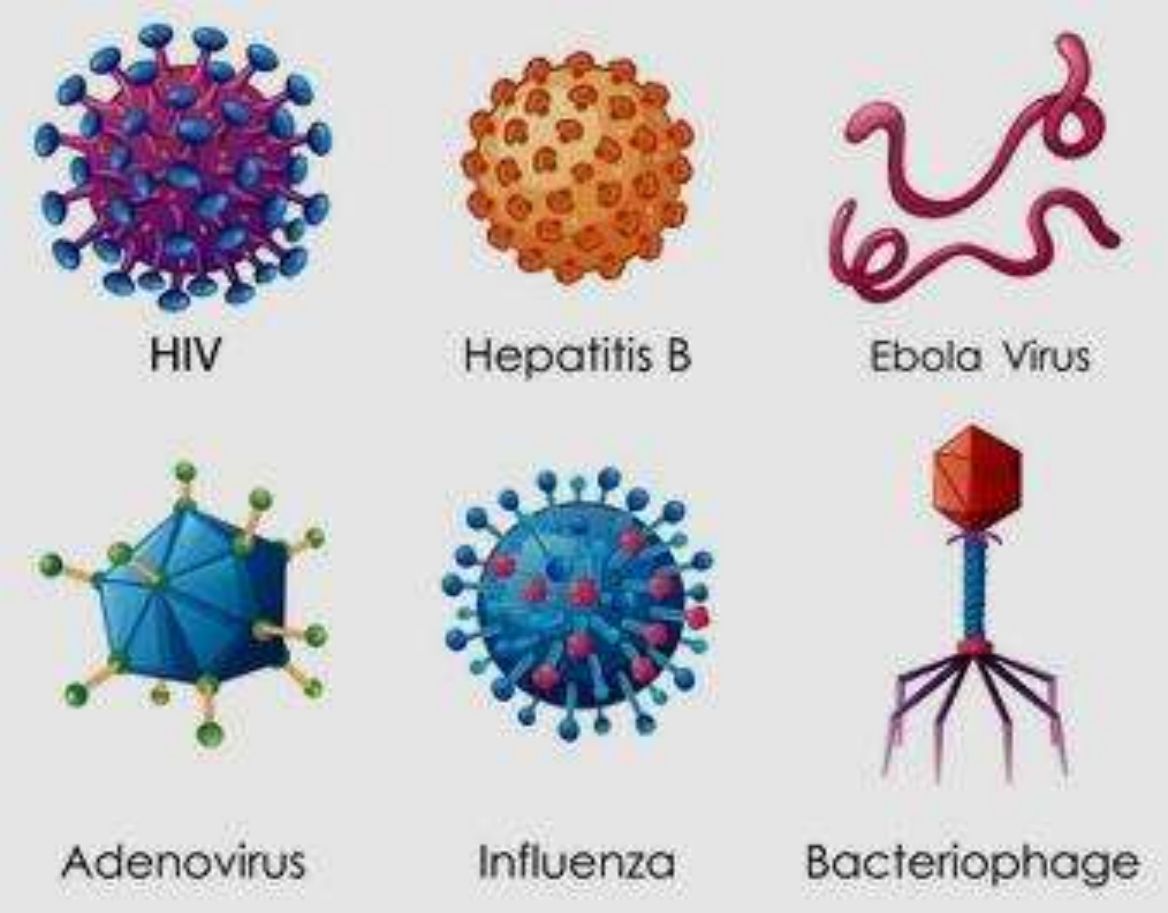
Ribonucliec Acid (RNA)

WHY VIRUSES NEED HOST?

- No enzymes of their own
- No metabolism
- Require host machinery

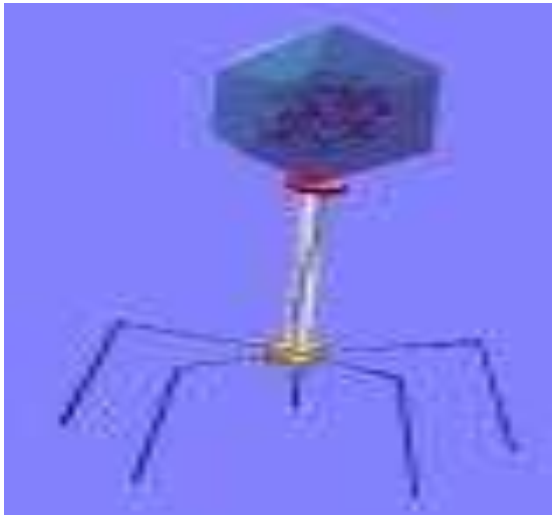
COMPLEX VIRUSES

- Additionally contain:
  - Lipid
  - Polysaccharides
  - Trace elements

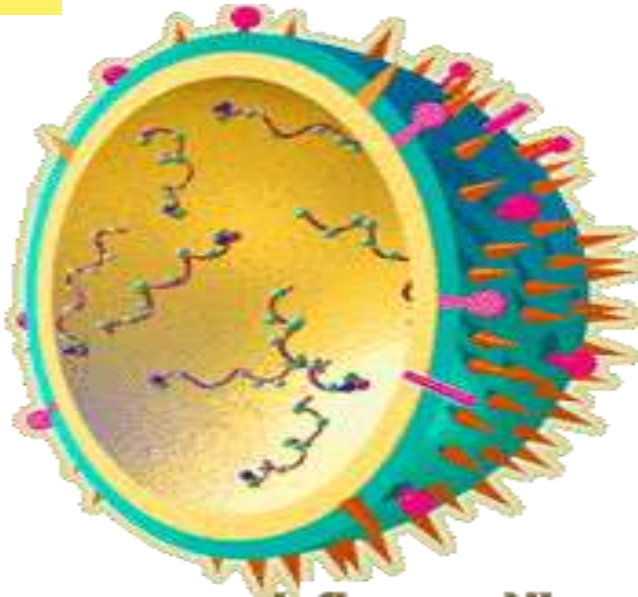


Ppt no. 09

GENERAL PROPERTIES OF VIRUSES



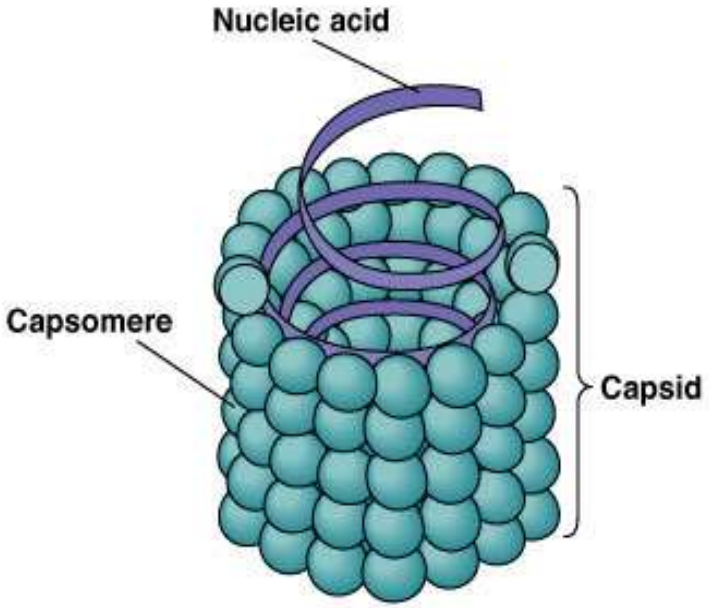
- Non living structures
- Non-cellular, Contain a protein coat called the capsid
- Have a nucleic acid core containing DNA or RNA (one or the other - not both)
- Capable of reproducing only when inside a HOST cell
- Some viruses are enclosed in an protective envelope. Some viruses may have spikes to help attach to the host cell, Most viruses infect only specific host cells



MORPHOLOGY

- Viruses come in a variety of shapes
- Some may be helical shape like the Ebola virus
- Some may be polyhedral shapes like the influenza virus
- Others have more complex shapes like bacteriophages

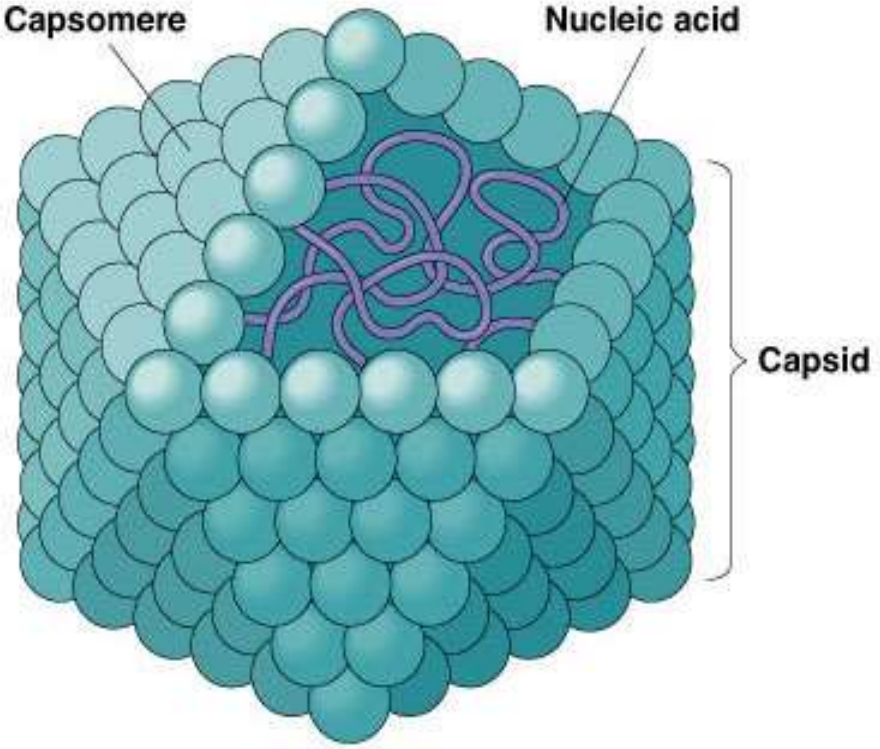
Types of Viruses: Helical Viruses



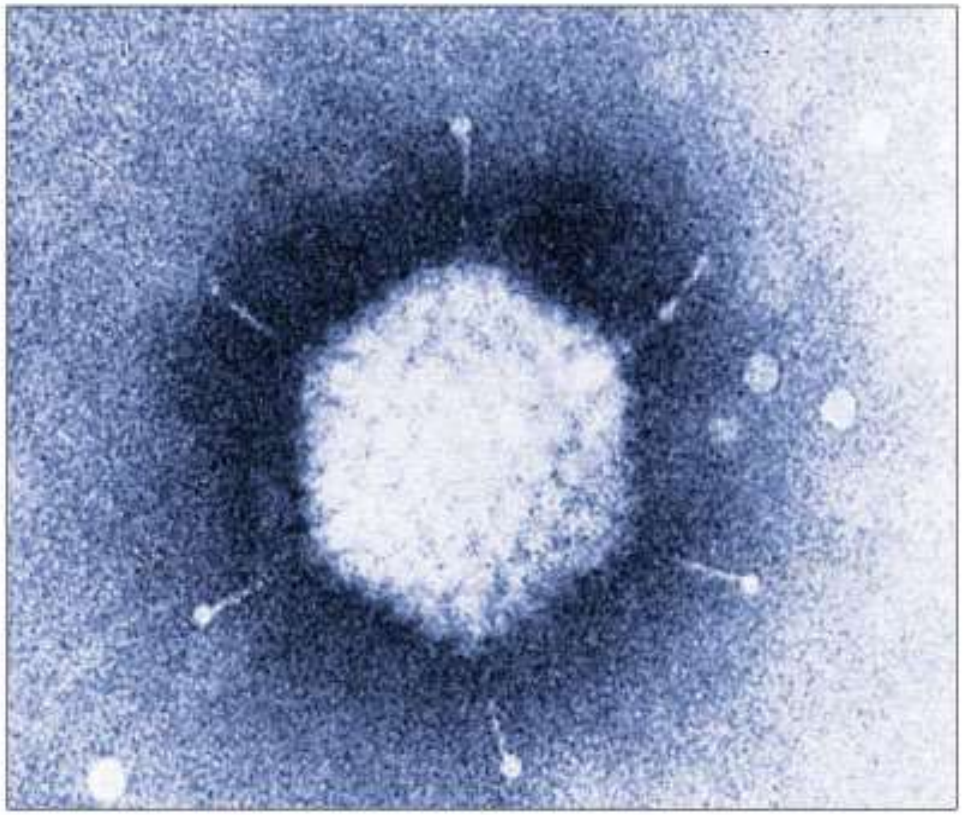
(a) A helical virus

(b) Ebola virus

Polyhedral Viruses

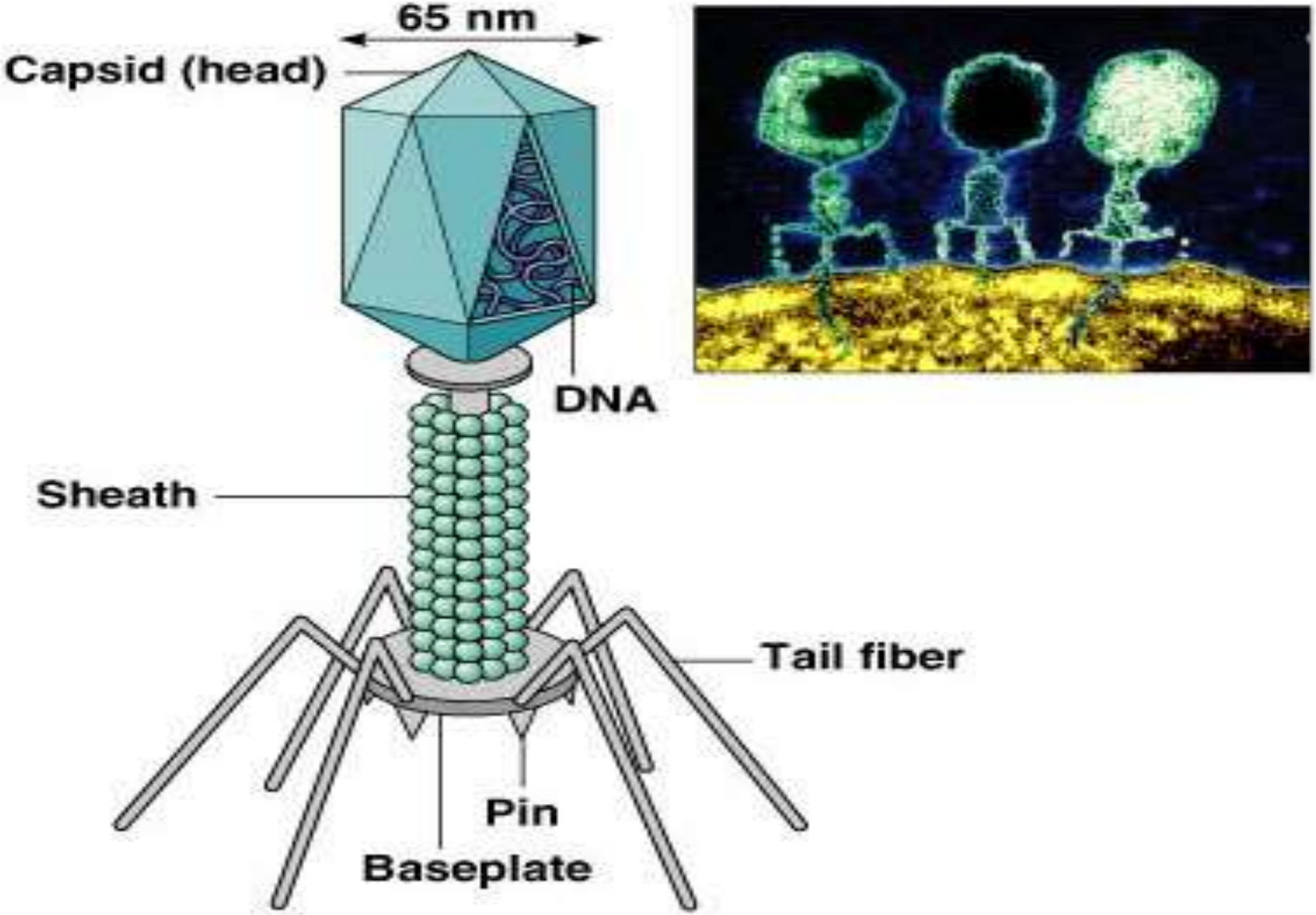


(a) A polyhedral virus



(b) A *Mastadenovirus*

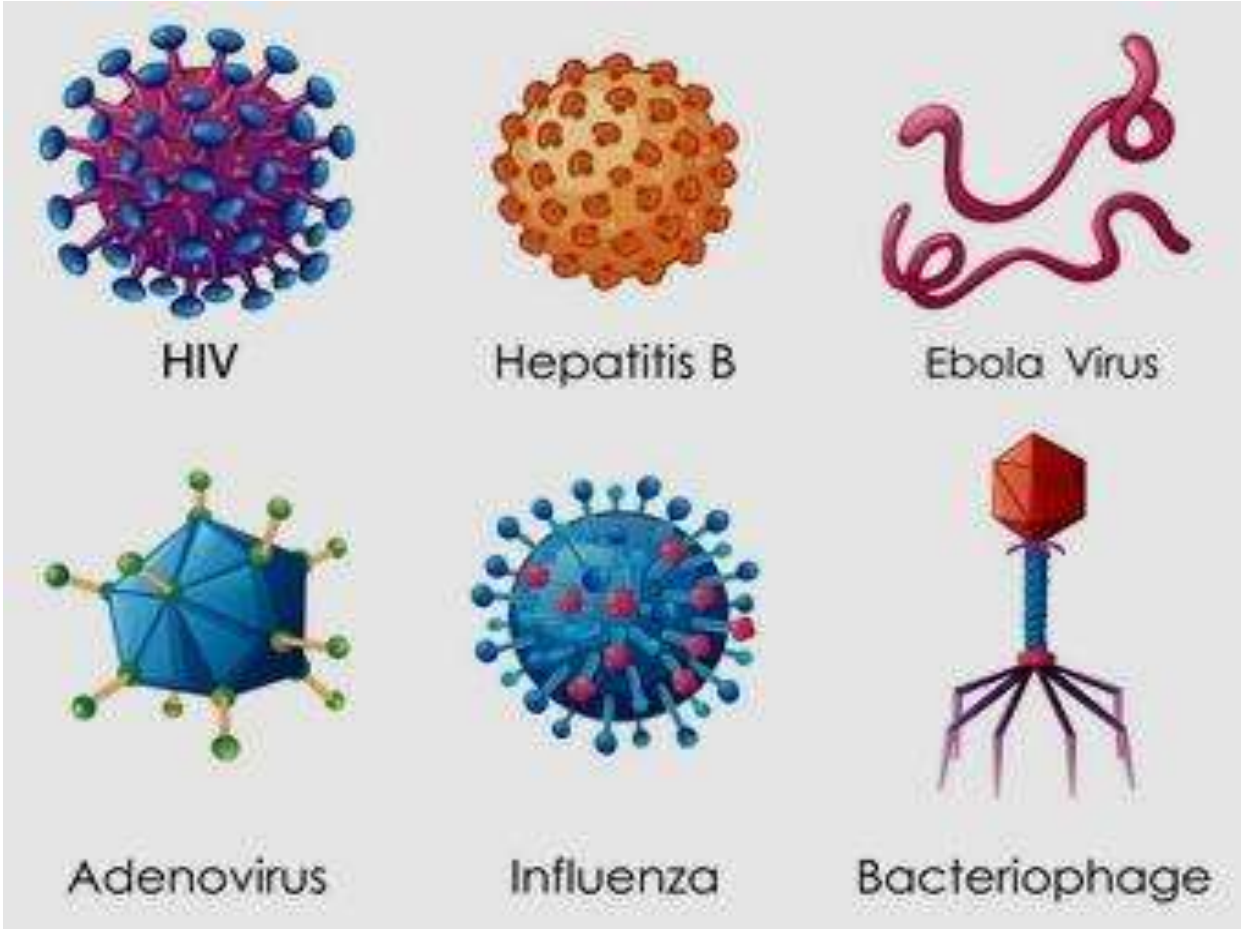
Complex Viruses



(a) A T-even bacteriophage

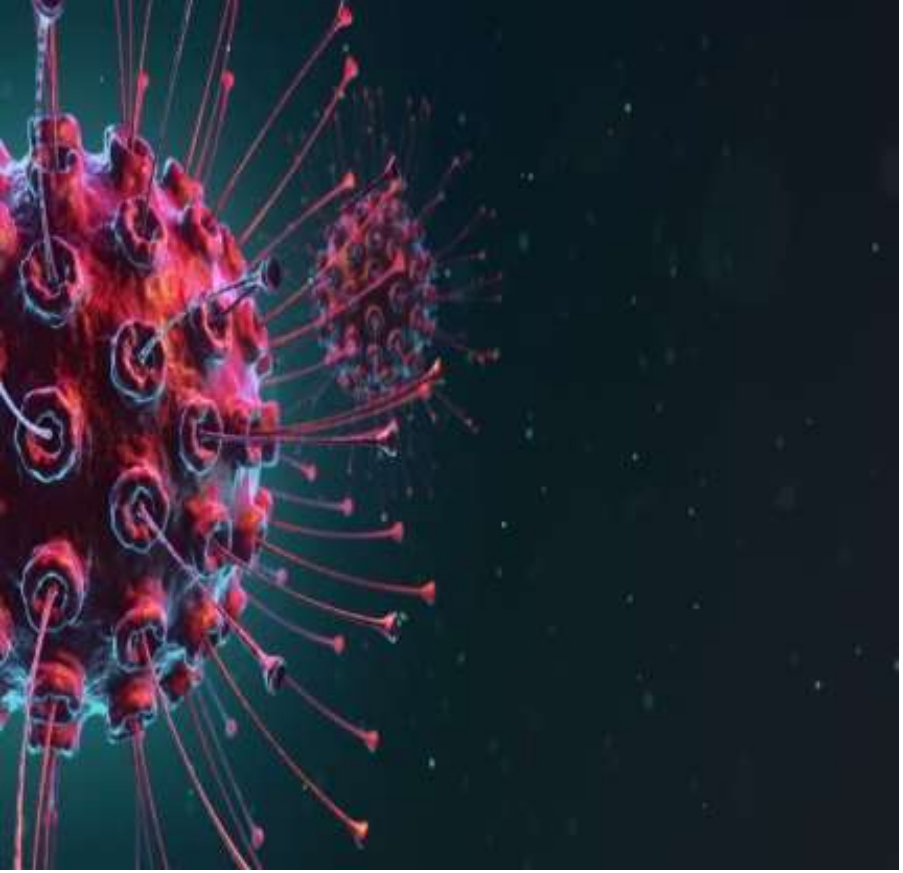
COMPLEX VIRUSES

- Additionally contain:
  - Lipid
  - Polysaccharides
  - Trace elements



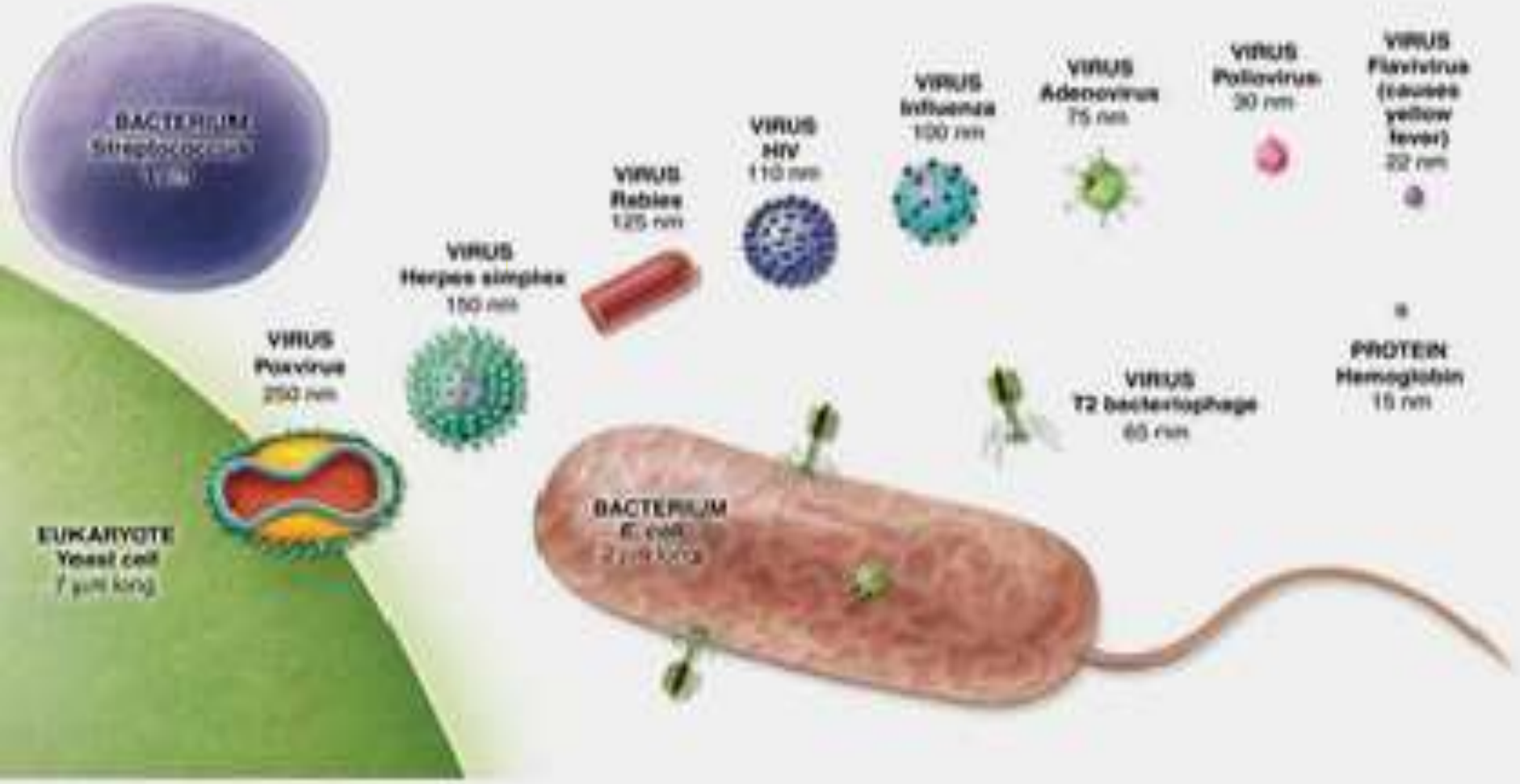
Ppt no. 10

Viral size

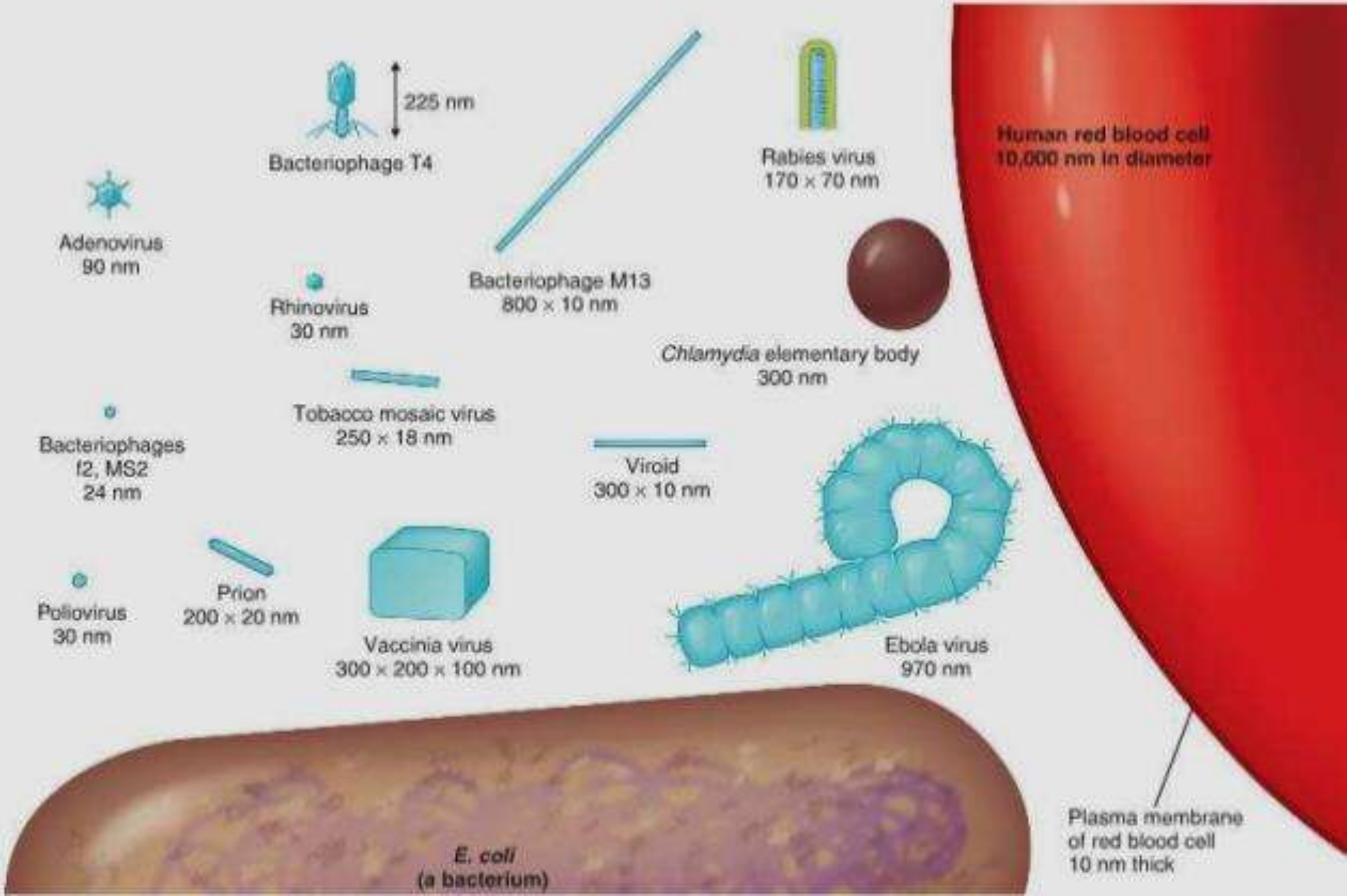


- Determined by electron microscopy
- Vary in size
- Most are smaller than bacteria
- Some are as the same size of bacteria (vaccinia)

Viruses vary in size, as well as in shape



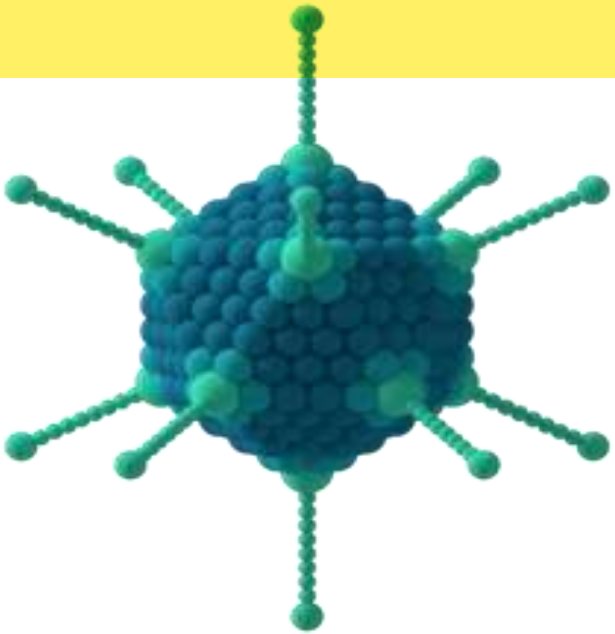
Size of Viruses



Ppt no. 11

FUNCTIONAL CHEMISTRY OF VIRUSES

CAPSID



Outer protein coat

Infectious component of virion

Infectivity depends on break down of protein capsid (Capsomeres)

Mostly host specific

SPECIFICITY

Most viruses infect specific host cells ie are host specific.

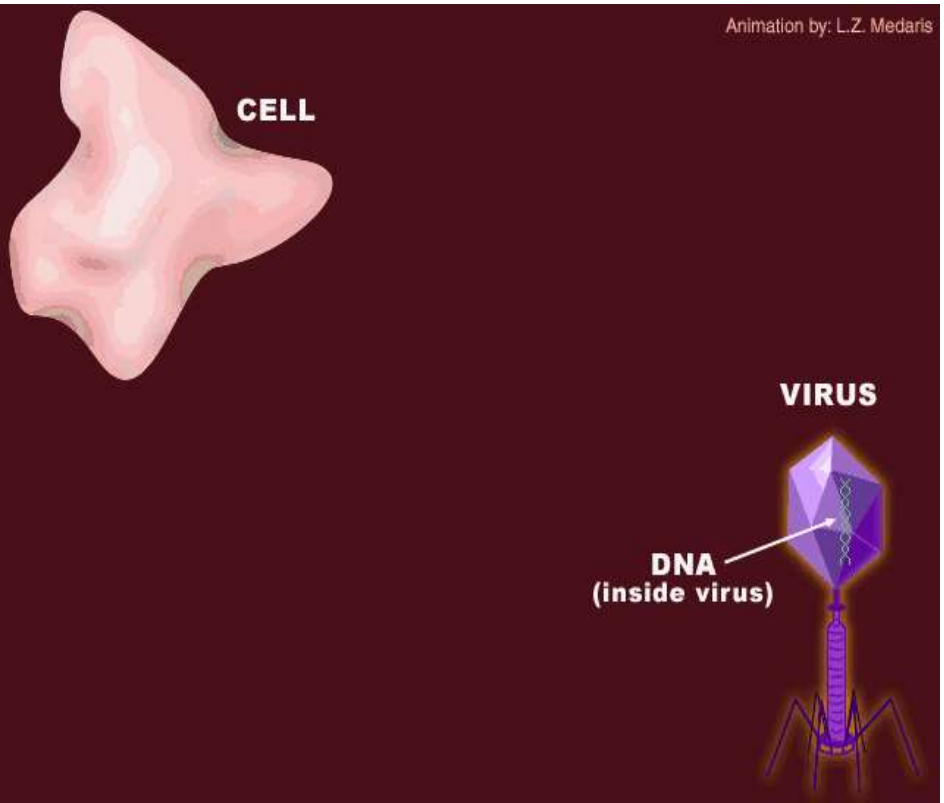
Host specificity is due to:

specific attachment sites on the host cells called receptors

Receptor sites for bacteriophage are found in bacterial cell walls or fimbrae or flagella

Animal cell membranes contain receptors for animal viruses

availability of cellular factors required for viral multiplication in the host cells



INFECTIVITY

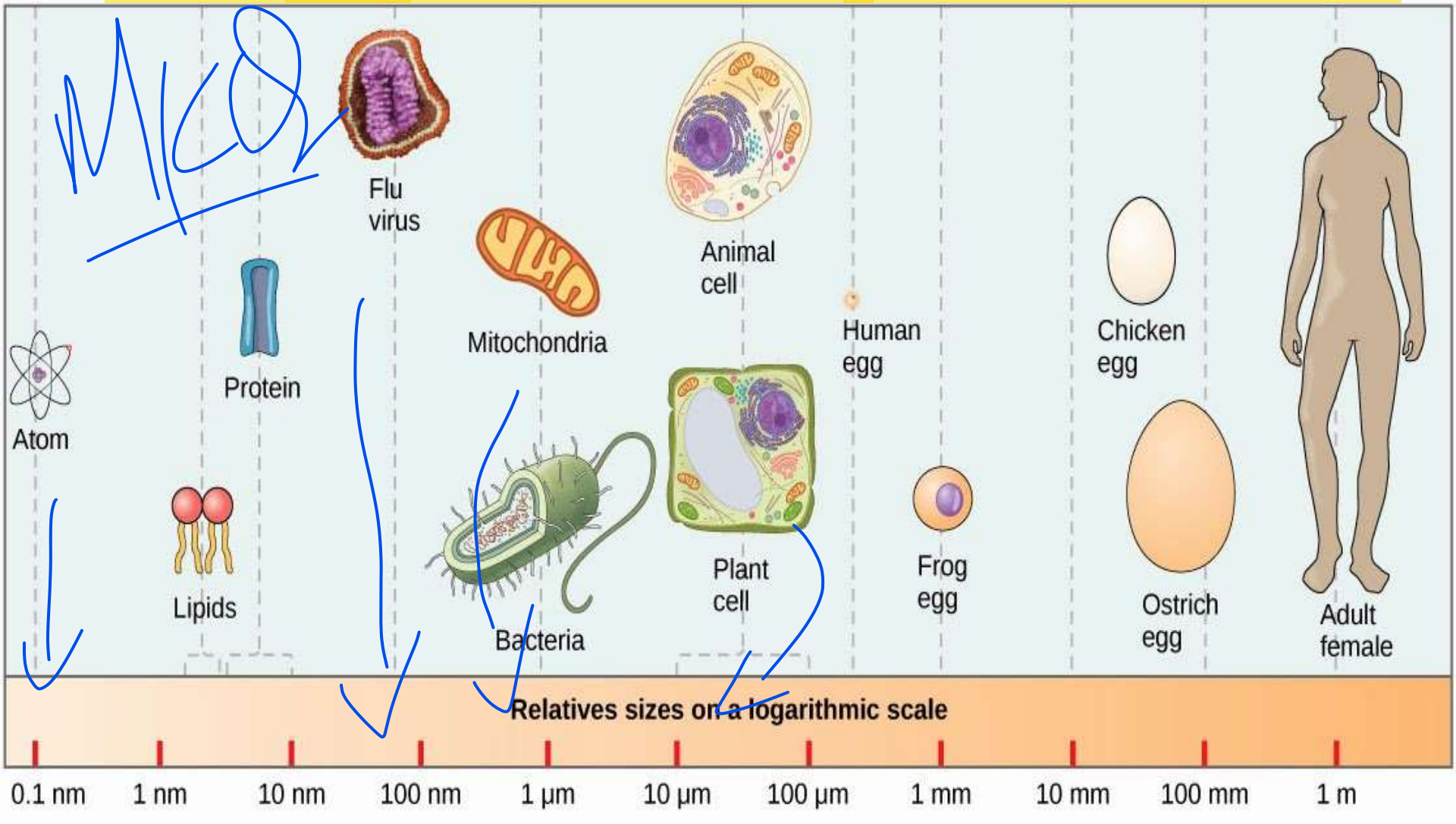
- For infectivity in host cells:

1. Viral nucleic acid
2. Viral structural sub components (spikes or glycoproteins)
3. Enzymes induced in host cells (polymerase enzymes)

Ppt no. 12

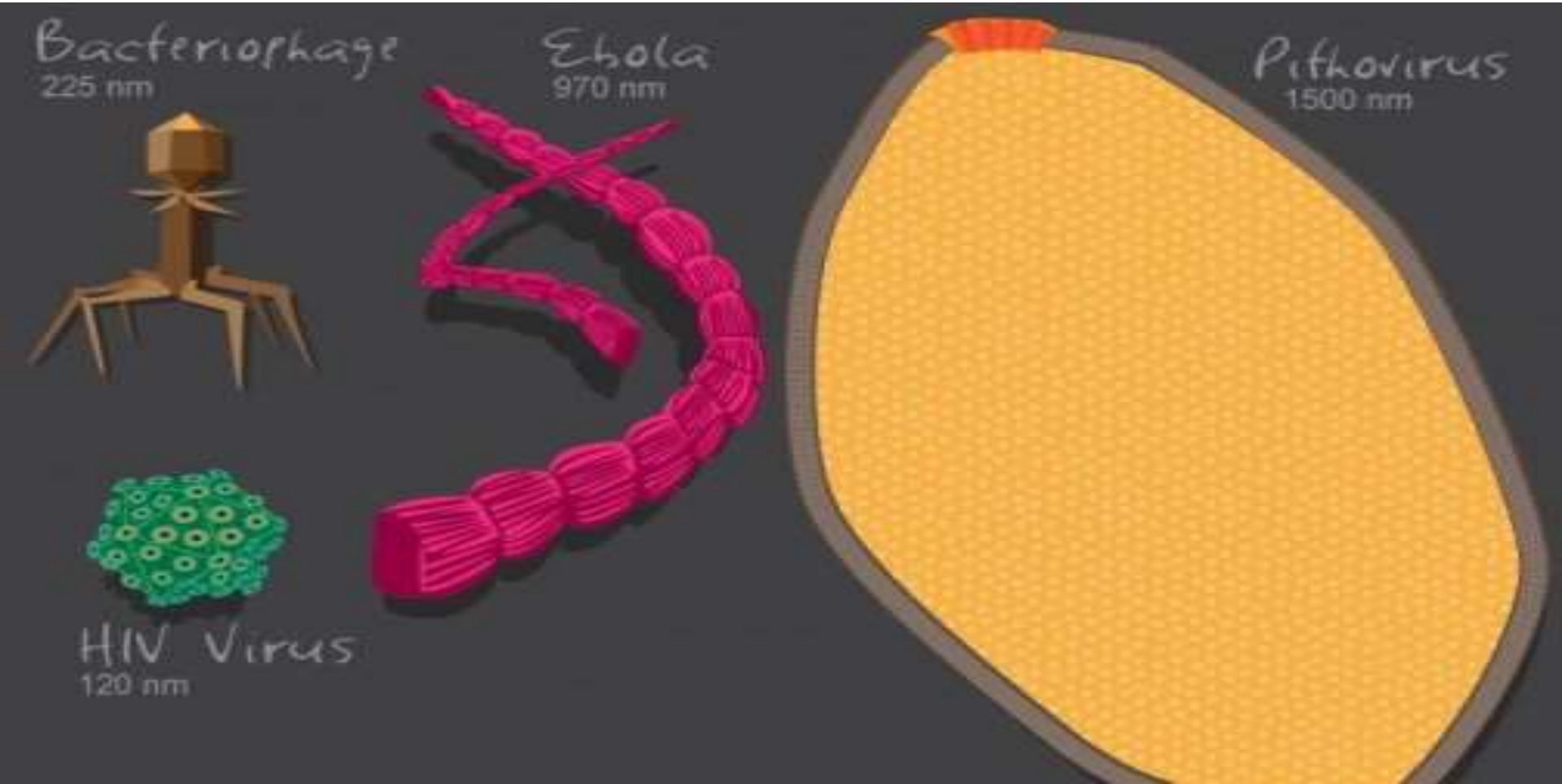
VISUALIZATION OF VIRAL PARTICLES

Size Small size Not visible through simple microscope



Variation

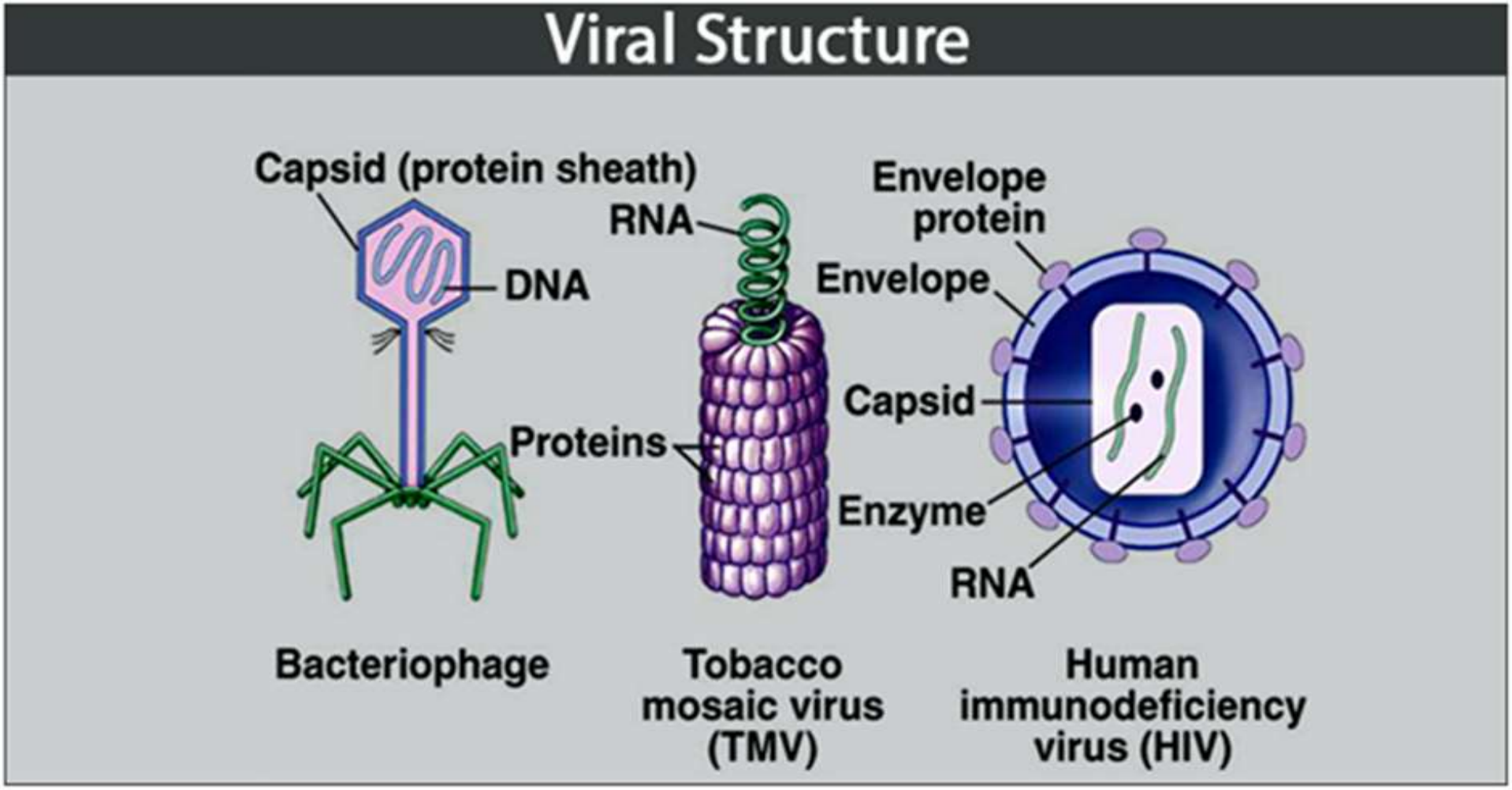
- Shape greatly varies
- Diversity
- Effects pathogenicity



General morphology

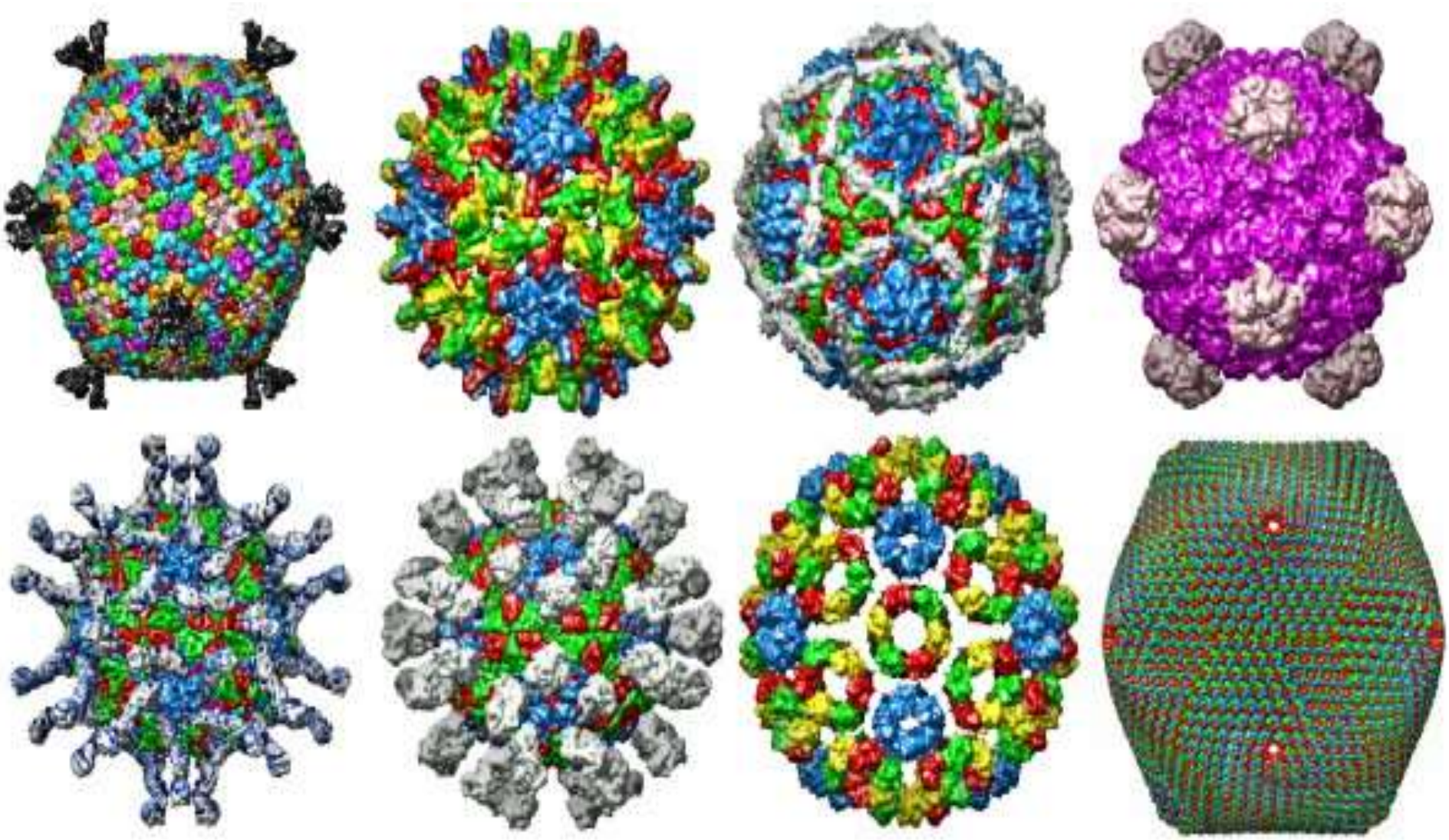
1. Icosahedral
2. Helical
3. Complex

General morphology

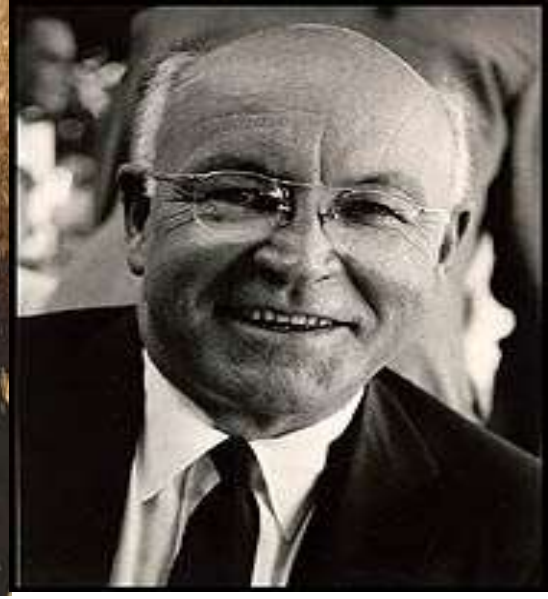


Ppt no. 13

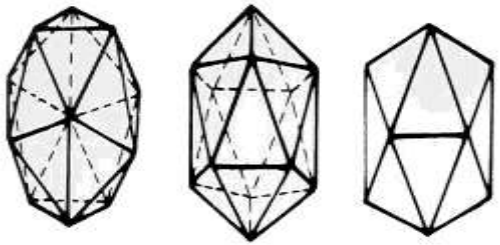
BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES



Wendell M. Stanley
1946 Nobel Prize in Chemistry
Crystallized Tobacco Mosaic Virus and Systematically Investigated its Biochemistry

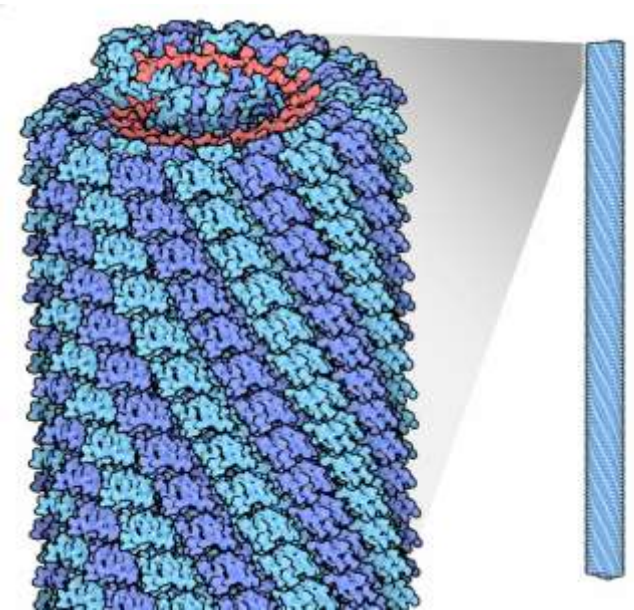


Symmetry of Viral Capsids

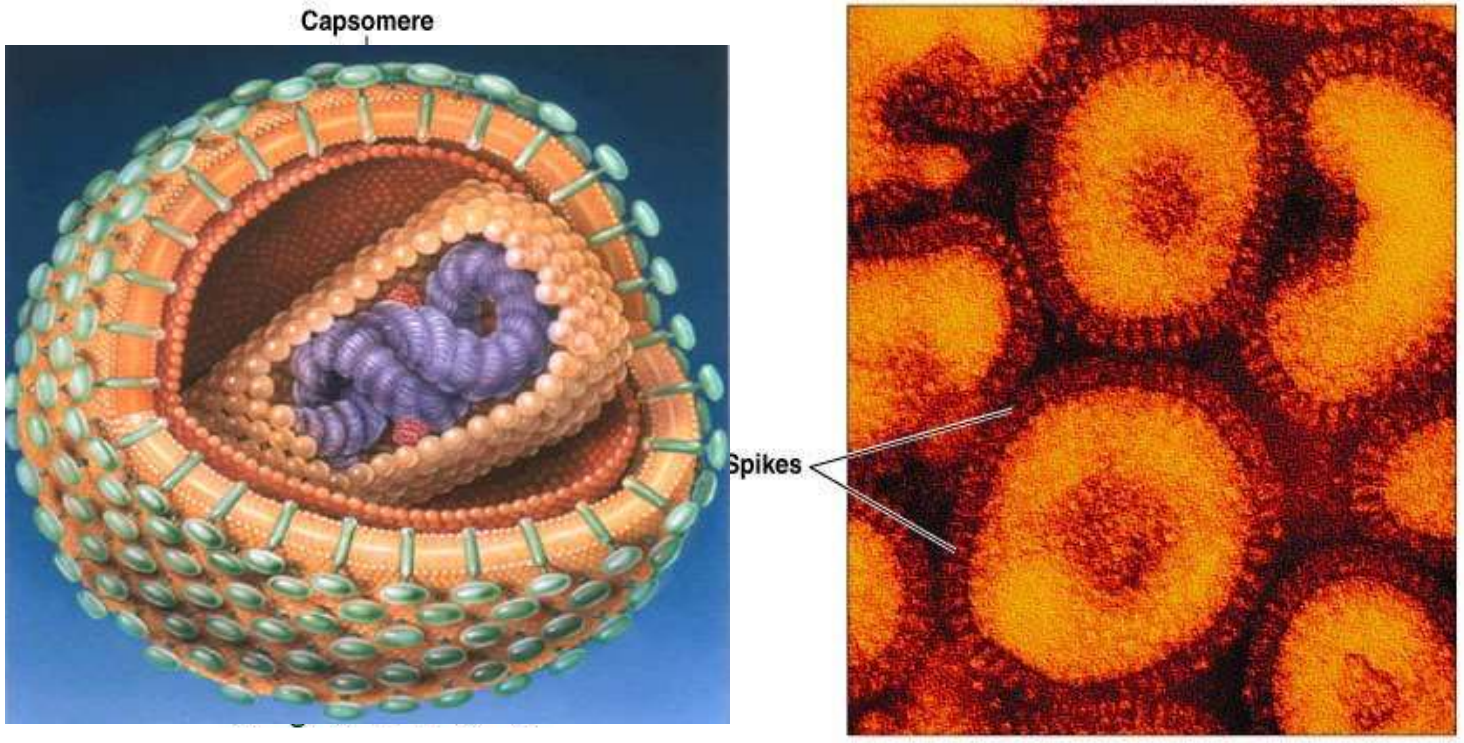


Icosahedral

Helical



ENVELOPED VIRUSES



(a) An enveloped helical virus

(b) *Influenzavirus*

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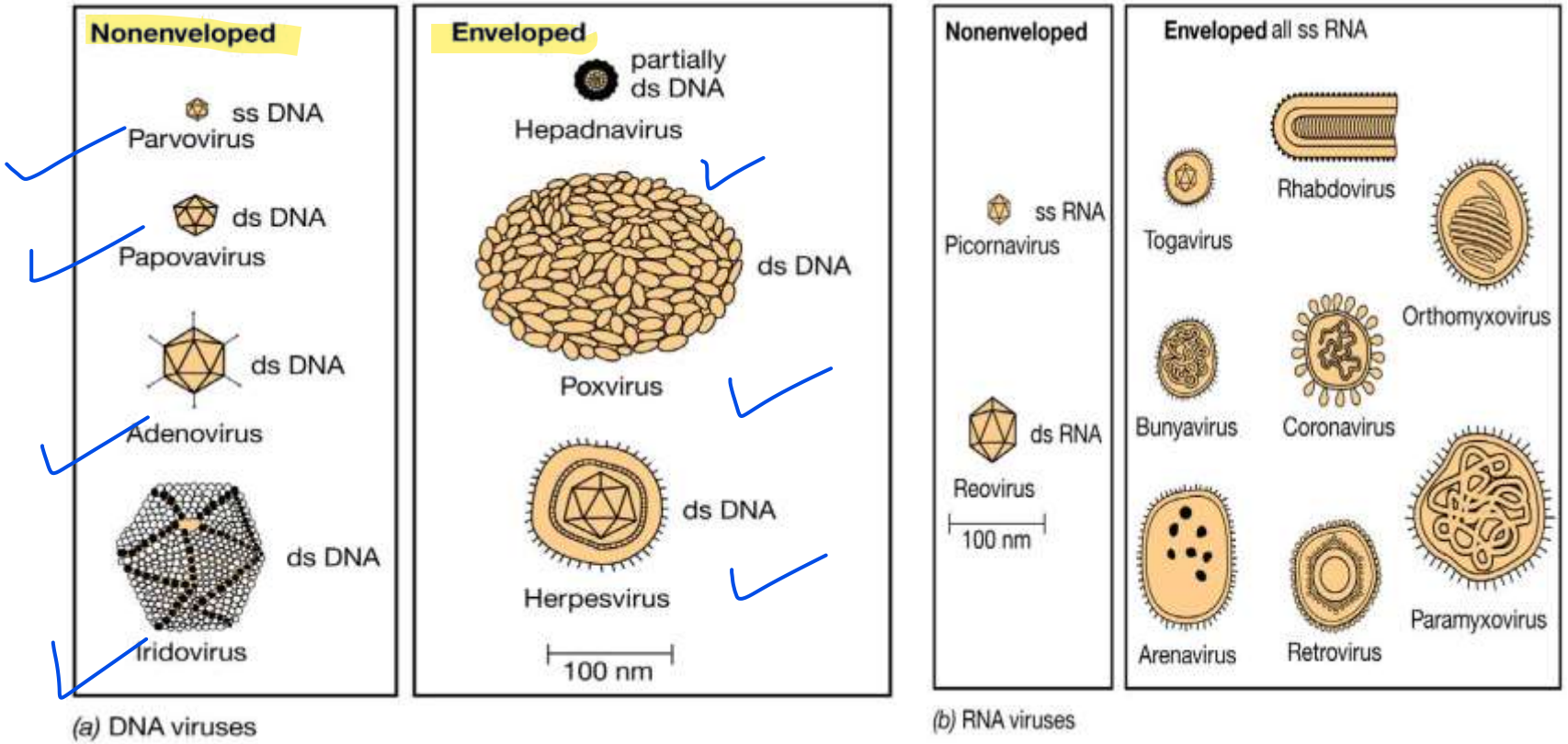
SURVEY OF HUMAN VIRUSES



Ppt no. 14

Living characters in host cells

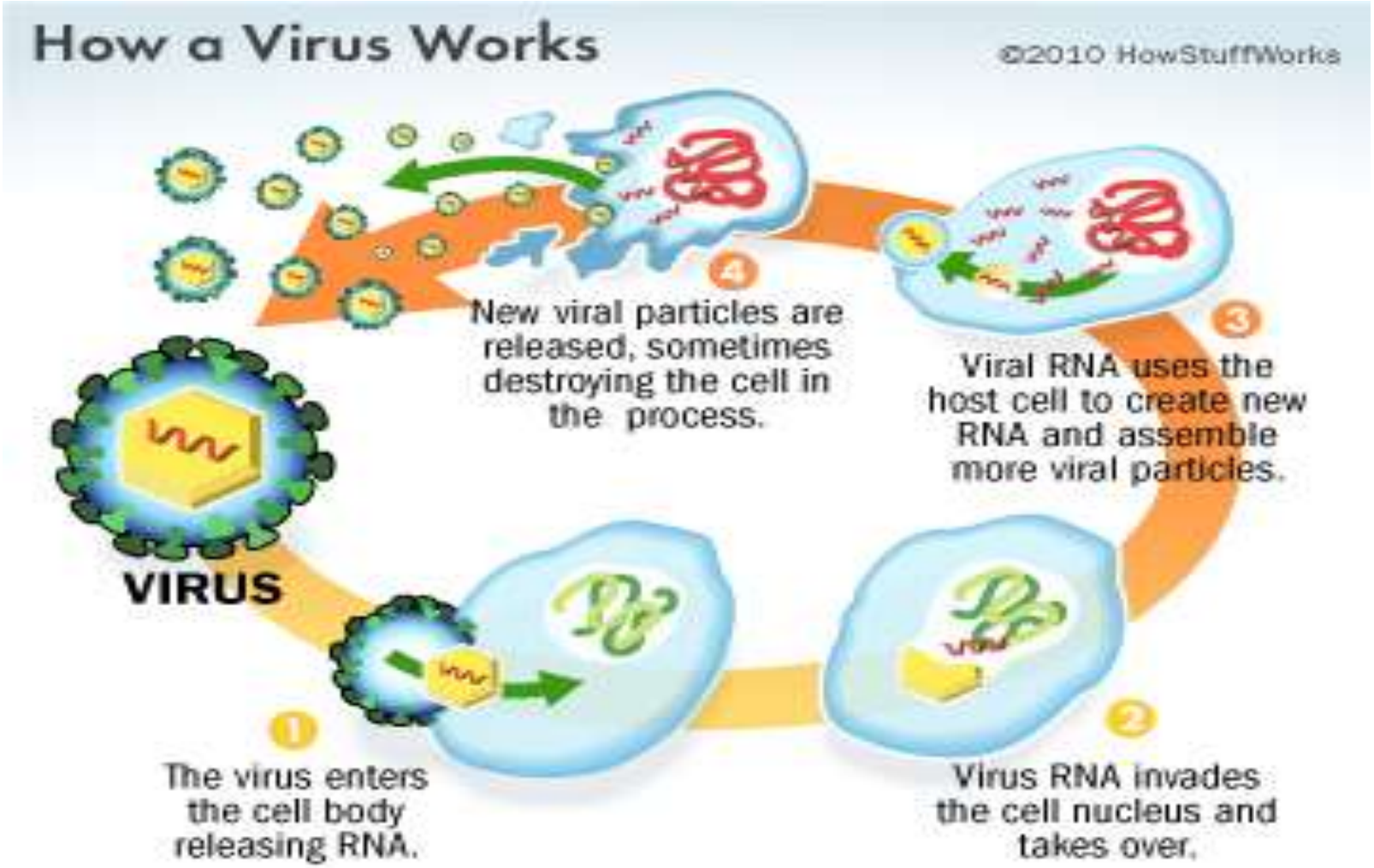
Molecular hijackers

Biologically inert

Dependent on host cells for replication and protein synthesis

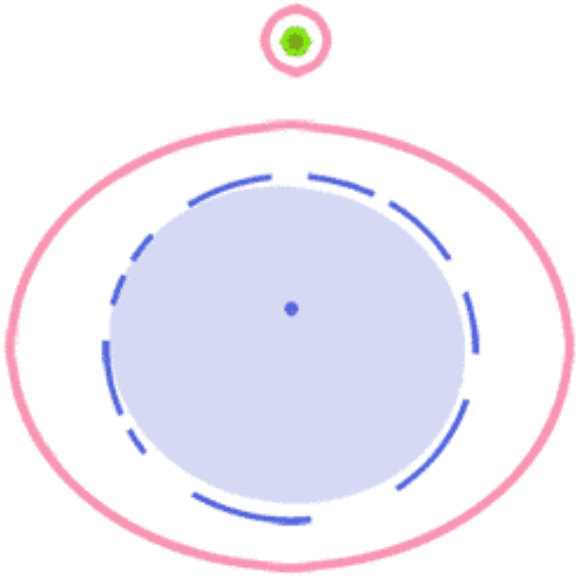
Lack in their own enzymes or organelles etc

Sometimes alter the host genome



Replication

- Using host machinery
- Force replication in host
- Increase in number
- Leave the cell simply or either by bursting the host cell



Effects on host

- None
- Genetic alterations
- Physical damage
- Oncogenesis

Shelter

- Take shelter in specific host cells
- Take nourishment
- Complete lifecycles
- Continuity of viral life

Ppt no. 15

HOST RANGE

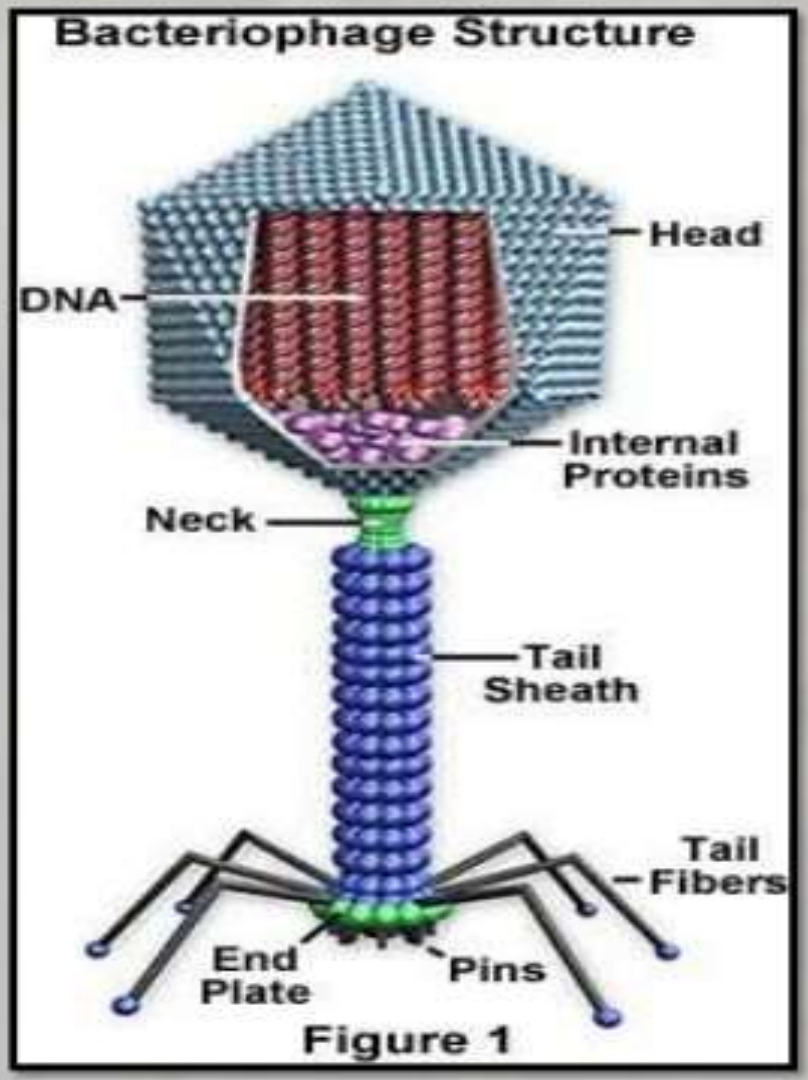
What is a host range?

- The spectrum of host cells that the virus can infect.

It can be narrow (infecting only one or a few species) or broad (infecting many species).

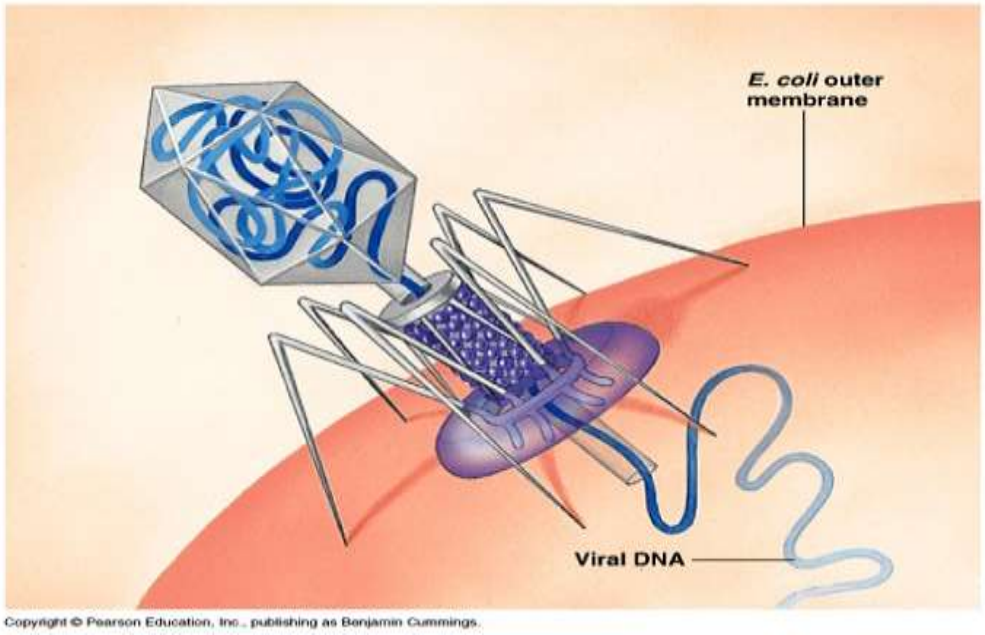
Host Range

1. Host range refers to the spectrum of host cells in which a virus can multiply.
2. Most viruses infect only specific types of cells in one host species.
3. Host range is determined by the specific attachment site on the host cell's surface and the availability of host cellular factors.
4. Viruses that infect bacteria are called bacteriophages, or phages.



Viral host range

- Plants
- Vertebrates
- Invertebrates
- Protists
- Bacteria
- Fungi



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- Most viruses are host specific
- Bind to specific receptors on cells
- Animal viruses: receptors on plasma membrane
- Form hydrogen bond with viral surface

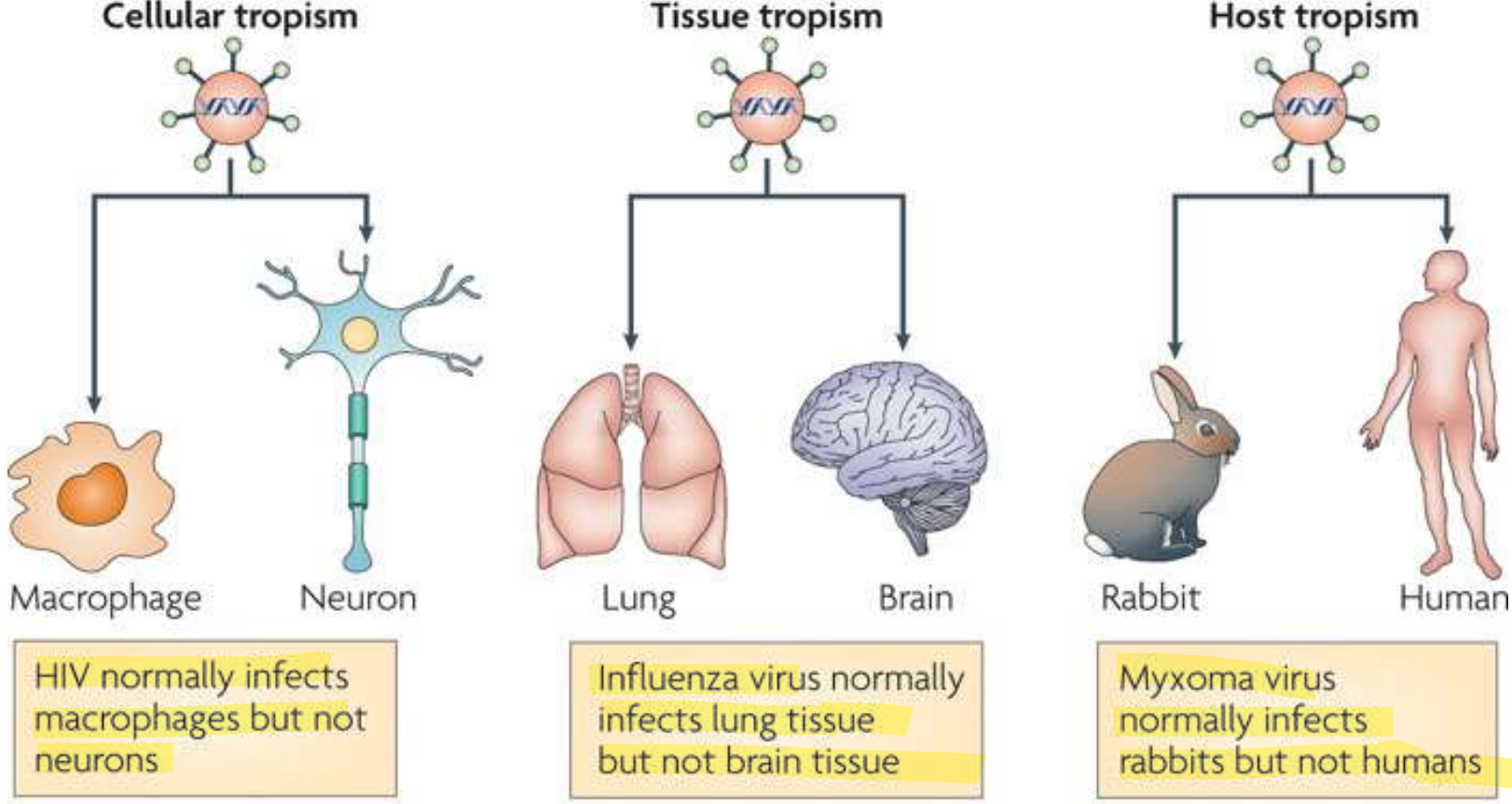
Ppt no. 16

Viral specificity

- All kingdoms can be infected by viruses
- Viruses that infect one kingdom are unable to infect another
- Viruses might infect closely related organisms
- Usually species specific

Examples

VIRUS	HOST
Tobacco Mosaic Virus	Tobacco plant
Rabies	Mammals
Swine flu	Pigs and humans
Foot and Mouth disease	Hooved animals
Bird Flu	Birds and human
Myxoma virus	Rabbits



Nature Reviews | Immunology

Ppt no. 17

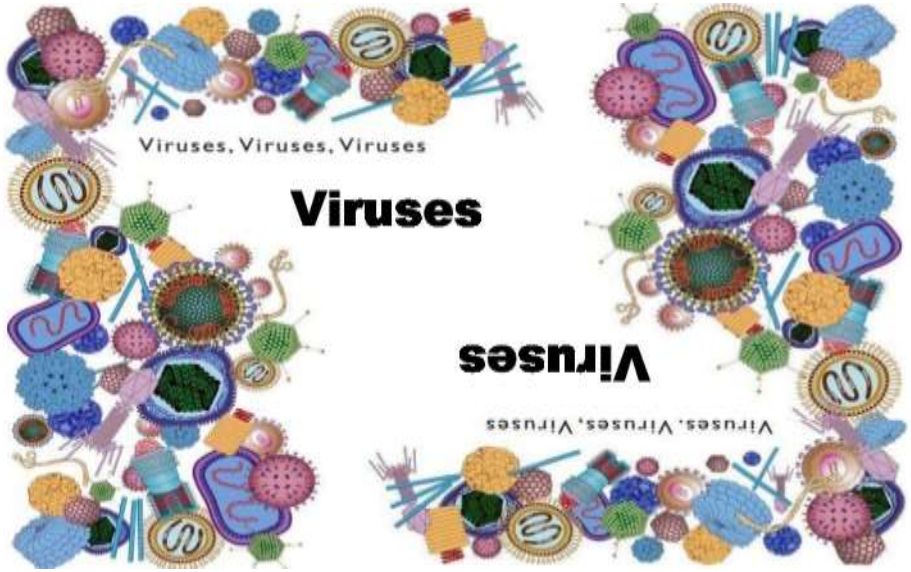
TAXONOMY I

WHAT IS TAXONOMY?

- Naming of something

OR

- Scheme of classification and nomenclature of organisms
- Viral taxonomy is the classification of viruses based on viral nature



BASED ON PHENOTYPE

- Morphology
- Structure of capsid
- Presence of envelope

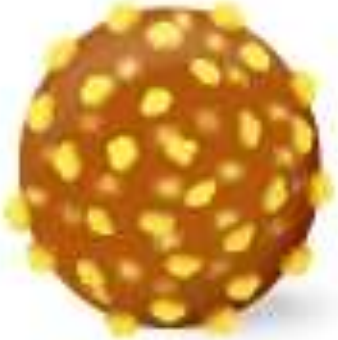
SCHEMES FOR CLASSIFICATION

- International Committee on Taxonomy of Viruses (ICTV) 1966
- Baltimore system of classification 1971
  - Type of nucleic acid
  - Mode of replication
  - Host organism
  - Diseases caused

VIRAL SHAPES

Polyhedral
(Adenovirus)

Spherical
(Influenza)



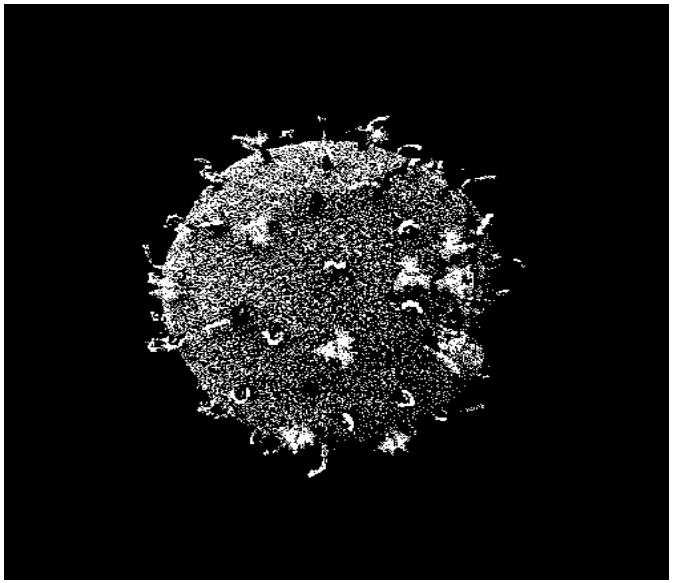
Helical
(Tobacco mosaic virus)

Complex
(Bacteriophage)

ICTV CLASSIFICATION

•A class where members have several properties in common e.g.

- Host
- Ecological niche
- Pathogenicity
- Mode of transmission
- Habitat



LEVEL OF CLASSIFICATION

ORDER	FAMILY	SUBFAMILY	GENUS	SPECIES
• -virales	• -viridae	• -virinae	• -virus	

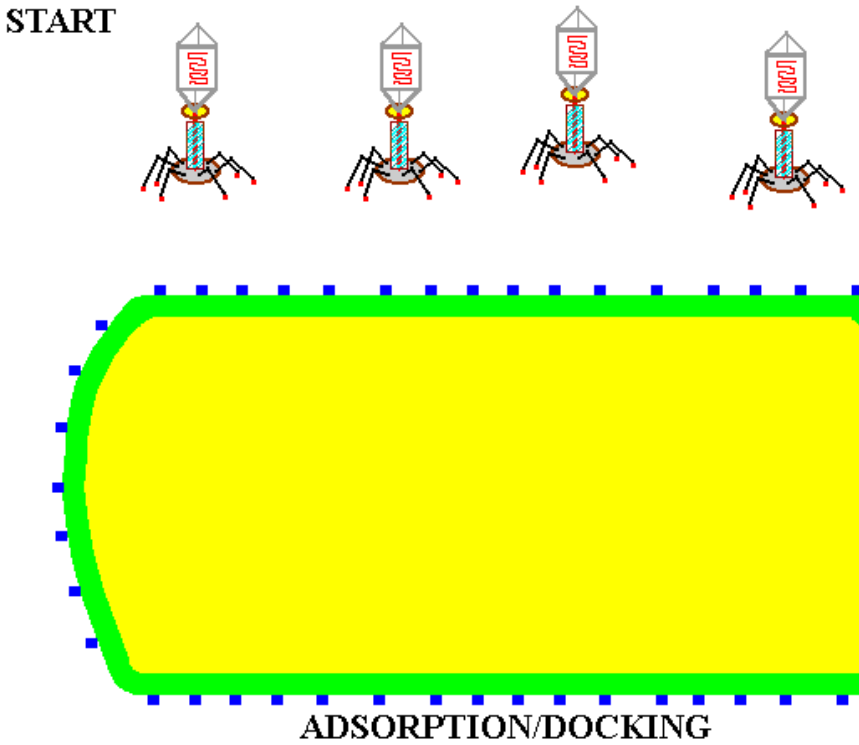
Ppt no. 18

TAXONOMY-II

BALTIMORE CLASSIFICATION

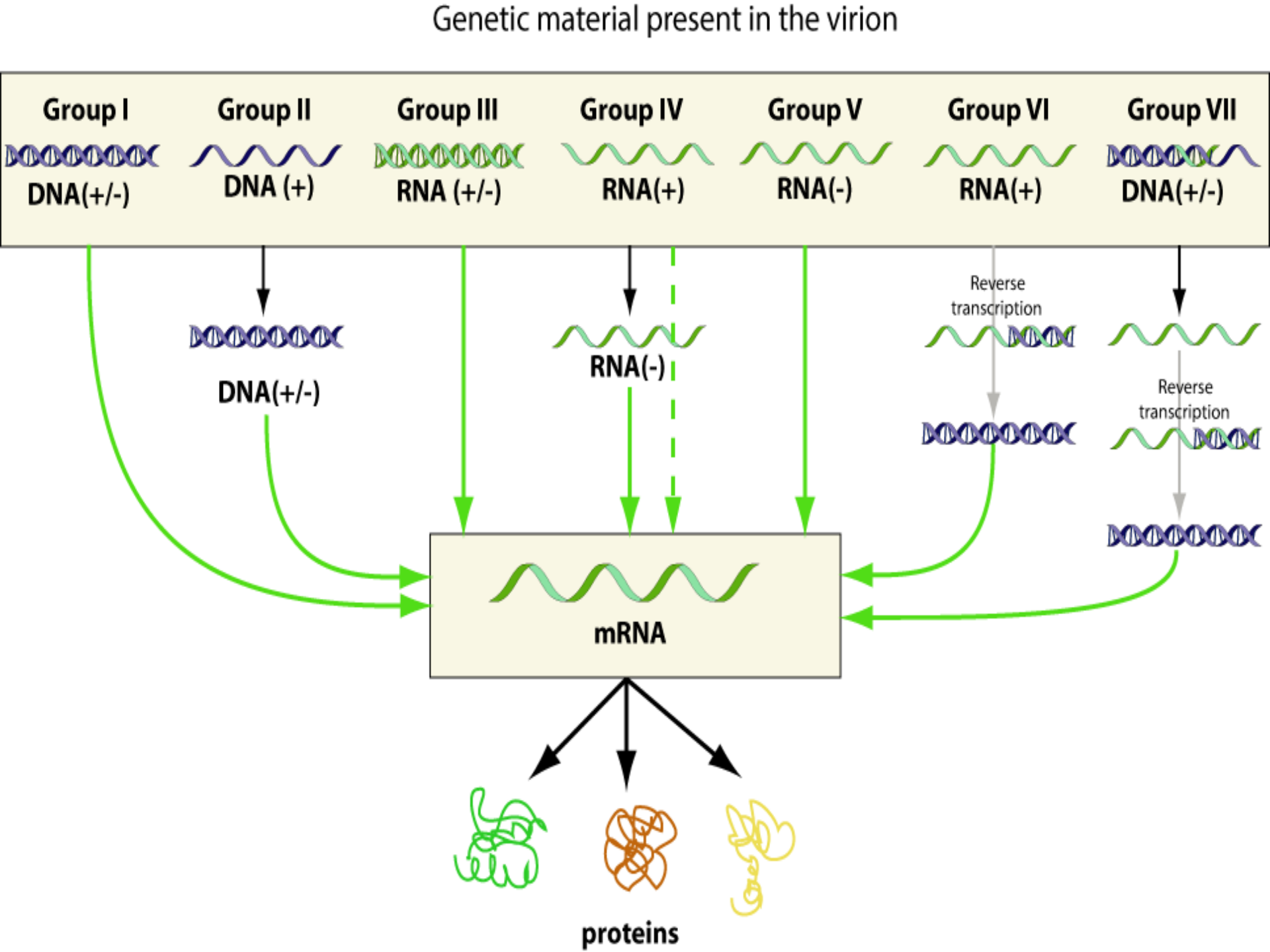
David Baltimore

1971



DEPENDS ON

1. Expression of nucleic acid
2. Stranded
3. Segmented/ Sense
4. Method of Replication


Genetic material present in the virion
Group I
DNA(+/-)
Group II
DNA (+)
Group III
RNA (+/-)
Group IV
RNA(+)
Group V
RNA(-)
Group VI
RNA(+)
Group VII
DNA(+/-)
DNA(+/-)
RNA(-)
Reverse transcription
Reverse transcription
mRNA
proteins

7 CLASSES:

GROUP I: double stranded DNA viruses (Adeno, Herpes)

GROUPII: single stranded DNA viruses (Parvo)

GROUP III: double stranded RNA viruses (Reo)

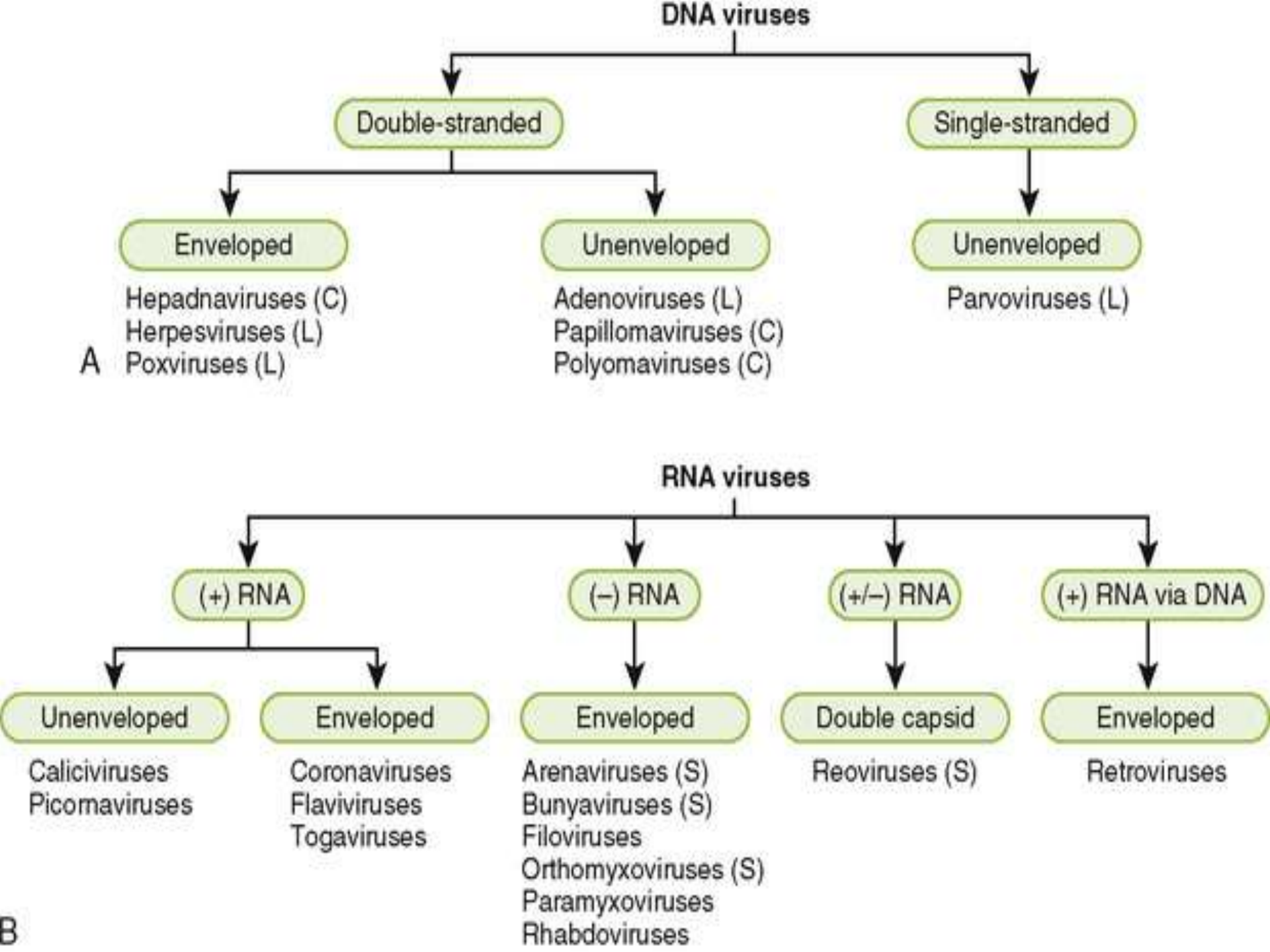
GROUP IV: positive sense single stranded RNA viruses (Polio,Picorna)

GROUP V: negative sense single stranded RNA viruses (ortho)

GROUP VI: Single stranded RNA viruses (Retero)

GROUP VII: Double stranded DNA viruses (Hepadna)

VIRAL CLASSIFICATION


DNA viruses
Double-stranded
Single-stranded
Enveloped
Unenveloped
Unenveloped
Hepadnaviruses (C)
Herpesviruses (L)
Poxviruses (L)
Adenoviruses (L)
Papillomaviruses (C)
Polyomaviruses (C)
Parvoviruses (L)
A
RNA viruses
(+) RNA
(–) RNA
(+/–) RNA
(+) RNA via DNA
Unenveloped
Enveloped
Enveloped
Double capsid
Enveloped
Caliciviruses
Picornaviruses
Coronaviruses
Flaviviruses
Togaviruses
Arenaviruses (S)
Bunyaviruses (S)
Filoviruses
Orthomyxoviruses (S)
Paramyxoviruses
Rhabdoviruses
Reoviruses (S)
Retroviruses
B

Ppt no. 19

ARTIFICIAL TRANSMISSION OF VIRUSES IN PLANTS

BARRIERS

External surface

Structural differentiation of host

Internal barriers

Direct transmission only through wounds or poorly protected tissues

MECHANICAL TRANSMISSION

Few plant viruses

TMV and WMV by root inoculation

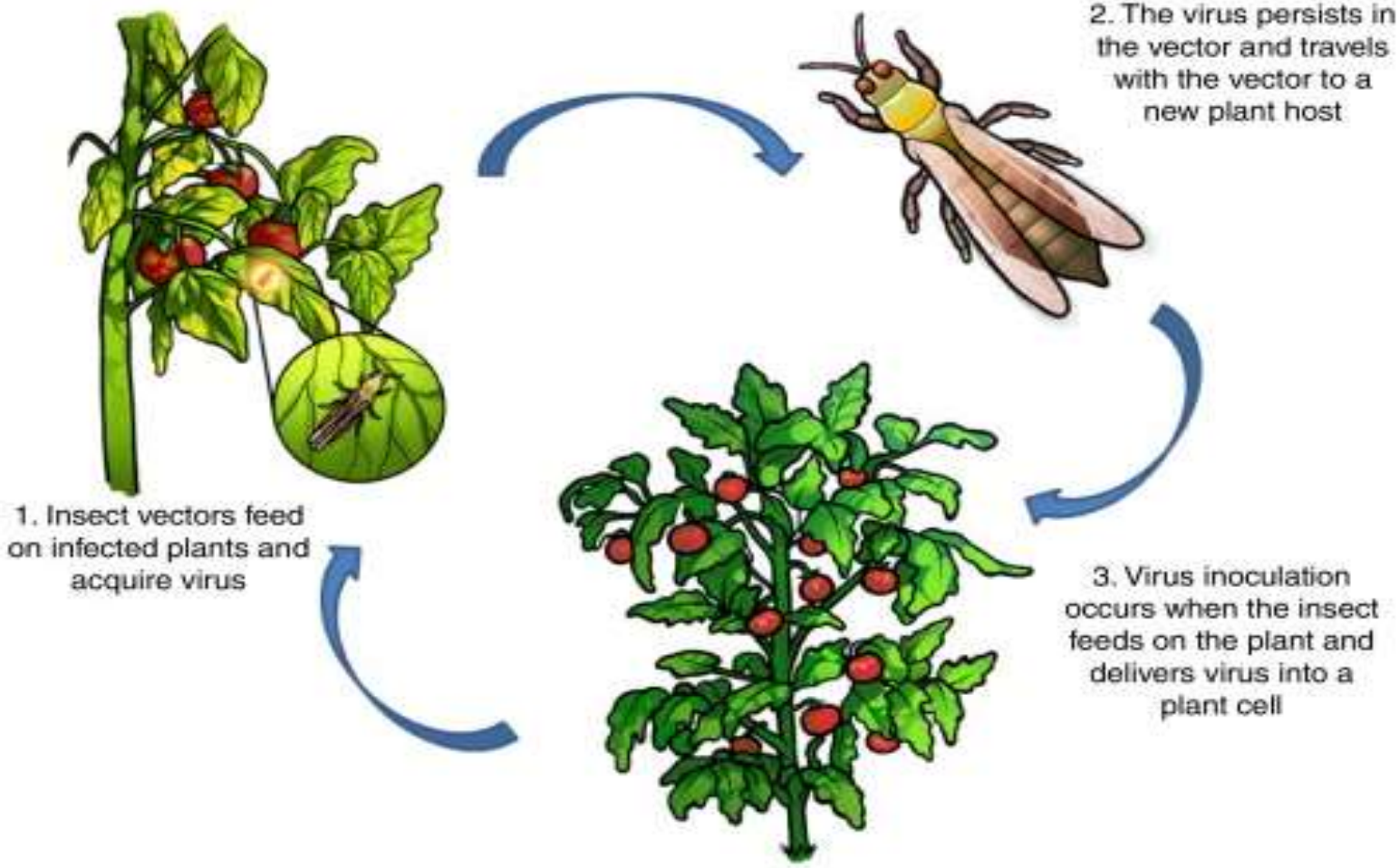
Potato viruses by infected tuber plantation

- **SEED TRANSMISSION**
- Rare
- But exists

DISEASES

- Rice dwarf virus
- Maize streak
- Tomato spoiled wilt virus
- Pigeon pea sterility virus
- VECTOR TRANSMISSION
- Most common
- Grass hopper
- Aphids
- Leaf hoppers
- White flies
- Mites

- Beetles



ARTIFICIAL TRANSMISSION OF VIRUSES IN ANIMALS

Ppt no. 20

REASON

- Viruses that cannot be cultivated on cell culture or embryonated eggs
- To experiment on animals rather human beings

ANIMAL INOCULATION

- Route: Subcutaneous, intracerbral, intraperitoneal, Intradermal or intraocular
- Virus multiplies in host and disease develops

ANIMALS USED

- Monkey
- Rabbit
- Guinea pig
- Rat
- Hamsters mice

GROWTH CONFIRMATION

- Pathogenesis
- Lesions
- Immune response
- Oncogenesis
- Death

ADVANTAGES

- Clinical manifestations are determined
- Primary isolation of viruses
- Used for study of immune response

Ppt no. 21

MICROSCOPY

MICROSCOPE

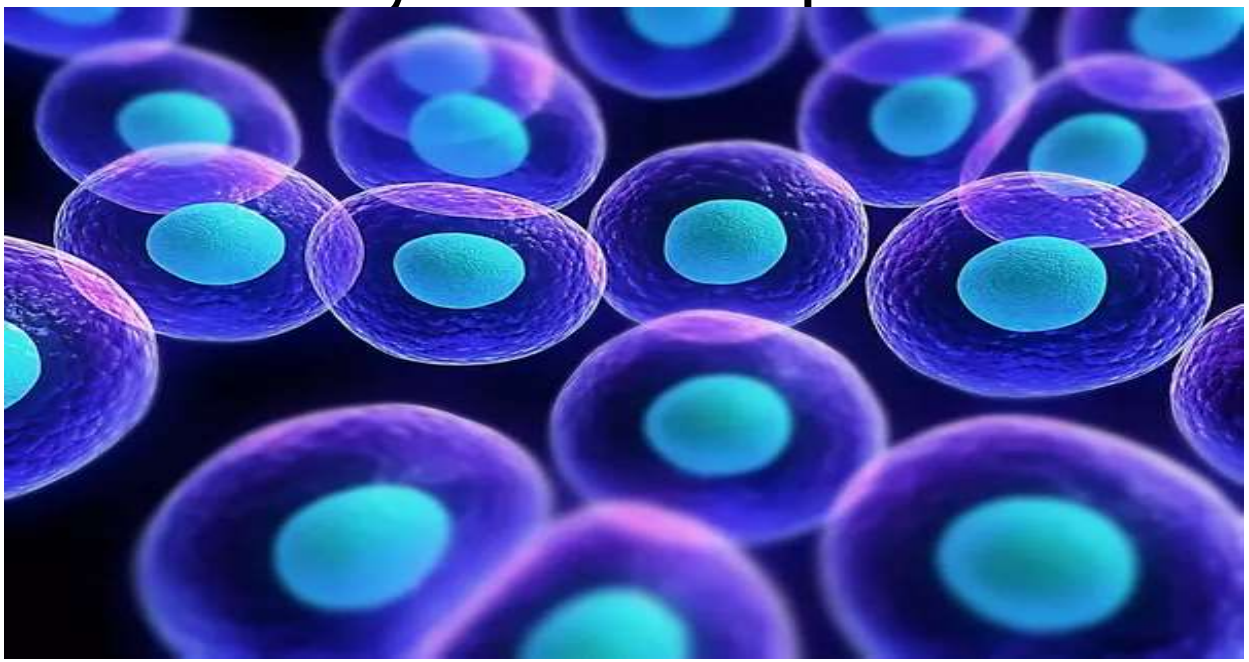
– An instrument used to see objects that are too small for the naked

MICROSCOPY

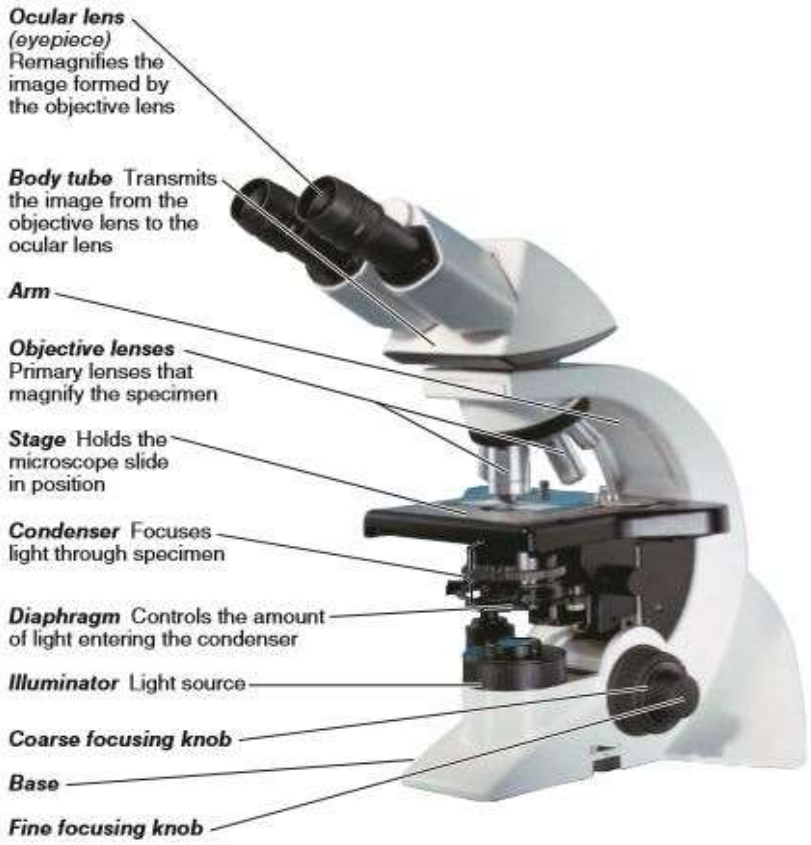
– The science of investigating small objects using such an instrument is called microscopy

MICROSCOPIC

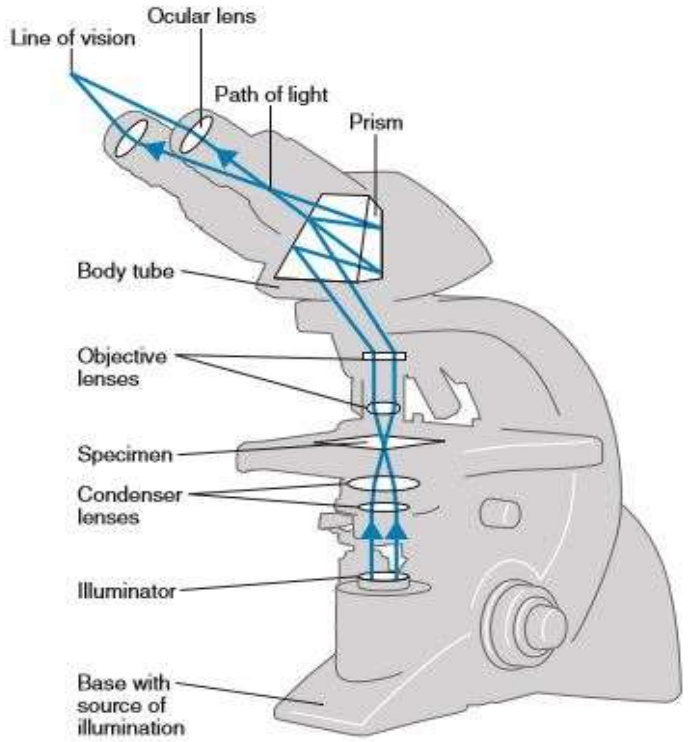
– Microscopic means invisible to the eye unless aided by a microscope



PARTS OF MICROSCOPE



(a) Principal parts and functions



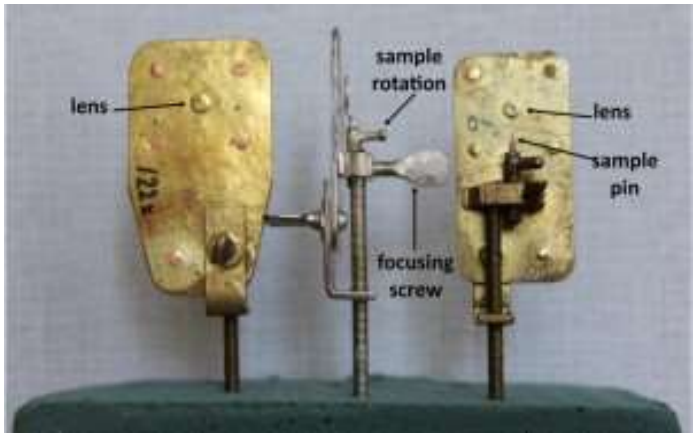
(b) The path of light (bottom to top)

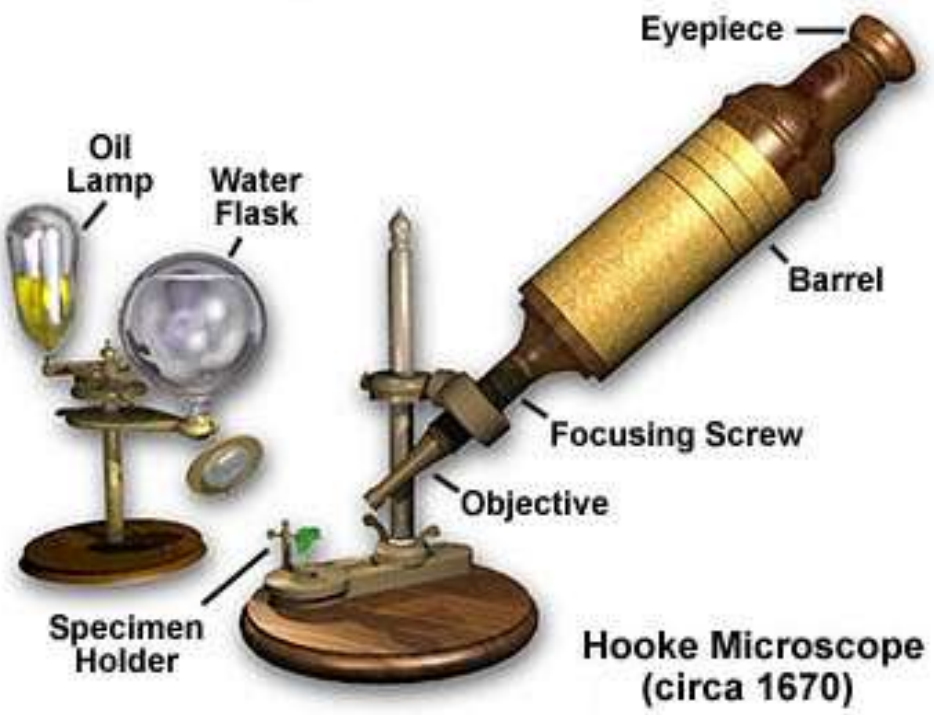
Ppt no. 22

HISTORY OF MICROSCOPY

FIRST MICROSCOPE

- First simple microscope by Leeuwenhoek
- 17th century





ROBERT HOOKE

- Robert Hooke in 1665 observed cork cells

COMPOUND MICROSCOPE

- Zaccharias Janssen made the first compound microscope

BETTER MICROSCOPE

- In 1830 a better microscope introduced by Joseph Lister

Microscopes Over the Ages

Leeuwenhoek Microscope (circa late 1600s)

British Microscope (circa 1865)

Hand-Held Microscope (circa early 1700s)

Winkel-ZEISS Dissecting Microscope (circa 1927)

ZEISS Primo Star (circa 2010)

Figure 1

Ppt no. 23

LIGHT MICROSCOPY

LIGHT MICROSCOPY

- Light microscopy refers to the use of any kind of microscope that uses visible light to observe specimens.

TYPES:

- Compound light microscopy
- Dark field Microscopy
- Phase-contrast Microscopy
- Differential interference contrast (DIC) Microscopy
- Fluorescence Microscopy
- Confocal Microscopy

COMPOUND LIGHT MICROSCOPY

- Light rays from an illuminator (the light source)
- Pass through a condenser, which has lenses that direct the light rays through the specimen
- From here, light rays pass into the objective lenses
- The image of the specimen is magnified again by the ocular lens (eyepiece)

MAGNIFICATION

- Objective lens X Eyepiece lens
- It is the ratio of the size of an object seen under microscope to the actual size observed with unaided eye.

RESOLUTION:

- refers to the ability of the lenses to distinguish two points that are a specified distance apart
- The resolving power of human eye is 0.25 mm
- The light microscope can separate dots that are 0.25µm apart.
- The electron microscope can separate dots that are 0.5nm apart.

Virus structure

In this lecture we will look at the structure of viruses

At the end of this lecture

Student must have fully understood:

Virus structure

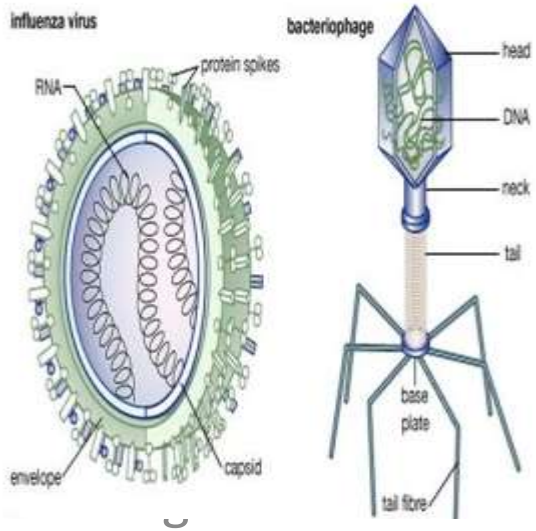
- Genetic materials
- Capsid
- Envelope

Virus structure

- Basic rules of virus architecture, structure, and assembly are the same for all families
- Some structures are much more complex than others, and require complex assembly and dissassembly

Nucleic acid

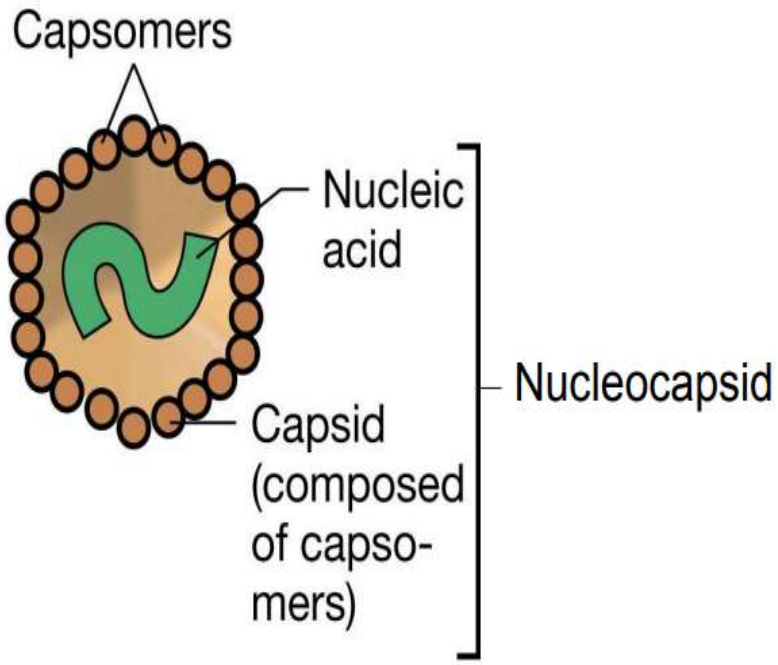
- A virus can have either DNA or RNA—but never
- Nucleic acid of a virus can be single-stranded or
- Can be linear or circular
- Influenza virus: the nucleic acid is in several sepa



Virus structure

Capsid

- The nucleic acid of a virus is protected by a protein coat called the capsid
- Each capsid is composed of protein subunits called capsomeres
- The arrangement of capsomeres is characteristic of a particular type of virus



Virus structure

Capsid

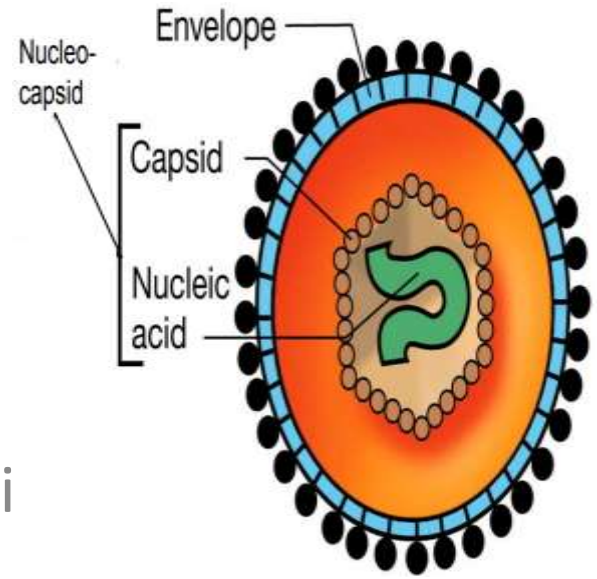
The capsid (coat) protein is the basic unit of structure; functions that may be fulfilled by the capsid protein are to:

- Protect viral nucleic acid
- Interact specifically with the viral nucleic acid for packaging
- Interact with vector for specific transmission
- Interact with host receptors for entry to cell
- Allow for release of nucleic acid upon entry into new cell
- Assist in processes of viral and/or host gene regulation

Virus structure

Envelope

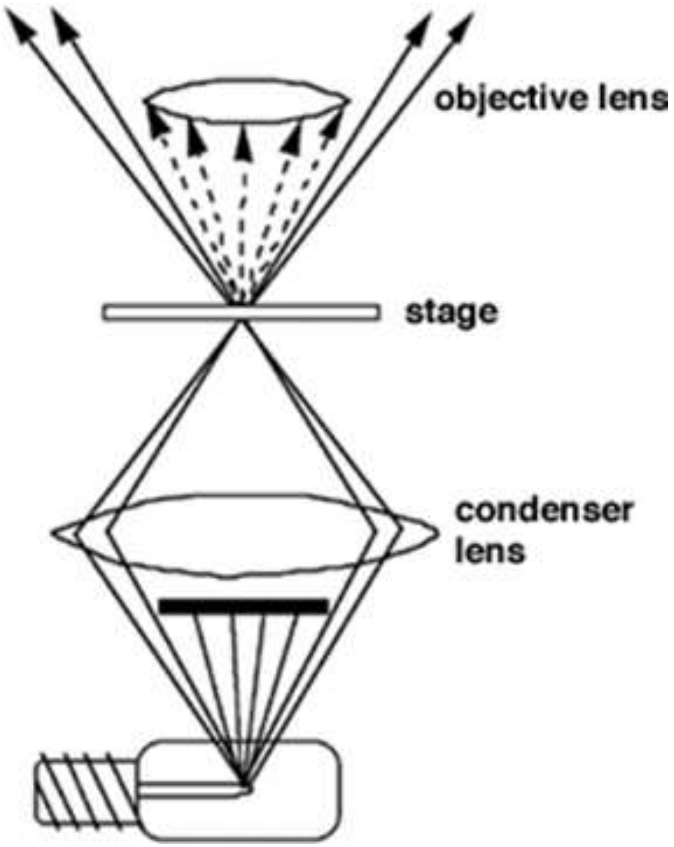
- In some viruses, the capsid is covered by an envelope
- lipids, proteins, and carbohydrates



- Spikes
- Projection from the surface of the envelope
- Viruses attach to host cells
- Influenza virus to clump red blood cells i associated with spikes
- Viruses whose capsids are not covered by an envelope are known as nonenveloped viruses

Ppt no. 24

DARK FIELD MICROSCOPY



DARK FIELD MICROSCOPY

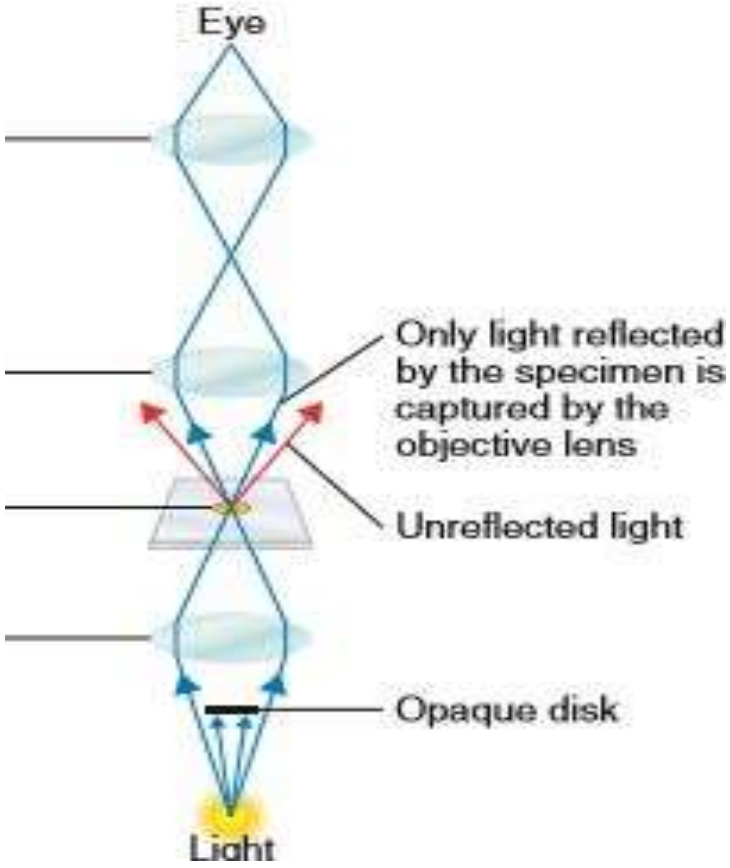
- Refers to the use of microscope that excludes the unscattered beam from the image
- As a result, the field around the specimen (i.e., where there is no specimen to scatter the beam) is generally dark

USE:

•Used for live microorganisms that either are invisible in the ordinary light microscope, cannot be stained by standard methods.

DARK FIELD MICROSCOPY

- Uses a darkfield condenser containing an opaque disk
- The disk blocks light that would enter the objective lens
- Only light that is reflected off (turned away from) the specimen enters the objective lens
- Because there is no direct background light, the specimen appears light against a black background–the dark field.


Eye
Only light reflected by the specimen is captured by the objective lens
Unreflected light
Opaque disk
Light

Ppt no. 25

DIFFERENTIAL INTERFERENCE CONTRAST MICROSCOPY

DIC MICROSCOPY

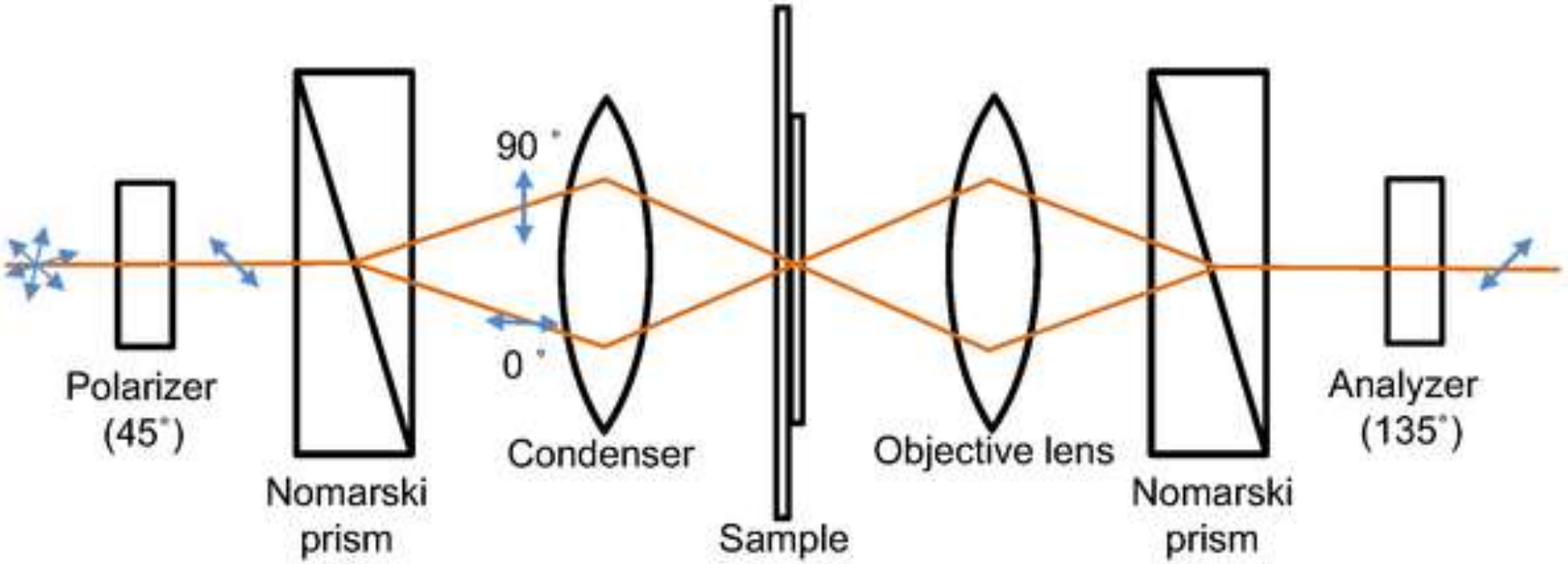
- Also known as Nomarski interference contrast microscopy
- used to enhance the contrast in unstained, transparent samples
- DIC works on the principle of interferometry to gain information about the optical

WORKING:

- DIC microscope uses two beams of light instead of one
- prisms split each light beam, adding contrasting colors to the specimen
- image is brightly colored and appears nearly three-dimensional

LIGHT PATH

- Unpolarized light enters and becomes polarized
- Then it enters the prism and gets separated into two beams
- These rays are focused by condenser
- Rays travel through specimen
- Then enter objective lens
- Second prism recombines the rays forming an image


90 °
0 °
Polarizer
(45°)
Nomarski
prism
Condenser
Sample
Objective lens
Nomarski
prism
Analyzer
(135°)

Ppt no. 26

FLUORESCENCE MICROSCOPY

FLUORESCENCE MICROSCOPE

- A microscope that uses fluorescence and phosphorescence instead of/or in addition to reflection and absorption to study properties of a microbe

PRINCIPLE:

– Fluorescence microscopy takes advantage of fluorescence, the ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength (visible

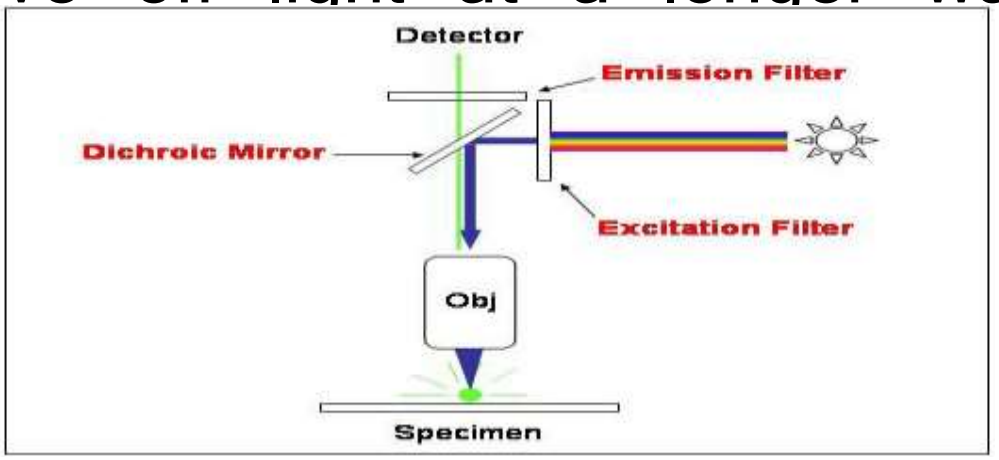
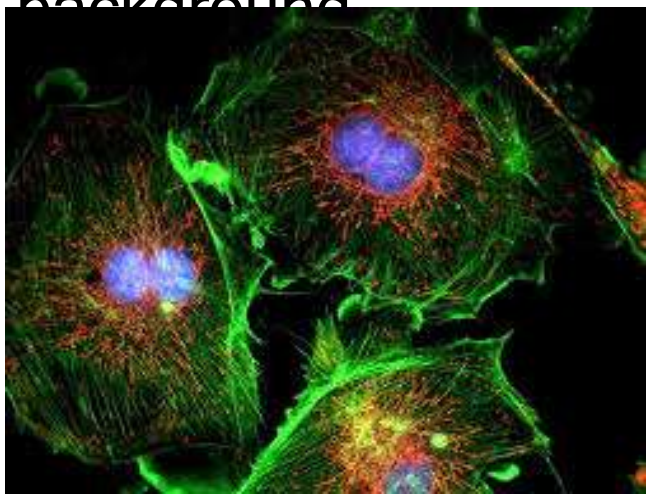
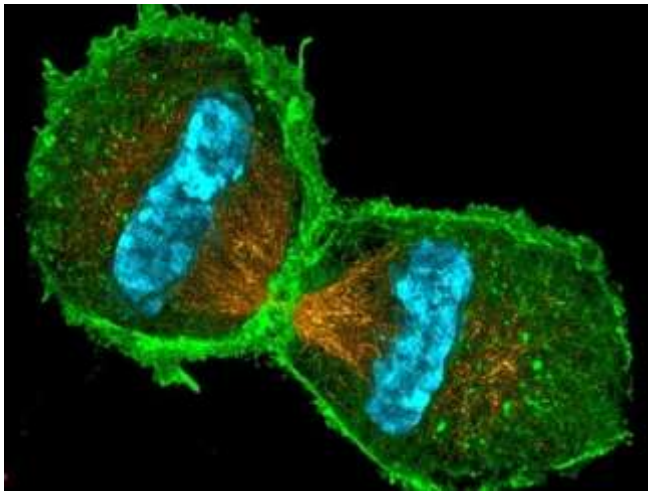
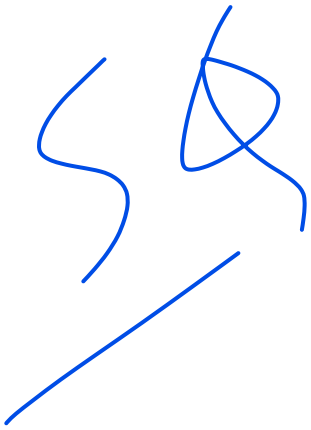


IMAGE:

- When microorganisms stained with a fluorochrome are examined under a fluorescence microscope with an ultraviolet or near-ultraviolet light source, they appear as luminescent, bright objects against a dark background.







LIGHT SOURCE:

- Monochromatic light source
- Four types:

1. Xenon arc lamps
2. Mercury vapor lamps
3. Super-continuum source
4. High power LED

Ppt no. 27

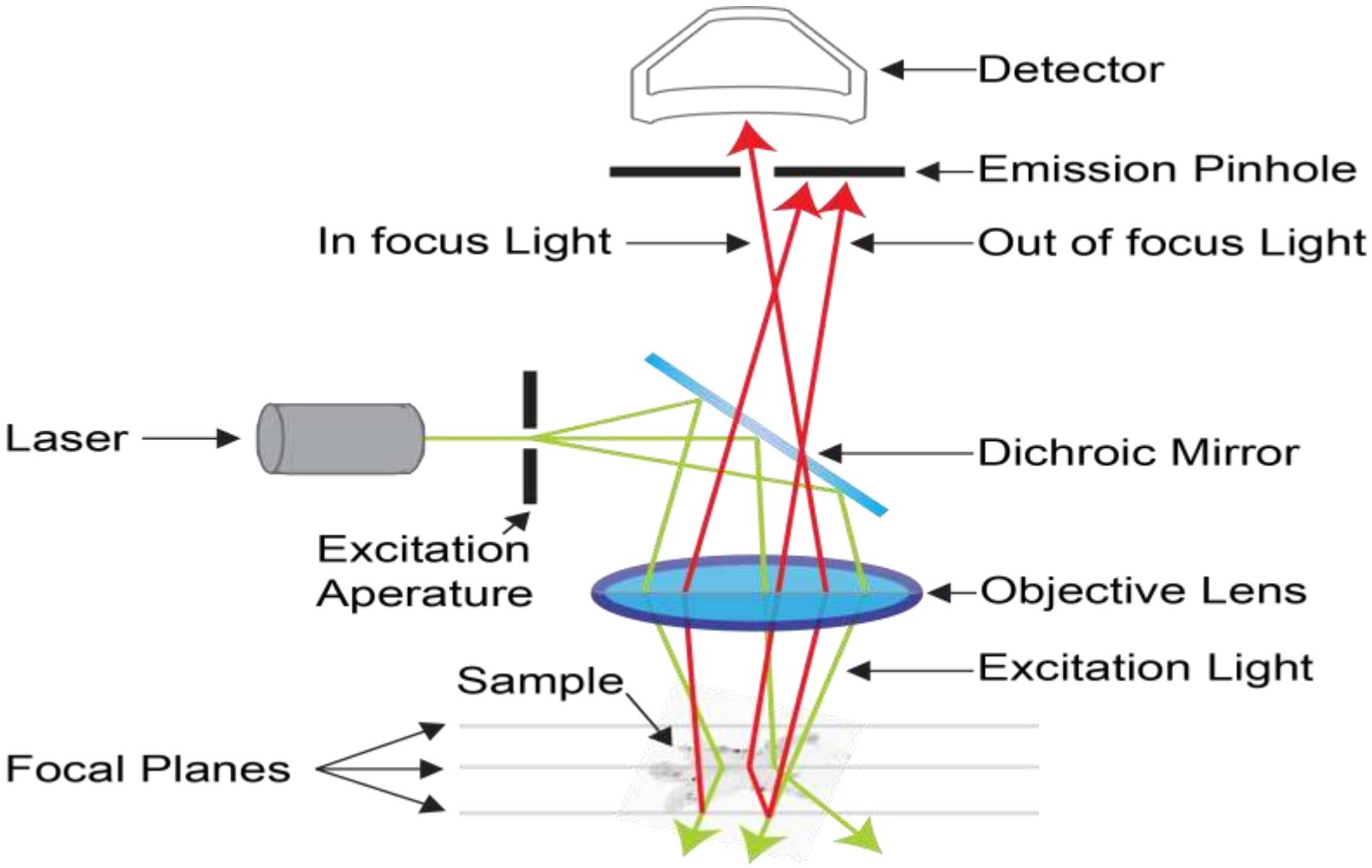
CONFOCAL MICROSCOPY

CONFOCAL MICROSCOPY

- An optical imaging technique for increasing optical resolution and contr ast of a micrograph
- by using a spatial pinhole to block out-of-focus light in image formation

PRINCIPLE:

- used to reconstruct three-dimensional images
- Like fluorescent microscopy, specimens are stained with fluorochromes so they will emit or return light
- But instead of illuminating the entire field, one plane of a small region of a specimen is illuminated with a short-wavelength (blue) light which passes the returned light through an aperture aligned with the illuminated region

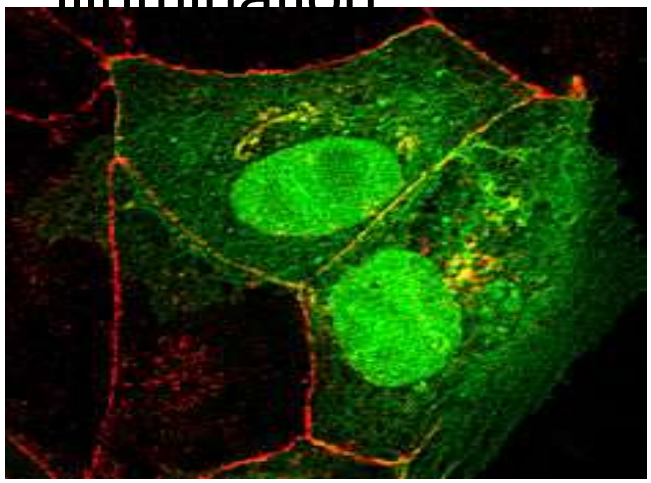

Detector
Emission Pinhole
In focus Light
Out of focus Light
Laser
Dichroic Mirror
Excitation
Aperature
Objective Lens
Excitation Light
Sample
Focal Planes

PROPERTIES:

- Each plane corresponds to an image of a fine slice that has been physically cut from a specimen
- Successive planes and regions are illuminated until the entire specimen has been scanned
- Confocal microscopy can image cells in detail only to a depth of less than 100 mm

IMAGE

- Point laser source
- 3D dimensional fraction pattern
- Better resolution of Wide field illumination



TYPES:

1. Confocal laser scanning microscope
2. Spinning disk confocal microscope
3. Microlens enhanced or dual spinning disk confocal microscope
4. Programmable array microscope (PAM)

Ppt no. 28

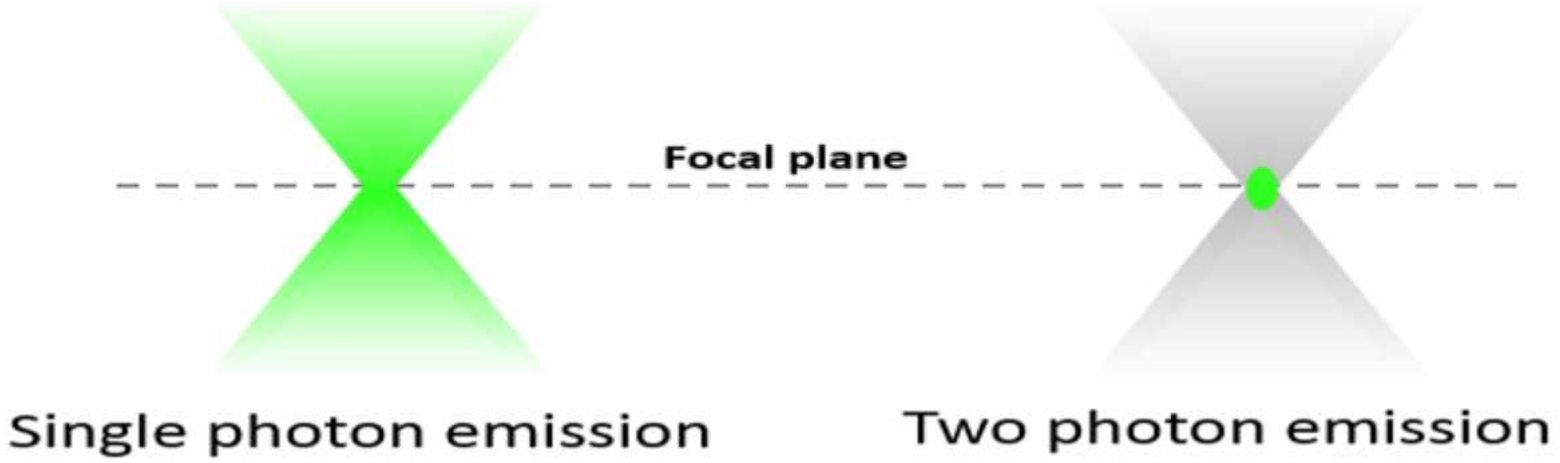
TWO PHOTON MICROSCOPY

TWO PHOTON MICROSCOPY

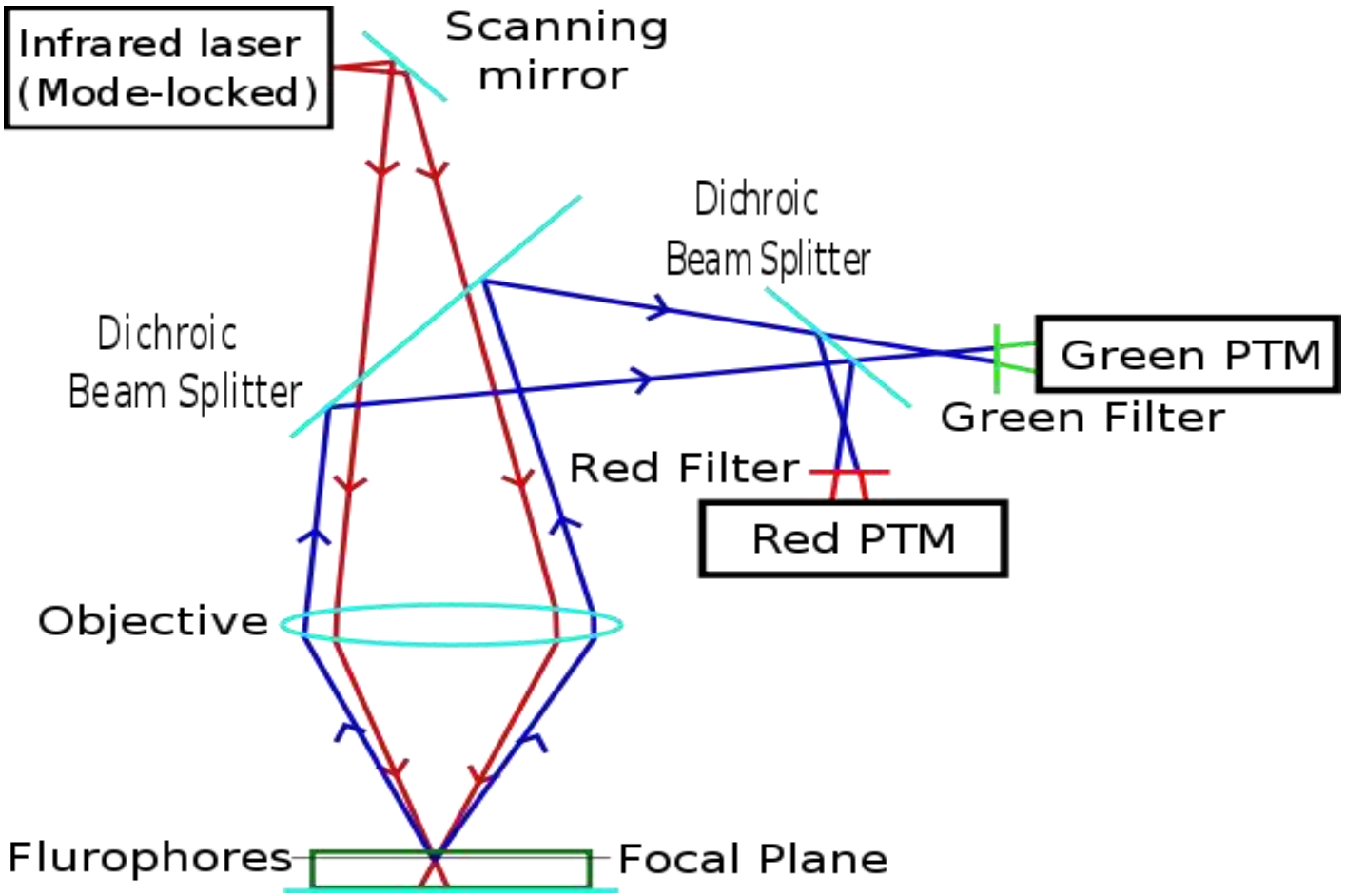
- It is a fluorescence imaging technique that allows imaging of living tissue up to about one millimeter in depth.

PRINCIPLE:

Specimens are stained with a fluorochrome
Two-photon microscopy uses long-wavelength (red) light, and therefore two photons, instead of one, are needed to excite the fluorochrome to emit light. The longer wavelength allows imaging of living cells in tissues up to 1 mm (1000 mm) deep

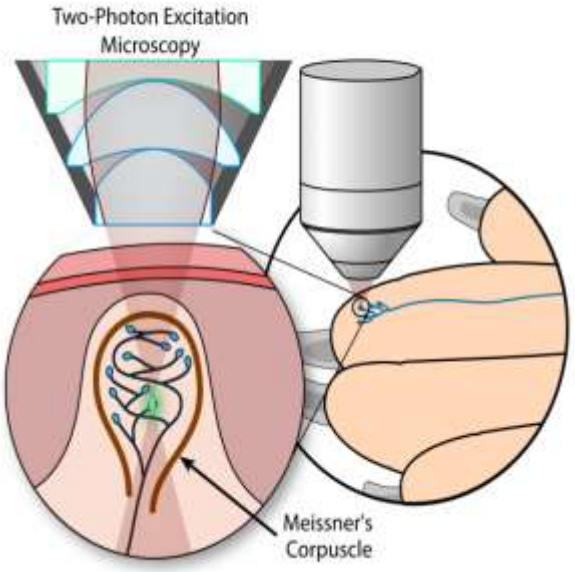


TWO PHOTON MICROSCOPY ILLUSTRATION



APPLICATIONS:

- Applicable for very thin transparent cells (skin)
- Non-invasive optical biopsy



Ppt no. 29

SCANNING ACOUSTIC MICROSCOPY

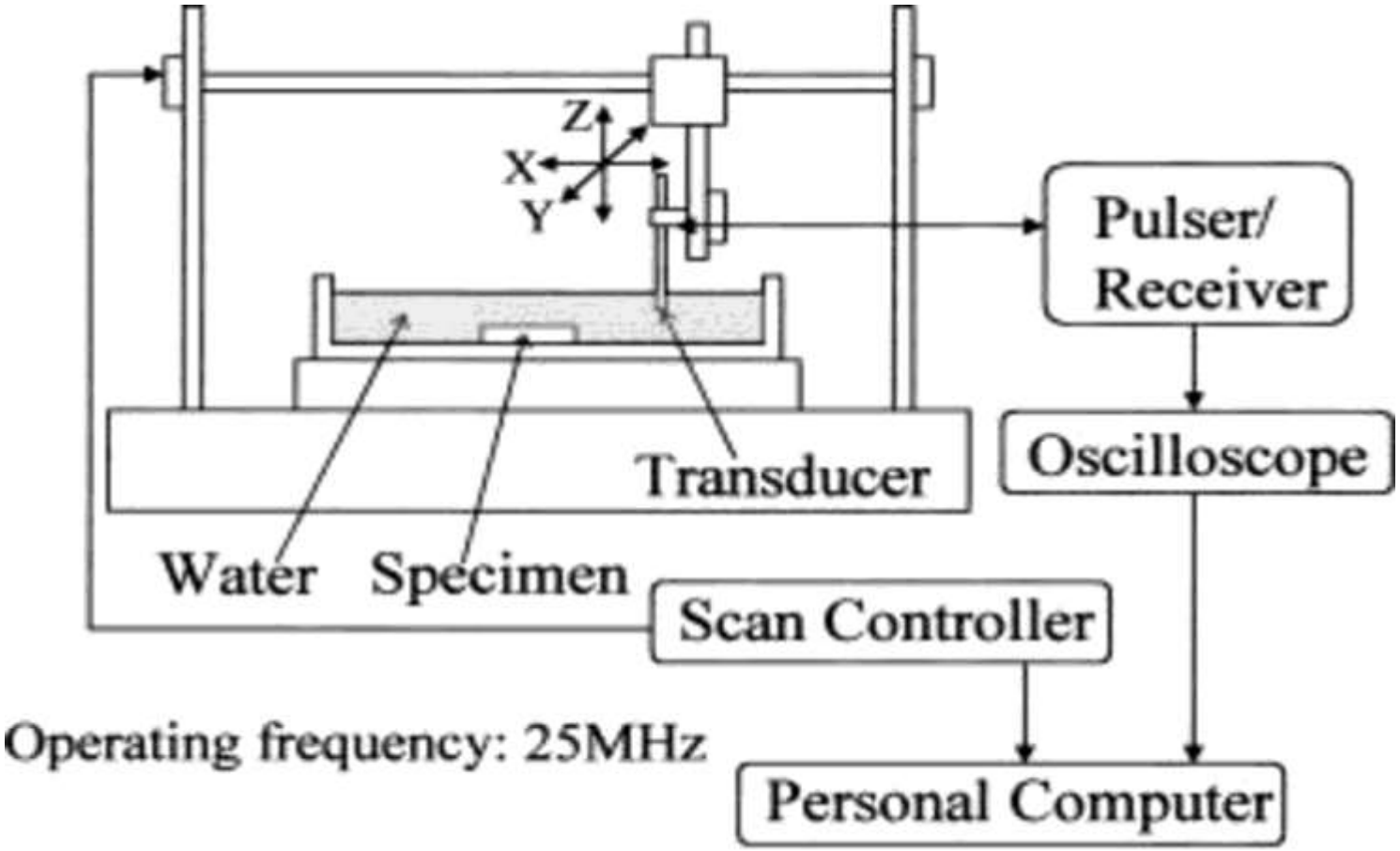
SCANNING ACOUSTIC MICROSCOPY

- A device that uses focused sound to image an object
- Applicable in biological and medical research
- Developed in 1974

PRINCIPLE:

- Scanning acoustic microscopy (SAM) basically consists of interpreting the action of a sound wave sent through a specimen
- A sound wave of a specific frequency travels through the specimen, and a portion of it is reflected back every time it hits an interface within the material

SAM ILLUSTRATION:



IMAGE

- Resolution= 1 µm
- Both 2D and 3D image

USE:

SAM is used to study living cells attached to another surface, such as cancer cells, artery plaque, and bacterial biofilms that foul equipment

Ppt no. 30

TRANSMISSION ELECTRON MICROSCOPY

TEM MICROSCOPY

A technique that uses a beam of electrons

- Transmitted through the specimen
- Ultra-thin section of specimen
- Image on imaging device

PRINCIPLE:

- Focused beam of electrons from an electron gun passes through ultrathin section of the specimen
- Beam gets focused on a small area of the specimen by an electromagnetic condenser lens
- Instead of being placed on a glass slide, the specimen is usually placed on a copper mesh grid

TEM ILLUSTRATION

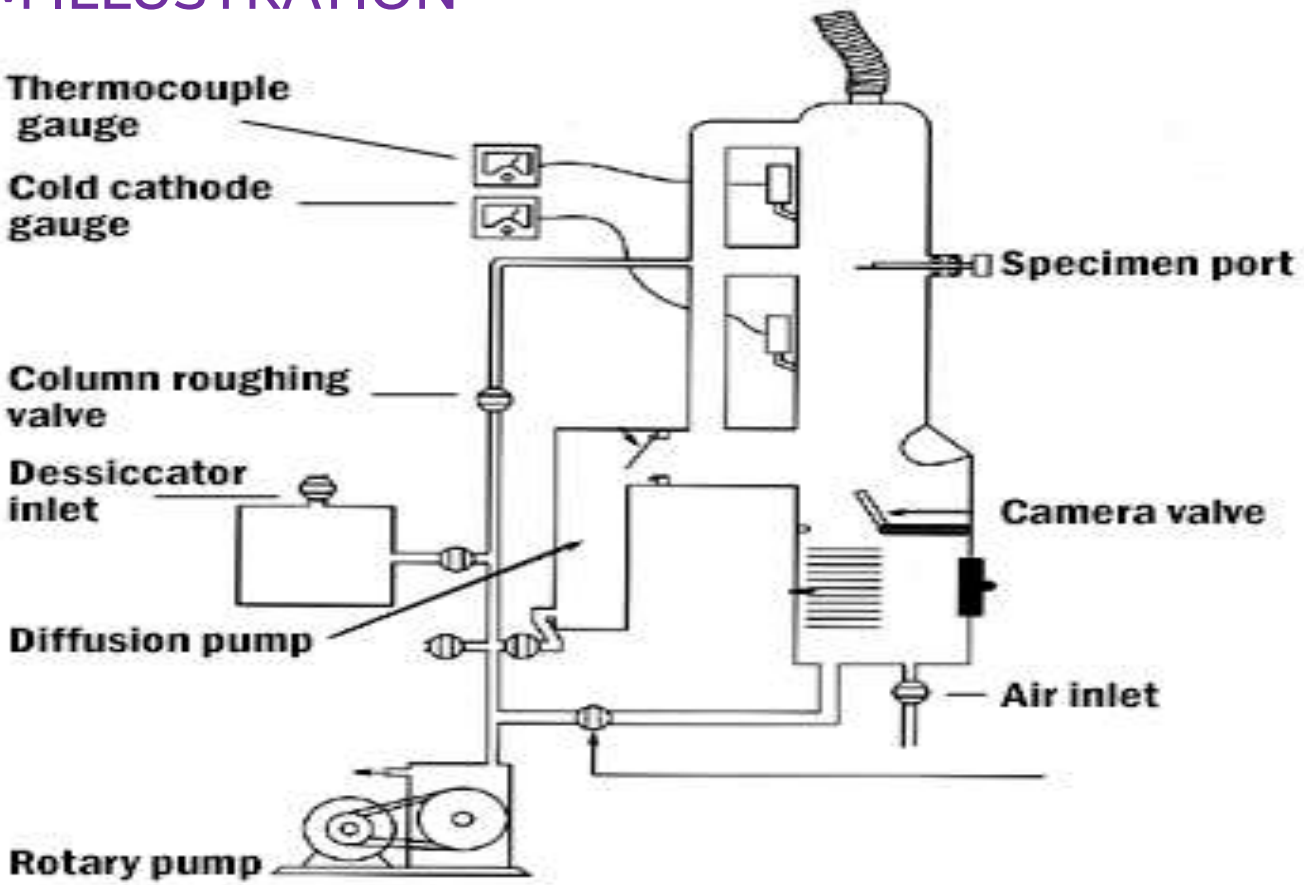
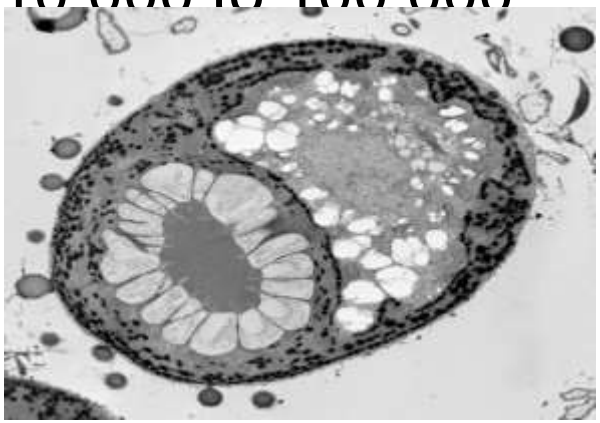


IMAGE FORMATION:

- Electrons passes through the specimen and then through an electromagnetic objective lens, which magnifies the image
- The electrons are focused by an electromagnetic projector lens onto a fluorescent screen or photographic plate
- The final image, called transmission electron micrograph, appears as many light and dark areas, depending on the number of electrons absorbed by different areas of the specimen

IMAGE

- The transmission electron microscope can resolve objects as close together as 10 pm
- The objects are generally magnified 10,000 to 100,000*



DYE:

- Contrast can be greatly enhanced by using a “dye” that absorbs electrons and produces a darker image in the stained region
- Lead, osmium, tungsten, and uranium (Stains)
- These metals can be fixed onto the specimen (positive staining)
- or used to increase the electron opacity of the surrounding field (negative staining)

LIMITATIONS:

- Only a very thin section of a specimen (about 100 nm) can be studied effectively
- Specimens must be fixed, dehydrated, and viewed under a high vacuum to prevent electron scattering
- These treatments not only kill the specimen, but also cause shrinkage and distortion

Ppt no. 31

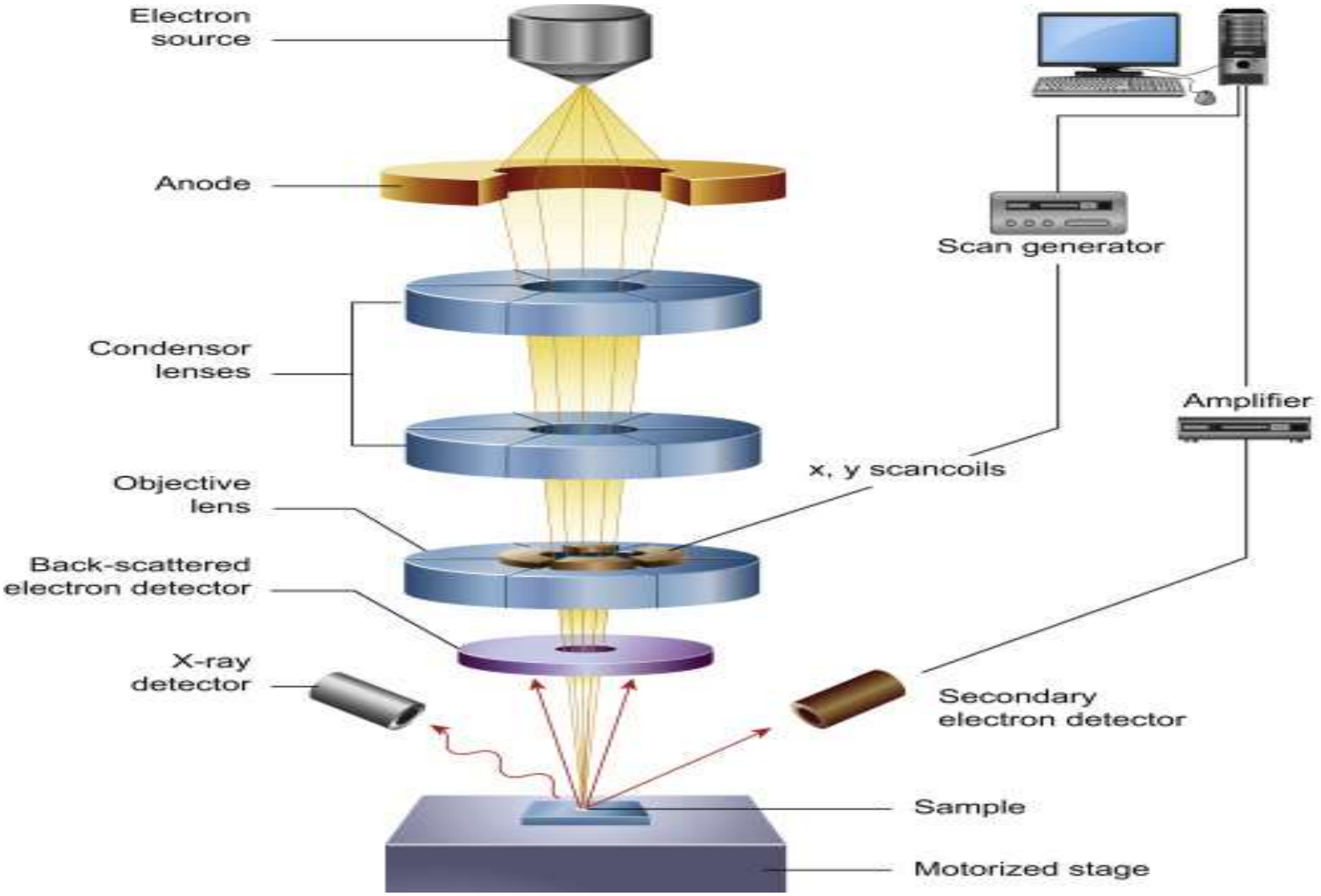
SCANNING ELECTRON MICROSCOPY

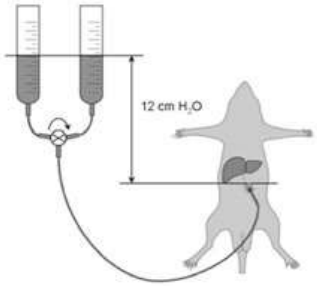
SEM MICROSCOPY

- A microscope that produces images of a sample by scanning the surface with a focused beam of electrons
- Produces various signals that contain information about the surface topography and composition of the sample

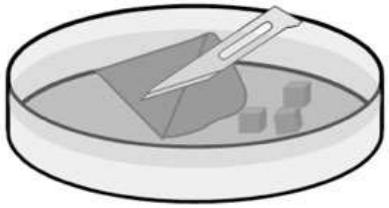
PRINCIPLE:

- An electron gun produces a finely focused beam of electrons called the primary electron beam
- These electrons pass through electromagnetic lenses and are directed over the surface of the specimen
- The primary electron beam knocks electrons out of the surface of the specimen, and the secondary electrons thus produced are transmitted to an electron collector, amplified, and used to produce an image on a viewing screen or photographic plate. The image is called a scanning electron micrograph

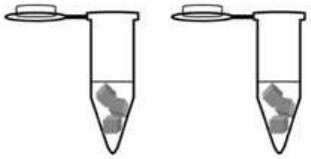

Electron source
Anode
Condensor lenses
Objective lens
Back-scattered electron detector
X-ray detector
Scan generator
x, y scancoils
Amplifier
Secondary electron detector
Sample
Motorized stage



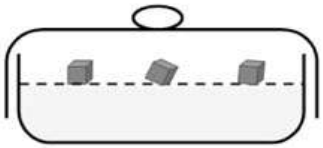
Step I (2-10 min)



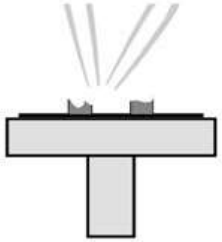
Step II (10 min)



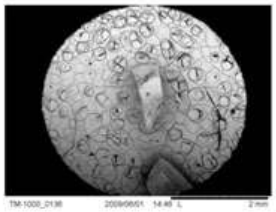
Step III (5×2 min & 10 min)



Step IV (10 min)



Step V (10 min)



Step VI

SAMPLE PREPARATION:

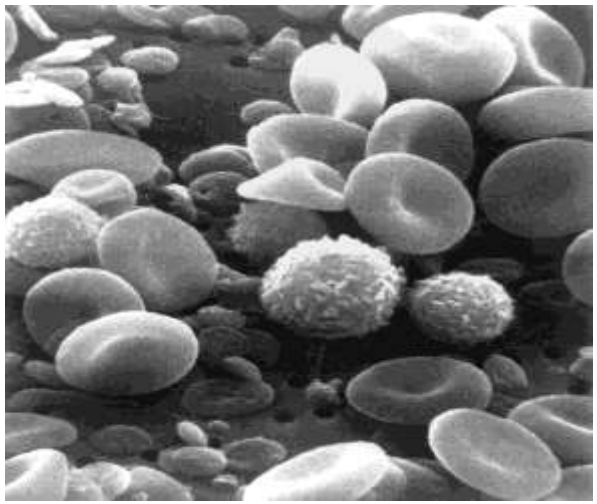
- Prepared in presence of vacuum and beam of electrons
- Mounted on specimen holder
- Specimens get conductive by beam of electrons

IMAGE

- Provides striking three-dimensional views of specimens
- it can resolve objects as close together as 10 nm, and objects are generally magnified 1000 to 10,000

USES:

- useful in studying the surface structures of intact cells and viruses
- Also used for fractography



Ppt no. 32

SCANNED PROBE MICROSCOPY

SCANNED PROBE MICROSCOPY

- Forms images of surfaces using physical probe that scans the specimen
- Founded in 1981

PRINCIPLE:

- The interaction between the sharp probe and surface provides 3D topographic image of surface at the atomic scale
- Two most popular modes:

1. Scanning Tunneling Microscopy (STM)
2. Atomic Force Microscopy (AFM)

SPM ILLUATRATION:

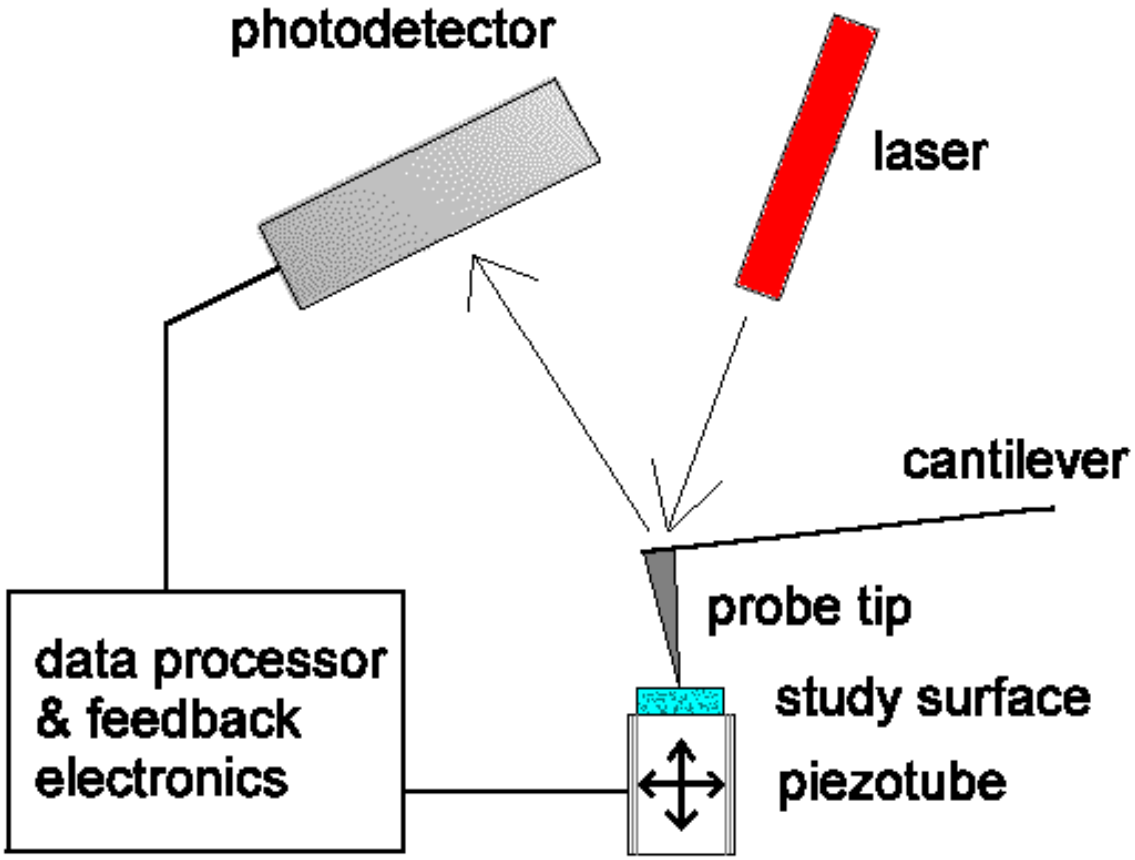
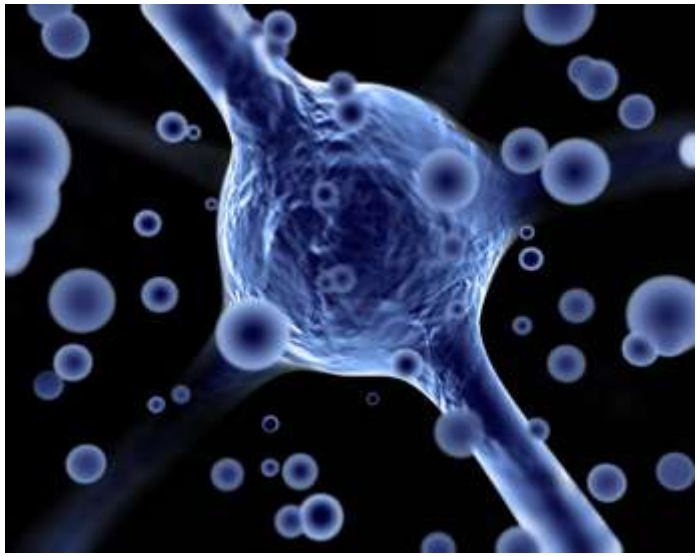


IMAGE FORMATION:

- Scans tip over surface
- A value is recorded
- Displayed as heat map



LIMITATIONS:

- Detailed shape of probe is difficult to determine sometimes
- Slower in image acquiring
- Maximum image size is smaller
- Cannot examine buried solid-solid or liquid-liquid interfaces

Ppt no. 33

STRUCTURE AND FORMATION OF VIRAL PARTICLES

Viral Capsids and Envelopes

CAPSID

Virus particles contain the viral genome packaged in a protein coat called the capsid.

ENVELOPE

Surrounding the capsid

composed of lipid bilayer that contains viral proteins

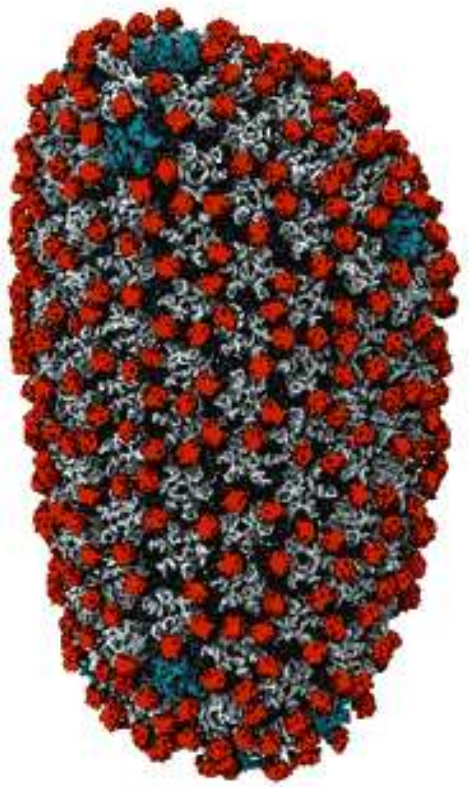
derived from the host cell membranes.

FUNCTIONS

The capsid and envelope play many roles in viral infection, including virus attachment to cells, entry into cells, release of the capsid contents into the cells, and packaging of newly formed viral particles. The capsid and envelope are also responsible for transfer of the viral genetic material from one cell to another. These structures also determine the stability characteristics of the virus particle, such as resistance to chemical or physical inactivation.

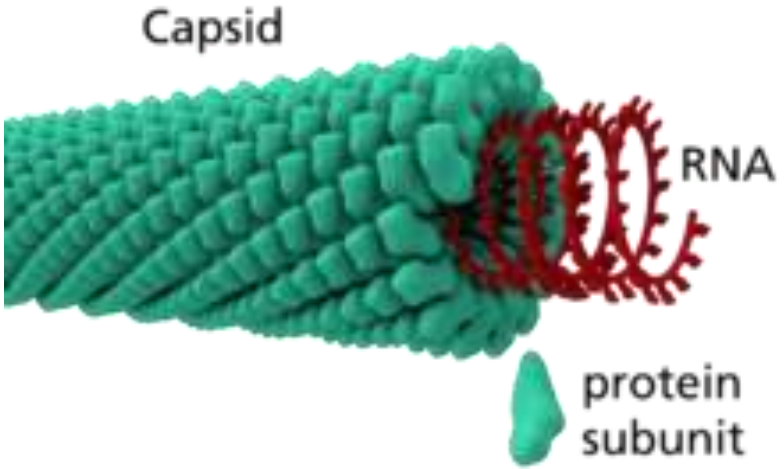
Ppt no. 34

VIRAL CAPSID



MEANING

- Protein shell of virus
- Made up of protein sub-units (protomers)
- Encloses genome

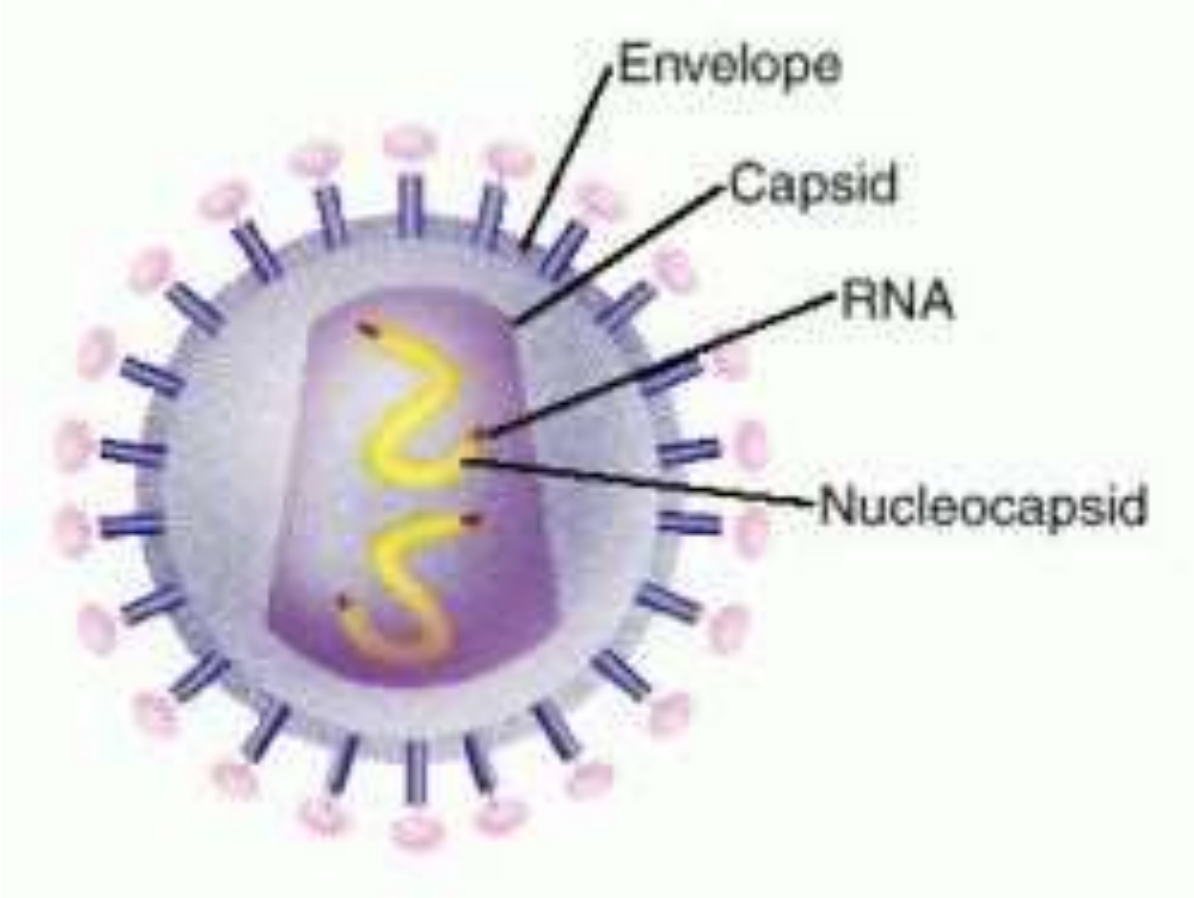


FUNCTION

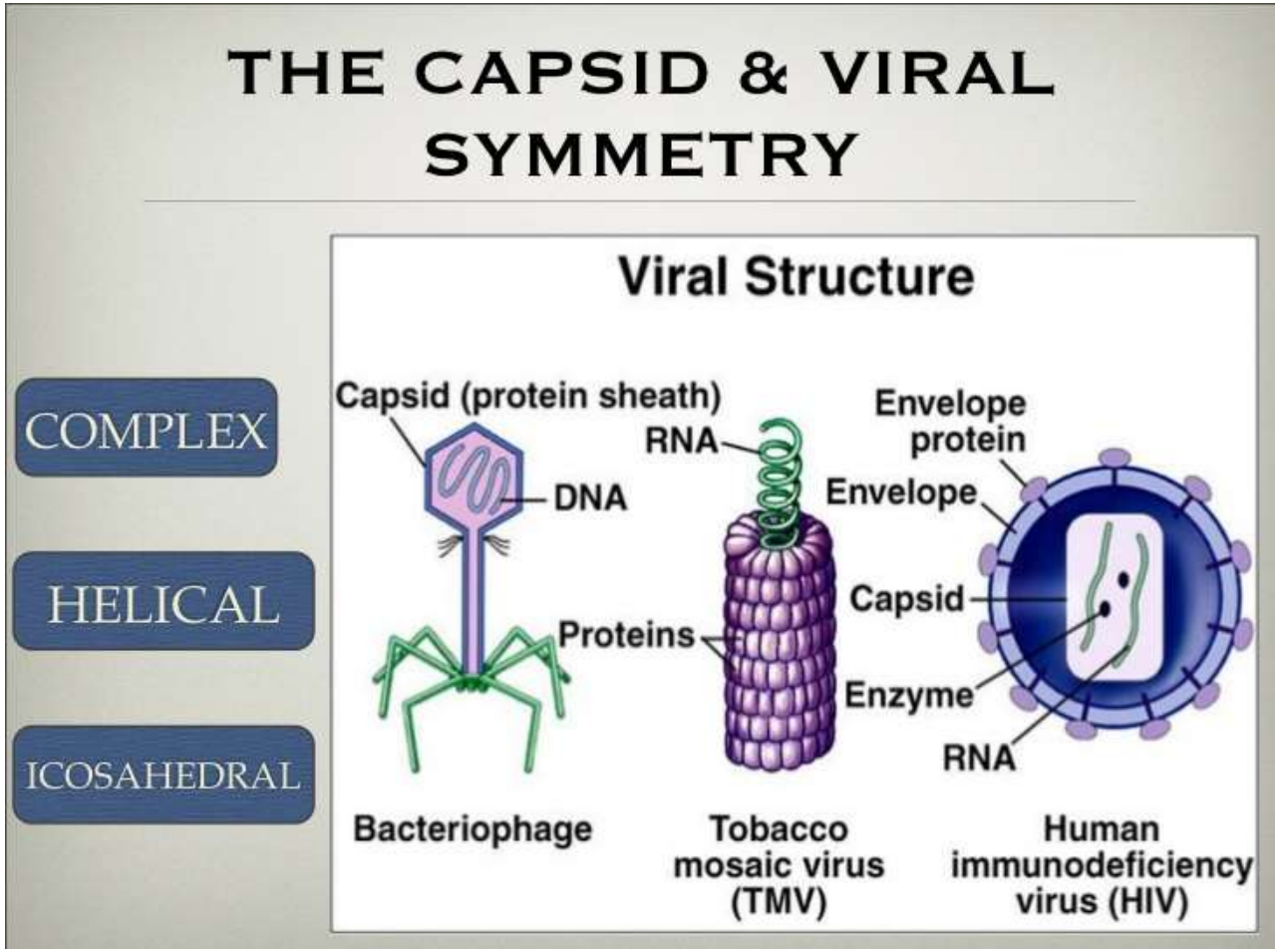
- Protection of genome
- Prevent digestion by cellular enzymes
- In cytoplasm Helps to introduce genome in host cell
- Bind to cell surface

NUCLEOCAPSID

Genome plus capsid Collective term



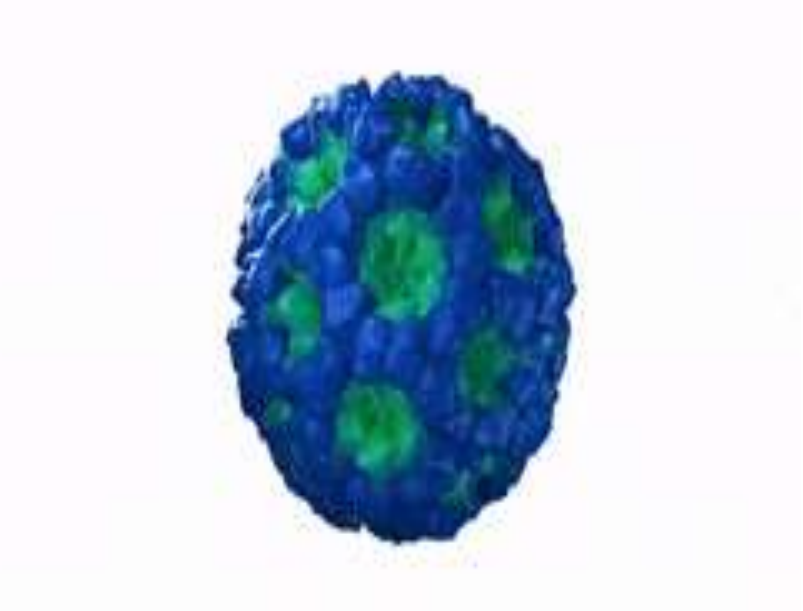
TYPES

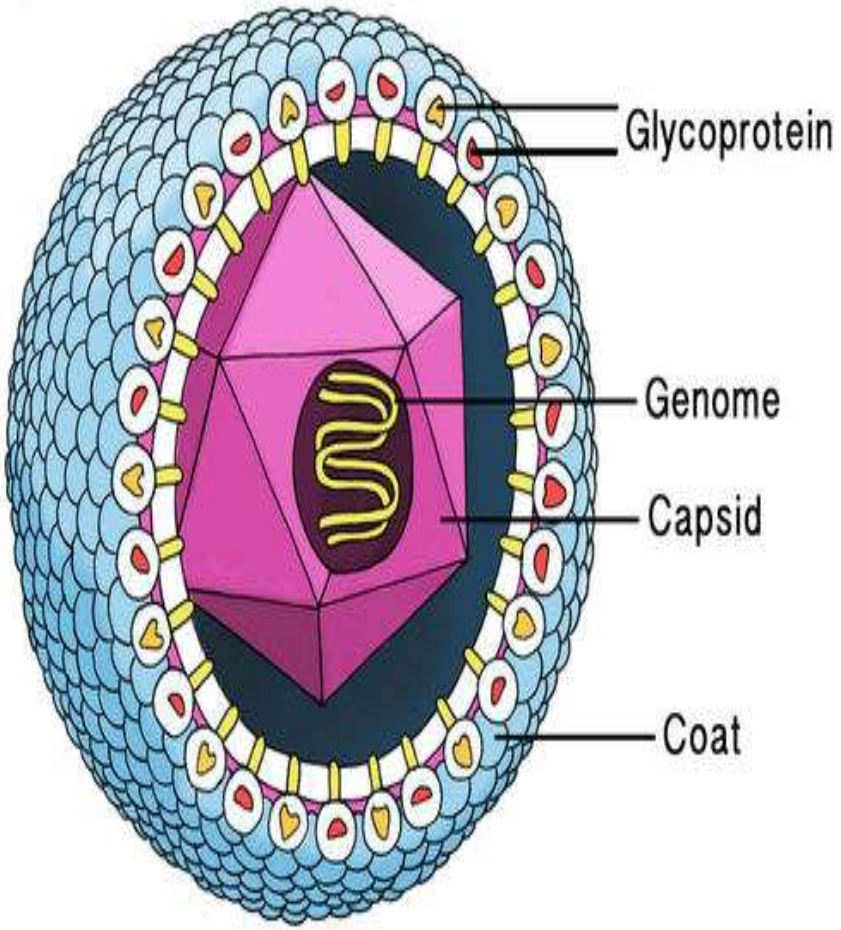

THE CAPSID & VIRAL SYMMETRY
COMPLEX
HELICAL
ICOSAHEDRAL
Viral Structure
Capsid (protein sheath)
DNA
Proteins
RNA
Envelope protein
Envelope
Capsid
Enzyme
RNA
Bacteriophage
Tobacco mosaic virus (TMV)
Human immunodeficiency virus (HIV)

Ppt no. 35

CAPSID

Symmetry and Architecture




Glycoprotein
Genome
Capsid
Coat

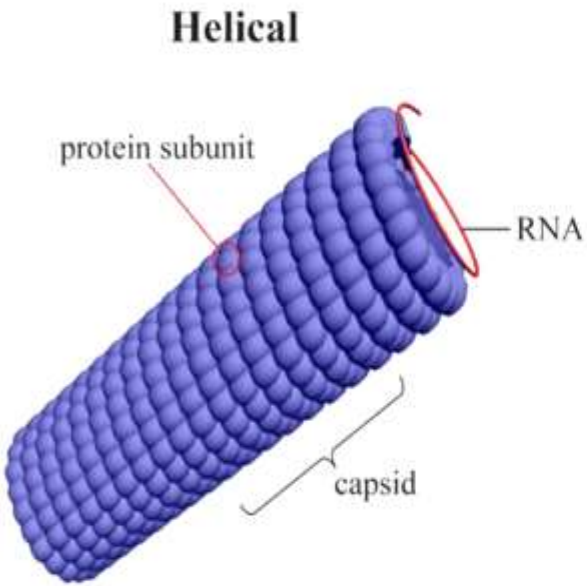
COMPOSITION

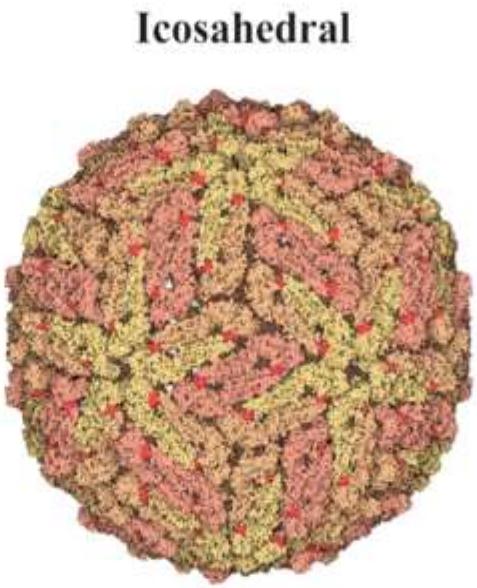
Capsid:

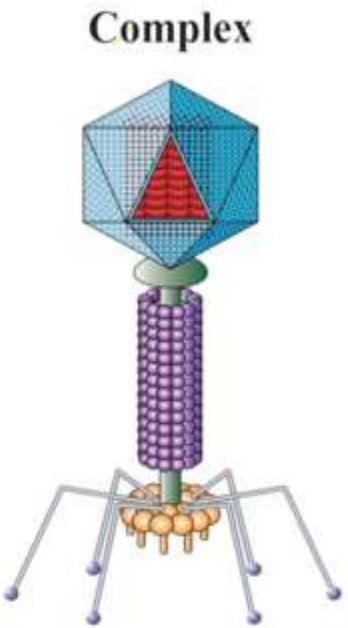
- Proteinic nature
- Composed of protomers and/or
- capsomers

TYPES

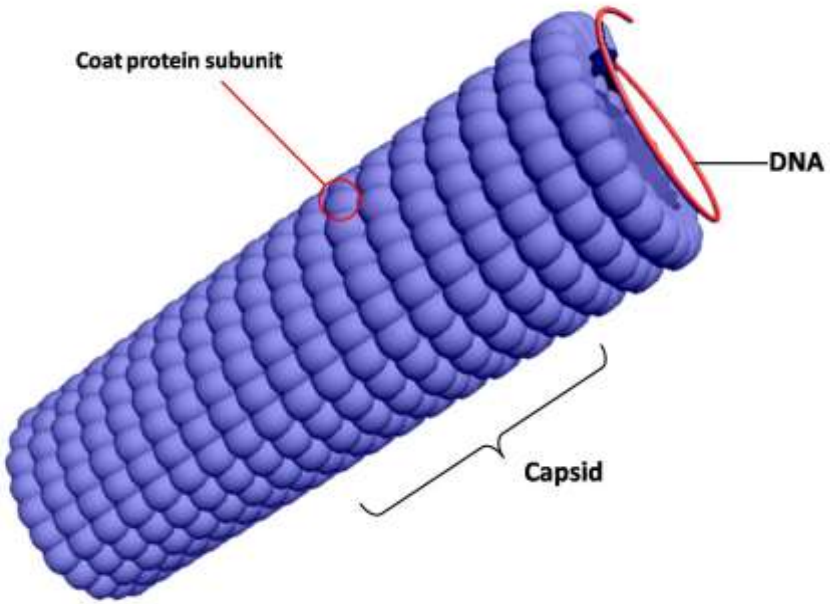
Helical Icosahedral Polyhedral







HELICAL SYMMETRY



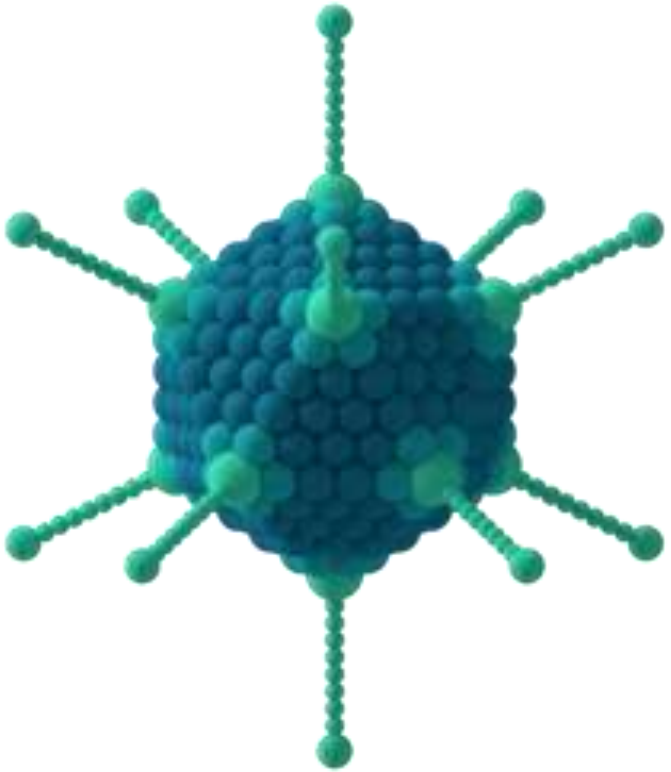
- Rod like structure
- Genome in the centre of helix
- Helix made by repeating units in a spiral
- Examples, Tobacco Mosaic Virus

HELICAL SYMMETRY

- Virus particles elongated or pleomorphic (not spherical)
- Nucleic acid also spiral
- Capsomeres arranged around Nucleic acid

ICOSAHEDERAL SYMMETRY

- Characteristic of many spherical viruses
- In adenovirus (icosahedral) composition consists of : Hexon, Pentone base, and fibre

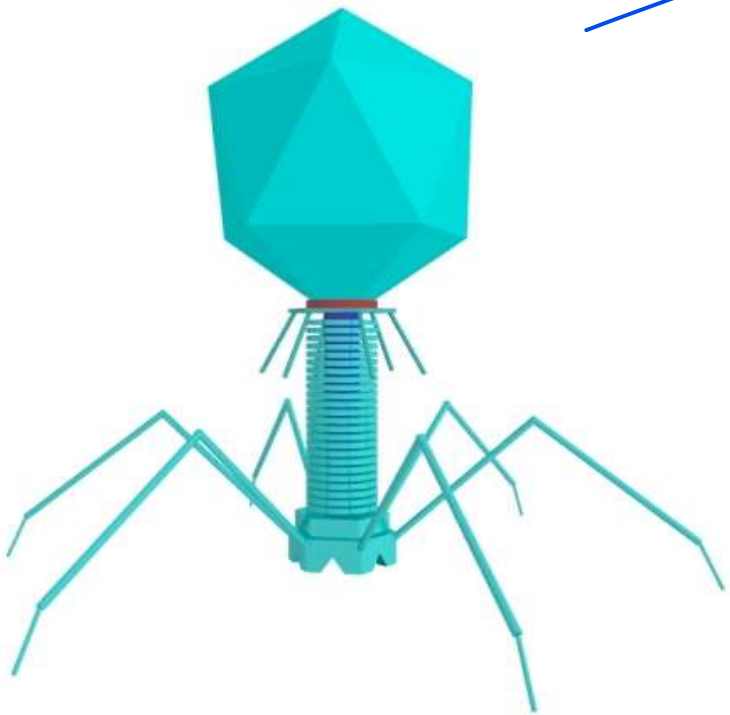


ICOSAHEDERAL SYMMETRY

- Also called cubic capsid
- Mostly spherical
- Particles composed of 20 equilateral triangles, 12 vertices and 2,3,5 rotational symmetry

POLYHEDERAL SYMMETRY

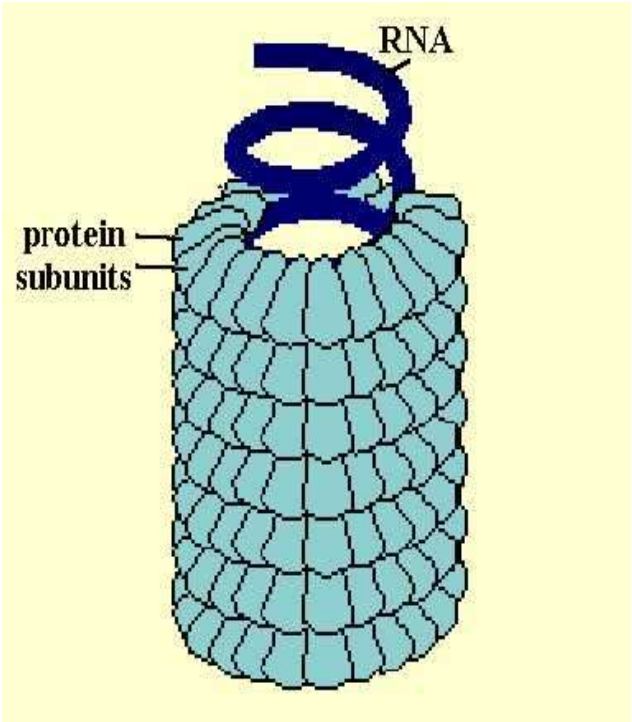
- Also called complex
- Neither helical nor icosahederal symmetry
- E.g Bacteriophage



Ppt no. 36

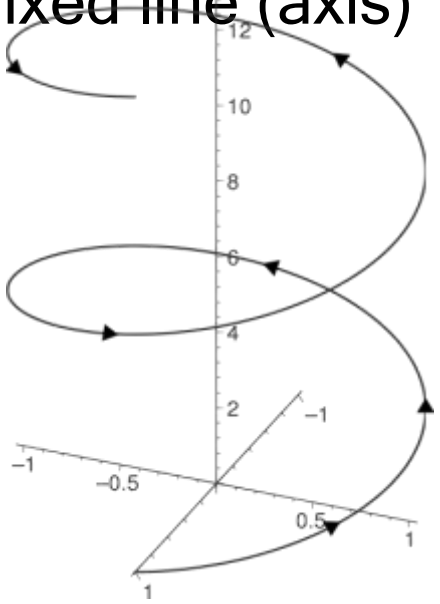
CAPSID SYMMETRY

Helical

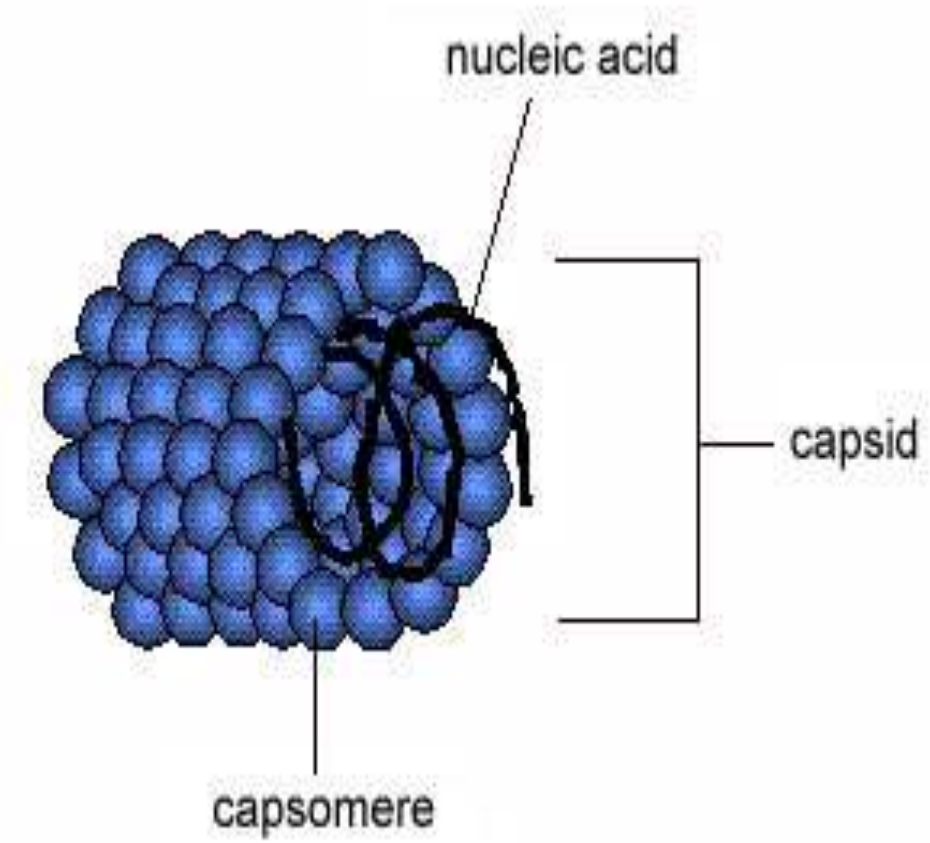


WHAT IS HELICAL SHAPE?

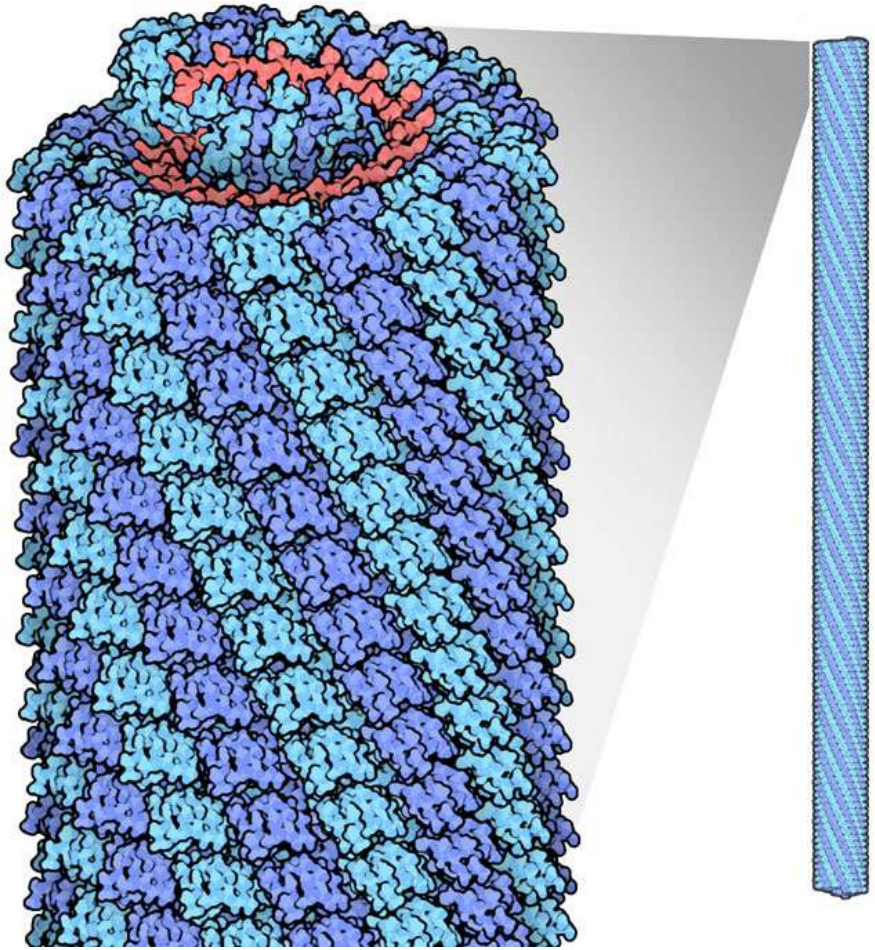
- Property that a tangent line at ay point makes a constant angle with a fixed line (axis)



HELICAL VIRUS



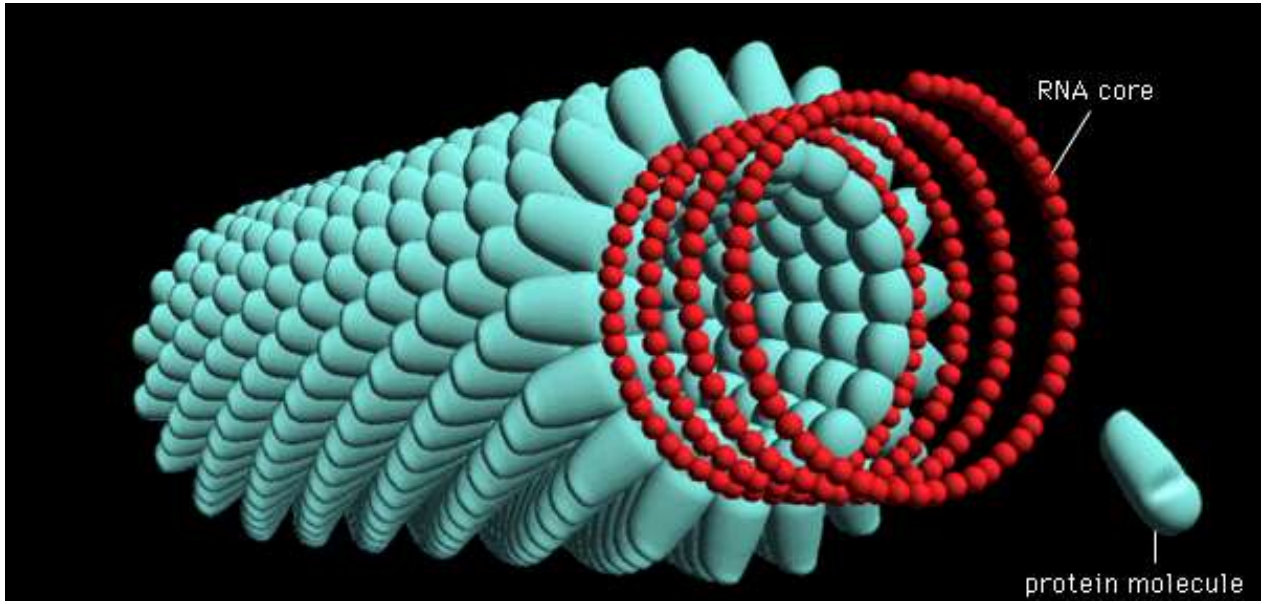
- Resembles long rods
- Ribbon-like structure
- Hollow tubes with protein walls



- A single protomer associates together in a helical arrangement to produce a long rigid tube

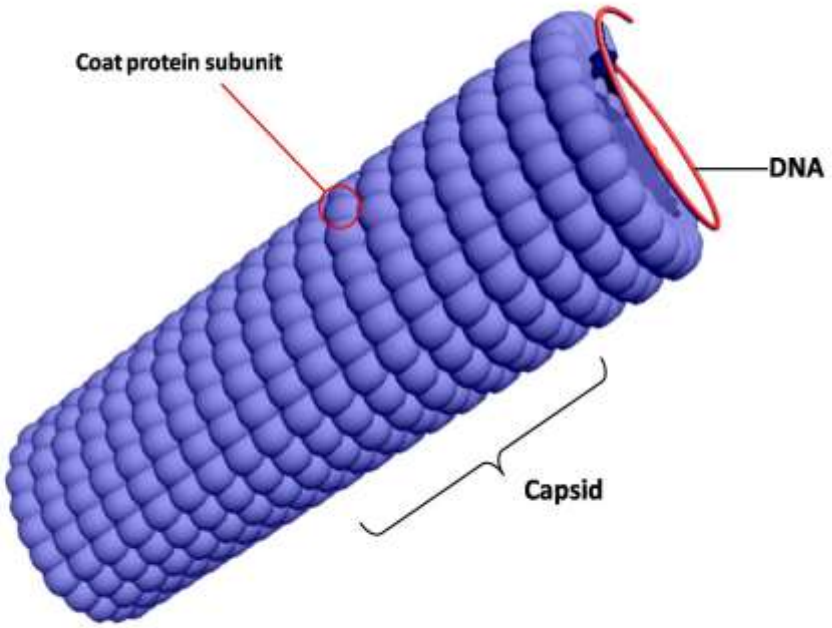
GENOME

- Genome is wound in a spiral
- Inside the capsid
- Within a groove



LENGTH

- The Nucleic acid determines the length of the helix
- Capsid doesn't extend much beyond the genome end



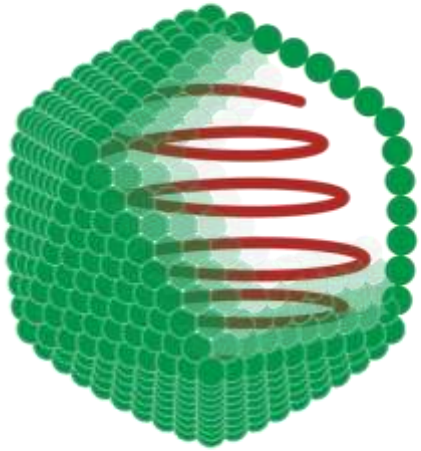
EXAMPLES

- Most are enveloped
- All are RNA viruses
- Tobacco Mosaic Virus (TMV)
- Influenza virus

Ppt no. 37

CAPSID SYMMETRY

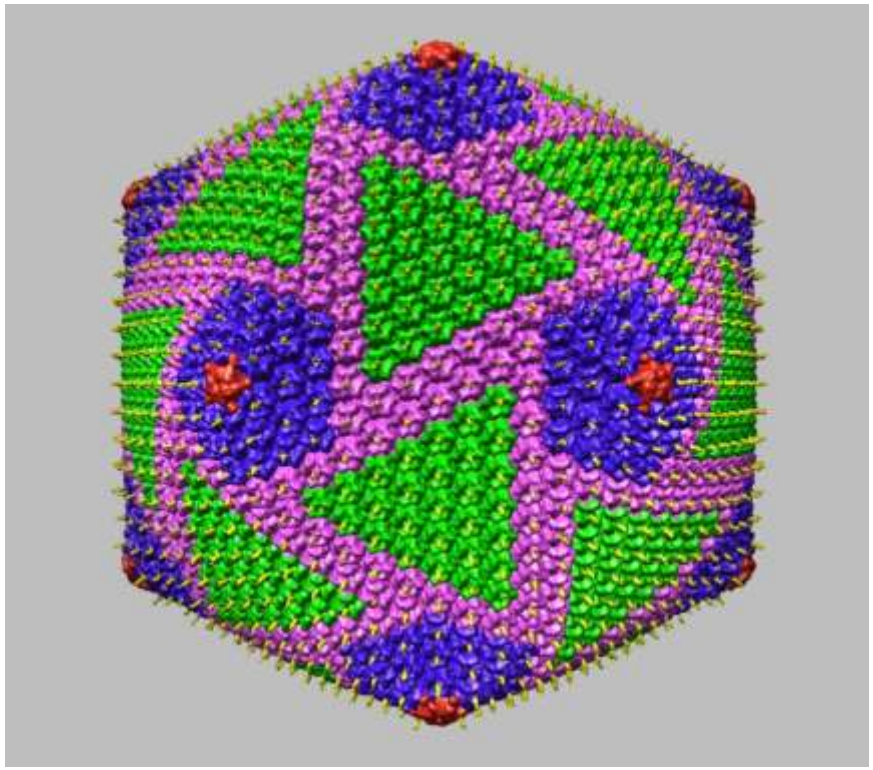
Icosahederal



WHAT IS ICOSAHEDERAL SHAPE?

- An icosahedral is a polygon with 12 vertices (corner), 20 facet (sides) and 30 edges

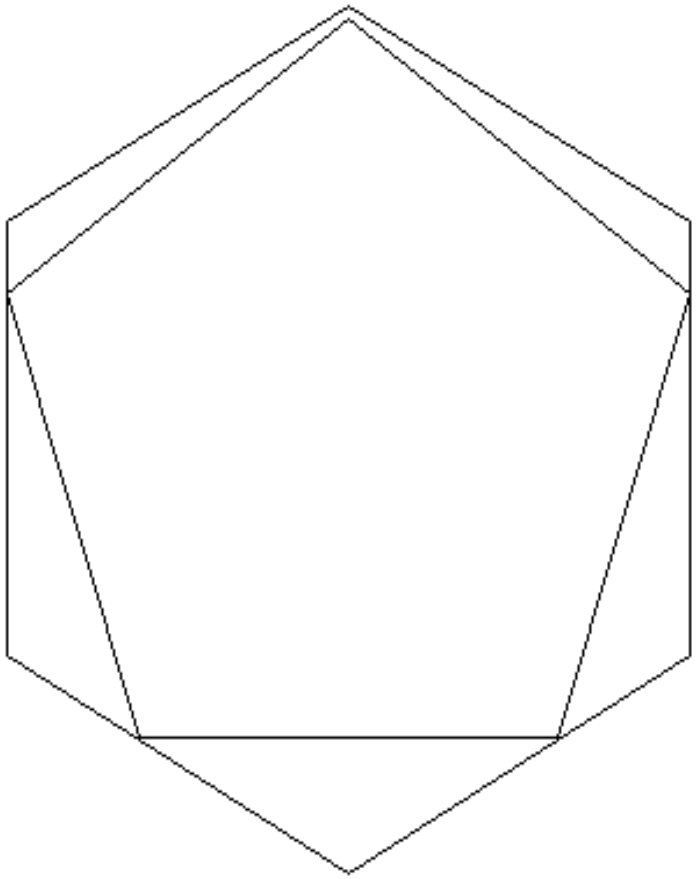
WHY ICOSAHEDERAL?



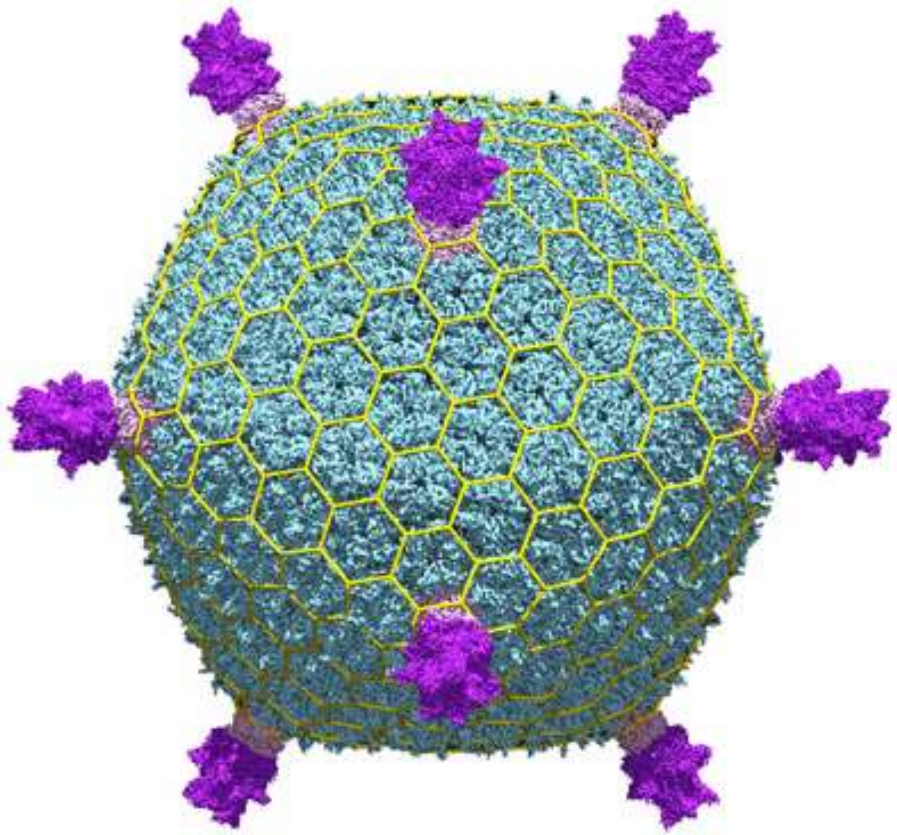
- Most efficient way to enclose a space
- small number of linear genes can specify large structure

TYPES

- Hexamers (6 units)
- Pentamers (5 units)



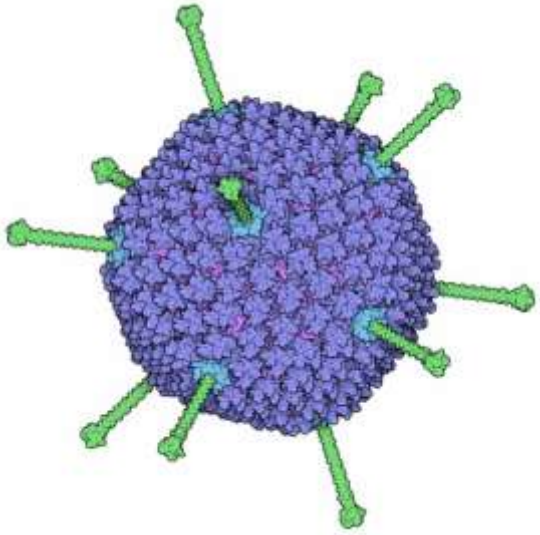
CAPSOMERE



- Composed of knob-shaped structures
- Called capsomeres
- Composed of 5 or 6 protomers

EXAMPLES

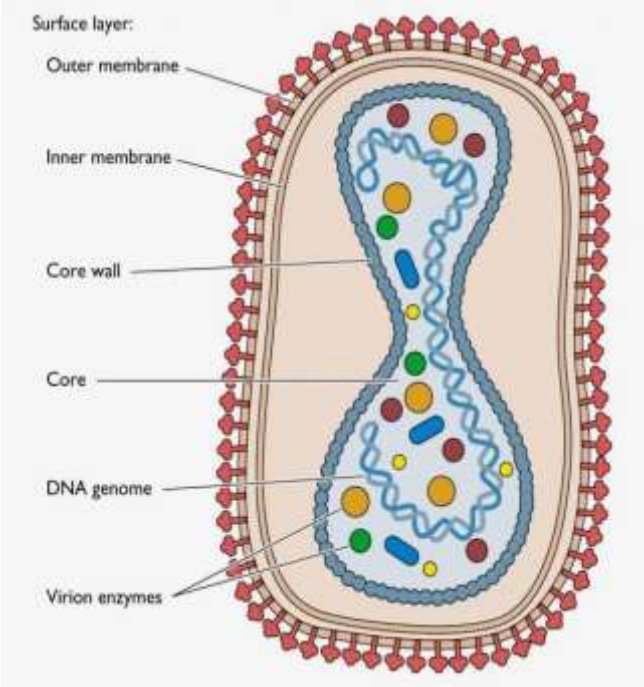
- most stable
- Adenovirus
- Picornavirus



Ppt no. 38

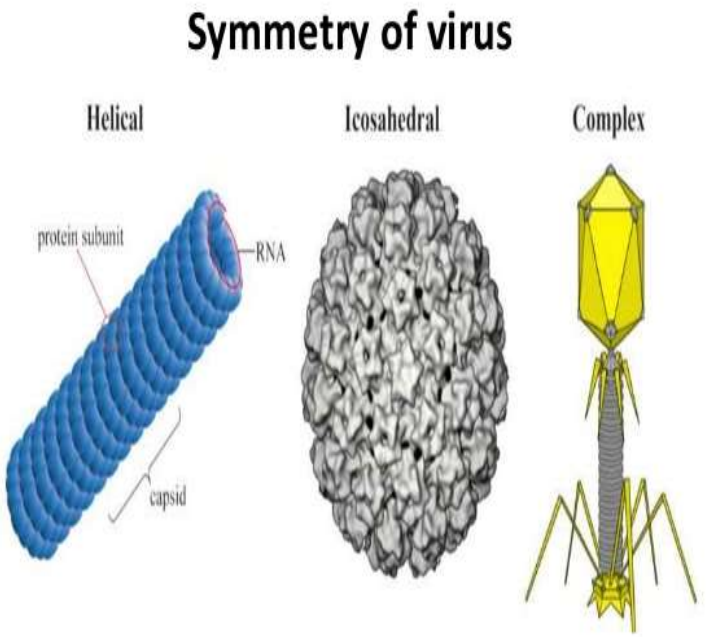
CAPSID SYMMETRY

Complex

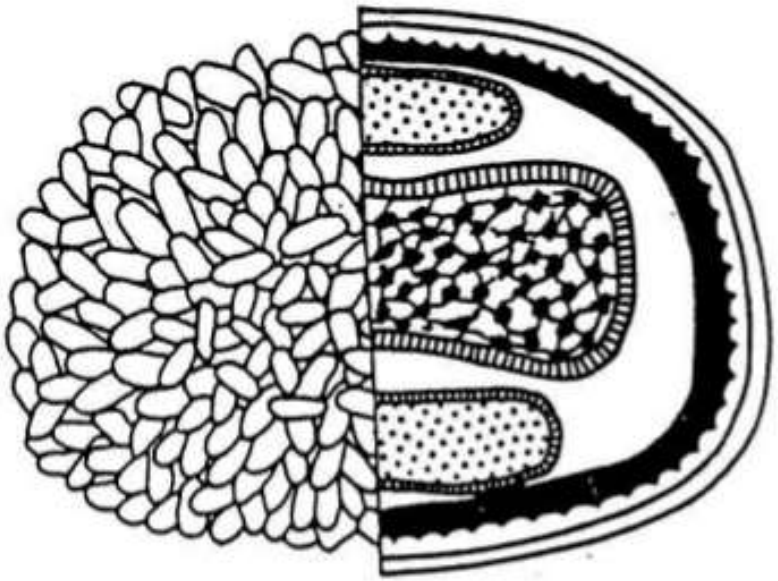


WHAT IS COMPLEX SHAPE?

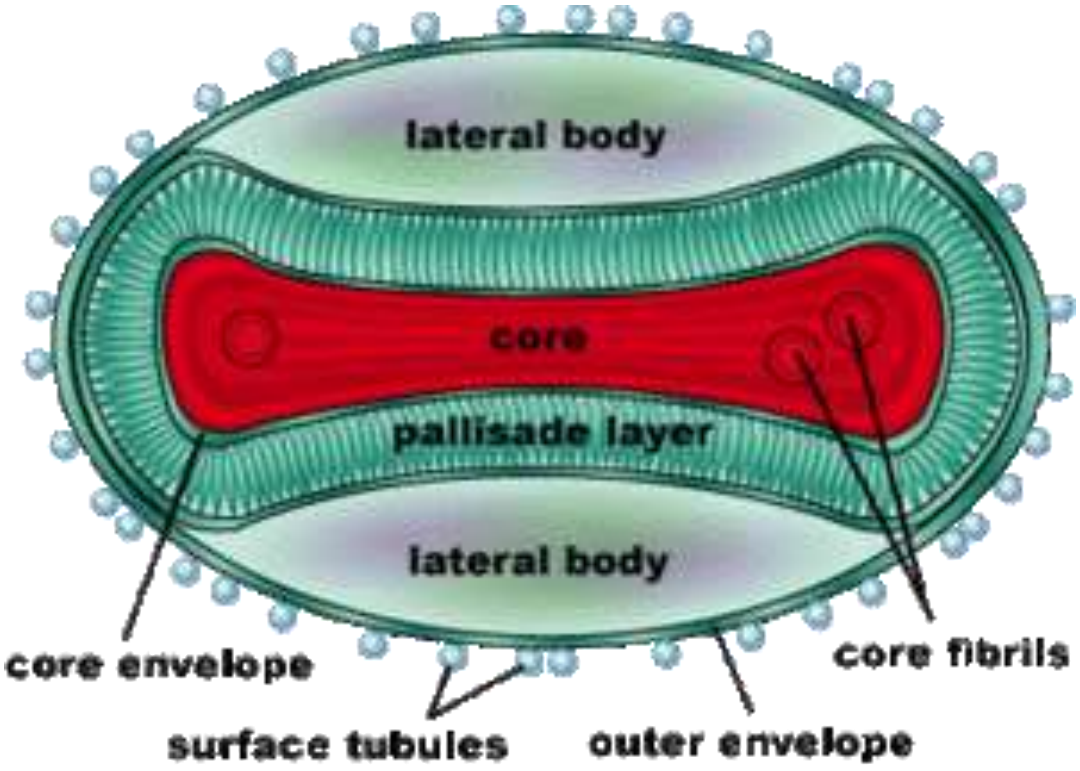
- Neither helical nor icosahederal



HELICAL VIRUS



- Complex
- Ovoid interior
- Brick-shaped exterior



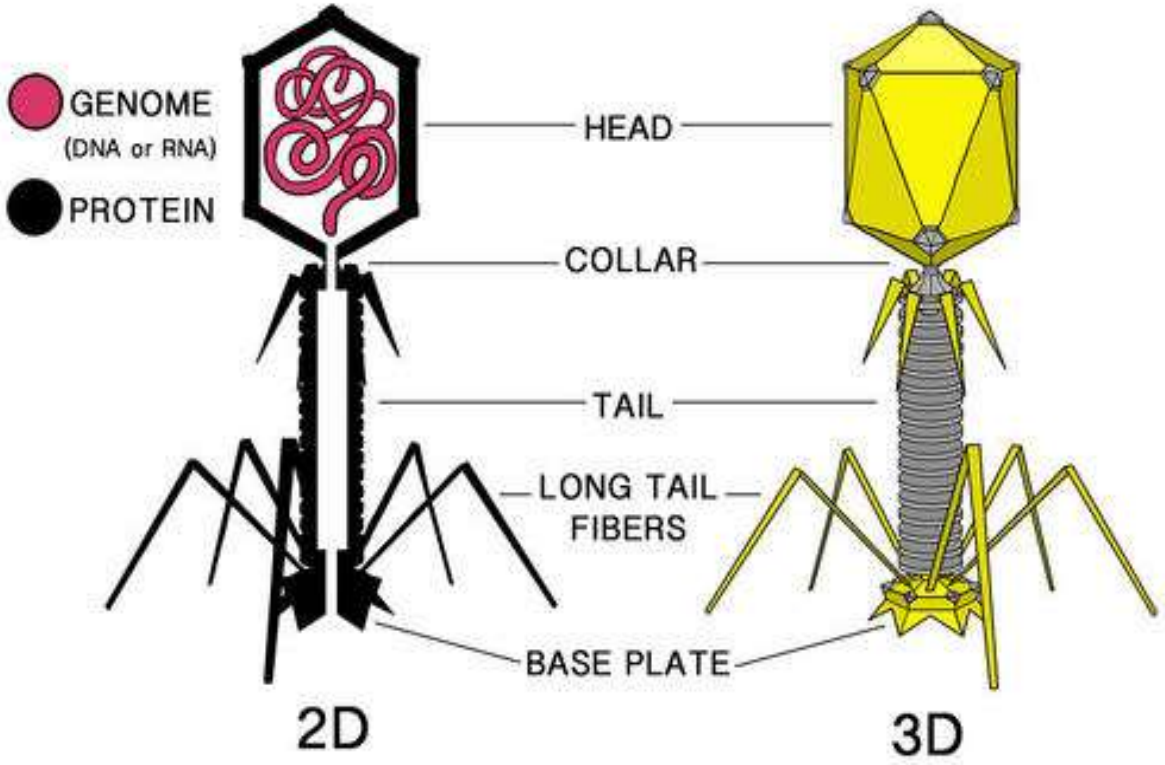
Several

separate

capsomeres

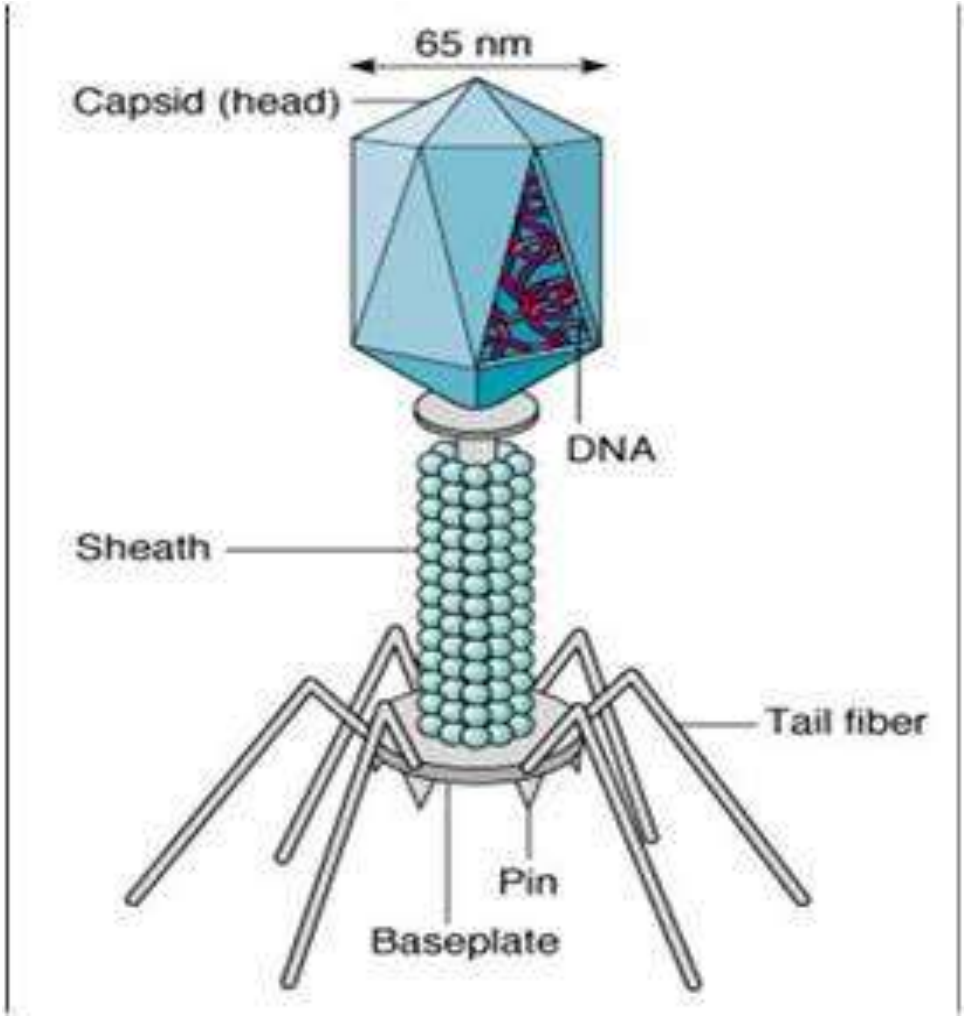
GENOME

- Genome associated with proteins
- Contained in nucleoid



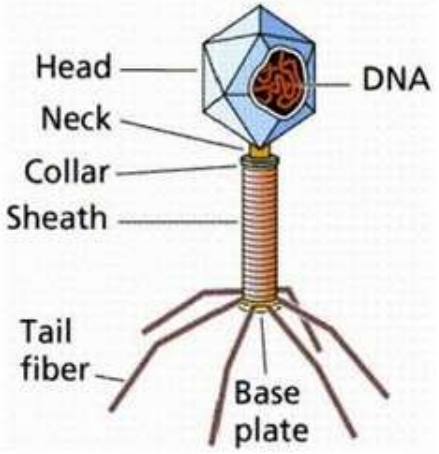
EXAMPLES

- Bacteriophages
- Poxviruses



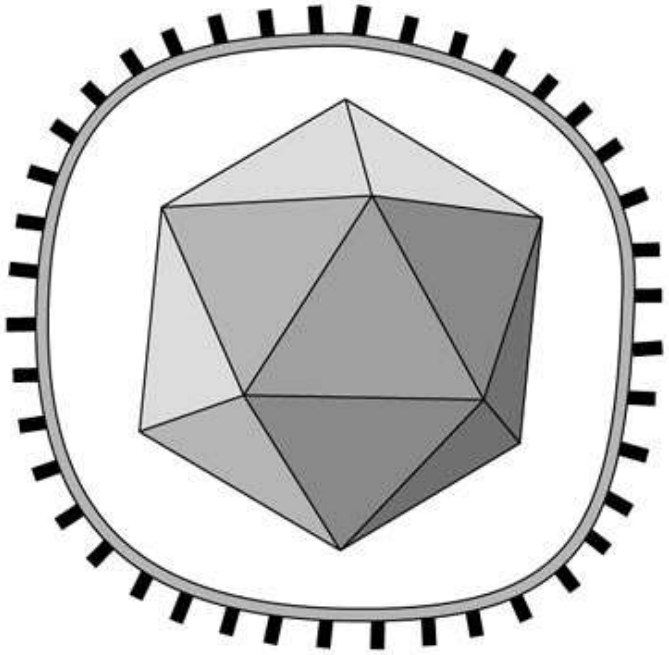
T-PHAGES

- Head and tail
- Called binal symmetry
- Icosahederal head and helical tail symmetry



Ppt no. 39

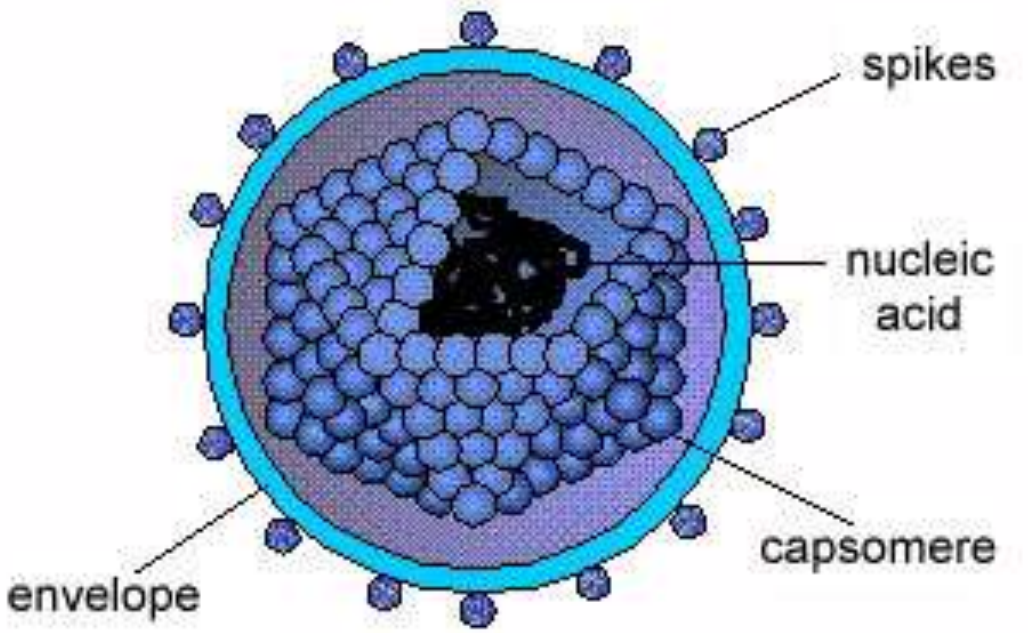
ENVELOPED VIRUSES



- Enveloped viruses are less virulent than non-enveloped viruses.
- These have lipid membrane outside the protein capsid.
- Sensitive to dryness, acid or heat.
- not toxic

EXAMPLES

- Many animal viruses
- HIV
- Influenza virus
- Herpesvirus
- Adenovirus
- Hepatitis B virus
- Parvovirus



CONCLUSION

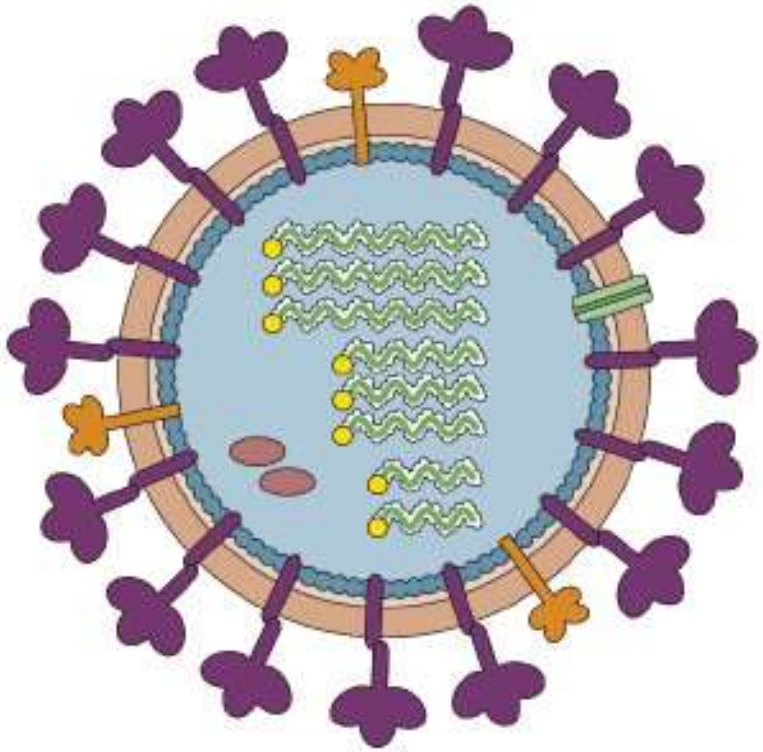
- Surrounds capsid
- Composed of glycoproteins and lipid bilayer

CONCLUSION

- Helps virus to enter host
- Not all viruses have it (Most animal viruses)

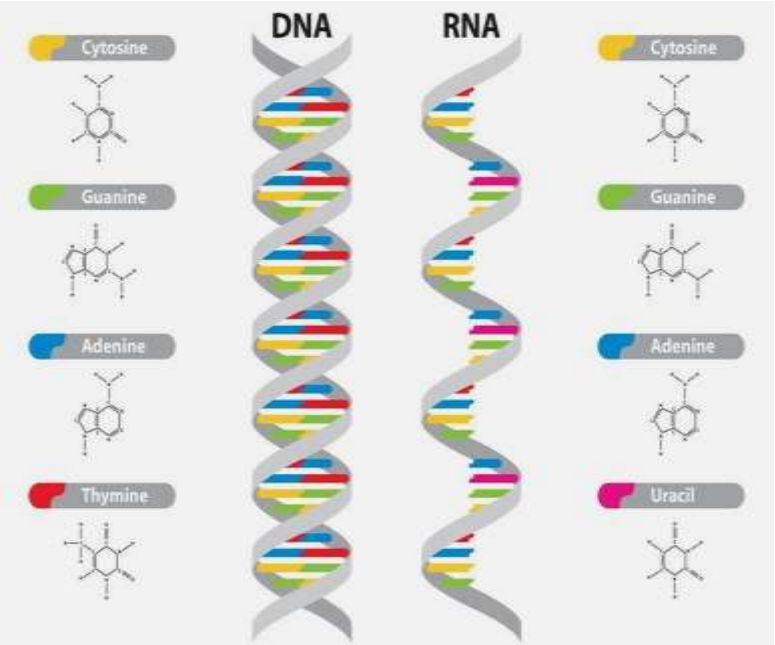
Ppt no. 40

VIRAL GENOME



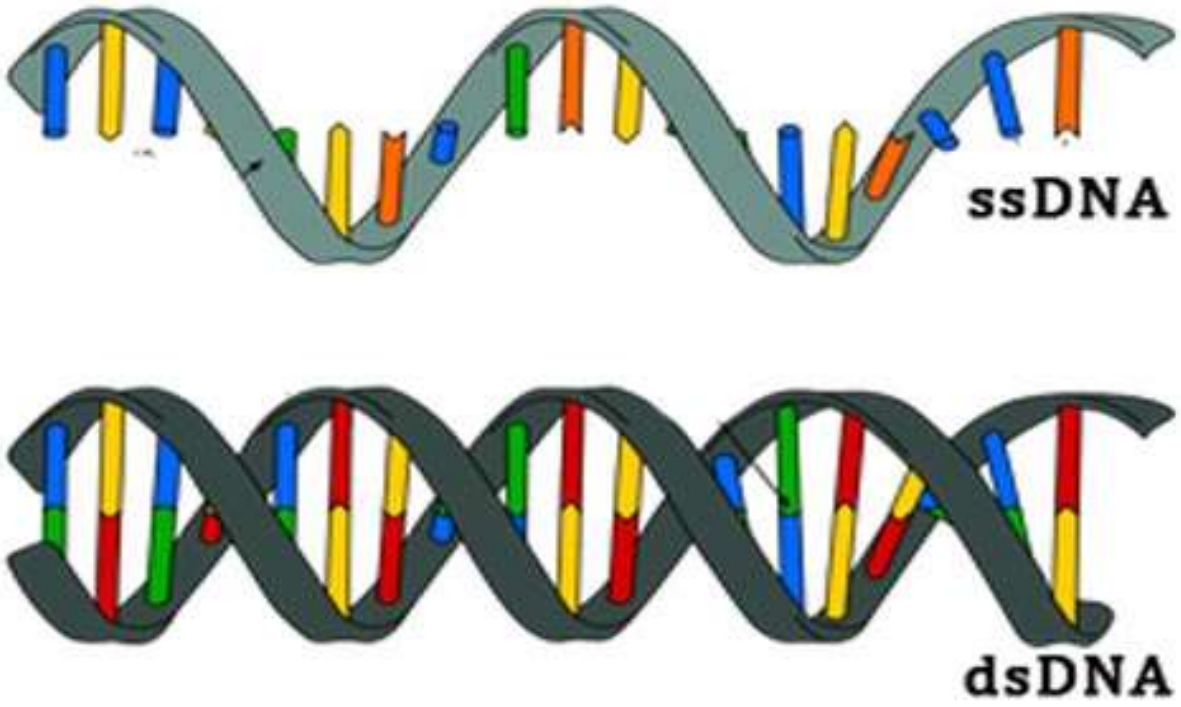
NUCLIEC ACID

- Either DNA
- Or RNA



STRANDEDNESS

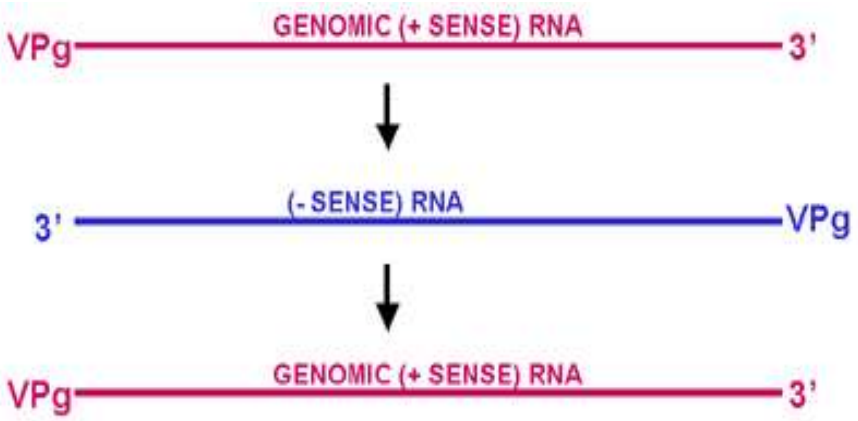
- Single stranded
- Double stranded



SHAPE

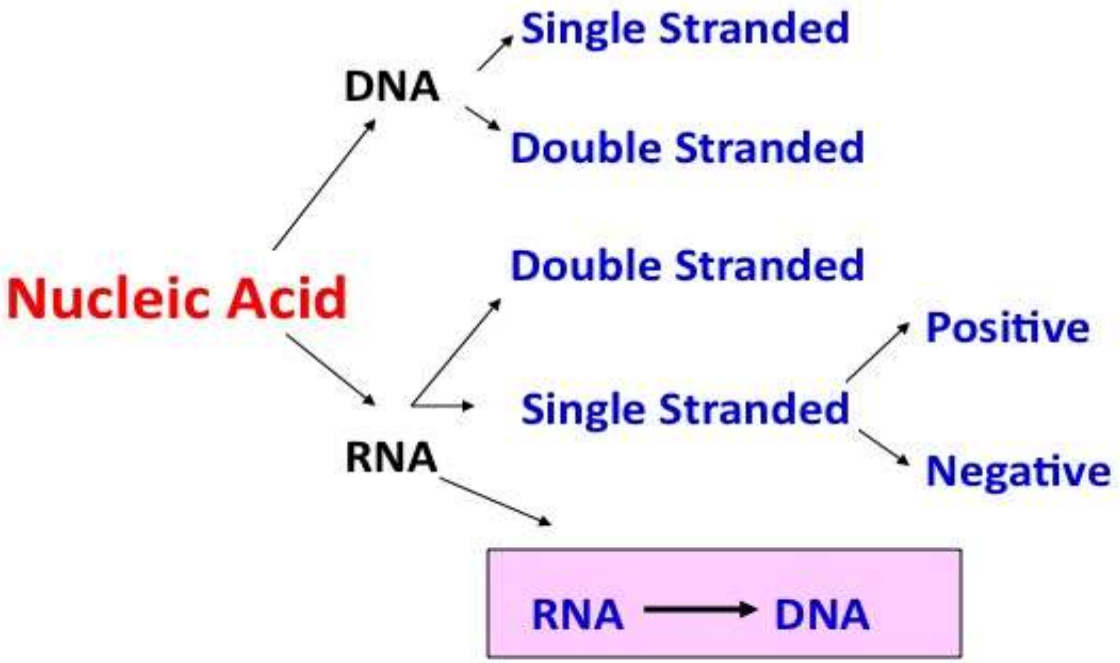
- Circular
- Linear
- Segmented

SENSE



- Positive sense (+)
- Negative sense (-)
- Ambisense (+/-)

Viral Genomes

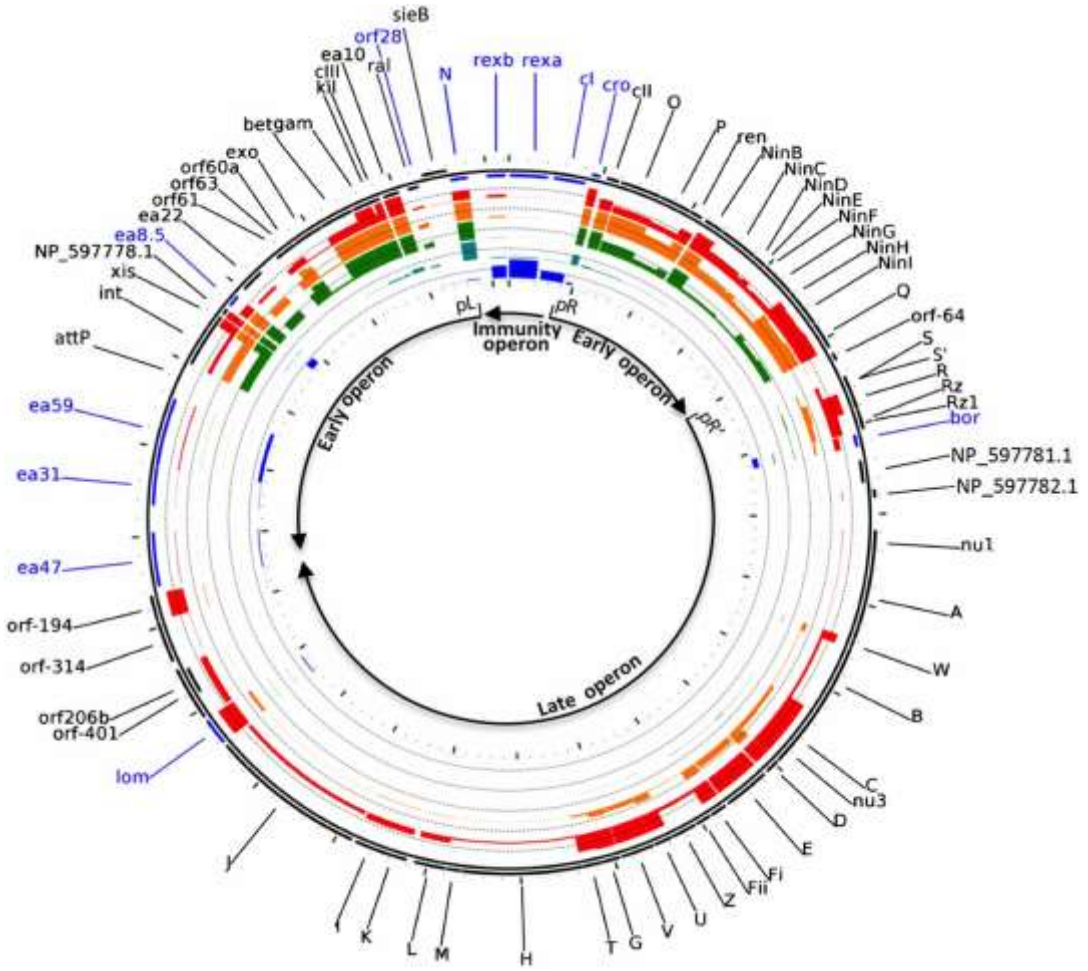


PLANT VIRUSES

- Single stranded
- RNA genome

BACTERIOPHAGES

- Double stranded
- DNA genome



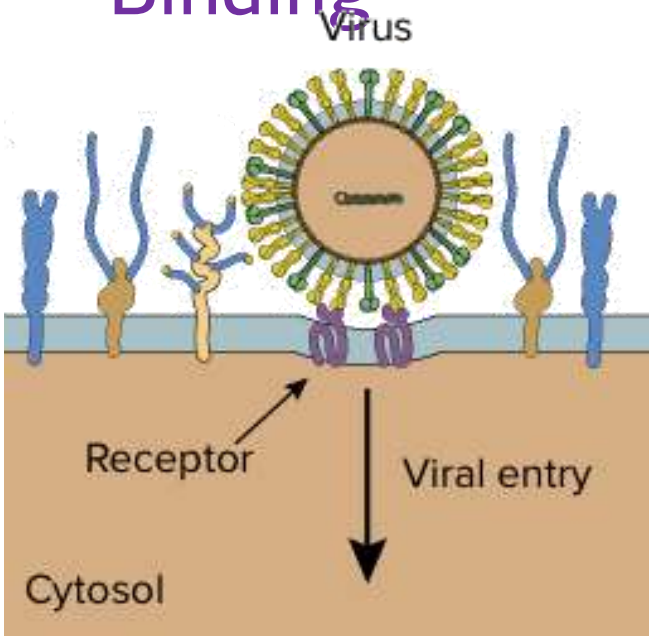
GENOME SIZE

- Varies
- Smallest genome of *Circovirus* (2 kilobases)
- Largest *Pandoravirus* (2 Megabases)

Ppt no. 41

VIRUS RECEPTORS

Recognition and Binding

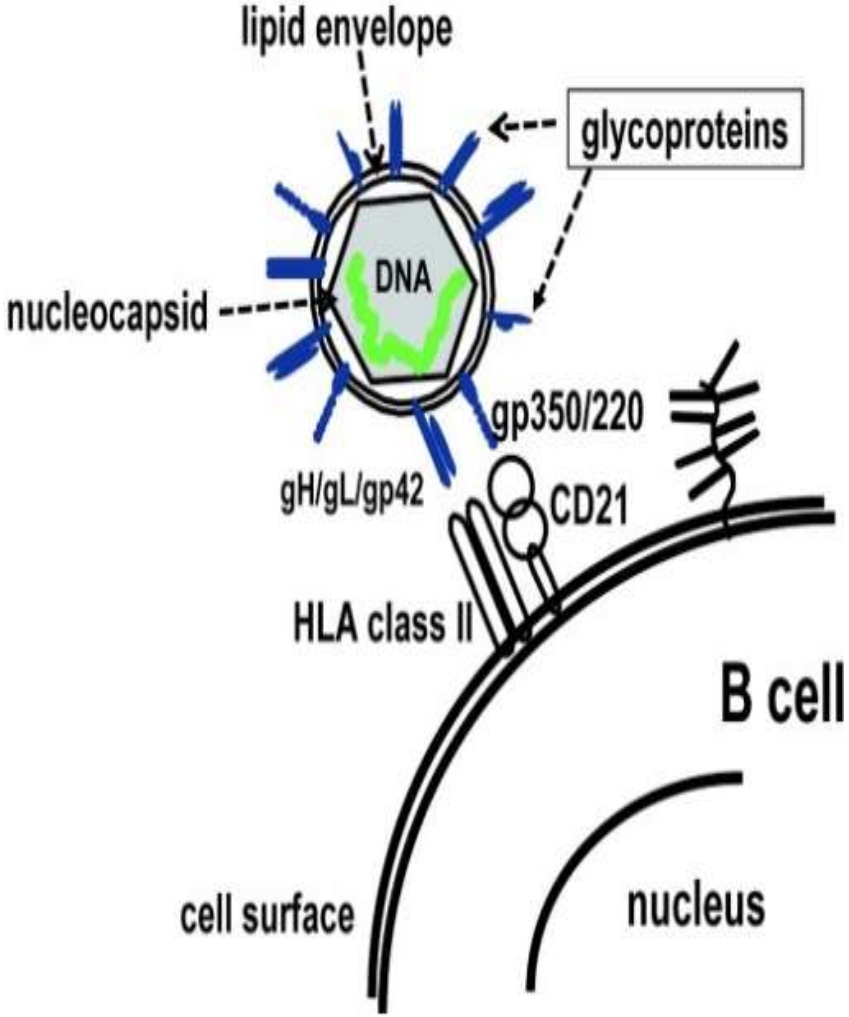


ENTRANCE

- Virus interacts with host receptors
- Cascade of events
- Leading entry of viruses
- Same for enveloped and nonenveloped

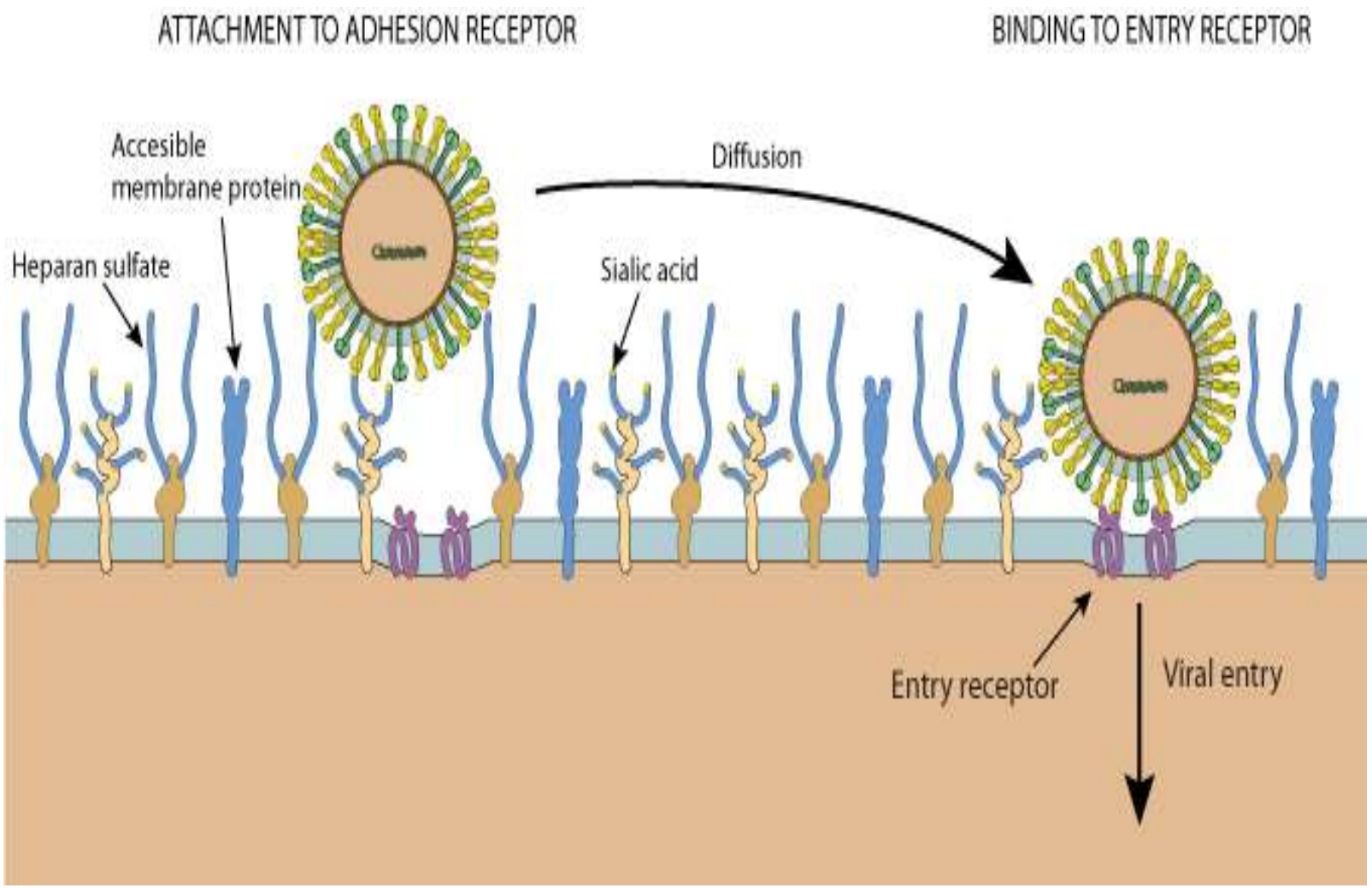
RECEPTORS

- Host and virus interaction through receptors
- Receptors can be protein, lipids or carbohydrates


lipid envelope
glycoproteins
DNA
nucleocapsid
gp350/220
gH/gL/gp42
CD21
HLA class II
B cell
cell surface
nucleus

RECEPTOR RECOGNITION

- Viruses have various receptor recognition structures on surface
- Frequently have spikes
- Bind on the receptors on host cell surface


ATTACHMENT TO ADHESION RECEPTOR
BINDING TO ENTRY RECEPTOR
Accesible
membrane protein
Diffusion
Heparan sulfate
Sialic acid
Entry receptor
Viral entry

ENTRANCE

Variable

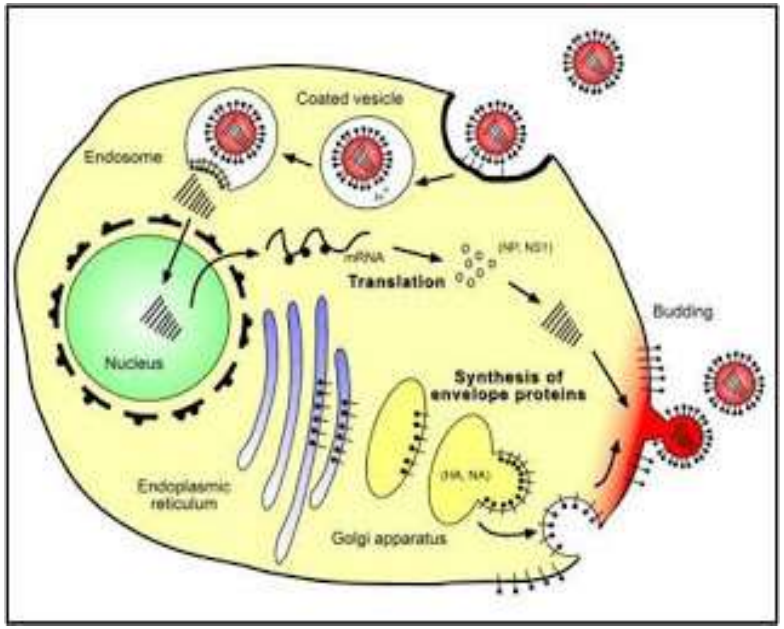
- Pinocytosis/endocytosis

  Membrane fusion

- Viral Penetration

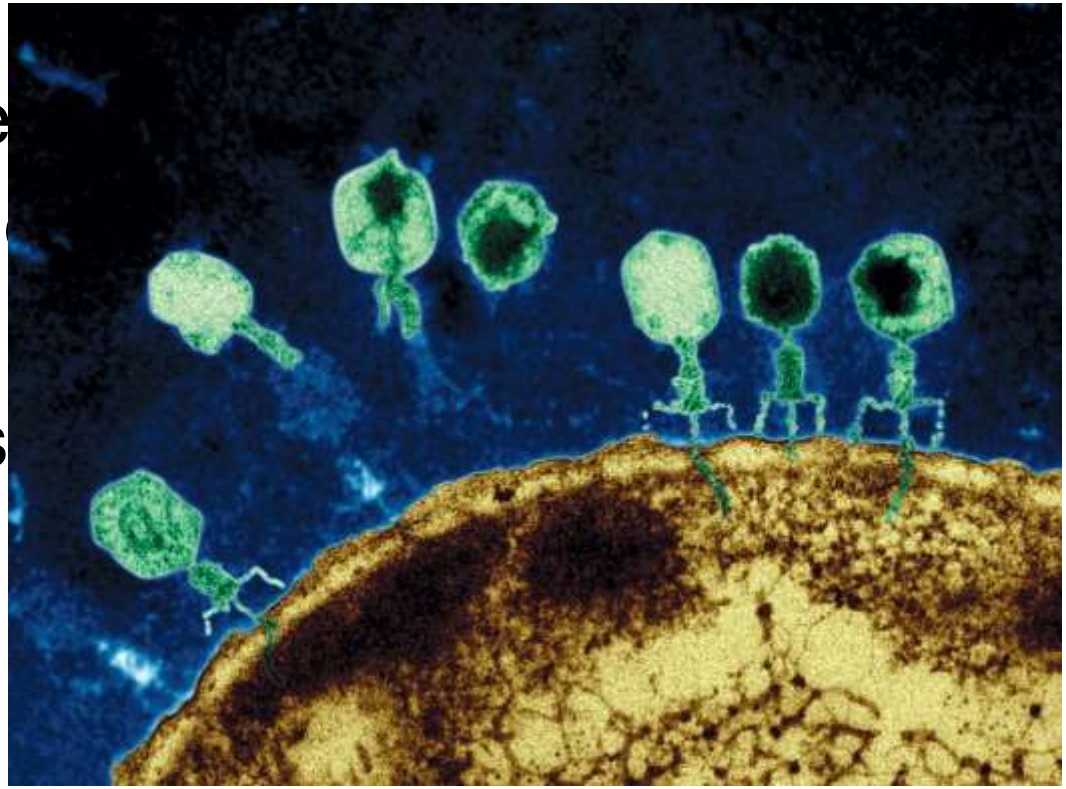
Ppt no. 42

Interaction of Virus and Host Cells

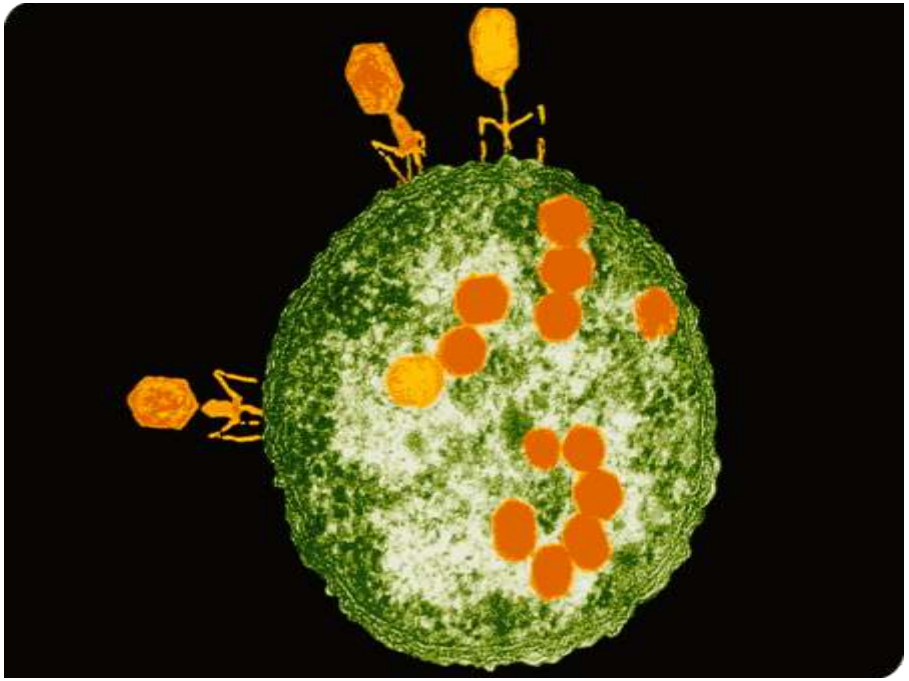


VIRUS-HOST

- Can cause
  - No damage
  - Cell damag
  - Cell death
  - Oncogenes



NON-IMMUNOGENIC RESPONSE



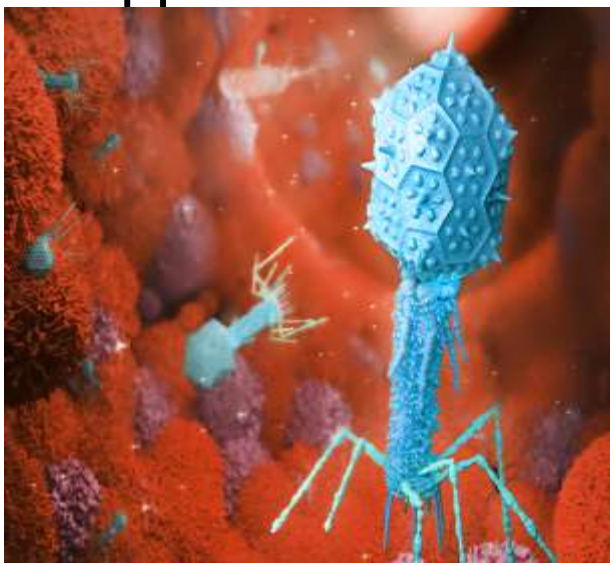
- Phagocytosis
- Raised body temp
- Hormones
- Malnutrition

BARRIERS

- Skin
- Lack of membrane receptors
- Mucus
- Epithelium
- Low pH
- Immune system

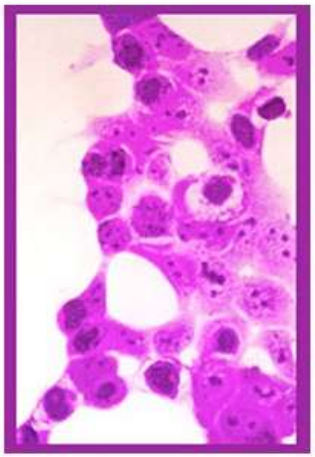
VIRAL INVASION

- Virus can escape immune system
- Cause generalized immuno-suppression

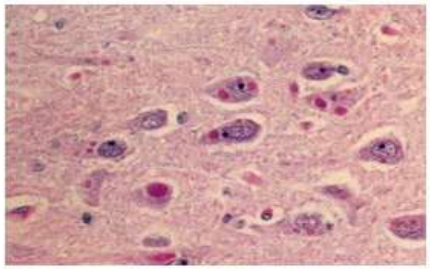


FORMATION OF INCLUSION BODIES

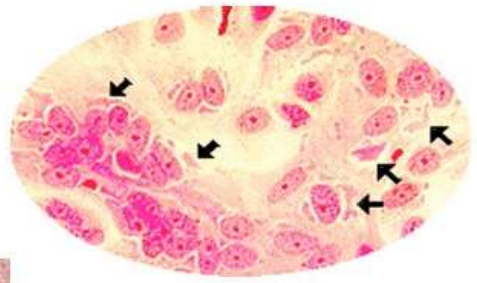
- Nuclear or cytoplasmic aggregates
- Stainable
- Sites of viral multiplication



Adenovirus
nuclear inclusions



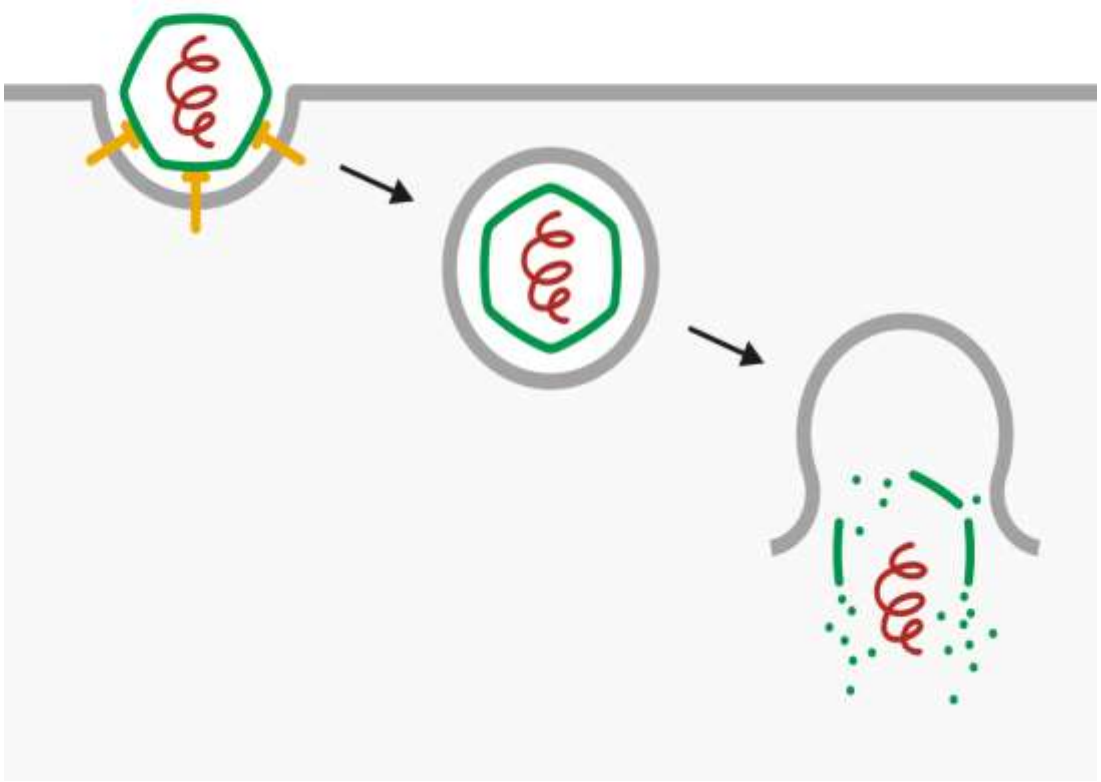
Rabies virus
cytoplasmic inclusions
Negri bodies



RSV
cytoplasmic inclusions

PATHOGENESIS

- Inapparent
- Apparent
- Acute
- Subacute
- Chronic
- Latency

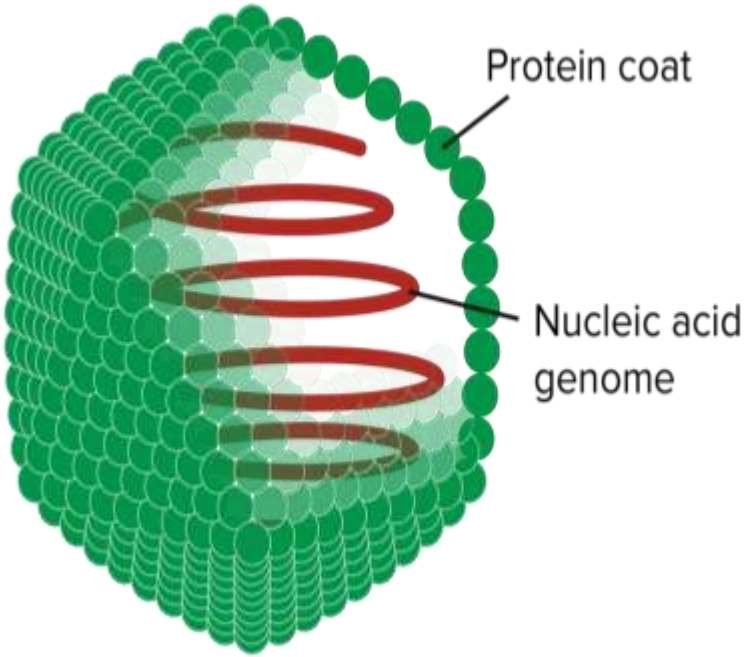


Entrance of virus: ENTRANCE

- Respiratory tract
- Digestive tract
- Skin
- Conjuctivae
- Sexual contact
- Vectors

Ppt no. 43

Protein-Nucleic acid Interactions

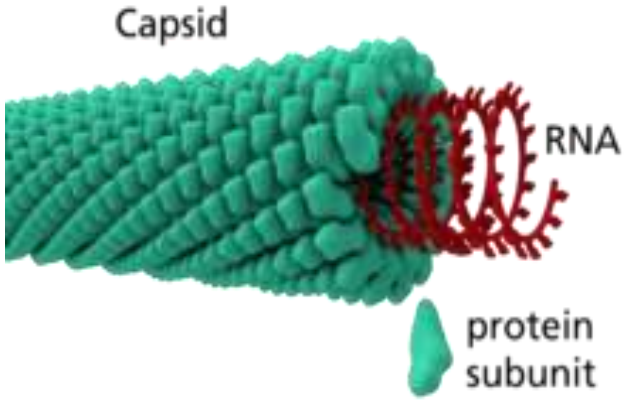


HISTORY

- Late 19th century, Microscopic observation
- Since then researchers have used variety of procedure to understand the relation
- Play a crucial role in biological processes

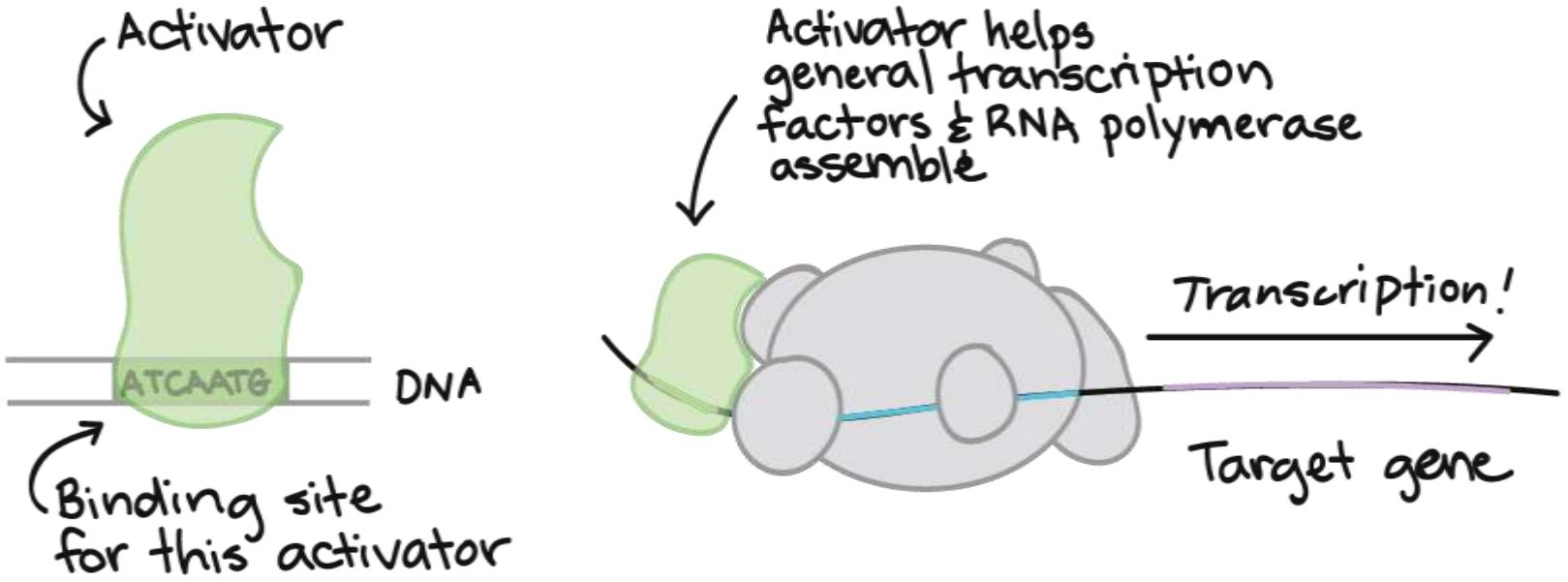
PREOTECTION OF GENOME

- Primary function of virus
- Capsid proteins protect the genome



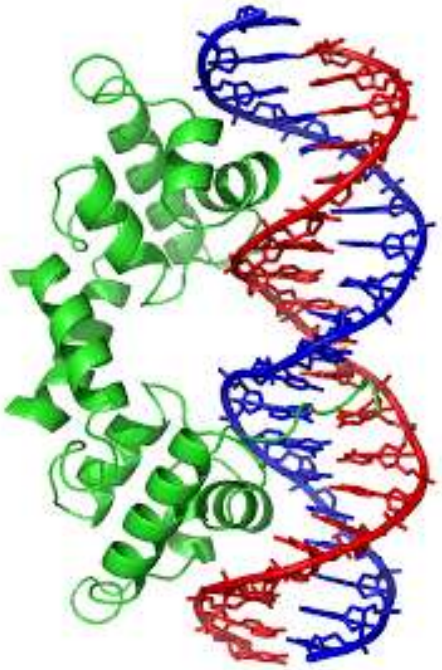
REGULATION OF BIOLOGICAL FUNCTIONS

- Usually expression of gene
- Usually transcription factors
- Gene regulation



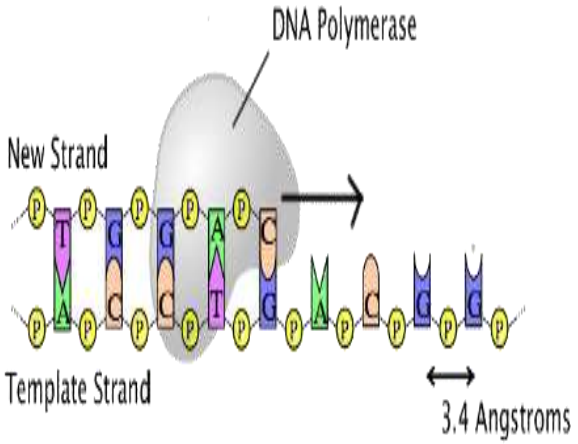
STRUCTURE

- DNA binding motif
- With the help of different ions



ENZYMATIC ROLE

- Polymerases and Restriction Endonucleases
- For replication and translation of proteins

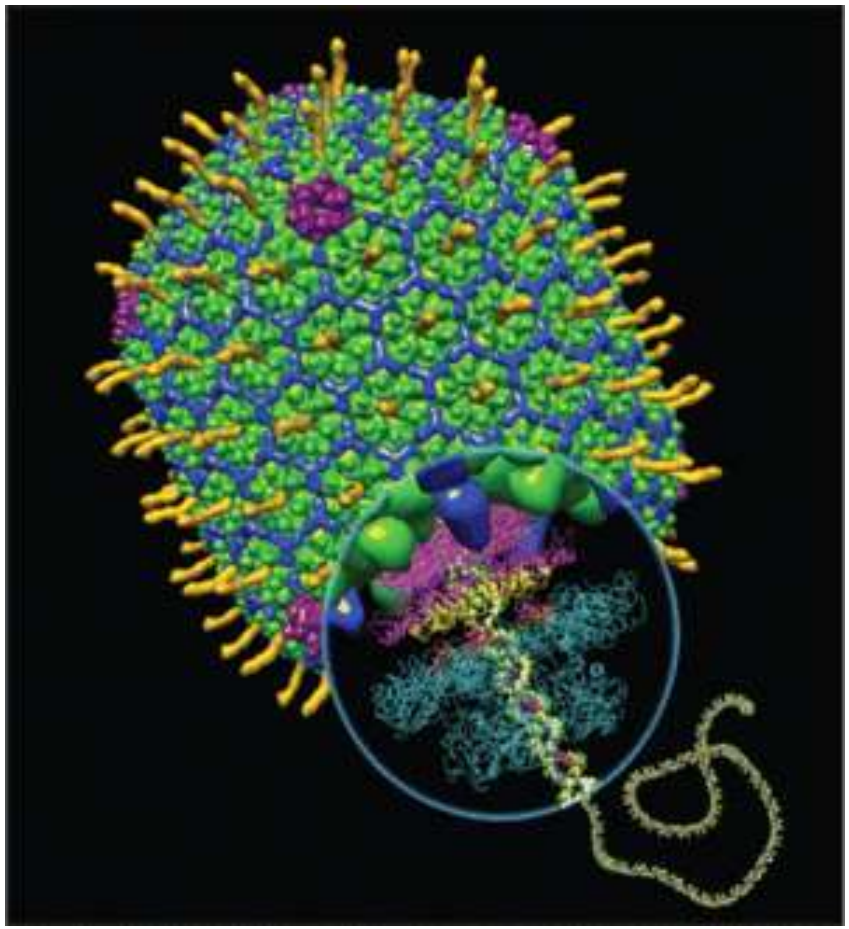


Ppt no. 44

Genome Packaging

- Fundamental step in viral lifecycle
- Usually preformed through capsids
- Capsids are reformed sometimes

PACKAGING MOTOR PROTEIN



- Driven by hydrolysis of ATP
- Condenses genome into a confined place

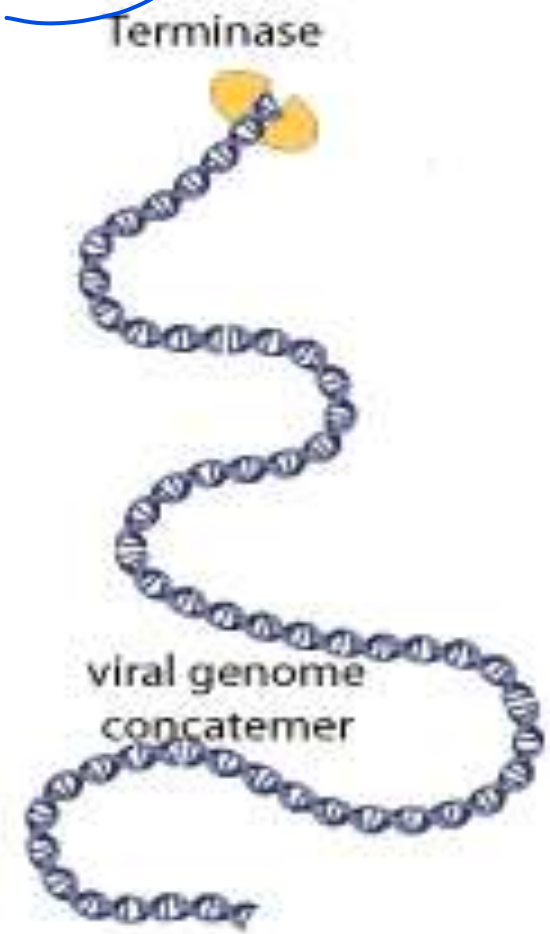
Steps of Genome packaging:

STEPS

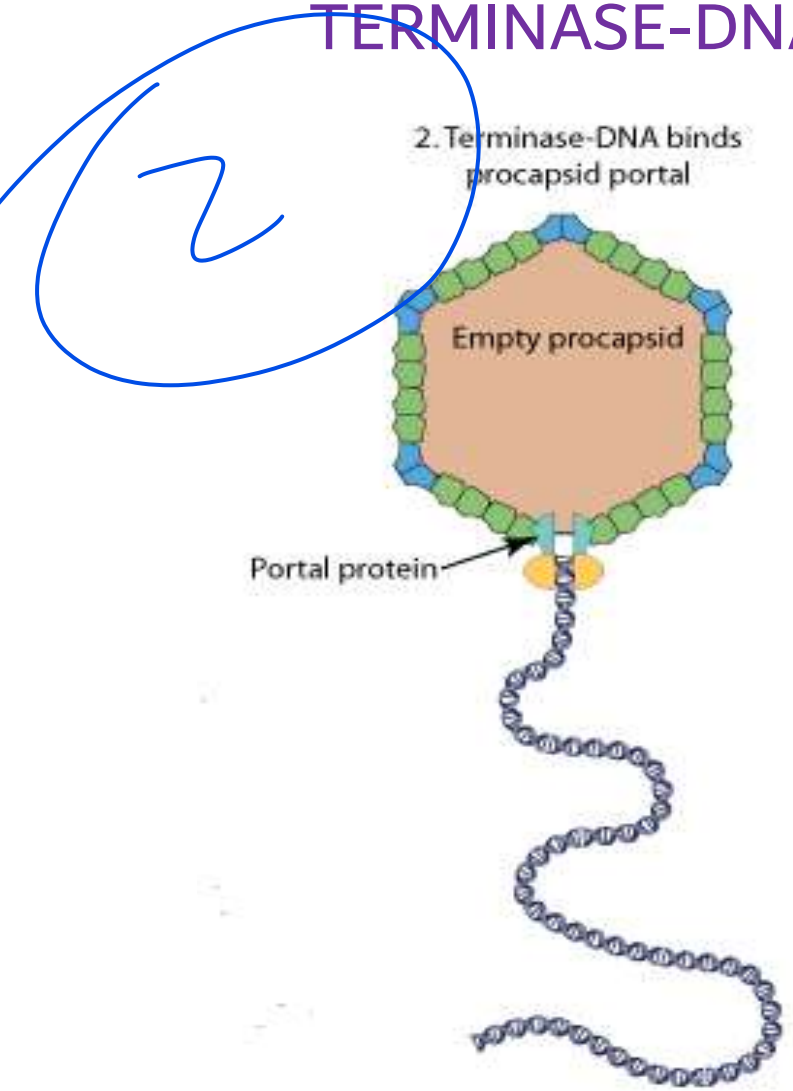
1. Binding of Terminase
2. Binding to Procapsid portal
3. DNA translocation
4. Contamer Cleavage

BINDING OF TERMINASE

- First step
- Terminase binds to viral genome

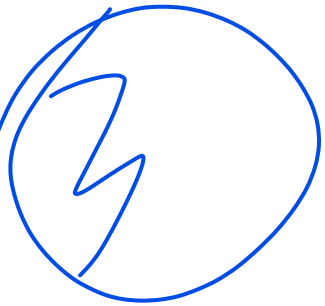


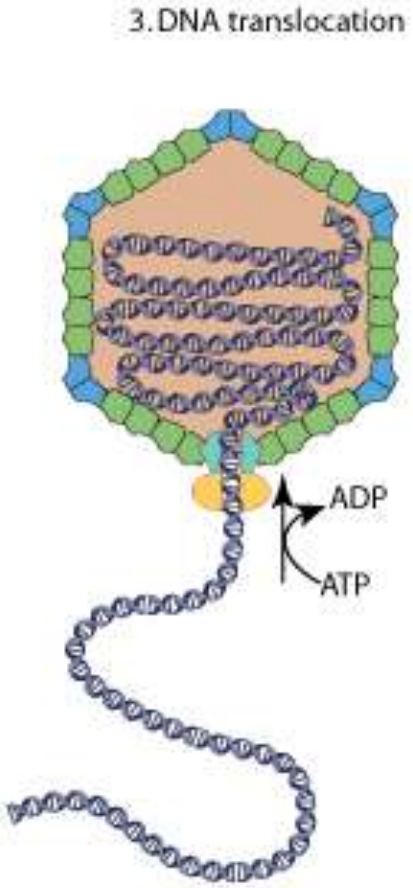
TERMINASE-DNA BINDS TO PROCAPSID PORTAL



- Preassembled capsid or procapsid usually
- Terminase-DNA complex binds to the portal of entry on the capsid

DNA TRANSLOCATION

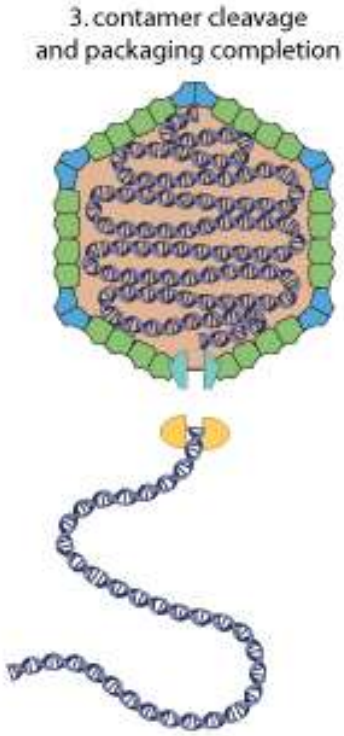




- Portal structure provides entry to DNA
- Packaging motor protein
- Large terminase

CONTAMER CLEAVAGE

- DNA cleavage
- Completion of genome packaging



Ppt no. 45

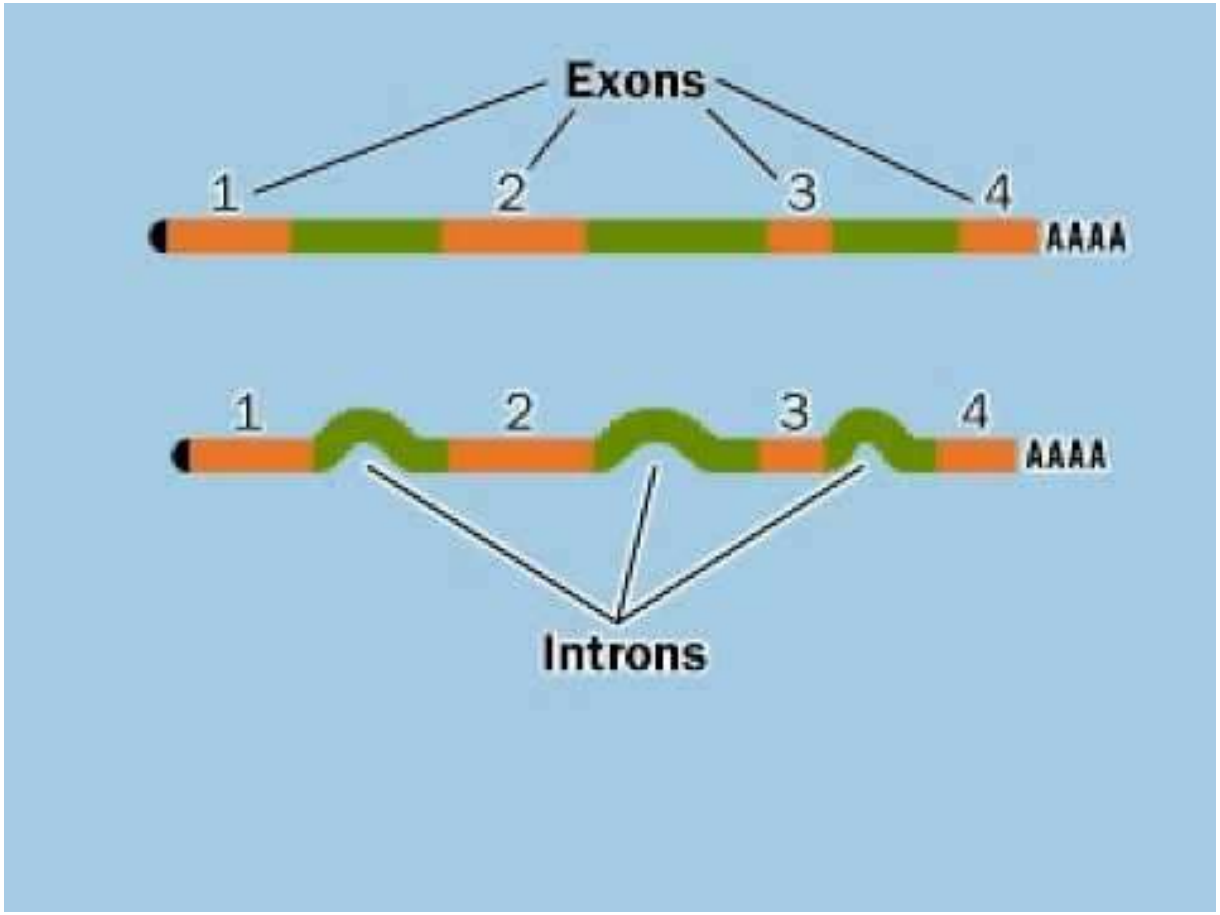
An Introduction to Viral Genomics

VIRAL GENOME

- More complex than any other species
- DNA or RNA
- Single stranded or double stranded
- Positive sense
- Negative sense

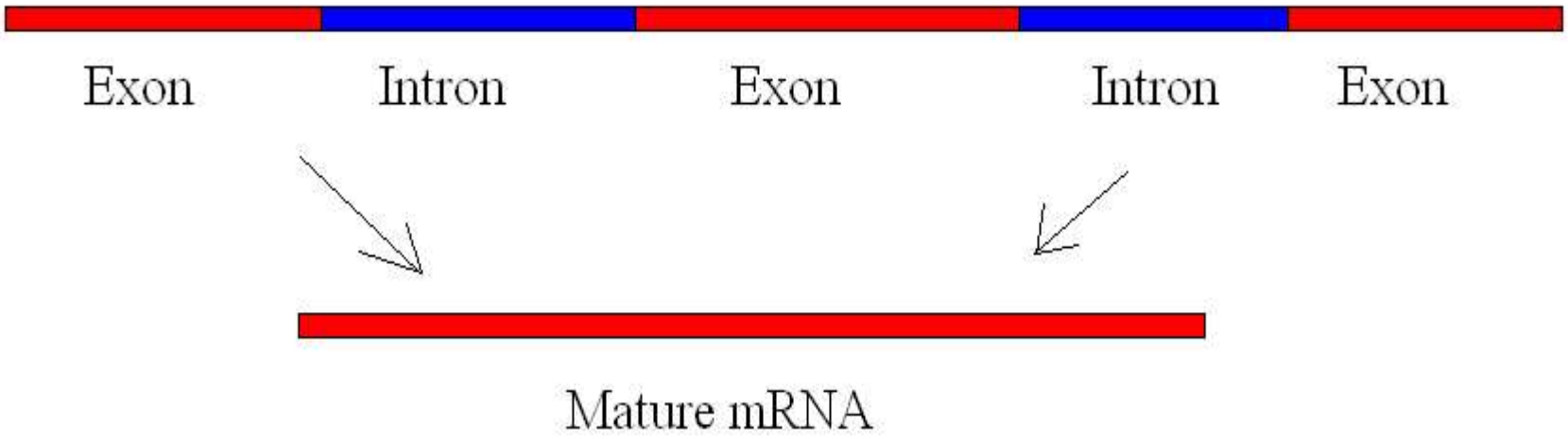
INTRONS

- Non-coding regions on viral genome



EXONS

- Coding regions on viral genome



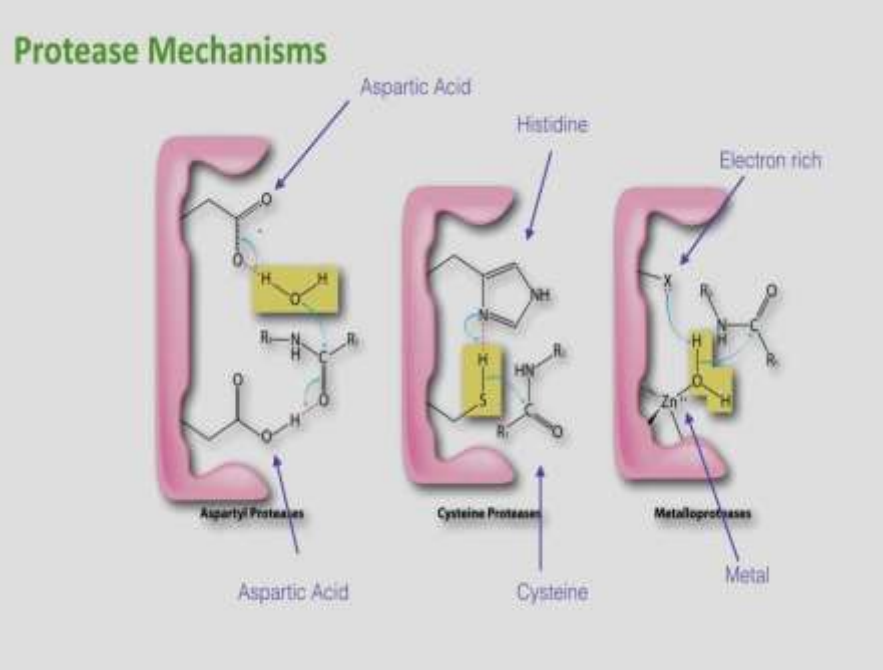
MAIN GENES

- All viruses have these:
- Replicase – an enzyme that replicates genome
- Capsid – a protein that protects the genome

OTHER PROTEIN GENES

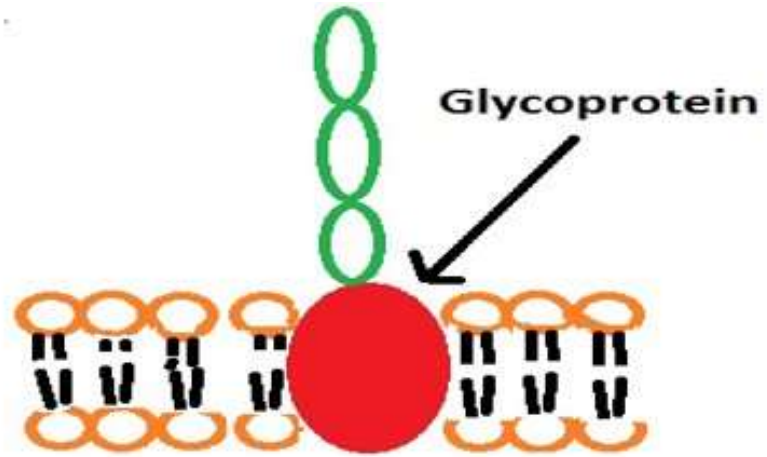
- Protease
- Glycoprotein
- Host-shut-off proteins
- Anti-host defence proteins

PROTEASE



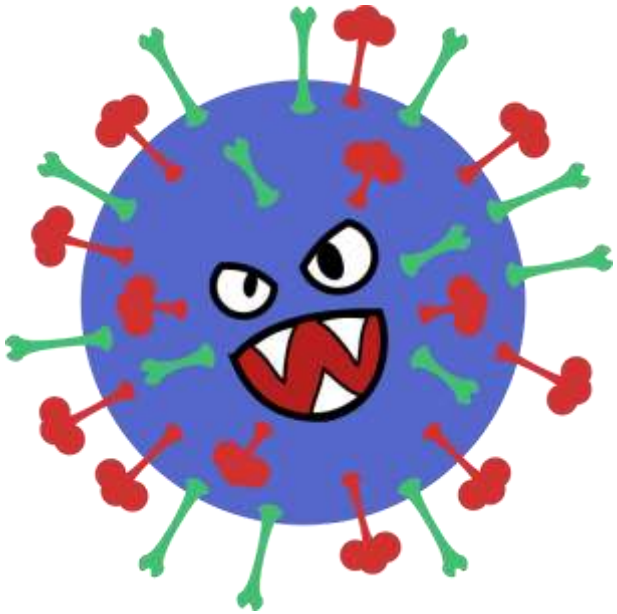
- Processes viral proteins
- Allows assembly
- Maturation to infect other cells

GLYCOPROTEIN



- Only in enveloped viruses
- Allows viral entry into host cell
- Targets specific type of host cells
- Aids in viral assembly

HOST-SHUT-OFF PROTEINS



- As name indicates
- Shut off host activities
- Only viral genes are processed

ANTI-HOST DEFENCE PROTEINS

- Number and type of these genes vary
- Prevent host defense mechanism from stopping viral replication

Ppt no. 46

Terminologies used in Viral Genetics

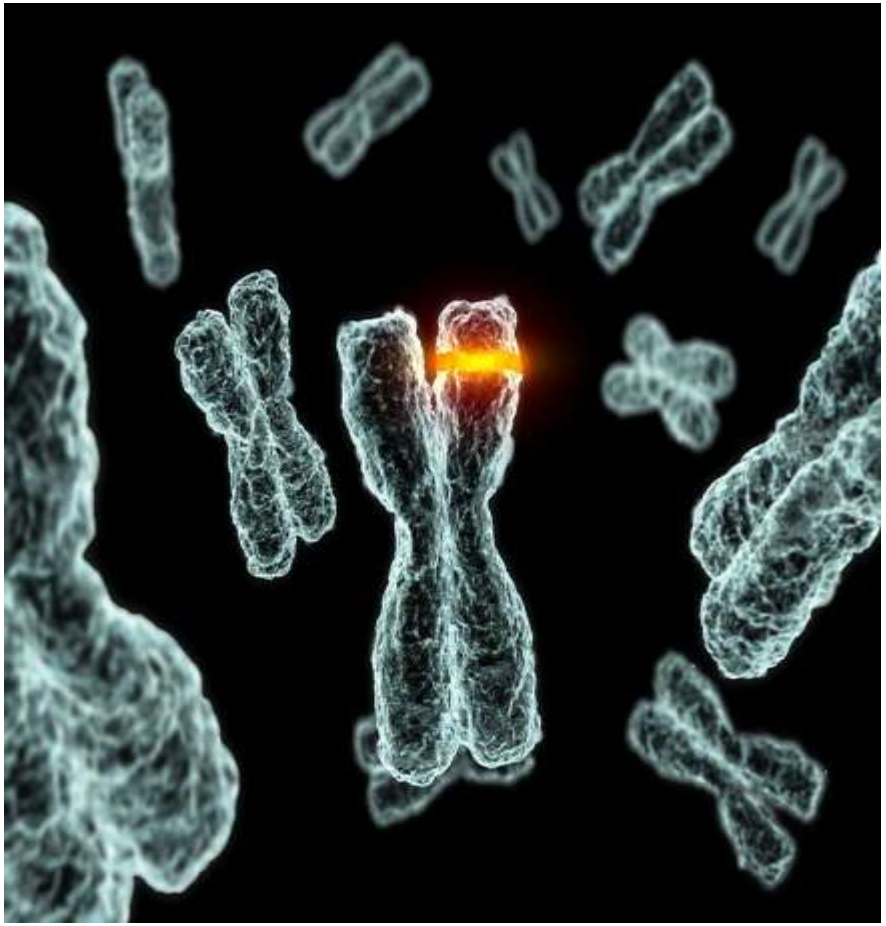
GENE

A segment of DNA molecule that encodes for synthesis of a single polypeptide

GENOME

- The entire DNA sequence of an individual/species

MUTATION



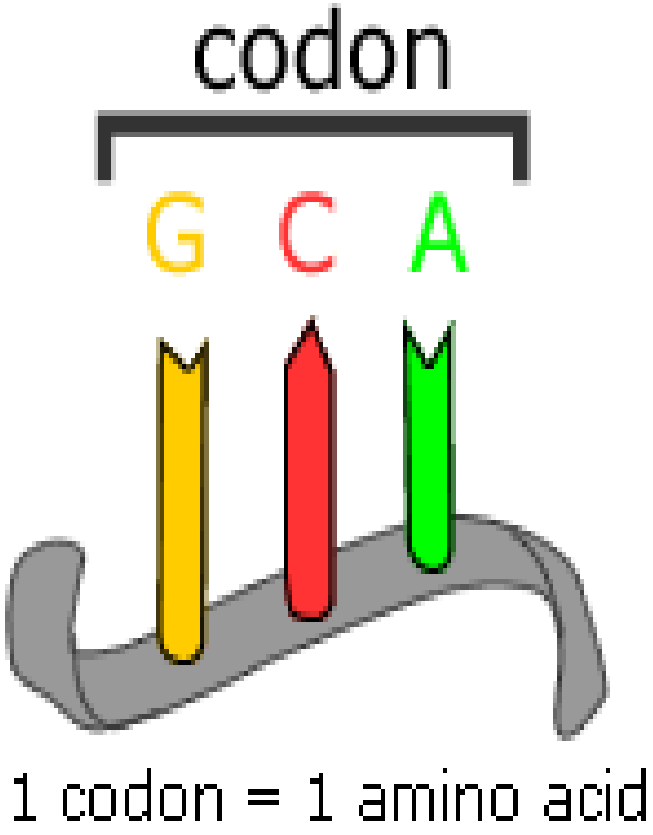
- A permanent inheritable change in a single gene that results in the existence of two or more alleles occurring at the same locus

PSEUDOTYPES

- viruses that have acquired the genotype of one parent and phenotype of another

RECOMBINANTS

- A new combination of genes not found together on the chromosome in either parent
- A recombinant results from crossing over

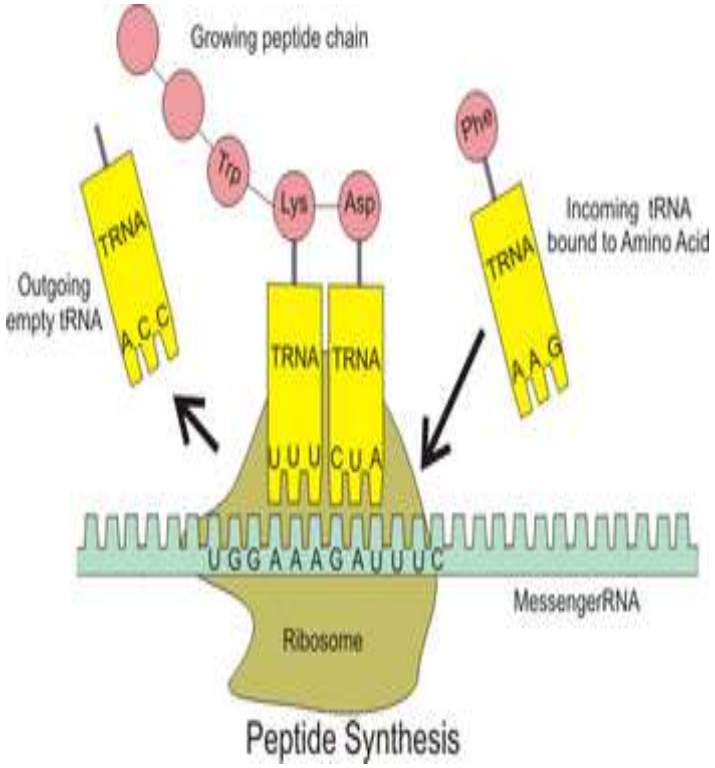


CODON

- A sequence of three bases in DNA or RNA that codes for a single amino acid
- Enables specific proteins to be made by specific genes

ANTI-CODON

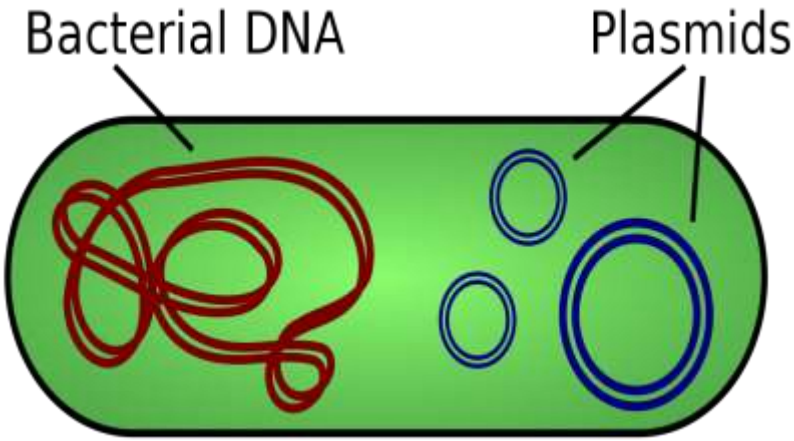
- A sequence of three bases in tRNA that is complementary to a codon in mRNA
- Enables tRNA to sequence amino acids in the order specified by mRNA



RIBOSOMES

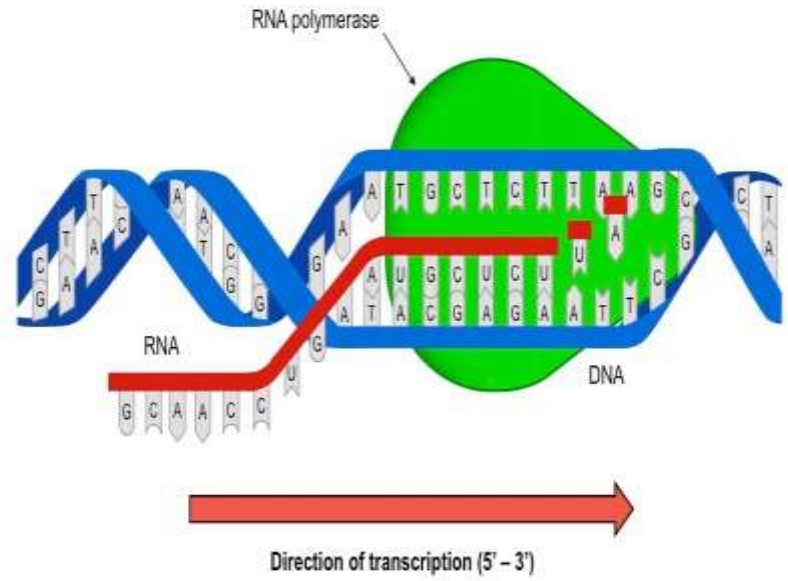
- Complexes of rRNA and protein in cytoplasm that serve as platforms for translation for mRNA into protein

PLASMID



- Extrachromosomal circular DNA in bacteria
- Independently replicate and encode a product

TRANSCRIPTION



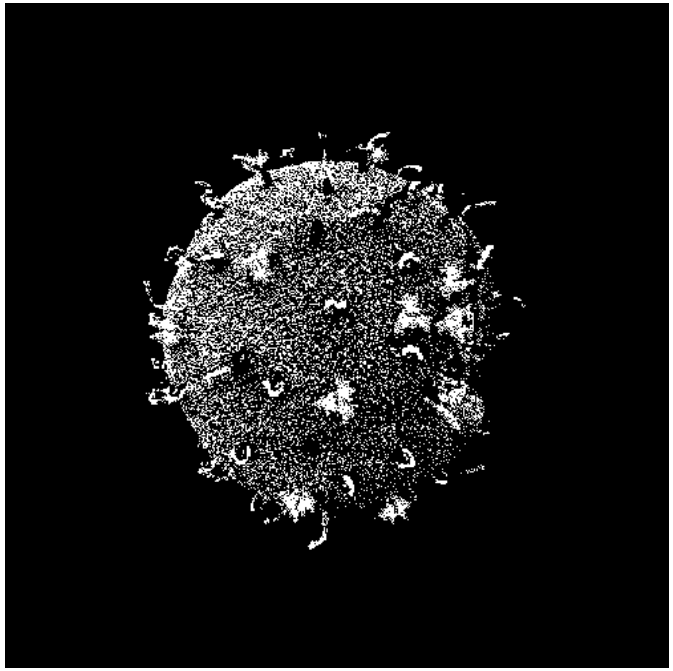
- Synthesis of single-stranded RNA by RNA polymerase using DNA as a template
- The process in the nucleus whereby DNA is transcribed into mRNA

TRANSLATION

- The process of translating the codon sequence in mRNA into polypeptides with the help of tRNA and ribosomes

Ppt no. 47

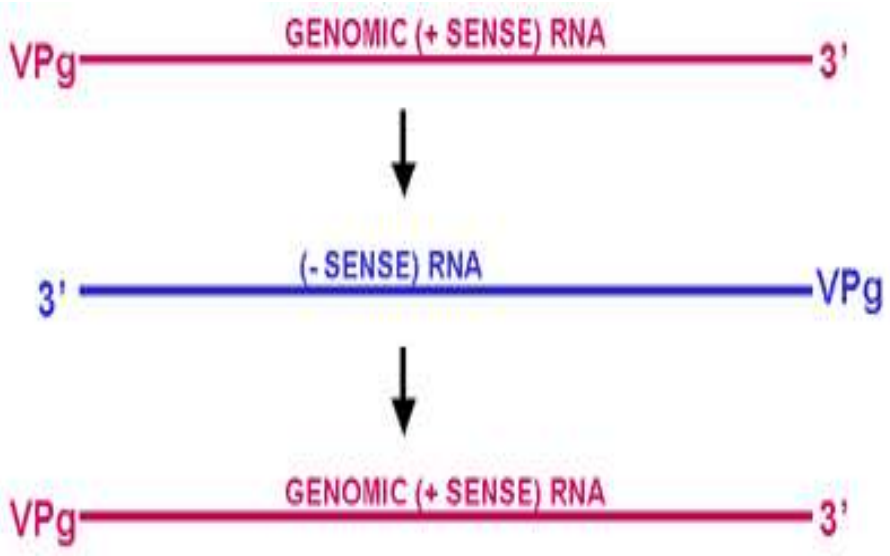
Molecular Genetics of Viruses



VIRAL GENETICS

- Viruses store their genetic info in six different types of genome
- Based on how nucleic acid is transcribed into viral mRNA

POSITIVE RNA STRAND

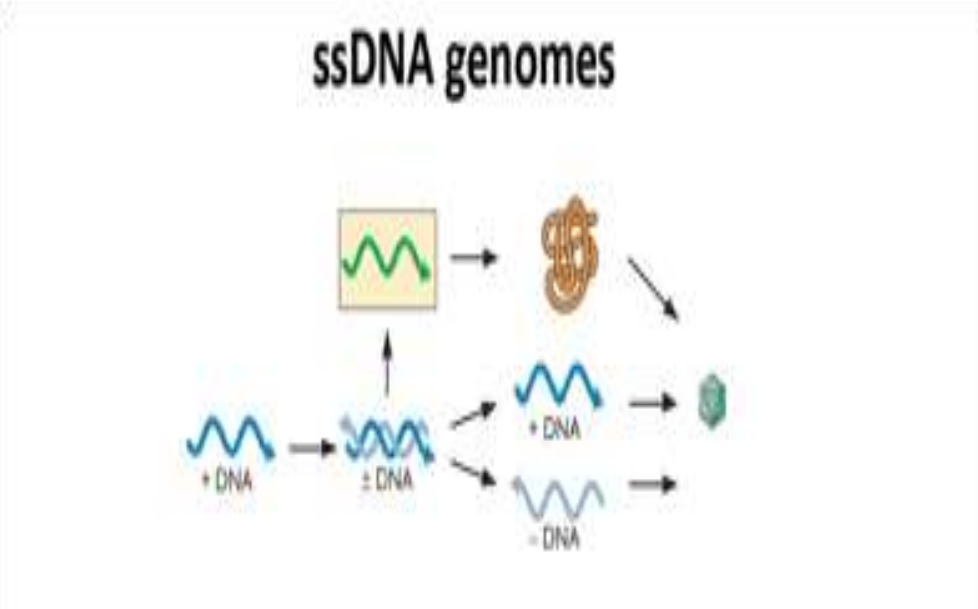


- RNA dependent RNA polymerase produces single stranded negative strand of RNA
- Then it produces positive strand of RNA
- Togavirus, Picornavirus etc.

POSITIVE/ NEGATIVE DNA STRANDS

- Transcribed into mRNA
- DNA dependent DNA polymerase copies the strand to DNA strands
- RNA dependent RNA polymerase converts RNA into mRNA
- Then translated into viral proteins

POSITIVE SINGLE STRANDED DNA

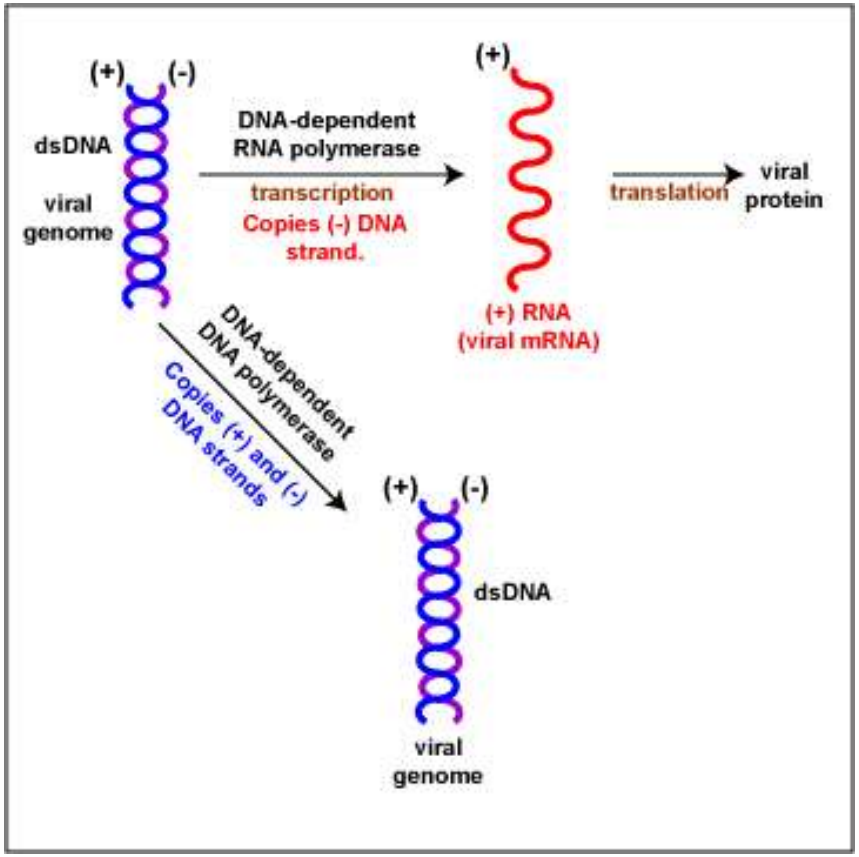


- DNA dependent DNA polymerases forms a double stranded DNA strand

  DNA dependent RNA polymerase forms the mRNA

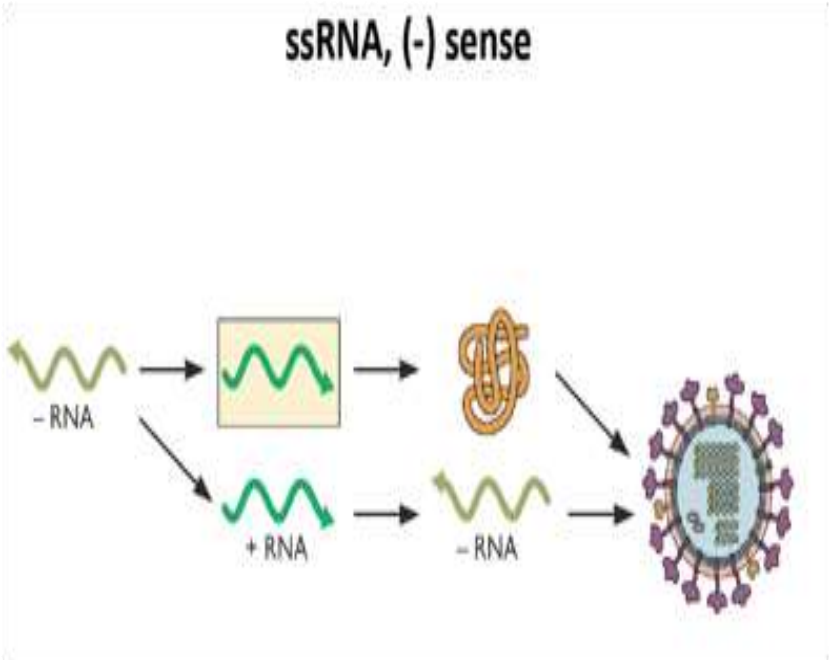
- Example, Parvovirus

POSITIVE/ NEGATIVE DOUBLE STRANDED DNA



- RNA dependent RNA polymerases forms positive and negative RNA strand
- RNA dependent RNA polymerase forms the mRNA
- Translated into proteins

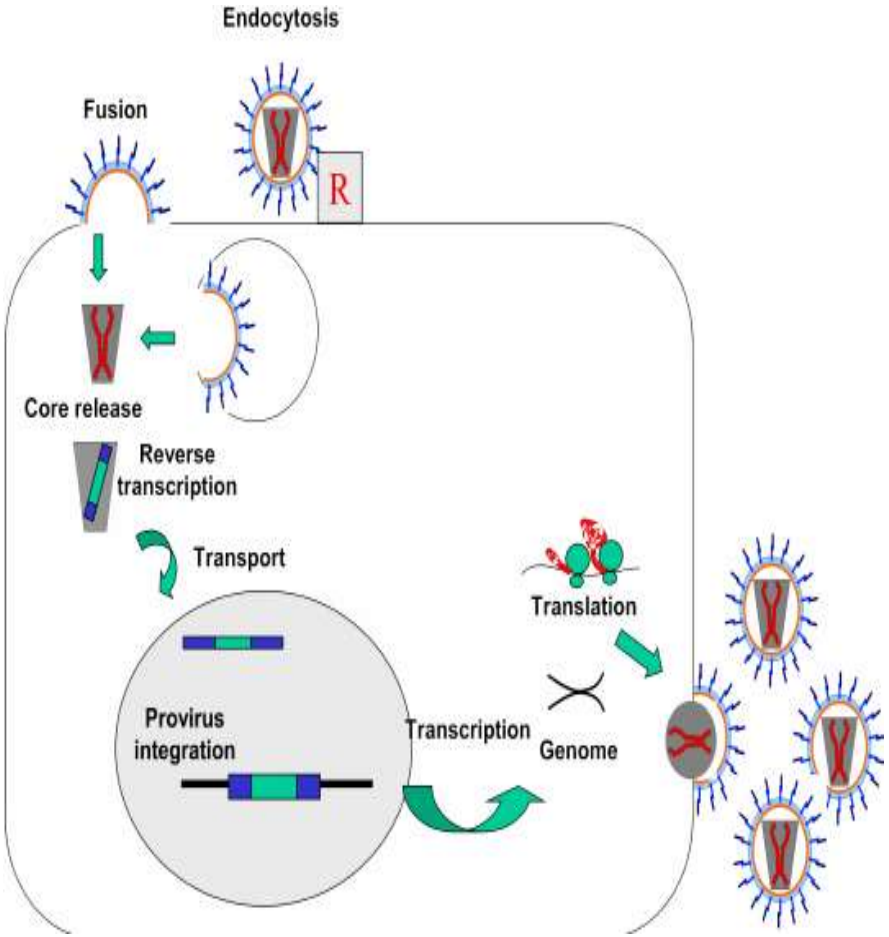
NEGATIVE SINGLE STRANDED DNA



- RNA dependent RNA polymerases forms positive RNA strand
- RNA dependent RNA polymerase forms the mRNA
- Translated into proteins

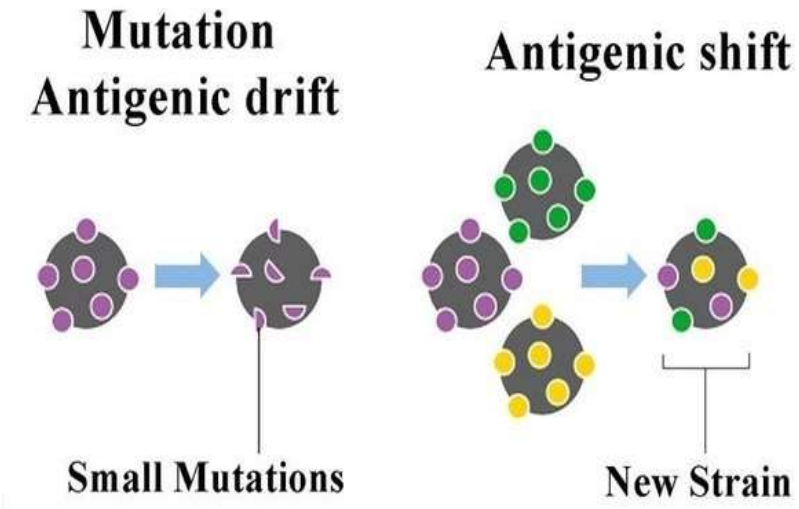
POSITIVE RNA RETEROVIRUSES

- Reverse transcriptase enzyme
- Produces single stranded negative sense DNA
- DNA dependent DNA Polymerase copies to form double stranded DNA



- DNA dependent RNA polymerase produces single stranded positive RNA genomes (mRNA)
- Examples, HIV-1, HIV-2, HTLV-1

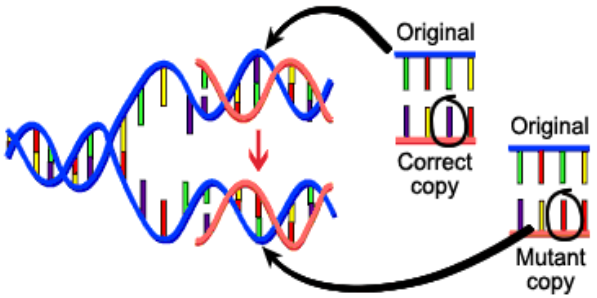
VIRAL GENETICS



- Viruses usually grow rapidly, producing more chance of mutation
- Genetic changes by various mechanisms:
  - Genetic drift (Bases change)
  - Antigenic Shift (Recombination)

MUTATION

- Spontaneous Mutation that arise naturally during replication
- Induced Mutation that is caused by physical or chemical means



Ppt no. 48

Genetic Techniques in Virology

There are two types of Genetic techniques in virology:

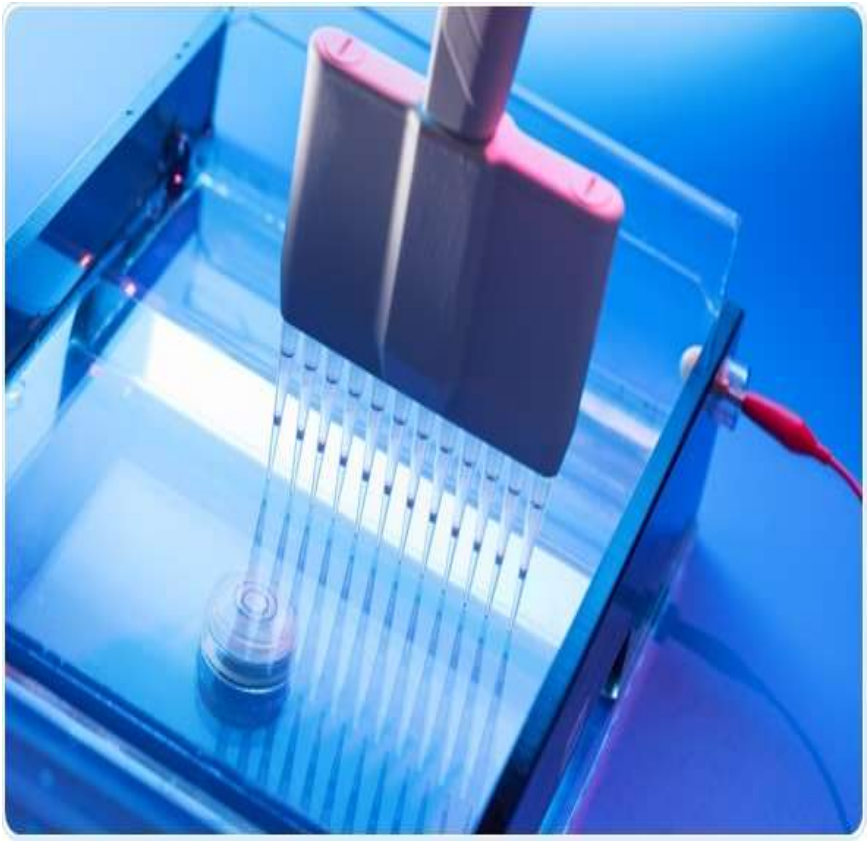
TYPES

- In-vitro (Physical analysis)
- In-vivo (viral genotype and phenotpe)

IN-VITRO

- Physical analysis
- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using Reverse transcriptase

MOLECULAR ANALYSIS



- Molecular cloning
- Gel electrophoresis
- Hybridization of complementary nucleotide sequences

(In-vivo) PHENOTYPIC ANALYSIS

- Standard technique in virology
- Cell-free extracts to translate mRNAs
- Transfection tests
- In-vitro protein formation
- In-vivo protein replication

Ppt no. 49

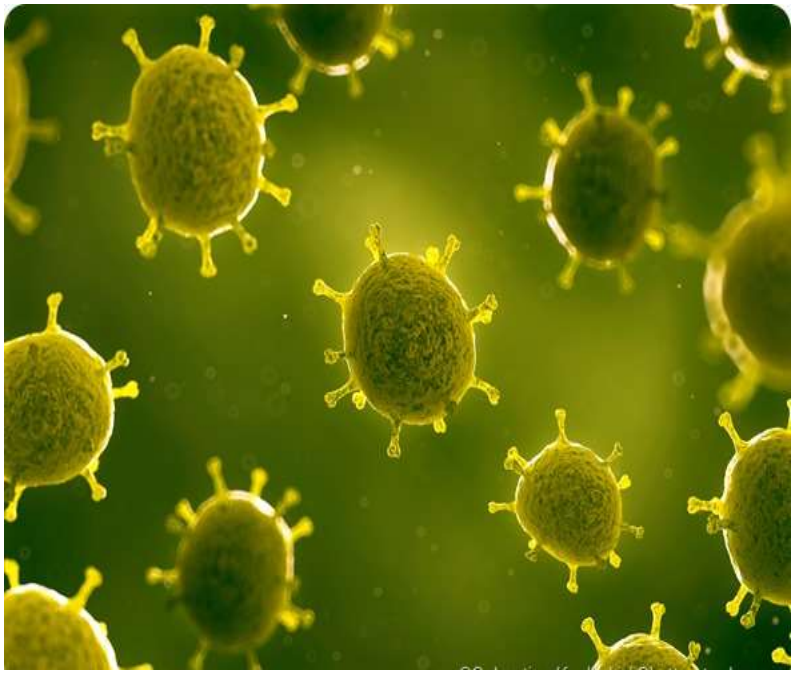
LATENT VIRUSES

MEANING

- The pathogenic viruses having the ability to lie dormant (or latent) within host cell
- Called lysogenic part of lifecycle

- Latency refers to a phase in certain viral infections during which the virus remains in the host's body in a dormant state, without causing active disease.
- During latency, the virus is present in the host cells but is not actively replicating or causing symptoms.

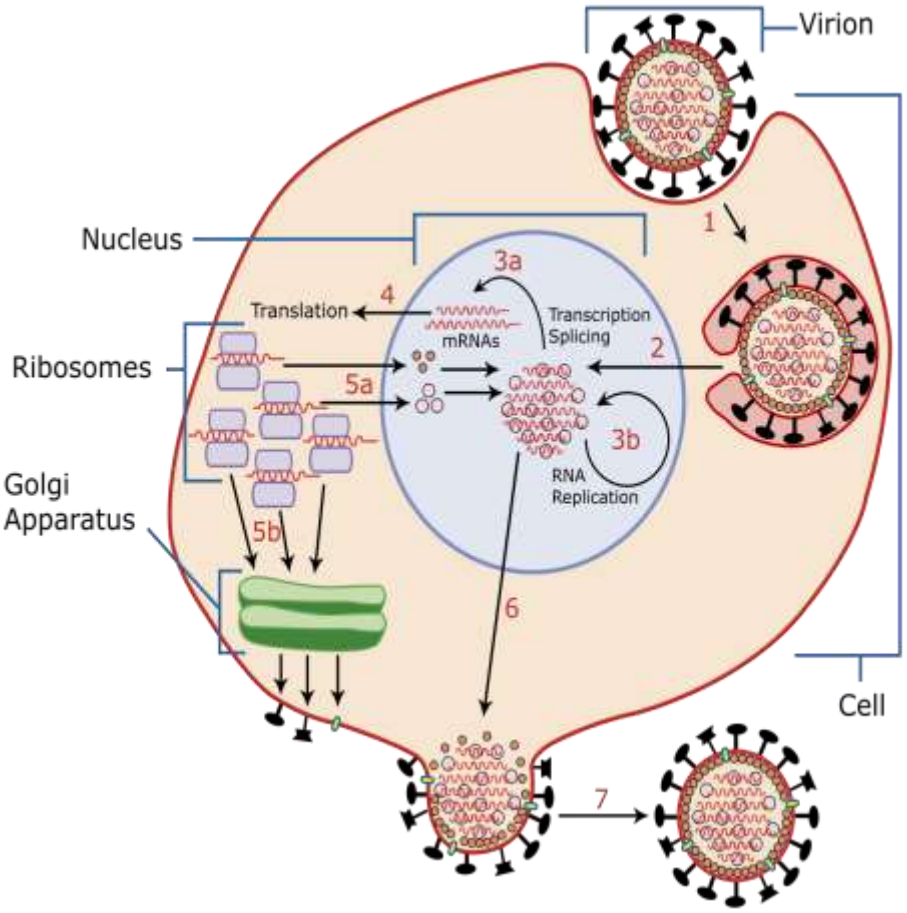
LATENCY



- The phase in which viral proliferation ceases after initial infection
- Viral genome not fully eradicated

Examples of Latent Viruses:
-Herpes Simplex Virus (HSV)
-Varicella-Zoster Virus (VZV)
-Human Immunodeficiency Virus (HIV)
-Epstein-Barr Virus (EBV)

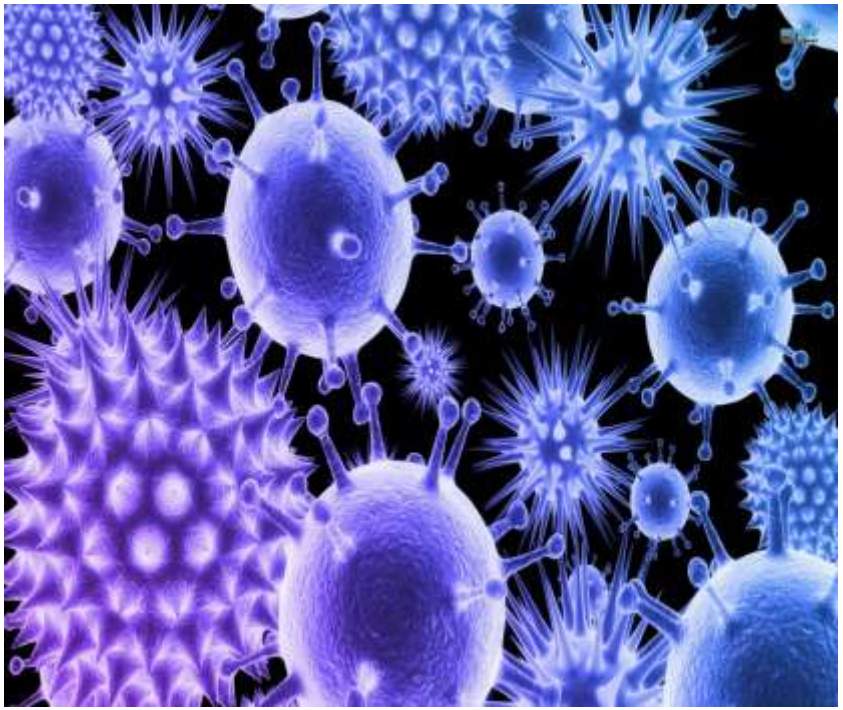
LYTIC PHASE



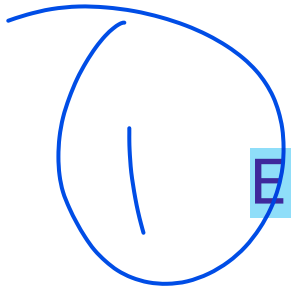
- The virus activates
- Continues to multiply increases its progeny
- Clinical signs appear

-The lytic phase is a stage in the viral lifecycle during which the virus actively replicates within a host cell, leading to the production of new virions (virus particles).
-This process often results in the lysis (destruction) of the host cell, releasing the new virions to infect additional cells.

MECHANISMS



1. Episomal Latency
2. Proviral Latency
3. Maintaining Latency



EPISOMAL LATENCY

- Use of genetic episomes
- Viral genes float in cytoplasm or nucleus
- More vulnerable to host immune system
- Example, chicken pox and herpes simplex viruses

PROVIRAL LATENCY

- Virus genome is integrated into the DNA of a host cell
- Automatic host cell division during viral replication
- Viral genome enters nucleus
- Example; HIV

MAINTAINING LATENCY

- For continuity of infection
- By genes expressed during latency
- To evade host immune systme
- To protect virus genomre to be digested by cellular enzymes

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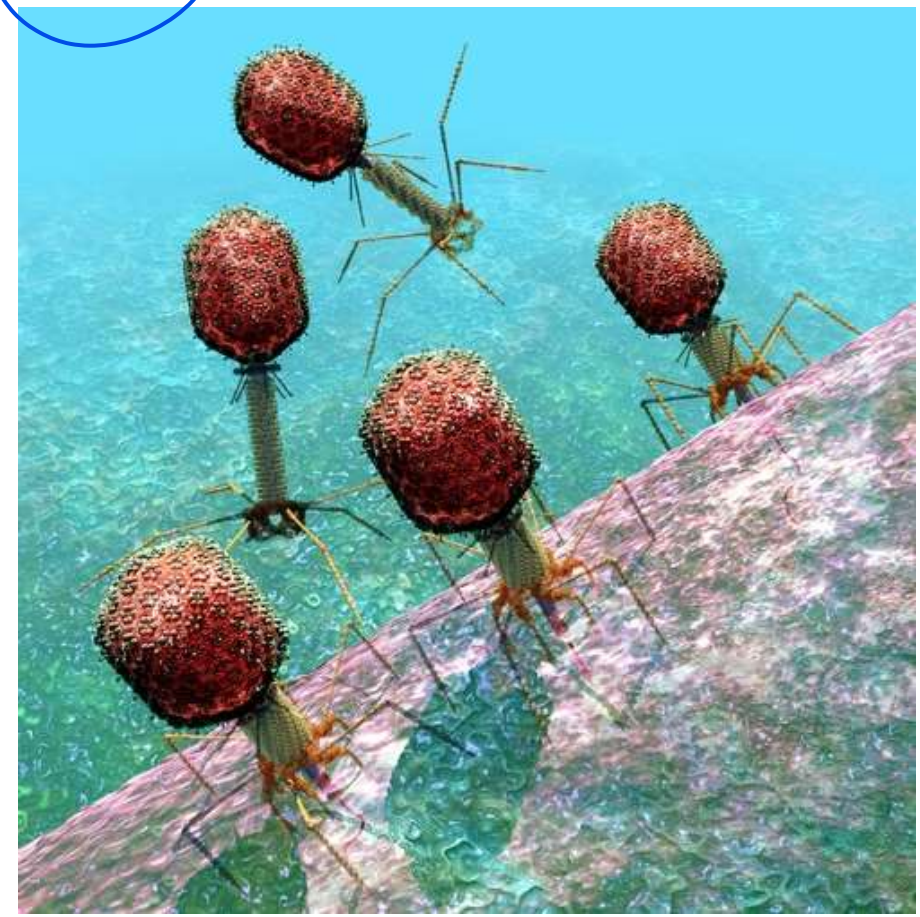
EXAMPLE OF LATENT VIRUS

LATENT VIRAL INFECTIONS

1. Chronic persistent infections
2. Latent Occult Viral infections

- The virus continuously replicates at low levels.
- The infection is typically detectable in the blood or tissues.
- It often causes ongoing, mild symptoms or damage over time.
-Example: Hepatitis B and C viruses.

CHRONIC PERSISTENT INFECTIONS



- Enveloped viruses
- e.g *Paramyxoviruses*, *Reteroviruses*, *Arenavirus*
- No disruption of celular functions
- Promotes shredding of viral Antigens from the cell surface

- The virus remains in a dormant state within the host cells.
- The infection is not detectable by standard methods because the virus is not actively replicating.
- The virus can reactivate under certain conditions, potentially causing symptoms again.
- Example: Herpes simplex virus.

LATENT OCCULT VIRAL INFECTIONS

Some viruses reappear to cause

acute disease

Three examples

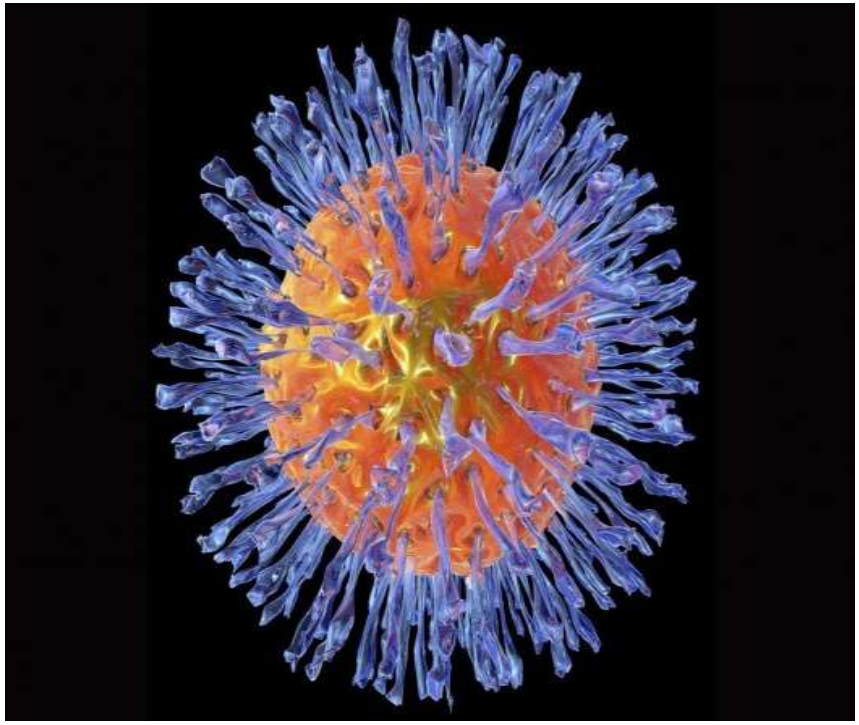
i. HSV

ii. Adenovirus

iii. SSPE

- Herpes Simplex Virus (HSV) is a common viral infection classified into two types: HSV-1 and HSV-2.
- HSV-1 primarily causes oral herpes, leading to cold sores or fever blisters around the mouth.
- HSV-2 primarily causes genital herpes, resulting in sores or lesions on the genital and anal areas.

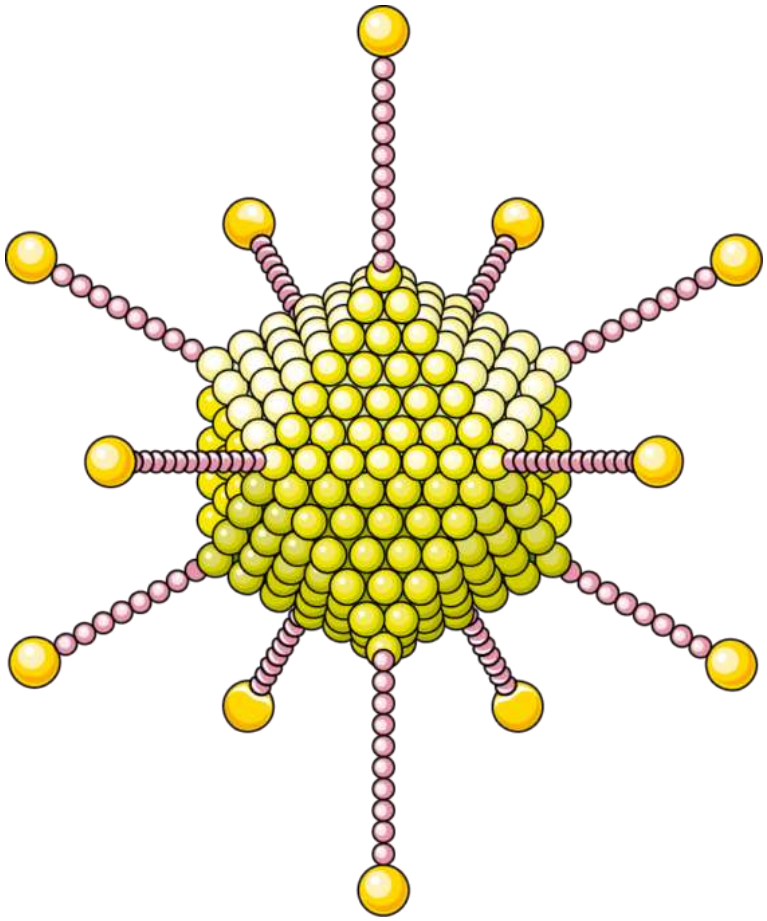
HSV



- Herpes Simplex Virus
- Primary infection in early age
- Found only during recurrent acute episodes
- Virus can be isolated from tissue homogenates

- Adenoviruses are a group of common viruses that can infect various parts of the body, leading to a range of illnesses.
- Non-enveloped viruses with a double-stranded DNA genome.
- Common in crowded environments like schools and military barracks.
- Can cause respiratory illnesses (e.g., common cold, bronchitis, pneumonia).

ADENOVIRUS



- Self-limitng
- Persistent infection of tonsils and adenoids
- Failure in viral isolation
- Absence of mature virion
- DNA in linear episomal form

MEASLES VIRUS

- The measles virus is a highly contagious virus that causes measles, a serious respiratory illness.
- An enveloped virus with a single-stranded RNA genome.
- Spread through respiratory droplets when an infected person coughs or sneezes.
- Initial symptoms include high fever, cough, runny nose, and conjunctivitis (red, watery eyes).
- Can lead to serious complications, especially in young children and immunocompromised individuals, such as pneumonia, encephalitis (brain inflammation), and death.
- Can cause long-term effects like subacute sclerosing panencephalitis (SSPE), a rare but fatal brain disorder.

- Incomplete viral production
- Immature nucleocapsids produced
- Disease depends on immune response
- May cause tumorigenesis

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VIRAL MUTANTS

Introduction

MUTANT

- A mutant is an organism or a new genetic character arising or resulting from an instance of mutation, which is generally an alteration of the DNA/RNA sequence of the genome or chromosome of an organism

RELEVANT TERMS

- Strain
- Type
- Variant
- Mutant

DIFFERENCE IN RELEVANT TERMS

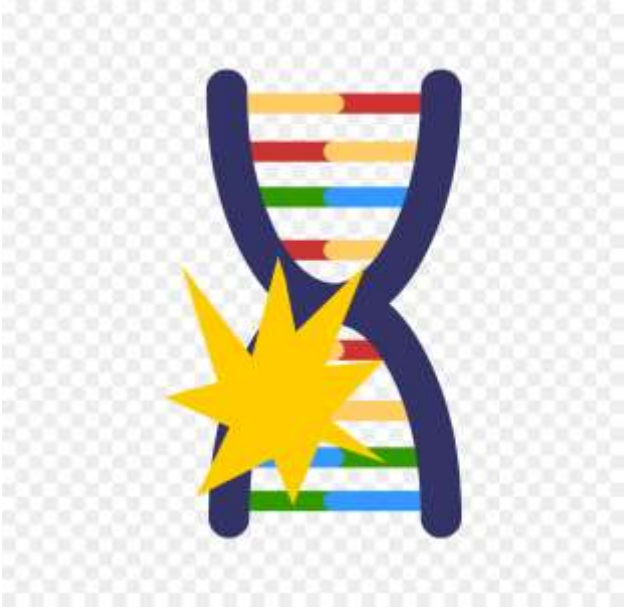
- Strain: lines or isolate of same virus
- Type: serotypes of same virus
- Variant: virus, whose phenotype varies from original virus
- Mutant: virus showing genetic alteration

ORIGIN OF MUTATION

- **Spontaneous mutation:**
  - Very high to low rate of mutation
  - Due to difference in type of genome
  - Error rate higher in RNA dependent RNA polymerase
- **Induced Mutation**
  - In-vivo
  - or in-vitro mutagens

MAIN REASON

- Replication Error
- Chemicals
- Radiations



TYPES OF MUTANT VIRUSES

- Biochemical markers: drug-resistance mutations
- Deletions: similar to non-sense mutations
- Host range: refers to whole host animal
- Non-sense mutation: alteration of coding sequence
- Plaque morphology
- Temperature sensitive
- Cold sensitive
- Reverents

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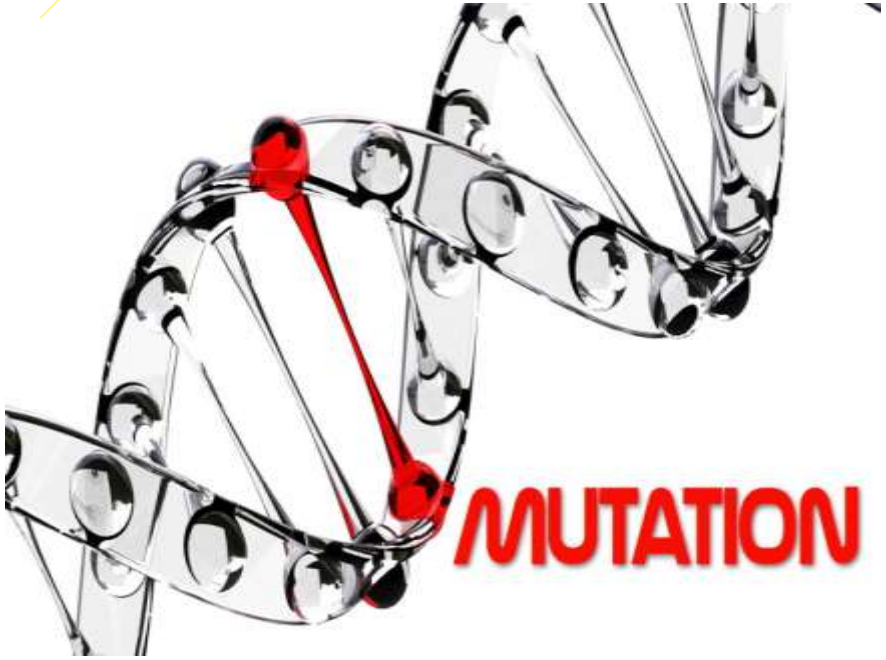
VIRAL MUTANTS

Spontaneous



RATE

In spontaneous mutation:

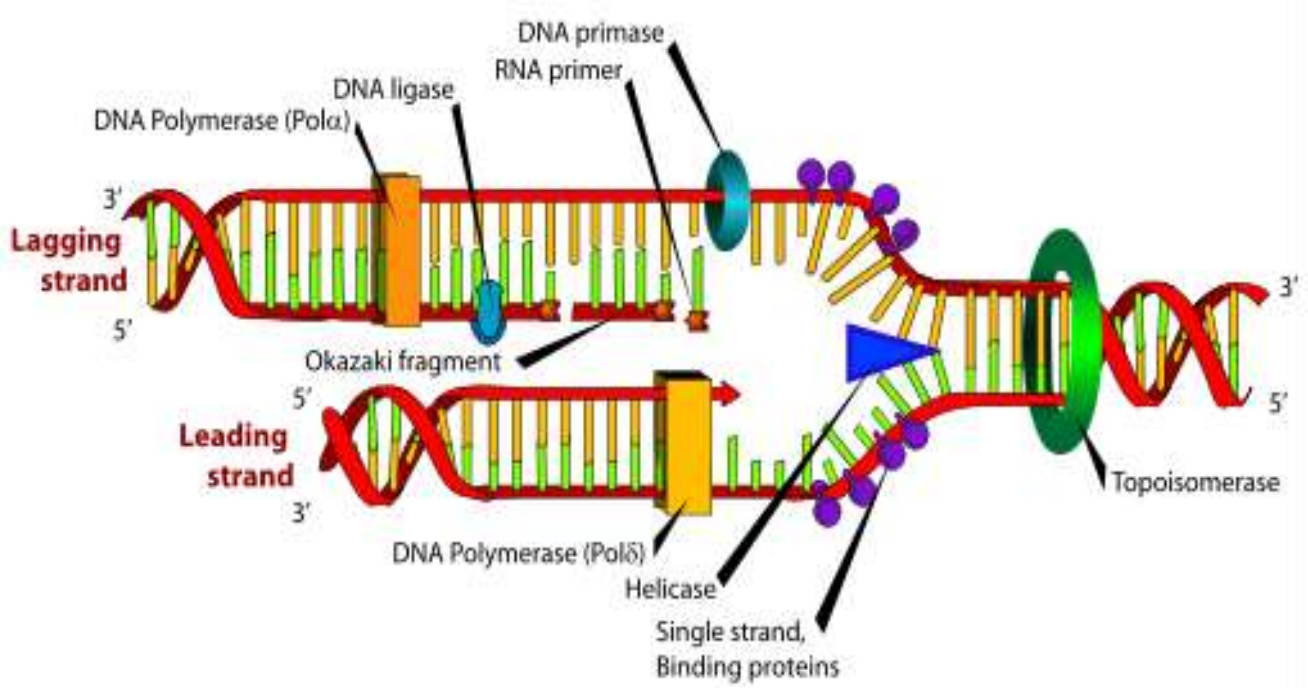
- High to low
- Very high in HIV
- Very low in Herpesvirus
- Resulting mutation in cellular DNA

REASON

- Mechanism of genome replication
- RNA dependent RNA polymerase shows higher error rate than DNA dependent DNA polymerase

DO RNA VIRUSES HAVE PROOF-READING?

- Yes!
- Yet they show higher rates of mutation than DNA viruses

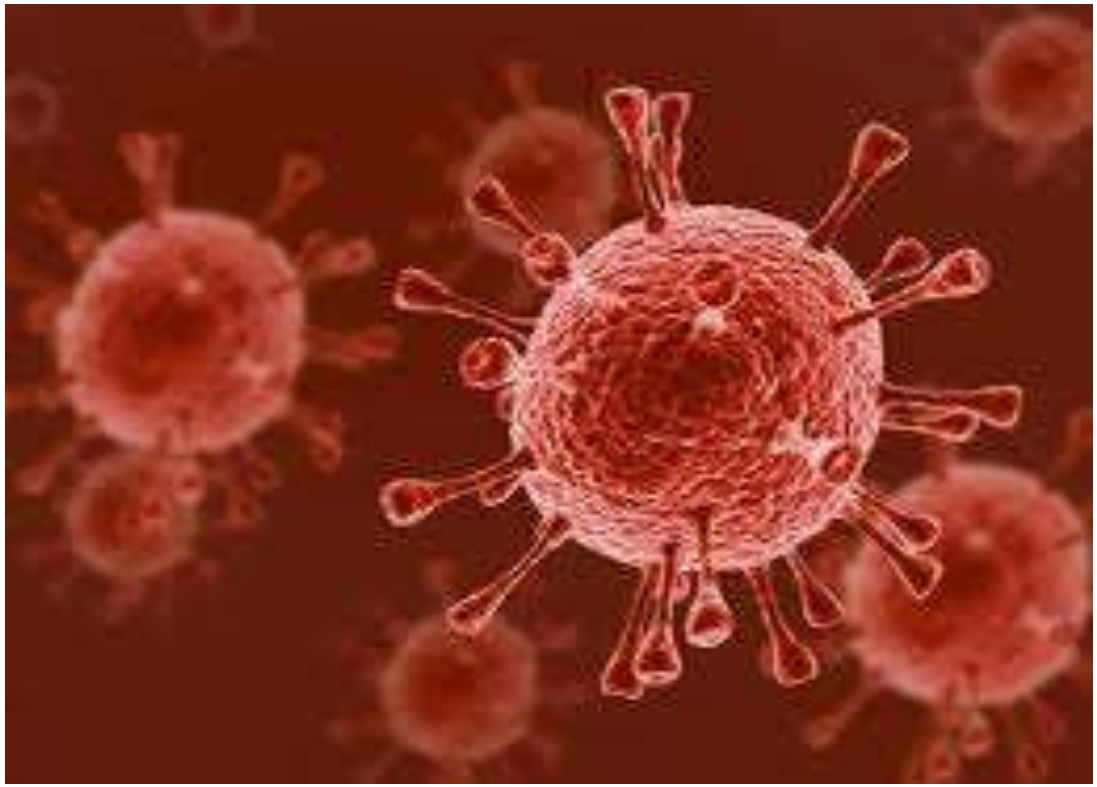


MUTATION: GOOD OR BAD?

– Sometimes good for the virus
– Helps escape the immune system
– But some mutations may be deleterious
– Due

HIV

- Genome is 9.7kb long
- Mutatuion rate can be 0.9 to 9.7 mutaations in every genome copied



QUASISPECIES

- Mixture of molecular variants
- Usually the wild-type virus dominates
- Mostly non-infectious
- Important in viral evolution

REASON?

- Tautomerism: base change due to repositioning of Hydrogen atom
- De-purination: Loss of purine base
- De-amination and hydrolysis: change of normal base to atypical base
- Slipped strand mis-pairing: Denaturation of new strand
- Replication slipping: error during replication process

Ppt no. 53

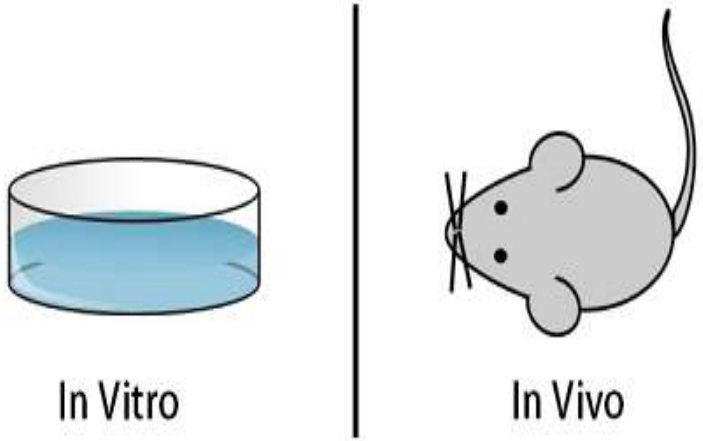
VIRAL MUTANTS

Induced Mutation

TYPES:

Commonly used chemical mutagens are of two types: in vitro mutagens and in vivo mutagens

- In-vitro
- In-vivo



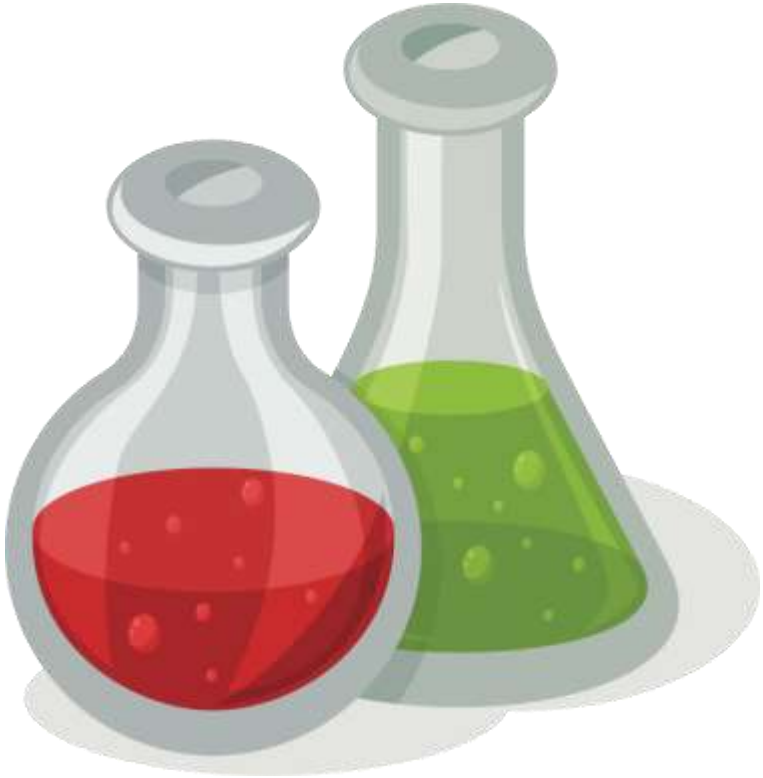
IN-VITRO

- Chemically modify nucleic acids
- Do not require replication
- E.g: nitrous acid, hydroxylamine and alkylating agents like nitrosoguanidine

IN-VIVO

- Requires Metabolically active nucleic acid that is replicating
- Compounds incorporated intoo newly synthesized nucleic acids
- E.g:5-bromouracil resulting erroneous base pairing, intercalating bases that stack between bases, UV irradiation

CHEMICAL REASON



- Hydroxylamine
- Base analogs
- Alkylating agents
- DNA intercalating substances
- DNA cross linkers
- Oxidative damage
- Nitrous acid

RADIATION REASON

- Ultraviolet rays
- Ionizing radiation

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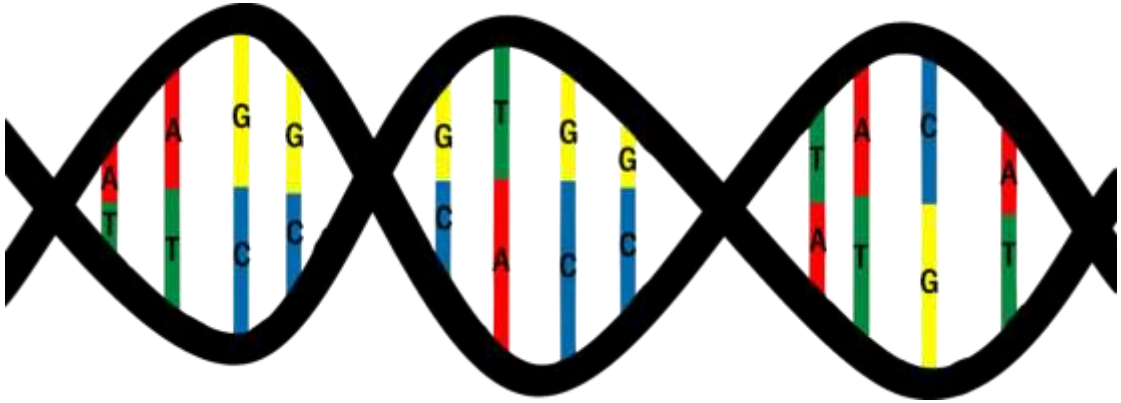
Types of Mutant Viruses

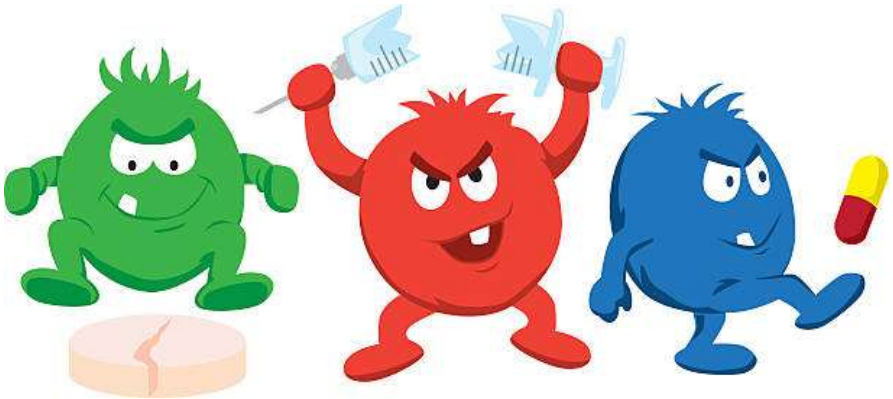


Types of Mutant Viruses

TYPES

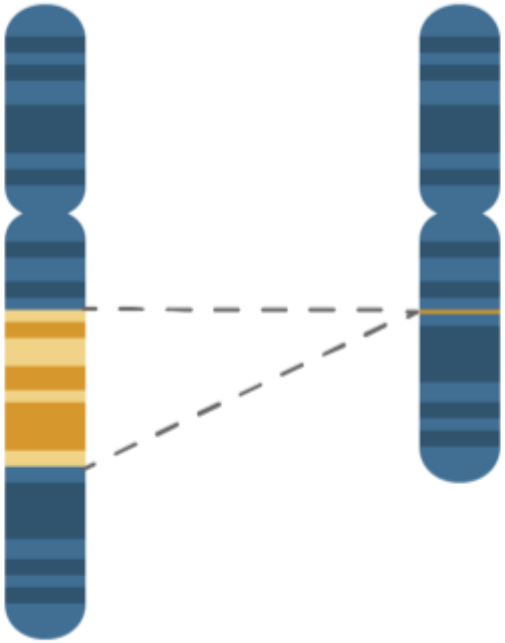
- Biochemical Markers
- Deletions
- Host Range
- Non-Sense
- Plaque morphology
- Temperature sensitive
- Cold Sensitive
- Reverents

BIOCHEMICAL MARKERS



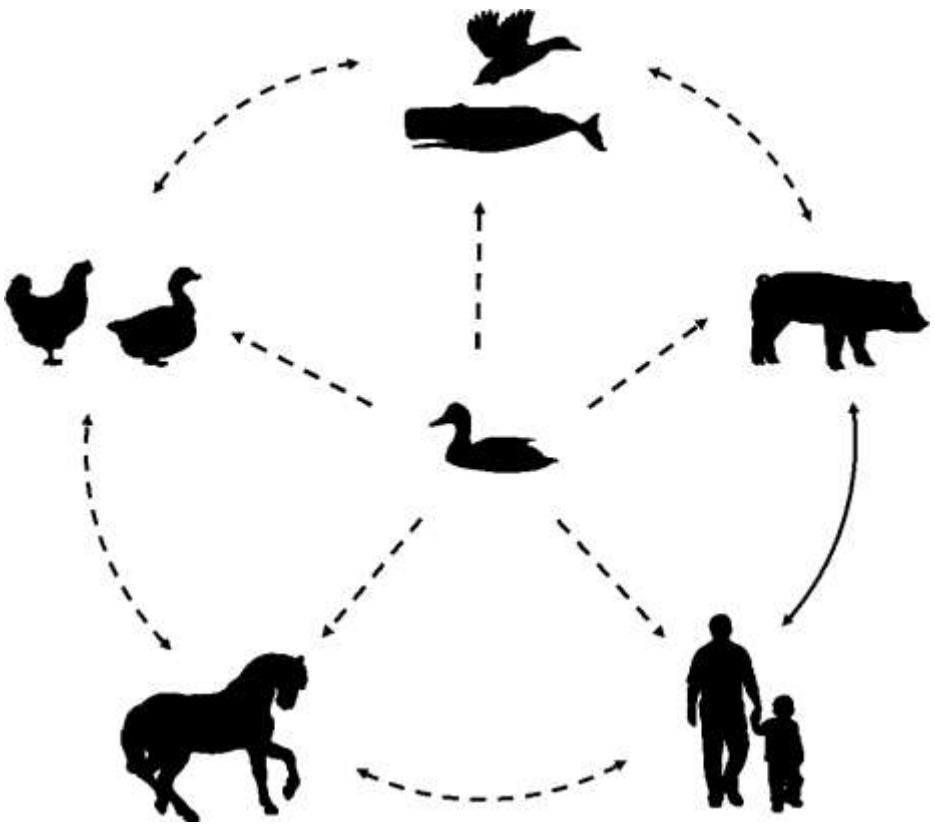
- Drug resistance mutations
- Altered virulence
- Polymorphism
- Altered proteins or nucleic acids
- Sensitivity to inactivating agents

DELETIONS



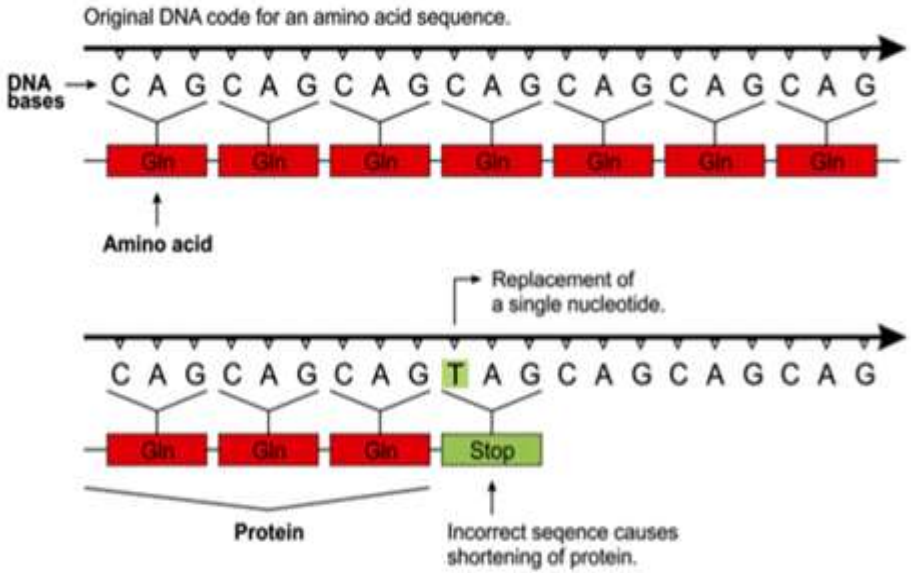
- Similar to nonsense mutation
- May include one or more viral genes
- Involve non-coding regions of genome
- Defective-interfering particles (DIP)
- Can only revert to wild-type by recombination

HOST RANGE



- Refers to whole host animal or cell type in vitro
- Conditional mutants

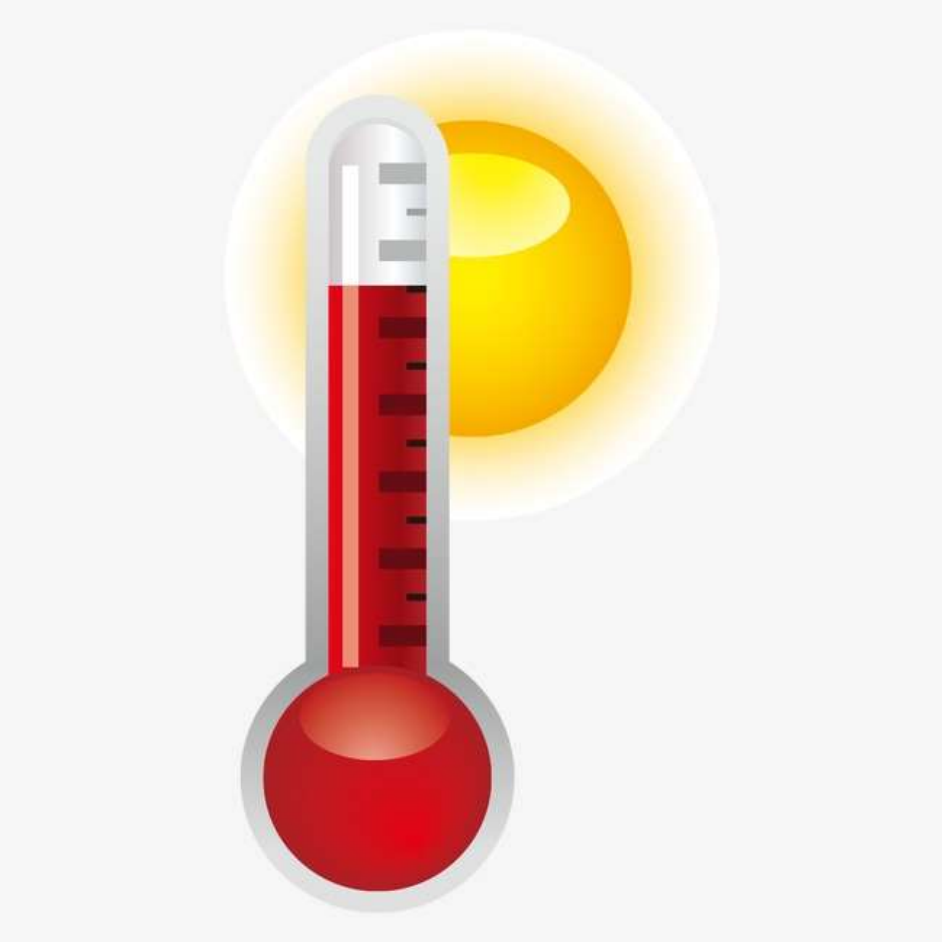
NON-SENSE



- Alteration of coding sequence of one of the three stop codons (UAG, UAA, UGA)
- Termination of translation
- Low reversion frequency

PLAQUE MORPHOLOGY

- Large plaque mutants or small plaque mutants
- Plaque size referred to temp. sensitivity
- Useful as unselected markers in multi-factorial crosses



TEMPERATURE SENSITIVE

- Useful for isolation of conditional-lethal mutations
- Usually occur by mis-sense mutation in proteins
- Some activities work on non-permissive temperatures

COLD SENSITIVE

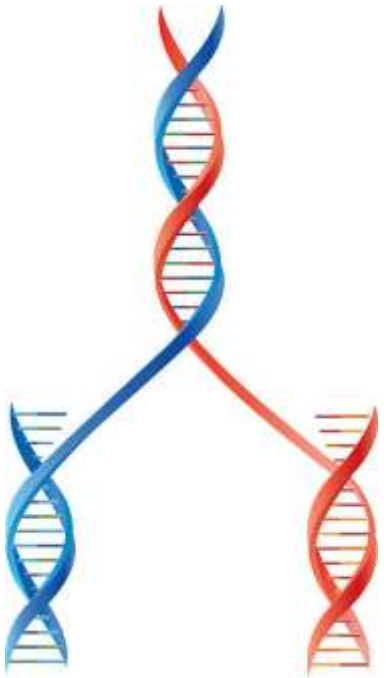
- Opposite to temperature sensitive
- Useful in bacteriophages and plant viruses as their cells can be propagated at low temperatures

REVERENTS

- Valid type of mutation
- Reverse mutations
- ‘back mutation’ or ‘compensatory mutation’

Ppt no. 55

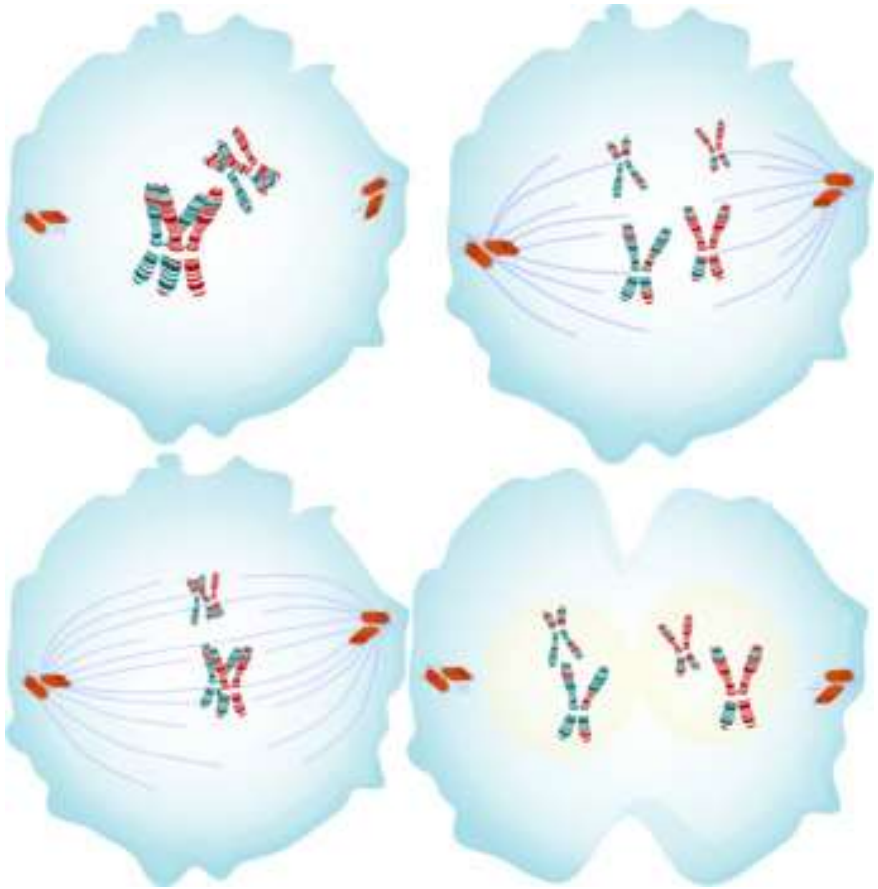
VIRAL RECOMBINATION



DEFINITION

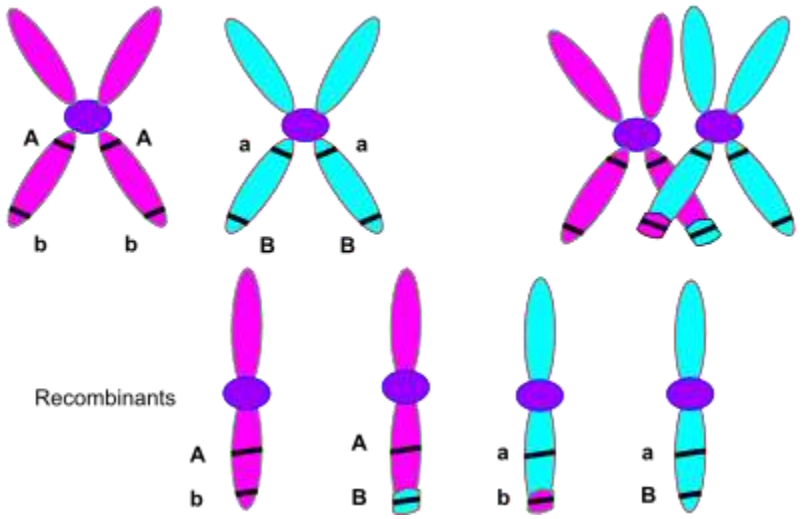
- The exchange or transfer of genetic material between different but closely related viruses infecting the same cell

HOW?



- Can also occur between virus and host
- Recombinants contain new genetic information

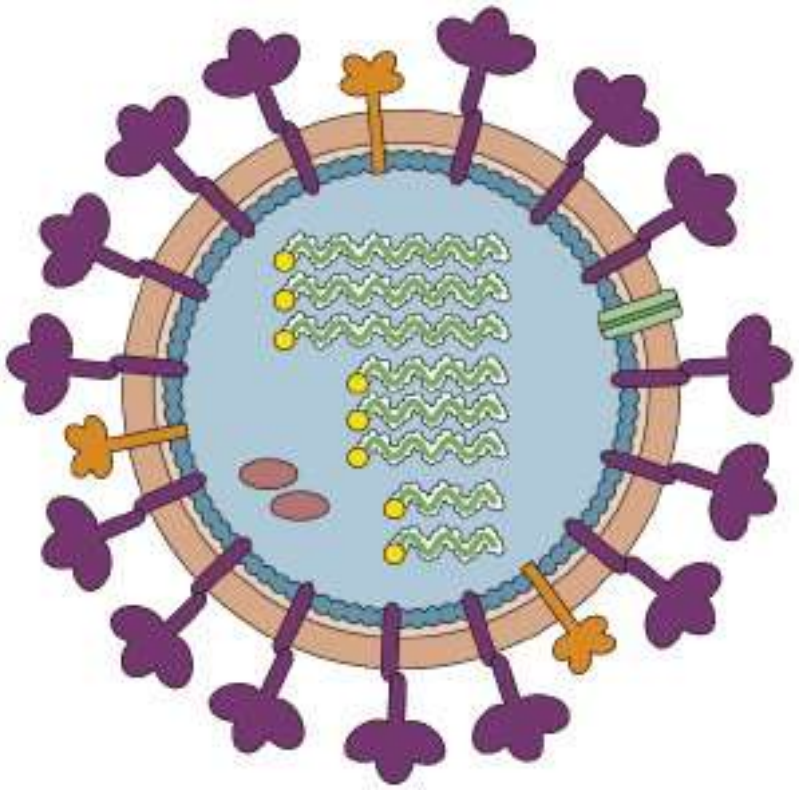
INTRAMOLECULAR RECOMBINATION



- Usually in DNA viruses
- Dissociation and re-establishment of covalent bonds with nucleic acids
- Template switching mechanism

EXAMPLES

- Picornavirus
- Togavirus
- Corona virus
- Influenza A virus



RNA VIRUSES

- Randomly with segmented genome
- E.g orthomyxovirus, reovirus, bunyavirus
- Genome segments show recombinations

GENETIC REACTIVATION

- Infectious progeny produced from parents, in which one or both are noninfectious
- Mixed infection of cell

ADVANTAGE

- Genetic variaility in nature
- Rapid adaptation of viruses
- Virulence changes
- Epidemiology of viruses (e.g. Influenza A)

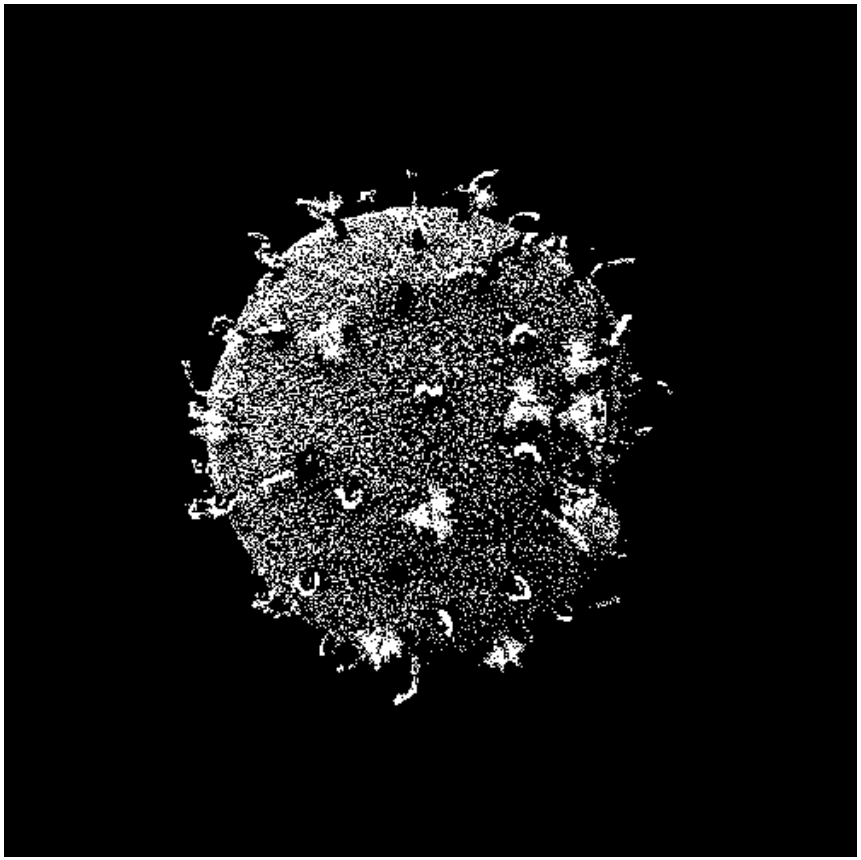
Ppt no. 56

SUPRESSION OF VIRAL GENOME

DEFINITION

- Inhibition of a mutant phenotype by a second suppressor mutation
- Which may be either in viral genome or in host cell

HOW?



- Mechanism different from suppression of chain-terminating amber mutations by host encoded suppressor tRNAs
- That is also referred as informational suppression

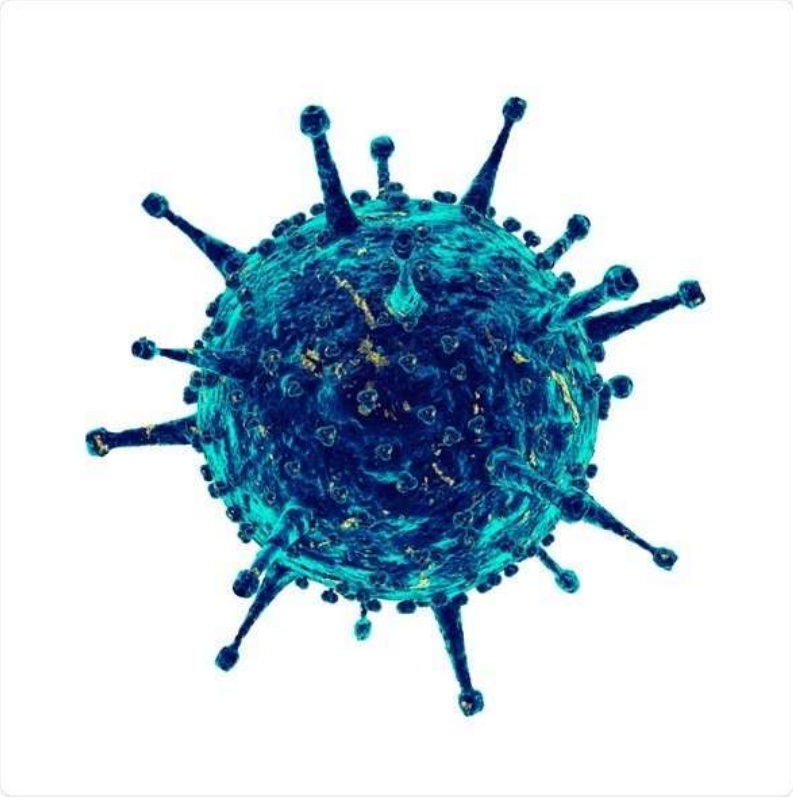
PSEUDOREVERENT

- Usually apparent wild-type strain
- Genetically mutant
- Prokaryotic cells
- Now discovered in reovirus, vaccinia, influenza

REVERSION

- Mutant viruses can revert by three ways:
- True reversion
- Intragenic reversion
- Extragenic Reversion

TRUE REVERSION



- Back mutation of original mutation
- Gives wild-type genotype or phenotype

INTRAGENIC SUPPRESSION

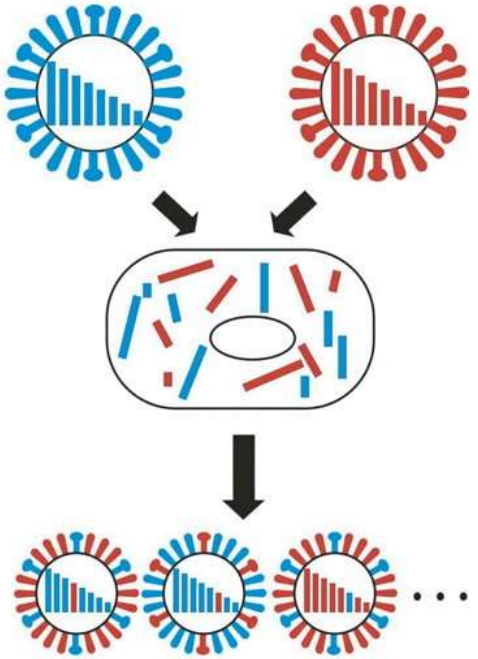
- Compensatory mutation, which occurs in same gene as originally mutated gene
- Restoring it

EXTRAGENIC SUPPRESSION

- A suppressor mutation in a different gene
- Virus gene or host gene

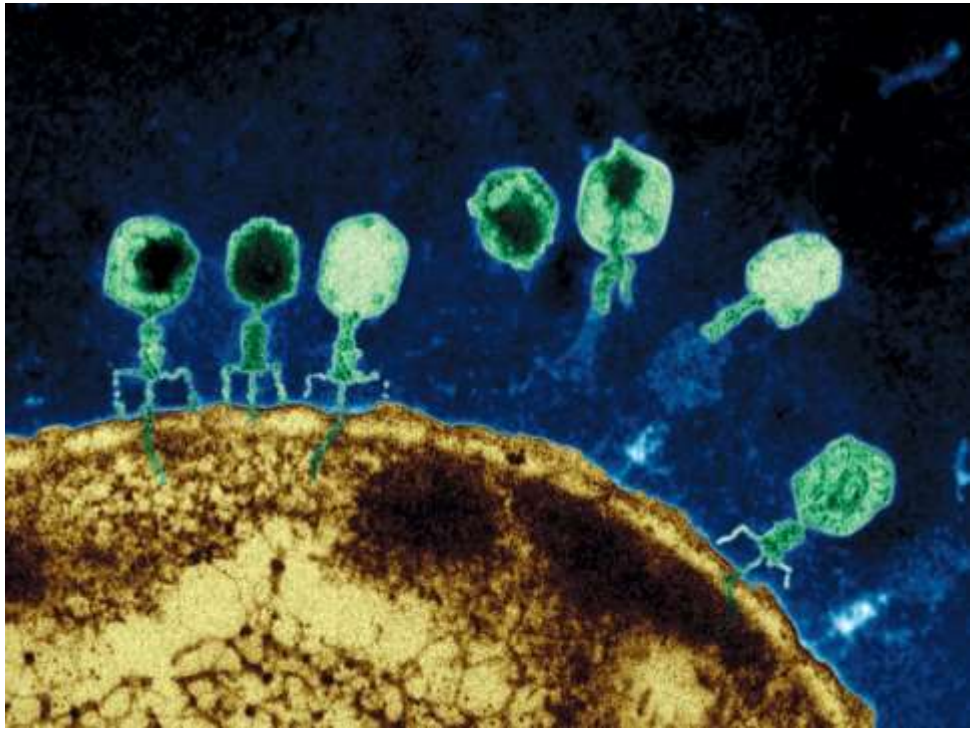
Ppt no. 57

GENETIC INTERACTIONS BETWEEN VIRUSES



WHEN?

- Naturally
- More than one infection in host



Super-infection

- Mixed infection in cell culture
- Two purposes:
- COMPLEMENTATION
- RECOMBINATION

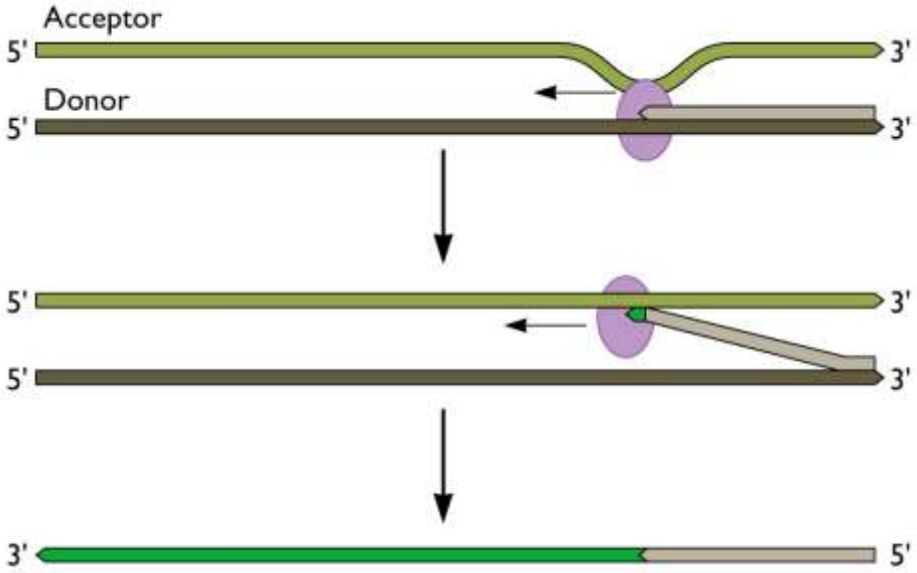
COMPLEMENTATION

- Assignment of mutants to functional groups
- Resulting production of one or both parent viruses
- One of the viruses provides functional gene product
- Allows functional analysis of unknown mutations

TYPES OF COMPLEMENTATION

- Allelic (intragenic) complementation where different mutants have defects in same protein
- Non-allelic (intergenic) results from mutants with defects in different genes

RECOMBINATION



- Physical interaction of viral genomes resulting gene combinations not present in either parents
- Three mechanisms: Intra-molecular recombination by strand breakage and re-ligation, intra-molecular recombination by copy-choice and Reassortment

INTRA-RECOMBINATION BY STRAND BREAKAGE AND LIGATION

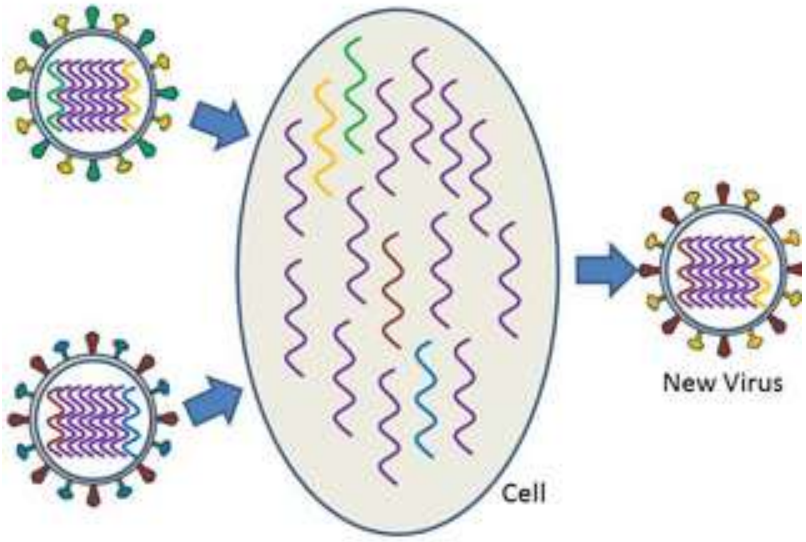
- Occurs in all DNA and RNA viruses that replicate via DNA intermediate
- Mediated by cellular enzymes

INTRA-MOLECULAR RECOMBINATION BY COPY CHOICE

- In RNA viruses
- During switching of viral polymerase
- Cellular enzymes involved
- Defective Interfering

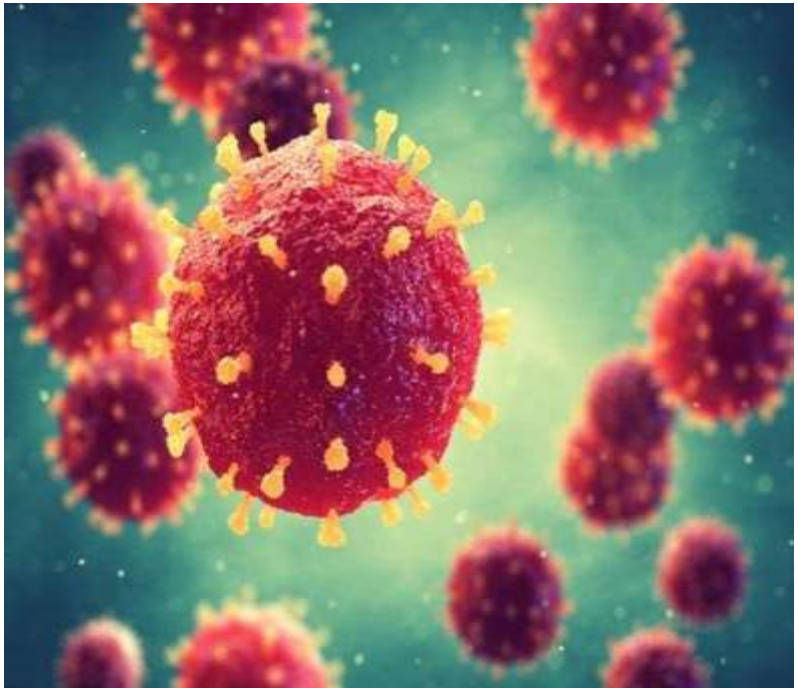
REASSORTMENT

- Random shuffling
- Progeny viruses receive atleast one of the parent gene segment
- Packaging mechanism not understood



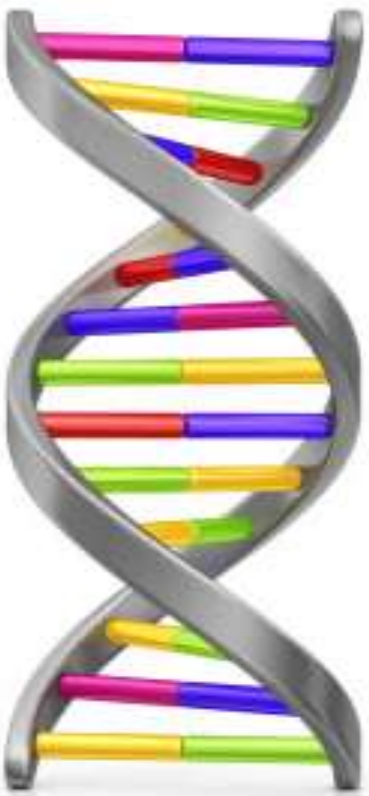
Ppt no. 58

NON-GENETIC INTERACTIONS BETWEEN VIRUSES



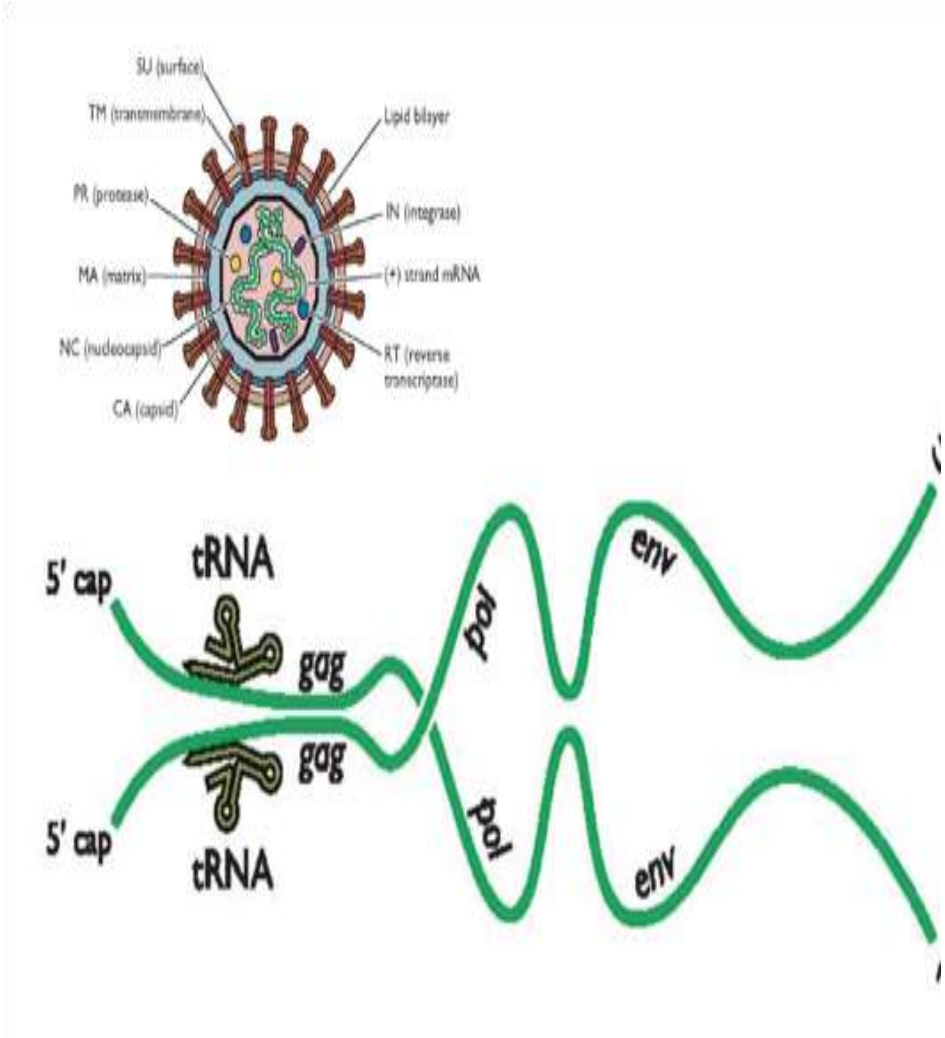
EUKARYOTIC CELLS

- Diploid genome
- Two copies of each chromosome
- Each having own allele of same gene
- Difference in allelic markers at loci



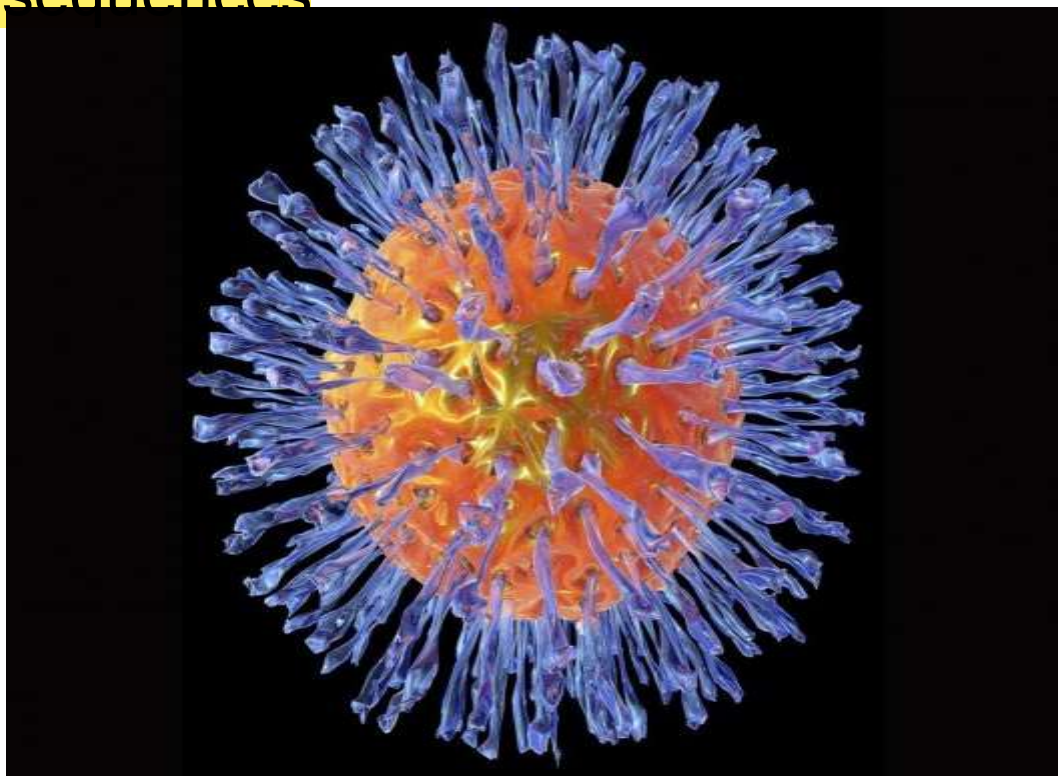
RETROVIRUS

- Truly diploid
- Two copies of entire genome



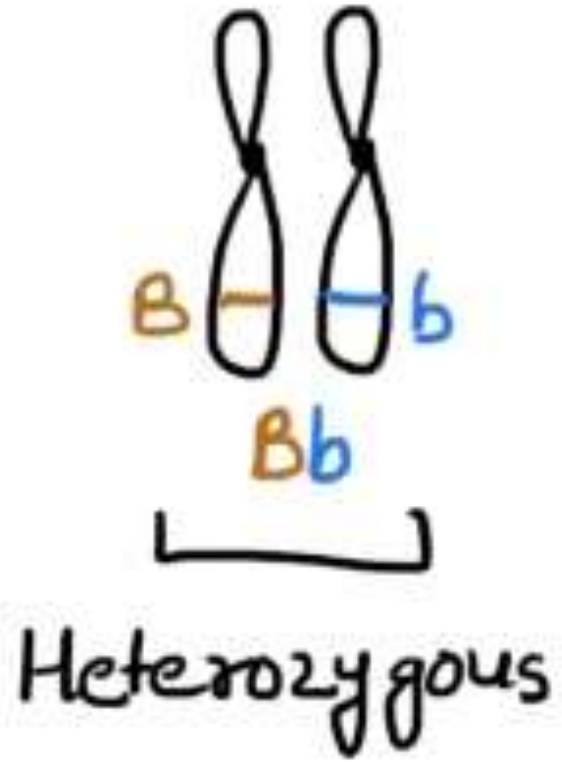
HERPESVIRUS

- Partially heterozygous
- Repeated sequences



HETEROZYGOSIS

- Aberrant packaging of multiple genomes
- May result in multiploid particles
- genetic complexity of population



SUPERINFECTION

- Interference as a result of superinfection
- Homologous interference against same virus from presence of particles which blocks replication
- Interference by mutation
- Interference by sequestration of virus receptors due to virus-attachment protein

PHENOTYPIC MIXING

Variable

Genome maybe enclosed in capsid/envelope of another (pseudo-typing)

capsid contains genomic particles of both viruses

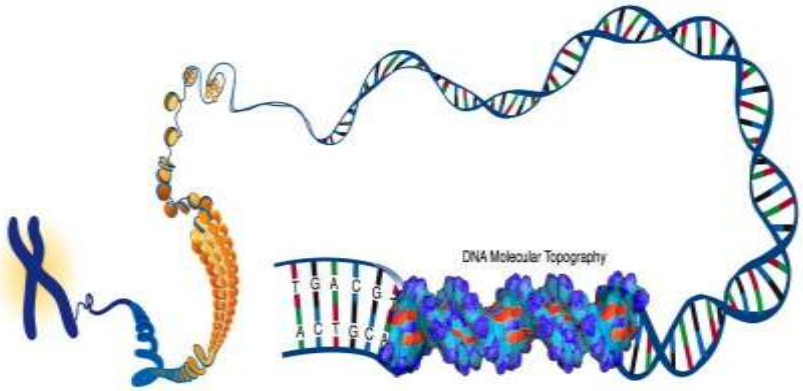
Non-specific incorporation of different viral glycoproteins

PHENOTYPIC MIXING AS A TOOL

- Examining biological properties of viruses
- Vesicular stomatitis virus forms pseudotypes containing retrovirus envelope glycoproteins
- Plaque forming properties of VSV, cell tropism properties of retrovirus
- Used for HIV and Retrovirus

Ppt no. 59

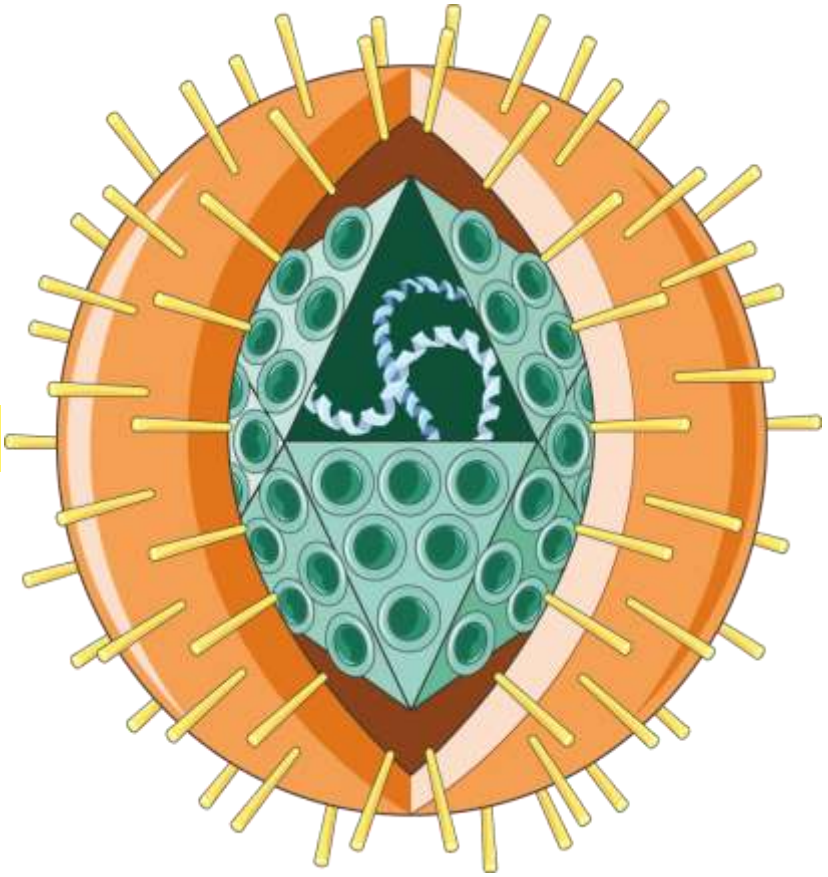
LARGE DNA GENOMES



- Viral genome could be considerably big e.g DS DNA genomes
- Generally, Genetically similar to host cells e.g *Adenovirus, herpes ctc.*

HERPESVIRIDAE

- Large family
- More than 100 members
- Eight human viruses, with similar overall structure
- Subdivided into 3 families



SUBFAMILIES

Alphaherpesvirinae

- Latent infections in sensory ganglion, genome size 120-180kbp
- Simplexvirus, Varicellovirus, Human herpes virus-I and II,
- Human herpes virus-III

Betaherpesvirinae

- Restricted host range, genome size 140-253 kbp
- Cytomegalovirus, Muromegalovirus, Roseolovirus, Human herpes virus-V, Mouse cytomegalovirus-I, Human herpes virus-VI and VII

Gammaherpesvrinae

- Infection of lymphoblastic cells, genome size 105-175 kbp
- Lymphocryptovirus, Rhadinovirus, Human herpesvirus-IV, Human herpes virus-VIII

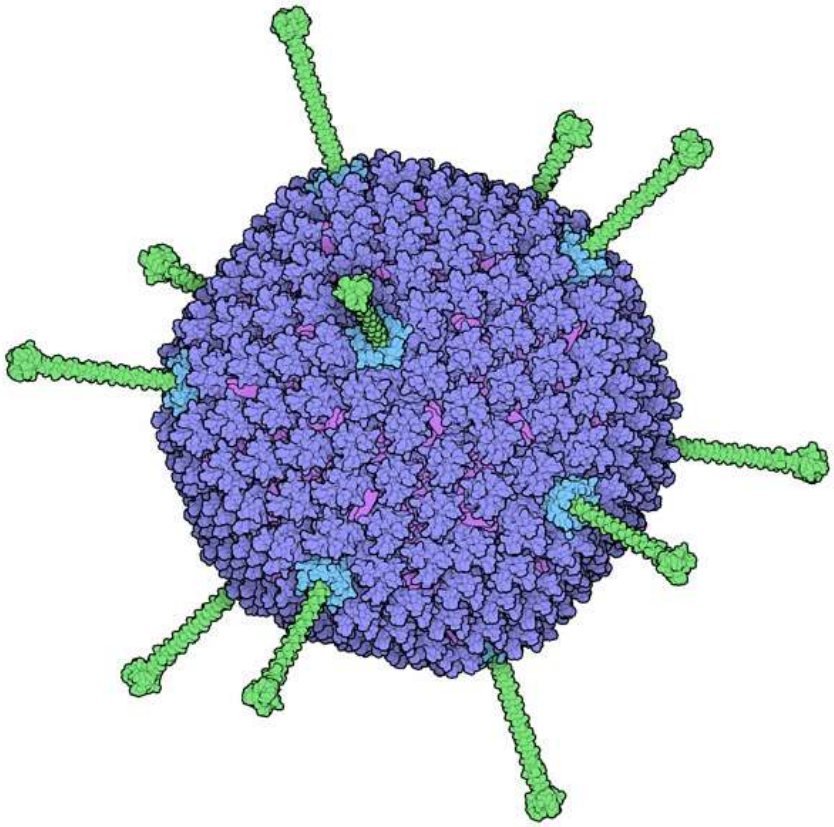
HERPESVIRUS

- large linear double stranded DNA genome containing 35 viral polypeptides encoding variety of enzymes
- Two covalently joined sections, Unique long (U_L)and Unique short regions(U_S) by inverted repeats
- Allows structural rearrangements of unique regions

HERPES SIMPLEX VIRUS

- 152 kbps ds DNA, 80 genes
- Densly packed with overlapping reading frames
- Each gene expressed by own promoter
- Extensively mapped by conventional genetic analysis
- Most intensively studied virus

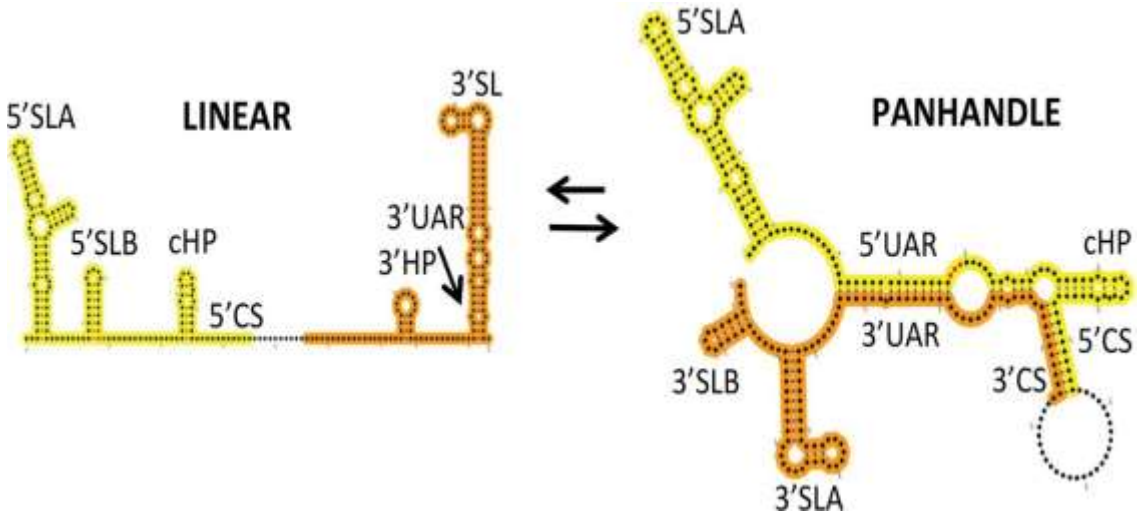
ADENOVIRUS



- Linear double stranded DNA genome of 30 to 38 kbp, 30-40 genes
- Terminal sequence is inverted repeat of 100 to 140bp
- Denatured single strands form panhandled structure

PANHANDLED STRUCTURES

- Important in replication
- 55kDA protein known as terminal protein
- Attached to 5’ end of each strand
- Acts as primer and initiates synthesis of new strands

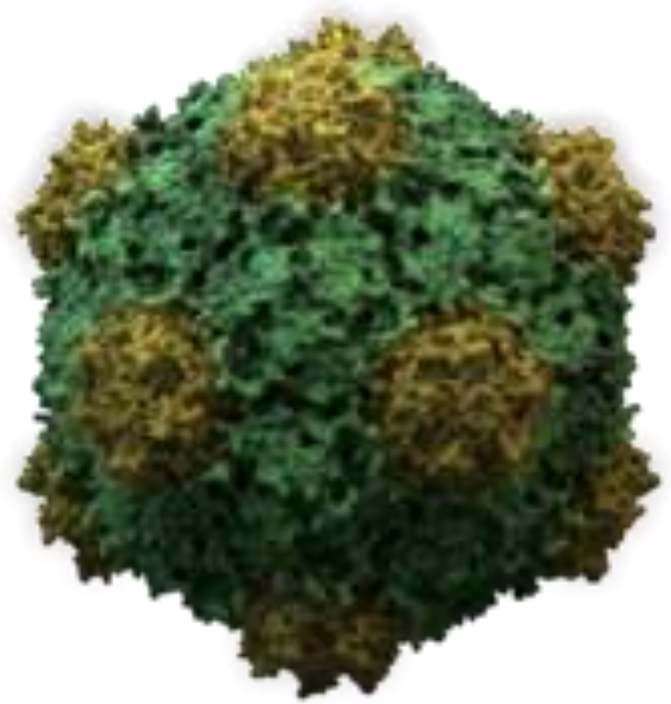


EXPRESSION

- Promoters
- Splicing patterns

Ppt no. 60

VIRUSES WITH SMALL DNA GENOMES



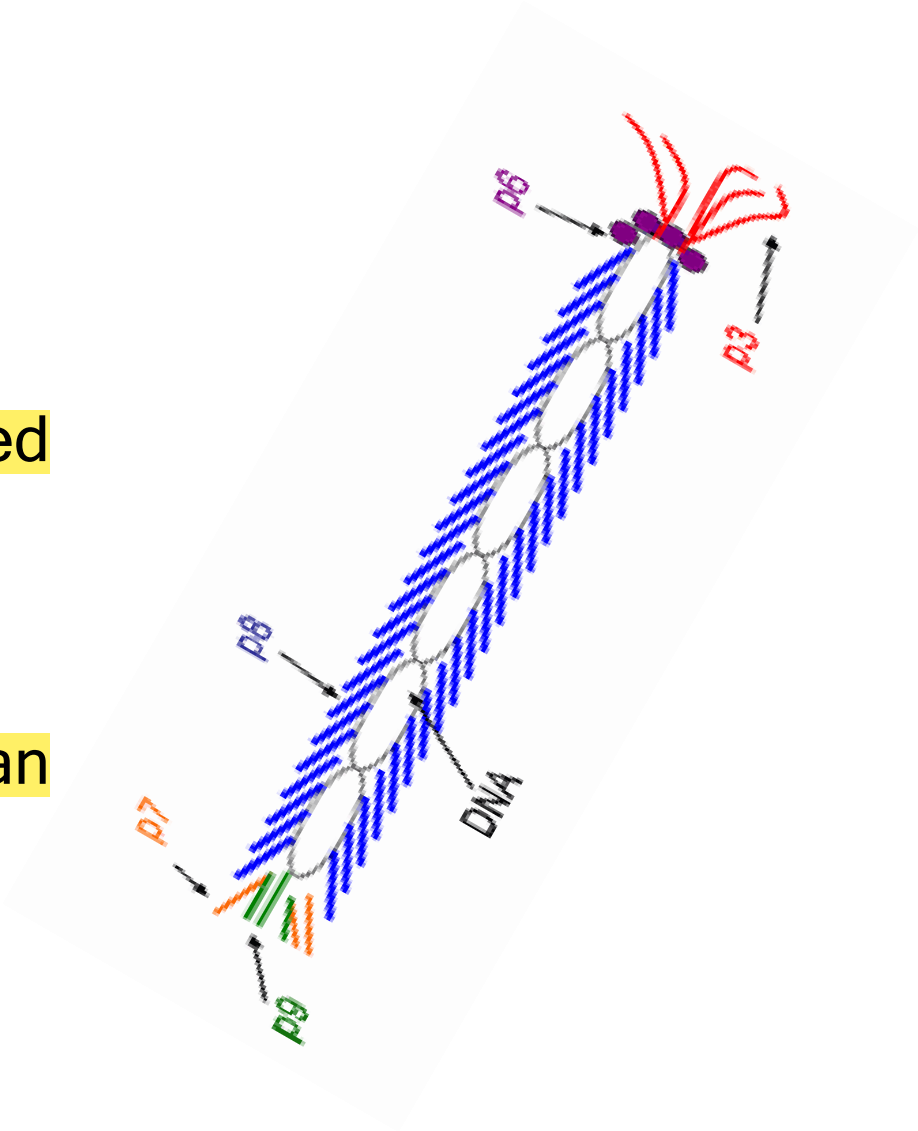
M-13 PHAGE

- Simple genome
- 6.4kb
- Single stranded
- Positive sense
- Circular DNA

encoding ten genes

M-13 PHAGE

- Icosahederal symmetry
- Capsid can be expanded by protein subunits
- Hence, genomic size can also be increased



LAMBDA PHAGE

- Rigid packaging restraints
- Only DNA of between 95 and 110% of normal genome size can be packaged into the capsid
- 49kbp
- Linear genome

LAMBDA PHAGE

- Substrate packaged into phage heads during assembly consist of long concatemers of phage DNA
- DNA apparently whirled by phage head
- Complete genome cleaved by phage coded endonuclease
- *Cos-site*
- DNA ligase

T-4 PHAGE

- 160 kbp
- Double stranded DNA
- Linear genome

T-4 PHAGE

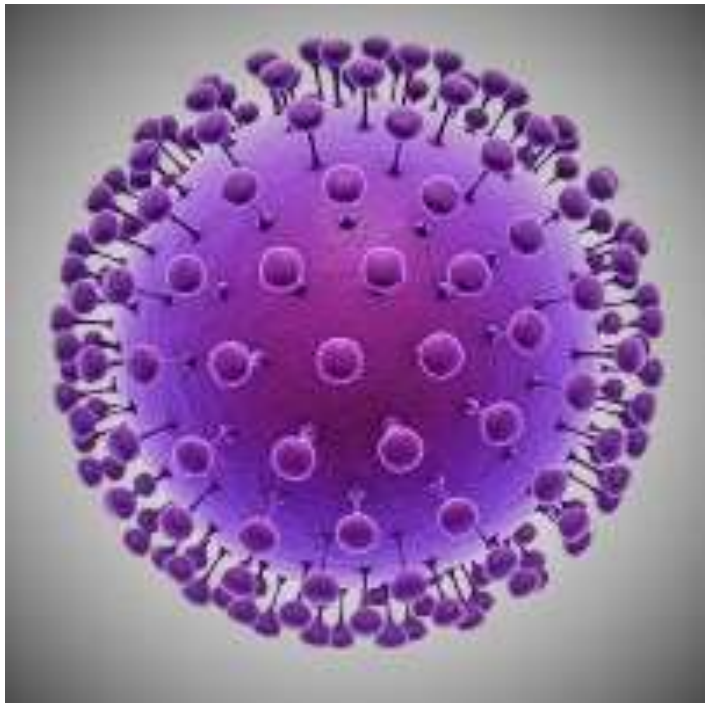
- Terminal redundancy
- Long Contactmers of DNA genome
- Longer than complete genome

PARVOVIRUS

- Linear, non-segmented, single stranded negative sense genome, 5kb
- Most packaged as negative sense
- A portion of positive sense strands , very small
- Two genes *cap* and *rep*

PARVOVIRUS

- Splicing pattern for each gene
- Pallindromic sequences for hair-pin structures



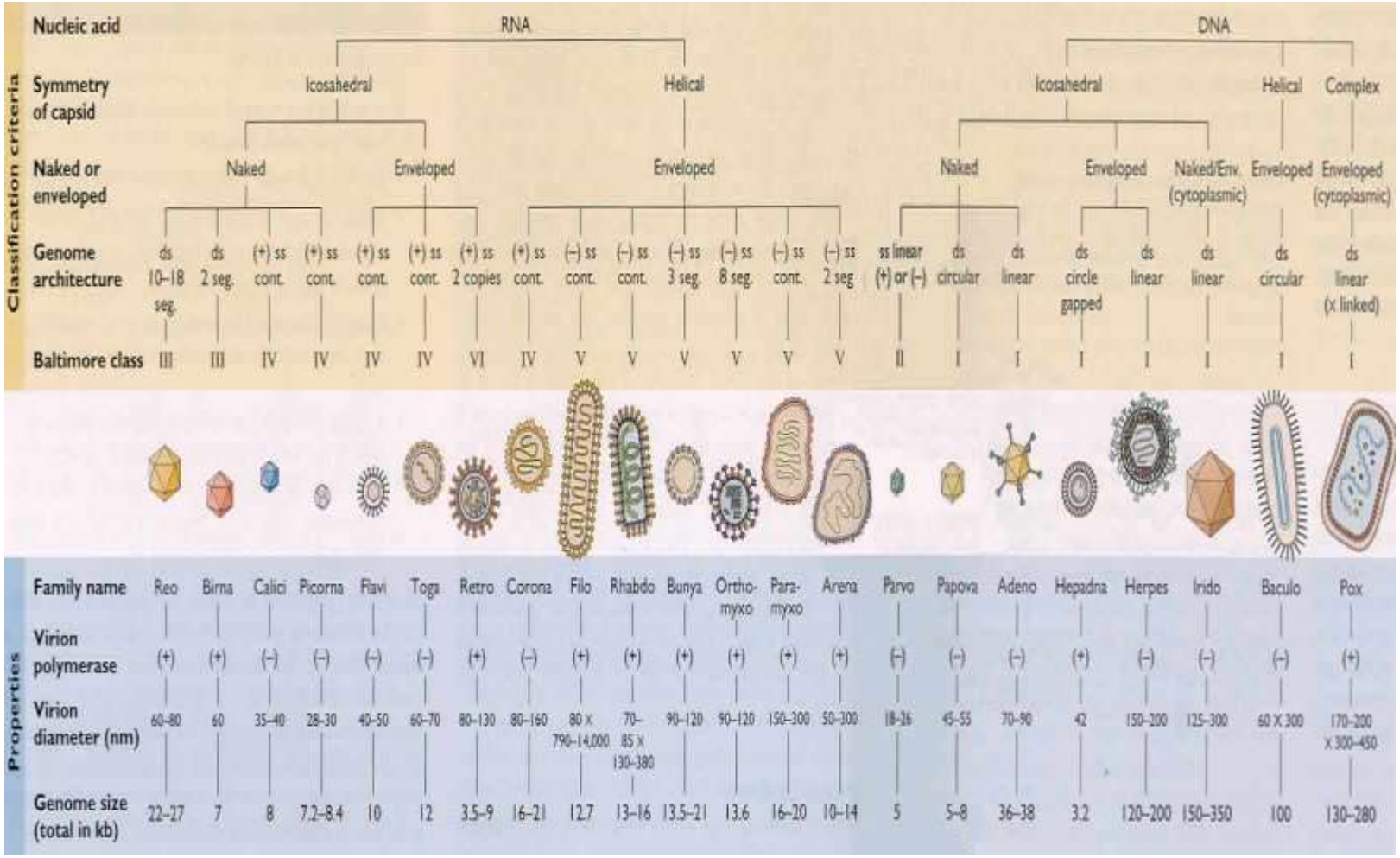
POLYOMAVIRUS

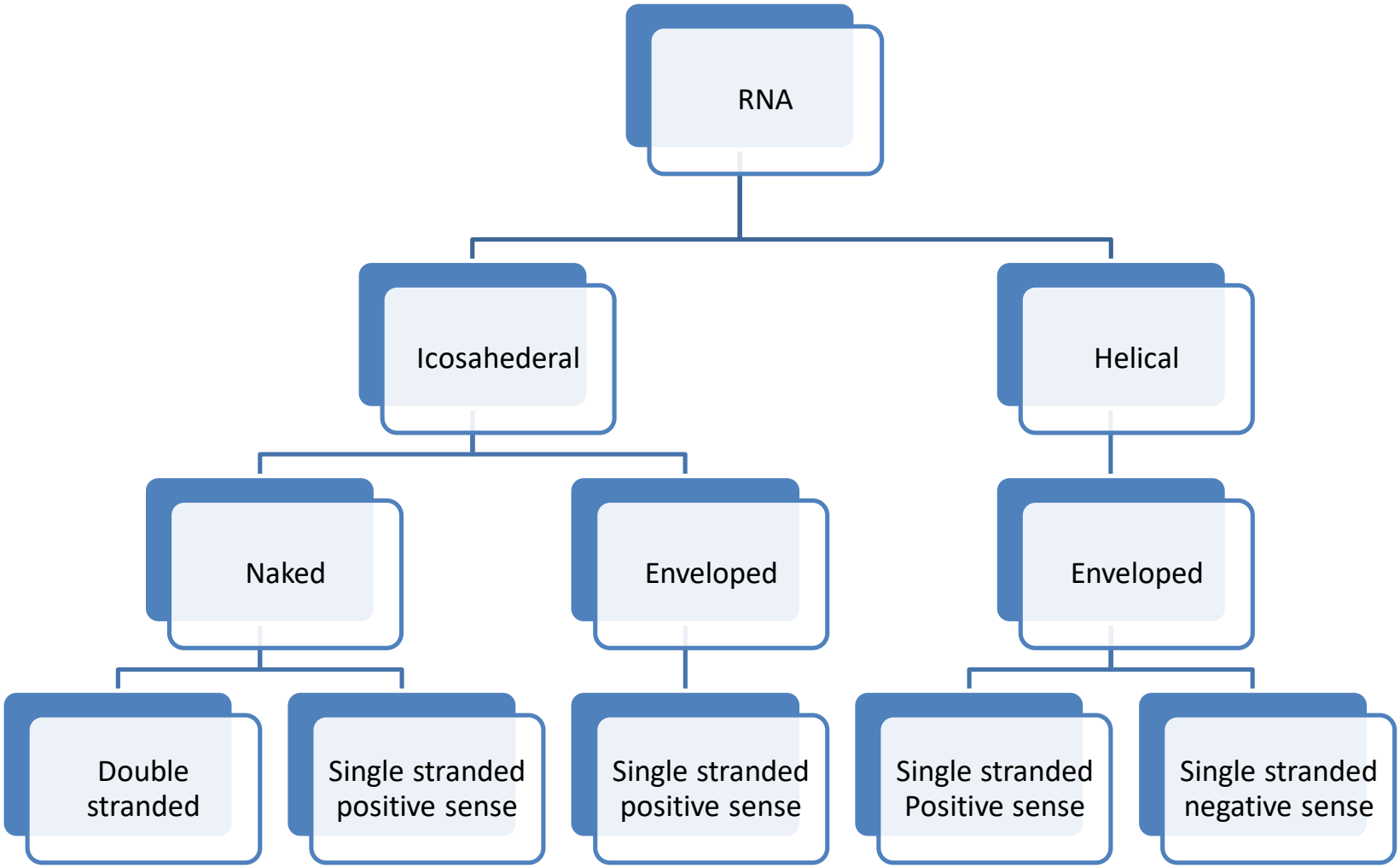
- Double stranded, circular DNA, 5 kb
- DNA in super-coiled form
- Four cellular histones
- Maximum information in minimum space

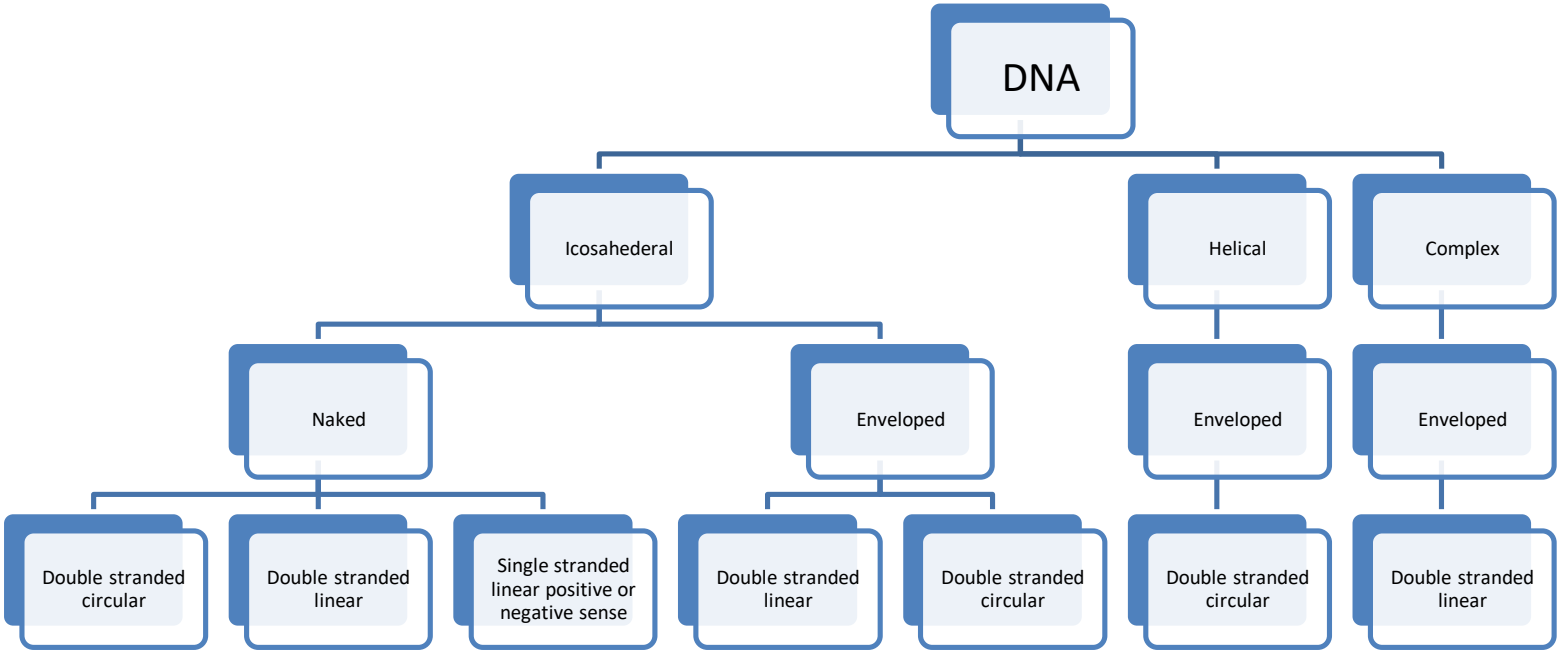
Ppt no. 61

GENERAL CLASSIFICATION OF VIRUSES

- Naming of organisms and categorizing them
- Based on a certain factors like phenotype, nucleic acid type, replication etc

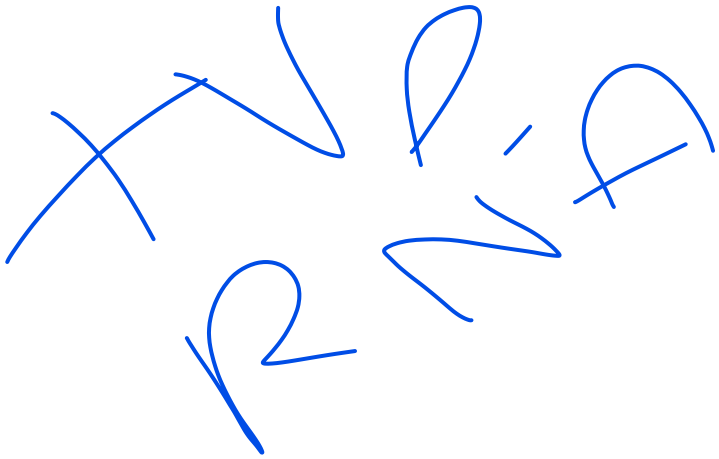

Classification criteria
Nucleic acid
RNA
DNA
Symmetry of capsid
Icosahedral
Helical
Icosahedral
Helical
Complex
Naked or enveloped
Naked
Enveloped
Enveloped
Naked
Enveloped
Naked/Env. (cytoplasmic)
Enveloped
Enveloped (cytoplasmic)
Genome architecture
ds 10–18 seg.
ds 2 seg.
(+) ss cont.
(+) ss cont.
(+) ss cont.
(+) ss cont.
(+) ss 2 copies
(+) ss cont.
(–) ss cont.
(–) ss cont.
(–) ss 3 seg.
(–) ss 8 seg.
(–) ss cont.
(–) ss 2 seg
ss linear (+) or (–)
ds circular
ds linear
ds circle gapped
ds linear
ds linear
ds circular
ds linear (x linked)
Baltimore class
III III IV IV IV IV VI IV V V V V V V II I I I I I I I
Properties
Family name
Reo Birna Calici Picorna Flavi Toga Retro Corona Filo Rhabdo Bunya Ortho-myxo Para-myxo Arena Parvo Papova Adeno Hepadna Herpes Irido Baculo Pox
Virion polymerase
(+) (+) (–) (–) (–) (–) (+) (–) (+) (+) (+) (+) (+) (+) (–) (–) (–) (+) (–) (–) (–) (+)
Virion diameter (nm)
60–80 60 35–40 28–30 40–50 60–70 80–130 80–160 80 × 790–14,000 70–85 × 130–380 90–120 90–120 150–300 50–300 18–26 45–55 70–90 42 150–200 125–300 60 × 300 170–200 × 300–450
Genome size (total in kb)
22–27 7 8 7.2–8.4 10 12 3.5–9 16–21 12.7 13–16 13.5–21 13.6 16–20 10–14 5 5–8 36–38 3.2 120–200 150–350 100 130–280


RNA
Icosahederal
Helical
Naked
Enveloped
Enveloped
Double stranded
Single stranded positive sense
Single stranded positive sense
Single stranded Positive sense
Single stranded negative sense


DNA
Icosahederal
Helical
Complex
Naked
Enveloped
Enveloped
Enveloped
Double stranded circular
Double stranded linear
Single stranded linear positive or negative sense
Double stranded linear
Double stranded circular
Double stranded circular
Double stranded linear

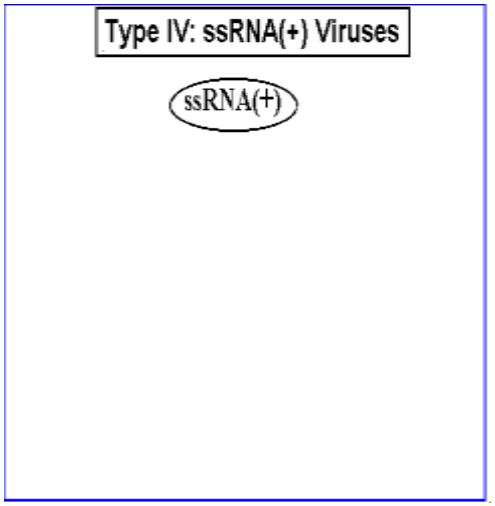
Ppt no. 62

CLASSIFICATION OF POSITIVE SENSE RNA VIRUSES



RNA VIRUS

- Ribonucleic acid as genome
- Single stranded or double stranded
- Belongs to Group III, Group IV or Group V of Baltimore Classification system



DOUBLE STRANDED RNA VIRUSES

FAMILY	GENUS	TYPE SPECIES
Birnaviridae	Avibirnavirus Aquabirnavirus	Infectious bursal disease virus Infectious pancreatic necrosis virus
Reoviridae	Orthoreovirus Orbivirus Rotavirus Coltivirus Aquareovirus	Mammalian orthoreovirus Bluetonue virus-1 Rotavirus A Colorado tick fever virus Aquareovirus A

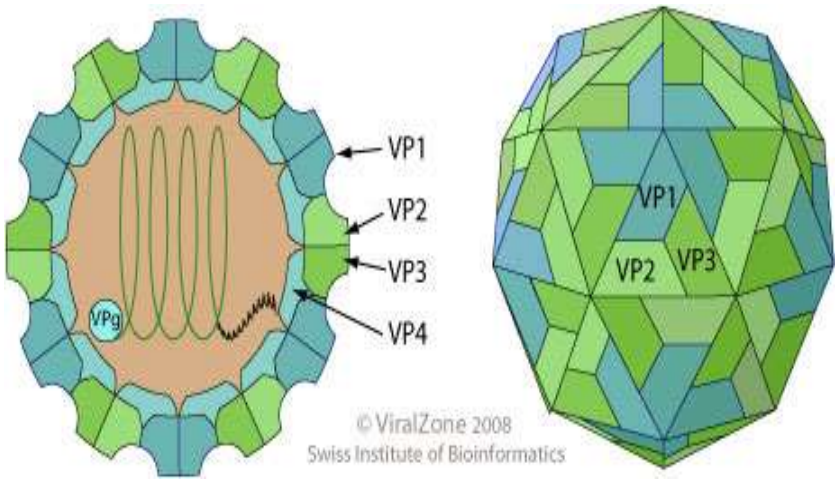
SINGLE STRANDED RNA VIRUSES

FAMILY	GENUS	TYPE SPECIES
Arteriviridae	Arterivirus	Equine arterivirus
Astroviridae	Astrovirus	Human astrovirus-1
Calciviiridae	Vesivirus Lagovirus Hepatitis E like virus	Swine vesicular exanthema virus Hepatitis E virus
Coronaviridae	Coronavirus Torovirus	Infectious bronchitis virus Equine torovirus
Flaviviridae	Flavivirus Hepacivirus pestivirus	Yellow fever virus Hepatitis C virus Bovine viral diarrhea
Picornaviridae	Enterovirus Rhinovirus Hepatovirus	Poliovirus Human rhinovirus A Hepatitis A virus
Togaviridae	Alphavirus Rubivirus	Sindbis virus Rubella virus

Ppt no. 63

PICORNAVIRUS

Properties



PROPERTIES

- Non-enveloped
- Icosahedral symmetry

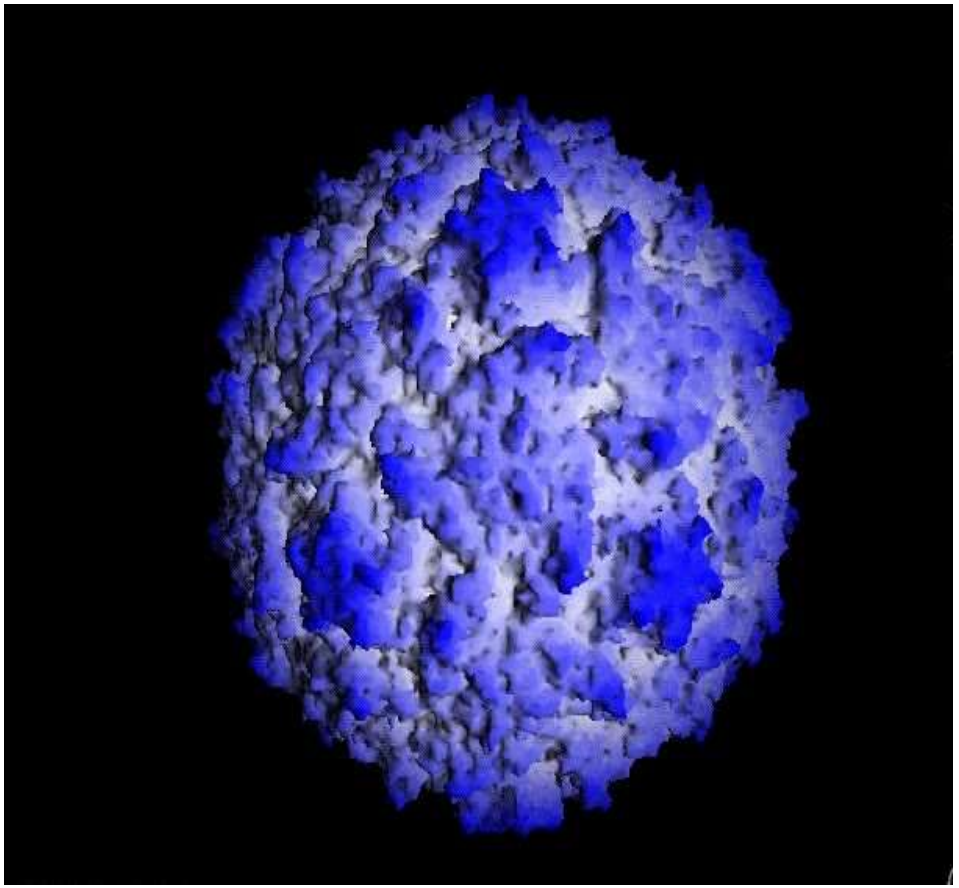
Biochemicaly,

- Ether- resistant
- Acid stable

- One of the largest families
- PICO RNA

Small sized RNA genome

- 9 genera
- 230 members



- A picornavirus is a virus belonging to the family *Picornaviridae*, and order Picornavirales.
- Vertebrates, including humans, serve as natural hosts.
- Birds also serve as host
- Picornaviruses are nonenveloped viruses that represent a large family of small, cytoplasmic, plus-strand RNA (~7.5kb) viruses .
- 30-nm icosahedral capsid.

- There are currently 80 species in this family, divided among 35 genera.
- These include the *Enterovirus*, *Aphthovirus*, *Cardiovirus*, *Rhinovirus* and *Hepatovirus* genera.
- The viruses in this family can cause a range of diseases including paralysis, meningitis, hepatitis and poliomyelitis.

Enteroviruses	Rhinoviruses	Other picornaviruses
• poliomylietis • Echovirus • Bovine enterovirus • Simian Enterovirus • Avian Enterovirus	• Human cold virus • Equine rhinovirus • Bovine rhinovirus • Foot and mouth disease • Vesicular exanthema	• Avian encephalomylietis • Mouse encephalitis • Feline picornavirus

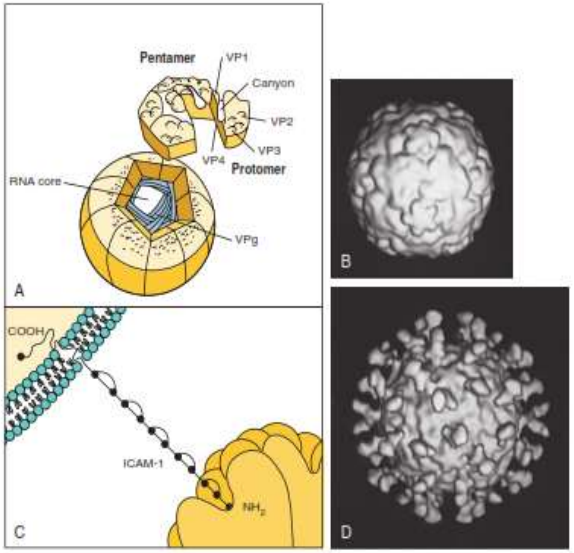
Ppt no. 64

PICORNAVIRUS

Genome

GENOMIC PROPERTIES

- Resembles mRNA
- Positive sense
- Single stranded RNA
- 7200 to 8450 bases



GENOME

- The genome is non-segmented and positive-sense.
- The geome RNA is unusual because it has a protein on the 5' end that is used as a primer for transcription by RNA polymerase.
- This primer is called VPg genome range between 2–3 kb.
- VPg contain tyrosine residue at the 3’ end.
- Tyrosine as a –OH source for covalently linked to 5’ end of RNA.

- Picornaviruses are non-enveloped, with an Icosahedral capsid.
- The capsid consists of 60 protomers.
- Each protomer consists of 4 polypeptides
- These peptides are called VP (viral protein) 1, 2, 3 and 4.
- VP2 and VP4 polypeptides originate from one protomer known as VP0
- VP0 is cleaved to give the different capsid components.

REPLICATION

- The mRNA encodes RNA dependent RNA polymerase.
- This polymerase makes complementary minus strands of RNA.
- RNA will act as template to make more plus strands.

REPLICATION

an overview of the steps in picornavirus replication attachment,

- entry
- translation
- transcription/genome replication
- Assemble
- exit

Ppt no. 65

PICORNAVIRUS

Important diseases

POLIOVIRUS

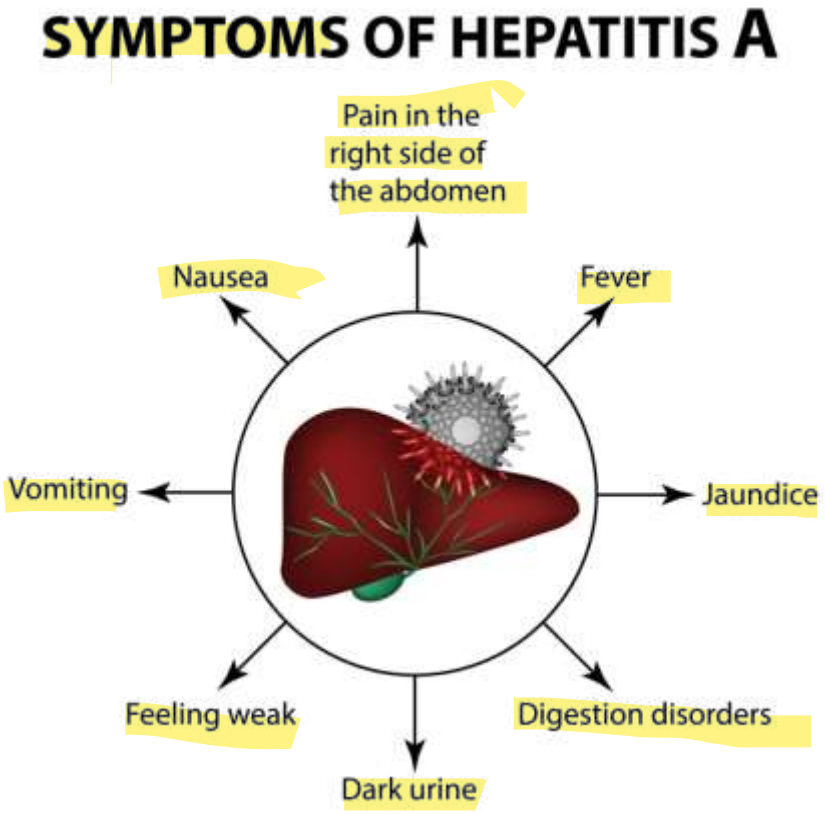
- Causes polio
- Deadly neurological conditions
- Mostly subclinical
- 1% suffer from paralytic legs
- Deadly respiratory failure

RHINOVIRUS

- Most common virus
- Common flu in humans
- Rhinovirus infection proliferates in temperatures between 33-35 °C (91-95 °F)

HEPATITIS A

- Loss of appetite
- Jaundice
- Dark urine
- No long term consequences



FOOT AND MOUTH DISEASE

- In hooved animals
- Lesions around foot and mouth
- Inability to walk
- Fast spread
- Significant economic damage

Lesions

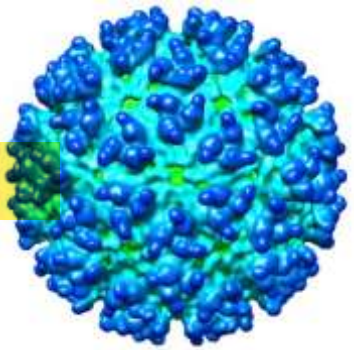
Ppt no. 66

TOGAVIRUS

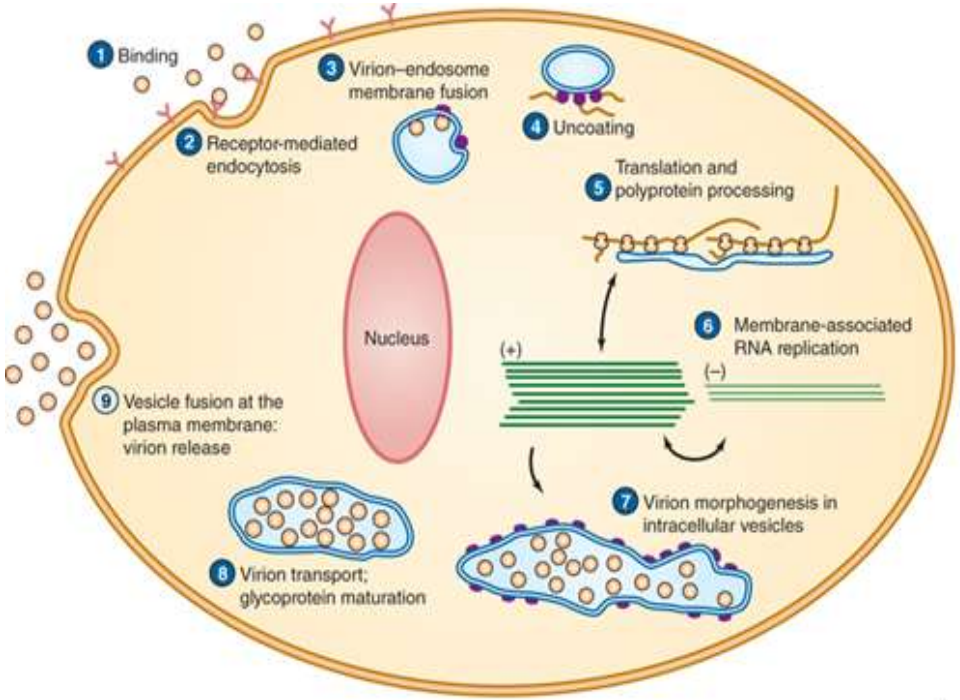
General properties

PROPERTIES

- Enveloped
- Positive single stranded RNA
- Icosahedral
- 2 genera having 32 species
  - Arbovirus
  - Rubula viru



- Early nonstructural and late structural protein synthesis
- Cytoplasmic replication
- Budding at plasma membrane



GENOME

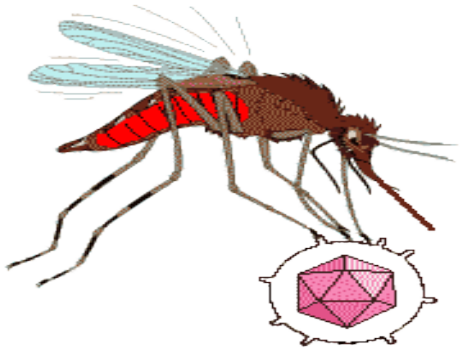
- Linear, non-segmented, single-stranded, positive sense RNA
- 10,000–12,000 Nucleotides
- The virus is enveloped and forms spherical particles (65–70 nm diameter)
- The capsid is icosahedral, constructed of 240 monomers, having a triangulation number of 4.

REPLICATION

- Entry into the host cell is achieved by attachment of the viral E glycoprotein to host receptors
- which mediates clathrin-mediated endocytosis.
- The receptors for binding are unknown
- Glycoprotein petal-like spikes act as attachment proteins.
- After virus attachment and entry into the cell, gene expression and replication takes place within the cytoplasm.

ARBOVIRUS

- Arthropod born virus (mosquitoes)
- Broad host range
- (Humans and plants)
- Zoonotic



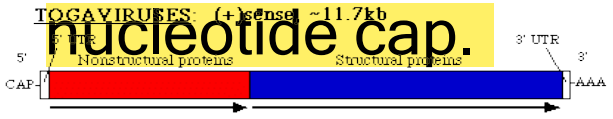
Ppt no. 67

TOGAVIRUS

Genome

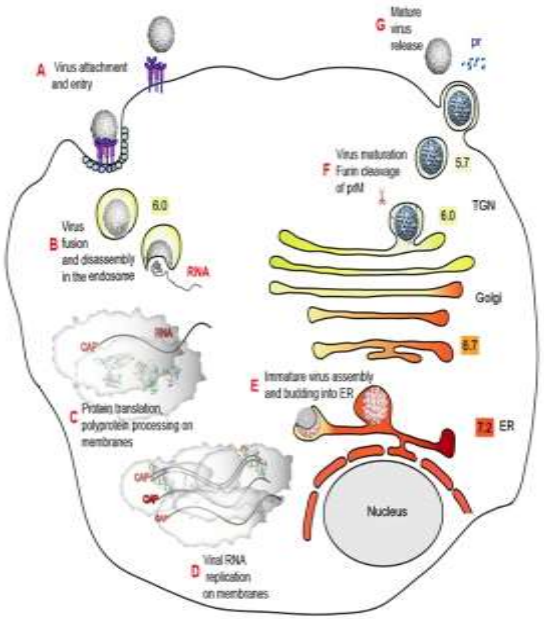
GENETIC CHARACTERS

- Positive sense
- Single stranded RNA
- Around 11.7kb
- The 5'-terminus has a methylated nucleotide cap.



REPLICATION

- Viral RNA acts as both genome as well as mRNA
- After fusion of viral capsid with cell receptor
- RNA released in cytoplasm



VIRION STRUCTURE

- Togavirus virions consist of an envelope and a nucleocapsid.
- The virions are spheical to pleomorphic in shape, and about 70 nm in diameter.
- Glycoprotein spikes evenly cover the surface of the virion.
- The genome is contained in the T=4 icosahedral capsid, which measures 40 nm in diameter.

REPLICATION

- Nonstructural proteins formed first
- Replication within cytoplasm

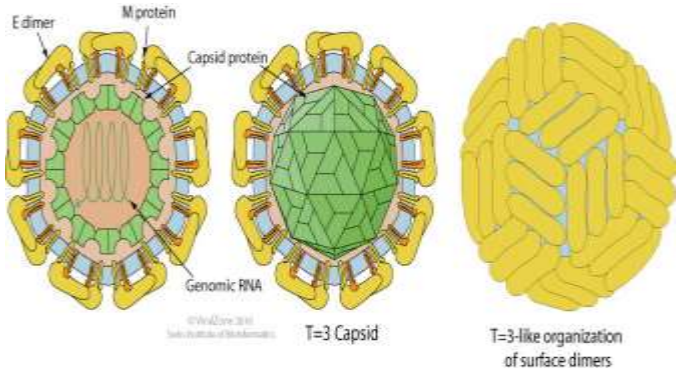
Ppt no. 68

FLAVIVIRUS

Properties

CHARACTERISTICS

- RNA virus
- Enveloped



DENGUE FEVER VIRUS

- Mosquito borne
  - *Aedes Egyptei*
- Infects bone marrow (break bone fever)
- Low platelet level
- Hemorrhegic fever
- Can lead to renal failure
- Septicemia and death

YELLOW FEVER

- *Aedes* mosquito
- Jaundice
- Backache
- Bloody diarrhea
- Bloody vomiting
- Vaccine available

WEST NILE VIRUS

- Birds are reservoir
- Mosquitoes Vector
  - Culex
- Encephalitis
- Meningitis
- Flaccid paralysis
- Seizures and coma

HEPATITIS C

- Transmission: Exposure to infected blood
- Highly antigenic variable
- Jaundice
- 60-80% become chronic cases
- Cirrhosis
- Primary cause of hepatocellular carcinoma
- Interferon treatment

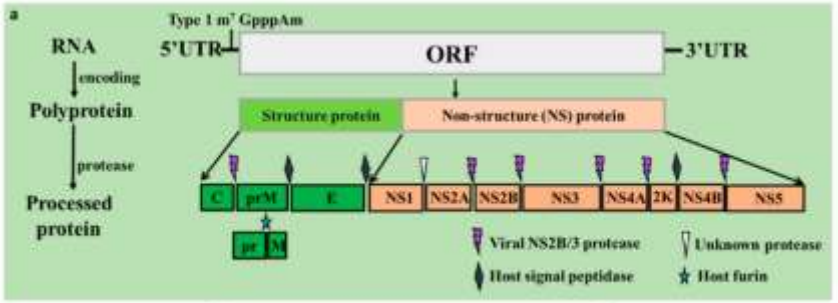
Ppt no. 69

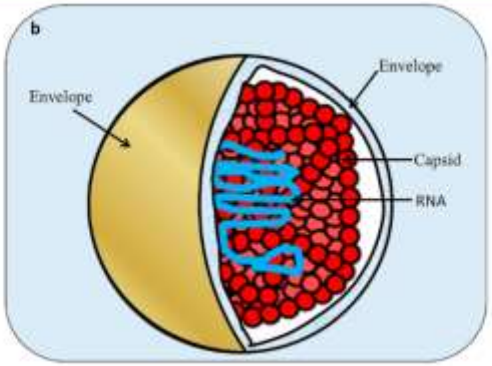
FLAVIVIRUS

Genome

GENOME

- RNA genome
- Positive sense
- Non-segmented





TWO GROUP HYPOTHESIS

On the basis of morphological information

Obtained using electron microscopy

supported the hypothesis of the existence of at least two virus groups

- One group includes non-enveloped virus
- Family Reoviridae (genera Orbivirus, Coltivirus and Seadornavirus)
- Viruses with an overall diameter of 60-80 nm
- Icosahedral symmetry
- Several concentric capsid layers that surround a segmented double-stranded RNA (dsRNA) genome

- Second group includes enveloped viruses (inactivated by ether and deoxycholate)
- 50-60 nm in diameter
- Infectious ssRNA genome of positive polarity.
- Two antigenically distinct subgroups i.e.
- Group A and Group B.

MEDICAL IMPORTANCE

- Mostly serologically related to each other
- Antibodies of one useful against others
- Antigenic variability of Hepatitis C proteins
- The 17D vaccine is one of the most efficient vaccines ever developed and was derived from a strain of Flavivirus isolated from a man who recovered from infection by the virus.

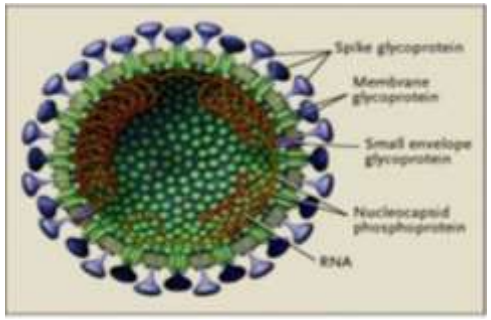
Ppt no. 70

CORONAVIRUS

Structure

STRUCTURE

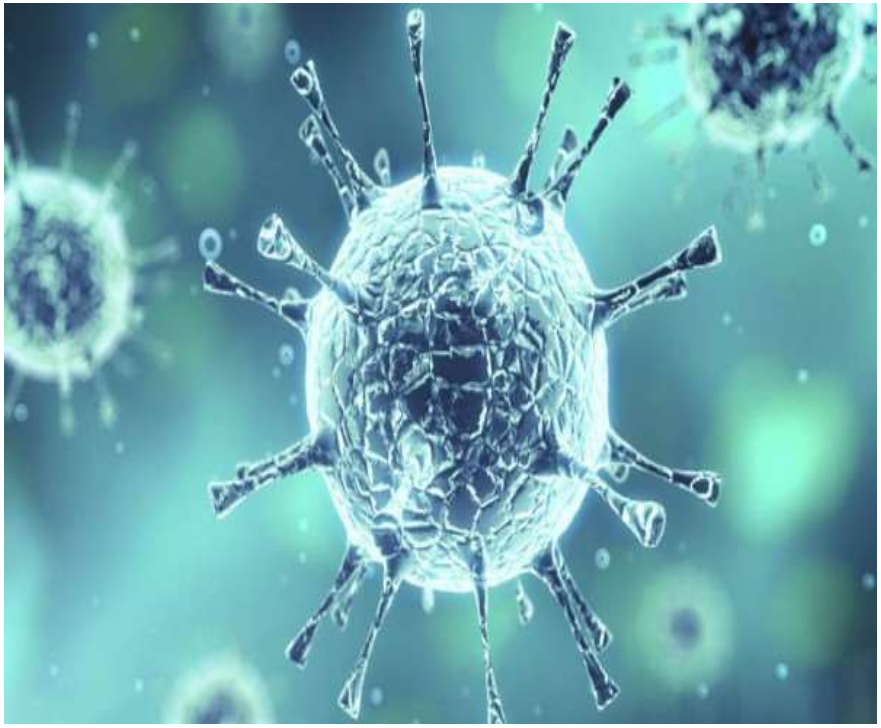
- Enveloped/ encapsulated
- Club shaped projections on envelope
- 80-160nm diameter
- Helical shaped



- *Coronaviruses* are species belonging to the subfamily Coronavirinae in the family Coronaviridae, in the order Nidovirales.
- Coronaviruses are enveloped viruses with a positive sense single-stranded RNA genome and with a nucleocapsid of helical symmetry.
- The Genomic size of coronaviruses ranges from approximately 26 to 32 Kb, the largest for an RNA virus

COMMON COLD

- Second most common cause of common cold
- SARS (Severe Acute Respiratory syndrome)
- GI tract infections in kids



SARS

- Zoonotic origin
- Acute respiratory form
- Outbreaks leads to enormous number of deaths
- Flu and Fever
- Lethargy
- Cough
- Pnemonia

MERS

- Middle east respiratory syndrome
- Fever
- Cough
- Shortness of breath
- Sometimes diarrhea
- Mortality rate: 35%
- Camels are major reservoir hosts

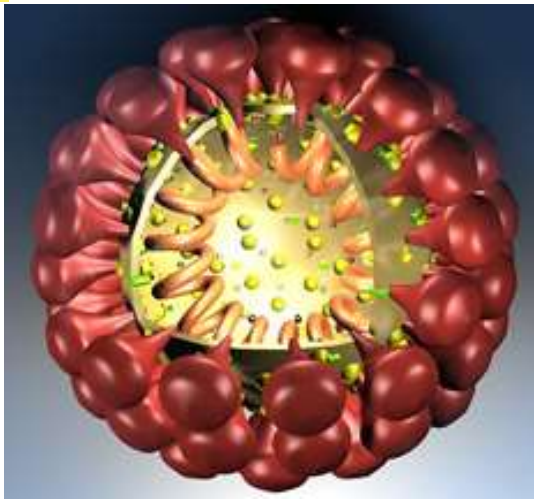
Ppt no. 71

CORONAVIRUS

Genome

GENOME

- Positive sense
- Single stranded
- RNA
- 27,000 -30,000 kb genome



- Enveloped
- Spherical
- About 120 nm in diameter
- The RNA genome is associated with the N protein to form the nucleocapsid (helical for the genus coronavirus, and tubular for the genus torovirus)
- Linear ssRNA

- 27-32kb in size (the largest of all RNA virus genomes)
- Capped
- Polyadenylated
- The leader RNA (65-89 bp) at the 5' end of the genome is also present at the end of each subgenomic RNAs.

REPLICATION

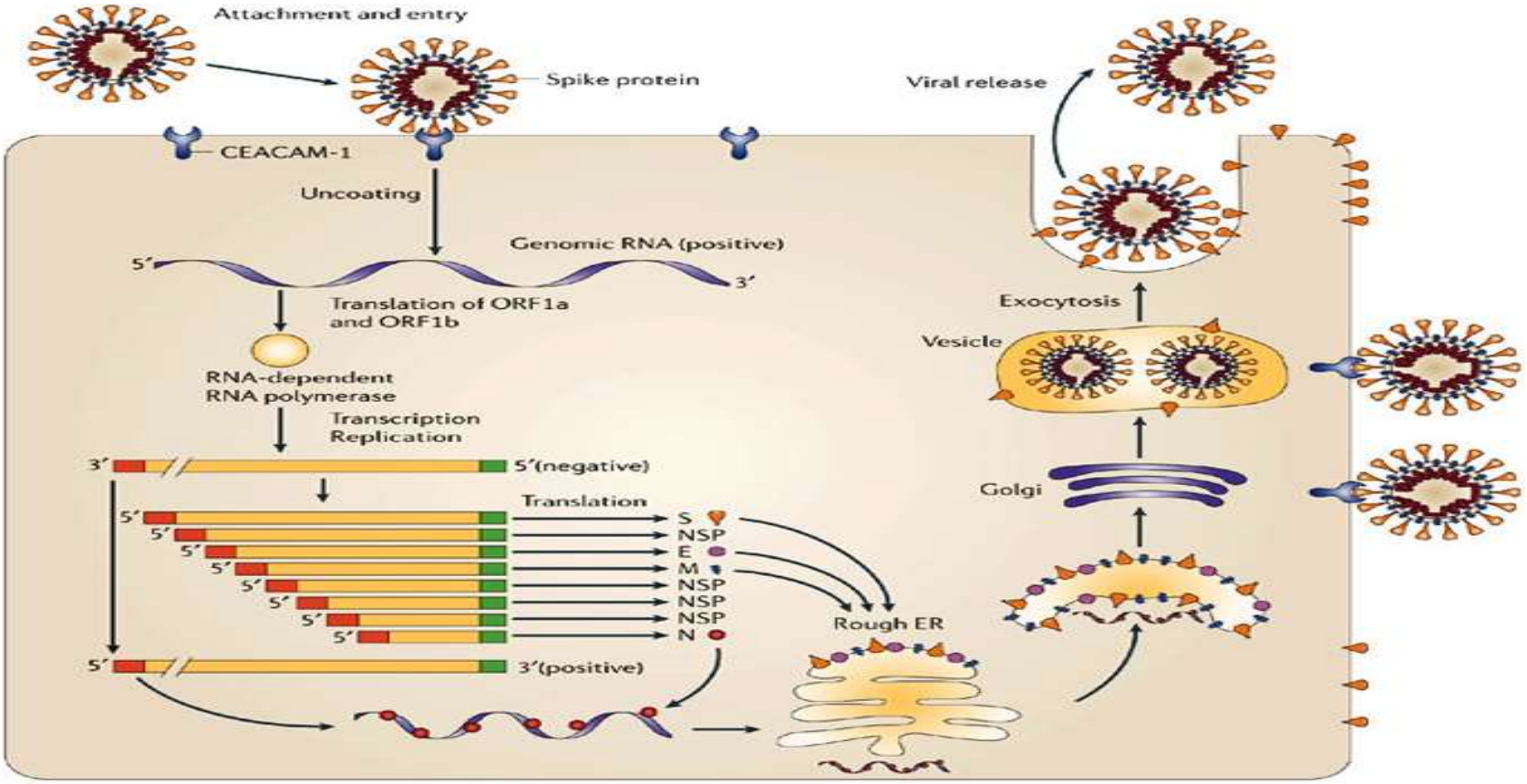
CYTOPLASMIC

- Attachment of the viral S protein receptors
- Endocytosis of the virus into the host cell
- Fusion of virus membrane with endosomal membrane
- Genome is released into the cytoplasm.
- Synthesis and proteolytic cleavage of the replicase polyprotein.

REPLICATION

- A dsRNA genome is synthesized from the genomic ssRNA(+).
- The dsRNA genome is transcribed by providing viral mRNAs/new ssRNA(+) genomes.
- Synthesis of structural proteins encoded by subgenomic mRNAs.
- Assembly and budding at membranes of the endoplasmic reticulum (ER), the intermediate compartments, and/or the Golgi complex.
- Release of new virions by exocytosis.

REPLICATION IN CYTOPLASM

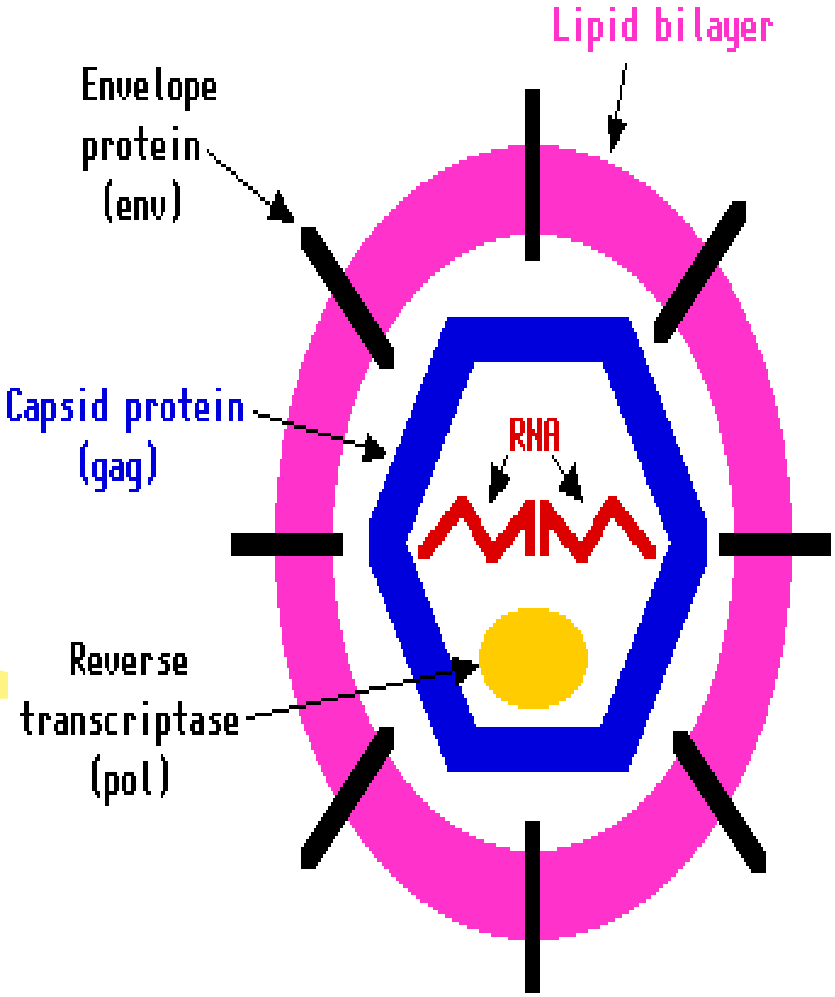


Ppt no. 72

RETROVIRUS

BASICS

- An envelope
- Envelope protein
- A capsid
- Two molecules of RNA
- Reverse transcriptase



REPLICATION PATTERN

Once inside the host cell cytoplasm, the virus uses its own RT enzyme to produce DNA from its RNA genome

The reverse of the usual pattern,
thus *retro* (backwards)

The new DNA is then incorporated into the host cell genome by an integrase enzyme referred to as a provirus

REPLICATION PATTERN

The host cell then treats the viral DNA as part of its own genome, transcribing and translating the viral genes along with the cell's own genes

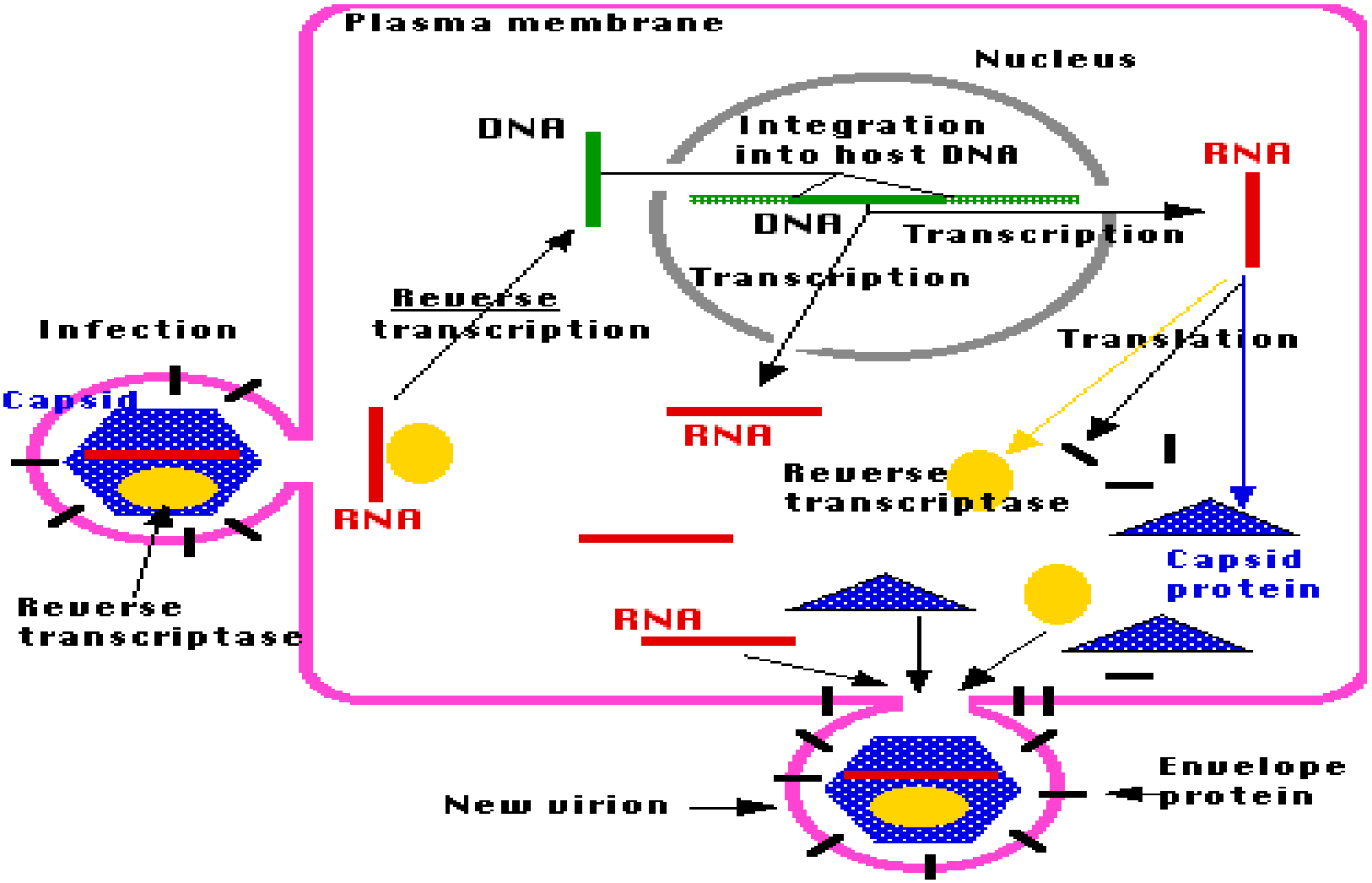
The proteins are produced which are required to assemble new copies of the virus.

It is difficult to detect the virus until it has infected the host.

Infection will persist indefinitely

REPLICATION PATTERN

- The gag gene --- capsid protein
- The pol gene --- molecules of reverse transcriptase
- The env gene --- molecules of the envelope protein


Plasma membrane
Nucleus
DNA
Integration
into host DNA
RNA
DNA
Transcription
Transcription
Reuerse
transcription
Infection
Translation
Capsid
RNA
RNA
Reuerse
transcriptase
Capsid
protein
Reuerse
transcriptase
RNA
Enuelope
protein
New virion

HUMAN IMMUNODEFICIENCY VIRUS

- Most important immunosuppressing virus
- Causes immuno-supression
- HIV-1 and HIV-2
- AIDS

DISCOVERY

- In February 1997
- Parkinson's disease
- Transplant of pig tissue to human beings
- Recipients immune system need to suppress to avoid rejection
- HIV-1 and HIV-2, retroviruses of primates

TRANSMISSION

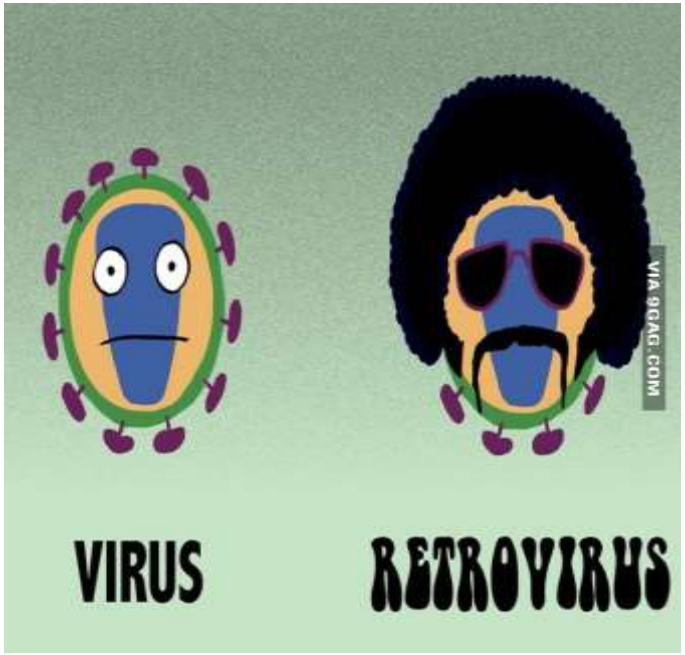
- blood
- Semen
- pre-seminal fluid
- rectal fluids
- vaginal fluids
- breast milk
- **Blood transfusion**
- **Syringes**

Ppt no. 73

RETROVIRUS

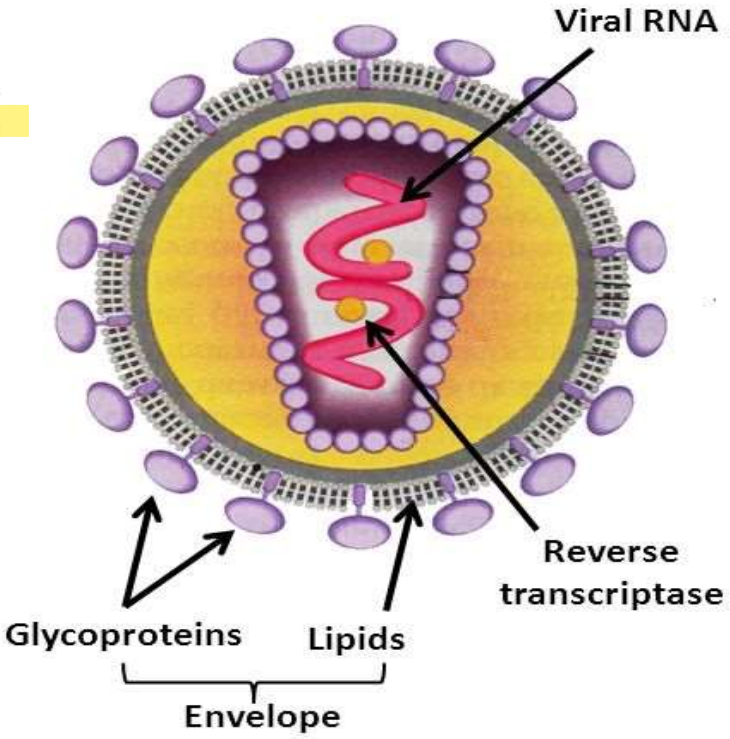
Characteristics

Most challenging virus for human beings so far.....



Retrovirus Structure

- All retroviruses:
  - Are surrounded by a viral envelope
  - Are icosahedral in shape
  - Contain 2 identical molecules of RNA
  - Contain the enzyme *reverse transcriptase*



- Virions of retroviruses consist of enveloped particles about 100 nm in diameter.
- The virions also contain two identical single-stranded RNA molecules 7-10kb in length.
- Although virions of different retroviruses do not have the same morphology or biology
- All the virion components are very similar.

Virion components

- Envelope
- Nucleic acid (RNA)
- Protiens

ENVELOPE

- composed of lipids
- obtained from the host plamsa membrane
- glycoprotein encoded by the env gene.

Functions

- protection from the extracellular environment via the lipid bilayer
- enabling the retrovirus to enter/exit host cells through endosomal membrane trafficking
- the ability to directly enter cells by fusing with their membranes.

NUCLEIC ACID

- dimer RNA
- cap at the 5' end and a poly(A) tail at the 3' end
- terminal noncoding regions for replication
- internal regions that encode virion proteins
- The 5' end includes four regions, which are R, U5, PBS, and L.

PROTEINS

- consisting of gag proteins, protease, pol proteins, and env proteins

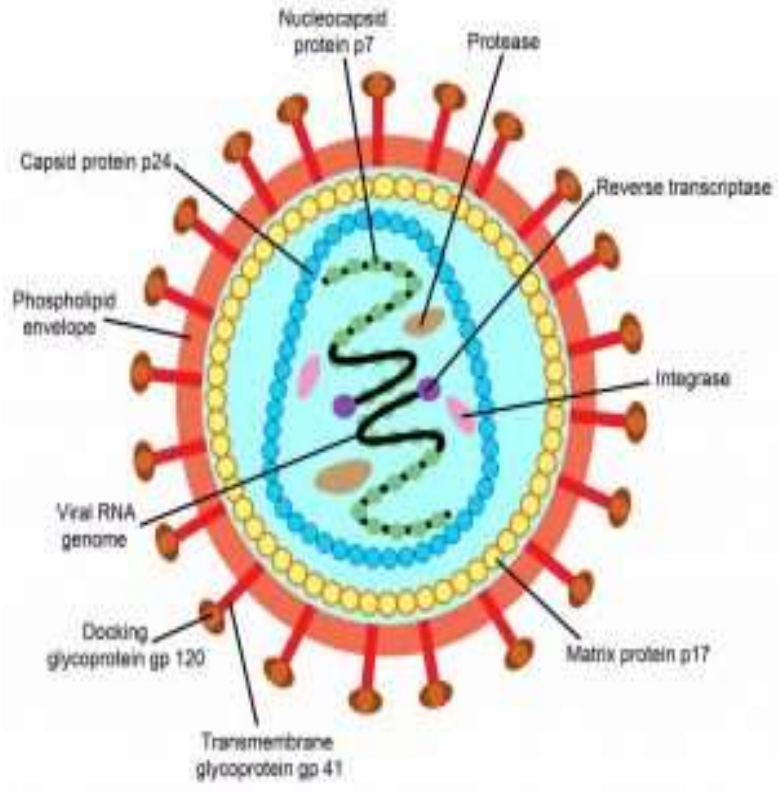
SALIENT CHARACTERISTICS

- Retroviruses are the only animal viruses that integrate into the host cell's genome
- The retroviral genomes can accommodate changes to its configuration.
- The retroviral life cycle begins in the nucleus of an infected cell.
- Retroviruses offer gene therapy researchers aid for delivering genes to target cells at high efficiency that allows for long-term, stable expression of introduced genetic elements

Ppt no. 74

RETROVIRUS

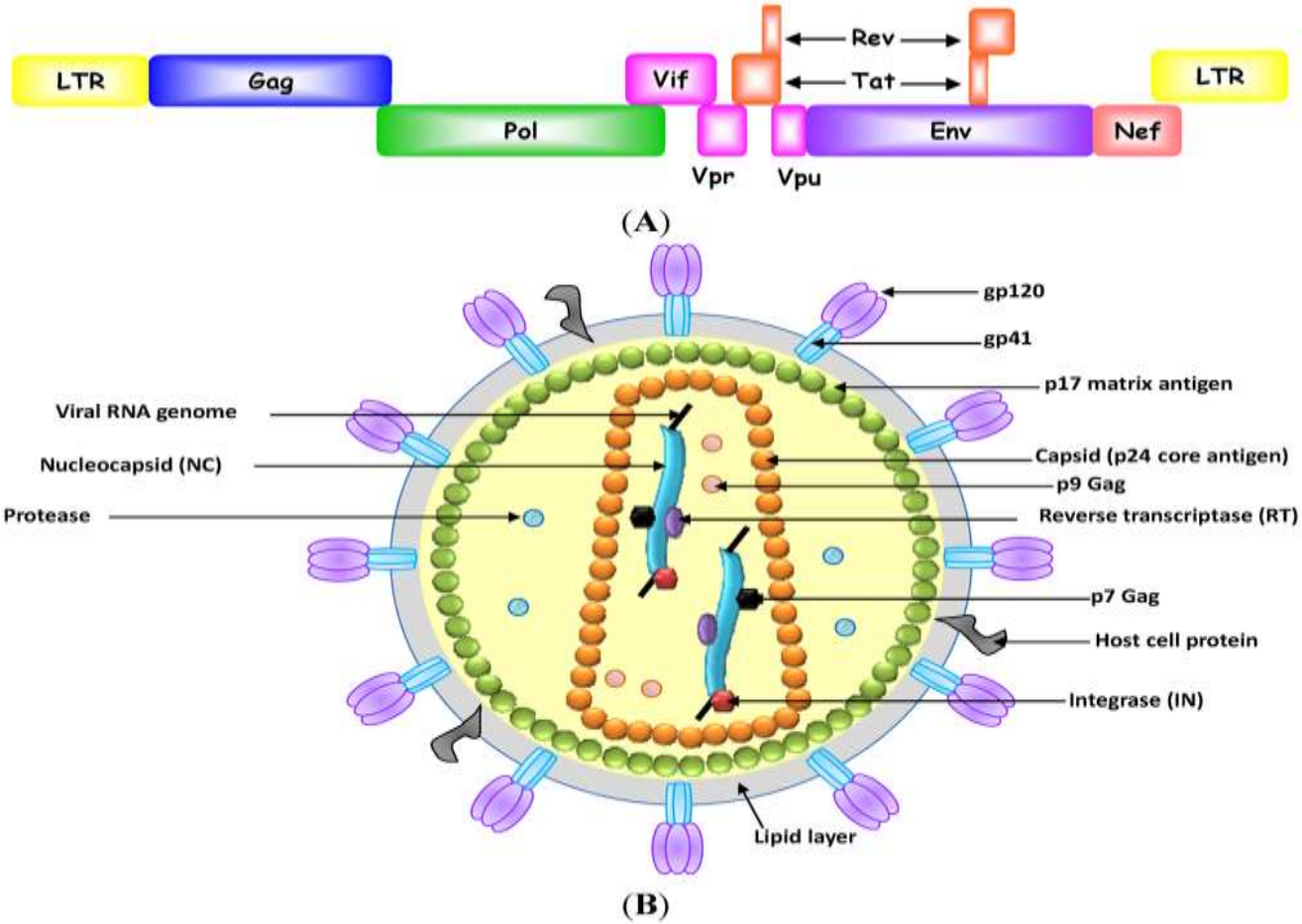
Genome



GENETIC CHARACTERS

- Single stranded RNA viruses
- Positive sense
- enveloped
- Changes to DNA by RT enzyme

Genomic Structure

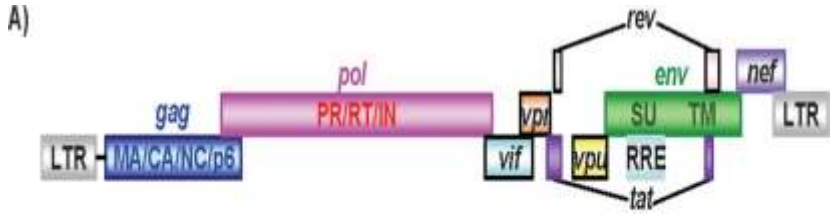


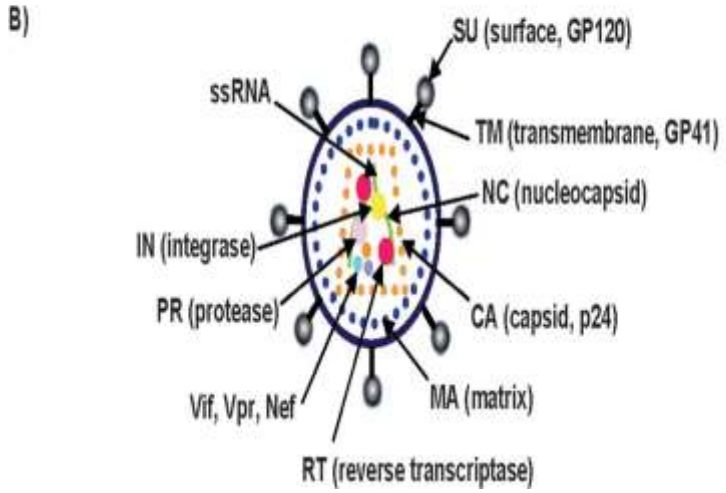
Genetic makeup

Diploid nature of virus

3 very imp. genes:

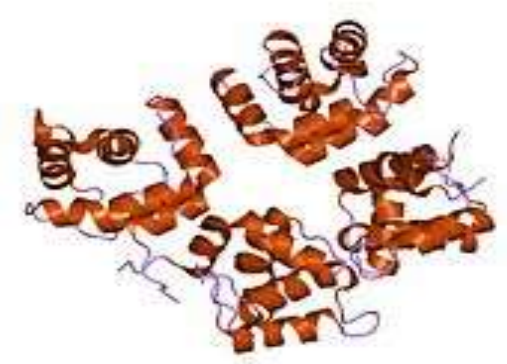
- GAG
- ENV
- POL





GAG

- Gag gene makes protein p24
- Which serves as capsule around RNA
- Cone shaped



ENV

- Envelope glcoproteins
- gp41 and gp120
- Gp41: transmembrane protein
- Gp120: outer glycoprotein
- Host receptor

POL

- Reverse transcriptase Polymerase
- Integrase activity
- DNA from RNA

Ppt no. 75

RETROVIRUS
Lifecycle

RETROVIRAL REPLICATION CYCLE

1- early steps

from the cell surface to the nucleus

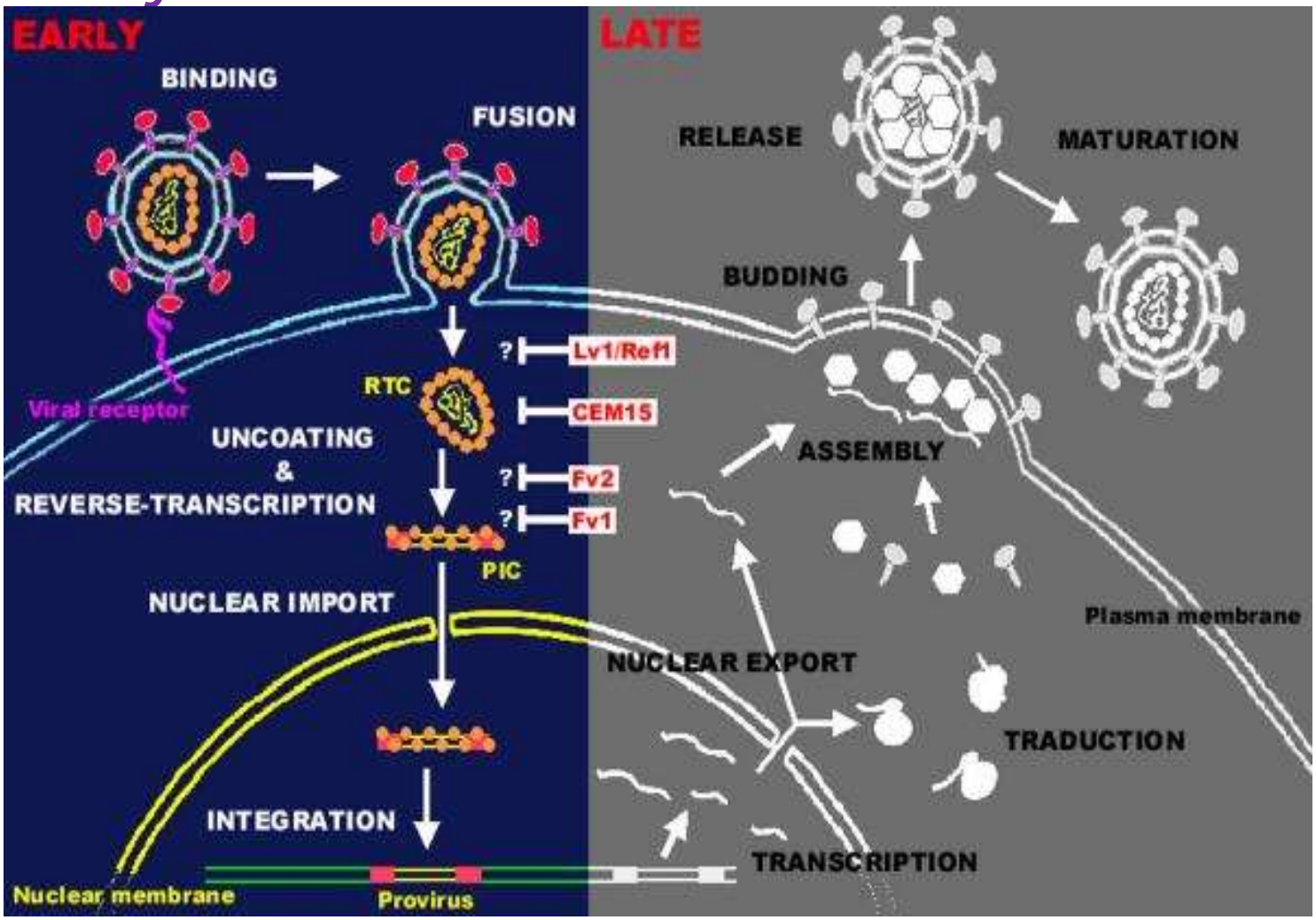
- Retroviral particles must bind specifically to their target cells
- cross the plasma membrane
- reverse-transcribe their RNA genome
- Uncoat the cores
- find their way to the nuclear membrane
- penetrate into the nucleus

Hijack the cellular machinery

2- LATER STAGE

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

- Lifecycle


EARLY
LATE
BINDING
FUSION
RELEASE
MATURATION
BUDDING
Viral receptor
RTC
? Lv1/Ref1
CEM15
UNCOATING
&
REVERSE-TRANSCRIPTION
? Fv2
? Fv1
PIC
ASSEMBLY
NUCLEAR IMPORT
Plasma membrane
NUCLEAR EXPORT
TRADUCTION
INTEGRATION
TRANSCRIPTION
Nuclear membrane
Provirus

Life Cycle of a Retrovirus - YouTube.MP4

Ppt no. 76

POSITIVE SENSE RNA PLANT VIRUS

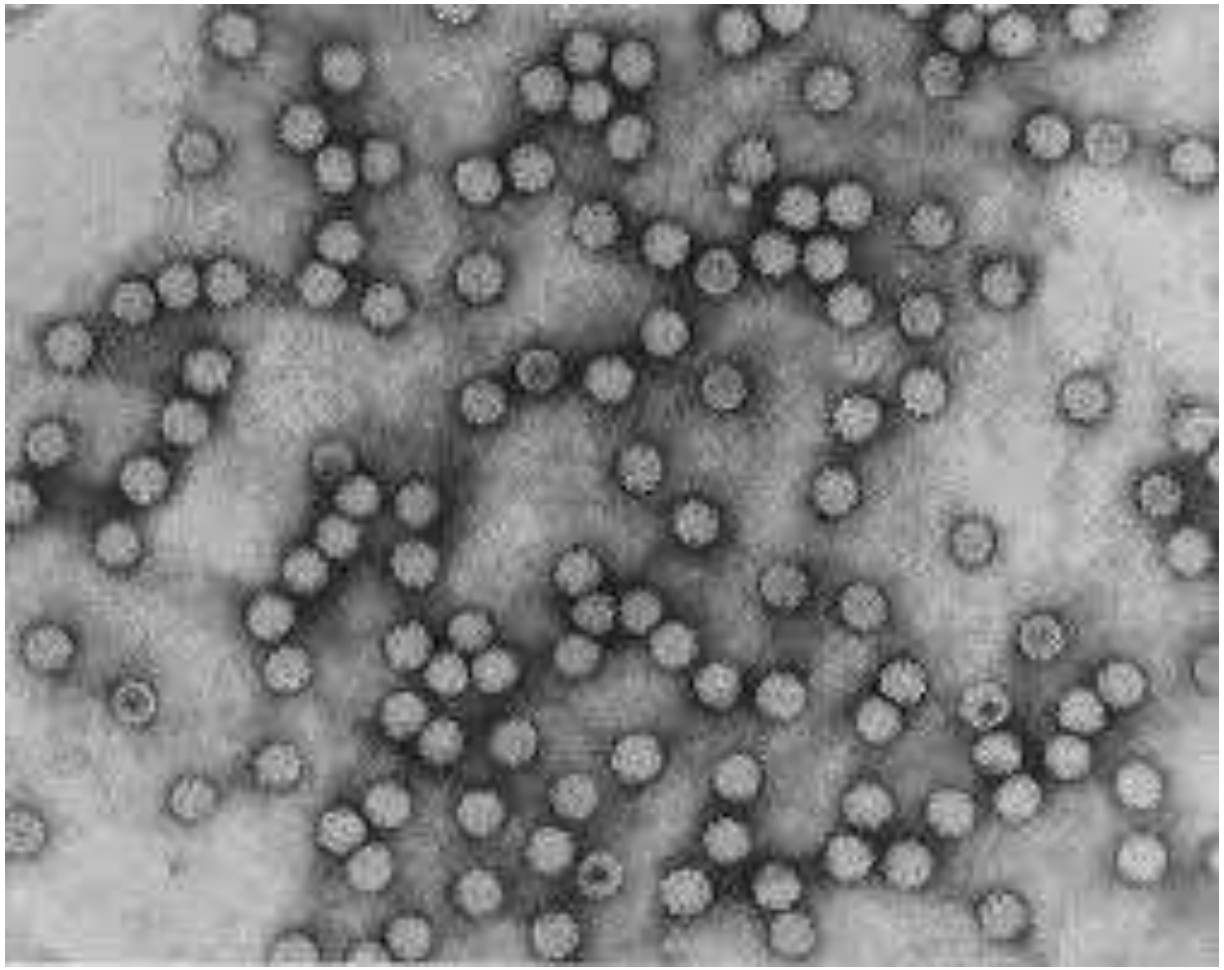
LUTEOVIRIDAE

- Sobemovirus

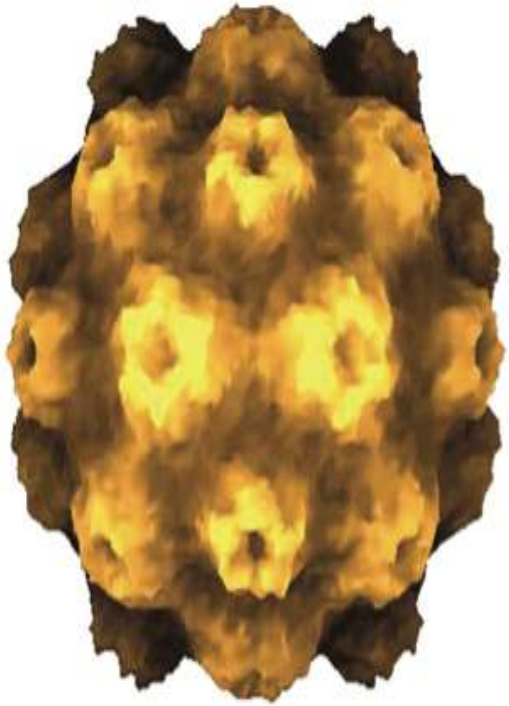
SEQUIVIRIDAE

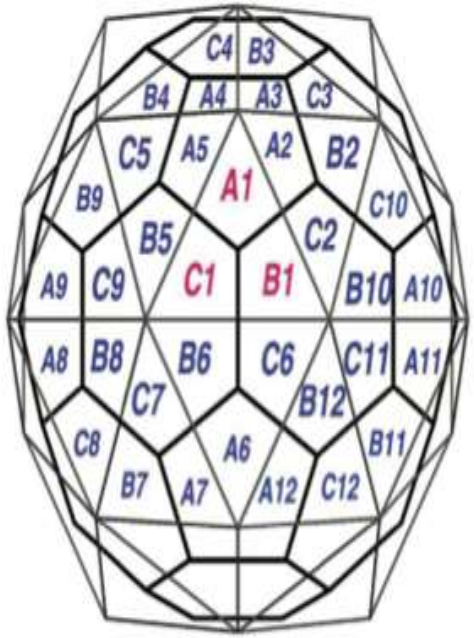
- Sadwavirus
- Chevavirus

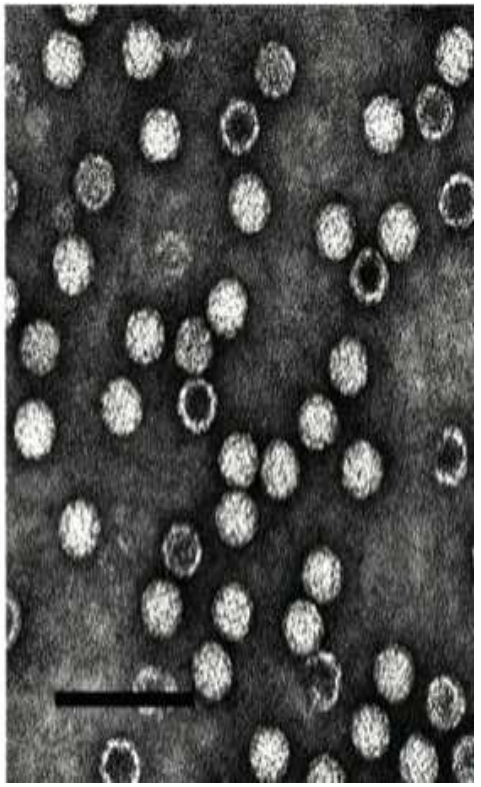
TOMBUSVIRIDAE



TYMOVIRIDAE







COMOVIRIDAE

- Idaeovirus

BROMOVIRIDAE

- Cucumovirus
- Bromovirus
- Ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordievirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus

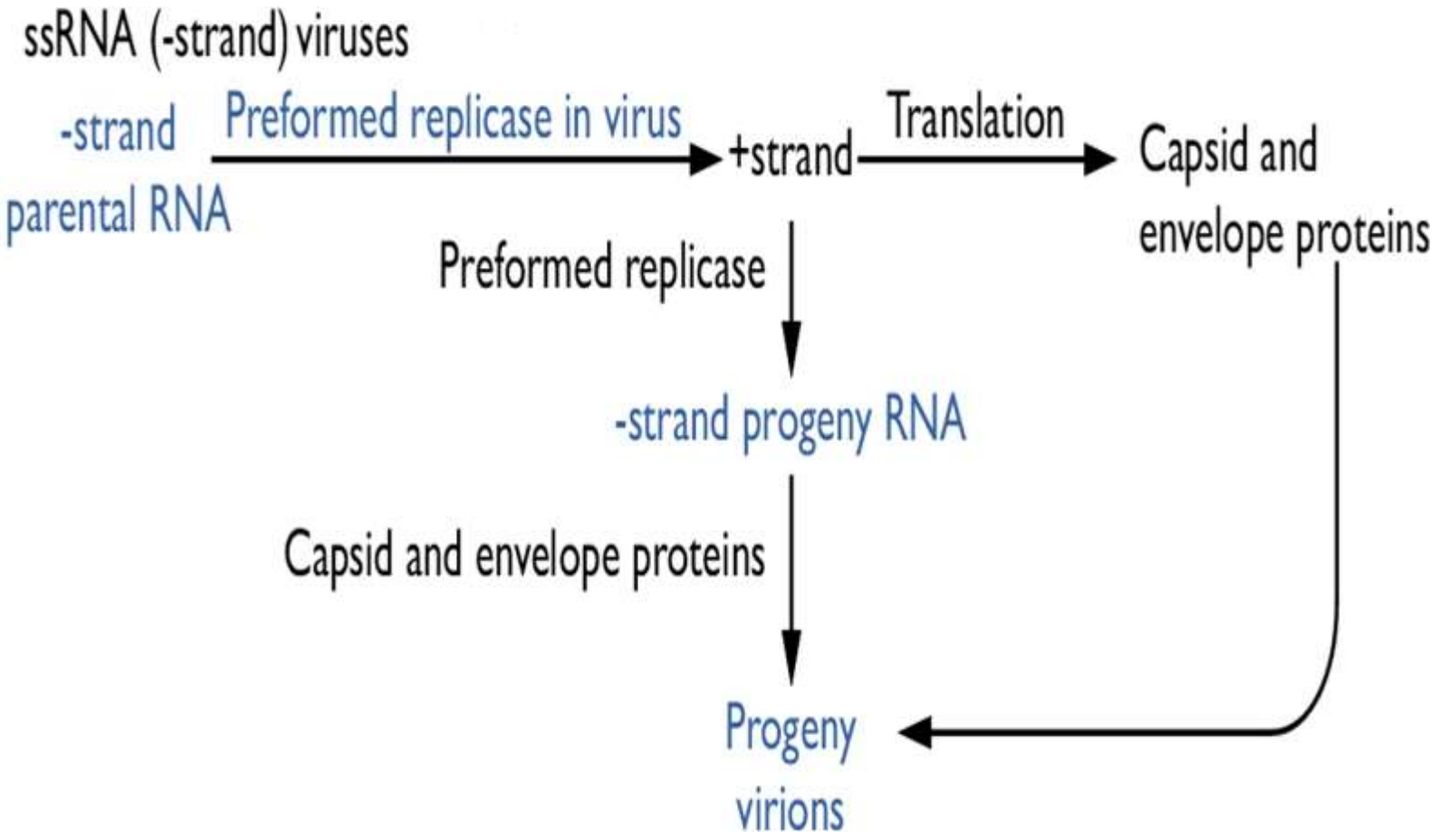
OTHERS...

- Flexiviridae
- Potyviridae
- Closteroviridae

Ppt no. 77

NEGATIVE SENSE RNA VIRUSES

REPLICATION



Family	Subfamily	Genus	Species
Bornaviridae		*Bornavirus*	Borna disease virus
Bunyaviridae		*Bunyavirus* *Hantavirus* *Phlebovirus*	Bunyamweravirus Hantaanvirus Rift valley fever
Orthoviridae		*Influenza virus A,B,C* *Thogotovirus*	Influenza A,B,C Thogotovirus
Paraviridae	*Paramyxovirinae* *Pneumovirinae*	*Respirovirus* *Morbillivirus* *Rubulavirus* *Pneumovirsu* *Metapneumovirus*	Sendai virus Measles Virus Mumps virus Human respiratory synctial virus
Rhabdoviridae		*Lyssavirus* *Vessiculovirus* *Ephemerovirus* *Novirhabdovirus*	Rabies virus Vessicular Stomatitis Virus Bovine Ephemeral fever virus

Ppt no. 78

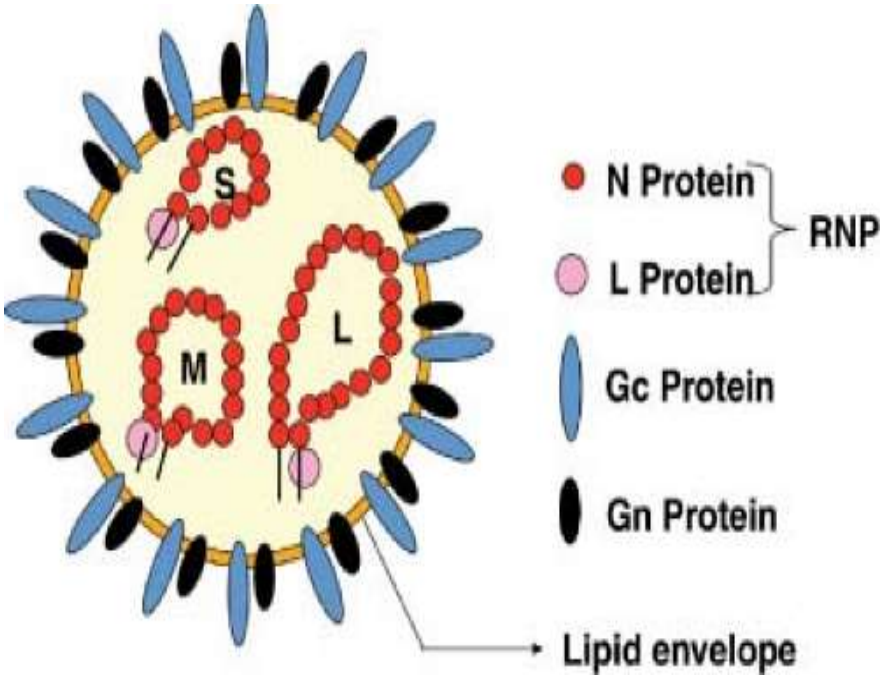
BUNYAVIRUS

Characteristics

GENERAL PROPERTIES

- Enveloped RNA viruses
- Found in Arthopods and rodents ; occasionally infecting humans
- Some of them also infect plants
- Vector-borne except Hantaviruses
- Vectors (mosquitos, tick, or sandfly)
- Hantaviruses are transmitted through contact with deer mice feces
- Vector activity, for example, mosquito-borne viruses are more common in the summer

MORPHOLOGY



- Spherical
- 90-100nm
- Lipid envelope
- Spikes
- Lack matrix proteins

CLASSIFICATION

- Largest family
- 5 genera:

1. Orthobunyavirus
2. Hantavirus
3. Nairovirus
4. Phleovirus
5. Tospovirus

DISEASES

Family: Peribunyaviridae

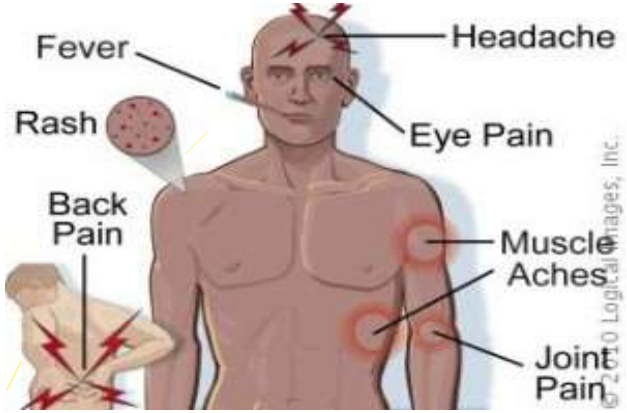
California encephalitis virus, La Crosse encephalitis virus, Jamestown Canyon virus and Snowshoe hare virus

vector: mosquitoes

Family: Nairoviridae

Crimmean Hemmorhegic Congo fever

Vector: ticks



DISEASES

Family:Phenuiviridae

- Rift velley fever
- Vector: mosquitoes

Family:Hantaviridae

- Hantavirus
- Vector: aerosolized excreta from these mammals

Ppt no. 79

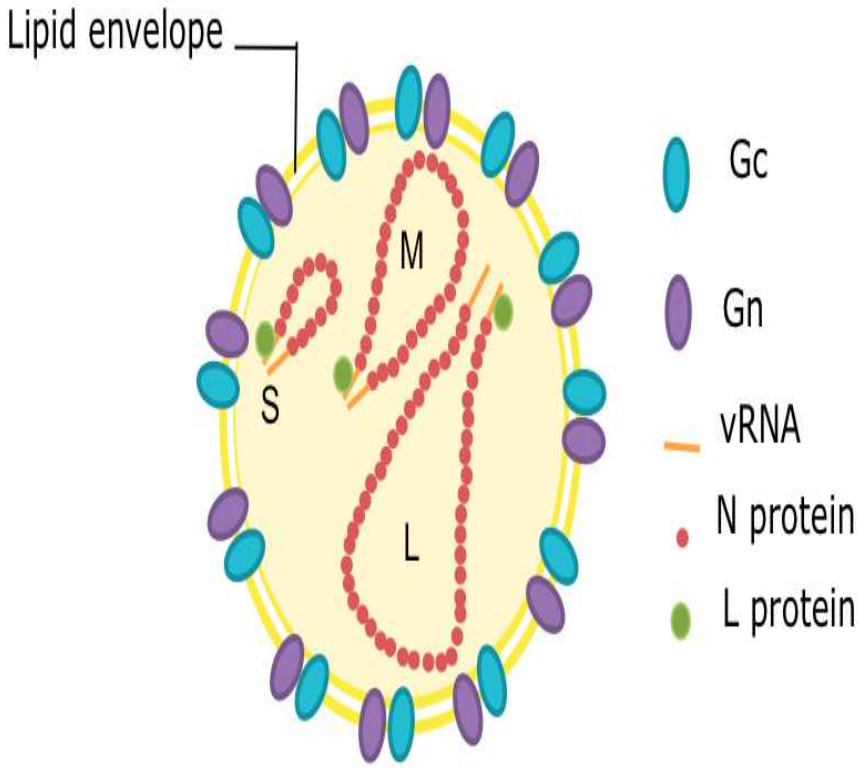
BUNYAVIRUS

Genome

GENOMIC PROPERTIES

- Negative sense RNA virus
- Three segments (segmented)
  - Large (L)
  - Medium (M)
  - Small (S)
- 11-19kbs

MORPHOLOGY



- Three segments (segmented)
  - Large (L)
  - Medium (M)
  - Small (S)

SEGMENT SIZE

- Vary

Segment	Size
L	6.3-12
M	3.5-6
S	1-2.2

FUNCTIONS

- The L segment encodes the RNA Dependent RNA-polymerase, necessary for viral RNA replication and mRNA synthesis.
- The M segment encodes the viral glycopro teins, which project from the viral surface and aid the virus in attaching to and entering the host cell.
- The S segment encodes the nucleocapsid protein (N).

- The L and M segment are negative sense.
- the S segment is ambisense (Phlebovirus and Tospov irus)
- The S segment codes for the viral nucleoprotein (N) in the negative sense and a non structural (NSs) protein in ambisense.

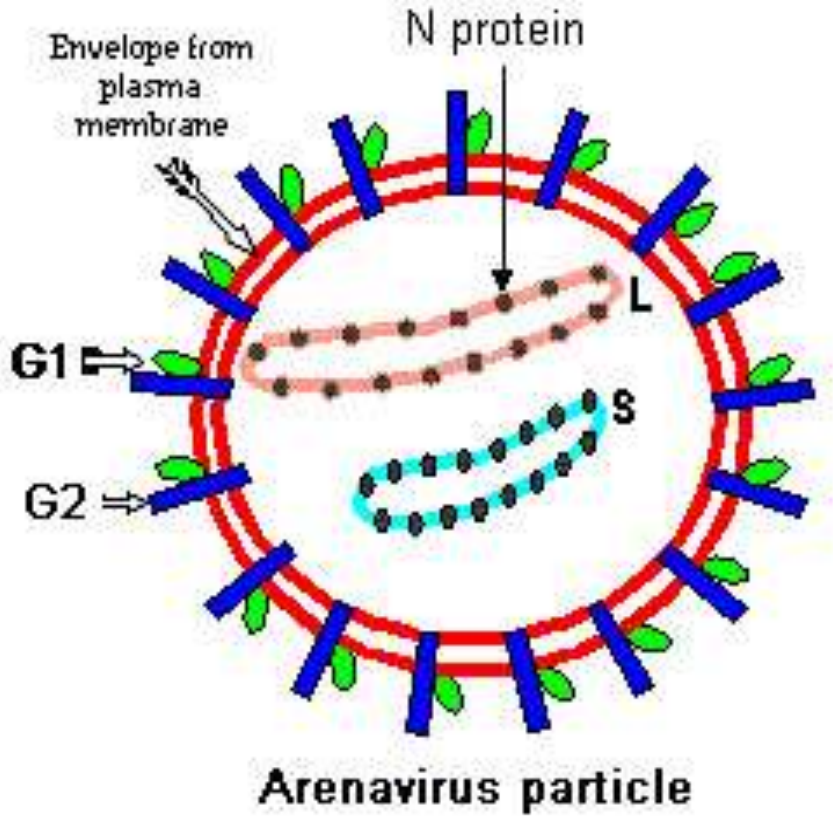
Ppt no. 80

ARENAVIRUS

Characteristics

GENERAL PROPERTIES

- Name from Latin "arenosus" meaning sand
- Sandy appearance of ribosomes
- two single-stranded RNA segments
- Virion Pleomorphic or spherical
- 60 to 300nm (variable)

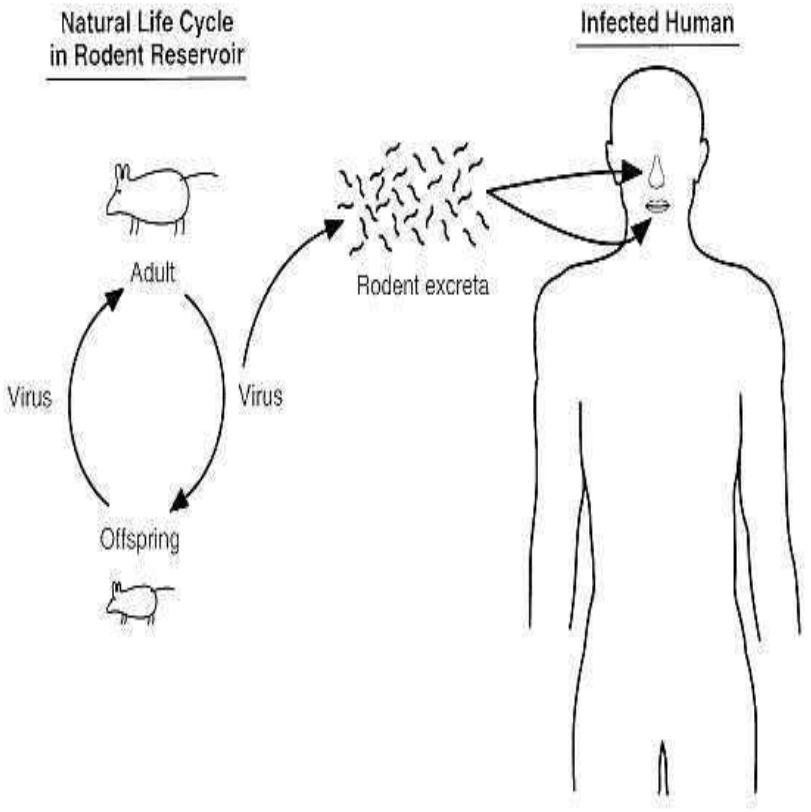


Arenavirus particle

- family *Arenaviridae*
- Hosts; rodents and occasionally humans
- Snakes
- At least eight arenaviruses are known to cause human disease
- The diseases derived from arenaviruses range in severity

DISEASES

- Transferred from rodents to humans
- Aseptic meningitis
- Viral hemorrhagic fever



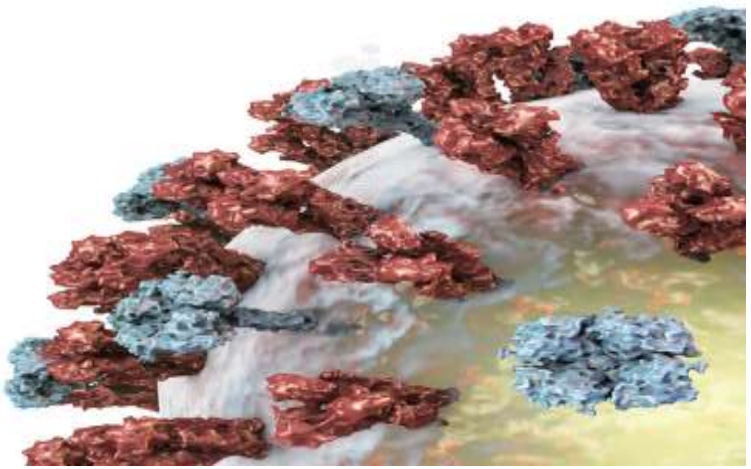
DISEASE

- Lymohocytic Choriomeningitis Virus infection
- Hemmorhegic fever from a number of virus belonging to this family

Ppt no. 82

ORTHOMYXOVIRIDAE

Characteristics



NOMENCLATURE

- Greek word
- Orthos meaning proper
- Myx meaning mucus

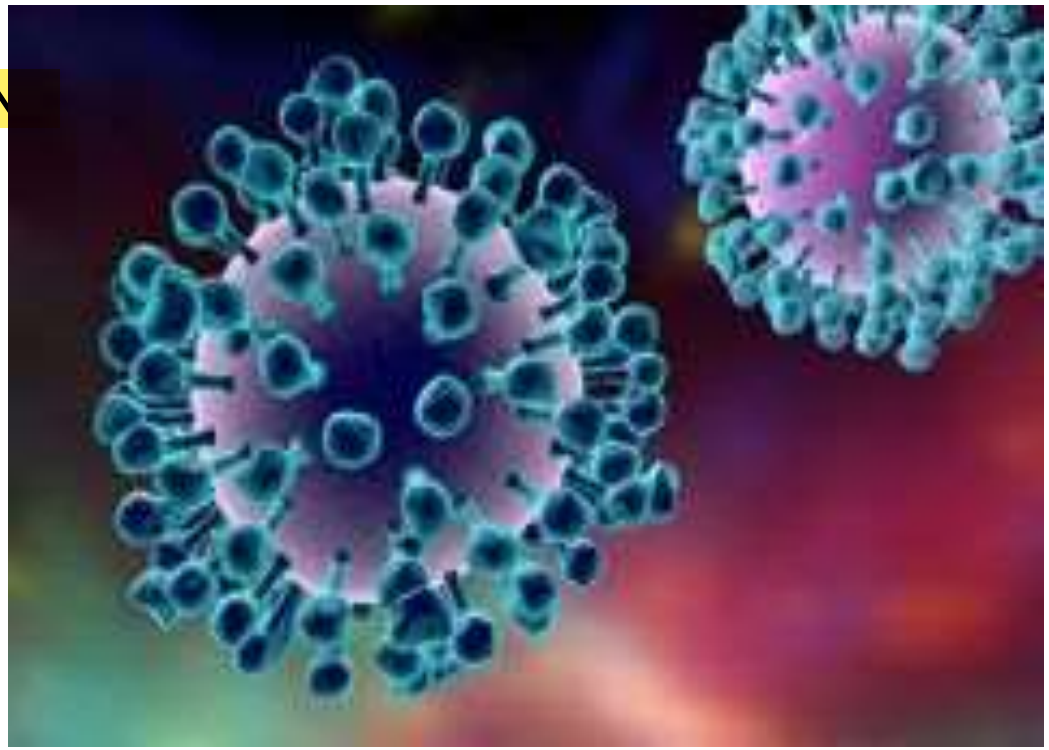
MORPHOLOGY

- Spherical/ pleomorphic
- Enveloped
- 80-120nm
- Spikes on Influenza A and B
- Labile to heat, pH variations, detergents

PROTEINS

- The hemagglutinin (HA)

- The neuraminidase (N



GENERA

- Influenza virus A
- Influenza virus B
- Influenza virus C
- Influenza virus D
- Isavirus
- Thogotovirus
- Quaranjavirus

DIEASES

- Influenza A infects humans, other mammals, and birds, and causes all flu pandemics
- Influenza B infects humans and seals
- Influenza virus C infects humans, pigs, and dogs
- Influenza virus D infects cattle and pigs

Ppt no. 83

ORTHOMYXOVIRIDAE

Genome

GENOMIC PROPERTIES

- Helical nucleopcapsid
- Segmented genome
- Negative sense
- Single stranded RNA
- 10 – 13.6kb
- 12000-15000kb

RNPs

- The segmented genome is encapsidated with each segment in a separate nucleocapsid called as *Ribonucleoprotein complexes (RNPs)*

- terminal sequences of RNA bound to trimeric RdRp.
- All these RNPs are surrounded by one envelope

segment	Protein
PB1	Polymerase
PB2	Polymerase
PA	Polymerase
HA	Hemagglutinin
NP	Nucleoprotein
NA	Neuraminidase
M	Membrane protein(s)
NS	non-structural protein(s)

REPLICATION

- Entrance through body fluids
- Virus binds through hemagglutinin-glycoprotein and sialic acid receptors
- Virus enters by endocytosis
- Proteins and polymerase forms complex
- New protein synthesis
- Replication in the nucleus
- Eight segments of genome formed

Ppt no. 84

PARAMYXOVIRIDAE

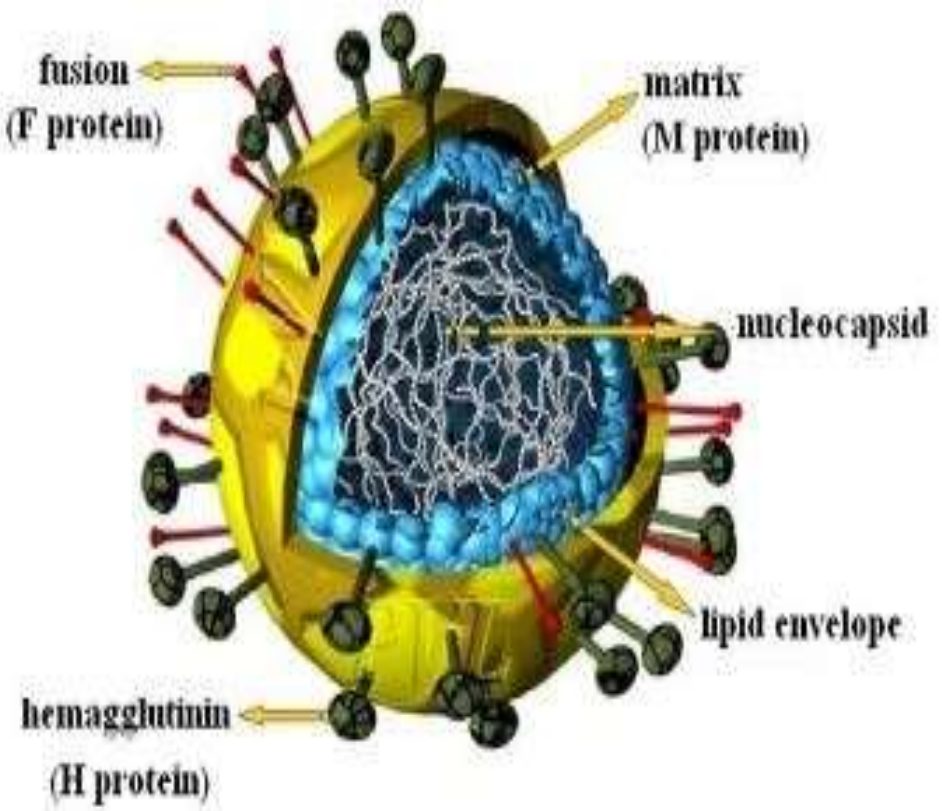
Characteristics

NOMENCLATURE

- Para means beyond
- Myxo means mucus
- Paramyxo and orthomyxo were formerly known as myxoviruses

PROPERTIES

- Pleomorphic
- 150-300nm diameter
- Enveloped
- Single stranded RNA



- Spikes on envelope
- Sensitive to heat, chemicals and radiations
- Host: Vertebrates
- Genera: 7
- Species: 49

PROTEINS

- N – the nucleocapsid protein associates with genomic RNA and protects the RNA from nuclease digestion
- P – the phosphoprotein binds to the N and L proteins and forms part of the RNA polymerase complex
- M – the matrix protein assembles between the envelope and the nucleocapsid core, it organizes and maintains virion structure

PROTEINS

- F – the fusion protein projects from the envelope surface as a trimer, and mediates entry
- H/HN/G – the cell attachment proteins span the viral envelope and project from the surface as spikes.
- L – the large protein is the catalytic subunit of RNA dependant RNA polymerase (RDRP)
- Accessory proteins – a mechanism known as RNA editing allows multiple proteins to be produced from the P gene.

Ppt no. 85

PARAMYXOVIRUS

Genome

GENOMIC CHARACTERS

- Nucleocapsid: helical
- 13-18nm
- Non Segmented linear
- Negative sense
- Single stranded RNA
- 15-16kb
- Contain 6-10 genes

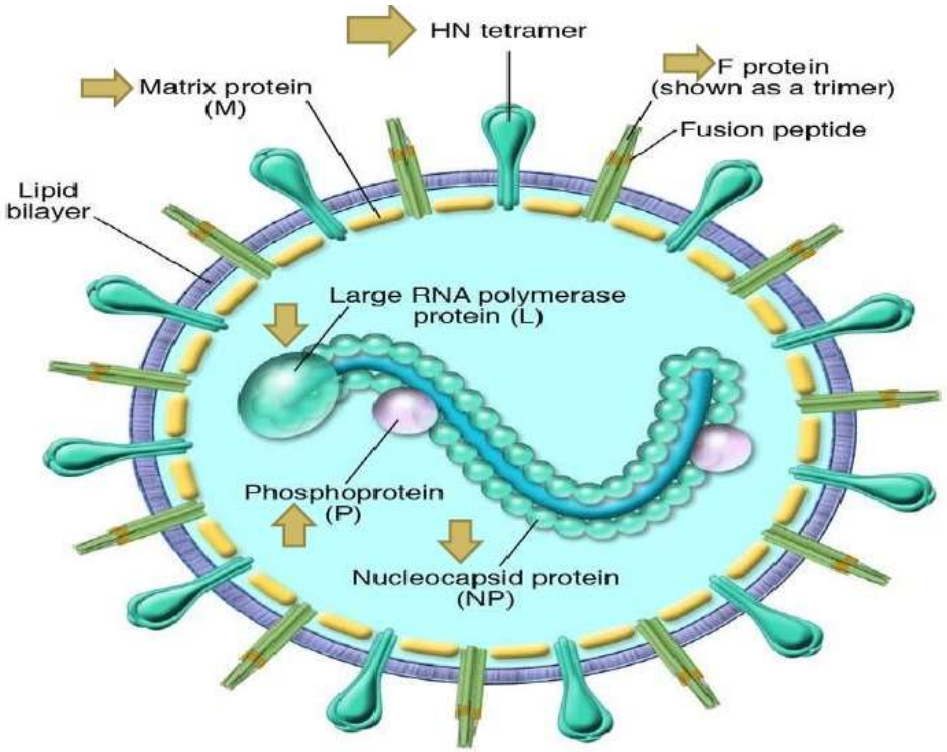
Extracistronic regions

- A 3’ leader sequence, , which acts as a transcriptional promoter.
- A 5’ trailer sequence, 50–161 nucleotides long

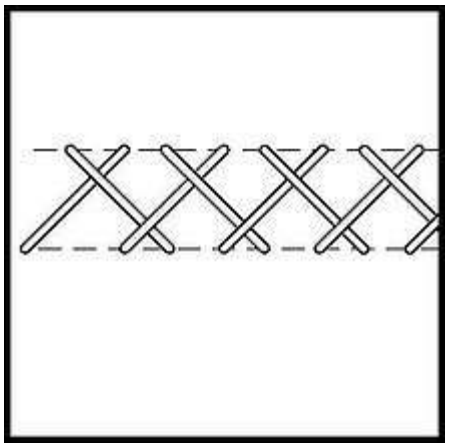
Intergenomic regions

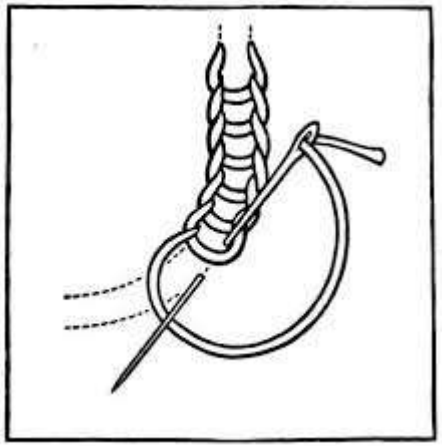
- between each gene
- which are three nucleotides long for morbilliviruses, respiroviruses and henipaviruses
- variable length (1-56 nucleotides) for rubulaviruses

- Fusion Protein
- Haemagglutinin-Neuraminidase
- Matrix protein
- Phosphoprotein
- Nucleoprotein
- Polymerase proteir



- Characteristic herring bone appearance
- Nucleocapsid core contains:
  - Genomic RNA
  - Nucleocapsid protein
  - Polymerase protein





Ppt no. 86

RHABDOVIRIDAE

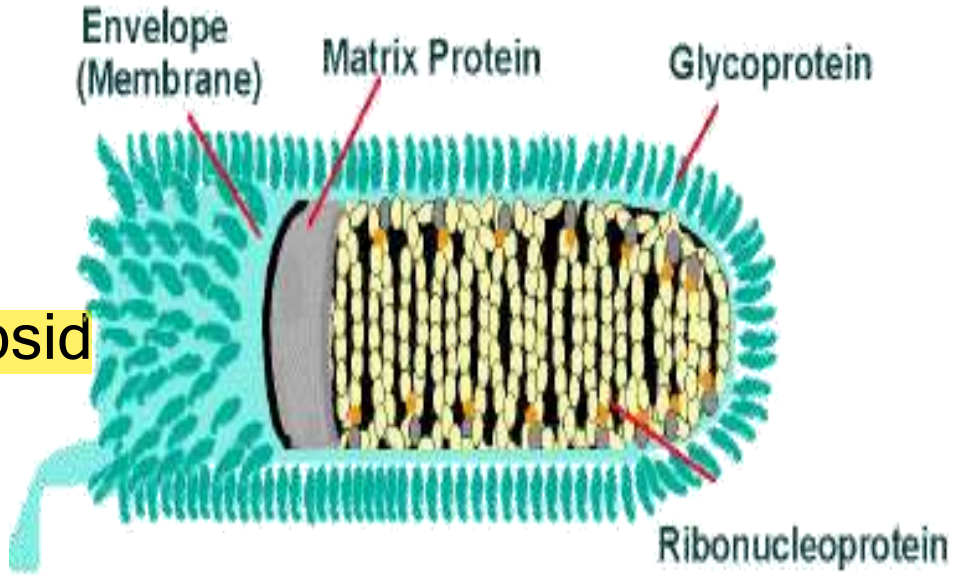
General Properties

MEANING

- Greek word
- Rhabdos meaning rod
- Referring to shape of virus
- Mononegavirales

MORPHOLOGY

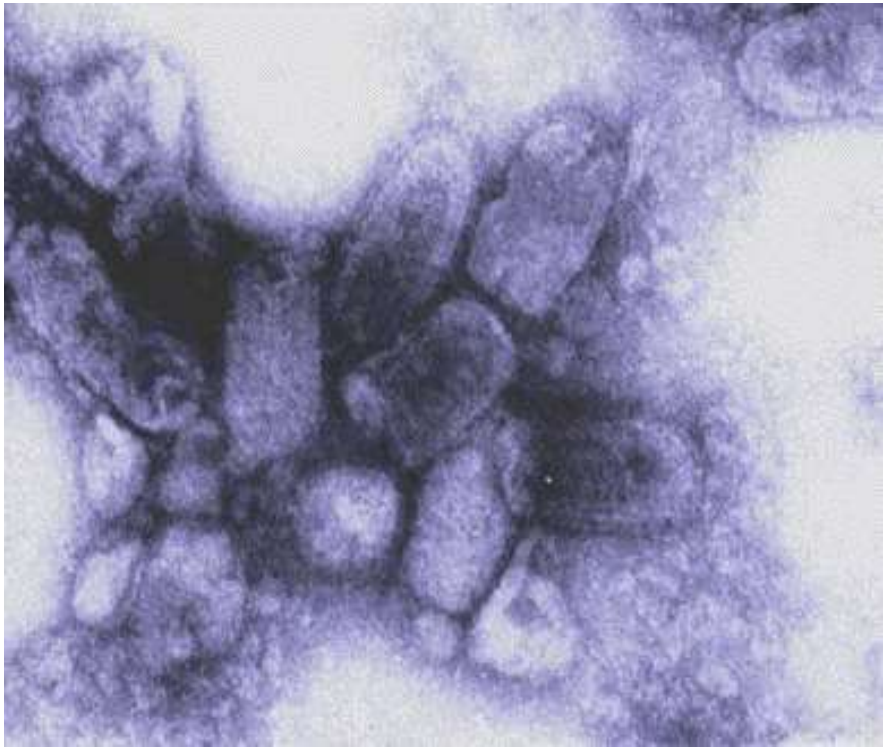
- Rod or bullet shaped
- Enveloped
- Bacilliform
- Helical neucleocapsid
- 75-180nm long
- RNA genome
- 18 genera



PROTEINS

- large protein (L)
- glycoprotein (G)
- nucleoprotein (N)
- phosphoprotein (P)
- matrix protein (M)

MEMBERS

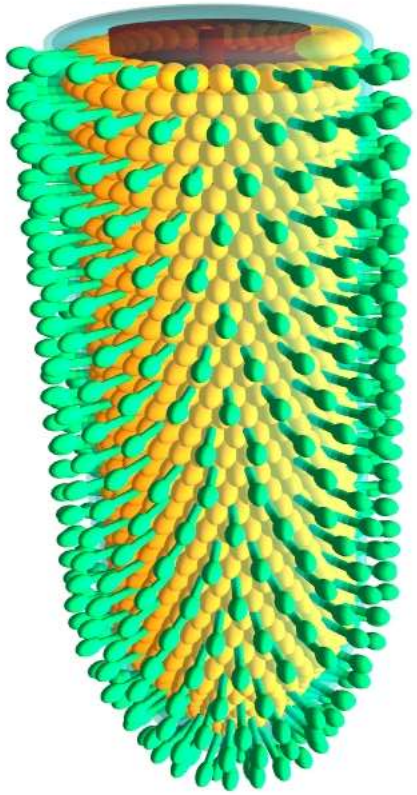


- Vessicular stomatitis Indiana Virus
- Lyssavirus
- Ephemerovirus
- Cytorhabdovirus
- Nucleorhabdovirus

VESICULOVIRUS

- Infects human, cattle, horses
- Flu-like illness
- Fever
- Siezures
- Diarrhea
- Vomiting
- Coma
- Death

LYSSAVIRUS



- Infects human
- Bat and rodent carrier
- Throughout world
- Causes rabies
- Anxiety, confusion, agitation, paranoia, hallucinations
- Death

EPHEMEROVIRUS

- Infects cattle primarily
- Causes Bovine Ephemeral Fever
- Fever, stiffness, lameness
- Nasal and ocular discharges
- High morbidity and low mortality

CYTORHABDOVIRUS

- Plant viruses
- Bacilliform
- Mature in cell cytoplasm
- lettuce necrotic yellow virus
- barley yellow striate mosaic virus

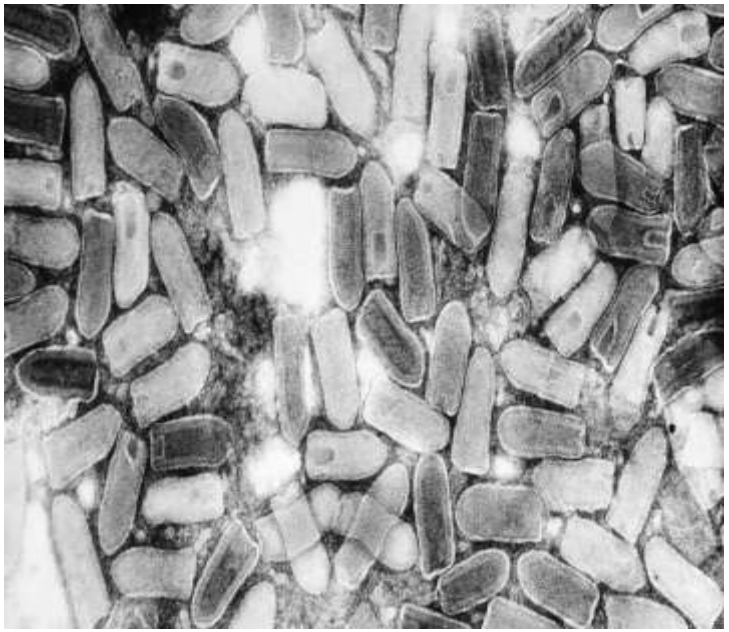
NUCLEORHABDOVIRUS

- Plant viruses
- Bacilliform
- Linear genomes encoding 5 to 6 proteins
- Nuclear viral replication transmitted by insect bite

Ppt no. 87

RHABDOVIRIDAE

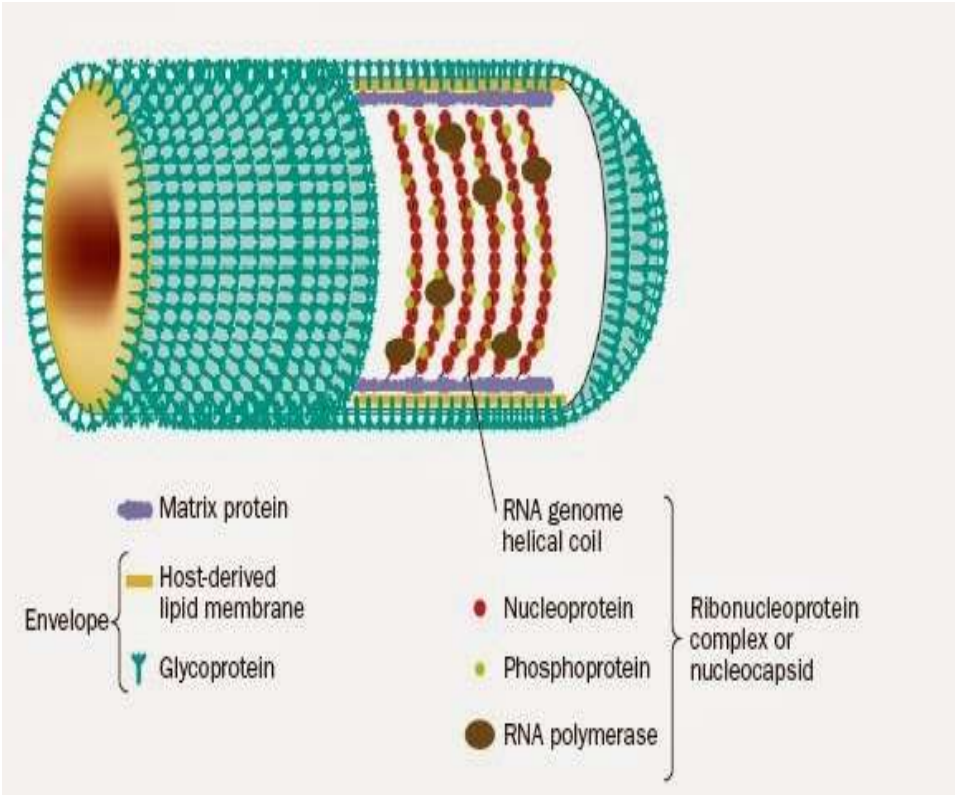
GENOME



Genomic Properties

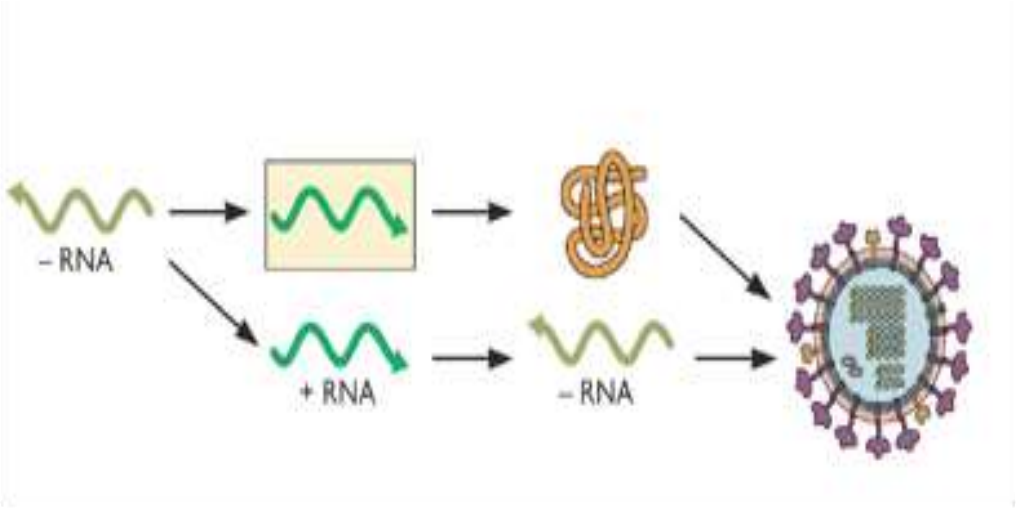
- Single stranded
- RNA

GENES



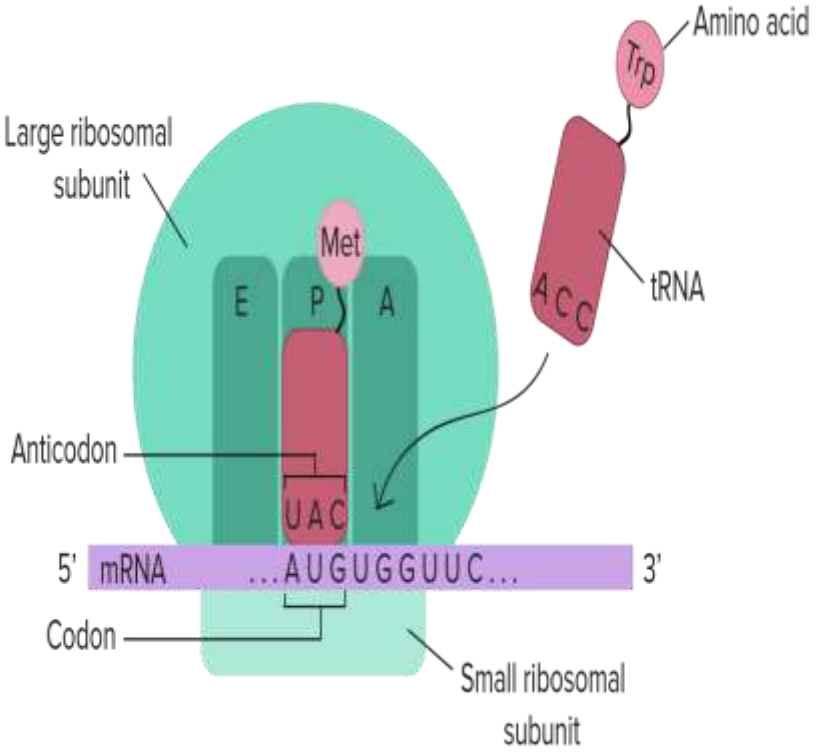
- Five proteins:
- Large protein (L)
- Glycoprotein (G)
- Nucleoprotein (N)
- Phosphoprotein (P)
- Matrix protein (M)

TRANSCRIPTION



- 1 L and 3 P proteins
- Right after entry
- From 3’ to 5’
- Produces mRNA

TRANSLATION



- Translated on free ribosomes
- G protein translated on RER
- Single peptides of G protein

REPLICATION

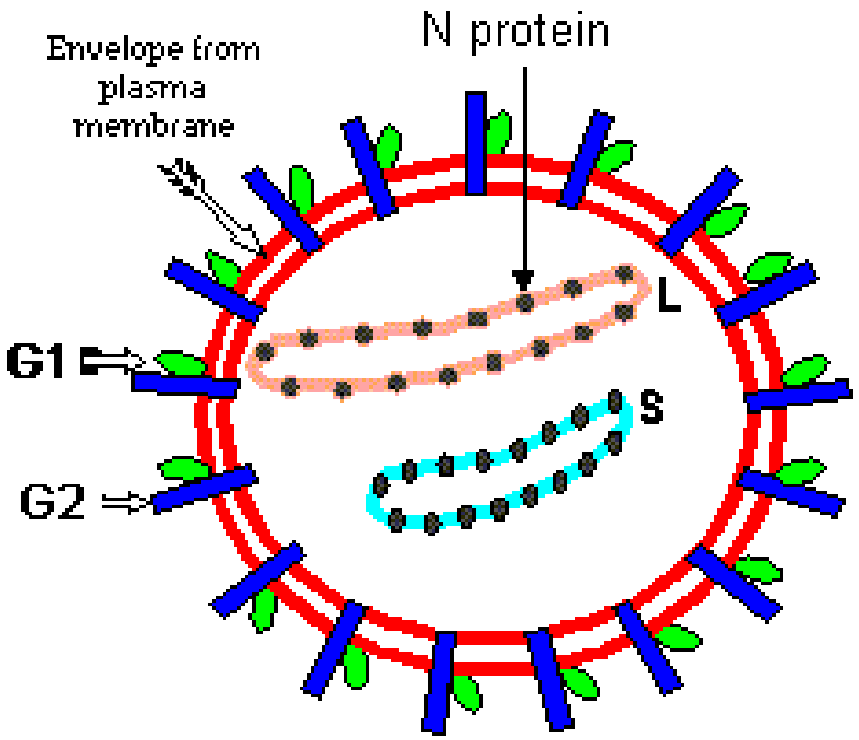
- Cytoplasmic
- Attachment of G protein
- Endocytosis
- Negative strand of RNA transcribed
- Virus leaves by budding

Ppt no. 88

SEGMENTED AND MULTIPARTITE GENOME

ARENAVIRUS

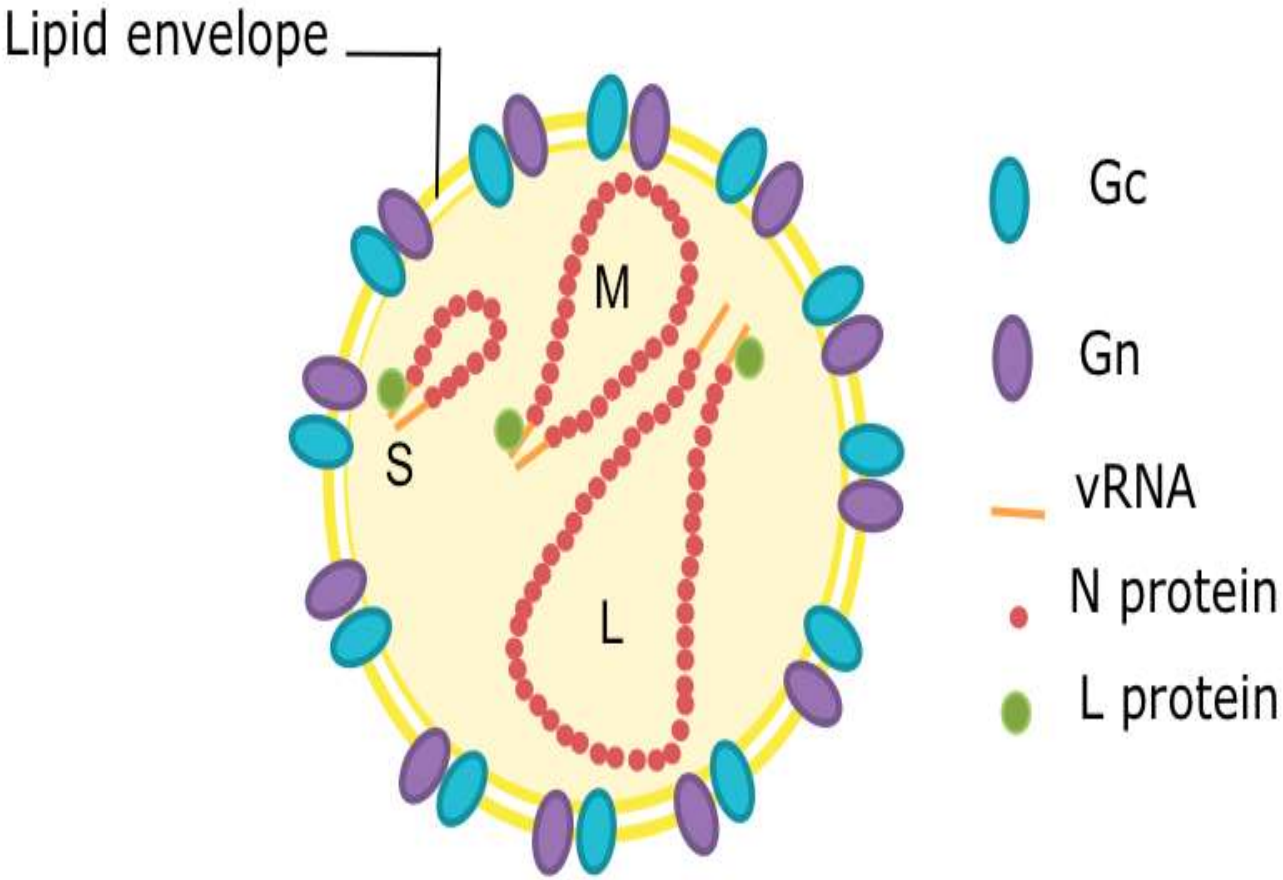
2 segments



Arenavirus particle

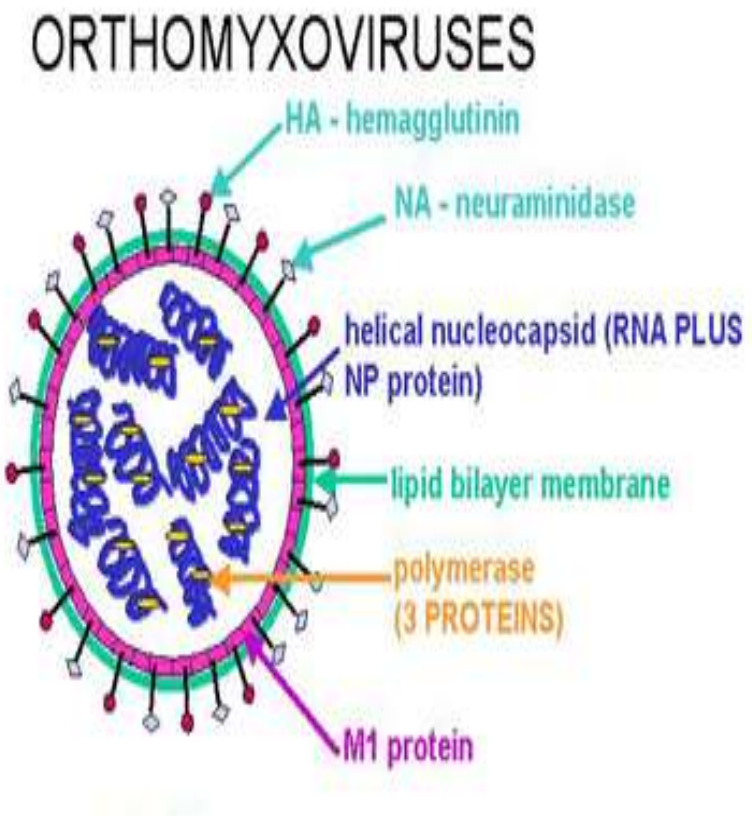
BUNYAVIRUS

3 segments



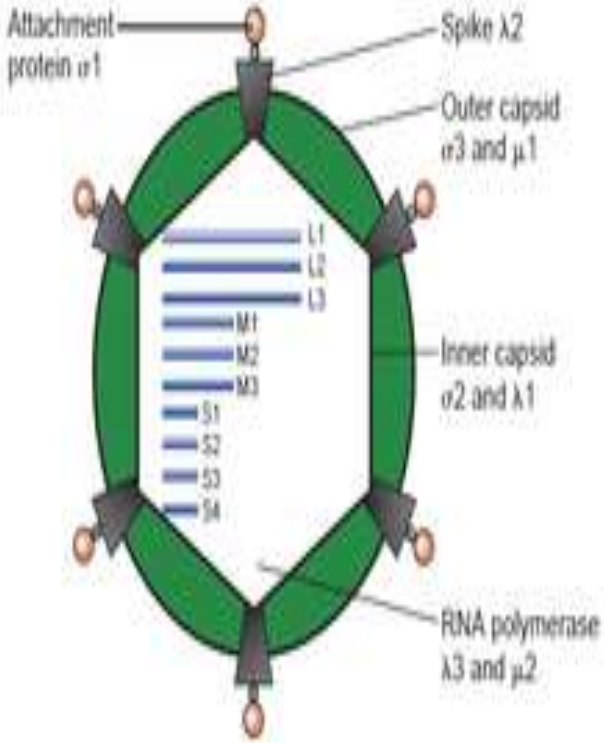
ORTHOMYXOVIRUS

8 segments



REOVIRUS

10 segments



Ppt no. 89

REVERSE TRANSCRIPTION

Reverse transcription is a process in which a double standard DNA molecules ae made from a single stranded RNA

The name of this method originated by tis opposite direction to transcription process

It involves the presence of

- Reverse transcriptase,

an enzyme

- A primer

some bases of nucleotides segments

- Reverse transcriptase inhibitor

leucine rich repeat protein

Reverse Transcription

Ppt no. 90

TRANSPOSONS

Importance

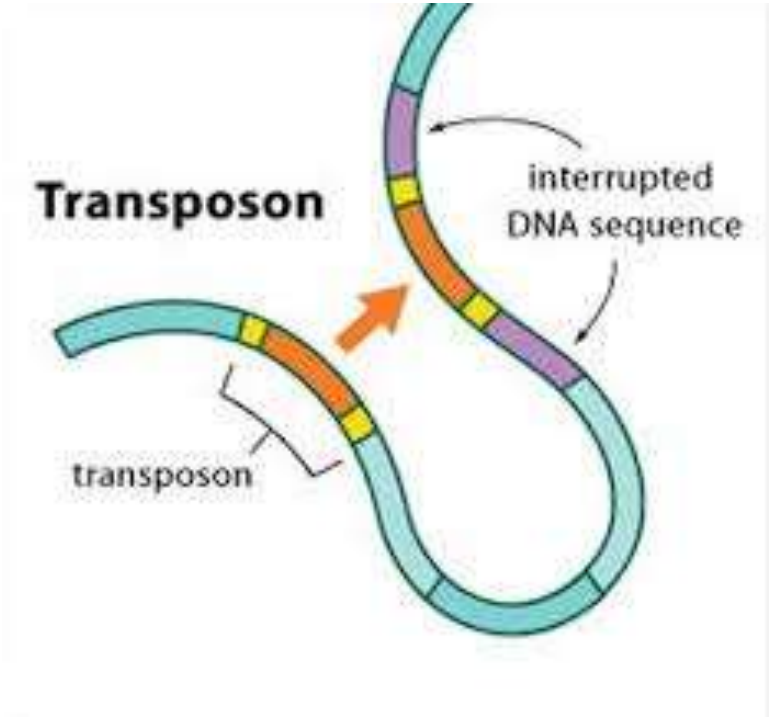
TRANSPOSONS

A sequence of DNA that can change its position, from place to place within a genome

DISCOVERY

– 1940s

– Barbera McClintock



TYPES

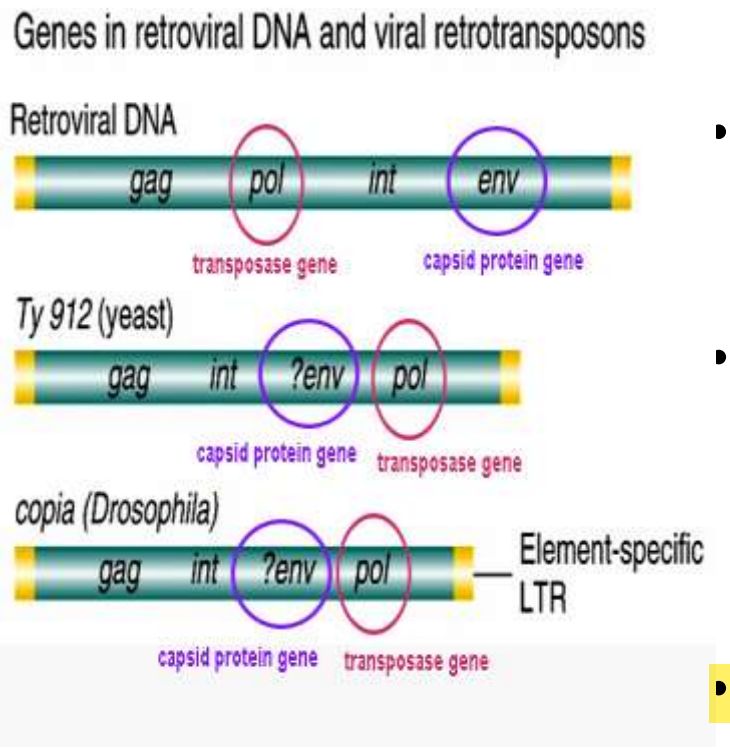
Two groups:

- Simple transposons
- Reterotransposons

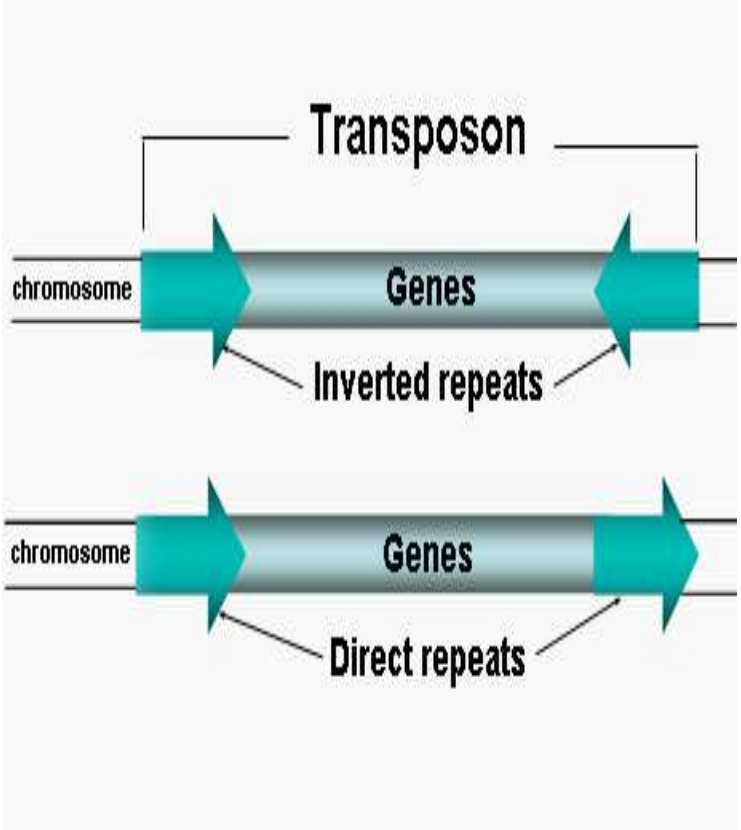
SIMPLE TRANSPOSONS

- In prokaryotes
- Which do not undergo reverse transcription
- e.g. genome of enterobacteria phage Mu

RETROTRANSPOSONS



- Closely resemble retroviral genome
- Bound by long repeat units (LTRs)
- Move by means of transcription/ reverse transcription/ integration mechanism
- Found in eukaryotes
- In *Metaviridae* and *Pseudoviridae*



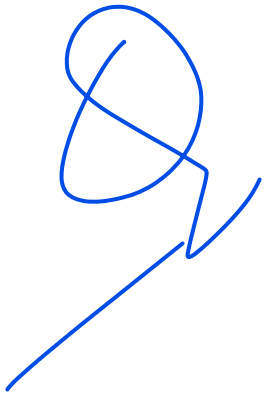
PROPERTIES

- Responsible for high mutation
- Promote genetic rearrangements like deletion, inversion, duplication etc
- Often accompanied by replication
- Control their own transposition functions

Ppt no. 91

Isolation, Cultivation and Identification of Viruses

ISOLATION



- Centrifugation
  - Differential centrifugation- high vs low speed to separate cells from viruses
  - Gradient centrifugation -

CULTIVATION

- To cultivate the virus in specific host

CULTIVATION

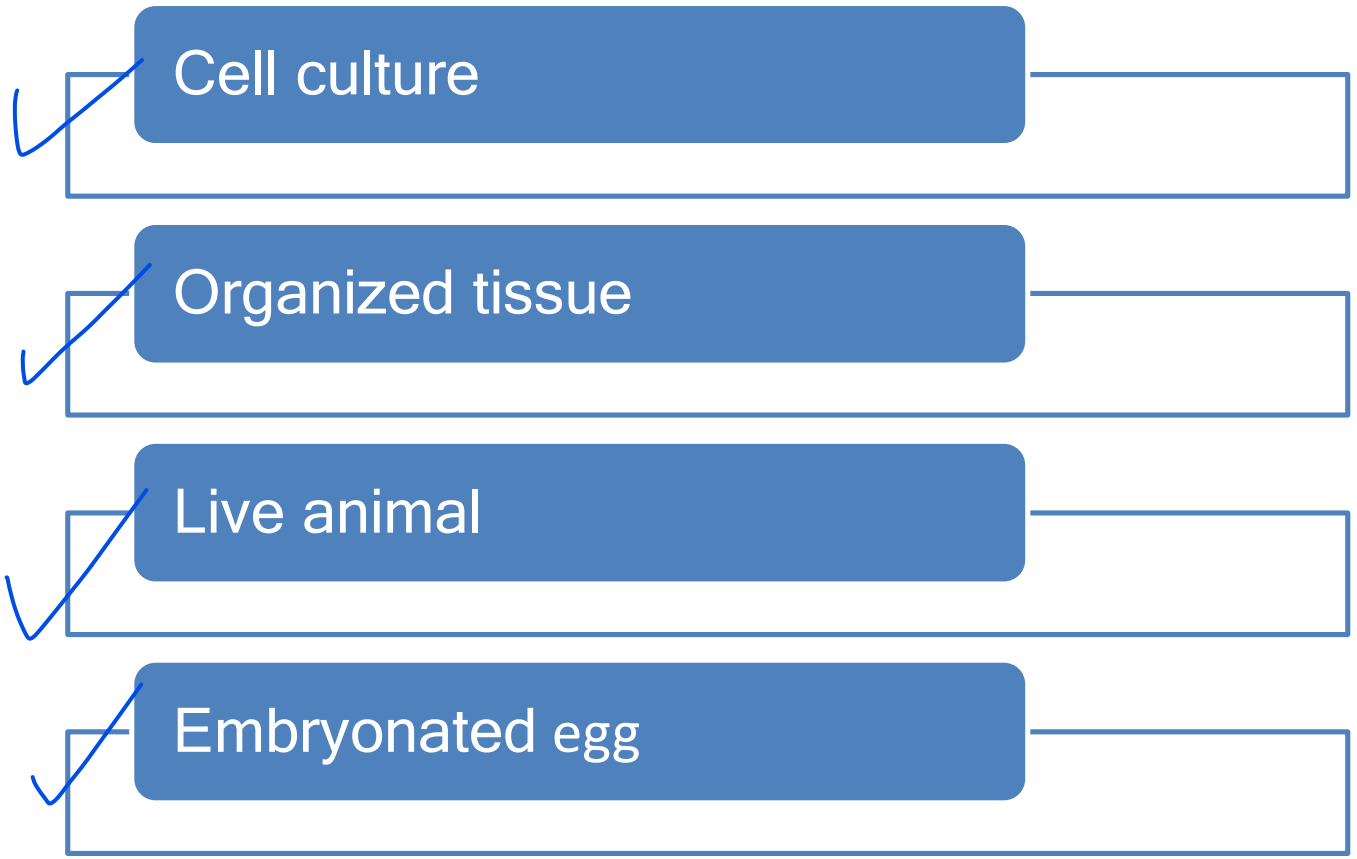
Animal

Plant

Bacteria

Insect

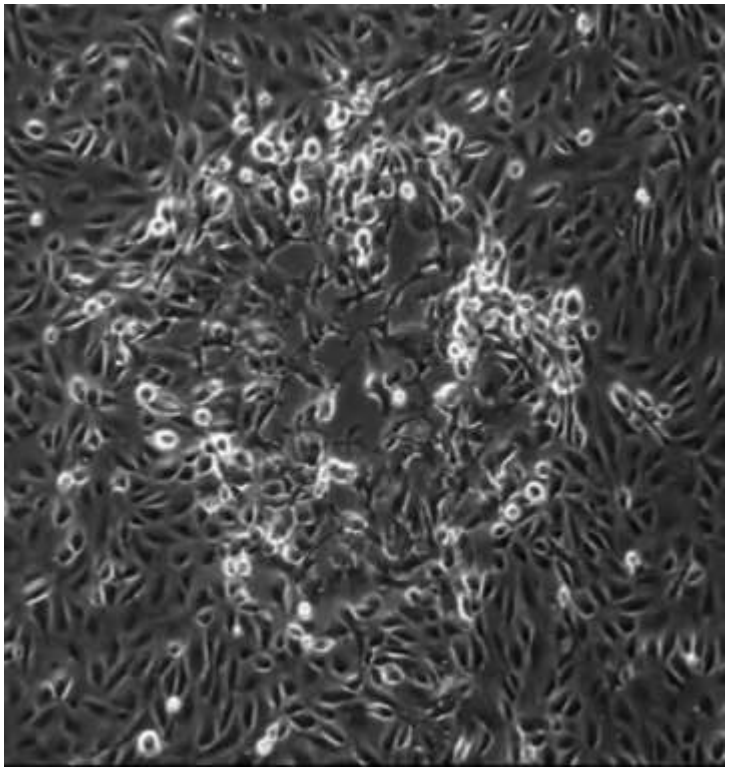
METHOD OF CULTIVATION



IDENTIFICATION

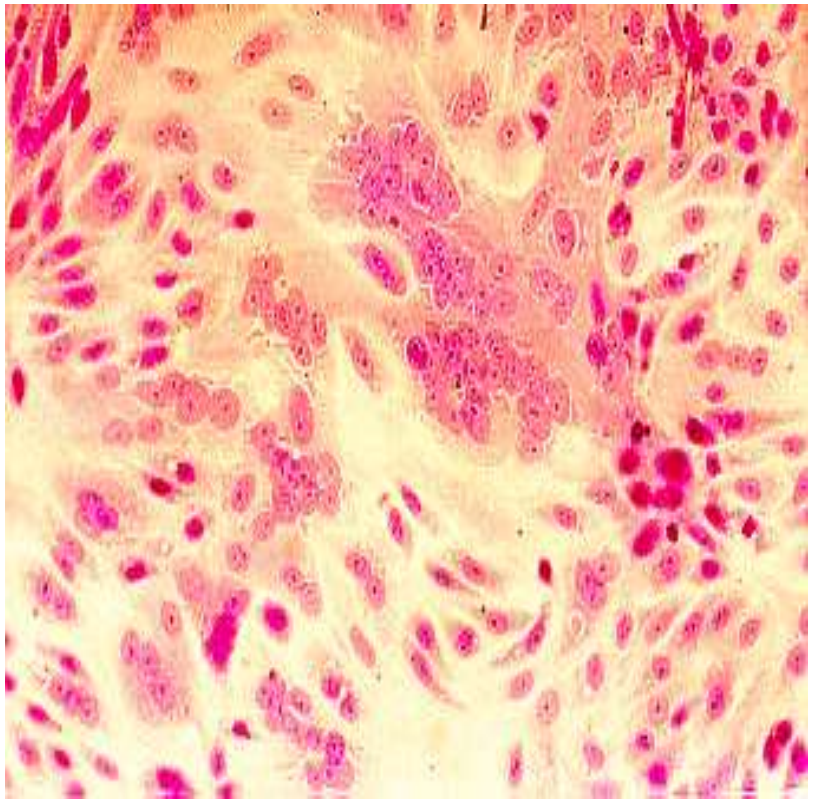
- Plaque formation
- Cytopathic Effects
- Fluorescent Antibody technique
- Nucleic Acid hybridization
- Electron Microscopy

PLAQUE FORMATION



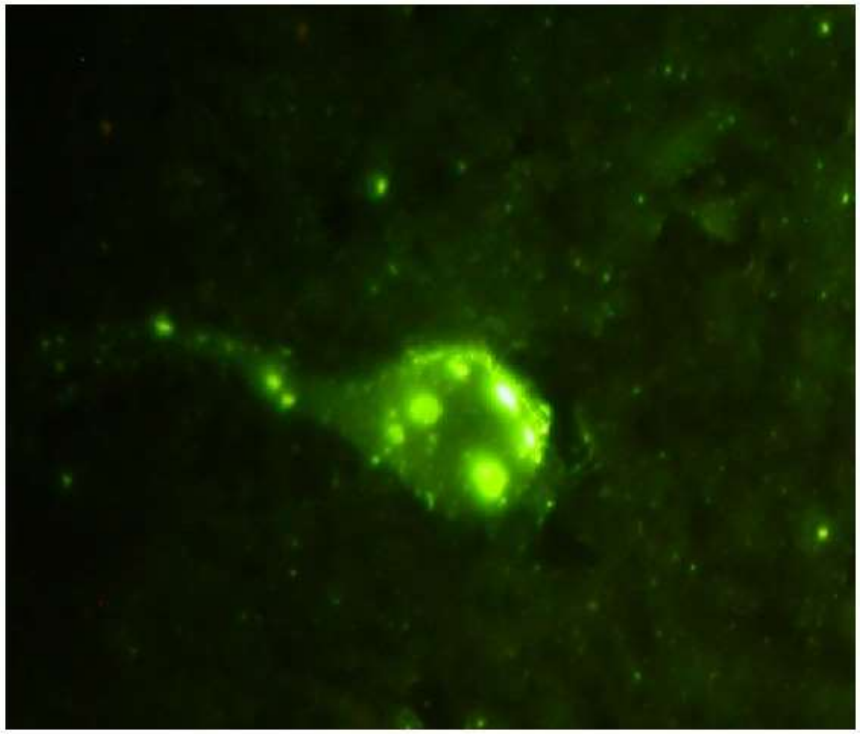
- a visible structure formed within a cell culture
- The bacteriophage viruses replicate and spread, thus generating regions of cell destructions known as plaques

CYTOPATHIC EFFECTS



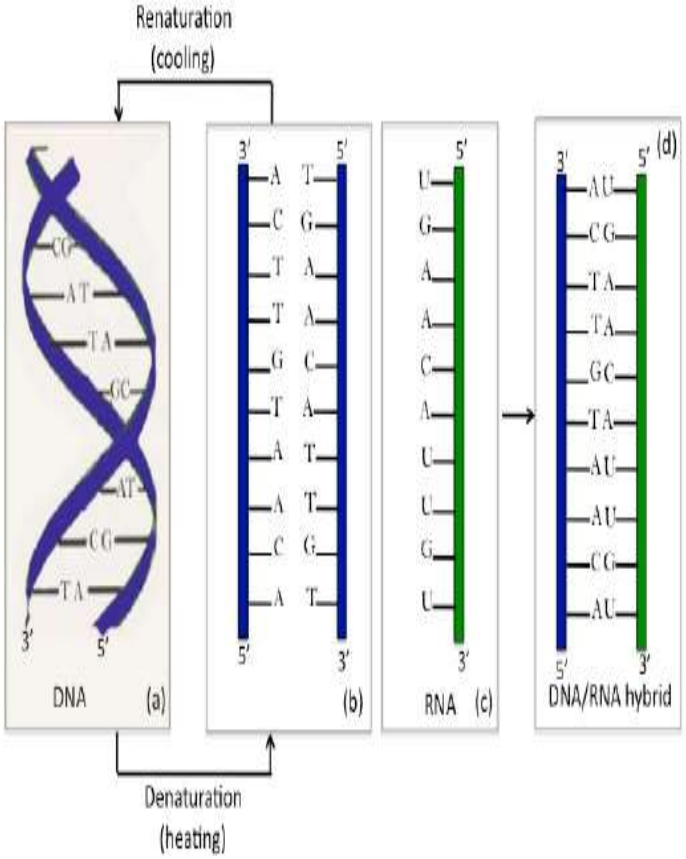
- Syncytia
- Cell rounding
- Membrane proliferations
- Vacuolization
- Inclusion bodies
- Focus (foci)
- Hemadsorption

FLUORESCENT ANTIBODY TECHNIQUE



- based on the observation that virus proteins are (antigen) present in the tissues
- e.g rabies virus present in nervous tissues

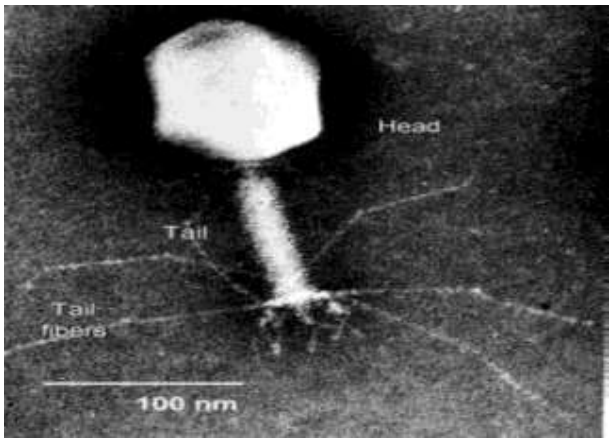
NUCLIEC ACID HYBRIDIZATION



A technique in which single-stranded nucleic acids (DNA or RNA) are allowed to interact so that complexes called hybrids are formed by molecules with similar, complementary sequences

ELECTRON MICROSCOPY

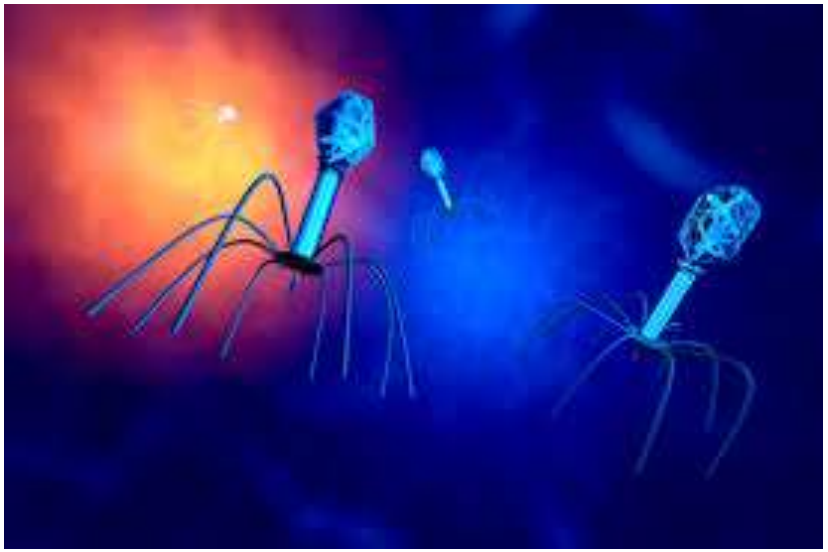
- uses a beam of **electrons** to create an image of the specimen
- Inclusion bodies formed



END

Ppt no. 92

Growing Bacteriophages in Laboratory

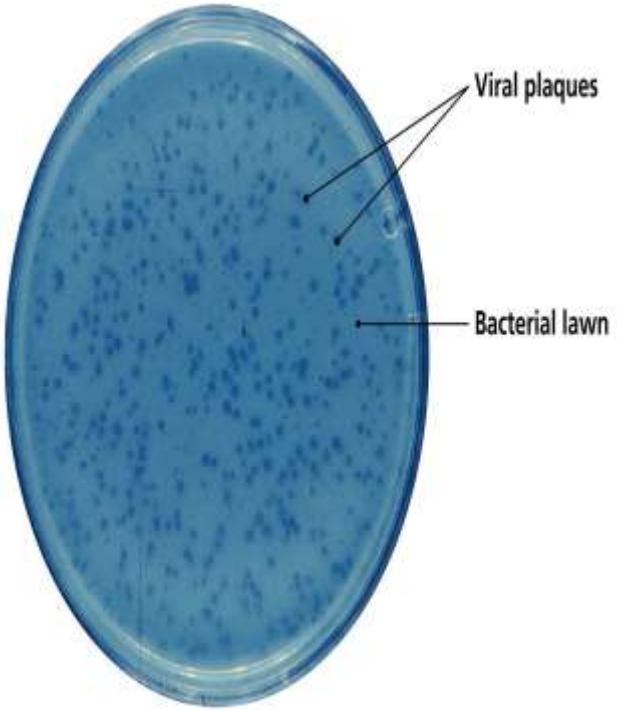


METHODS

Two methods

- In liquid media
- On solid media

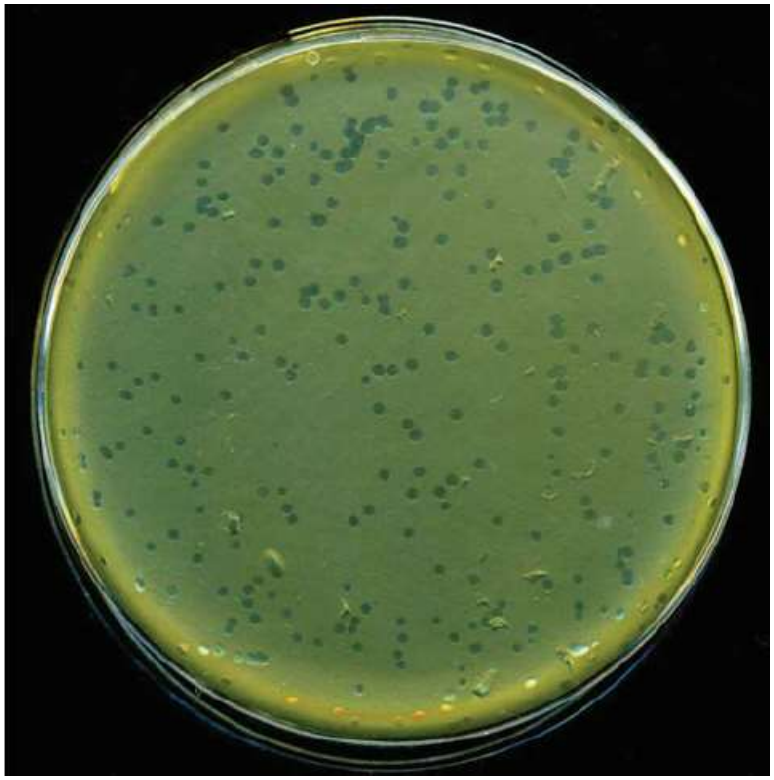
ON SOLID MEDIA



- Like bacteria
- Helps in plaque test to detect and count viruses
- Bacteriophage sample mixed with host bacteria and melted agar

INNOCULATION

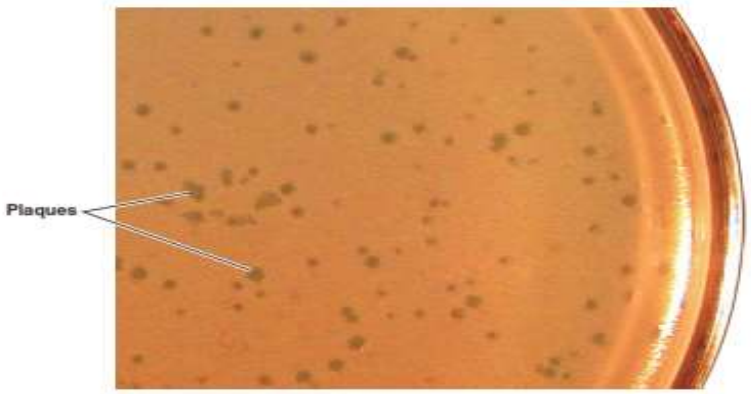
- Mixed agar poured into petri plate
- Petri plate already containing hardened layer of agar media
- Mixture solidifies into top thin layer



- Each virus infects bacteria
- Multiplies
- Releases new viruses
- New viruses infect other bacteria
- All bacteria get destroyed

PLAQUE FORMATION

- Produces a number of clearings or plaques
- Visible by naked eye
- Uninfected bacteria



END

Ppt no. 93

Growing Animal Viruses in Embryonated Eggs

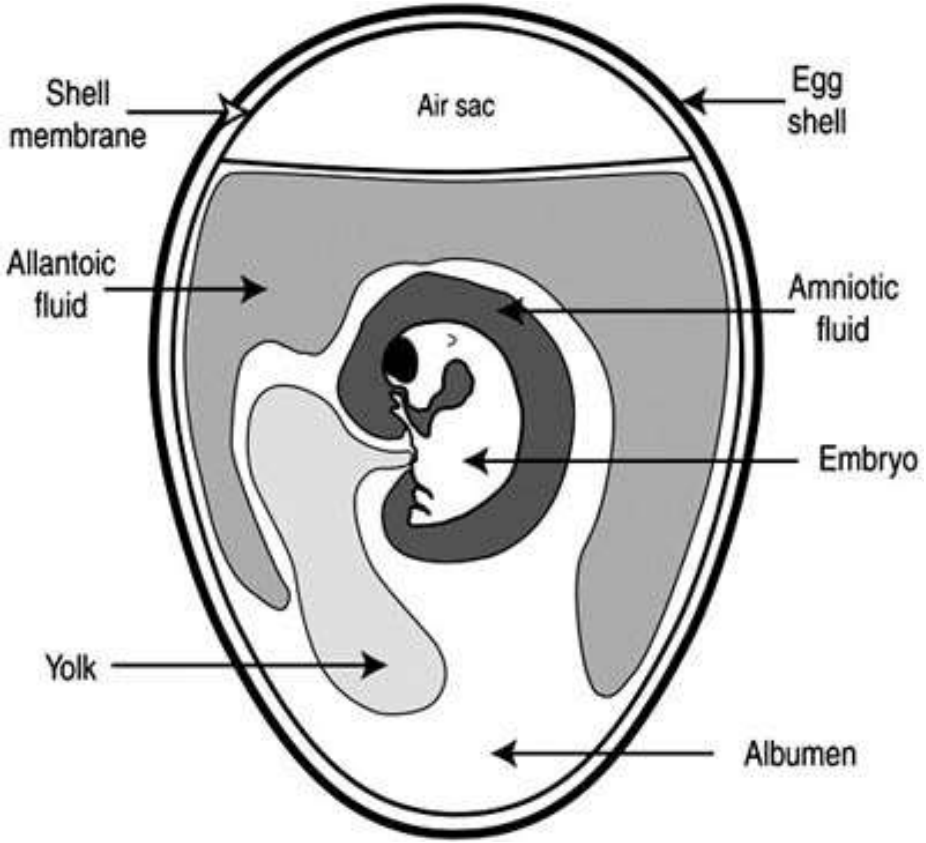
METHODS

1. Animals
2. Cell culture
3. **Embryonated eggs**

ADVANTAGE

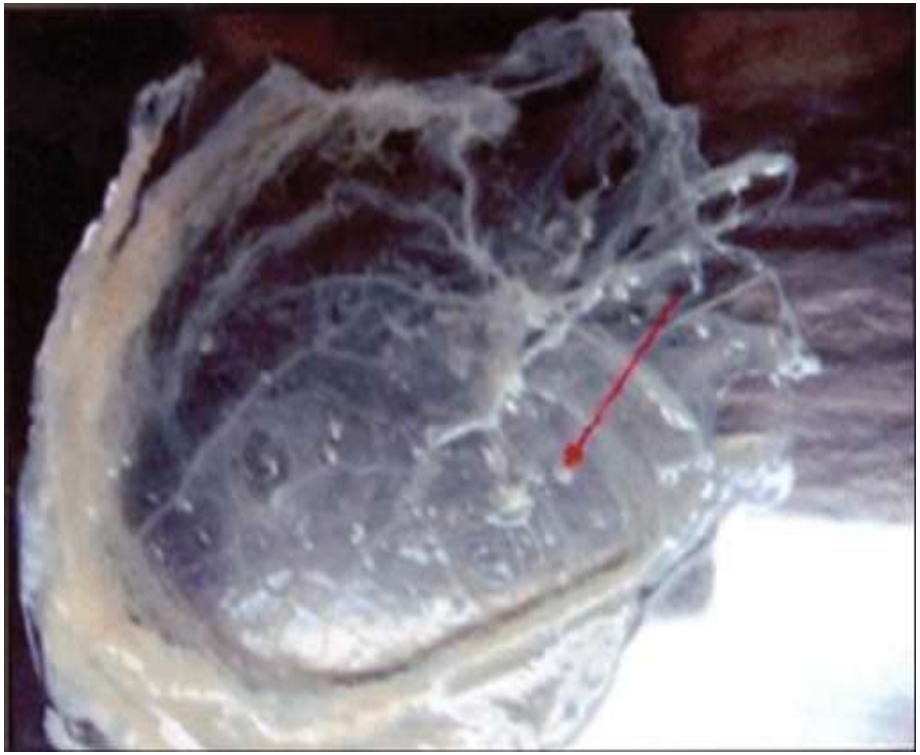
- Convenient
- Inexpensive
- Host for many viruses

PROCEDURE



- Hole is drilled in the shell
- Viral suspension inoculated
- Many routes

VIRAL GROWTH



- Death of embryo
- Damage to embryo
- Pocks or lesions on egg membrane

PURPOSE

- Isolation and growth of virus
- Growth for vaccination

ROUTES

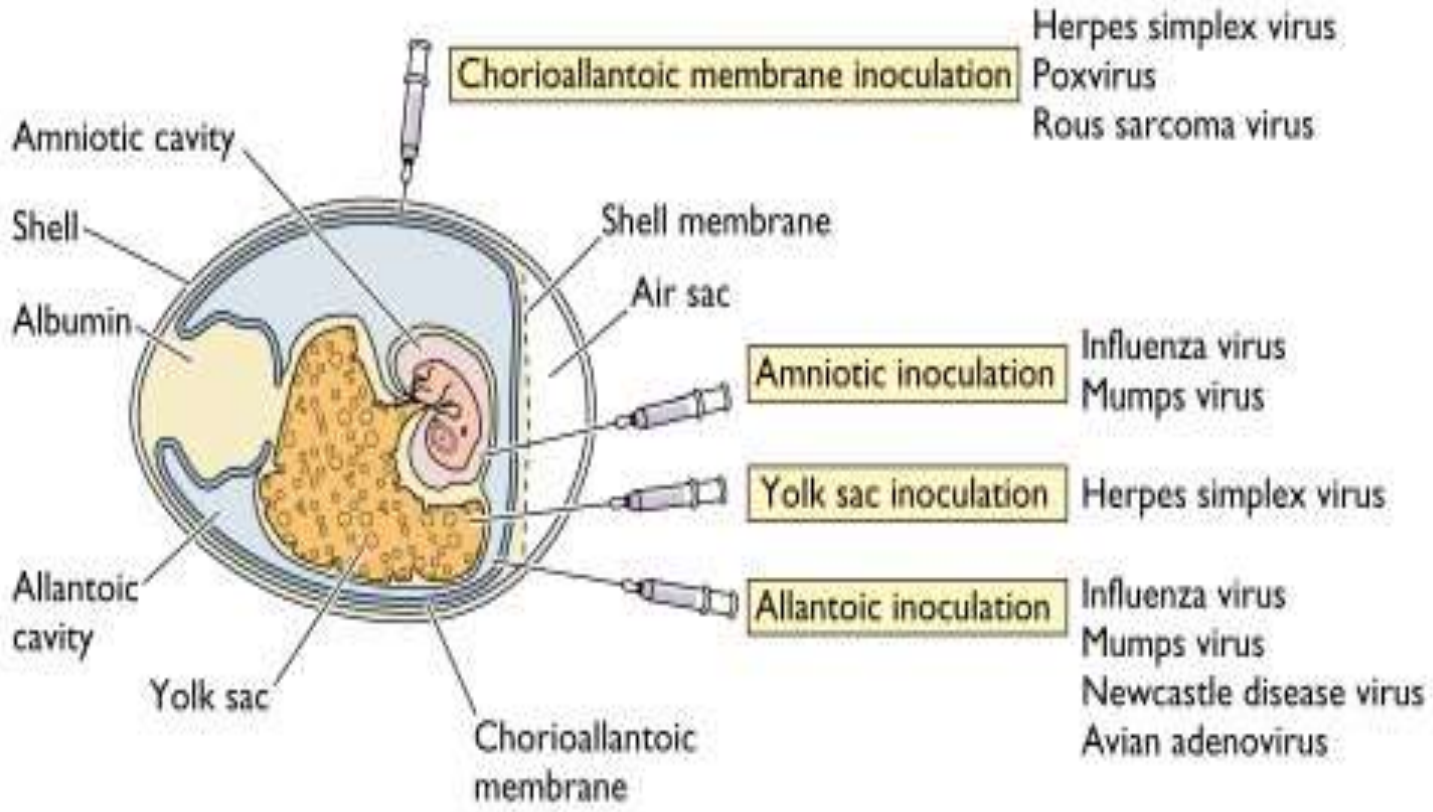
1. Chorioallantoic membrane inoculation
2. Amniotic inoculation
3. Allantoic inoculation
4. Yolk sac inoculation

END

Ppt no. 94

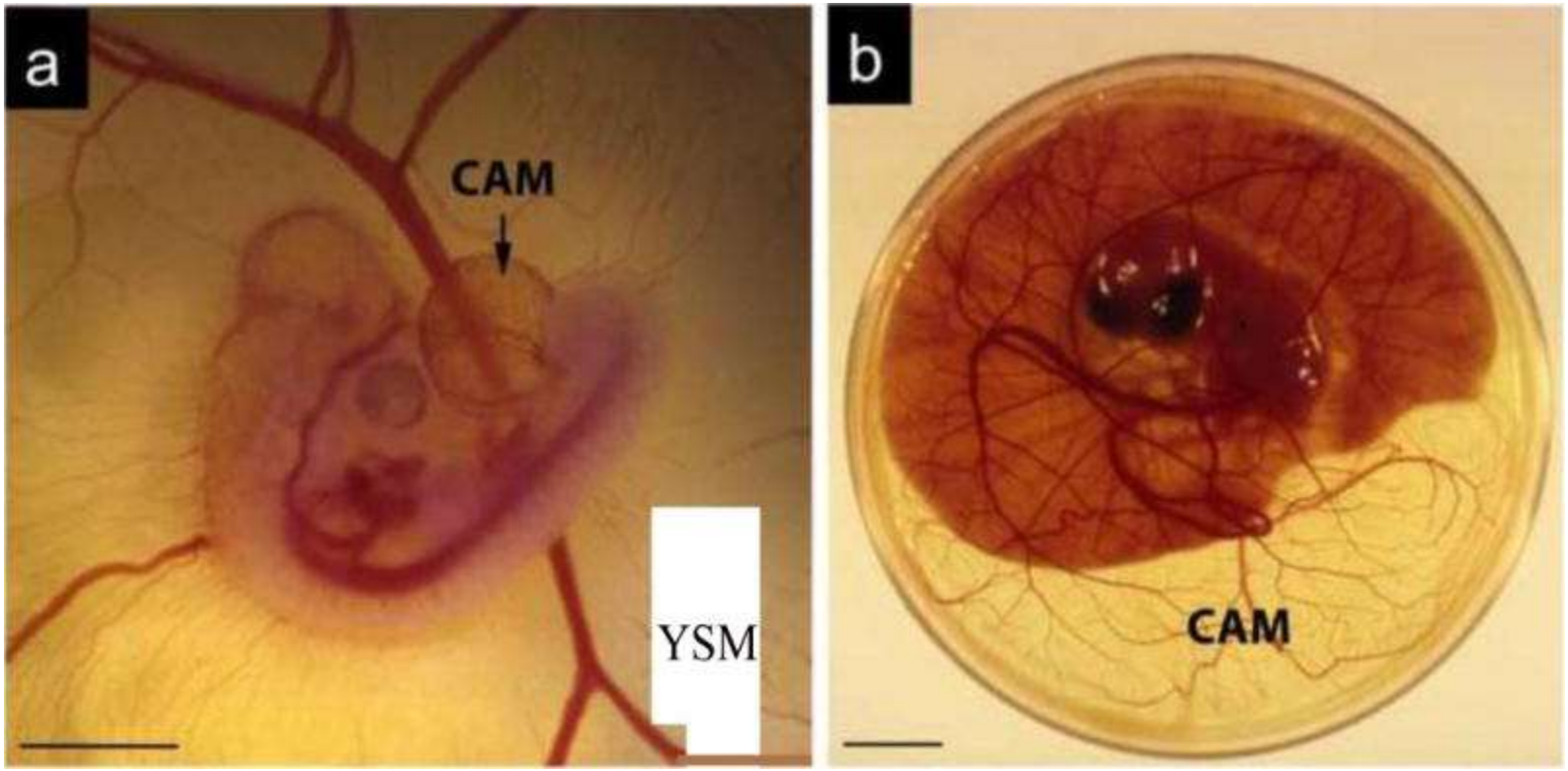
Chorioallantoic Membrane Inoculation

ROUTES OF INOCULATION

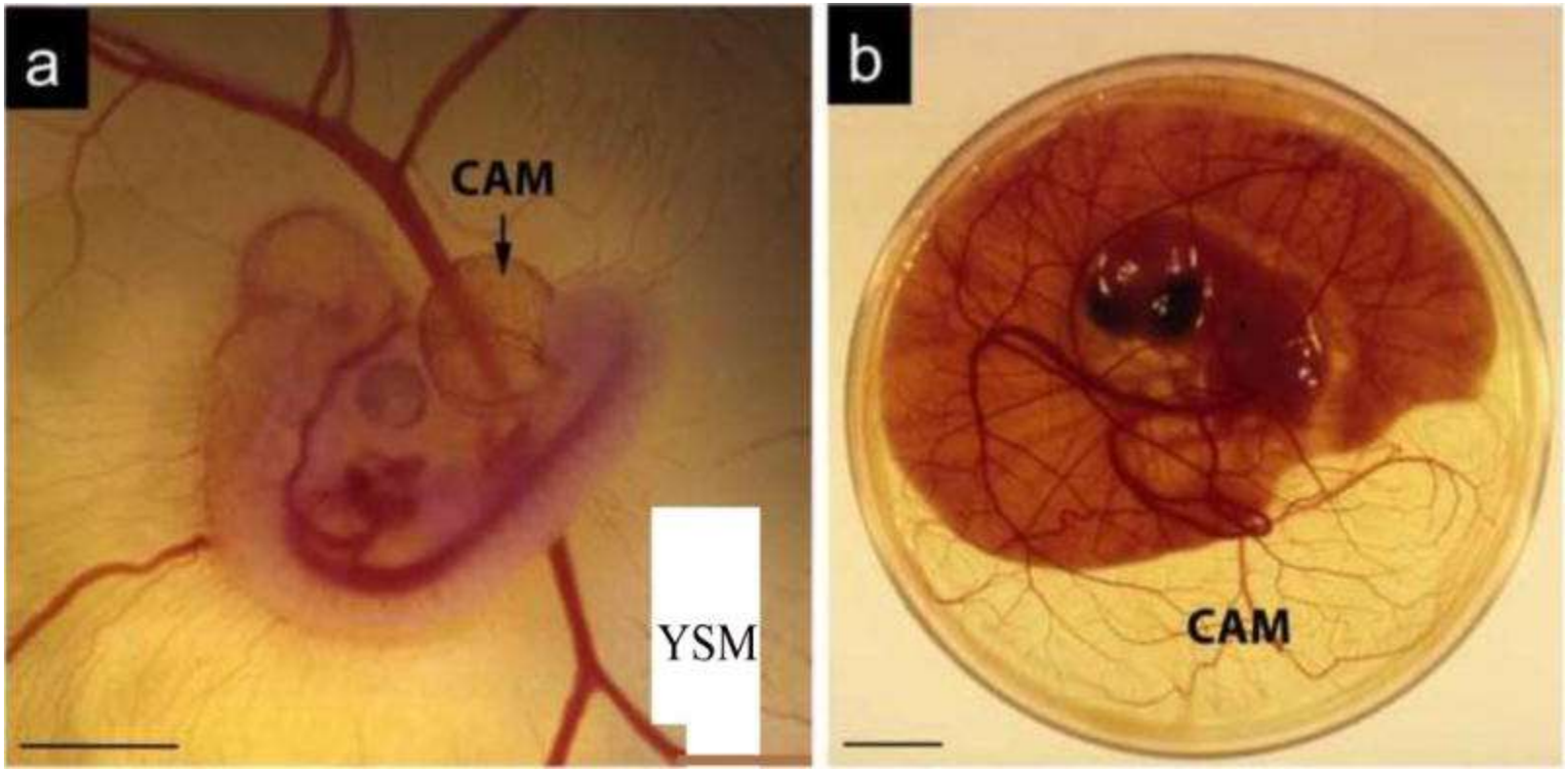


The different routes of inoculation into the egg are shown, as well as the different compartments in which viruses replicate.

- The chicken chorioallantoic membrane (CAM) is a simple, highly vascularized extraembryonic membrane, which performs multiple functions during embryonic development. Over the last two decades, interest in the CAM as a robust experimental platform to study blood vessels has been shared by specialists working in bioengineering, development, morphology, biochemistry, transplant biology, cancer research and drug development



Chicken embryo and associated extra-embryonic structures. a, Chicken embryo at day 4 post-fertilization the highly vascularized chorioallantoic membrane (CAM) that expands from the hind-gut. The yolk sac membrane (YSM), also highly vascularized, is seen in the background; b Embryo at day 12 post-fertilization into a 10 cm petri dish.



Chicken embryo and associated extra-embryonic structures. a Chicken embryo at day 4 post-fertilization. Note the highly vascularized chorioallantoic membrane (CAM) that expands from the hind-gut. The yolk sac membrane (YSM), also highly vascularized, is seen in the background; b Embryo at day 12 post-fertilization into a 10 cm petri dish.

Ppt no. 95

Allantoic Cavity Inoculation

- Taking Flu virus as an example
- Virus inoculated in allantoic cavity

Ppt no. 96

Intraembryonic, Intracranial and Intravenous Inoculation

- Inoculated
- Incubated for few days
- Lesions or death observed

- These inoculations of embryos are not widely used.
- Embryonic inoculation:
- Embryos are inoculated and incubated for a period of one to six days during which they are observed twice a day for the evidence of death of embryo. Some viruses, such as equine encephalomylietis virus kill the embryo within 24 hours while others kill the embryos in 2 to 3 days. Some viruses readily multiply within the embryonic tissue and are seldom lethal.

- Embryo death is accompanied by collapse of the large blood vessels of the chorioallantois so that they are difficult to see upon candling the egg, whereas they are quite distinct while the embryos are alive. Normal movements of the embryo are no longer observed if the embryo is moribund or dead.
- If the embryo is not dead 5 or 6 days of incubation, it is killed or examined for lesions.
-

II. Preparation and Setup for Stereotaxic Delivery of Viral Vectors

Ppt no. 97

SEROLOGICAL METHODS

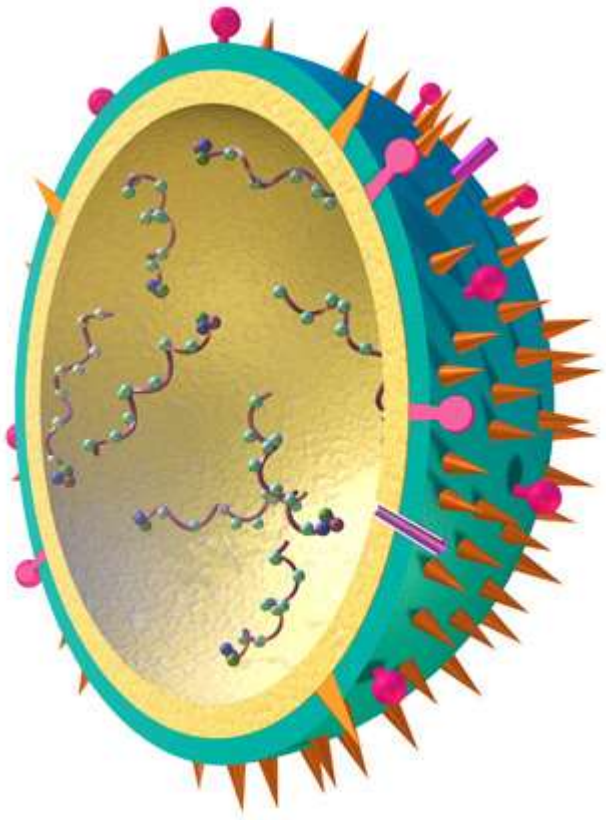
Agglutination and Precipitation

PRINICPILE

- Viruses maybe sensitized by specific antibody
- In the presence of electrolyte the virus particles cling to each other forming aggregates
- Observed through examinations
  - microscopic
  - macroscopic

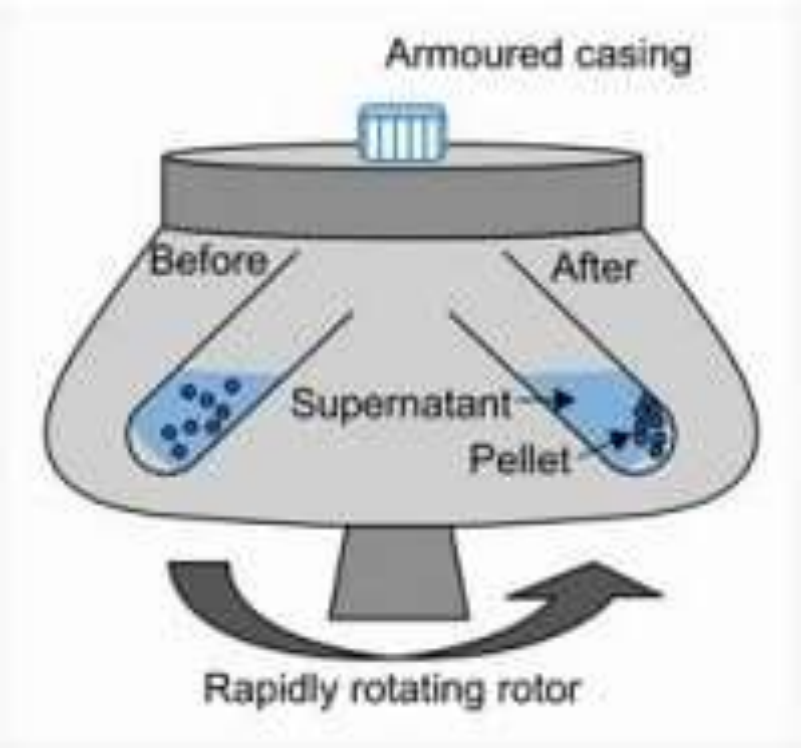
- Flocculation is a technique that can be used for the aggregation and removal of viruses from solutions.
- This process is particularly useful in vaccine production, purification of viral particles, and wastewater treatment where viral contamination is a concern.

FLOCCULUATION



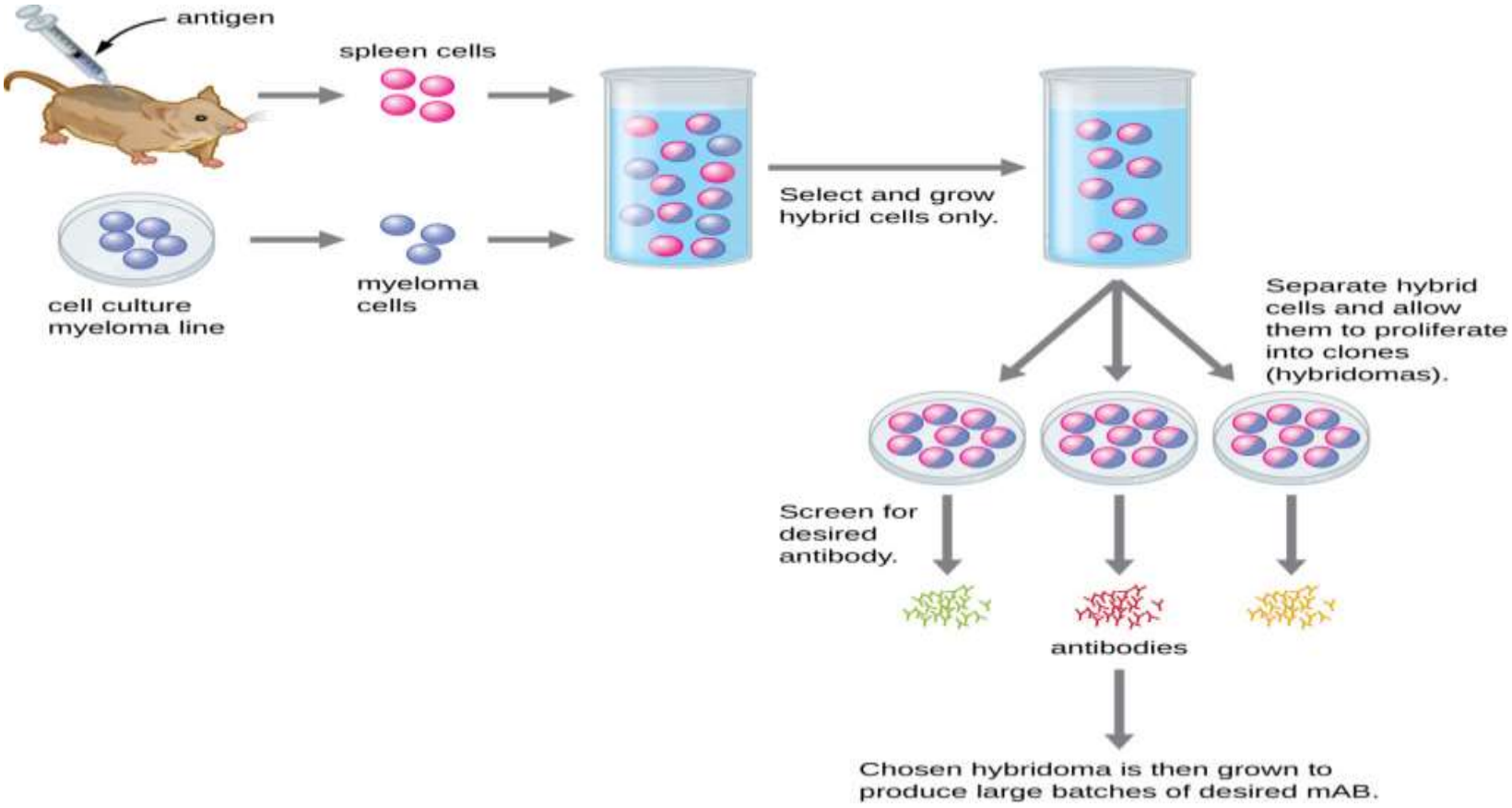
- Size of virus is intermediate between bacteria and protein molecule
- Hence the term, Flocculation is more appropriate

ANTIGEN PREPARATION



- Centrifugation
- Washing in buffered solution
- Recentrifugation

LINE OF ACTION



APPLICATIONS

- Two ways:

1. If virus is known, antibodies can be detected
2. If known serum is available, an unknown virus can be detected

CHALLENGES

- Preparation of antigen is difficult
- Viruses must be grown in in the animal tissues

END

Ppt no. 98

SEROLOGICAL METHODS

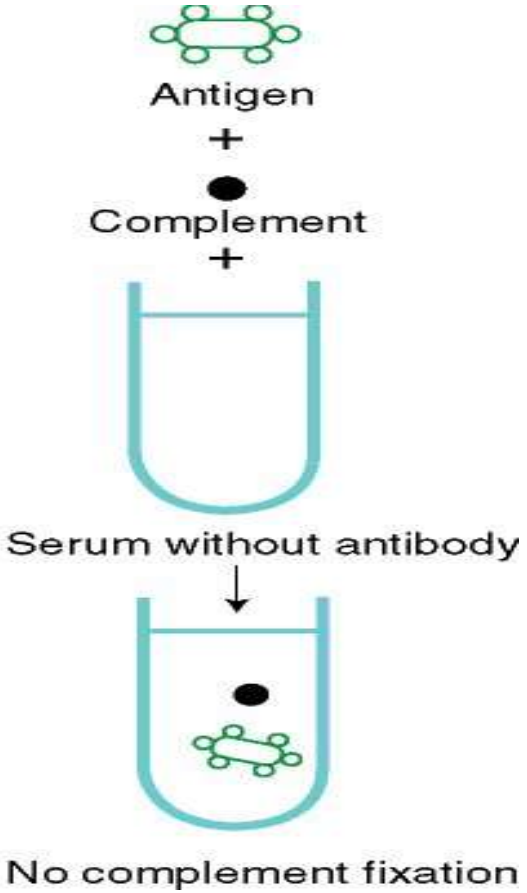
Complement Fixation

- The complement fixation test is an immunological test used to detect the presence of either specific antibody or specific antigen on the basis of complement fixation
- The complement fixation is a system of serum proteins that react with antigen-antibody complexes.
- If this reaction occurs on a cell surface, it will result in the formation of trans-membrane pores and therefore destruction of the cell.

HOW IT WORKS

- Viruses as antigen
- Virus antigen for this test obtained from infected animal tissues
- Tissues such as, lungs or brain

APPLICATIONS



- Detection of presence of antibodies in serum
- Identification of unknown virus by using known antibody

SHORT PROCEDURE

- Serum is separated from the patient.
- Patients naturally have different levels of complement proteins in their serum. To negate any effects this might have on the test, the complement proteins in the patient's serum must be destroyed and replaced by a known amount of standardized complement proteins.
  - The serum is heated in such a way that all of the complement proteins—but none of the antibodies—within it are destroyed.
  - A known amount of standard complement proteins are added to the serum. (These proteins are frequently obtained from guinea pig serum.)
- The antigen of interest is added to the serum.
- Sheep red blood cells (sRBCs) which have been pre-bound to anti-sRBC antibodies are added to the serum.

RESULT INTERPRETATION

- If the patient's serum contains antibodies against the antigen of interest, they will bind to the antigen in step 3 to form antigen-antibody complexes.
- The complement proteins will react with these complexes and be depleted.
- Thus when the sRBC-antibody complexes are added in step 4, there will be no complement left in the serum.
- However, if no antibodies against the antigen of interest are present, the complement will not be depleted and it will react with the sRBC-antibody complexes added in step 4, lysing the sRBCs and spilling their contents into the solution, thereby turning the solution pink

Ppt no. 99

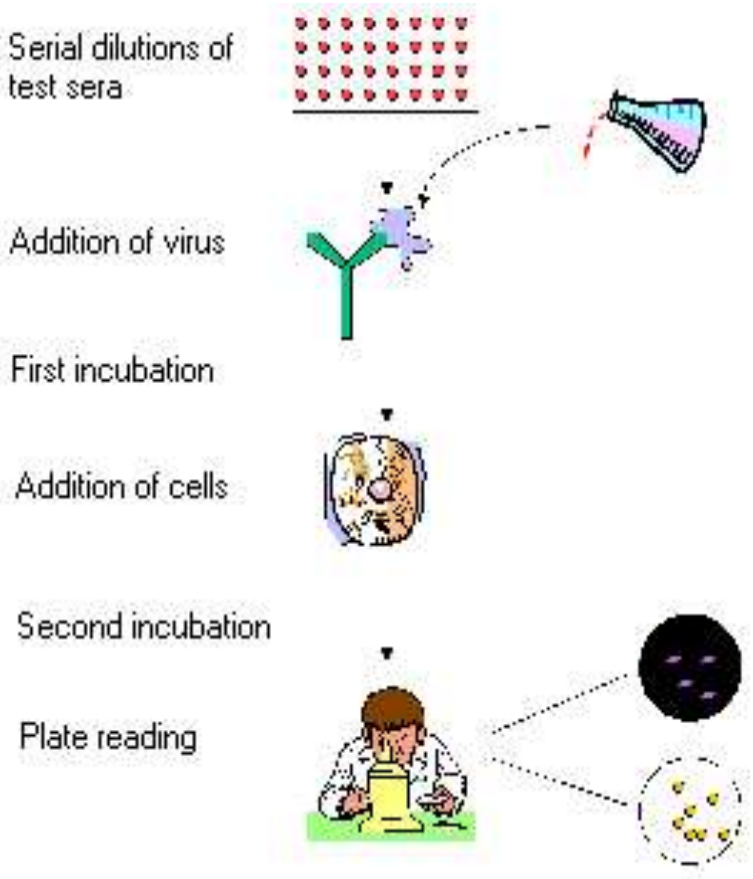
SEROLOGICAL METHODS

Serum Neutralization Test

HOW IT WORKS

- Used to confirm identity of an unknown viral isolate
- Use of known dilution of specific antiserum mixed with multiple dilution
- Observation of virus to produce Cytopathic effects

NECESSITIES



- Amount of virus
- Amount of serum
- Test animal
- Route of inoculation

BASIC PRINCIPLES

- Wide range of dilutions used
- Equal volumes of viral dilution
- Equal volume of virus dilution and immune serum used
- Inoculation of 5 to 6 animals
- Equal number of controls
- Observation of animals for signs, lesions or deaths

For comparisons

- L.D50 (Lethal dose 50)
- Viral dilutions checked for causing deaths and infectivity
- Titer noted

CREATED USING
POWTOON

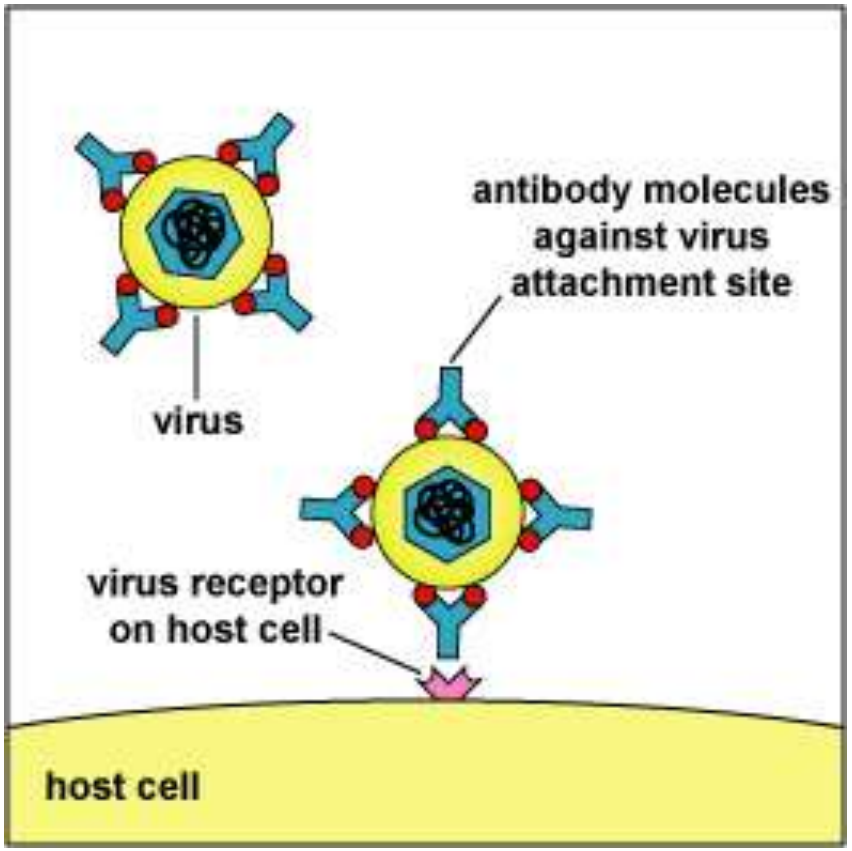
Ppt no. 100

SEROLOGICAL METHODS

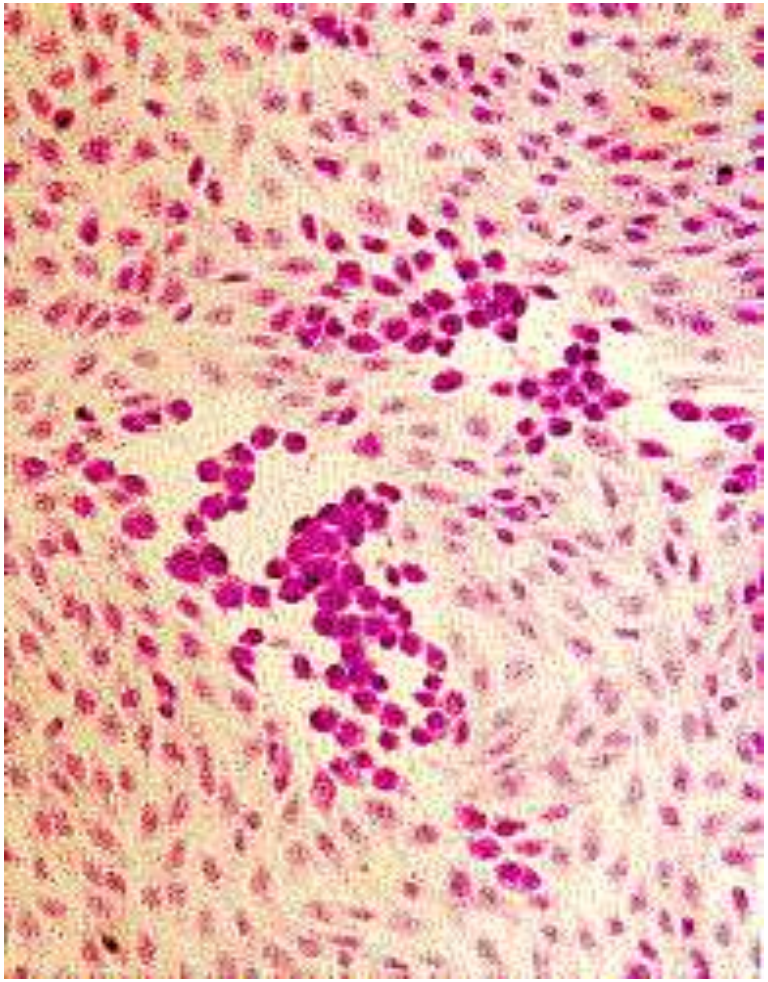
Virus Neutralization Test

INTRODUCTION

- Serological method that detects the presence of viral neutralizing antibodies
- The antibodies bind to viral particles
  - Monoclonal and polyclonal antibodies
- Block viral infectivity – no cell infection



- Conducted by:
  - Mixing dilutions of antibodies with standardized amount of virus , incubating them and cultured into cells, eggs or animals to have clear cytopathic observation

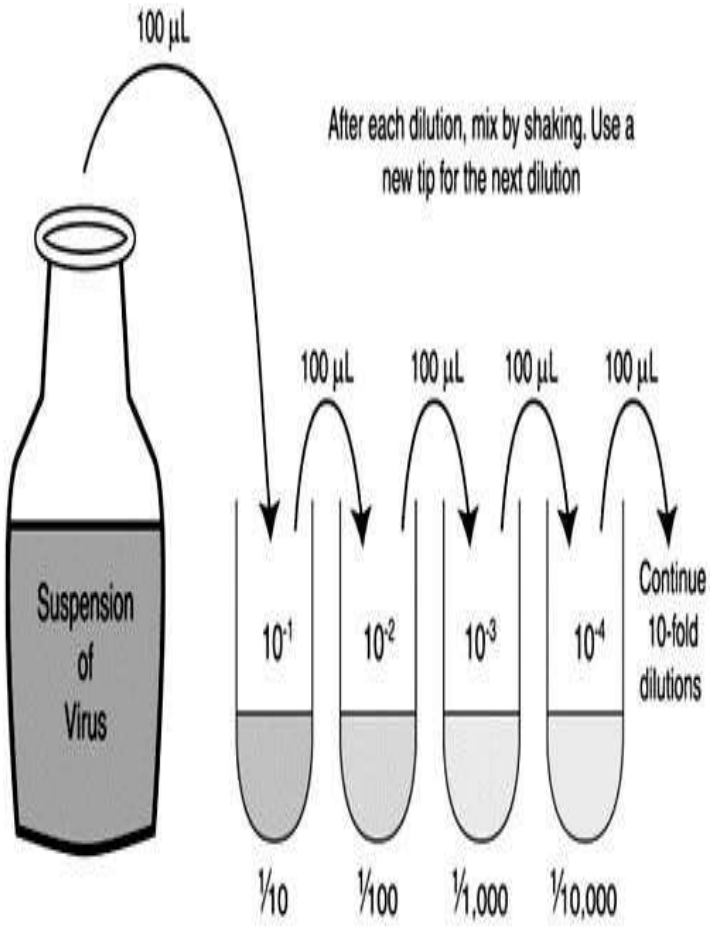


CYTOPATHIC EFFECTS

- Rounding of cells
- Change in texture
  - Granular
  - Hyaline
  - Glassy
- Formation of synctium

STEPS

- First step:
  - Serial dilution of serum
  - Known virus
  - Incubated for 1-2 hours at 37°C
- Second step
  - Cell culture inoculated with mixture
  - Incubated at 37°C
  - Direct microscope



PROS

- Can be specific
- Sensitive
- Vaccine production/ immunological status

CONS

- Very slow
- Intensive
- Required skilled people
- Depends on cell lines

END

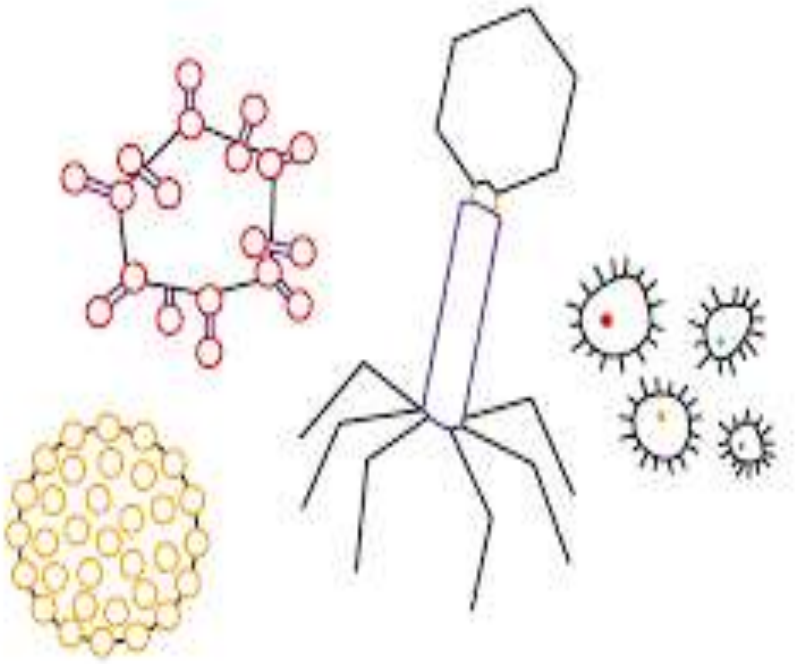
Ppt no. 101

SEROLOGICAL METHODS

Protection Tests

INTRODUCTION

- Not strictly serological
- For identification of those viruses
- Used to produce successful vaccines
- Procedure also employed for testing commercial serum and vaccine products



- Production of either active or passive immunity in an animal followed by a challenge dose of virulent virus
- It is also called in-vivo neutralization test

- Vaccine prepared for all types of immunity to different lab animals
- When satisfactory immunity is produced, all are challenged with a known virus

EXAMPLES

- Used commercially for Rabies and hog cholera vaccines
- Used by injecting in graded doses

END

Ppt no. 102

SEROLOGICAL METHODS

Result interpretation of Protection Tests

HYPOTHETICAL TEST

- Suppose we have three types of vaccines
- We inject them in three different guinea pigs
- Observe for immunity

Immunized guinea pigs	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
Eastern type vaccine	5	5	2	1	0
Western type vaccine	1	0	0	0	0
Venezuelan type vaccine	5	5	3	0	0
Controls	5	5	3	1	0

- Disease: equine encephalomylietis
- Control also set

- Vaccine that provides best immunity and no death proves to be the best

END

Ppt no. 103

SEROLOGICAL METHODS

Haemagglutination /Haemagglutination inhibition Test

INTRODUCTION

- Few viruses possess haemagglutinin protein that binds to Sialic Acid receptors on RBCs

SAMPLE

- Blood
- Using anticoagulant like EDTA or Heparin

1% RBC'S PREPARATION

- Preparation of 1% RBCs
- Equal amount of blood and Normal Saline (NS) centrifuged in Falcon tube
- Remove supernatant and add the same volume of NS again and centrifuge
- Repeat the procedure until clear supernatant
- Remove the supernatant and add ten times of its volume of NS

HAEMAGGLUTINATION TEST

- A ninety six well plate marked according to sample
- Each row assigned to one sample
- Add 50uL NS to all wells
- Add antigen to first well and make two fold dilution till second last well
- Leave last well as control
- Add 25uL RBCs solution to all wells

HAEMAGGLUTINATION HAEMINHIBITION TEST

- 96 well plate labelled according to samples
- 25uL antiserum added to first well and make two fold dilution till end
- 0.5uL RBCs added to all wells
- Incubation for 30 minutes

Ppt no. 104

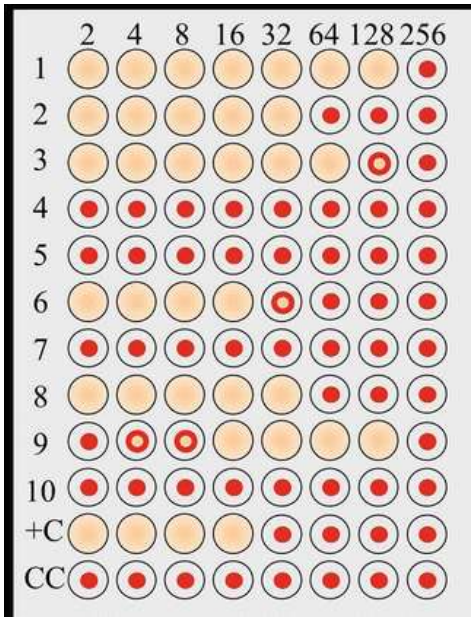
SEROLOGICAL METHODS

Result interpretation of HA/HI test

INTERPRETATION OF HA TEST

- Button formation showed no hemagglutination
- Hazy wells showed agglutination

- One HA calculates on the basis of last well showing button formation
- Button formation at controls

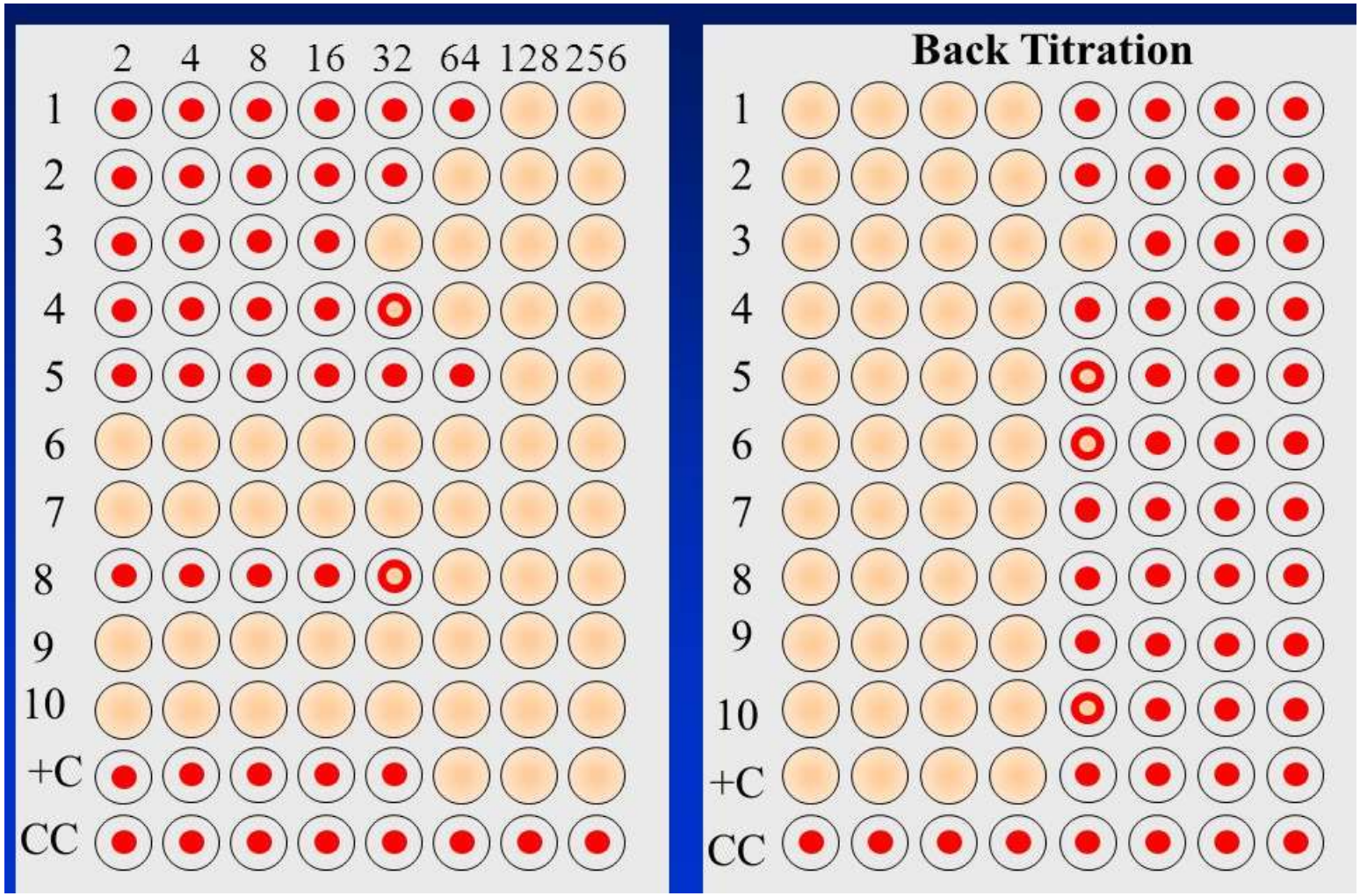


- Dilution value shows us 4HA unit value
- To determine one HA unit the value is divided by four

Well	1	2	3	4	5	6	7	8	9	10	11	12
Dilution	1/10	1/20	1/40	1/80	1/160	1/320	1/640	1/1280	1/2560	1/5120	1/10240	Control
Pattern												

INTERPRETATION OF HI TEST

- Shows button formation at the start
- Hazy wells at the end
- 8HA units were calculated per 25uL
- By dividing endpoint by 16


2 4 8 16 32 64 128 256
1
2
3
4
5
6
7
8
9
10
+C
CC
Back Titration
1
2
3
4
5
6
7
8
9
10
+C
CC

Ppt no. 105

CYTOPATHOGENECITY IN TISSUE CULTURE

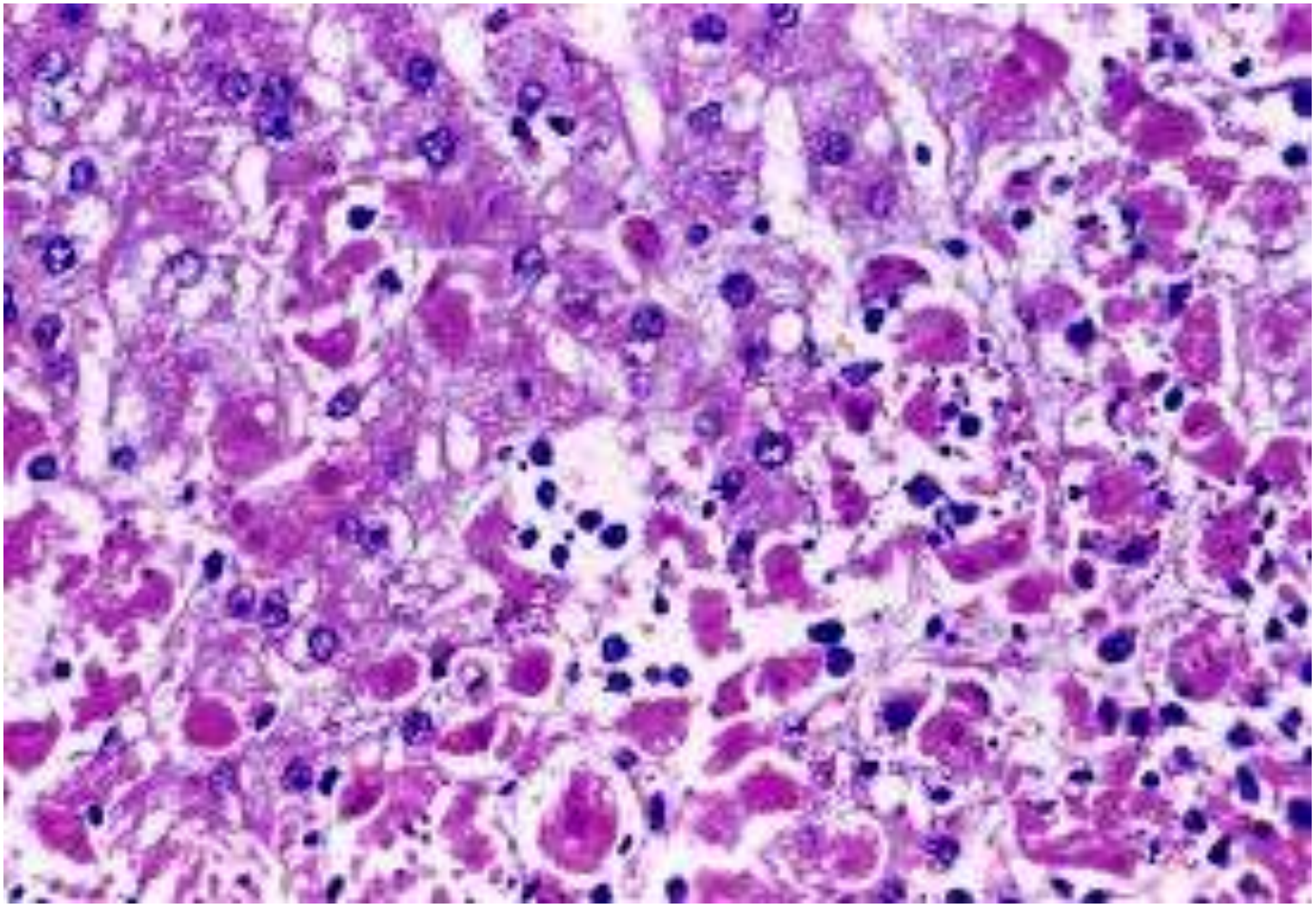
Basic Ingredients

1. Tissue cells
2. Lactalbumin hyrolysate
3. Serum
4. Balanced salt solution

Other Ingredients

1. Ascitic fluid
2. Amniotic fluid
3. Serum ultrafilterate
4. Amino acids
5. Vitamins
6. Glucose
7. Inorganic salts
8. Penicillin and streptomycin

- Glassware used must be absolutely clean and free film and toxic substances
- Chemicals and distil water must be free from toxic ions



Cytopathogenecity

- Abnormalities observed in cells
- Inclusion bodies observed
- Synctia formation

Ppt no. 106

SEROLOGICAL METHODS

Haemadsorption Test

Introduction

- Technique similar to haemagglutination test
- Used for detection of certain viruses in tissue culture

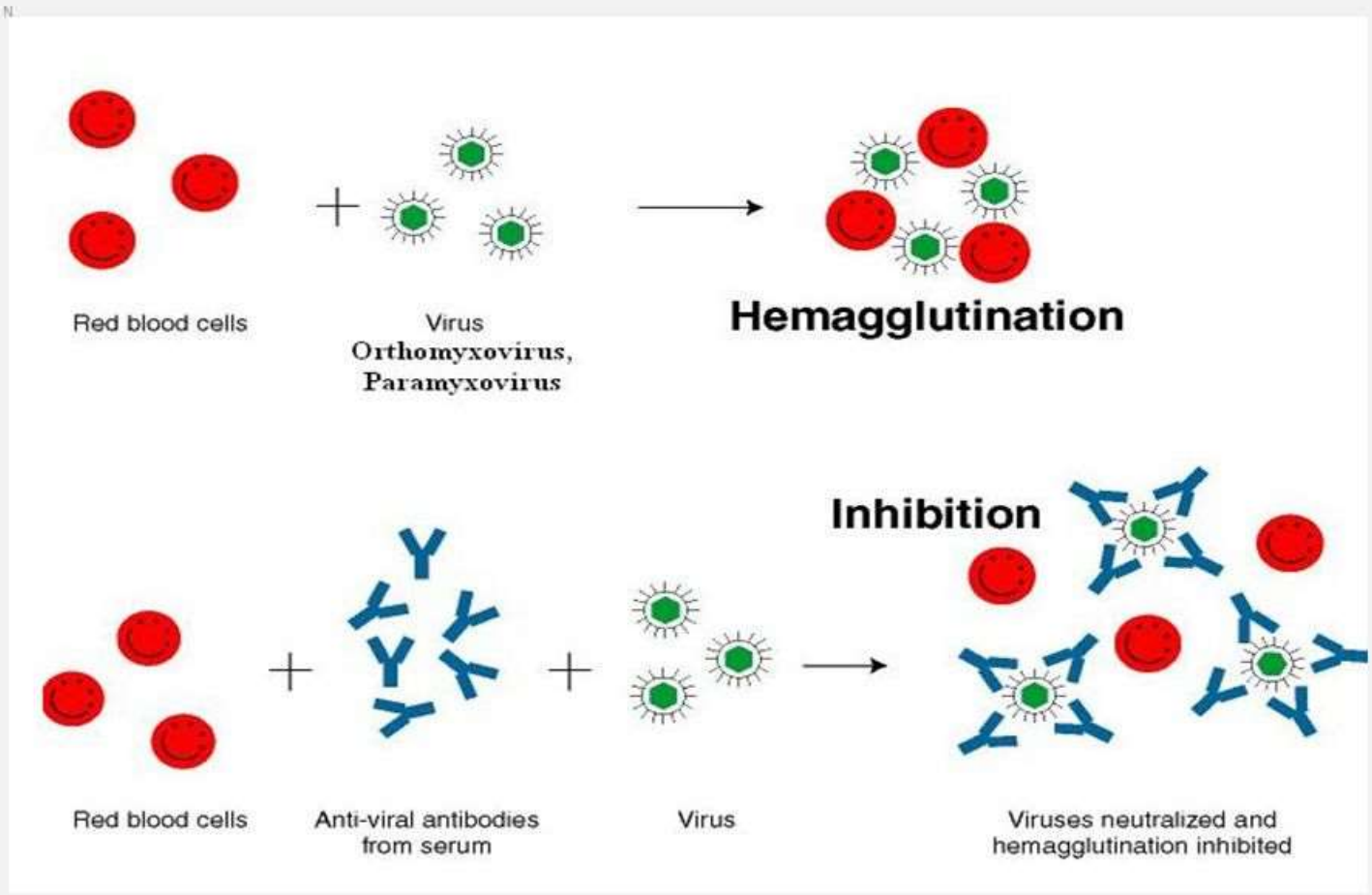
Haemagglutinin

- Viruses producing haemagglutinin will adsorb erythrocytes to the surface of cells in which the virus has grown

PROCEDURE

- A suspension of washed RBCs is added to tissue culture tubes
- Presences of haeadsorbing virus, results in clustering of RBCs
- Rosettes around the infected cells of the culture
- Can be observed microscopically

Haemadsorption and agglutination inhibition:

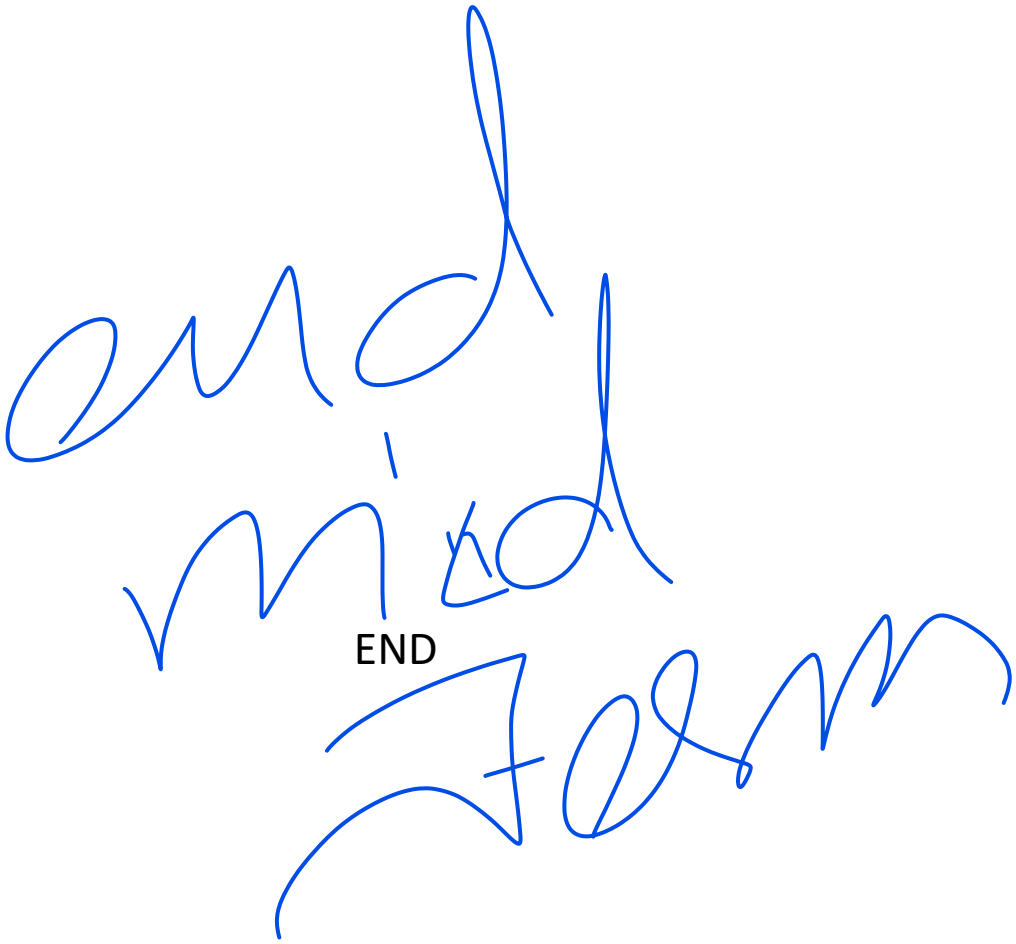


USE

- Useful for detection of viruses producing haemagglutinins

Inhibition

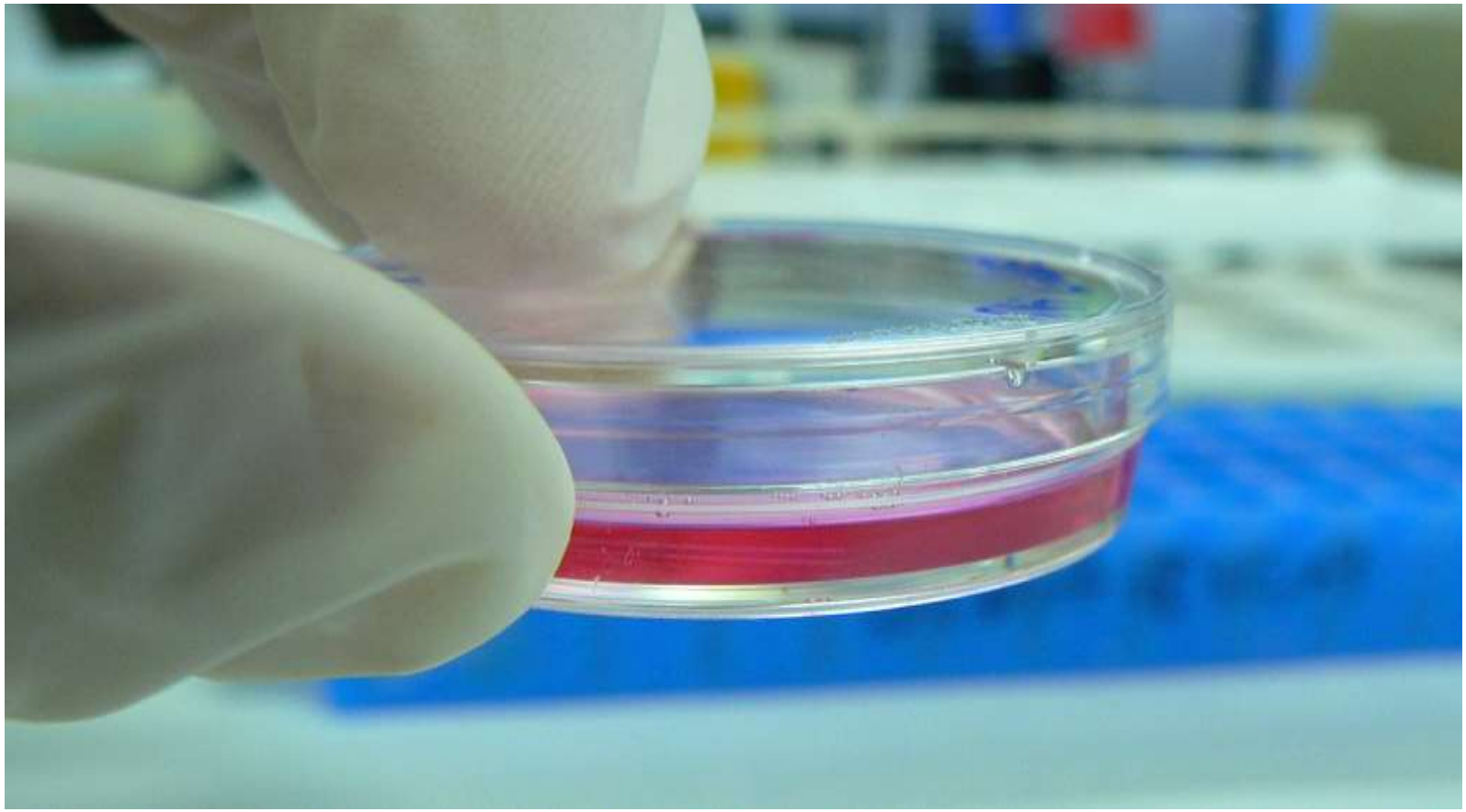
- Can be inhibited by specific antibody
- Hence haemadsorption-inhibition test is devised for antibody titer



Ppt no. 107

Growing Animal Viruses in Cell Culture

THE PROCESS BY WHICH CELLS ARE GROWN UNDER CONTROLLED CONDITIONS



CONTROLLED CONDITIONS

Vary for each cell type

- Essential nutrients (amino acids, carbohydrates etc.
- Growth factors
- Hormones
- Gases (CO_2 or O_2)
- Physio-chemical environment regulators (pH buffer, osmotic pressure, temperature etc.

History

Cell culture was first successfully undertaken by Ross Harrison in 1907

- Roux in 1885 for the first time maintained embryonic chick cells in a cell culture
- 1878: Claude Bernard proposed that physiological systems of an organism can be maintained in a living system after the death.
- 1885: Roux maintained embryonic chick cells in a saline culture.
- 1897: Loeb demonstrated the survival of cells isolated from blood and connective tissue in serum and plasma.
- 1903: Jolly observed cell division of salamander leucocytes *in vitro*.

1907

Harrison cultivated frog nerve cells in a lymph clot held by the hanging drop' method and observed the growth of nerve fibers *in vitro* for several weeks. He was considered by some as the father of cell culture.

APPLICATIONS

Areas where cell culture technology is currently playing a major role

- Model systems for

  Studying basic cell biology, interactions between disease causing agents and cells, effects of drugs on cells, process and triggering of aging & nutritional studies
- Toxicity testing

  Study the effects of new drugs
- Cancer research

  Study the function of various chemicals, virus & radiation to convert normal cultured cells to cancerous cells

APPLICATIONS

Areas where cell culture technology is currently playing a major role

- Virology

  Cultivation of virus for vaccine production, also used to study there infectious cycle.

- Genetic Engineering

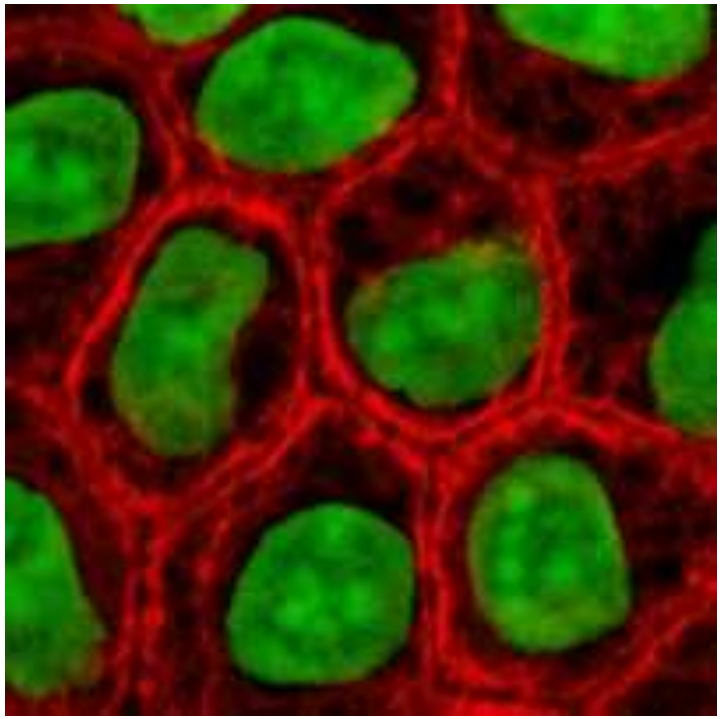
  Production of commercial proteins, large scale production of viruses for use in vaccine production e.g. polio, rabies, chicken pox, hepatitis B & measles

- Gene therapy

  Cells having a functional gene can be replaced to cells which are having non-functional gene

Culture of Mammalian Cells

Culture of Non-mammalian Cells



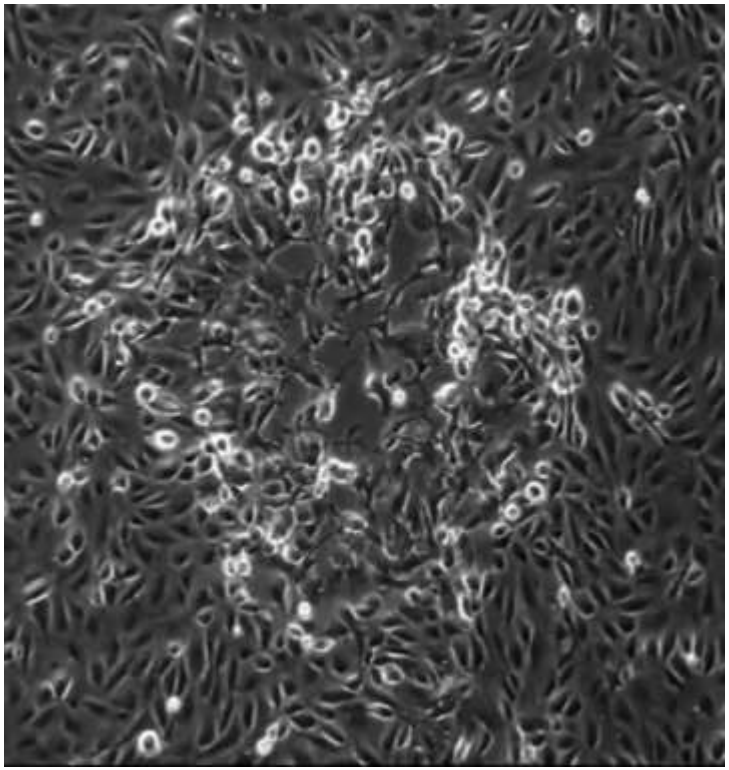
Ppt no. 108

Viral Identification After Growing

IDENTIFICATION

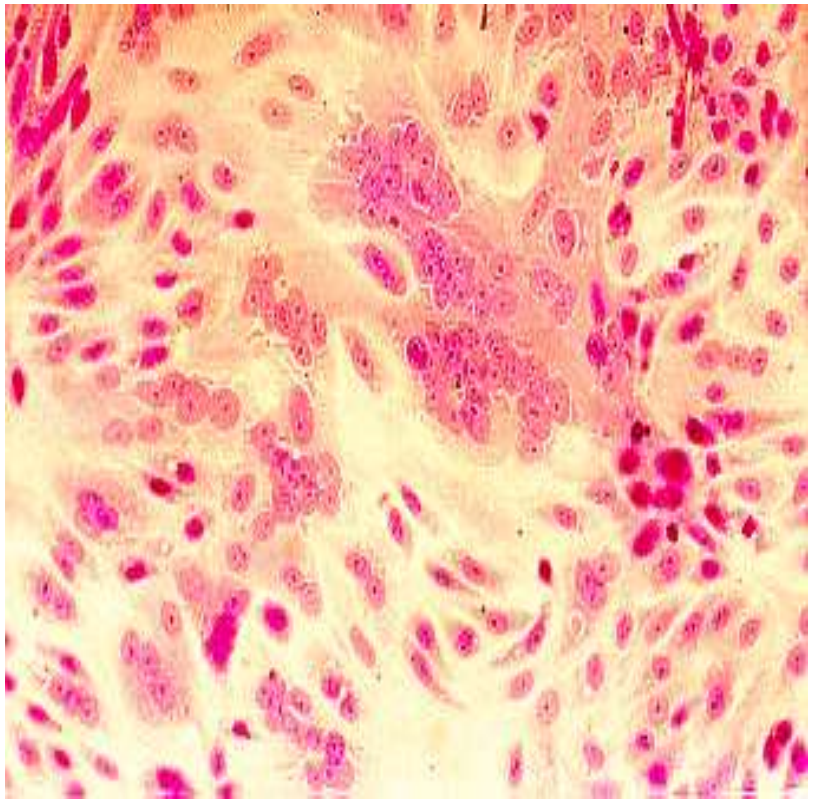
- Plaque formation
- Cytopathic Effects
- Fluorescent Antibody technique
- Nucleic Acid hybridization
- Electron Microscopy

PLAQUE FORMATION



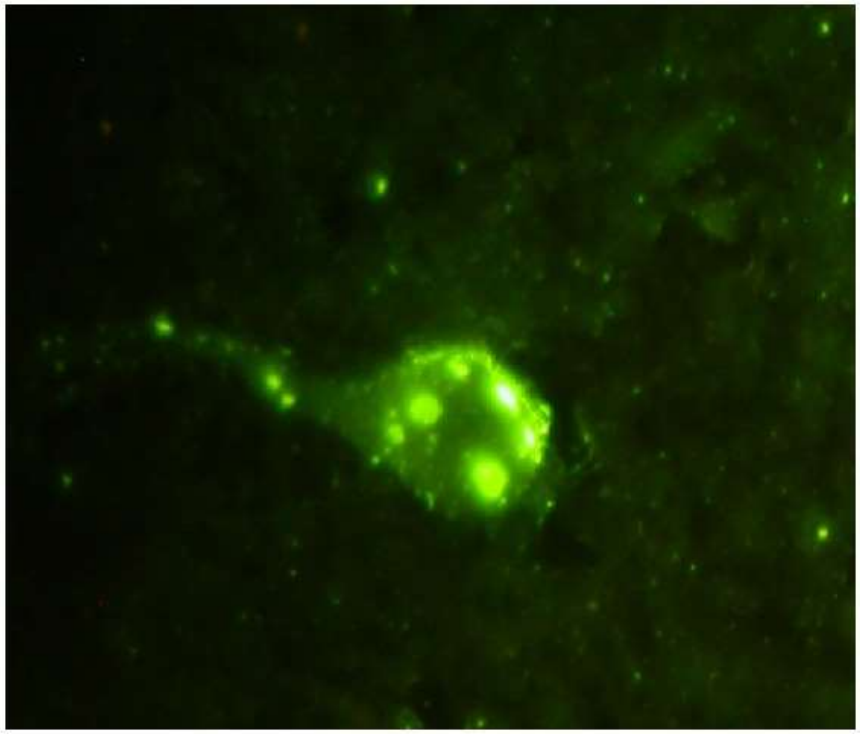
- a visible structure formed within a cell culture
- The bacteriophage viruses replicate and spread, thus generating regions of cell destructions known as plaques

CYTOPATHIC EFFECTS



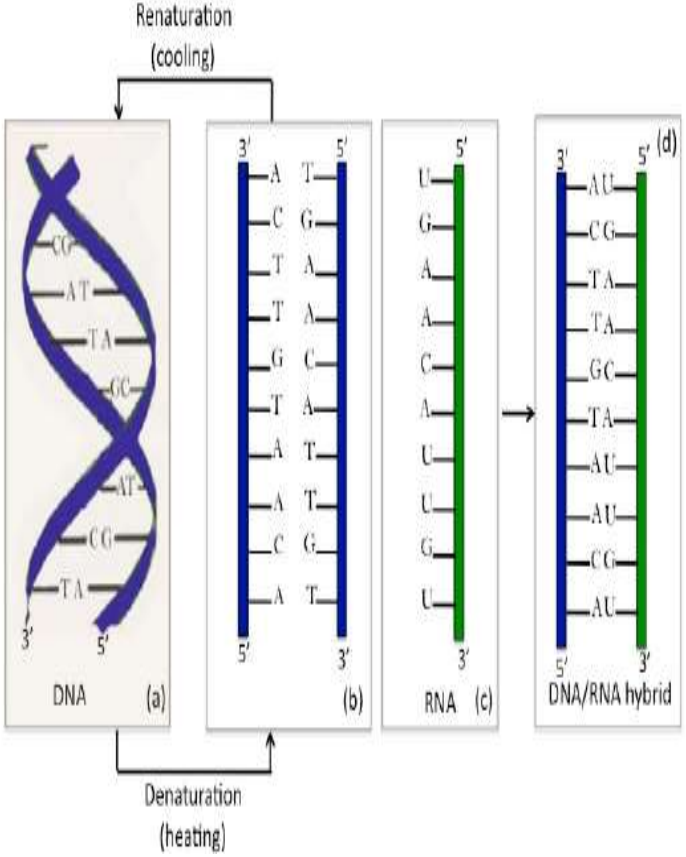
- Syncytia
- Cell rounding
- Membrane proliferations
- Vacuolization
- Inclusion bodies
- Focus (foci)
- Hemadsorption

FLUORESCENT ANTIBODY TECHNIQUE



- based on the observation that virus proteins are (antigen) present in the tissues
- e.g rabies virus present in nervous tissues

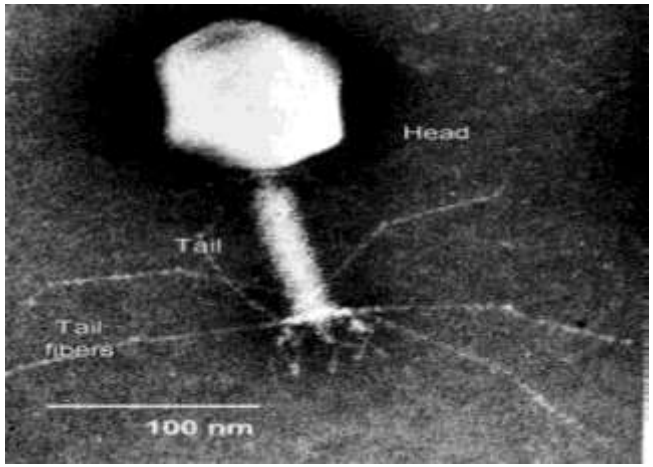
NUCLIEC ACID HYBRIDIZATION



A technique in which single-stranded nucleic acids (DNA or RNA) are allowed to interact so that complexes called hybrids are formed by molecules with similar, complementary sequences

ELECTRON MICROSCOPY

- uses a beam of **electrons** to create an image of the specimen
- Inclusion bodies formed

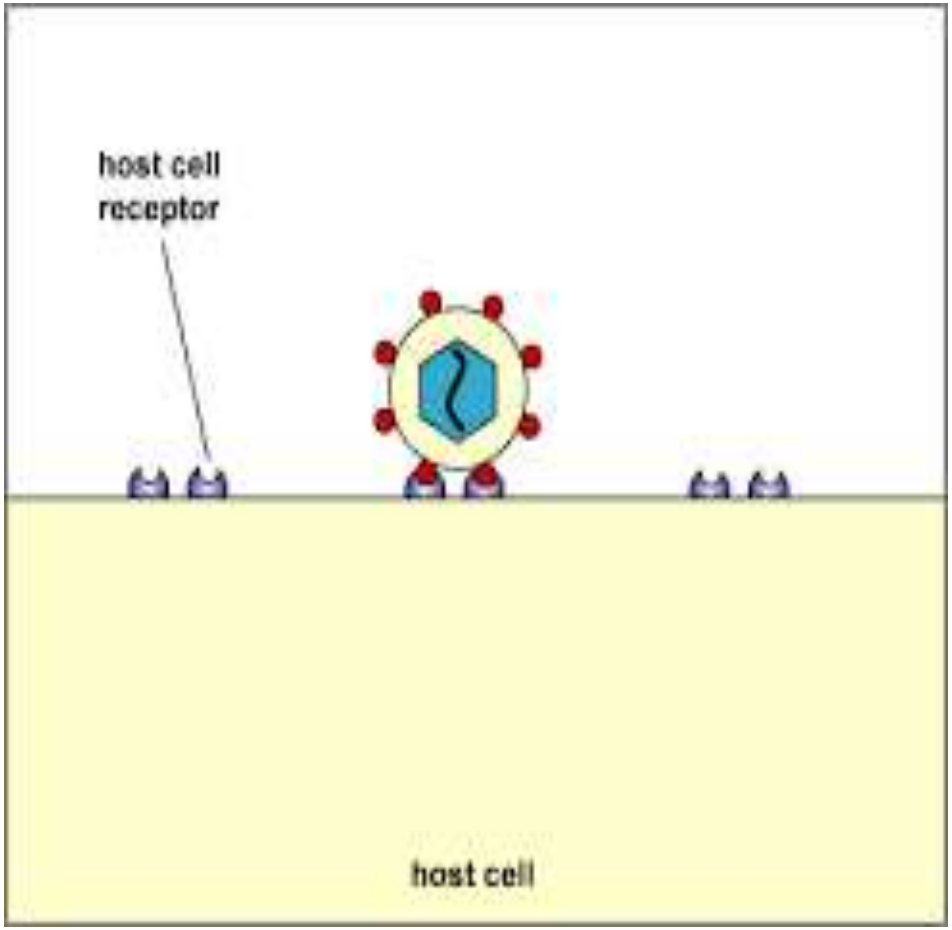


Ppt no. 109

MULTIPLICATION OF VIRUSES

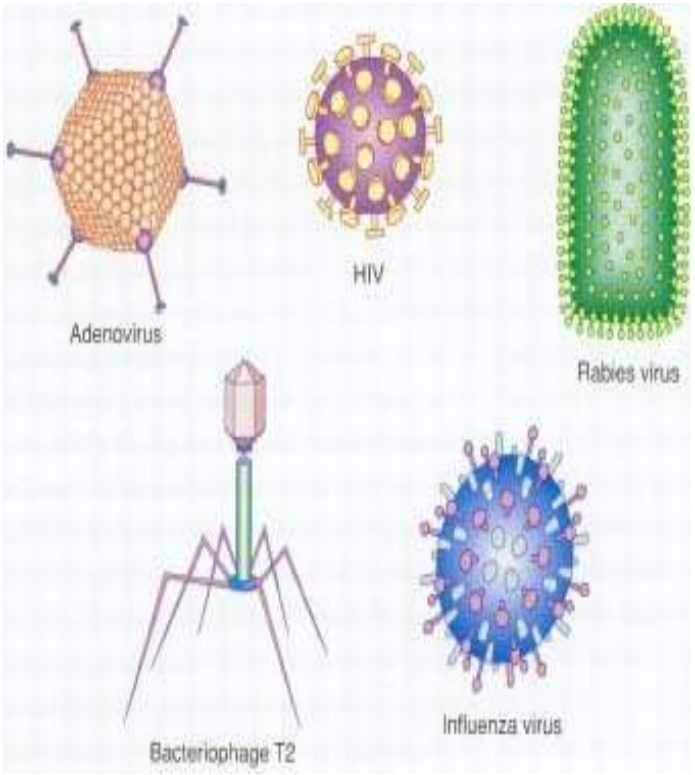
Attachment

- Electrostatic phenomenon
- Divalent cations involved
- Viral particles and cell surface have negative charges
- Cations reduce the repulsion



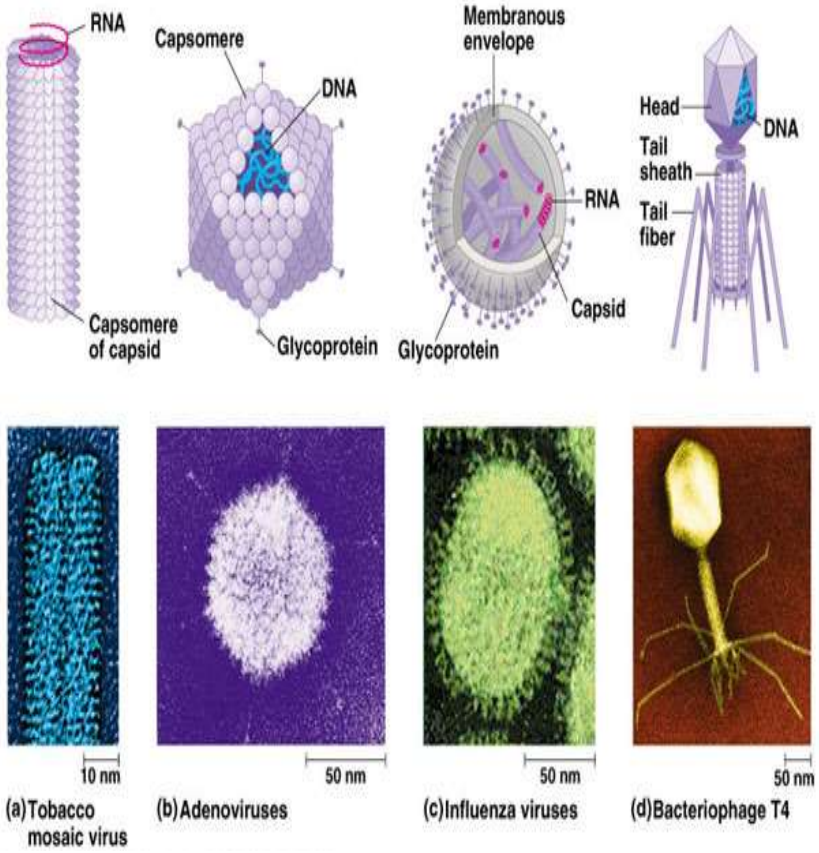
- Viral particles and cell surface have negative charges
- Cations reduce the repulsion

SPECIFICITY

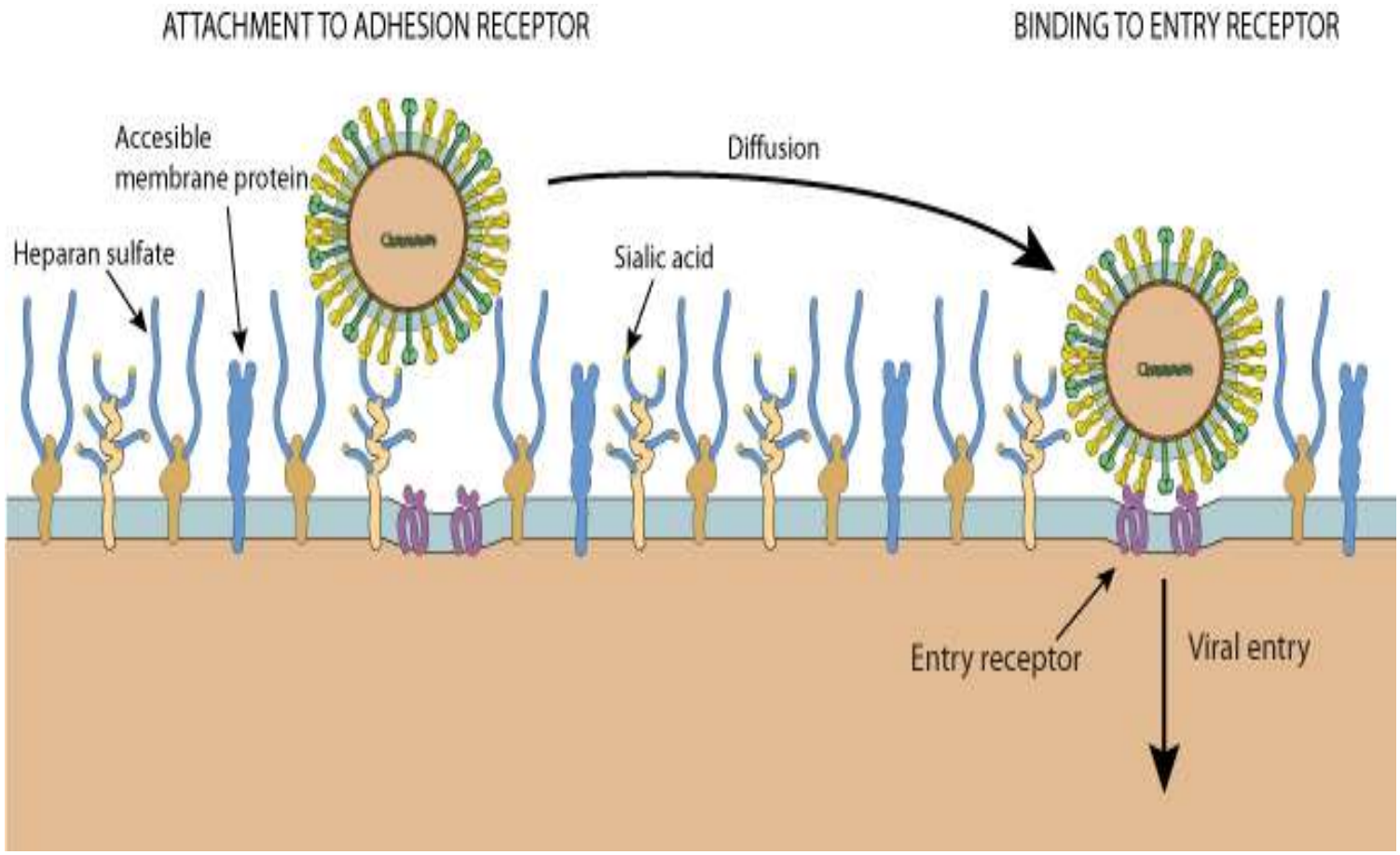


- Specific receptors of cell surface
- Host tropism- Tissue tropism- Cell tropism

SPECIFICITY



- Receptor has electrostatic charge arrangement complementary to virus
  - Weakening of capsid
  - Release of nucleic acid
  - Initiation of infection


ATTACHMENT TO ADHESION RECEPTOR
BINDING TO ENTRY RECEPTOR
Accesible
membrane protein
Diffusion
Heparan sulfate
Sialic acid
Entry receptor
Viral entry

Ppt no. 110

MULTIPLICATION OF VIRUSES

Entry

- After attachment
- Nucleic acid components releases in cytoplasm inside a vacuole
- Active, unspecific, temperature dependent

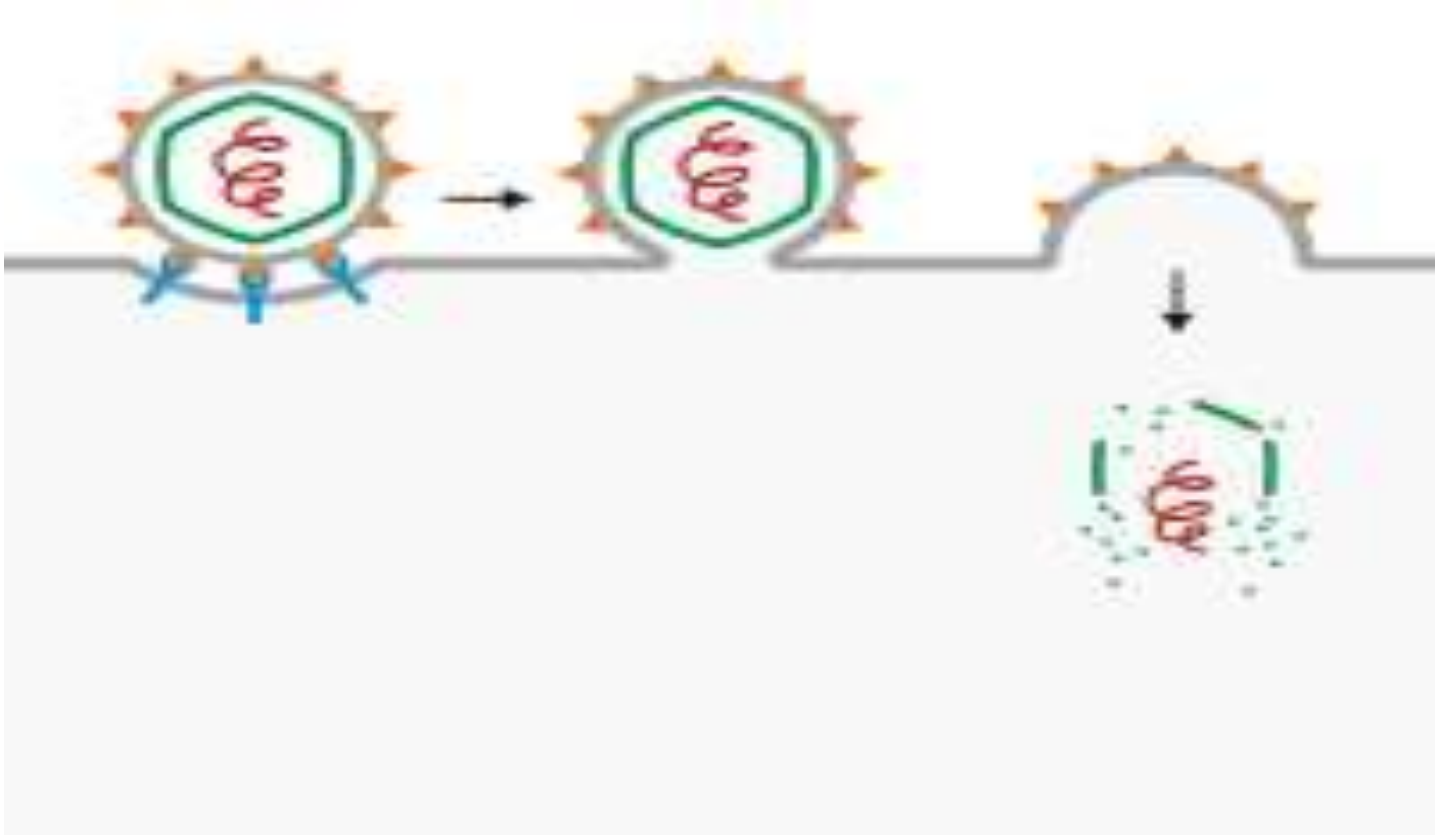
VIRAL ENTRY

- ***Membrane Fusion*** or ***Hemifusion State***: The cell membrane is punctured and made to further connect with the unfolding viral envelope.
- ***Endocytosis***: The host cell takes in the viral particle through the process of endocytosis, essentially engulfing the virus like it would a food particle.
- ***Viral Penetration***: The viral capsid or genome is injected into the host cell's cytoplasm.

VIRAL ENTRY VIA MEMBRANE FUSION

- In viruses with a envelope
- Viral receptors attach to the receptors on the surface of the cell
- The viral envelope fuses with the host cell membrane
- Now-bare virus enters into the cell
- In essence, the virus's envelope "blends" with the host cell membrane, releasing its contents into the cell. Examples: HIV and herpes simplex virus

VIRAL ENTRY VIA MEMBRANE FUSION



Viral entry via endocytosis

- Viruses with no viral envelope
- Ingested by the host cell through the cell membrane
- Knocking at the door
- Cell will engulf the virus
- Once inside the cell, the virus must now break out of the vesicle by which it was taken up in order to gain access to the cytoplasm.
- Examples include the poliovirus, Hepatitis C virus, FMDV

- Many enveloped viruses also enter the cell through endocytosis

- Entry via the endosome guarantees low pH and exposure to proteases which are needed to open the viral capsid and release the genetic material inside.
- Further, endosomes transport the virus through the cell and ensure that no trace of the virus is left on the surface, which could be a substrate for immune recognition

ENTRY VIA GENETIC INJECTION

- Attaching to the surface of the cell via receptors
- Injecting only its genome into the cell
- Leaving the rest of the virus on the surface
- This is restricted to viruses in which only the gene is required for infection of a cell

Example includes the Bacteriophages; for example, when the tail fibers of the T2 phage land on a cell, its central sheath pierces the cell membrane and the phage injects DNA from the head capsid directly into the cell.

Ppt no. 111

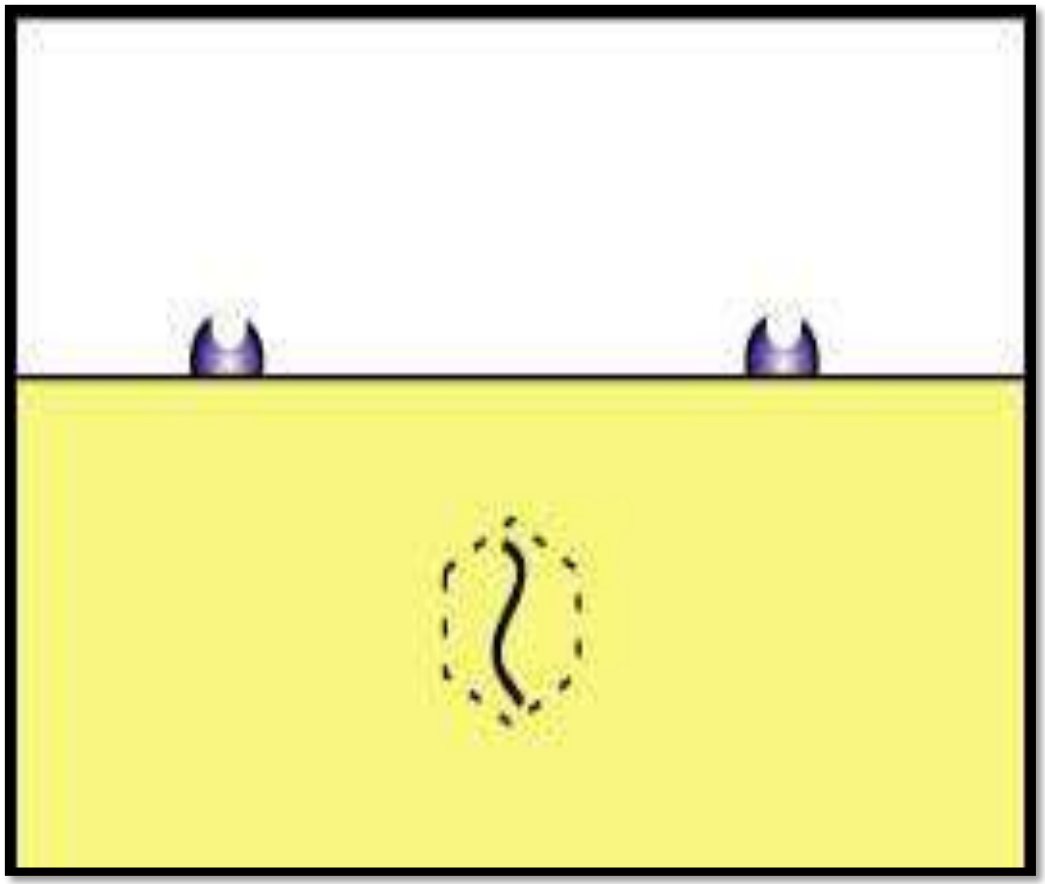
MULTIPLICATION of VIRUSES

Uncoating

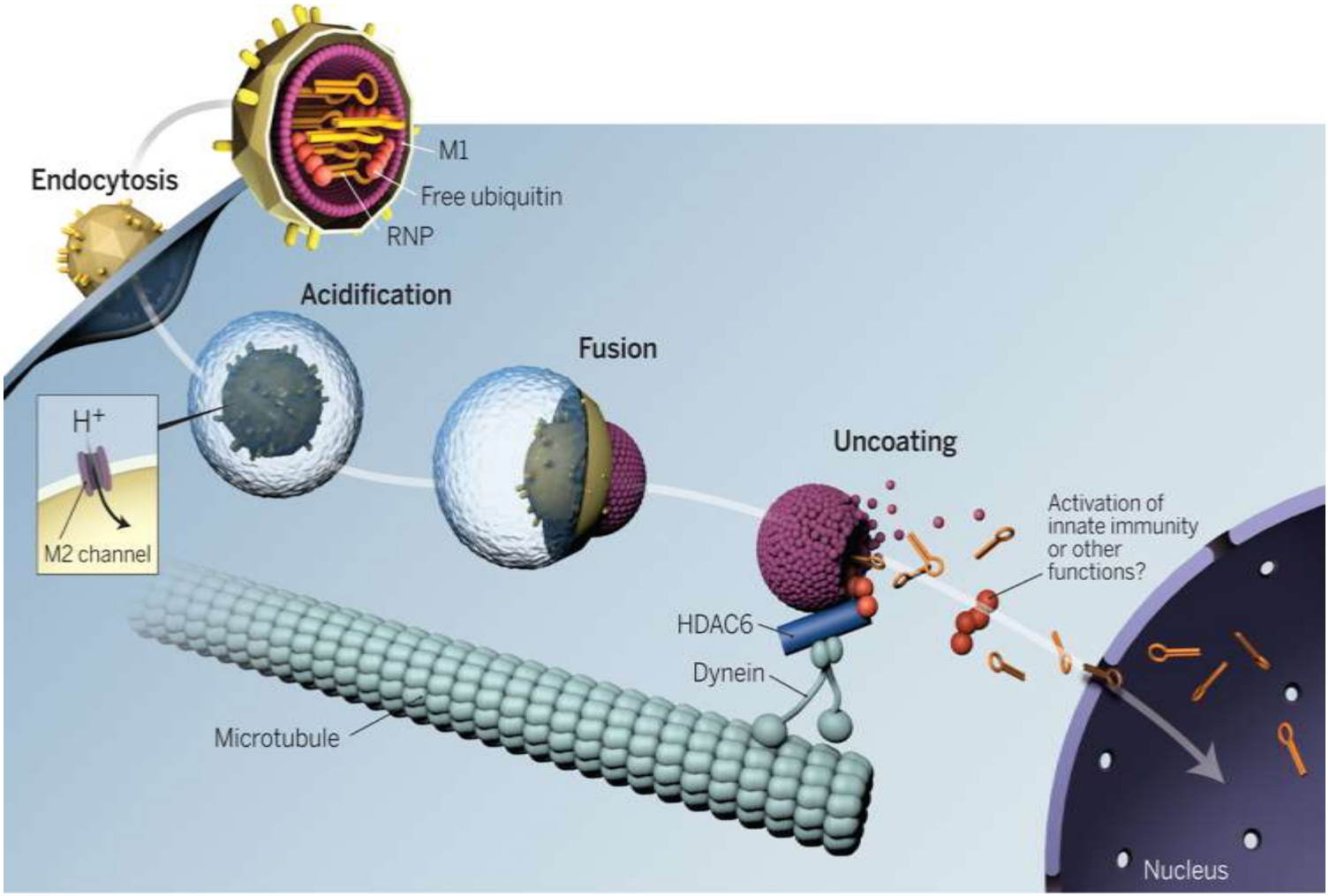
Uncoating

- Uncoating means removal of capsid
- Exposing the viral genome in the host cell

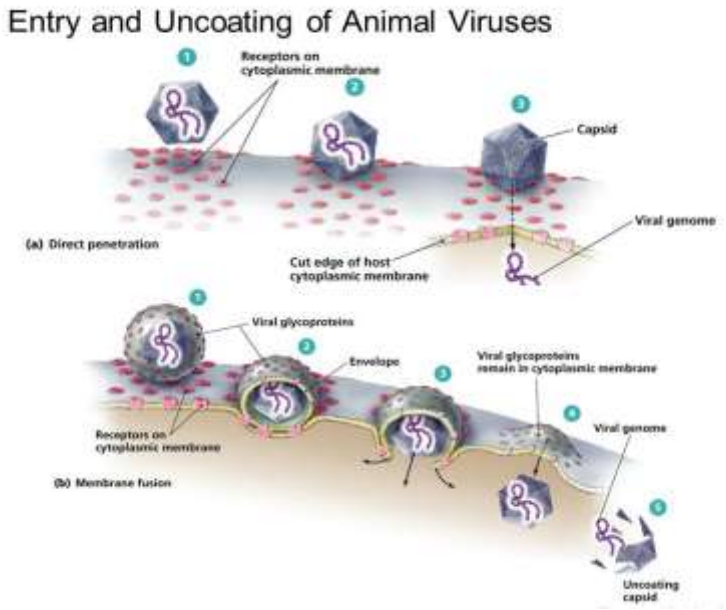
- The Protein coat around genome gets disintegrated
- Cytoplasm
- Cellular enzymes



- Vital important step
- Makes genome ready to takeover the host machinery
- Activates cascade of the steps required for viral replication


Endocytosis
M1
Free ubiquitin
RNP
Acidification
Fusion
H+
M2 channel
Uncoating
Activation of innate immunity or other functions?
HDAC6
Dynein
Microtubule
Nucleus

- Genome becomes ready to invade the host machinery



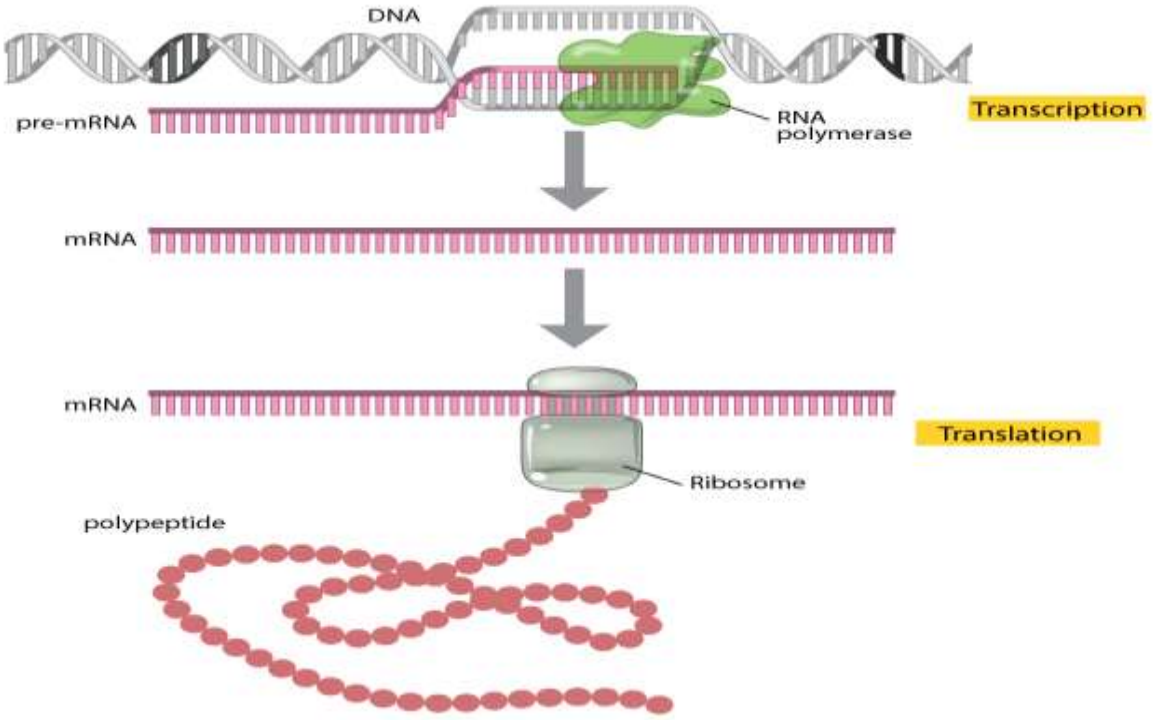
Ppt no. 112

MULTIPLICATION oF VIRUSES

Biosynthesis in DNA viruses

TRANSCRIPTION

- All viruses make mRNA and then proteins



Differences in mRNA

- Single stranded
- Thymine replaced by Uracil
- mRNA is exactly similar to positive strand

Translation

- Information on mRNA changed to proteins
- Viruses that replicate in nucleus

  Use host polymerase
- Viruses that replicate in cytoplasm

  Use own polymerase

POLYMERASES

1. Cellular DNA dependent RNA polymerases
2. Viral DNA dependent RNA polymerases

SYNTHESIS OF VIRAL PROTEINS

Viral protein synthesis: virus mRNA translated on cell ribosomes into two types of virus protein

a. **Structural:** proteins which make up the virus particle are manufactured and assembled

b. **Non – structural:** not found in particle, mainly enzymes for virus genome replication

Synthesis of Viral Genome

Viral nucleic acid synthesis

new virus genome synthesized, templates are either the parental genome or with single stranded nucleic acid genomes, newly formed complementary strands

https://www.youtube.com/watch?v=nR9e7eXm7gk&t=55s

Ppt no. 113

MULTIPLICATION of VIRUSES

Biosynthesis of RNA viruses

Replication

- RNA viruses usually replicate in cytoplasm

Positive sense RNA virus

- Itself acts as mRNA
- They need their own viral RNA dependent RNA polymerase

Negative sense RNA virus

- They are a copy of mRNA
- They first make a positive strand with RNA dependent RNA polymerase
- It acts as mRNA
- It will further make copies using RNA dependent RNA polymerase

- https://www.youtube.com/watch?v=ZGE4BLuAkuU

Ppt no. 114

VIRAL DIAGNOSTIC TESTS

Immunofluorescence Assay

APPLICATIONS ON

tissue sections

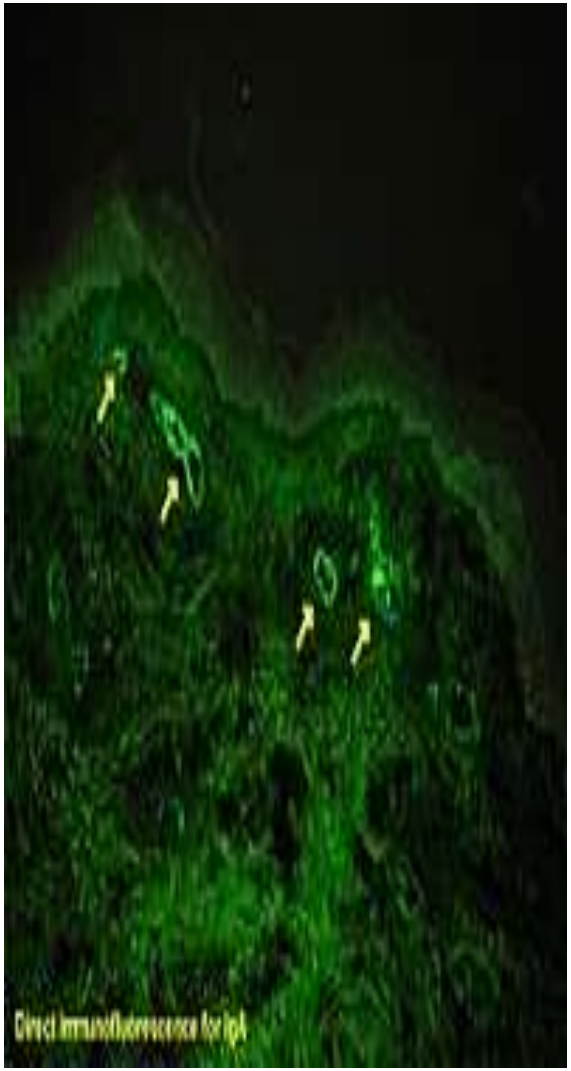
Cultured cell lines

individual cells

Analyze the distribution of

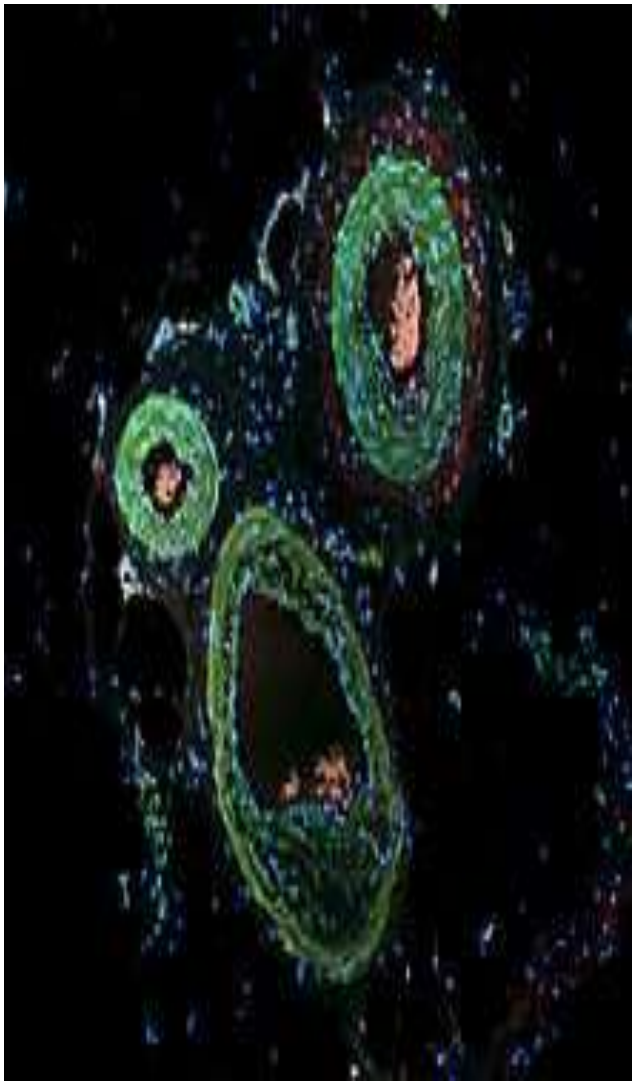
proteins, glycans and small biological and non-biological molecules.

This technique can even be used to



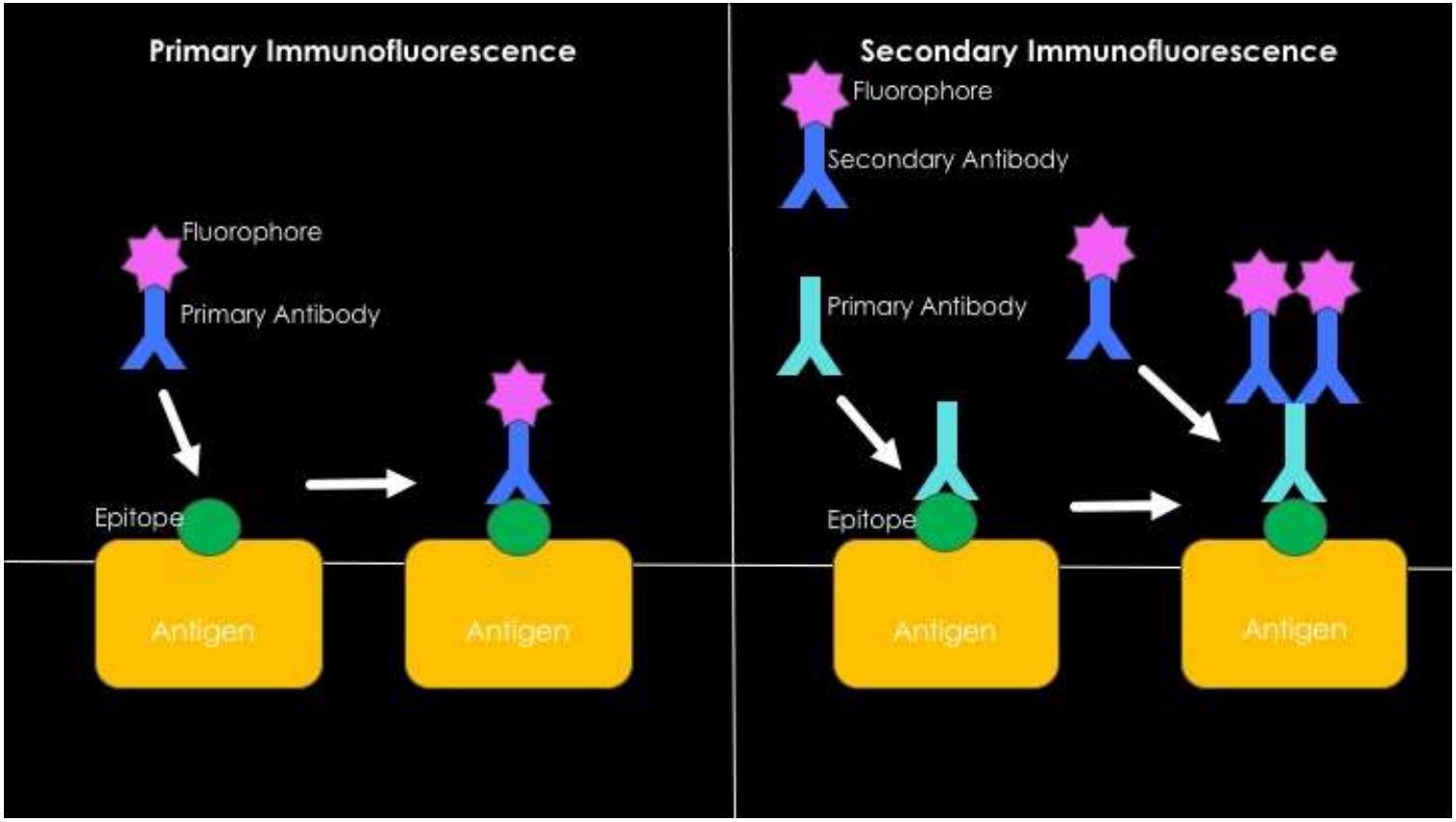
Primary (direct)

- A single, primary antibody, chemically linked to a Fluorophone
- The primary antibody recognizes the target molecule (antigen)
- Binds to a specific region called the epitope.



Secondary (indirect)

- Two antibodies
- The unlabeled first (primary) antibody specifically binds the target molecule
- The secondary antibody, which carries the fluorophore, recognizes the primary antibody and binds to it

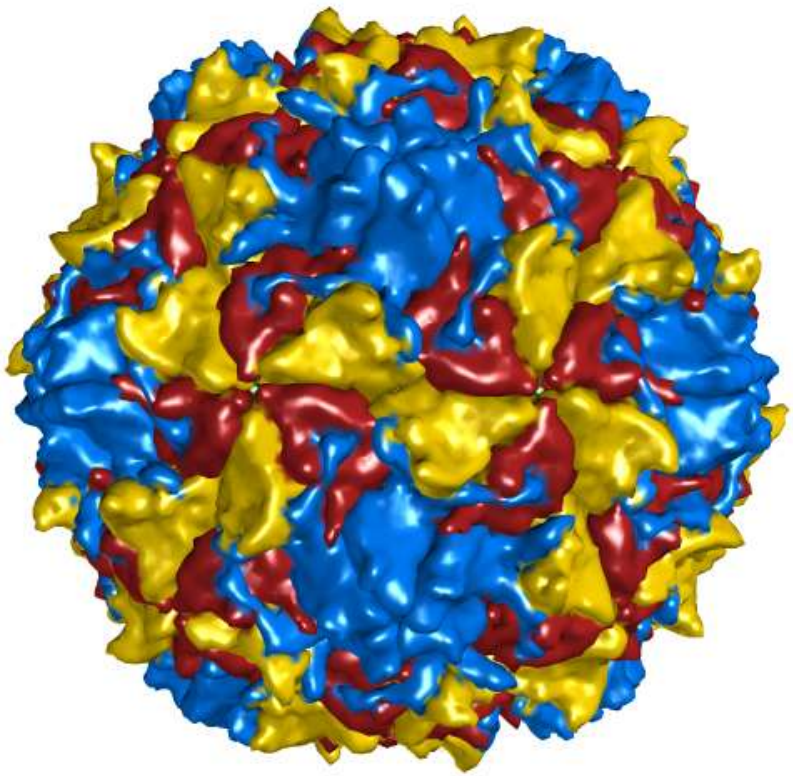

Primary Immunofluorescence
Fluorophore
Primary Antibody
Epitope
Antigen
Antigen
Secondary Immunofluorescence
Fluorophore
Secondary Antibody
Primary Antibody
Epitope
Antigen
Antigen

Ppt no. 115

Biochemical Changes Induced in Virus-Infected Cell

BASIC QUESTION?

- Virus induced changes leading to cell death or accelerated growth
- Answer hidden in different proteins coded by viral genome

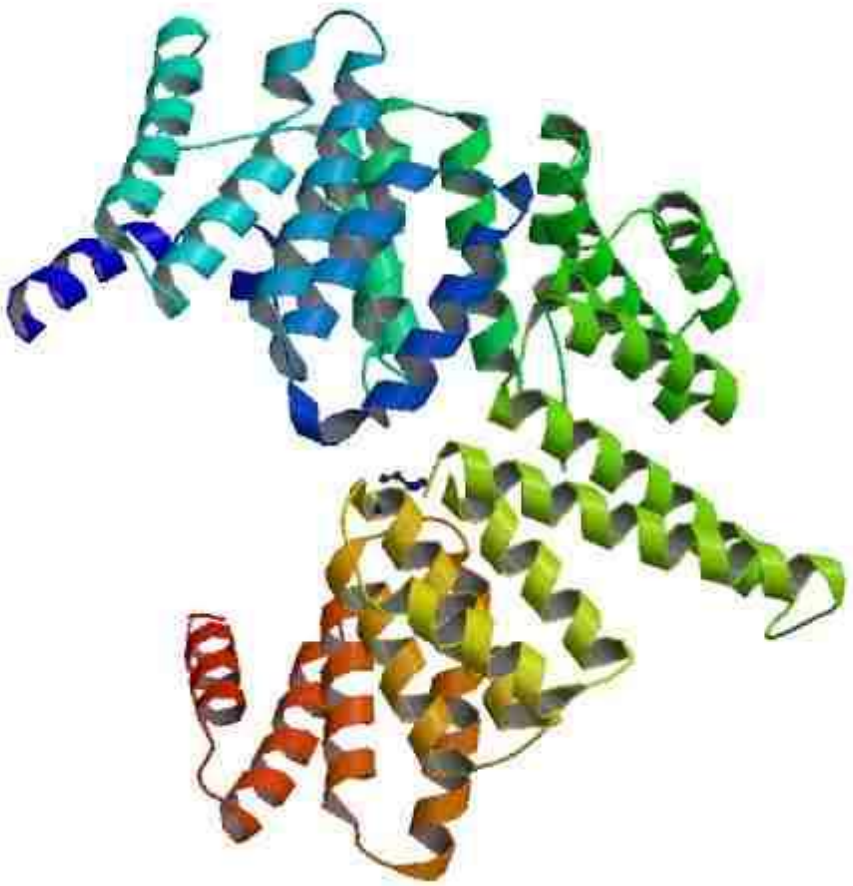


- e.g the smallest virus i.e. Picornavirus
- Contains 4000-6000 nucleotides
- Genetic information codes 1500 to 2000 amino acids or 10 to 15 proteins

- DNA of vaccinia virus contains 250,000 nucleotide pairs which provide information for 400 to 600 proteins

- Two types of proteins:

1. Early proteins
2. Late proteins



- Early protein synthesis usually due to genetic functioning of parental viral genomes
- Include those which inhibit host cellular DNA replication, host cell RNA formation, host cell protein synthesis

- Late proteins include viral structural proteins and proteins that function in repression or switching off the formation of virus induced enzymes

- Actinomysin prevents transcription of RNA from DNA template
- Puromycin or parafluoroalanine inhibts translation of genetic message from mRNA into protein

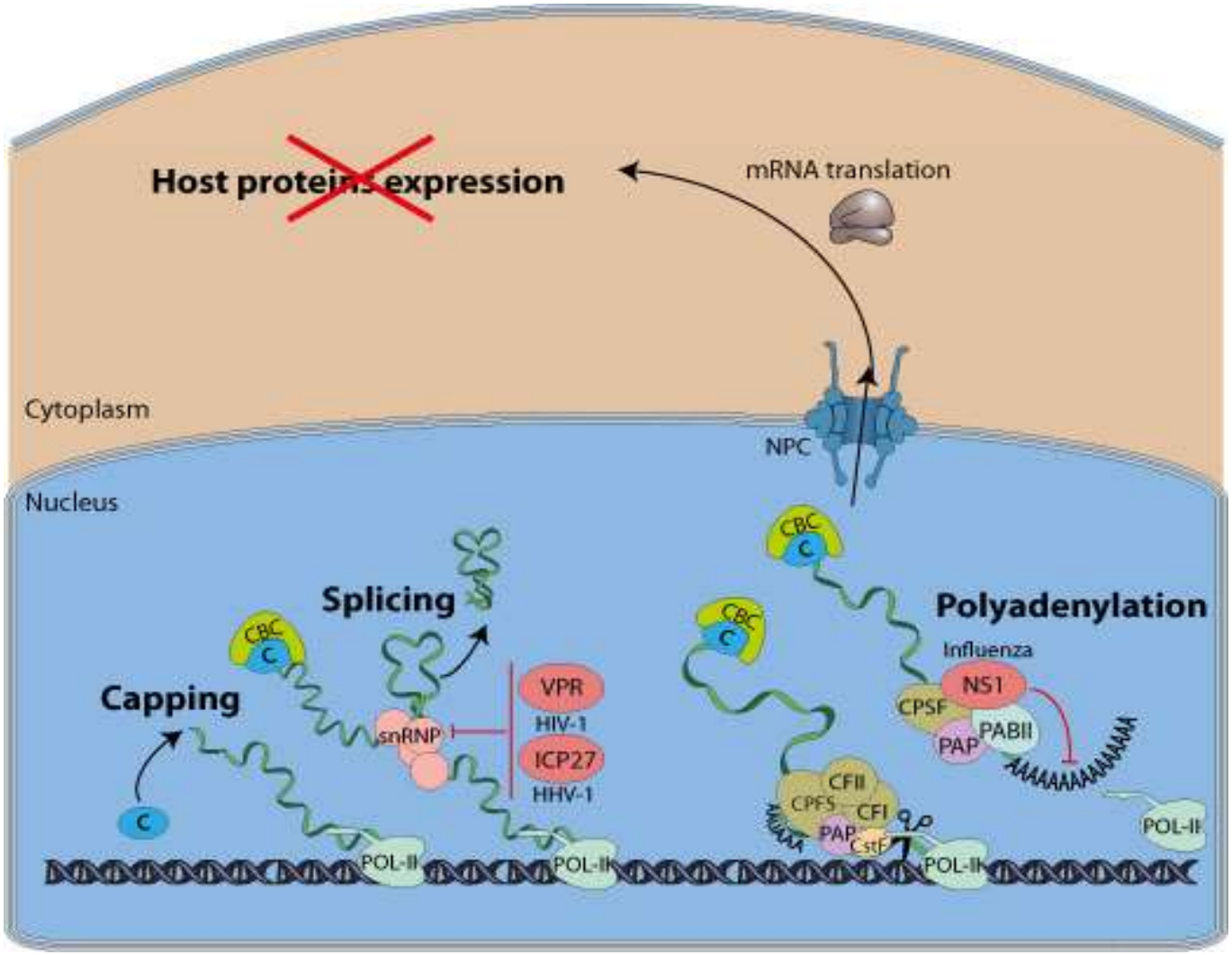
END

Ppt no. 116

Inhibition of Host Cell RNA Synthesis

- Poliovirus inhibits protein synthesis approximately two hours post infection

- e.g the smallest virus i.e. Picornavirus
- Contains 4000-6000 nucleotides
- Genetic information codes 1500 to 2000 amino acids or 10 to 15 proteins


Host proteins expression
mRNA translation
Cytoplasm
NPC
Nucleus
Splicing
Capping
Polyadenylation
CBC
C
snRNP
VPR
HIV-1
ICP27
HHV-1
Influenza
NS1
CPSF
PAP
PABII
CFII
CPFS
CFI
CstF
POL-II

- DNA of vaccinia virus contains 250,000 nucleotide pairs which provide information for 400 to 600 proteins

- When menangioma virus infects mouse origin L cells:
- Marked decrease in cellular enzyme which polymerizes RNA from DNA template

END

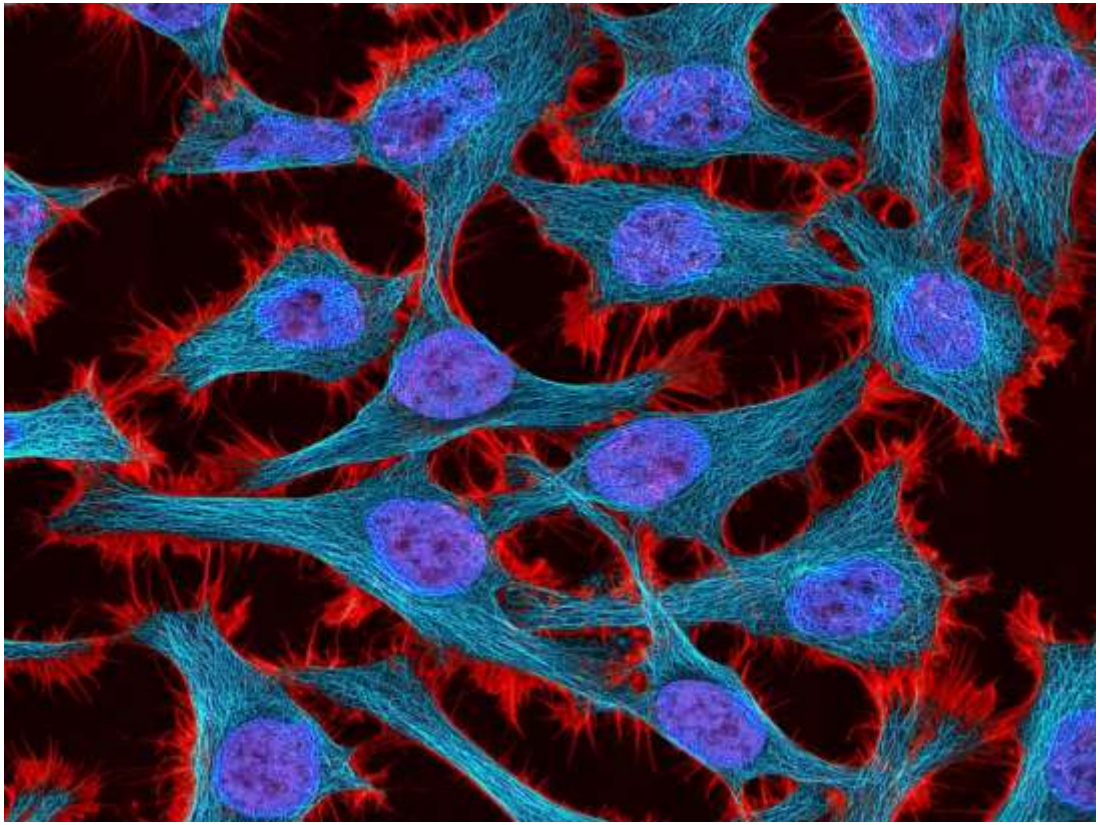
Ppt no. 117

Inhibition of Host Cell Protein Synthesis

- Most cellular protein is assembled and synthesized on aggregates of ribosomes called polyribosomes

- Cellular protein synthesis blocked by interference with synthesis and function of mRNA
- If cellular mRNA synthesis is blocked, the rate of decrease of protein synthesis will increase

- Half life of mRNA in actinomycin treated HeLa and L cells is approximately three hours



- Cellular protein inhibitory effect can be blocked by a few substances
- E.g the protein inhibitory effect of Newcastle disease virus and Hela cells is blocked by puromycin or ppara-fluorophenyl alanine

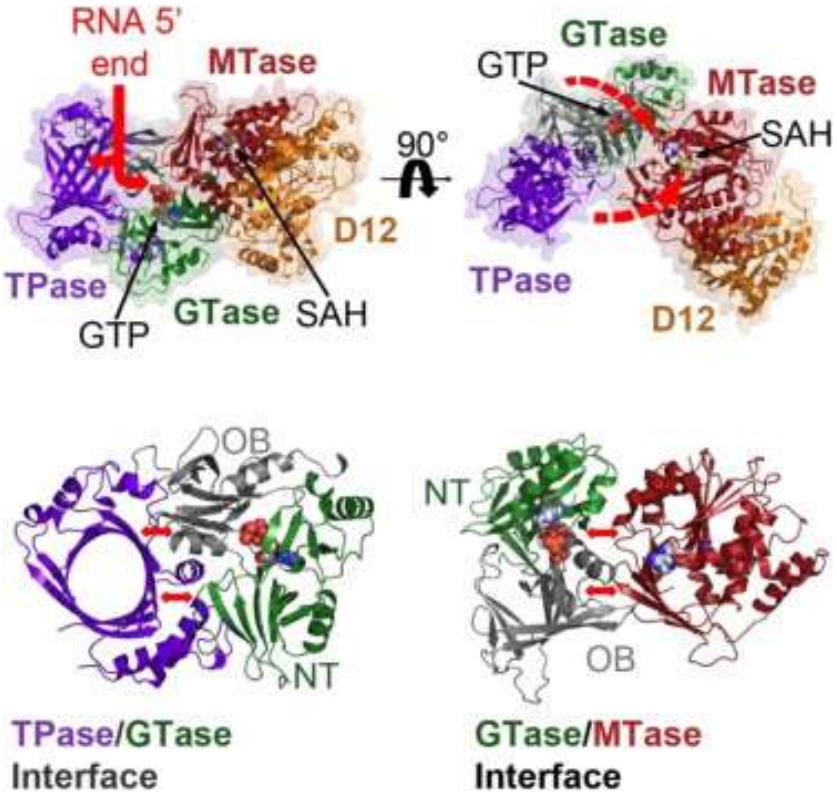
Ppt no. 118

Enzyme Metabolism of Virus Infected Cells

- Enzymes normally speed up different biochemical reactions within host cell
- They help to break down protein, fat or carbohydrate molecules

- Induction of new enzymes is followed by cellular infection by virus
- These enzymes not found in normal uninfected cells
- Or have an additive effect with the already present enzymes

- Some enzymes are induced *de novo* from information encoded by viral genome



- Observed enzyme changes may be secondary from overall disorganization of cell brought by viral development

END

- Particular enzyme activities thought to be regulated by information encoded in the viral genome:

1. Changes in RNA-synthetase
2. Changes in structural protein

Ppt no. 119

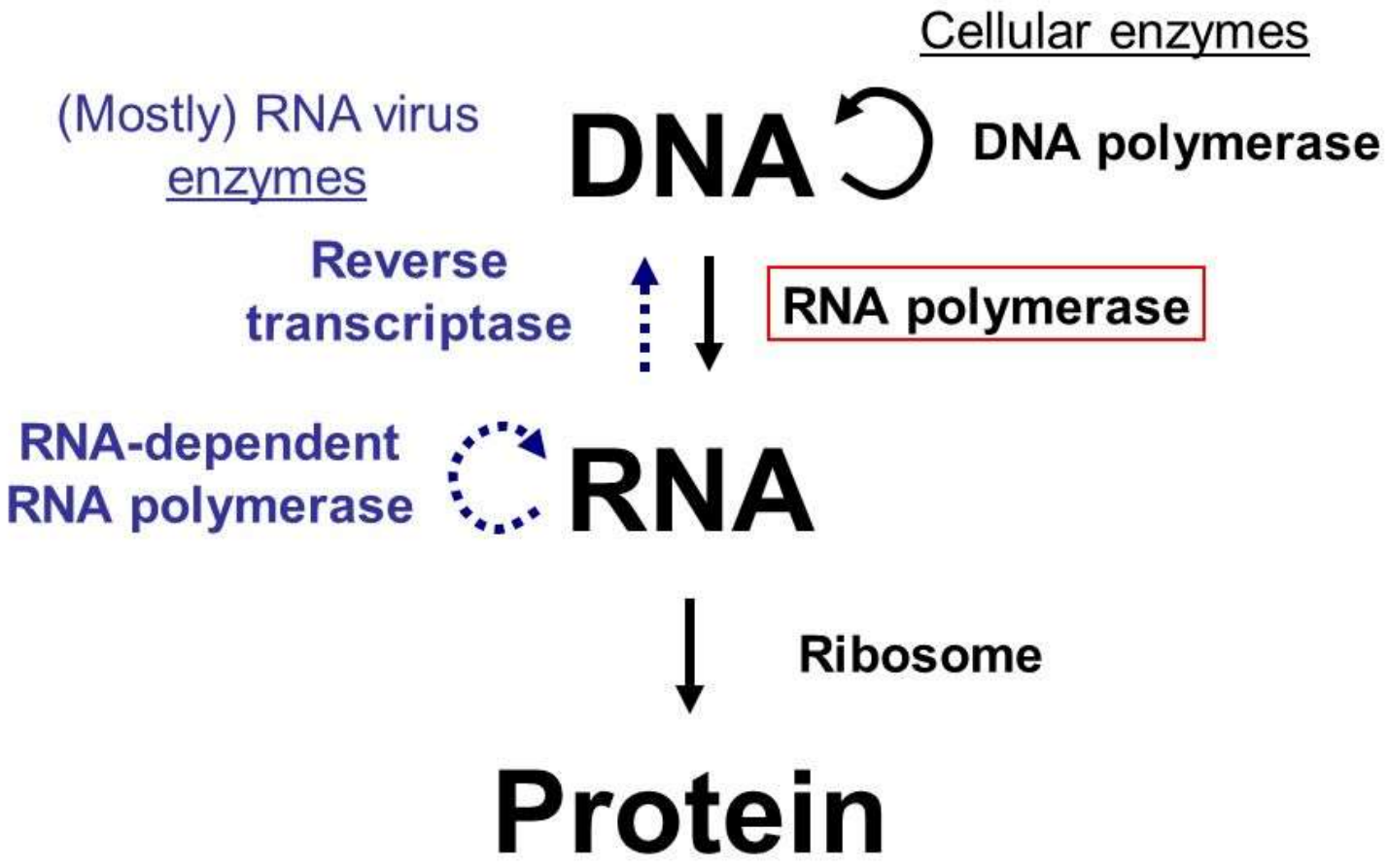
Enzymatic Changes in Virus-infected Cells

RNA-dependent RNA Polymerase

RNA dependent RNA polymerase

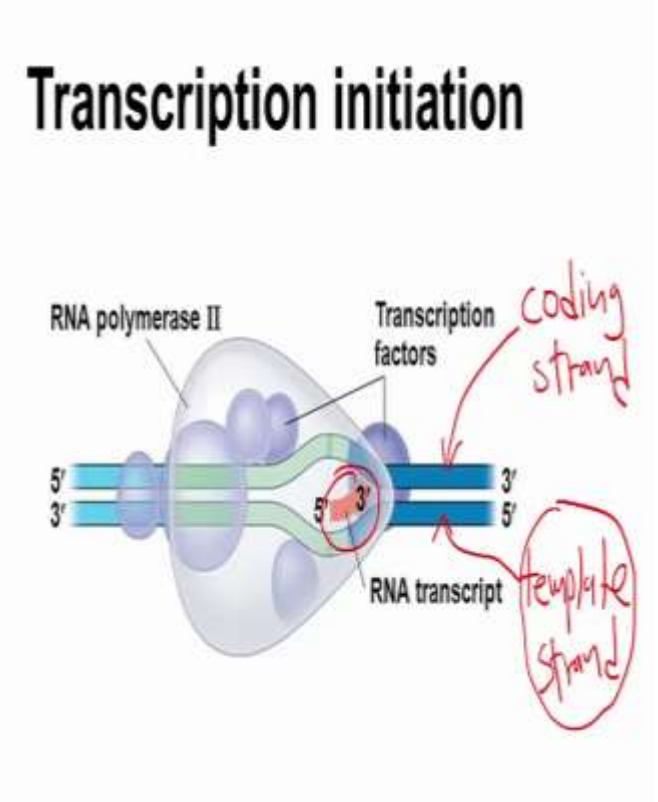
- RNA-dependent RNA polymerase (RdRP), (RDR), or RNA replicase, is an enzyme that catalyzes the replication of RNA from an RNA template
- This is in contrast to a typical DNA-dependent RNA polymerase, which catalyzes the transcription of RNA from a DNA template

Enzymes in the central dogma



NORMAL CELL

- All RNA transcribed from cellular DNA
- Direct transcription
- Instead of formation of mRNA



Cell Infected With Rna Virus

- Which replicate in cytoplasm
- Polymerization of viral RNA from cellular RNA template
- This enzyme not found normally

Ppt no. 120

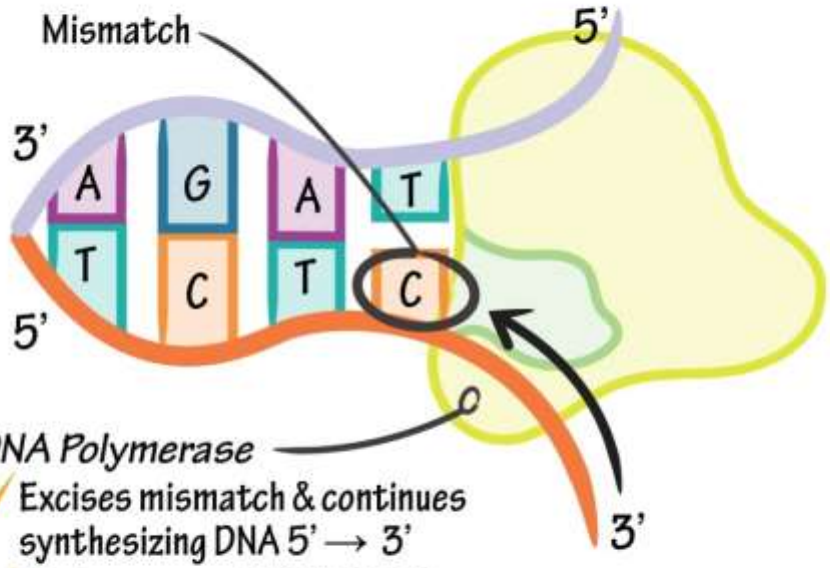
Enzymatic Changes In Virus-Infected Cells

DNA Polymerase

DNA Polymerase

- DNA polymerase is an enzyme that synthesizes DNA molecules from deoxyribonucleotides, the building blocks of DNA
- These enzymes are essential for DNA replication and usually work in pairs to create two identical DNA strands from a single original DNA molecule

DNA Polymerase: Proofreading



DNA Polymerase

- ✓ Excises mismatch & continues synthesizing DNA 5' → 3'
- ✓ *Prokaryotes*: DNA Pol I & III are exonucleases
- ✓ *Eukaryotes*: Polδ & Polε are exonucleases

POXVIRUS

- Poxvirus replicates in cytoplasm
- DNA polymerase does not exist in cytoplasm
- Survival advantage in viral genome is obvious

Ppt no. 121

Enzymatic Changes in Virus-infected Cells

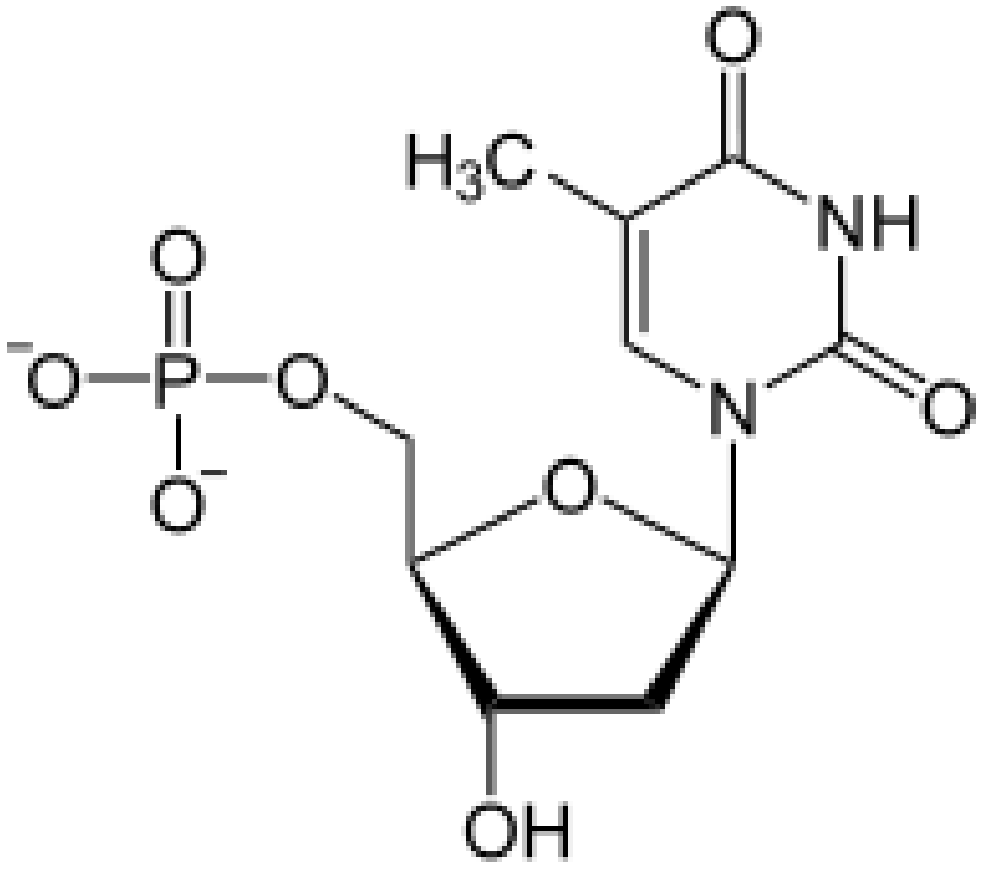
Thymidine Kinase

Thymidine Kinase

- Thymidine kinase is an enzyme, a phosphotransferase (a kinase): 2'-deoxythymidine kinase, ATP-thymidine 5'-phosphotransferase
- It can be found in most living cells
- Certain viruses also have genetic information for expression of viral thymidine kinases
- Thymidine kinase catalyzes the reaction:

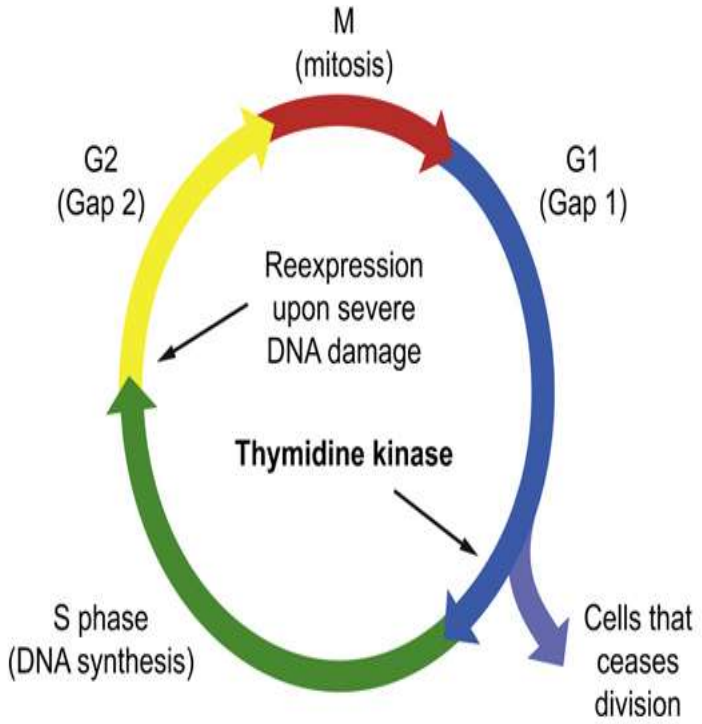
Thd + ATP → TMP + ADP

STRUCTURE



- Infection with herpesvirus and poxvirus
- Causes increase in thymidine kinase

Normal function in cell



- Not known until now
- It may phosphhorylate the free deoxyribonucleoside to a usable DNA precursor
- End result may be free thymidine which may be involved in DNA synthesis

In infected cell

- In vaccinia virus infection
- Enzyme production switched off after ten hours
- Can be inhibited by actinomycin or puromycin
- 40 fold increase in enzyme
- Mechanism yet unknown

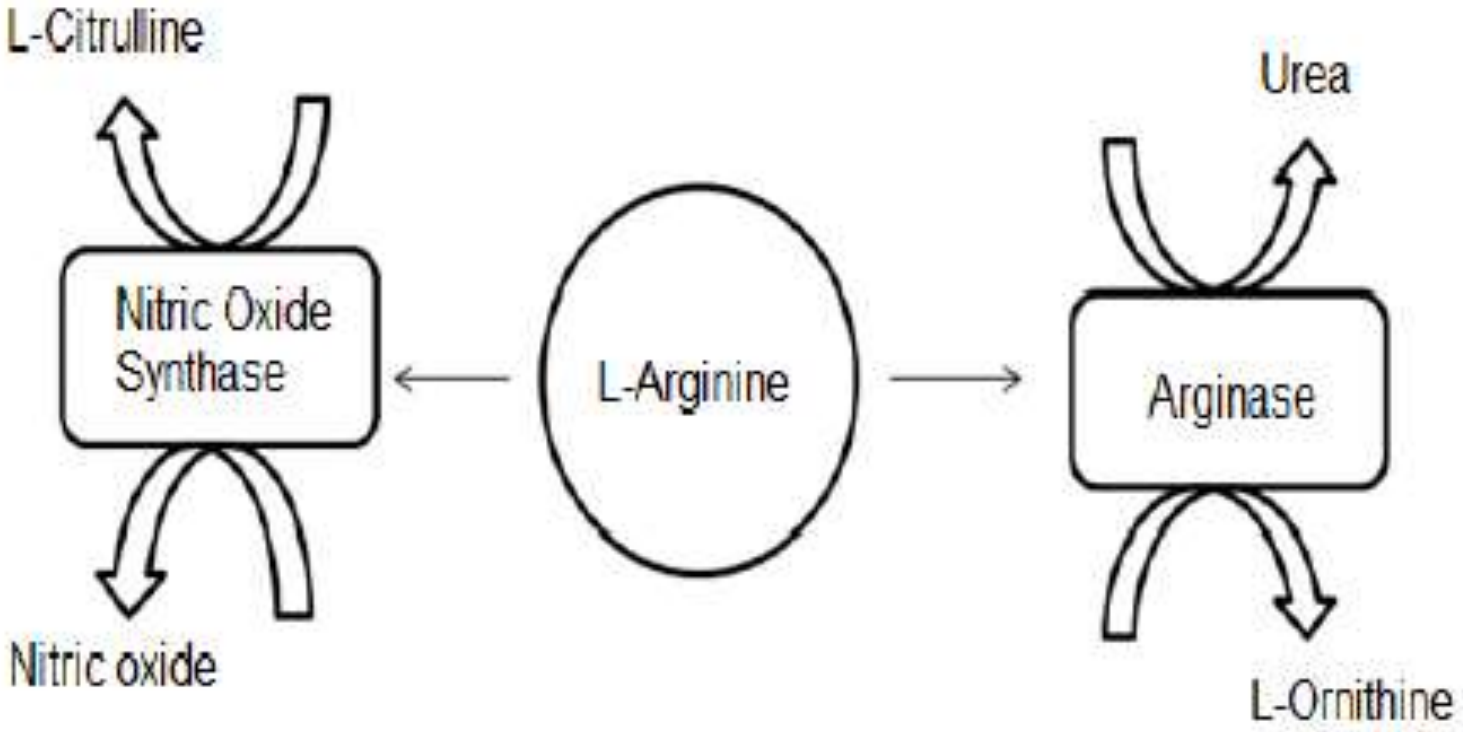
Ppt no. 122

Enzymatic Changes in Virus-Infected Cells

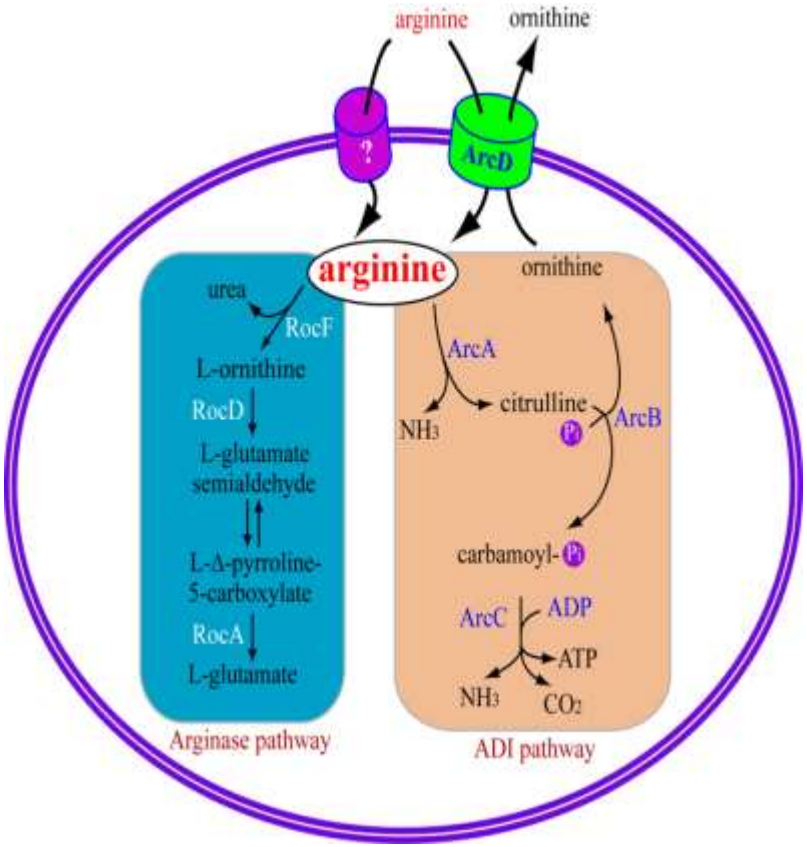
Arginase

Arginase

- Arginase (EC 3.5.3.1, arginine amidinase, canavanase, L-arginase, arginine transamidinase) is a manganese-containing enzyme
- The reaction catalyzed by this enzyme is: arginine + $H_2O \rightarrow$ ornithine + urea
- It is the final enzyme of the urea cycle


L-Citrulline
Nitric Oxide Synthase
Nitric oxide
L-Arginine
Urea
Arginase
L-Ornithine

- Rabbit papilloma virus induces high levels of arginine in squamous epithelium
- Squamous epithelium normally contains arginine



- The induced enzyme differs immunologically and physiochemically from normal enzyme
- Released from normal cellular control mechanism

Speculation: net effect is the depletion of cellular arginine which results in reducing the amount of arginine rich histones synthesized

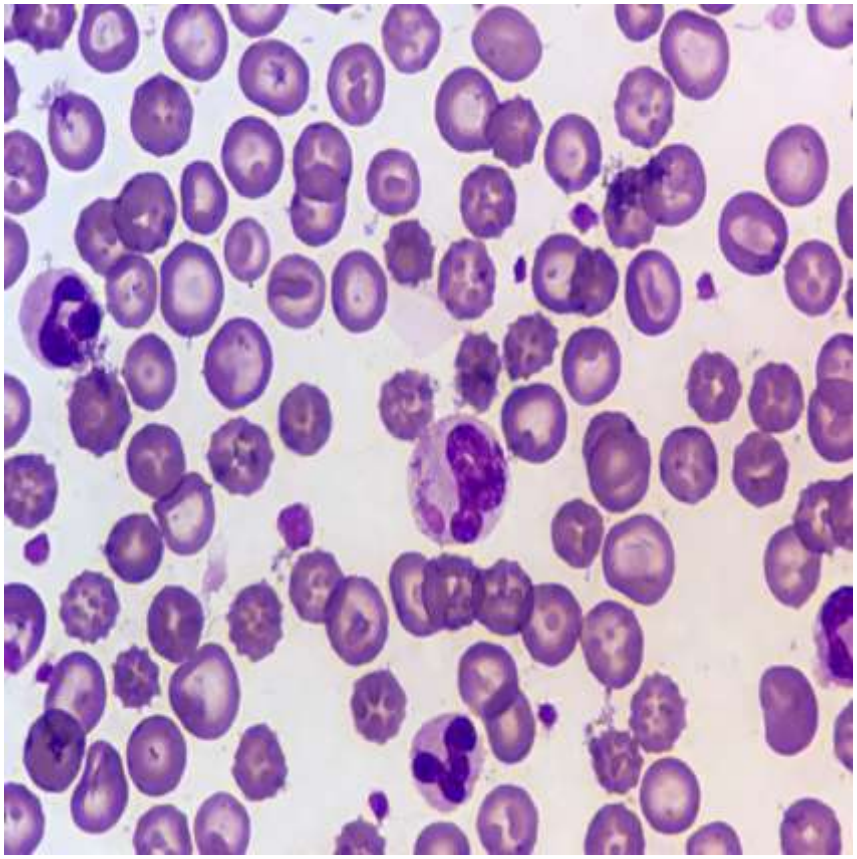
Ppt no. 123

Changes in Structural Proteins

INCLUSIONS

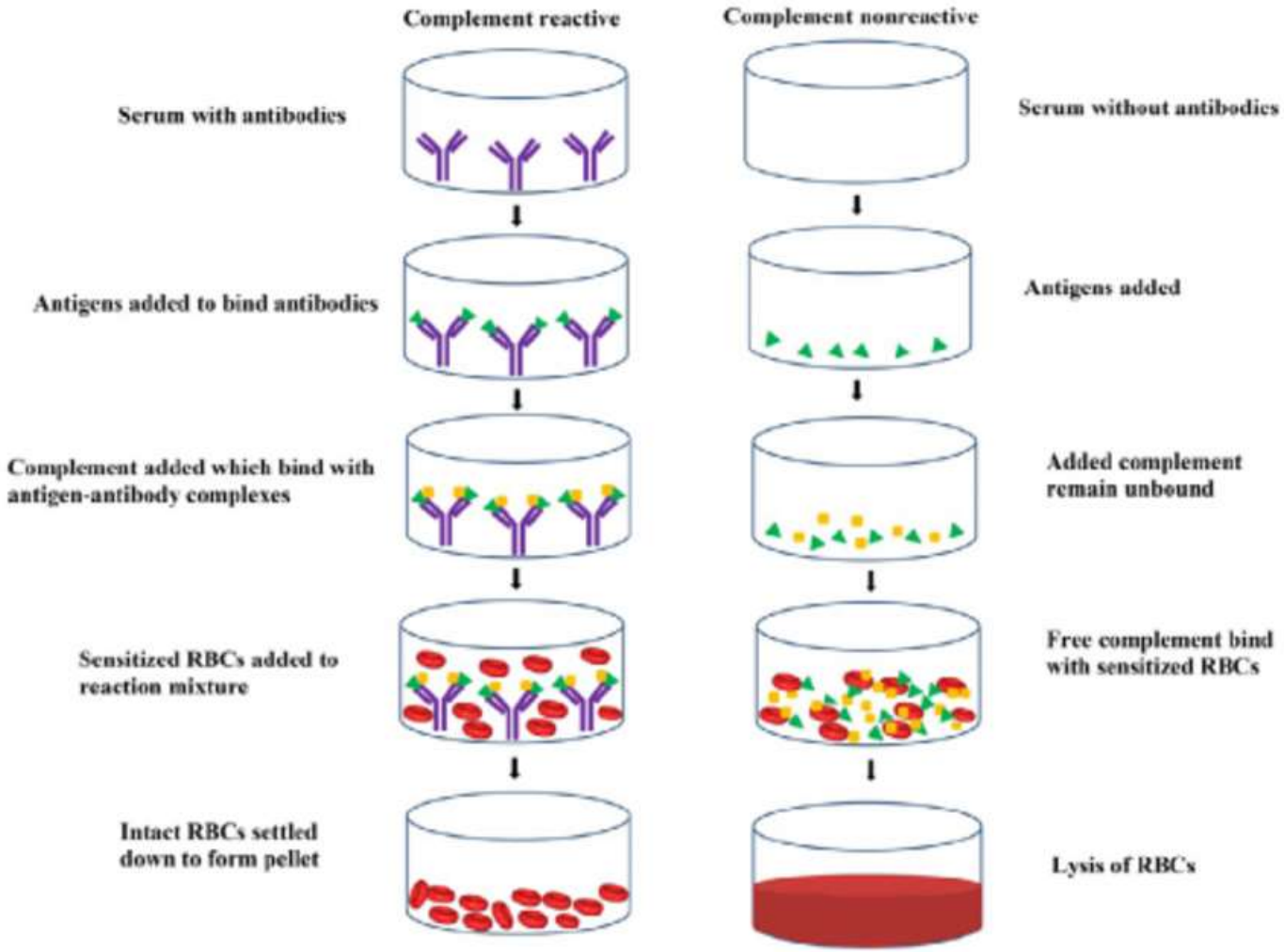
- Examination of virus infected cells by light or electron microscope shows viral multiplication and maturation referred as ‘inclusions’ or ‘factories’

EXAMPLES



- Inclusion may be large, in case of poxvirus infection
- Or small incase of picornavirus infection
- Or maybe limited by a membrane

- New proteins are usually associated with tumors
- Property of oncogenic viruses


Complement reactive
Complement nonreactive
Serum with antibodies
Serum without antibodies
Antigens added to bind antibodies
Antigens added
Complement added which bind with antigen-antibody complexes
Added complement remain unbound
Sensitized RBCs added to reaction mixture
Free complement bind with sensitized RBCs
Intact RBCs settled down to form pellet
Lysis of RBCs

DETECTION

- Complement fixation
- Fluorescent antibody detection
- Graft detection

CONTROL

- Most probably under control of viral genome
- Possibility of cellular response products to virus

Ppt no. 124

Factors Influencing Viral Multiplication

FACTORS

1. Viral concentration
2. Virus-cell systems
3. Elevated temperatures
4. Incomplete viral genomes

Viral Concentration

- One infectious unit equals to 10^3 to 10^5 noninfectious particles
- Low ratio associated with slow uncoating of viral particles
- Exception of vaccinia virus

Virus-cell Systems

- Blocks in viral development after viral genome starts to function
- E.g herpes simplex virus in dog kidney cells after development of cytopathic effects, inhibition of cell division, synthesis of viral

VIRUS-CELL SYSTEMS

- Failure of virus to replicate in dog kidney cells
- Possible that viral DNA acted upon cellular nuclease or that sufficient DNA polymerase not present

Elevated Temperatures

- At 40°C, poliovirus inhibits cellular RNA synthesis
- But viral RNA fails o replicate and no viral progeny produced
- Polymerization fails at this temp.

Incomplete Genomes

- Incomplete viral genomes
- Like influenza virus
- Produce nucleoproteins, hemagglutinin
- But no mature virus

Ppt no. 125

Viral Interference

Principle

- Demonstrated *invitro* and *invivo*
- By infecting cell with a virus followed by a second virus
- Demonstrated when the socnd virus fails to replicate or is inhibited

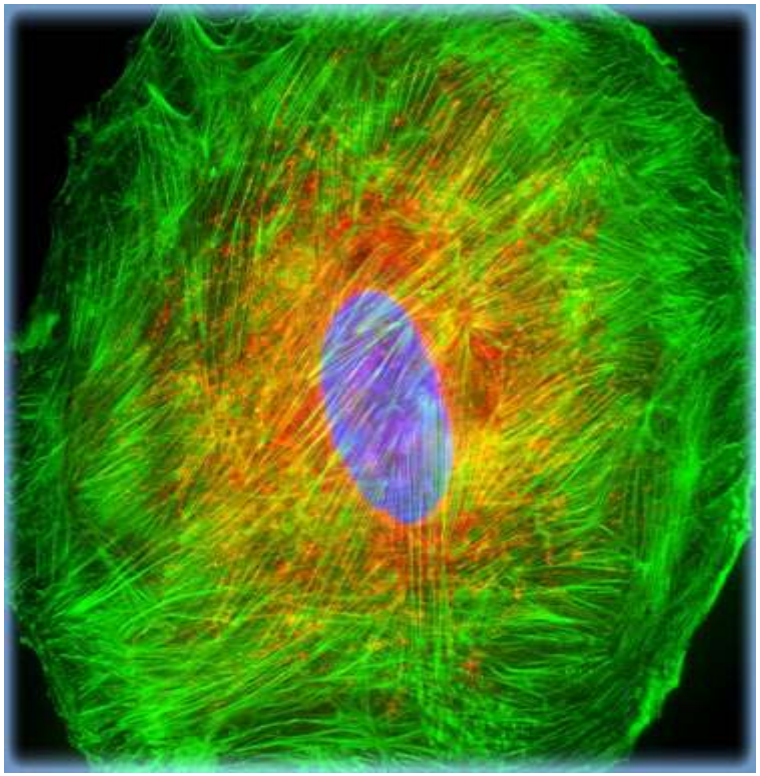
Groups

1. Interference mediated by virus induced cellular protein called interferon
2. Interference that is non interforn mediated

Interferon

- Interferons are 20,000 molecular weight proteins
- Produced by cells in response to infection
- Interferon producing information resides in host cell genome

INTERFERON



- Not virus-specific
- E.g chicken fibroblast cell interferon is inactive in mouse cells but inhibitory for viruses grown in chicken fibroblast cell cultures

ROLE

- Interferons block translation of viral genome in early stages
- Basic mechanism however remains speculative

- Interferon does not bind to naked viral RNA to reduce infectivity
- Suggests inhibitory action takes place at cellular site
- Role of interferons in recovering is being studied

Ppt no. 126

Collection of Samples for Viruses

CONSIDERATIONS

1. What to collect?

✓ Type of specimen

- e.g septicemic infection → blood
- Urinary tract infection → urine
- Respiratory infection → nasal swabs

✓ Quantity of sample

2. When to collect?

✓ Before treatment with antibiotics

✓ Site may vary with time

TRANSPORT MEDIA

- Allows organisms (pathogens and contaminants) to survive

✓UTM (Universal Transport Medium)

✓PBS/Glycerol transport Medium (NaCL, KCl, Na2HPO4 in distilled water)

Specimen	Transport media	Storage condition		Purpose/ Lab investigation
		Transport	Pending test	
Throat swab	VTM	2-8 ^{0}C	-20 ^{0}C	Isolation
NPA/ swab	VTM	2-8 ^{0}C	-20 ^{0}C	Isolation
CSF	No	2-8 ^{0}C	-20 ^{0}C	Isolation, serology
Stool	No	2-8 ^{0}C	-20 ^{0}C	Isolation
Urine	No	2-8 ^{0}C	-20 ^{0}C	Isolation
Serum/ Clotted blood	No	2-8 ^{0}C	-20 ^{0}C 2-8 ^{0}C	Isolation, serology
Whole blood	No	2-8 ^{0}C	2-8 ^{0}C	Isolation, serology

Ppt no. 127

Preservation of Samples of Virus

Types of Samples

- Tissue samples
- Extracted DNA
- Extracted RNA
- cDNA
- PCR products

Tissue Samples

- can be kept at -80°C for years
- Temperature should be maintained
- Thawing repeatedly may contaminate the sample

Extracted DNA

- can be kept at -80^{o}C
- Aliquots of sample
- Properly labeled and sealed

Extracted RNA

- Very sensitive due to presence of RNA everywhere
- Stores at -80^{o}C for a long period of time
- Can also be kept at -20^{o}C for short time periods
- Refreezing not recommended

cDNA

- Comparatively more stable than RNA
- Can be kept at -20^{o}C for several weeks
- Can be kept at -80^{o}C for many years

Specimen	Storage condition
	Pending test
Throat swab	-20 ^{0}C
NPA/ swab	-20 ^{0}C
CSF	-20 ^{0}C
Stool	-20 ^{0}C
Urine	-20 ^{0}C
Serum/	-20 ^{0}C
Clotted blood	2-8 ^{0}C
Whole blood	2-8 ^{0}C

Transport of Samples for Viruses

Ppt no. 128

Purpose Of Sop

- Safety from bio-hazardous material
- Results depend on sample collection and handling

Biosafety consideration

- Zoonotic or transmissible
- Latex gloves, Lab coats, Masks
- New syringes and containers
- Sealed container
- Wash hands after removing gloves
- In event of injury, wash area with soap and water
- Always have a first aid safety kit

Handling and transport of blood sample

- ✓ Collect in EDTA or Heparin containing containers
- ✓ Store at 4 ^{o}C
- ✓ Transport upright with cushion (to prevent hemolysis)
- ✓ Never freeze blood

Nasopharyngeal swabs transport

- ✓ All respiratory specimens should be transported in proper medium
- ✓ Transport as quickly as possible to the laboratory to reduce overgrowth by oral flora
- ✓ For transit periods up to 24 hours
  - – ambient temperature for bacteria
  - – 4-8°C for viruses

POSTMORTEM SAMPLES

- Collection
  - ✓Biopsy relevant tissues
    - in formalin for histopathology
    - in transport medium for microbiological testing
    - in sterile saline for isolation of viral pathogens
- Handling and Transport
    - in transport media within 24h at ambient temperature
    - in sterile saline at 4-8^{o}C within 48h

FILTERATION METHOD

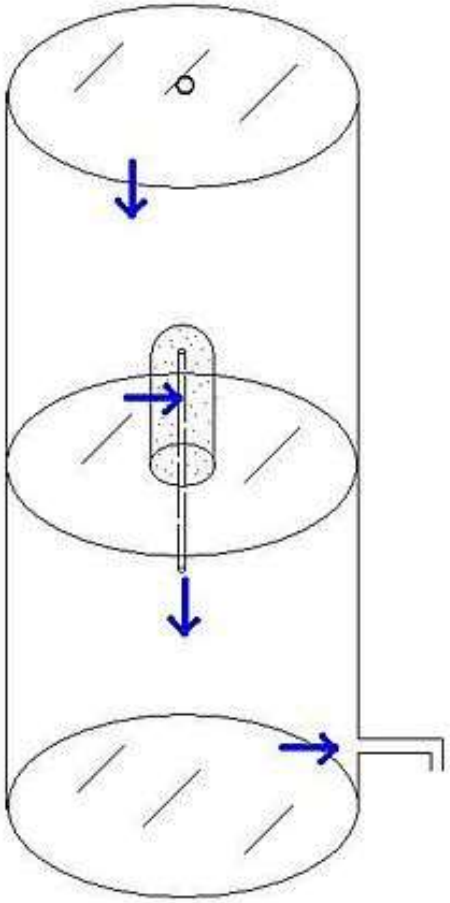
Berkefeld, Mandler and Pasteurchamberland filter

Ppt no. 129

The Berke Filter

- Manufactured of diatomaceous Earth
- Graded into three porosities
- The V (viel) grade is coarse
- Used for preliminary filtration to remove bacteria or large particles from a solution
- The N (normal) grade is most useful since it retains most bacteria, while all viruses will pass through
- The W (Weing) grade has very small pores that will retain all bacteria and large viruses

The Berke Filter



Schematic diagram of
a Berkefeld filter

The Mandler Filter

- Made of diatomaceous Earth
- Similar to berkefeld filters
- Three grades
  - Preliminary
  - Regular
  - Fine

The Mandler Filter

Pasteur chamberland Filter

- Made of porcelain
- Molded in candle form
- Graded in porosities
- L1 to L13
- L13 is the finest grade

Pasteur Chamberland Filter

FILTERATION METHOD

Selas, Seitz and Boerner filter

Ppt no. 130

The Selas Filter

- Made up of porcelain
- Seven grades of porosity
- Graded as XF, XFF, No. 10, and No. 01 (coarse grade)
- Graded as No. 015, No. 02, No. 03 which retain bacteria

The Selas Filter

Seitz Filter

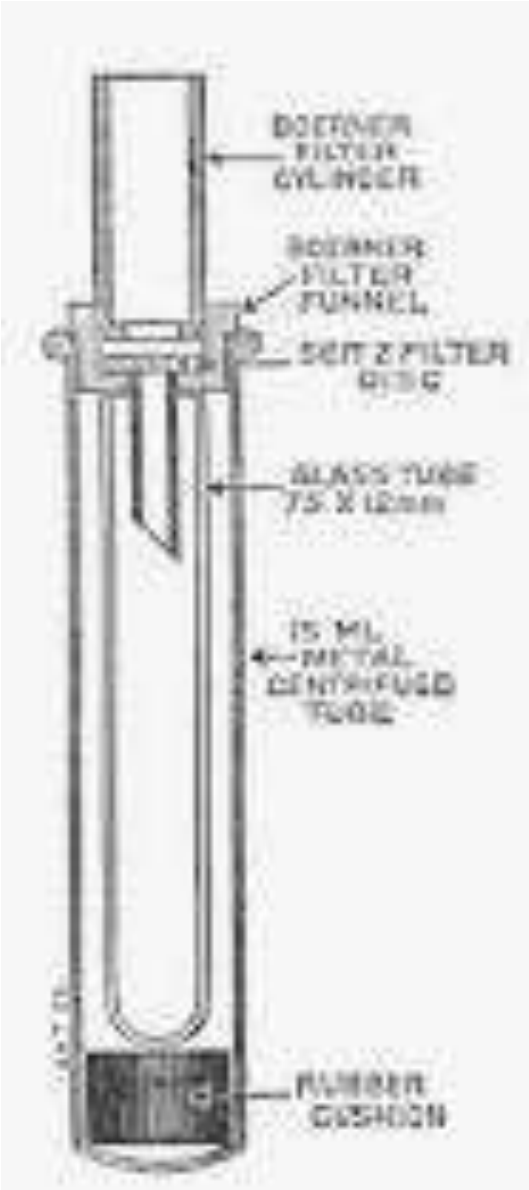
- Asbestos pads
- Manufactured in two grades
- K and EK (coarse)
- Pads in metal holder
- Discarded after one use

The Seitz Filter

Boerner Filter

- Same as Seitz
- Except they are small and fitted in small metal receptacle
- Adapts into small test tube
- Centrifuged to force liquid through them

Boerner Filter



FILTERATION METHOD

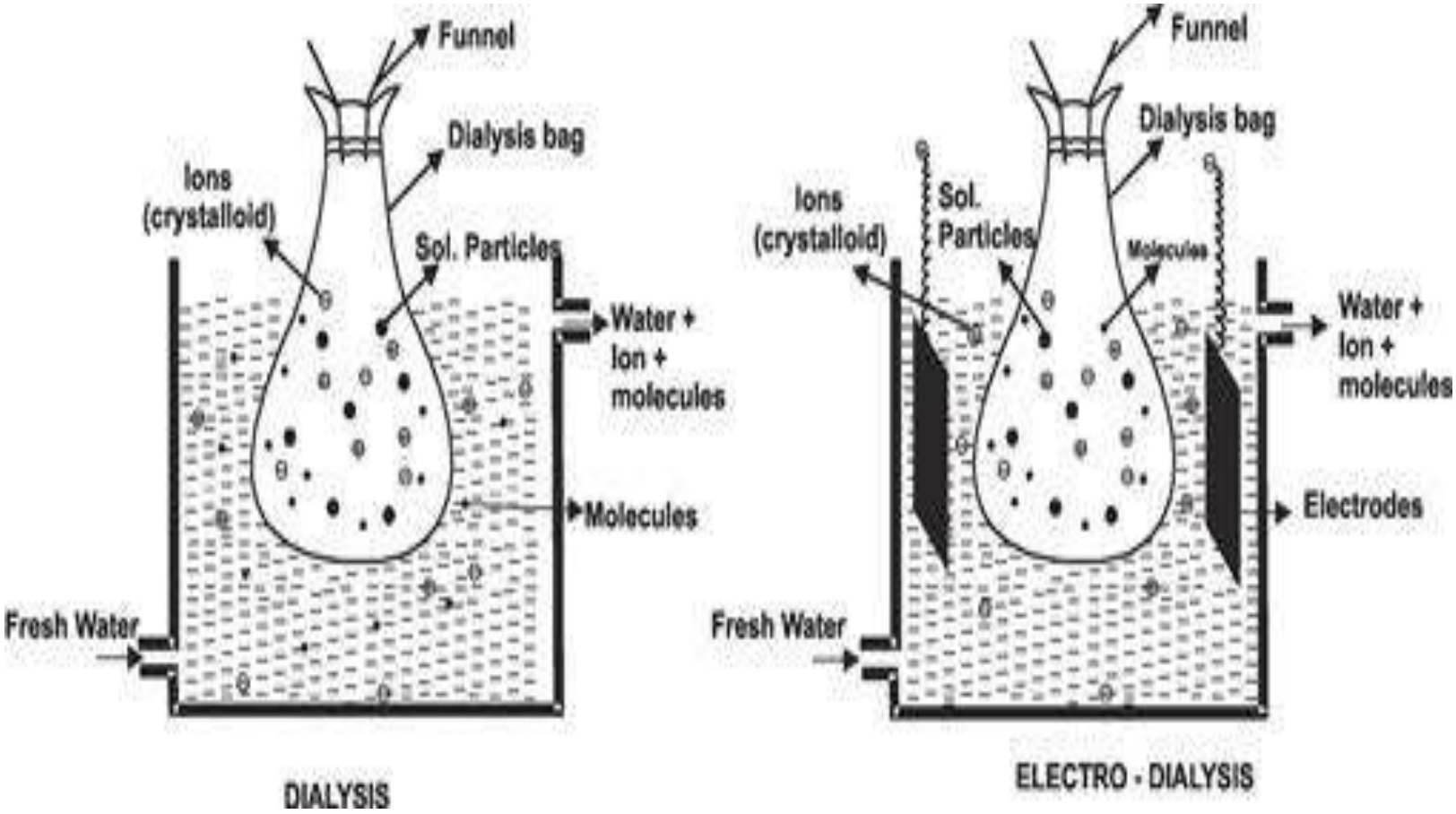
Collodian filter and
cellulose filter

Ppt no. 131

Collodion Filter

- Made up of nitro-cellulose
- Pore diameter varies
- Variation by addition of various amounts of alcohol, ether, acetone and acetic acid
- Pore diameter: 3um- 10um
- Most useful in determining viral size

Colloidan Filter



Cellulose Filter

- Pure cellulose or cellulose derivatives
- Coarse with pore diameter 3um to 0.75 um
- Medium with pore diameter: 0.75um to 0.7um
- Dense with 0.2um
- Very dense below 0.2um

The Cellulose Filter

Filteration Technique

Ppt no. 132

Filtration

A technique used either to remove solid impurities from an organic solution or to isolate an organic solid

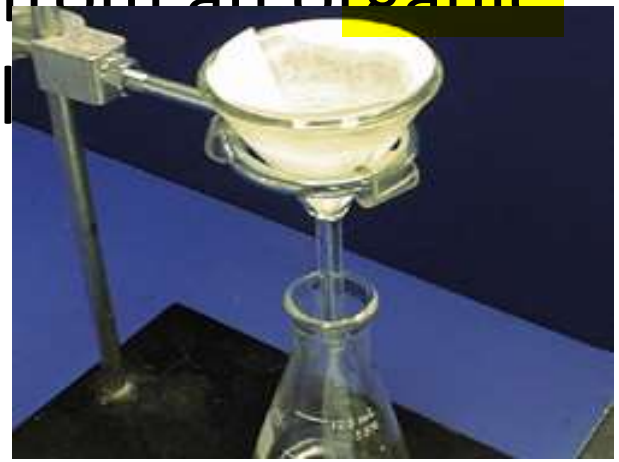
The two types of filtration commonly used

Gravity filtration

Vacuum or suction filtration

Gravity Filtration

- Gravity filtration is the method of choice to remove solid impurities from an organic



Vacuum Filtration

Vacuum filtration is used primarily to collect a desired solid, for instance, the collection of crystals in a

- https://www.youtube.com/watch?v=r4GYLuqJV7Y

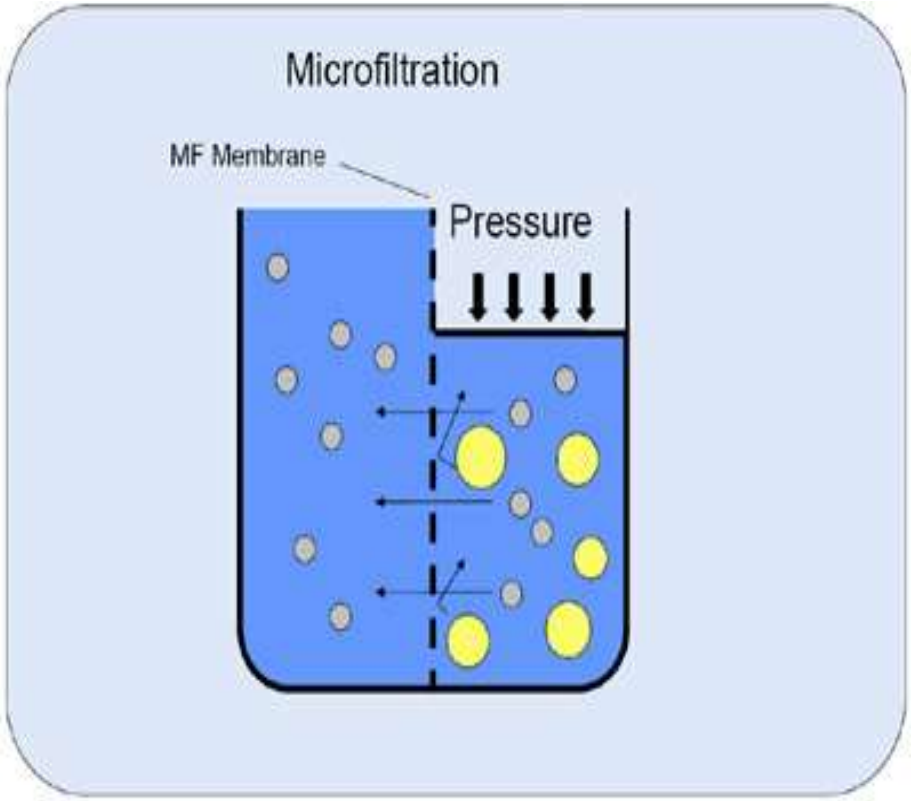
Determination of Viral Size by Filteration

Ppt no. 133

Membrane Technology

- Principle is physical separation
- The extent of removal depends on the size of the particles
- Microfilteration and ultracentrifugation

MICROFILTERATION



- Pore size 0.1 to 1um
- It removes all bacteria
- Only a part of viral contamination is removed

APPLICATIONS

Cold sterilization of beverages and pharmaceuticals

Clearing of fruit juice, wines and beer

Separation of bacteria from water

Separation of oil/ water emulsions

Reverse Osmosis

Solid-liquid separation for pharmacies or food industries

ULTRACENTRIFUGATION

Pore size: 0.001- 0.1 um from liquids

Examples

The dairy industry (milk, cheese)

The food industry (proteins)

The metal industry (oil/ water emulsions separation, paint treatment)

The textile industry

Centrifugation Techniques

Ppt no. 134

Centrifugation

A technique which involves the application of centrifugal force to separate particles from a solution according to their size, shape, density, viscosity of the medium and rotor speed

Principle

More-dense components of the mixture migrate away from the axis of the centrifuge (move to the outside), while less-dense components of the mixture migrate towards the axis,

i. e., move to the center.

Microcentrifuges

High-speed centrifuges

Ultrcentrifugations

Density

Gradient Centrifugation

https://www.youtube.com/watch?v=KEXWd3_fM94

Determination of Essential Lipids

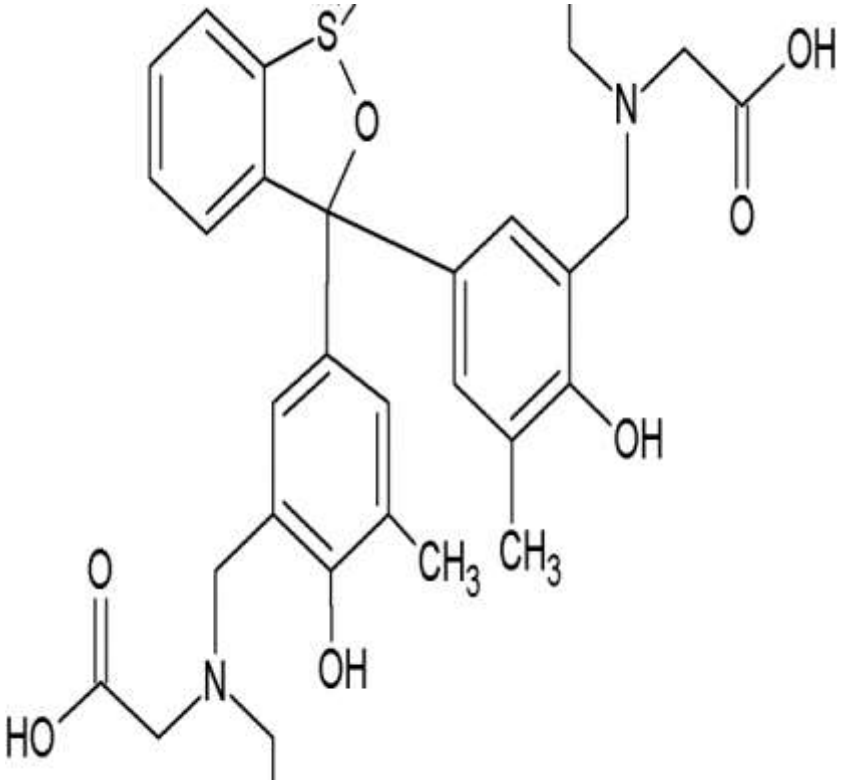
Ppt no. 135

Why Essential Lipids

- Used for identification and classification of animal viruses
- On the basis of presence or absence of essential lipids

Principle

Presence of essential lipids determined by mixing virus preparations with either ether or chloroform



Procedure

- Sufficient ethyl ether is added to virus preparation to make a 20% solution.
- Mixture is incubated at 18-24hrs at 4 ^{o}C

- The resultant is poured into a petri dish.
- Ether is allowed to evaporate.

- To expose a virus to chloroform, 0.05ml chloroform is added to 1ml of virus preparation.
- Mix for ten minutes at 4ºC.
- Chloroform is removed by centrifugation.

- Essential lipids can be determined by various methods used in labs.

Determination of Nucleic Acid Types

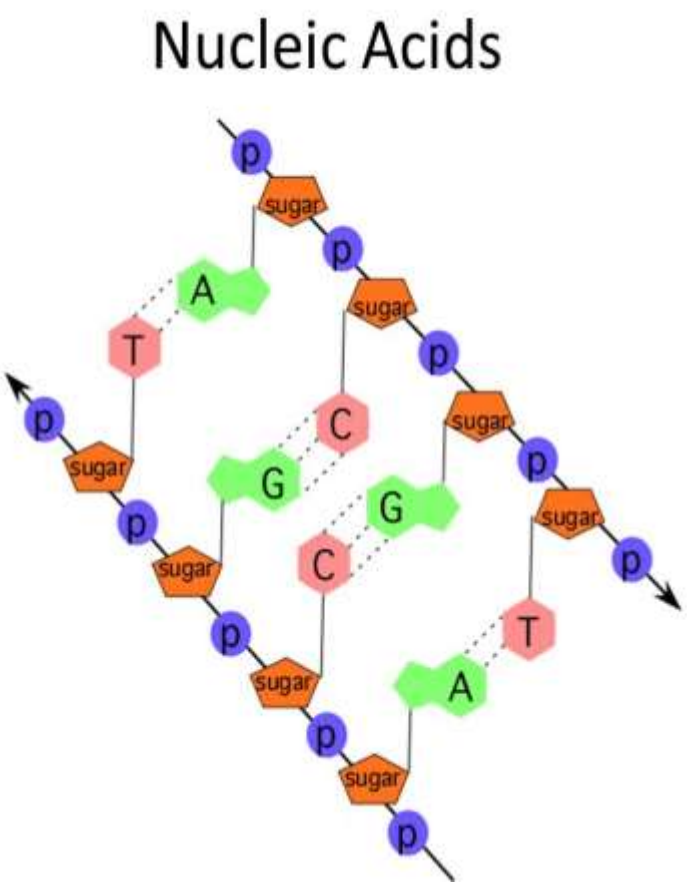
Ppt no. 136

Nucleic acid types

- Double stranded DNA.
- Positive Sense single stranded RNA
- Negative sense single stranded RNA
- Double stranded RNA

Principle

Use of DNA inhibitors determines type of Nucleic Acid a virus possesses



DNA inhibitors

- e.g 5-floro-2’

  -deoxyuridine (FUDR)

- 5-bromo-2’

  -deoxyuridine (BUDR)

- 5-iodo-2’

  -deoxyuridine (IUDR)

Action of DNA inhibitors

- Specific inhibition of viral DNA synthesis when added to cell culture medium

Interpretation

- If growth of virus is inhibited, it is presumed to contain DNA
- If not, it possesses RNA

END

Ppt no. 137 not available

VIRAL DIAGNOSTIC TESTS

Enzyme Linked ImmunoSorbent Assay (ELISA)

Ppt no. 138

1. Direct Examination of Specimen

- Electron Microscopy morphology / immune electron microscopy
- Light microscopy histological appearance - e.g. inclusion bodies
- Antigen detection immunofluorescence, ELISA etc.
- Molecular techniques for the direct detection of viral genomes

2. Indirect Examination

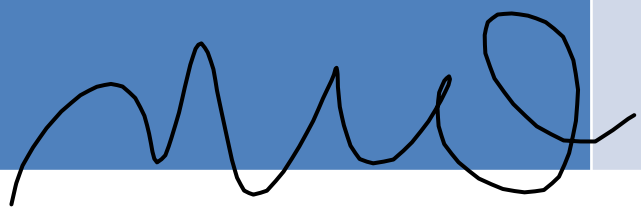
- Cell Culture - cytopathic effect, haemadsorption, confirmation by neutralization, interference, immunofluorescence etc.
- Eggs pocks on CAM - haemagglutination, inclusion bodies
- Animals disease or death confirmation by neutralization

3. Serology

- Detection of rising titres of antibody between acute and convalescent stages of infection, or the detection of IgM in primary infection.

Other Techniques

Classical Techniques	Newer Techniques
1. Complement fixation tests (CFT)	1. Radioimmunoassay (RIA)
2. Haemagglutination inhibition tests	2. Enzyme linked immunosorbent assay (EIA)
3. Immunofluorescence techniques (IF)	3. Particle agglutination
4. Neutralization tests	4. Western Blot (WB)
5. Single Radial Haemolysis	5. Recombinant immunoblot assay (RIBA), line immunoassay (Liatek) etc.



Enzyme Linked ImmunoSorbent Assay (ELISA)

ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique designed for detecting and quantifying substances such as peptides, proteins, antibodies and hormones.

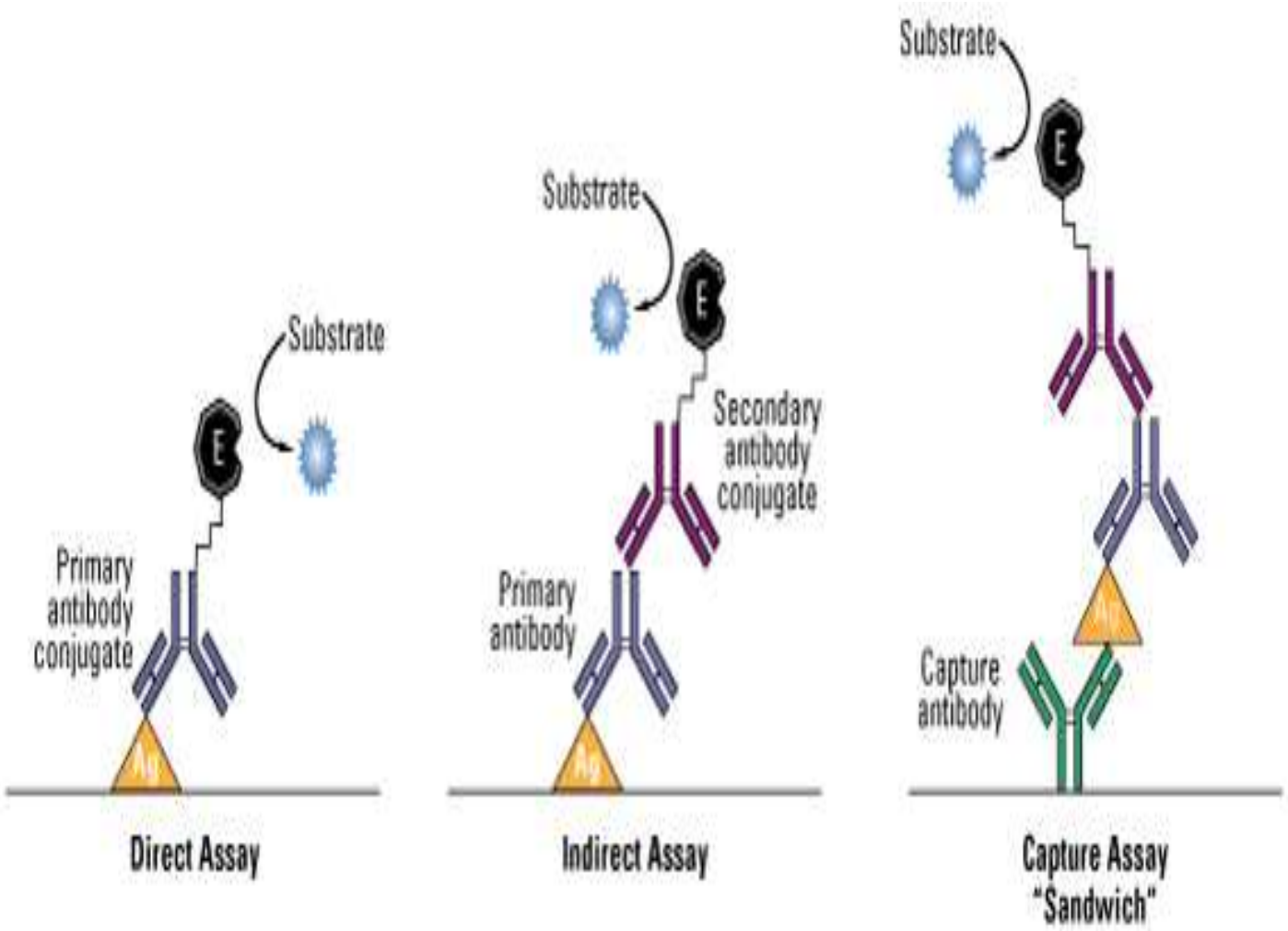
- In an ELISA, an antigen must be immobilized on a solid surface and then complexed with an antibody that is linked to an enzyme.
- Detection is accomplished by assessing the conjugated enzyme activity via incubation with a substrate to produce a measureable product.
- The most crucial element of the detection strategy is a highly specific antibody-antigen interaction.

The key step, immobilization of the antigen of interest, can be accomplished by direct adsorption to the assay plate or indirectly via a capture antibody that has been attached to the plate.

The antigen is then detected either directly (labeled primary antibody) or indirectly (labeled secondary antibody).

The most powerful ELISA assay format is the sandwich assay.

This type of capture assay is called a “sandwich” assay because the analyte to be measured is bound between two primary antibodies – the capture antibody and the detection antibody. The sandwich format is used because it is sensitive


Substrate
E
Primary antibody conjugate
Ag
Direct Assay
Substrate
E
Secondary antibody conjugate
Primary antibody
Ag
Indirect Assay
Substrate
E
Capture antibody
Ag
Capture Assay "Sandwich"

ELISA kits contain pre-coated antibody-plates, detection antibodies, buffers, diluents, standards, and substrates

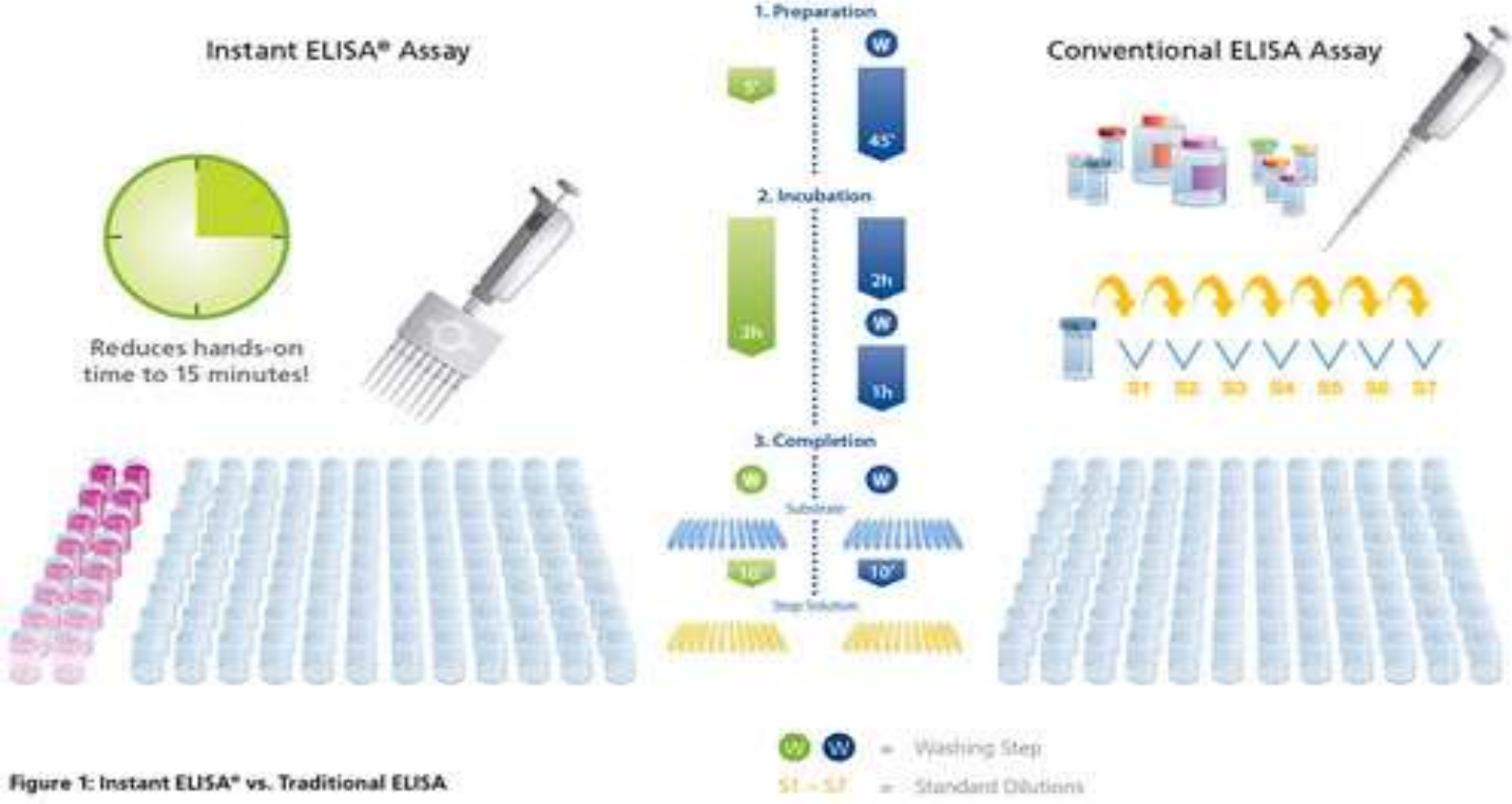


Figure 1: Instant ELISA® vs. Traditional ELISA

VIRAL DIAGNOSTIC PROCEDURES

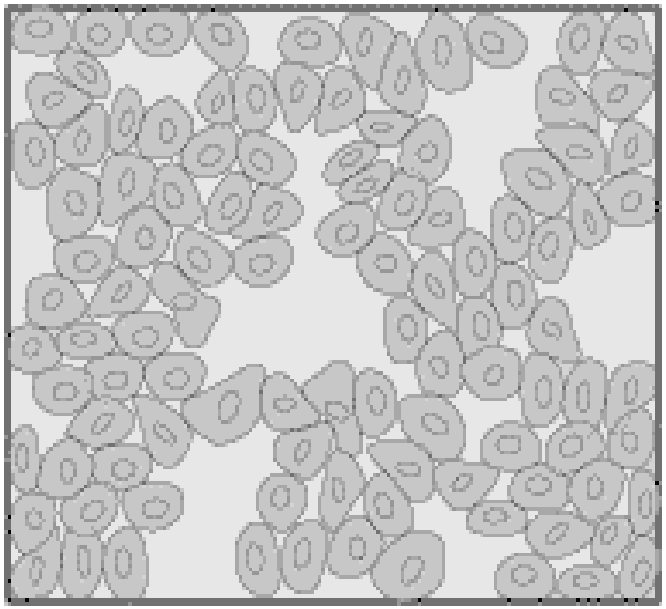
Histological Examination of Virus infected cells

Ppt no. 139

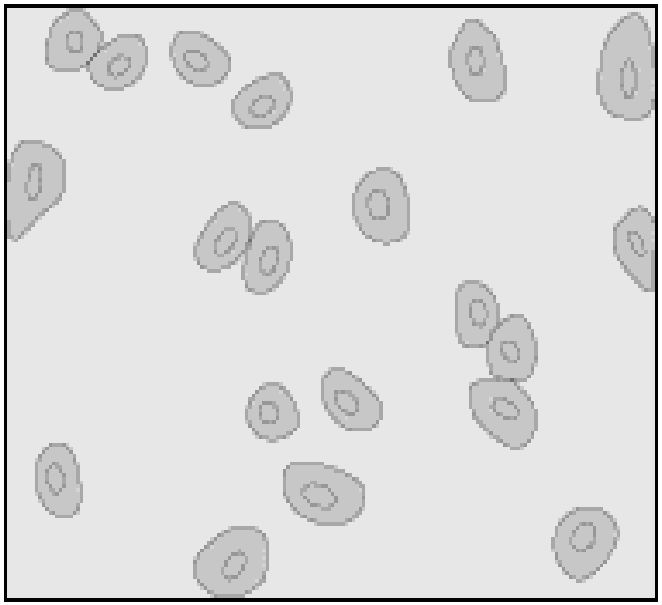
CULTURE

Grown in :

1. Primary cells
2. Semi-continuous cells
3. Continuous Cells



Before Infection

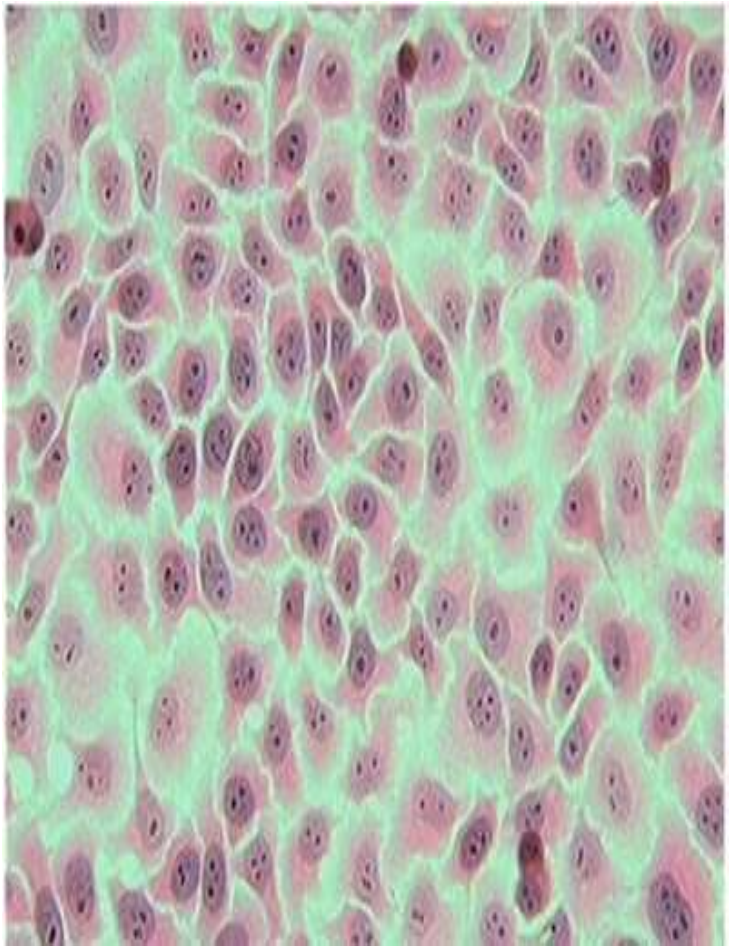


After Infection

Identification of growing viruses

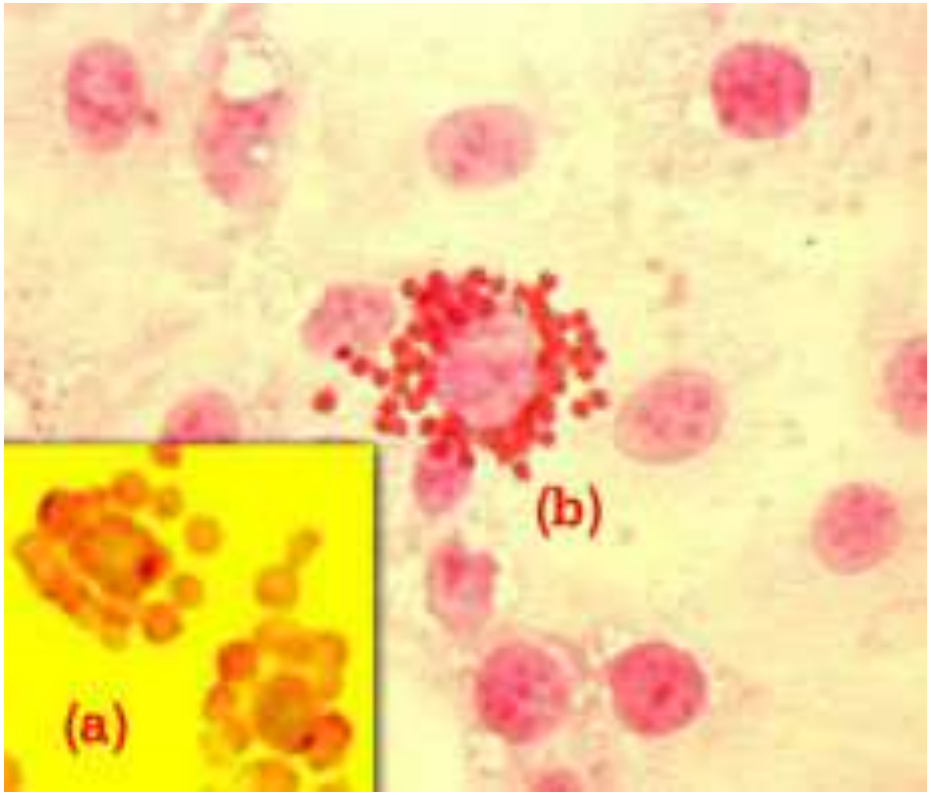
- Cytopathic effects (CPE)
- Haemadsorption

Cytopathic Effects

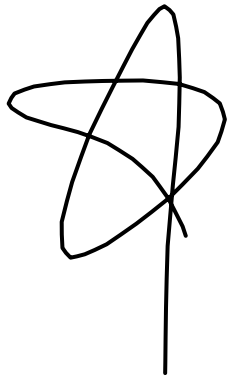


- Specific or non-specific signs
- Herpes Simplex Virus produces specific CPE
- Enterovirus don't produce CPE

Haemadsorption



- Cells acquire ability to stick to mammalian RBCs
- Mainly used for influenza and parainfluenza



Confirmation of virus

- Neutralization
- Haemadsorption-inhibition
- Immunofluoresence
- Molecular tests
  - PCR
  - ELISA

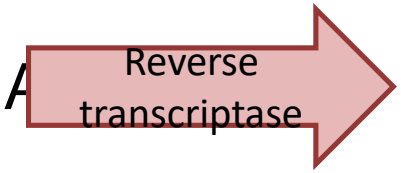
END

VIRAL DIAGNOSTIC PROCEDURES

Polymerase Chain Reaction

Ppt no. 140

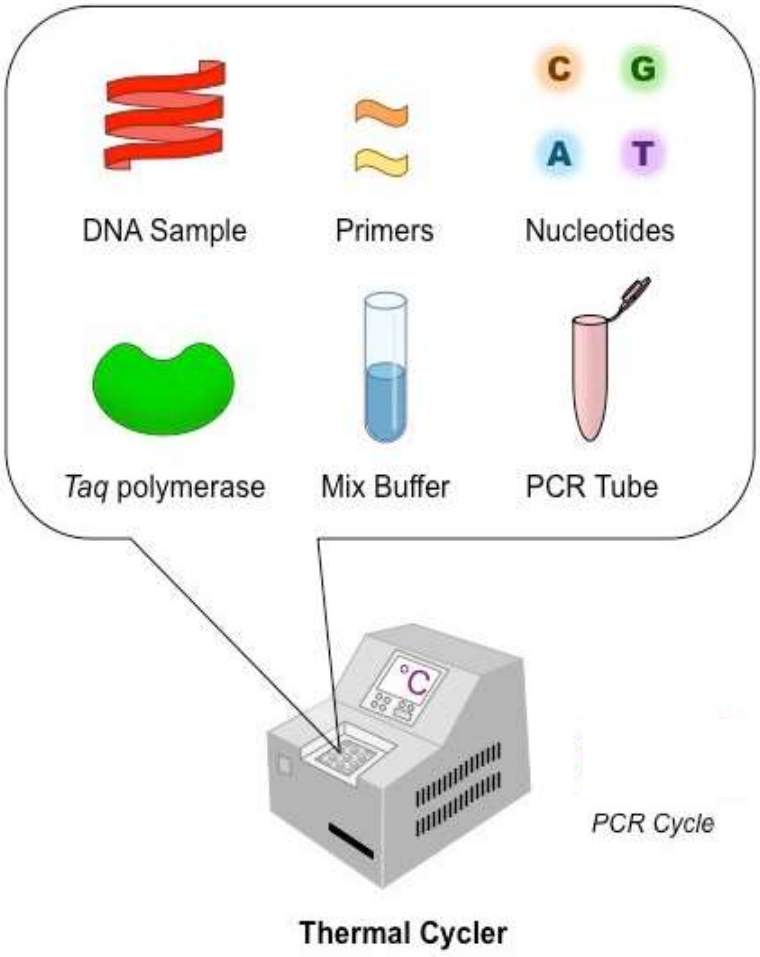
PCR

- Polymerase chain reaction is a laboratory technique used to amplify segments of DNA or RNA
- RNA 

cDNA
- Cycles: 25-30

ESSENTIALS

1. DNA sample
2. Primers (Reverse and forward)
3. Nucleotides (dNTP's)
4. Buffer
5. MgCl2
6. Heat stable Taq Polymerase
7. Distilled water
8. PCR tubes and therma cycler

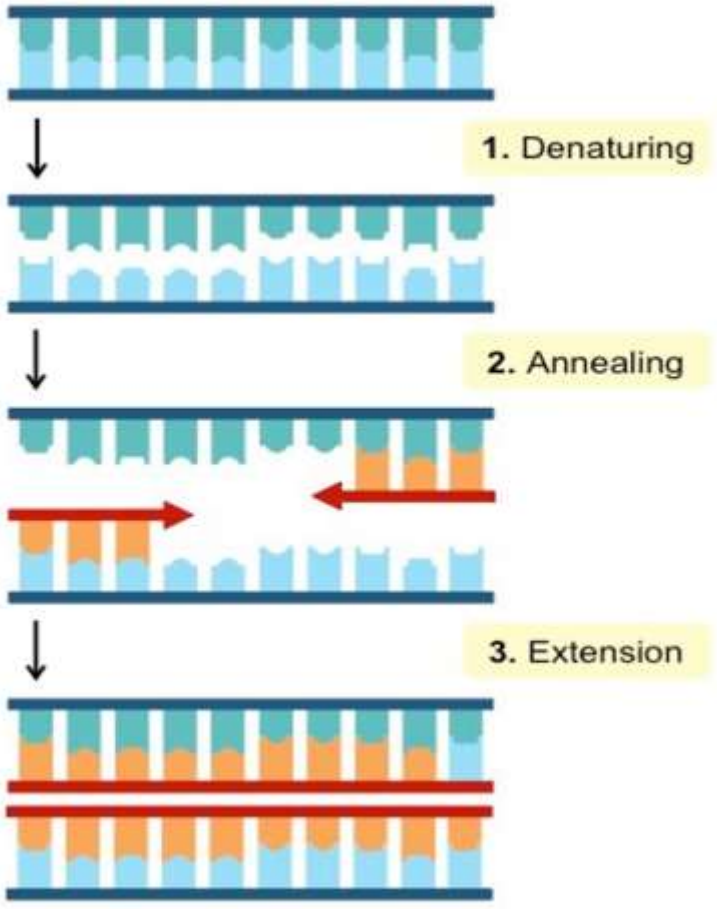


PROCEDURE

1. Step 1:
- Pre-denaturation

2. Step 2:
- Denaturation
- Annealing
- Extension

3. Step 3:
- Final polymerization

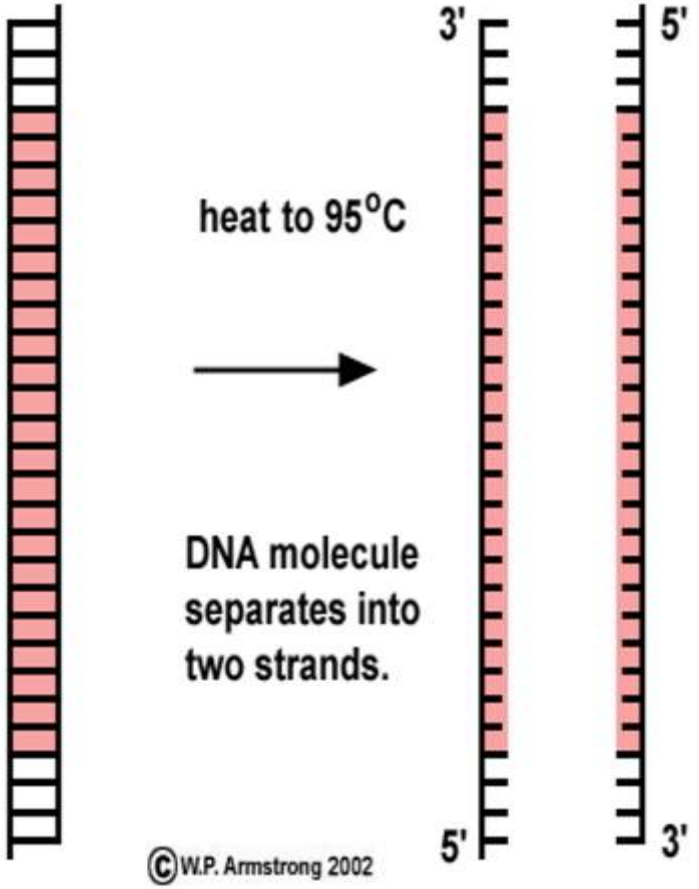


DENATURATION

Predenaturation: DNA boiled at 95°C for 5 minutes

Denaturation: 95°C for 15-30 seconds

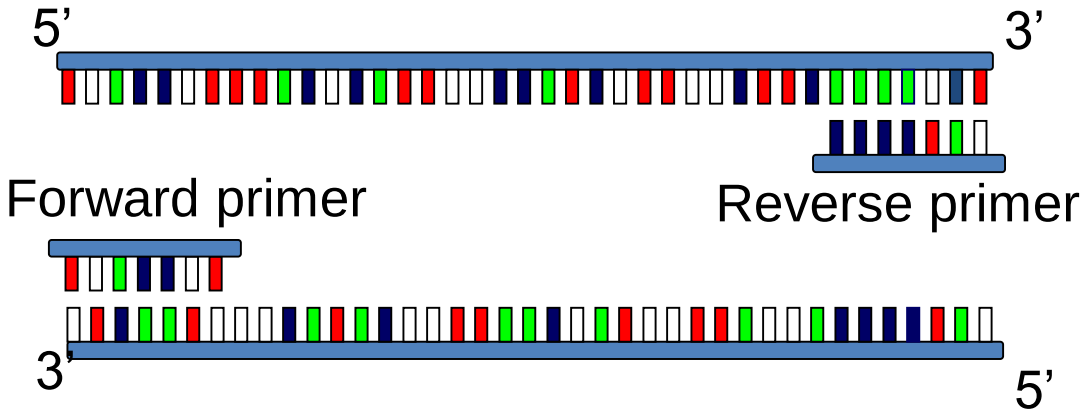
Hydrogen bonds break & the complementary strands of the double-stranded DNA separate



ANNEALING

Both primers (i.e. Reverse and forward) bind to their complementary sequences.

Temperature: 40-60°C



DENATURATION

- Temperature: 72°C
- MgCl2 facilitates Taq in adding up the nucleotides to the strand.
- Taq polymerase binds to the annealed primers and extends DNA at the 3’ end of the chain.
- Final polymerization: Fills gaps

Types of PCR

1. Real-Time PCR
2. Nested PCR
3. Multiplex PCR
4. Arbitrary Primed PCR
5. Hot start PCR
6. Inverse PCR

END

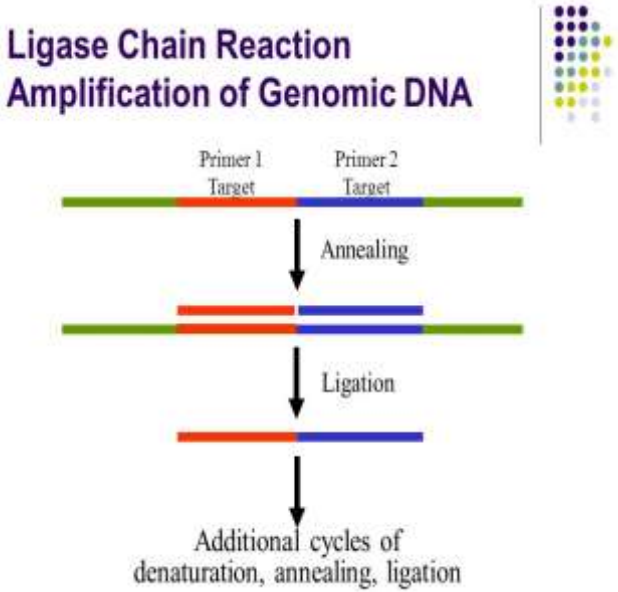
VIRAL DIAGNOSTIC TESTS

Ligase Chain Reaction

Ppt no. 141

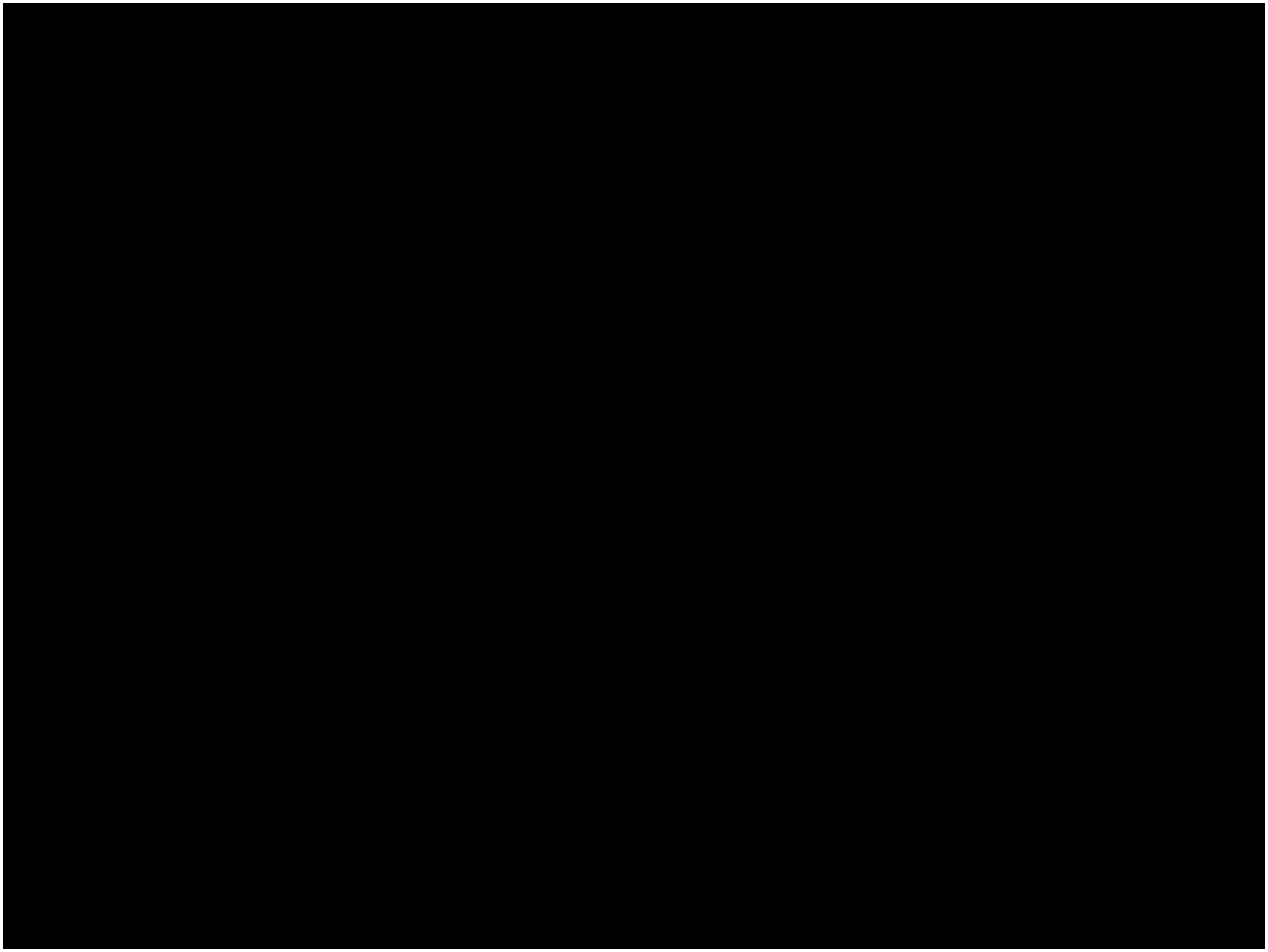
Ligase Chain Reaction

A method of DNA amplification.



- Differs from PCR
- Greater specificity
- Requires two different enzymes
  - ❖ Ligase
  - ❖ Taq polymerase enzyme

- https://www.youtube.com/watch?v=dvW-kznCzy0



VIRAL DIAGNOSTIC PROCEDURES

Nucleic acid sequence based amplification

Ppt no. 142

1. Direct Examination of Specimen

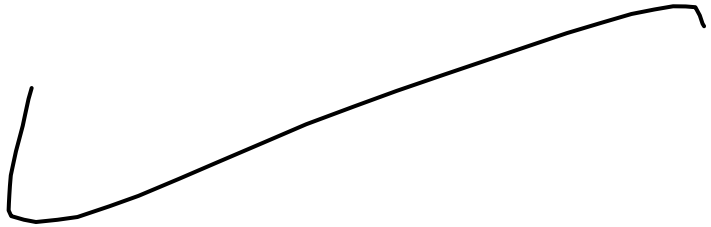
- Electron Microscopy morphology / immune electron microscopy
- Light microscopy histological appearance - e.g. inclusion bodies
- Antigen detection immunofluorescence, ELISA etc.
- Molecular techniques for the direct detection of viral genomes

2. Indirect Examination

- Cell Culture - cytopathic effect, haemadsorption, confirmation by neutralization, interference, immunofluorescence etc.
- Eggs pocks on CAM - haemagglutination, inclusion bodies
- Animals disease or death confirmation by neutralization

3. Serology

Detection of rising titres of antibody between acute and convalescent stages of infection, or the detection of IgM in primary infection.



Classical Techniques	Newer Techniques
1. Complement fixation tests (CFT)	1. Radioimmunoassay (RIA)
2. Haemagglutination inhibition tests	2. Enzyme linked immunosorbent assay (EIA)
3. Immunofluorescence techniques (IF)	3. Particle agglutination
4. Neutralization tests	4. Western Blot (WB)
5. Single Radial Haemolysis	5. Recombinant immunoblot assay (RIBA), line immunoassay (Liatek) etc.

- Strand displacement Amplification
- Polymerase Chain Reaction (PCR)
- Ligase Chain Reaction (LCR)
- Transcription based amplification

VIRAL DIAGNOSTIC PROCEDURES

Strand Displacement amplification

Ppt no. 143

Strand/Multiple displacement amplification (MDA)

- A non-PCR based DNA amplification technique
- This method can rapidly amplify minute amounts of DNA samples to a reasonable quantity for genomic analysis.

https://www.youtube.com/watch?v=j8YxNBw2CiM&t=4s

VIRAL DIAGNOSTIC TESTS

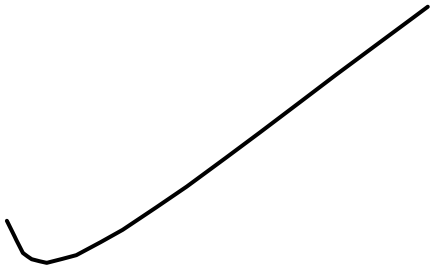
Branched Chain DNA

Ppt no. 144

Branched DNA assay

Signal amplification assay (as opposed to a target amplification assay) that is used to detect nucleic acid molecules

METHOD



Base

Capture Probe

Extender

Target

label extender

pre-amplifier

amplifier

BRANCHED DNA DETECTION

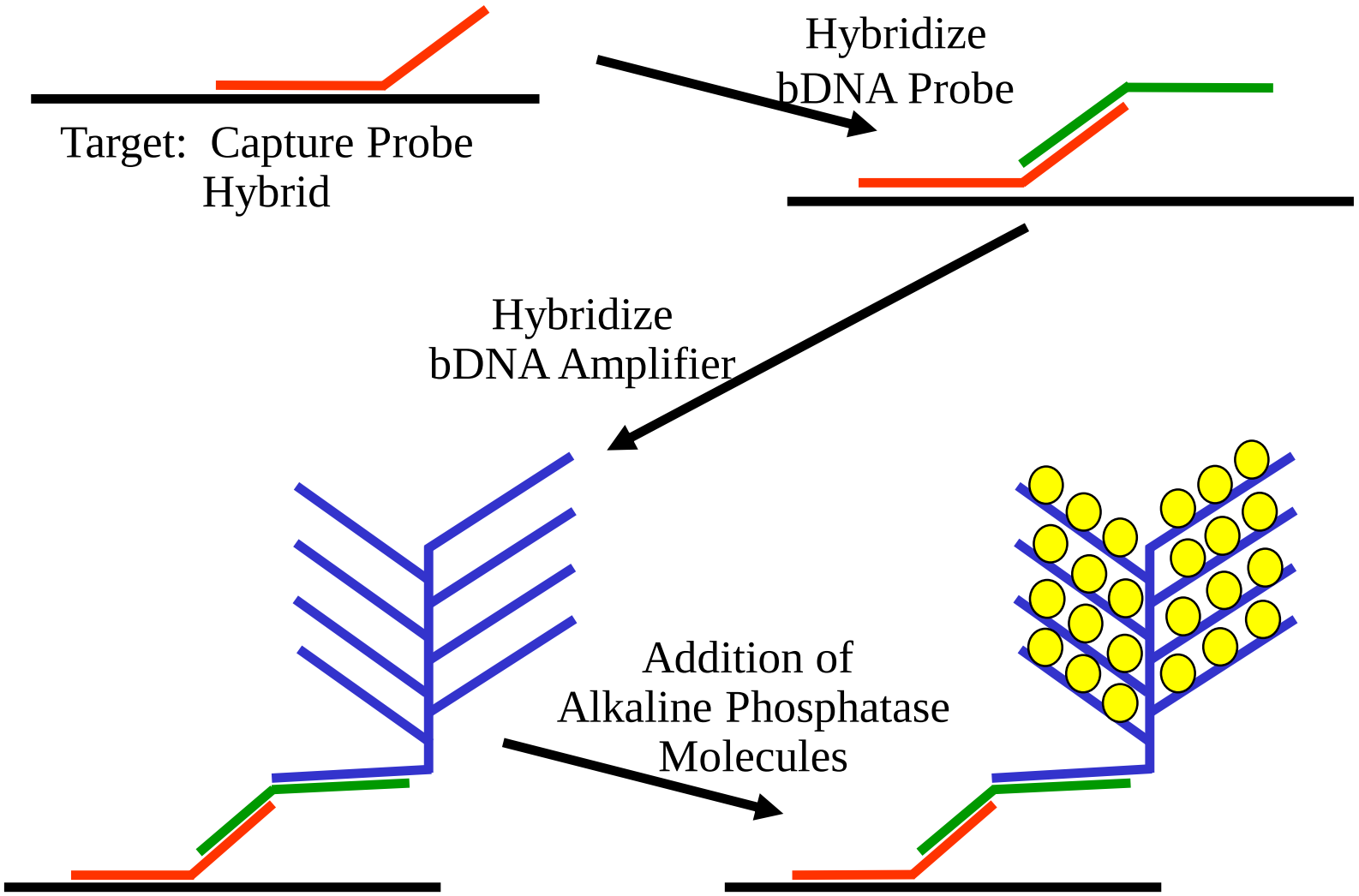
- Target nucleic acid sequences are not replicated through C.
- Uses “capture probes,” “bDNA probes” and “bDNA amplifier probes.”
- Assay is based upon microtiter plate technology.

- Solid phase signal amplification system
- Multiple sets of synthetic oligonucleotide probes

– Capture probes bound to well

– Target specific probes

– Amplifier molecule with 15 identical branches, each of which can bind to 3 labeled probes


Target: Capture Probe Hybrid
Hybridize bDNA Probe
Hybridize bDNA Amplifier
Addition of Alkaline Phosphatase Molecules

VIRAL DIAGNOSTIC PROCEDURES

Solid phase immunoassay

Ppt no. 145

Solid phase immuno assay

Most common used tests

Cheap

Less complex

For both composition and antigen detection

INCLUDES

1. ELISA (Enzyme linked immunosorbent Assay)
2. ELIspot (Enzyme Linked Immunospot)
3. GCSPR (grating-coupled surface plasmon resonance)
4. Solid phase radio immunoassay

APPLICATIONS

highly sensitive means to measure the presence of antigen in defined and homogeneous samples such as purified proteins in buffer, as well as in undefined heterogenous biological samples such as cell lysates, tissue culture supernatants, blood, and other clinical samples

RADIO IMMUNO ASSAY

https://www.youtube.com/watch?v=70TPrfL_8-M

VIRAL DIAGNOSTIC TESTS

Immunodiffusion

Ppt no. 146

IMMUNODIFFUSION

A diagnostic test which involves diffusion through a substance such as agar which is generally soft gel agar (1%) or agarose(1%), used for the detection of antibodies or antigen.

TYPES

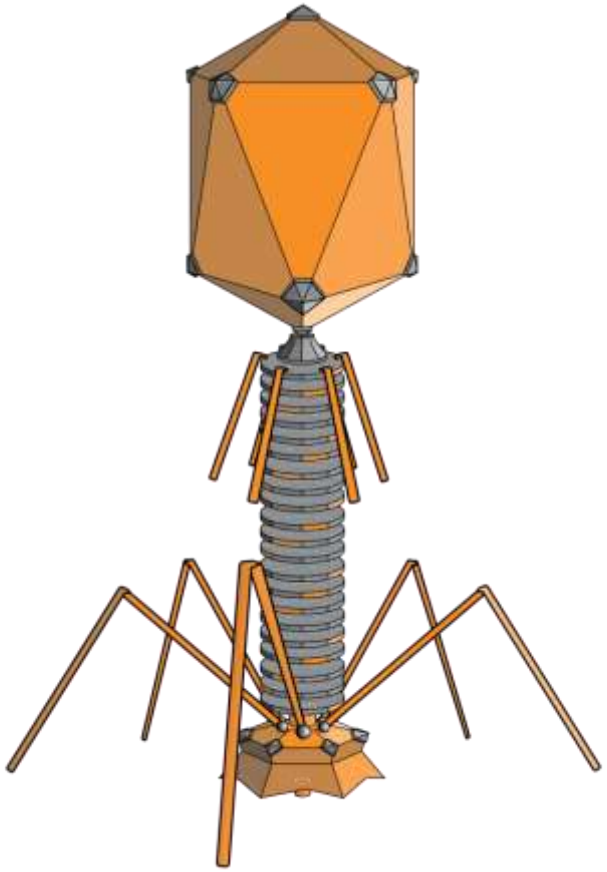
- Single diffusion in one dimension (Oudin procedure)
- 2. Double diffusion in one dimension (Oakley Fulthorpe procedure)
- 3. Single diffusion in two dimension (radial immunodiffusion or Mancini method)
- 4. Double diffusion in two dimensions (Ouchterlony double immunodiffustin)

https://www.youtube.com/watch?v=ADtT2dA1lsc

Ppt no. 147 not available

BACTERIOPHAGES
Properties

Ppt no. 148



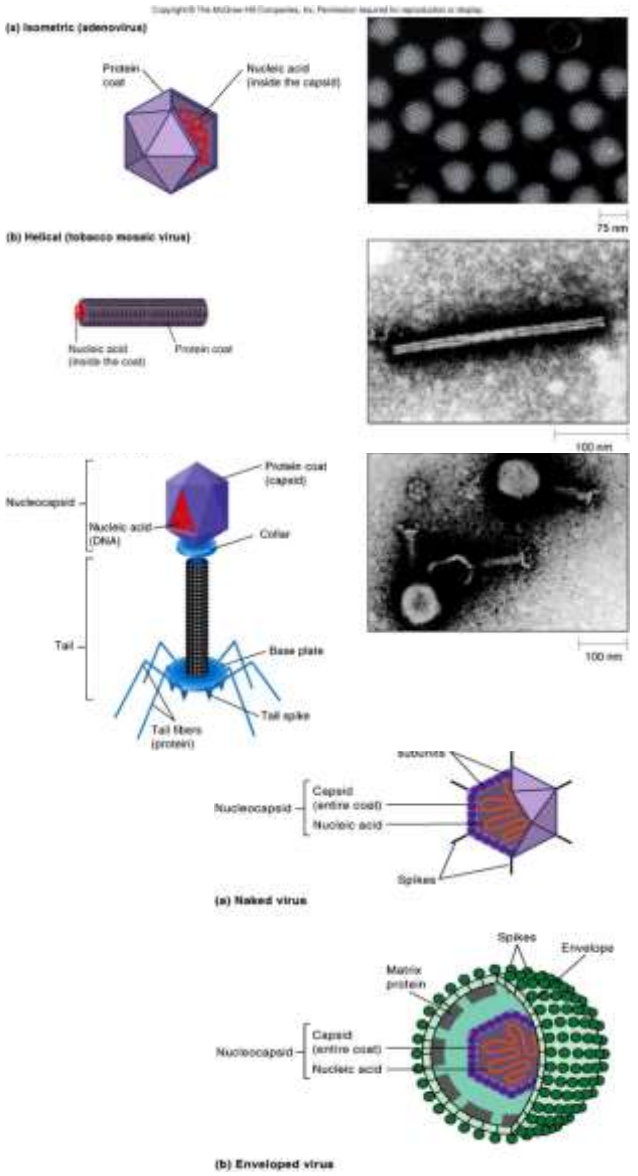
A **bacteriophage**

also known informally as a ***phage*** is a virus that infects and replicates within bacteria and archaea

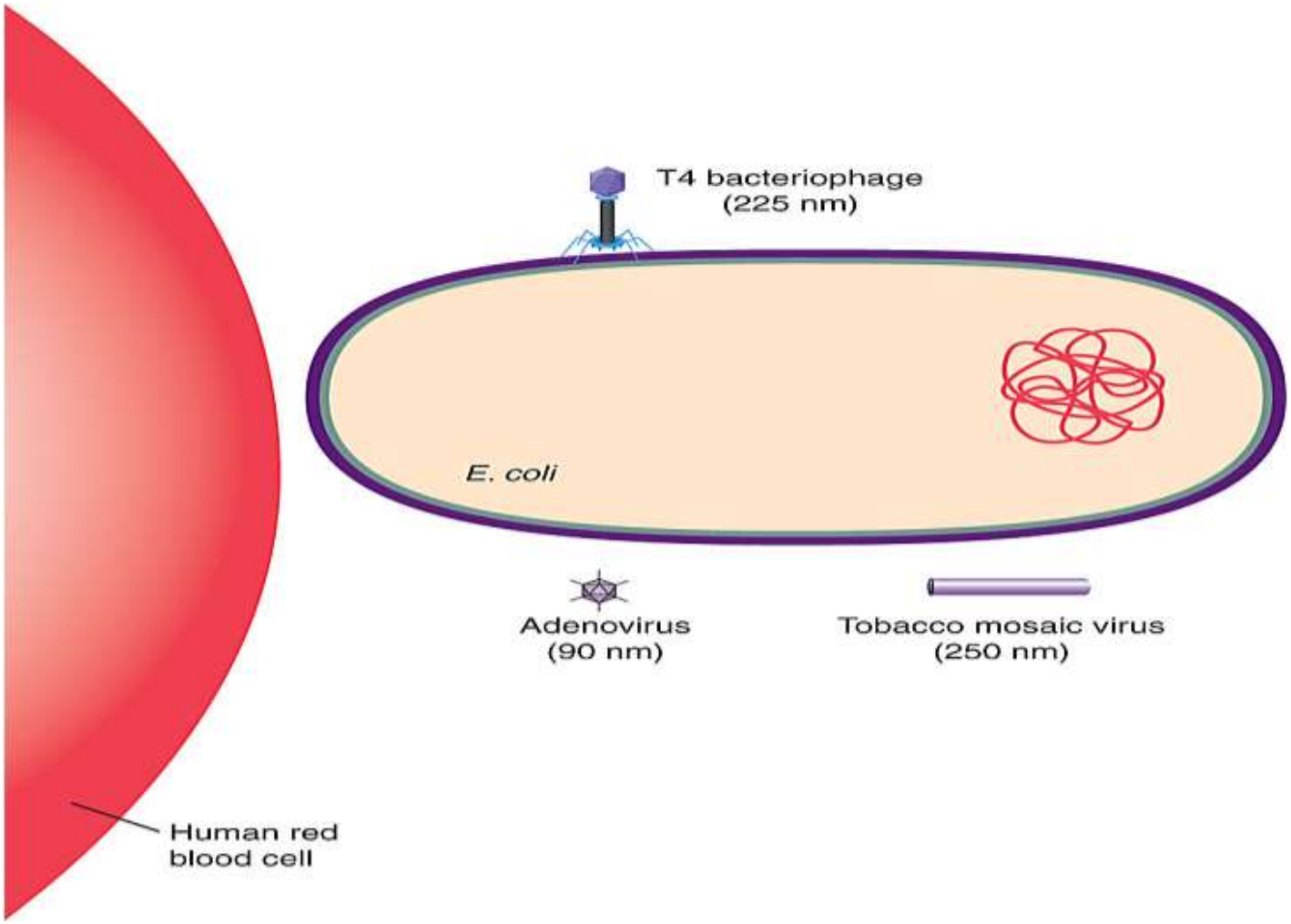
General Characteristics

Virus architecture

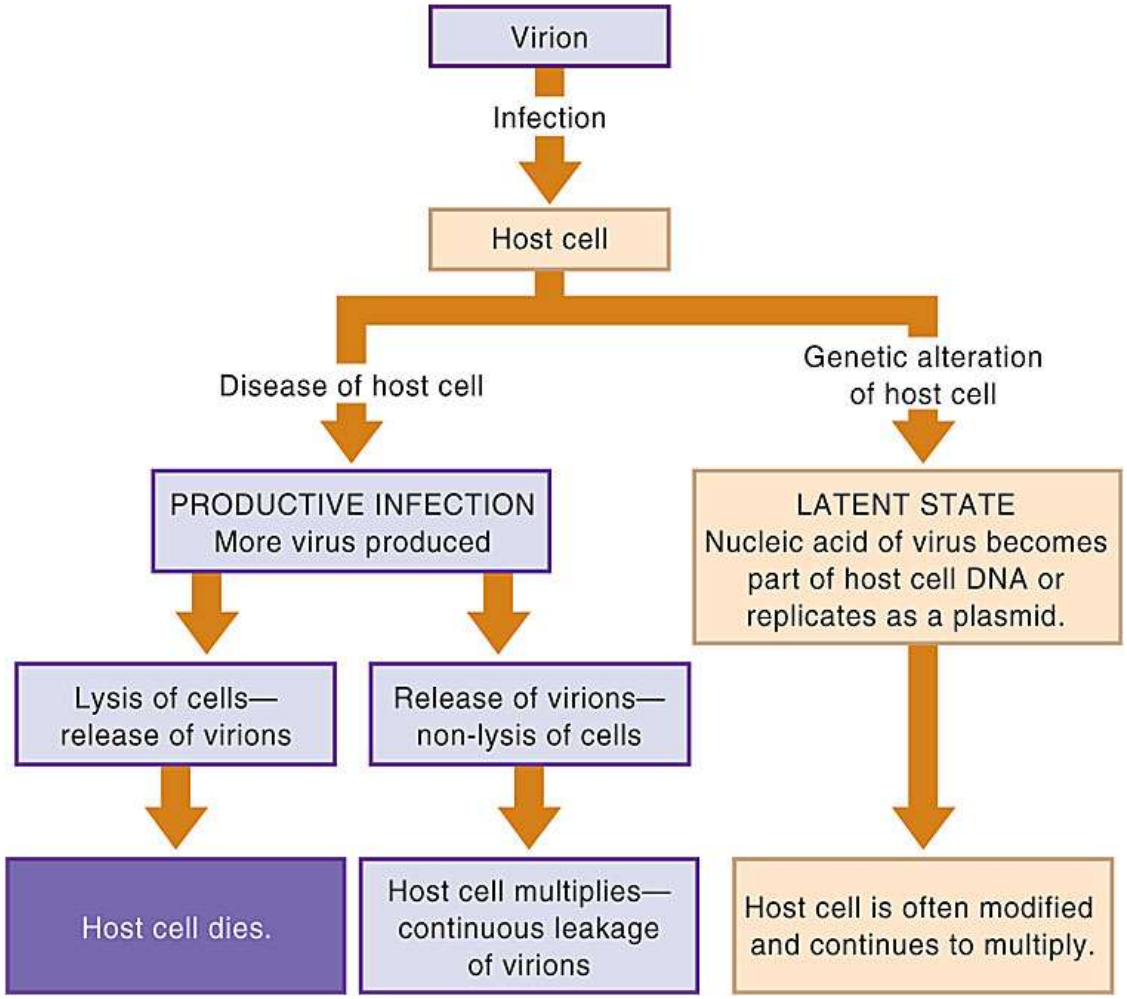
- Virus particle called virion
  - Consists of nucleic acid surrounded by protein coat
    - Capsid
- Shapes
  - Isometric
  - Helical
  - Complex
- Two types of virion
  - Naked – without envelope
  - Enveloped – surrounded by lipid membrane



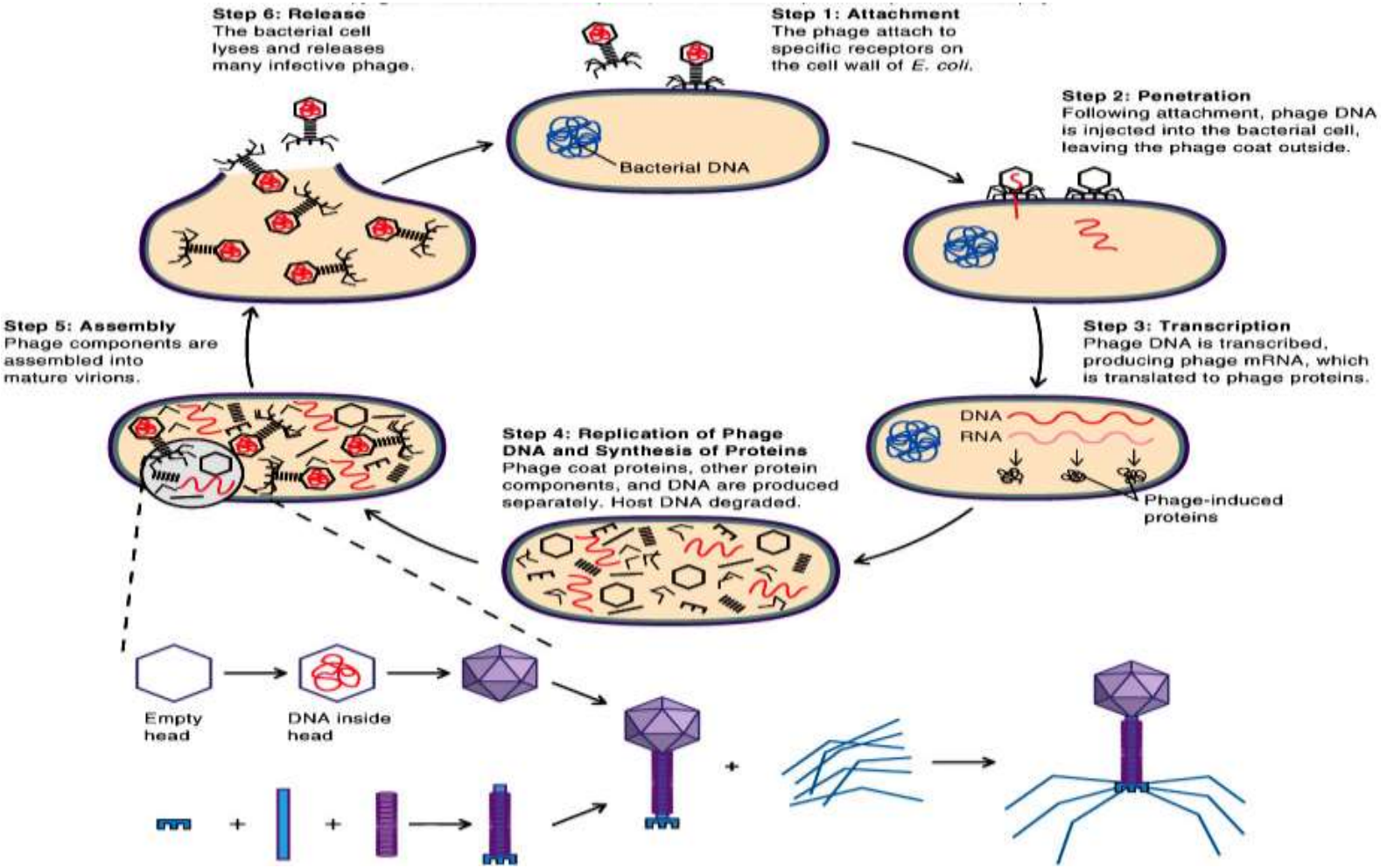
SIZE


T4 bacteriophage
(225 nm)
E. coli
Adenovirus
(90 nm)
Tobacco mosaic virus
(250 nm)
Human red
blood cell

VIRUS INTERACTIONS WITH HOST CELLS



LYTIC CYCLE


Step 6: Release
The bacterial cell lyses and releases many infective phage.
Step 1: Attachment
The phage attach to specific receptors on the cell wall of E. coli.
Bacterial DNA
Step 2: Penetration
Following attachment, phage DNA is injected into the bacterial cell, leaving the phage coat outside.
Step 5: Assembly
Phage components are assembled into mature virions.
Step 3: Transcription
Phage DNA is transcribed, producing phage mRNA, which is translated to phage proteins.
DNA
RNA
Phage-induced proteins
Step 4: Replication of Phage DNA and Synthesis of Proteins
Phage coat proteins, other protein components, and DNA are produced separately. Host DNA degraded.
Empty head
DNA inside head

https://www.youtube.com/watch?v=i9L-RoQ1frc

Relication of T-Phages

Ppt no. 149

Replication

- Attachment
- Penetration
- Uncoating
- Biosynthesis
- Release

Attachment

Specific receptors

Techoic acid

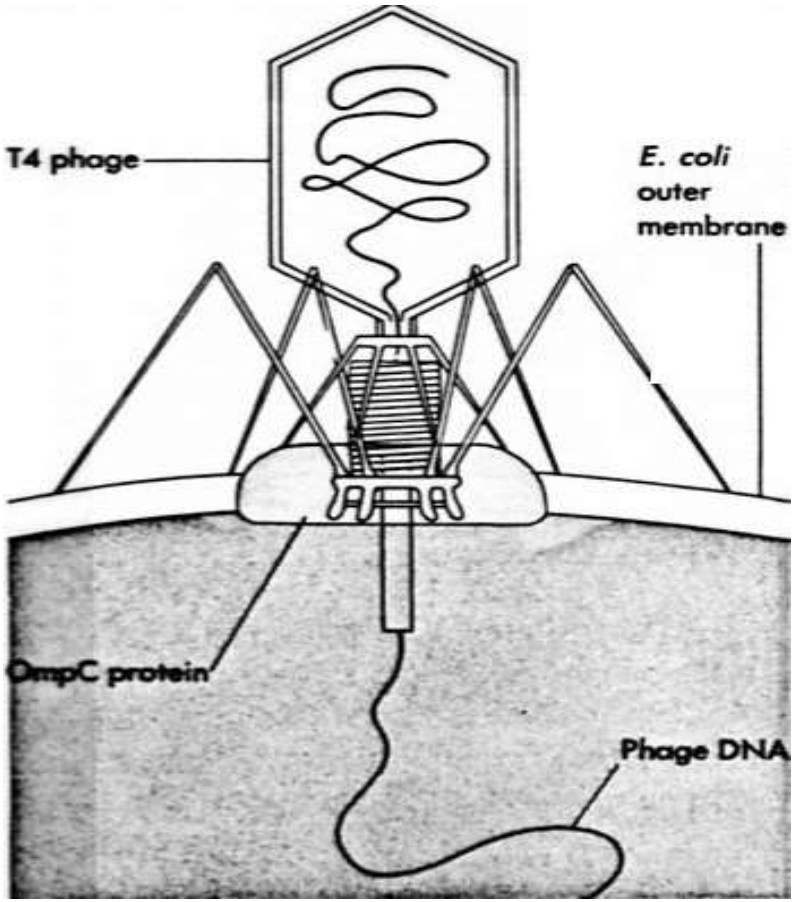
Lipids

Proteins

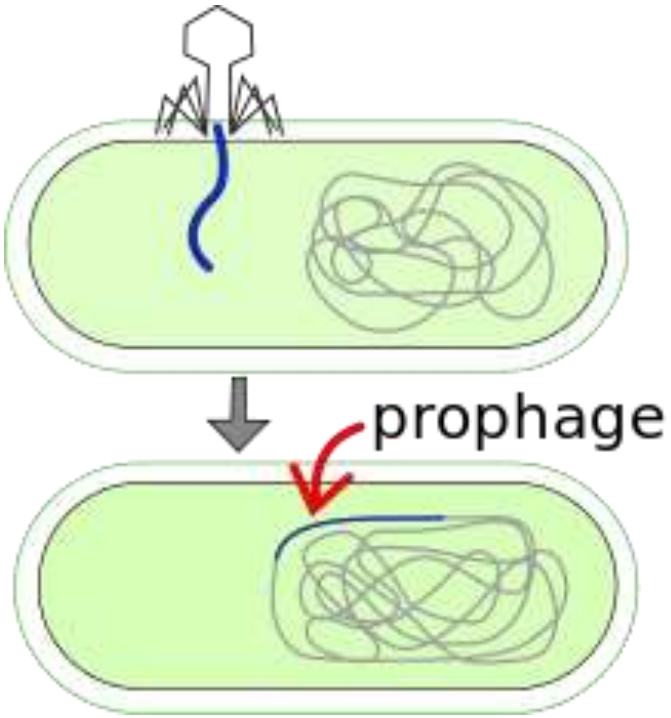
flagella

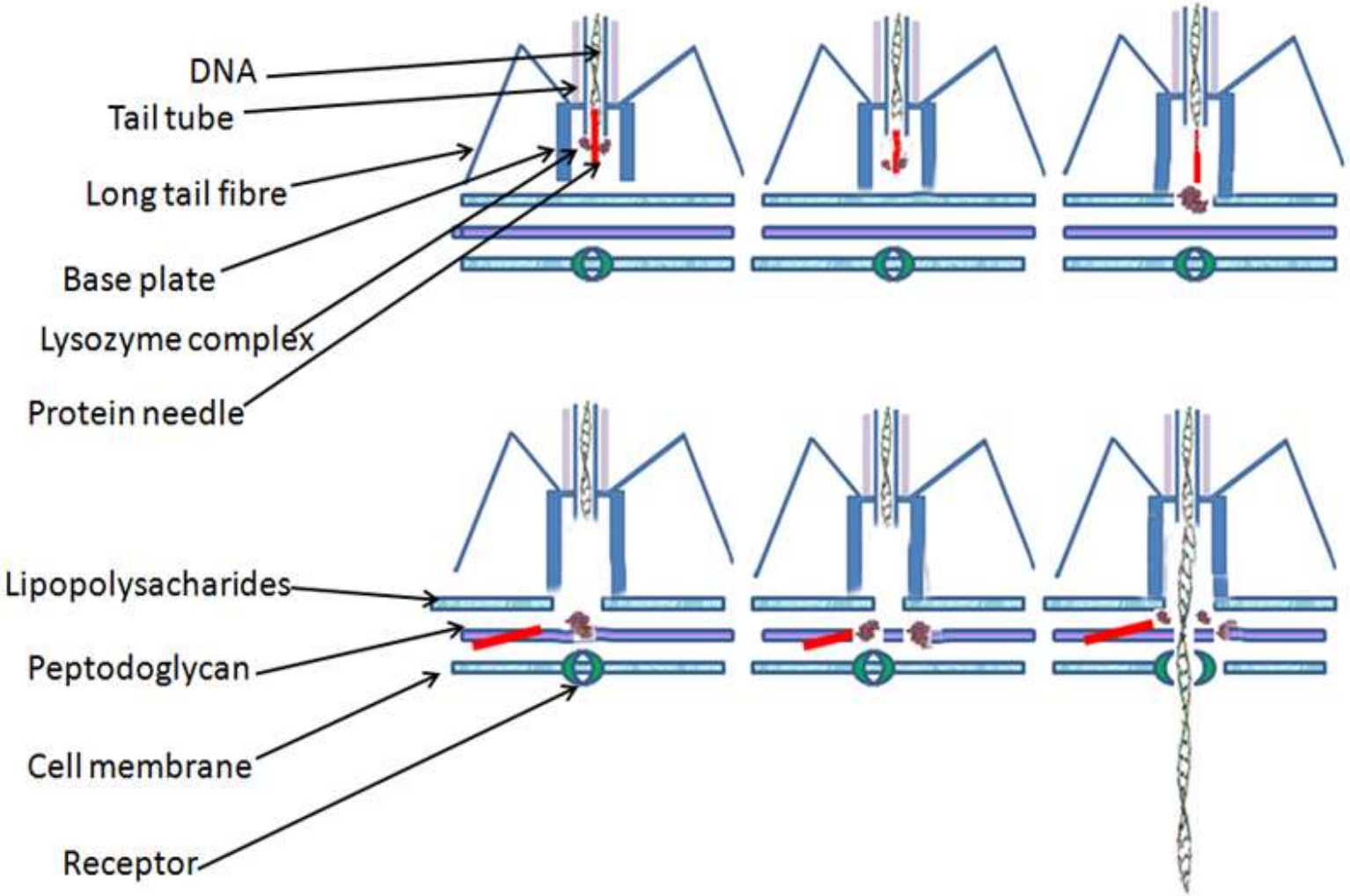
On specific bacterial hosts

Penetration

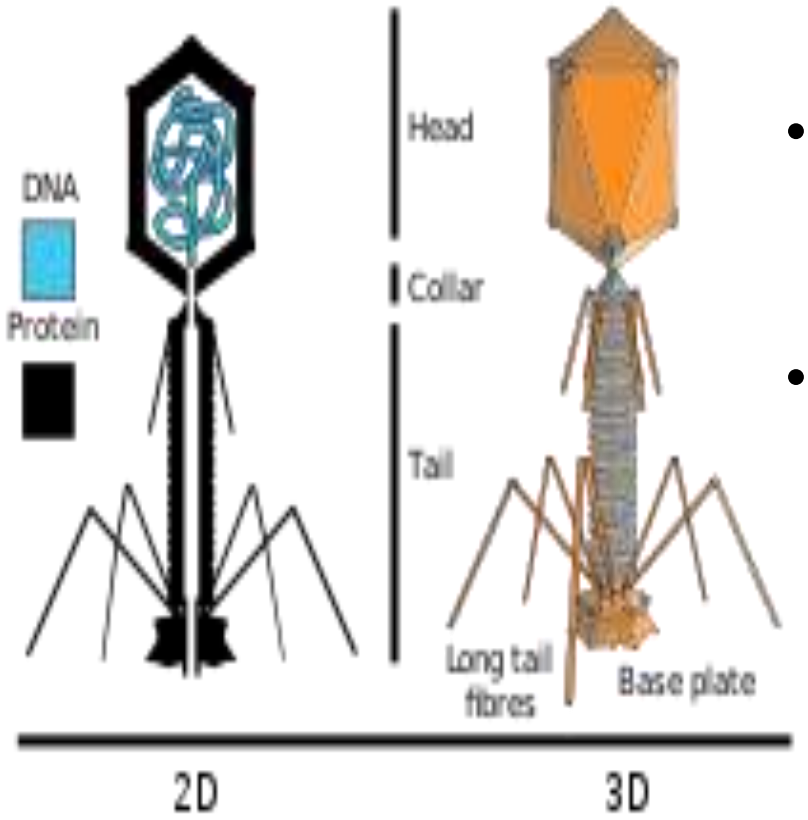


- Leave capsid outside




DNA
Tail tube
Long tail fibre
Base plate
Lysozyme complex
Protein needle
Lipopolysacharides
Peptodoglycan
Cell membrane
Receptor

Biosynthesis



- Bacterial ribosomes translate viral proteins
- Proteins modify bacterial RNA polymerase
- Proteins for new virions

Assembly

- T4: helper proteins required
- Base plates assembled first
- Tails built afterwards
- Heads constructed separately
- Whole process takes 15 minutes

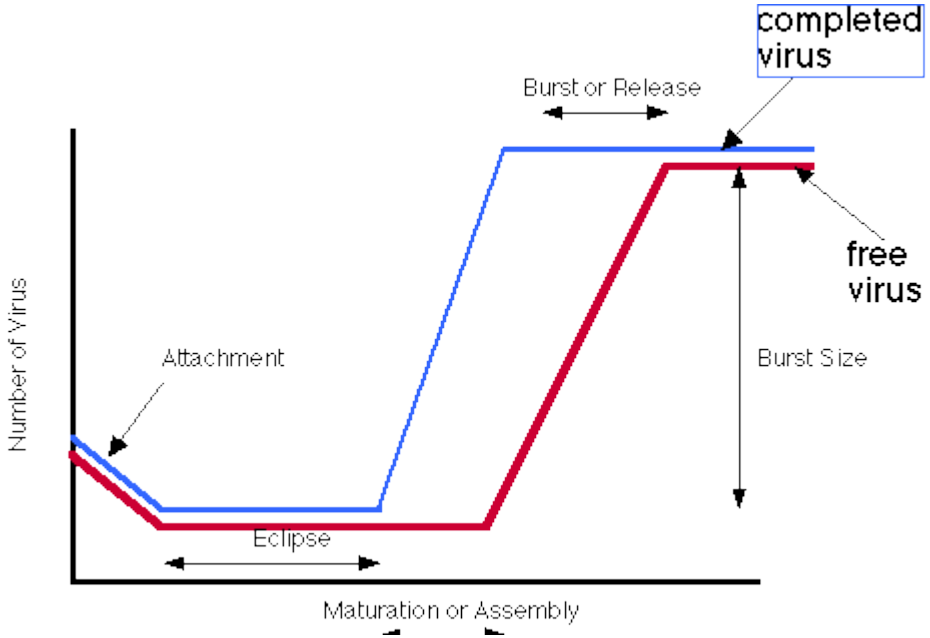
Release

- Lysis
  - ❖Tailed phages
  - ❖endolysin
- Extrusion
  - ❖enveloped
- Budding
- Secretion

END

Ppt no. 150

Phage Growth Curve

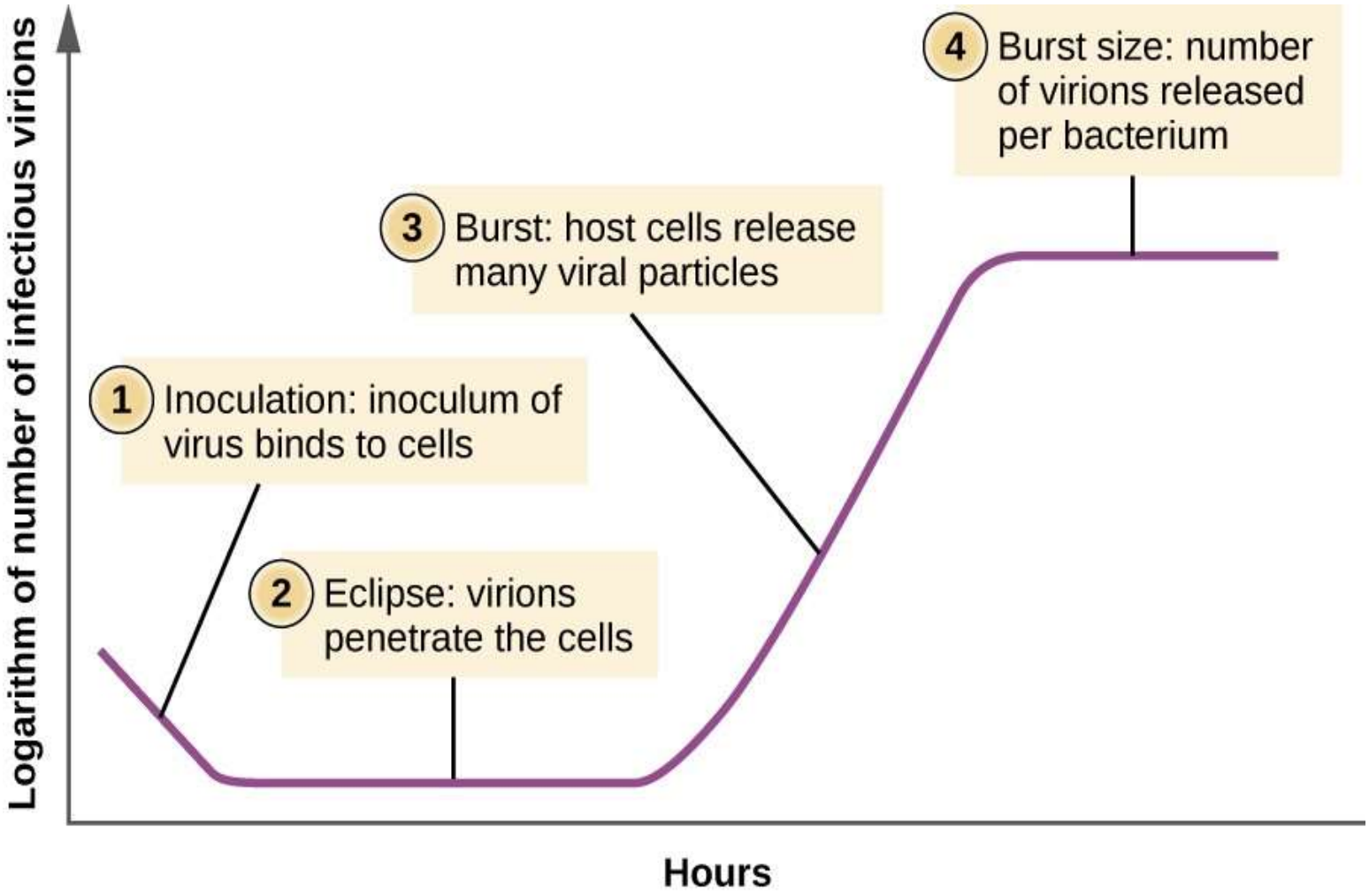


GRAPH

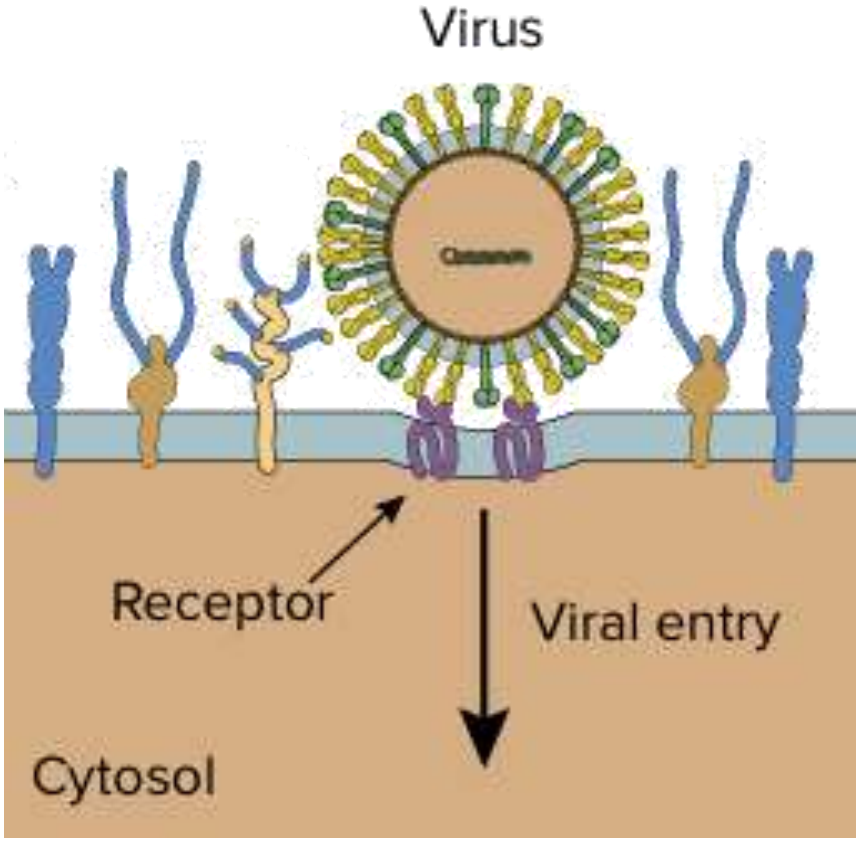
- Different from bacterial growth
- Not sigmoidal curve
- Consists of four steps

STEPS

1. Inoculation
2. Eclipse
3. Burst
4. Burst size


Logarithm of number of infectious virions
Hours
1 Inoculation: inoculum of virus binds to cells
2 Eclipse: virions penetrate the cells
3 Burst: host cells release many viral particles
4 Burst size: number of virions released per bacterium

INOCULATION/ ATTACHMENT



- Virus in blood stream attaches to host cell
- Beginning to decrease in number (negative gradient)

Eclipse/ Latent Period

More and more viruses replicate

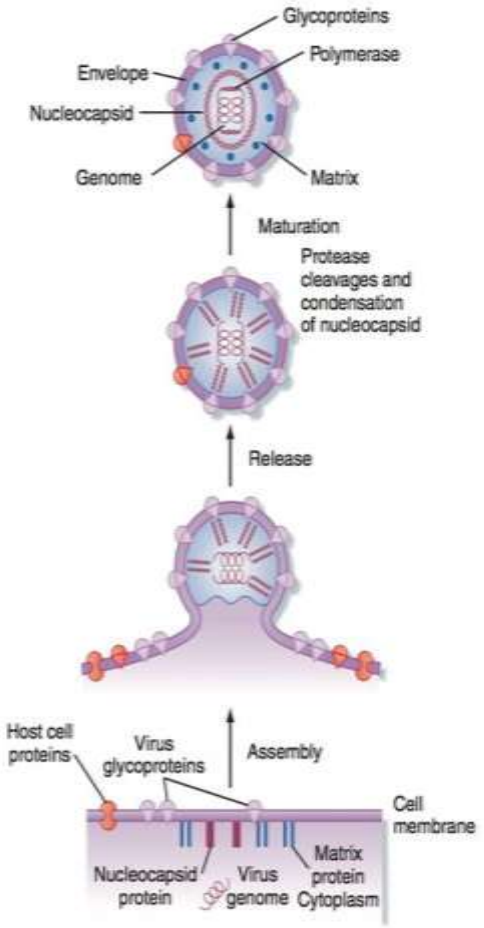
Thus virus titre decreases

Reaches at 0 gradient

No virus in external medium

BURST

Release

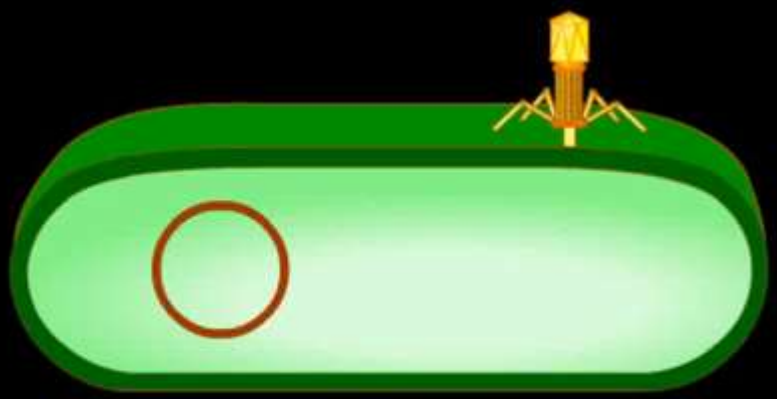


- Virions released
- Host cell releases many viral particles
- Host cells lyse
- Very steep rise in viral titre

Burst Size

- Number of virions released per bacterium
- Becomes constant

Bacteriophages



nt-end
YouTube

Ppt no. 151

Phage Growth

Bacteriophage Growth

- Phage collision with an adsorption-susceptible bacterium,
- Virion attachment,
- Virion nucleic acid uptake,

Bacteriophage Growth

- Synthesis of phage proteins and nucleic acid,
- A period during which virions mature,
- A virion release step

Phage Relationship with bacteria

The Red Queen hypothesis

- The interaction of phage with bacteria is the primary concern of phage community
- Bacteria have developed mechanisms that prevent phages from having an effect on them, which has led to this evolutionary arms race between the phages and their host bacteria
- Bacterial resistance to phages puts pressure on the phages to develop stronger effects on the bacteria.

The Red Queen hypothesis

- The Red Queen hypothesis describes this relationship, as the organisms must constantly adapt and evolve in order to survive.
- This relationship is important to understand as phages are now being used for more practical and medicinal purposes.

Restriction-Modification System

- Bacteria have developed multiple defense mechanisms
- Foreign DNA trying to enter the bacterial host is restricted by endonucleases that recognize specific base pairs within the DNA

Restriction-Modification System

- DNA of the cell is protected from restriction due to methylase
- RM systems have evolved to keep up with the ever-changing bacteria and phage.

- In general, these RM types differ in the nucleotide sequences that they recognize.
- There is an occasional slip where the endonuclease misses the DNA sequence of the phage and the phage DNA is able to enter the cell anyway, becoming methylated and protected against the endonuclease.

Clustered Regularly Interspersed Palindromic Repeats

- CRISPR means that the immunity to phages by bacteria has been acquired via adding spacers of DNA that are identical to that of the DNA from the phage.
- Some phages have been found to be immune to this mechanism as well.
- In some way or another, the phages have managed to get rid of the sequence that would be replicated.
- Phages are also capable of interacting with species other than bacteria, e.g., such as phage-encoded exotoxin interaction with animals
- Phage therapy is an example of applied phage community ecology.

Ppt no. 152

Phage Therapy

Phage Therapy

- ❖ Therapeutic use of bacteriophages for bacterial infections
- ❖ Phage therapy has many potential applications in
  - Human medicine
  - Dentistry
  - Veterinary science
  - Agriculture
- ❖ Biocontrol

Benefits

- Bacteriophages are much more specific than antibiotics
- They are typically harmless not only to the host organism but gut flora
- They have a high therapeutic index
- Few side effects
- A smaller dose will be effective

Obstacles

- The high bacterial strain specificity
- Different cocktails for treatment of the same infection
- Bacteria can evolve different receptors either before or during treatment

Obstacles

- In the case all bacteria is eliminated, the specific phage might evolve and attack good bacteria
- The need for banks of phages makes regulatory testing for safety harder and make it more expensive

Collection.

- The simplest method of phage treatment involves collecting local samples of water likely to contain high quantities of bacteria and bacteriophages
- for example effluent outlets, sewage and other sources.[7]

Collection.

- The samples are taken and applied to the bacteria that are to be destroyed which have been cultured on growth medium.
- If the bacteria die, as usually happens, the mixture is centrifuged; the phages collect on the top of the mixture and can be drawn off

Obstacles

- The high bacterial strain specificity
- Different cocktails for treatment of the same infection
- Bacteria can evolve different receptors either before or during treatment

Obstacles

- In the case all bacteria is eliminated, the specific phage might evolve and attack good bacteria
- The need for banks of phages makes regulatory testing for safety harder and make it more expensive

Administration

- Application is pill form
  - ❖ Temperature stability up to 55 °C
  - ❖ shelf lives of 14 months
- Application in liquid form
  - ❖ possible, stored preferably in refrigerated vials
- Applicatioin in Oral form
  - ❖ when an antacid is included, as this increases the number of phages surviving passage through the stomach.

Ppt no. 153

Estimation of Growth Numbers/Phage Ecology

- The most numerous "organisms" on Earth
- The viruses of bacteria (more generally, of prokaryotes
- Phage ecology is the study of the interaction of bacteriophages with their environments.

- Phages are Obligate intracellular parasites meaning that they are able to reproduce only while infecting bacteria
- Phages therefore are found only within environments that contain bacteria.
- Most environments contain bacteria, including our own bodies (called normal flora).
- Often these bacteria are found in large numbers. As a consequence, phages are found almost everywhere.

- As a rule of thumb, phage population will exceed bacterial densities by a ratio of 10-to-1 or more
- As there exist estimates of bacterial numbers on Earth of approximately 1030, there consequently is an expectation that 1031 or more individual virus (mostly phage) particles exist
- Making phages the most numerous category of “Orgamnisms" on our planet.

- Bacteria (along with archaea) appear to be highly diverse and there possibly are millions of species.
- Phage-ecological interactions therefore are quantitatively vast: huge numbers of interactions.
- Phage-ecological interactions are also qualitatively diverse
- There are huge numbers of environment types, bacterial-host types, and also individual phage types

Phage Ecology

- Phage population ecology
- Phage community ecology
- Phage ecosystem ecology

Phage population ecology

- A population is a group of individuals which either do or can interbreed interbreeding
- Population ecology considers characteristics that are apparent in populations of individuals
- These characteristics include so-called intraspecific interactions, that is between individuals making up the same population
- For example competition as well as cooperation

Phage community ecology

- A community consists of all of the biological individuals found within a given environment (more formally, within an ecosystem),
- Particularly when more than one species is present
- Community ecology studies those characteristics of communities that either are not apparent or less apparent
- Community ecology thus deals with interspecific interactions.
- Interspecific interactions, like intraspecific interactions, can range from cooperative to competitive but also to quite antagonistic (as are seen, for example, with predator-prey interactions).
- An important consequence of these interactions is coevolution.

Phage ecosystem ecology

- An ecosystem consists of both the biotic and abiotic components of an environment.
- Ecosystem ecology naturally differs from community ecology in terms of the impact of the community on these abiotic entities, and vice versa.
- In practice, the portion of the abiotic environment of most concern to ecosystem ecologists is inorganic nutrients and energy.

END

Ppt no. 154

LYSOGENY

Properties

Definition

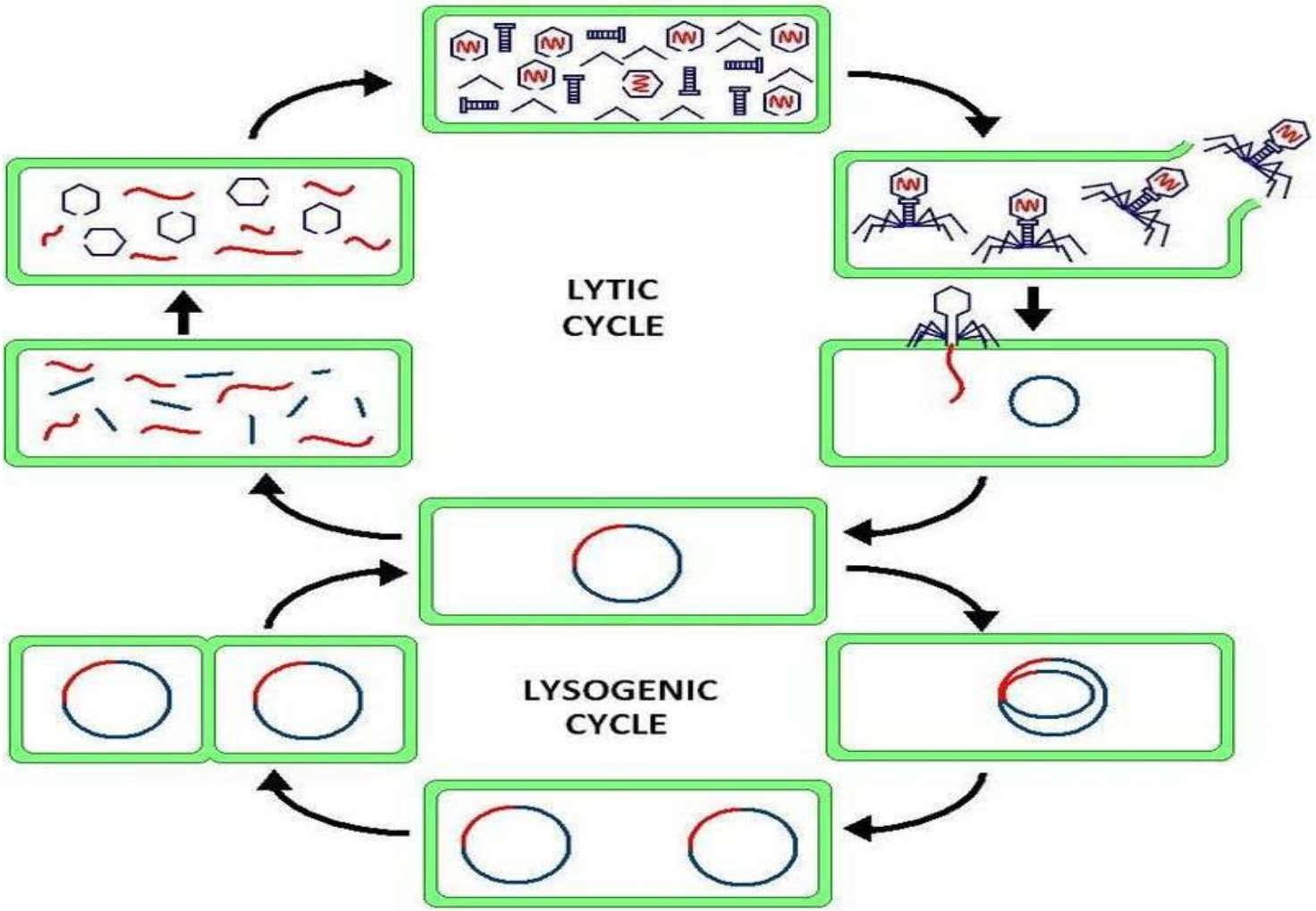
Lysogeny is characterized by integration of the bacteriophage nucleic acid into the host bacterium's genome or formations of a circular replicon in the bacterial cytoplasm

Properties

- Slow replication of viruses
- Mostly subclinical infection
- No harm to host
- Number of virions increases slowly

Advantages

- Genome incorporated in prophage
- Repressor prevents viral replication
- Lyse bacteria for release


LYTIC
CYCLE
LYSOGENIC
CYCLE

Lysogenic conversion

A phage induces a change in the phenotype of the infected bacteria that is not part of a usual phage cycle.

Bacterial survival

Lysogenic conversion has shown to enable biofilm formation in Bacillus anthracis

Bacterial virulence

Non-virulent bacteria have also been shown to transform into highly virulent pathogens through lysogenic conversion with the virulence factors carried on the lysogenic prophage

Ppt no. 155 not available

Ppt no. 156

SATELLITES

Definition

Satellite viruses are particles that contain nucleic acid genomes encoding a structural protein that encaptivates the satellite genome.

Properties

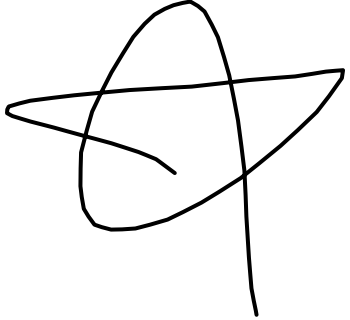
- Subviral agents composed of nucleic acid
- Satellites are dependent on the presence of a helper virus for their propagation

Properties

Commonly associated with plants

- Could be found in mammals, arthropods, and bacteria
- Have basic components to make their own protein shell to enclose their genetic material

Satellite Virus Vs. Virus



Virus	Satellite Virus	
Replication	Able to direct host cell to replicate genome	Depends on presence of helper virus for replication of genome
Nucleic acid	Contain DNA or RNA or both at different points in life cycle	Contain DNA or RNA
Genome Size	<10kbp to >2000kbp	0.22-1.5kbp
Structure	Contain protein shell or capsidPackaged genome with a capsid Envelope-not specific to all viruses	Satellite viruses contain the protein to encode own capsid with aid of helper virusSatellite RNA's and DNA's do not have capsids, rely on helper virus to enclose their genome
Host Range	Can infect all types of organism; animals, plants, fungi, bacteria, archaea	Plants (most common), mammals, arthropods, bacteria

Satellite

- **Satellite RNAs** do not encode capsid protein, but are packaged by a protein encoded in the helper virus genome. Satellite genomes may be single-stranded RNA or DNA or circular RNA, and are replicated by enzymes provided by the helper virus

Examples

- Satellite viruses may infect plants, animals, or bacteria
- An example of a satellite virus is satellite tobacco necrosis virus, which encodes a capsid protein that forms an icosahedral capsid that packages only the 1,260 nucleotide satellite RNA
- The helper virus, tobacco necrosis virus, encodes an RNA polymerase that replicates its genome and that of the satellite

In Plants

- may attenuate or exacerbate disease caused by the helper virus
- Examples: necrosis and systemic chlorosis, or reduced chlorophyll production leading to leaves that are pale, yellow, or yellow-white
- Symptoms by satellite RNAs are thought to be a consequence of silencing of host genes

Ppt no. 157

Viroids

Viroids

- The smallest infectious pathogens known so far
- They are composed solely of a short strand of circular, single-stranded RNA

Viroids

- RNA that has no protein coating
- All known viroids are inhabitants of plants in which they are causing diseases

Transmission

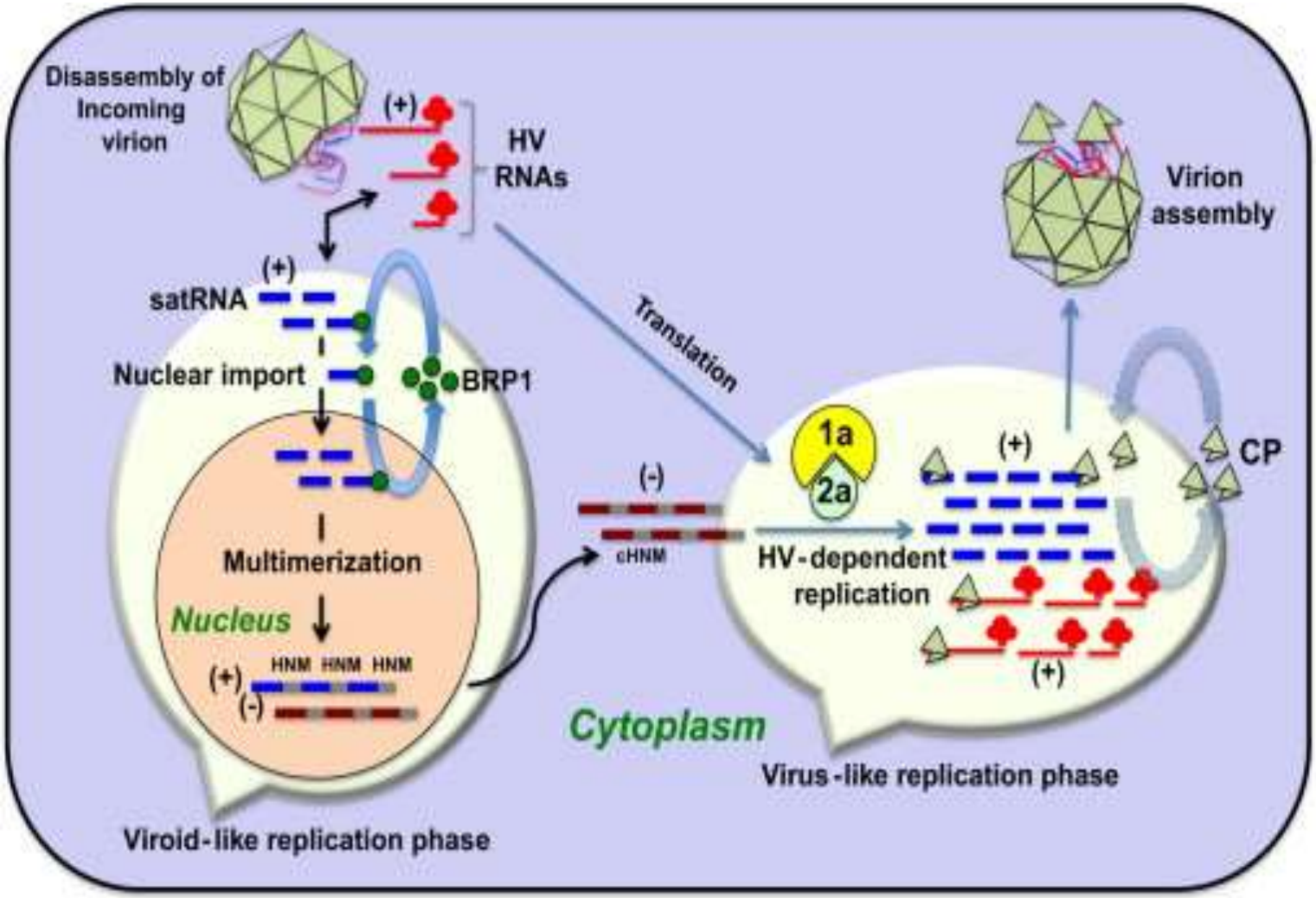
- By aphids
- By cross contamination
- Due to mechanical damage to plants as a result of horticultural or agricultural practices
- From plant to plant by leaf contact

Replication

- Viroids replicate
  - in the nucleus (Pospiviroidae)
  - or chloroplasts (Avsunviroidae)

Replication

- Three steps
- Through an RNA-based mechanism. They require RNA polymerase II, a host cell enzyme normally associated with synthesis of messenger RNA from DNA, which instead catalyzes "rolling circle" synthesis of new RNA using the viroid as template


Disassembly of Incoming virion
(+)
HV RNAs
Virion assembly
satRNA
(+)
Translation
Nuclear import
BRP1
1a
2a
(+)
CP
(-)
cHNM
Multimerization
HV-dependent replication
Nucleus
HNM HNM HNM
(+)
(-)
(+)
Cytoplasm
Virus-like replication phase
Viroid-like replication phase

Ppt no. 158

Prions

Prions

- Misfolded proteins
- Prions for proteinaceous infectious particle
- Characterize several fatal neurodegenerative diseases in animals and humans

Importance

Cause of

- Transmissible spongiform encephalopathies (TSEs)
- Scrapie in sheep
- Chronic wasting disease (CWD) in deer
- Bovine spongiform encephalopathy (BSE) in cattle (commonly known as "mad cow disease")
- Creutzfeldt-Jakob disease in humans

Structure

The protein that prions are made of (PrP) is found throughout the body, even in healthy people and animals.

PrPC

PrPC is a normal protein found on the membranes of cells.

It has 209 amino acids (in humans)

One disulfide bond,

A molecular mass of 35-36 kDa

A mainly alpha-helical structure.

The normal protein is not sedimentable; meaning that it cannot be separated by centrifuging techniques.

Structure

PrPres

Protease-resistant PrPSc-like protein (PrPres) is an isoform of PrPc which is structurally altered and converted into a misfolded proteinase K-resistant form in vitro.

PrPSc

The infectious isoform of PrP, known as PrPSc, is able to convert normal PrPC proteins into the infectious isoform by changing their conformation, or shape; this, in turn, alters the way the proteins interconnect. PrPSc always causes prion disease.

Ppt no. 159

Difference Between Satellites, Prions and Viroids

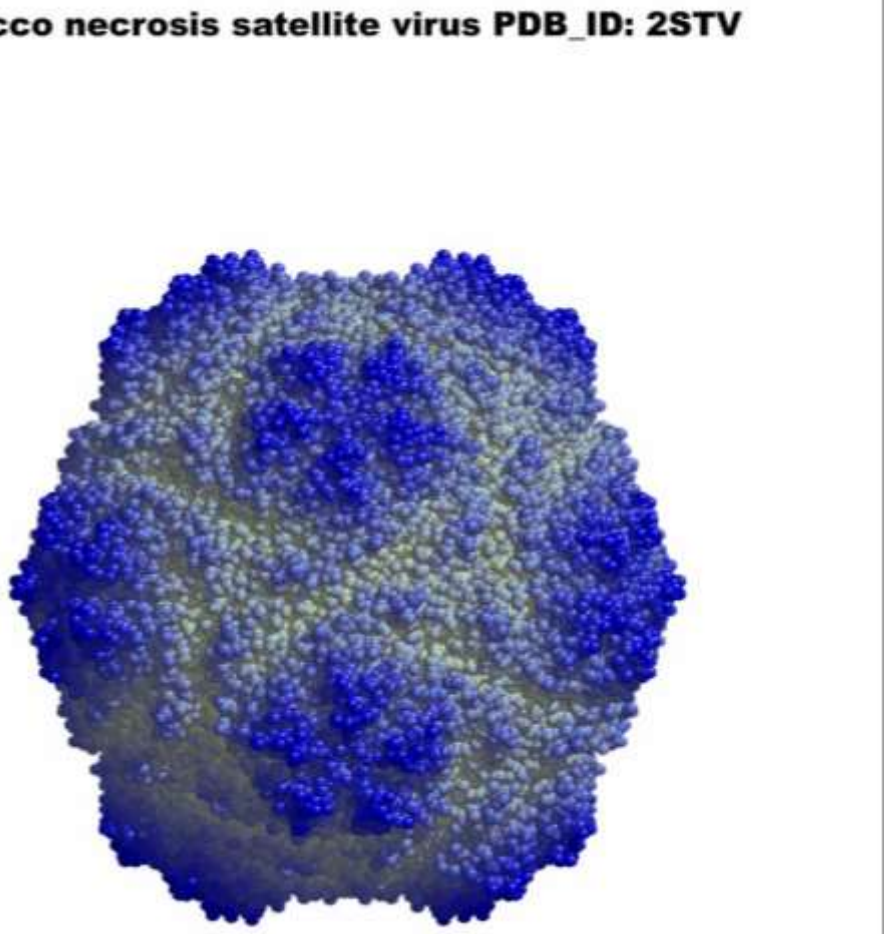
Satellite

- Smallest
- Most simple infectious
- RNA molecules
- Parasites of viruses

Satellite

- Satellites require helper viruses for replication
- Satellites do encode protein
- Circular RNA genome absent
- Satellites range between 475-1700 bases

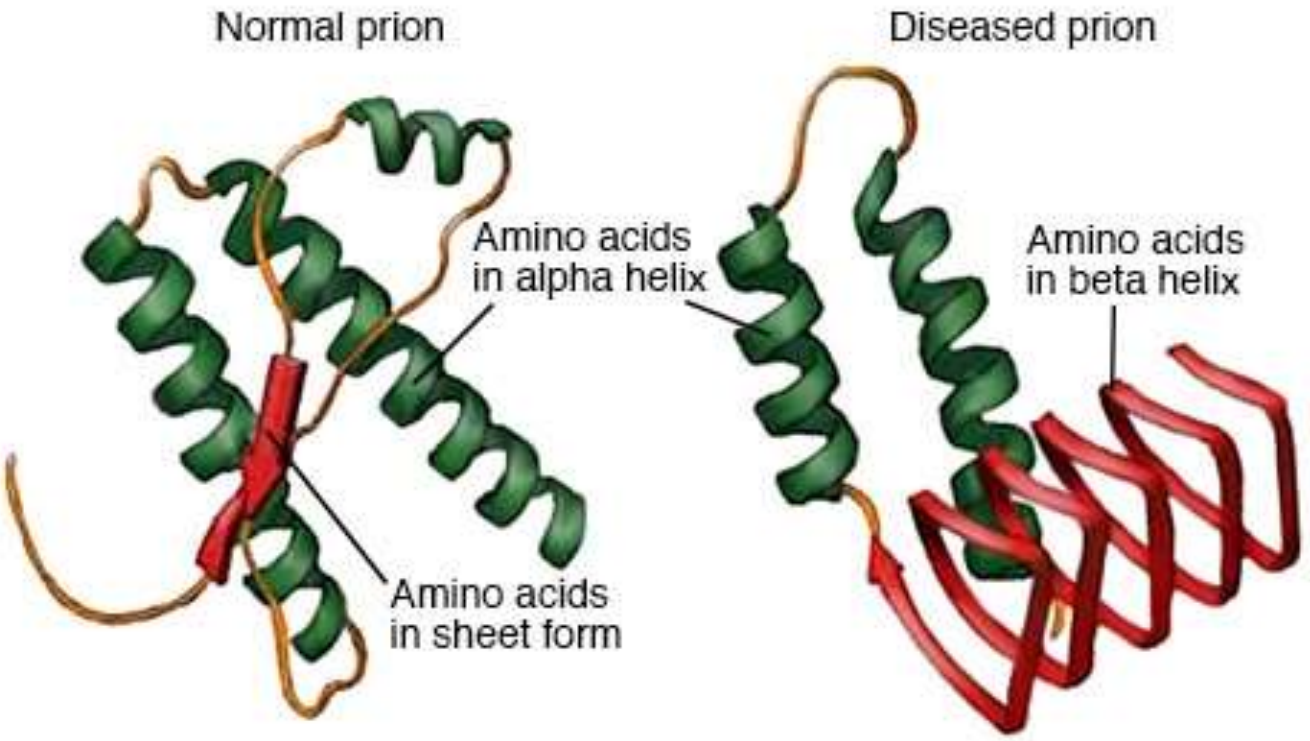
Satellite



Prions

- Neither have DNA or RNA
- Abnormal or mutated form of protein
- Causes several lethal diseases

Prions


Normal prion
Diseased prion
Amino acids
in alpha helix
Amino acids
in sheet form
Amino acids
in beta helix

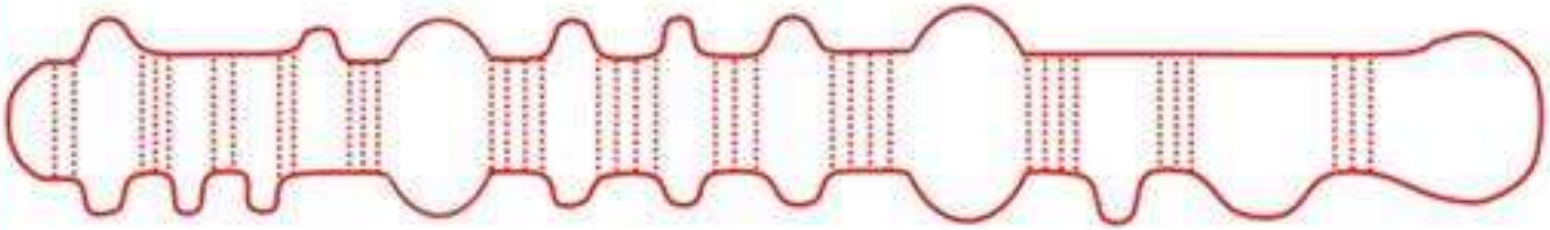
Viroids

- Autonomous
- Circular RNA genome present
- 250 to 400 bases
- Positive sense RNA
- Smaller than virus

VIROIDS

- Do not need helper virus for replication
- Two families: *Pospoviridae* and *Avsunviridae*
- Usually affects plants

Viroids



Structure of a viroid – circular single-stranded RNA with some pairing between complementary bases and loops where no such pairing occurs

Ppt no. 160

SATELLITES

Properties

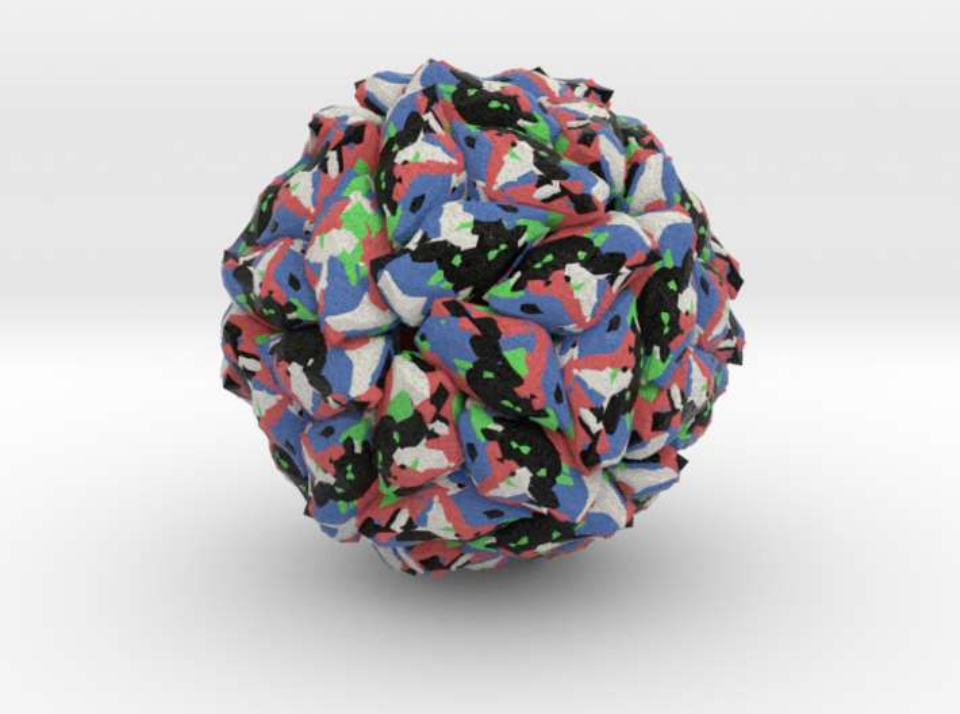
Satellite

- Smallest subviral agent
- Most simple infectious
- RNA molecules
- Parasites of viruses

Satellite

- Require helper viruses for replication
- Satellites do encode protein
- Satellites range between 475-1700 bases

Satellite



- Present in plants, mammals, arthropods
- Symbiotic relation with helper virus
- 0.22- 1.5kbp genomic size

Classification

- Single stranded RNA satellite virus
- Double stranded DNA satellite virus
- Single stranded satellite DNAs
- Double stranded satellite RNAs

Applications

- Application of mild strains to prevent plant viral diseases
- Transgenic plant production
- To study cell to cell movements in plants

Advantages To Study Satellites

- Ease in manipulation
- High in-vivo stability
- High expression level

END

Ppt no. 161

Satellites

Examples

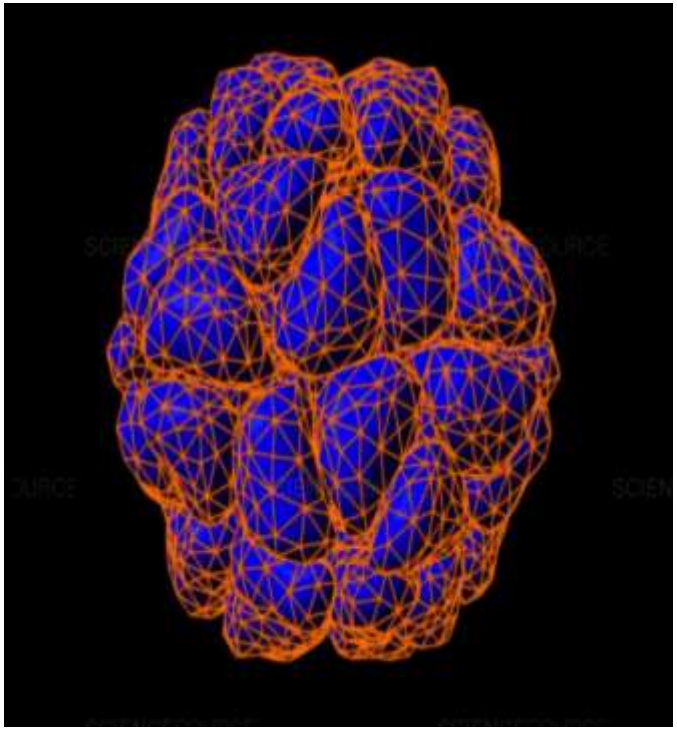
Necrosis and Systemic Chlorosis

- Reduced chlorophyll production
- Pale, yellow or yellowish white leaves

Hepatitis Delta Virus

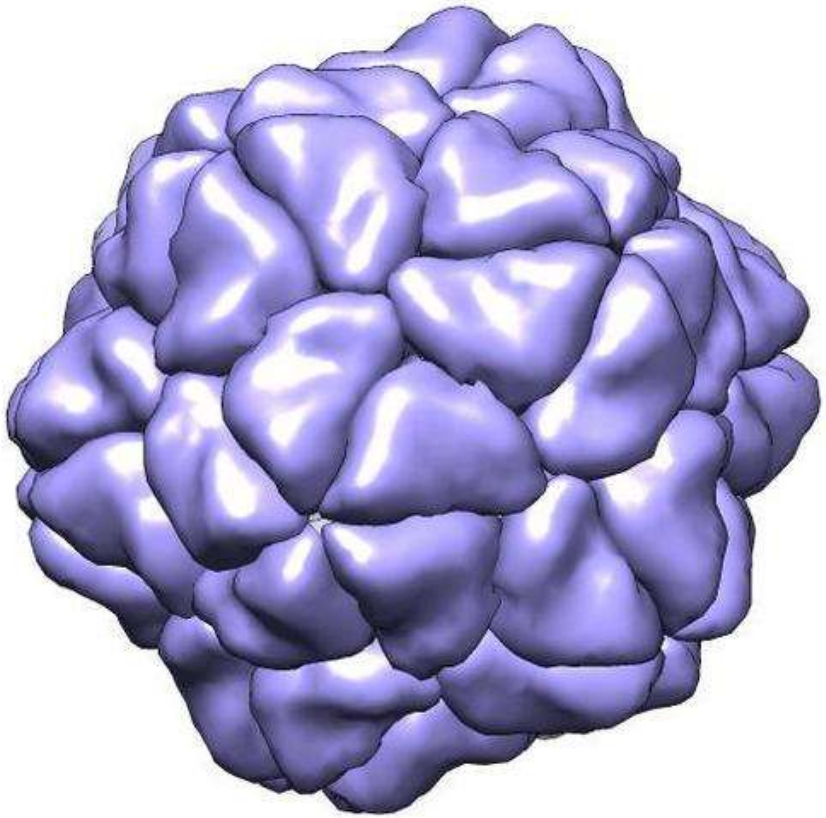
- Associated with human helper virus, hepatitis B virus
- 1.7 kb genome
- Circular single stranded RNA
- Replicated by cellular RNA polymerase

Tobacco Necrosis Virus



- First identified satellite
- 18nm isomeric particles
- Linear positive single stranded RNA genome
- Icosahederal
- 60 identical copies of identical proteins

Satellite Tobacco Mosaic VIRUS



- 60 identical copies of identical proteins
- 1063 nucleotides
- Single stranded RNA genome

Satellite Panicum Mosaic Virus

- Only affects grasses
- Icosahederal virus
- Eight stranded coat protein
- Induces mild mottle in millet plant

Ppt no. 162

VIROIDS

Properties

Properties

- The smallest infectious pathogens known so far
- They are composed solely of a short strand of circular, single-stranded RNA

Properties

- RNA that has no protein coating
- All known viroids are inhabitants of plants in which they are causing diseases

Nomenclature

- Family Pospiviroidae
- Genus *Pospiviroid* species: *Potato spindle tuber viroid* 361 nucleotides
- Genus *Pospiviroid* another species *Citrus exocortis viroid* 368-467 nt
- Genus *Hostuviroid* species: *Hop stunt viroid*, 294- 303 nt

Nomenclature

- Family Pospiviroidae
- Genus *Cocadviroid* species: *Coconut cadang-cadang viroid* 246-247 nt[1]
- Genus *Apscaviroid* type species *Apple scar skin viroid* 329-334 nt
- Genus *Coleviroid* type species: *Coleus blumei viroid*

END

Ppt no. 163

VIROIDS

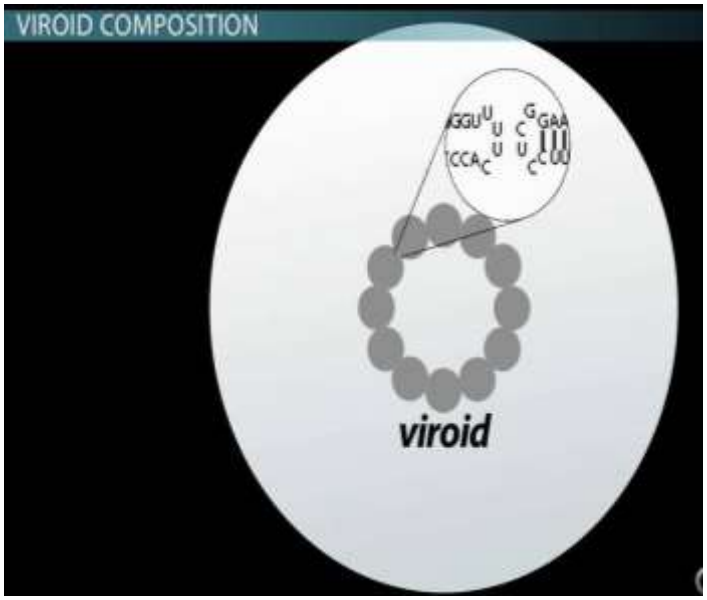
Structure

Structure

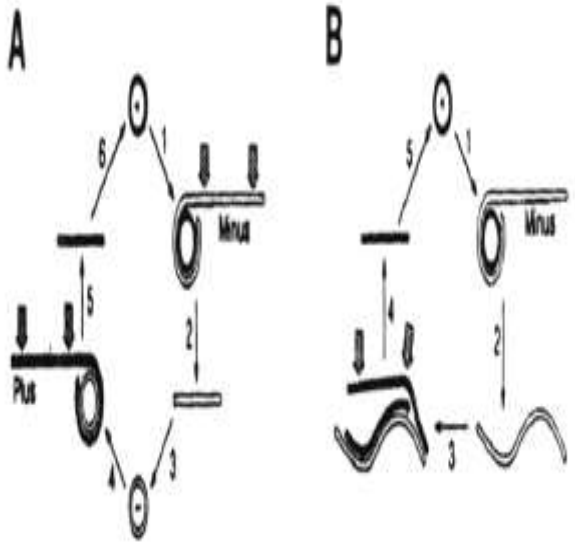
- Low molecular weight
- Unique structure
- Cause of disease

Structure

- Not encapsidated
- Replicate autonomously



Genome



- Single stranded
- Covalently closed, circular
- As well as, linear RNA molecules
- Intramolecular complimentarity

- Do not manufacture any proteins
- Infectious
- Highly structured
- Produce single specific RNA molecule

Replication

- Rolling circle mechanism in-vivo
- Of two types
- Non-hydrolytic reaction

END

Ppt no. 164

VIROIDS

Transmission

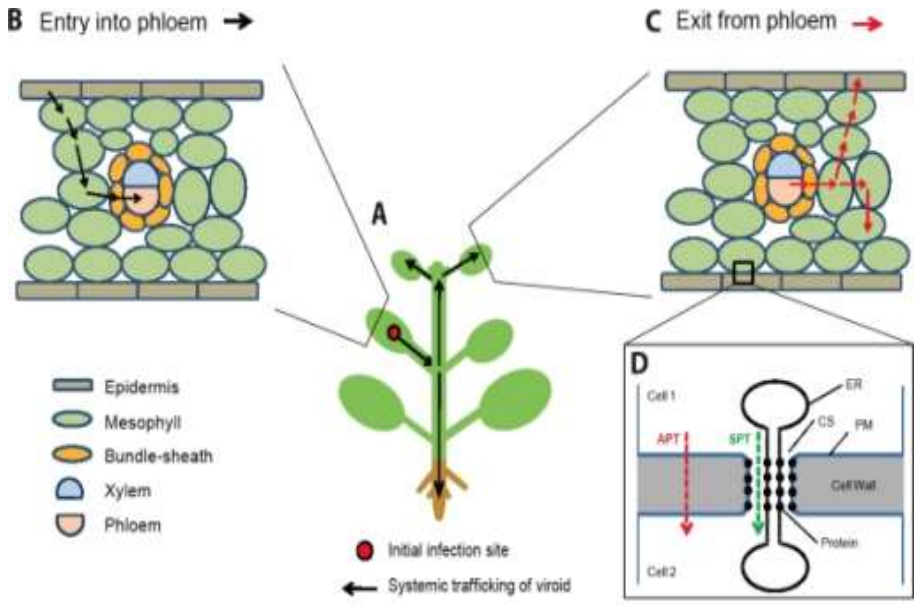
Transmission

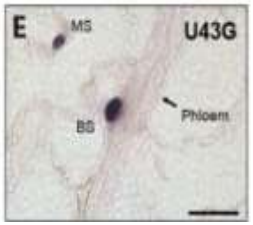
- Mechanical means
- Contaminated machinery, tools, pruning and mechanic agriculture techniques

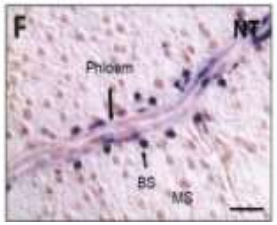
Transmission

- Transmitted sometimes through seeds and pollens
- No important natural vectors

Through plants





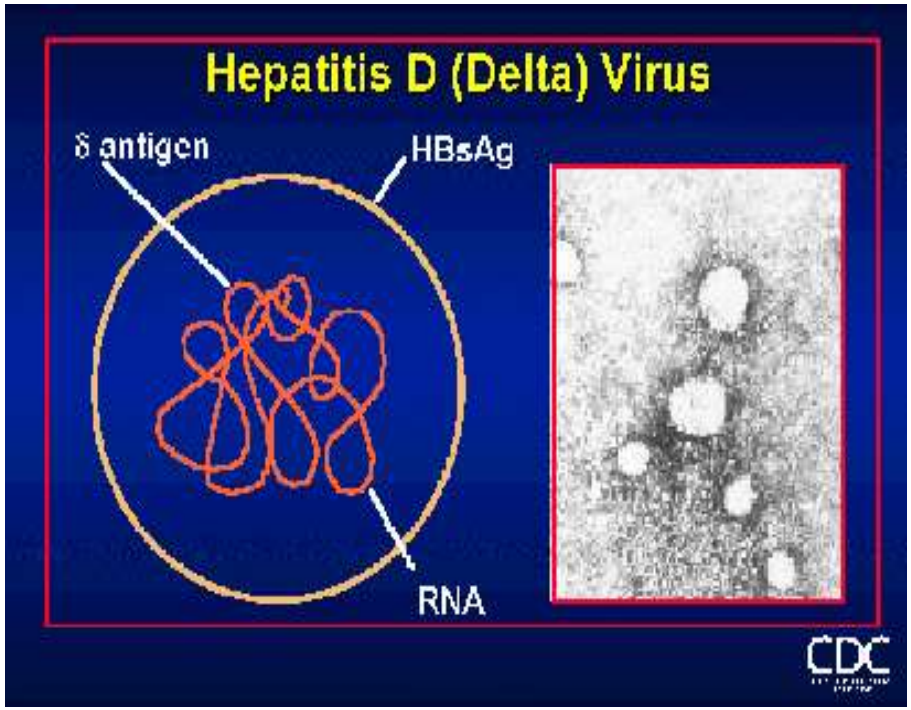


- Viroids enter pollen and ovule
- Transmitted through seed
- When seed germinates, the new plant gets infected

- Transmitted through aphids
- Mechanical damage
- Leaf to leaf contact

- Often spread by vegetative propagation
- Agriculture and horticulture practices
- Both natural and human driven transmission

- Both horizontal and vertical spread
- Horizontal: parasitic fungi, nematodes, insects
- Vertical: seed transmission,



- Hepatitis D viroid enter human liver cells, if binding protein is present
- Obtains it from hepatitis B virus
- Delta virus enters blood stream and spreads by blood transfusion

Ppt no. 165

VIROIDS

Examples

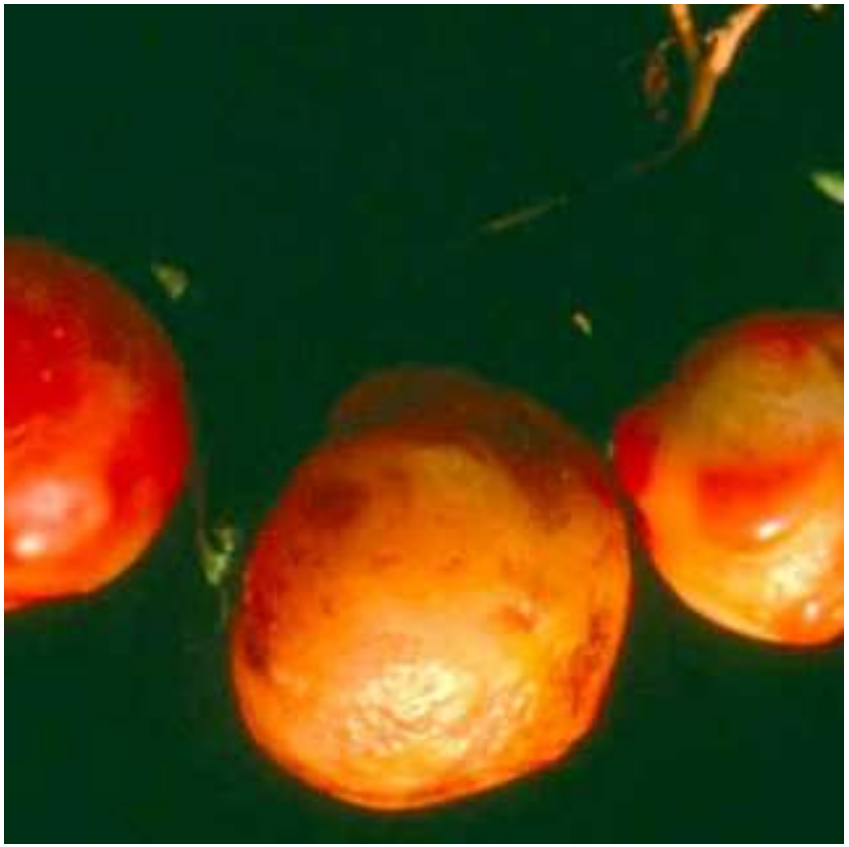
Cucumber Green mottle virus

- Cucumber, melons and watermelons
- Affects young leaves with green and light green spots
- Veins remain green
- With heavy infection, leaves are deformed
- Small fruits1

Carnation mottle virus

- Carnation flower plants
- Mild mottling of young leaves
- Faint chlorosis
- Yellow ring spots in heavy infection
- Vein chlorosis 8 to 10 days post infection

Tomato bushy stunt virus



- Economically significant plant pathogen
- Stunted growth, leaf mottling, deformed or absent fruits
- Broad host range
- Necrotic or chlorotic lesions

Potato Spindle Tuber Viroid

- Small circular RNA molecule
- Natural host: potato and tomato
- Transmission by seeds and aphids
- Necrosis of leaves and veins

Chrysanthemum stunt viroids

- Affects chrysanthemum and pinks
- Light colored flowers
- Leaves reduce in number and size
- Wavy or crumpled leaf surface

Hepatitis D viroids

- Only human disease
- Simultaneous infection with hepatitis B virus
- Causes liver cell death
- Sequestering 7S RNA and/or cleaving it

END

Ppt no. 166

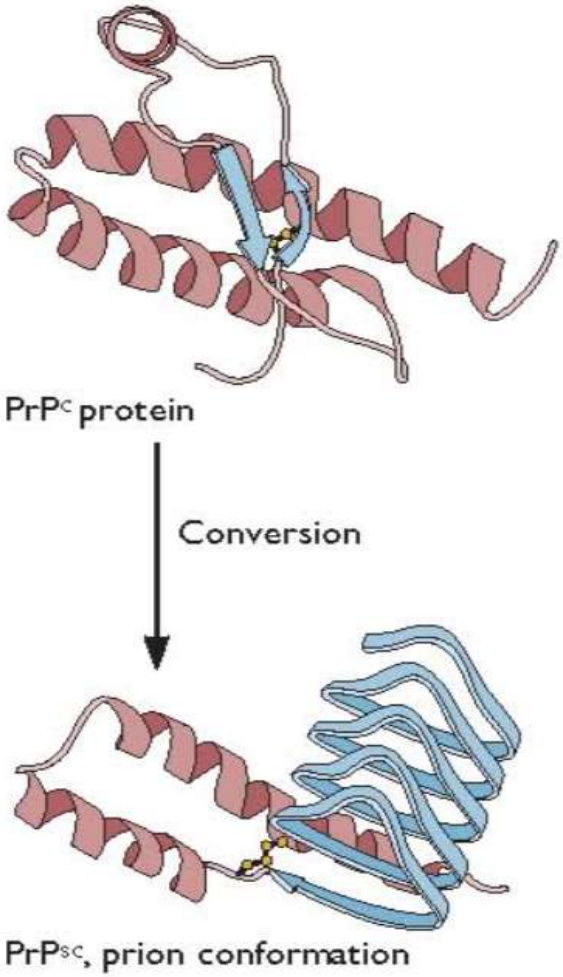
PRIONS

Structure

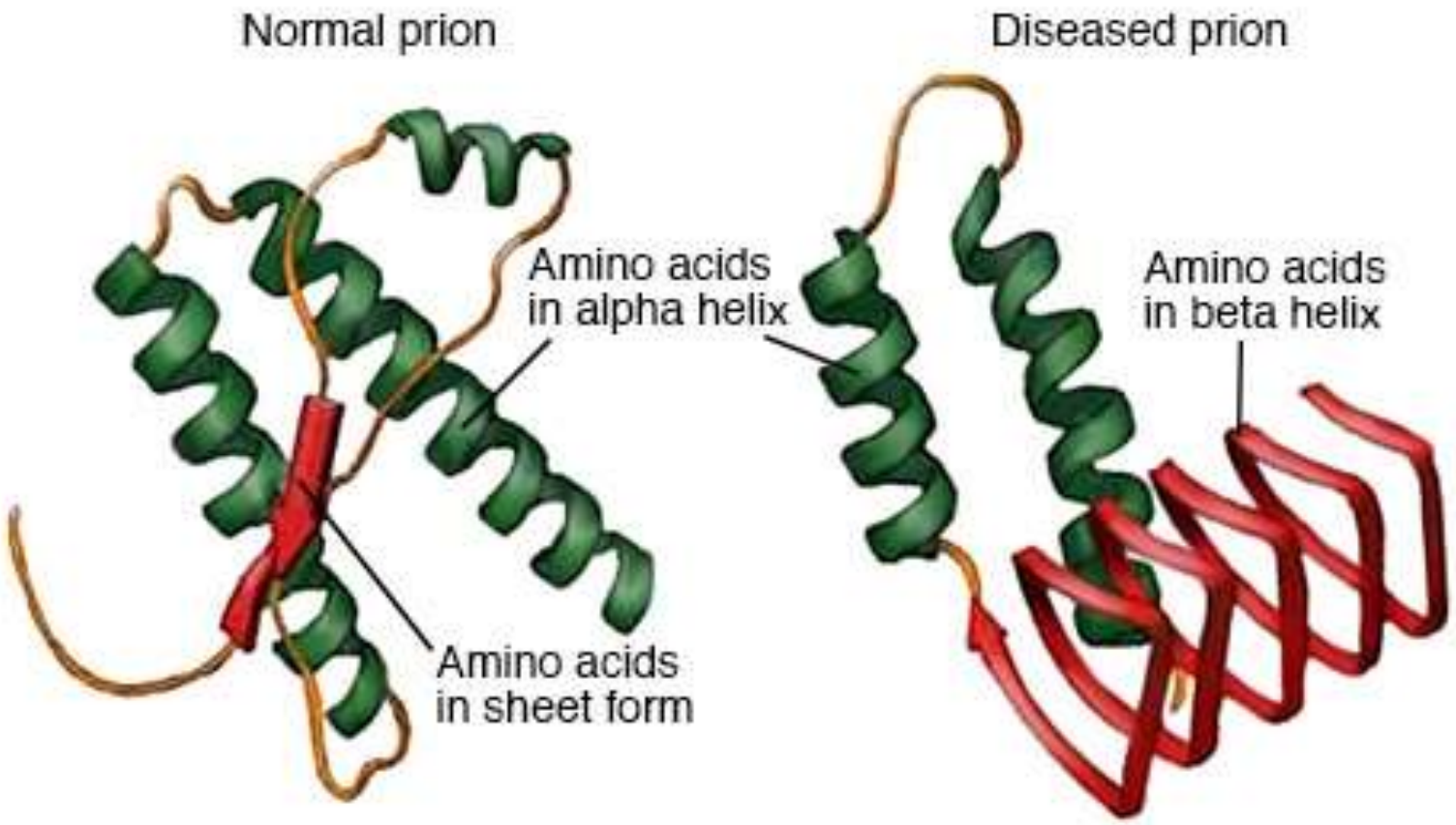
Normal proteins

- 200 to 250 amino acids
- Twisted into three telephone cord-like coils
- Named as HELICES
- With tails of more amino acids

Mutated proteins



- Built from same amino acids
- But take different shape
- 100 times smaller than smallest virus


Normal prion
Diseased prion
Amino acids
in alpha helix
Amino acids
in sheet form
Amino acids
in beta helix

Difference from Bacteria and Viruses

- Prions don’t contain nucleic acids
- Resistant to heat and chemicals
- Difficult to decompose biologically
- Survive in soil for many years

Structure

- Misfolded/ abnormally folded protein
- Pathogenic
- Causes similar proteins to misfold also

Replication

- Unique
- Replicate on their own and become infectious
- Occur in two forms: PrP-sen and PrP-res

END

Ppt no. 167

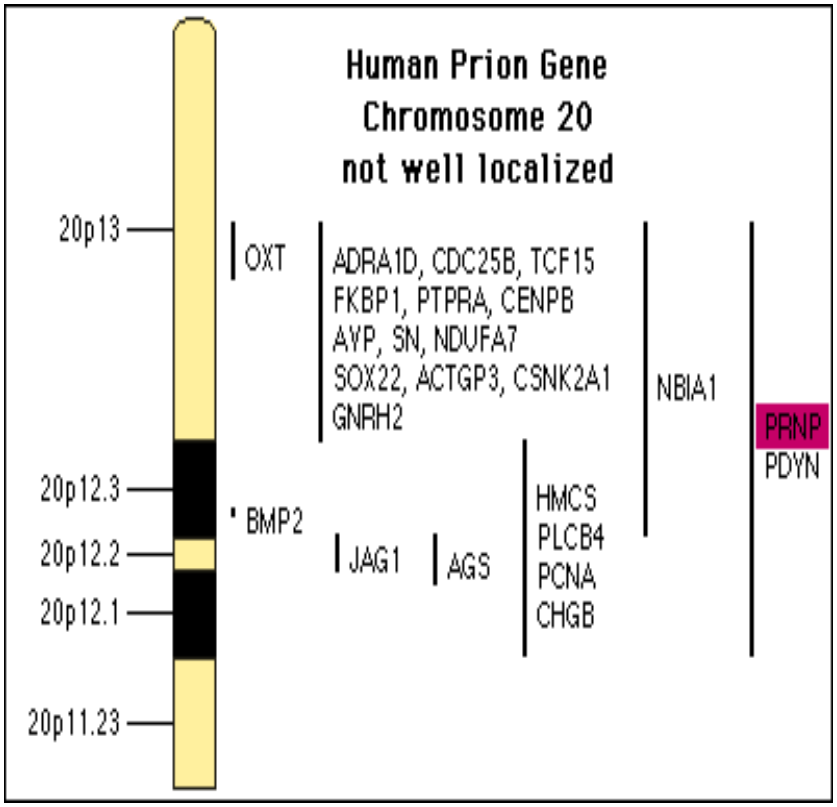
PRIONS

Genome

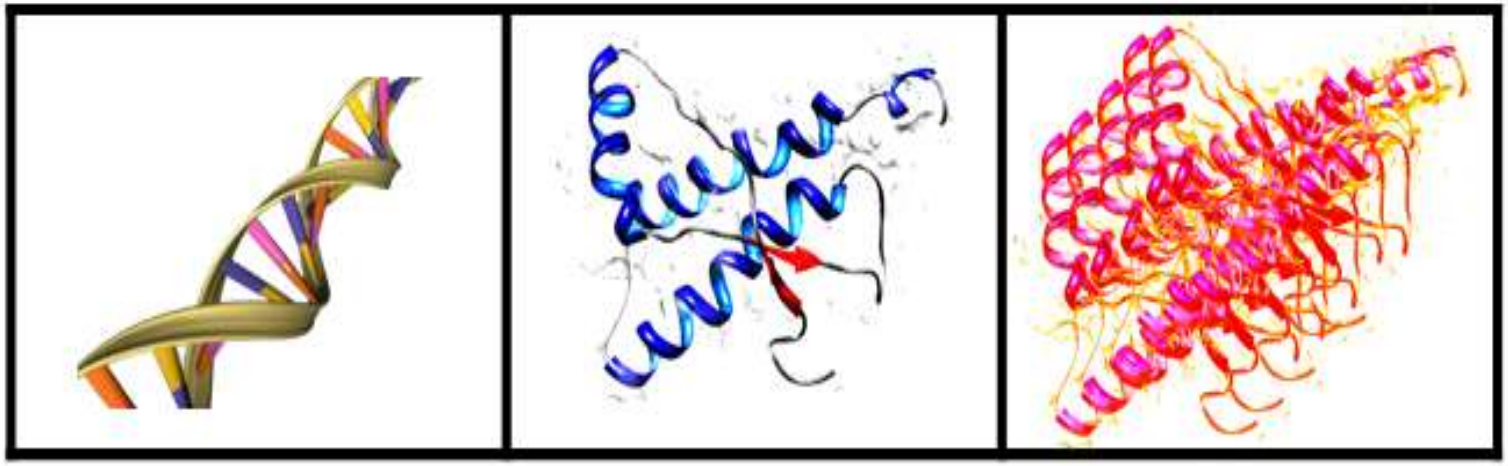
Genome

- Genome is absent
- No DNA
- No RNA

Gene



- single exon of a single-copy chromosomal gene
- PrP gene in mammals contain one or two 5’-non-coding exon



PRNP is a gene in your DNA which encodes for prion protein

Prion protein or **PrP** is a protein on the surface of your cells

A **prion** is an infectious particle made up of misfolded prion proteins

Protein

- Itself a protein
- Misfolded
- That makes other alike proteins misfolded
- causes disease condition in host

Structure

- Misfolded/ abnormally folded protein
- Pathogenic
- Causes similar proteins to misfold too

END

Ppt no. 168

PRIONS

Pathology of Disease

- TSE are caused by abnormal folding of prions
- The normal prion protein, which is designated as PrPc, is a 35kD membrane glycoprotein, which is water-soluble and proteinase-sensitive.
- Abnormal prions, designated as PrPSc or PrPTSE

Pathogenesis Of Prion Diseases

- The unique feature of prion diseases is that they are self-propagating and transmissible.
- Once PrPSc is generated endogenously or introduced into the body from the environment, it converts normal prions into abnormal ones.
- This conversion begins with the initial production of a small polymer of misfolded prions, (a seed), perhaps no more than 28 molecules.

Pathogenesis Of Prion Diseases

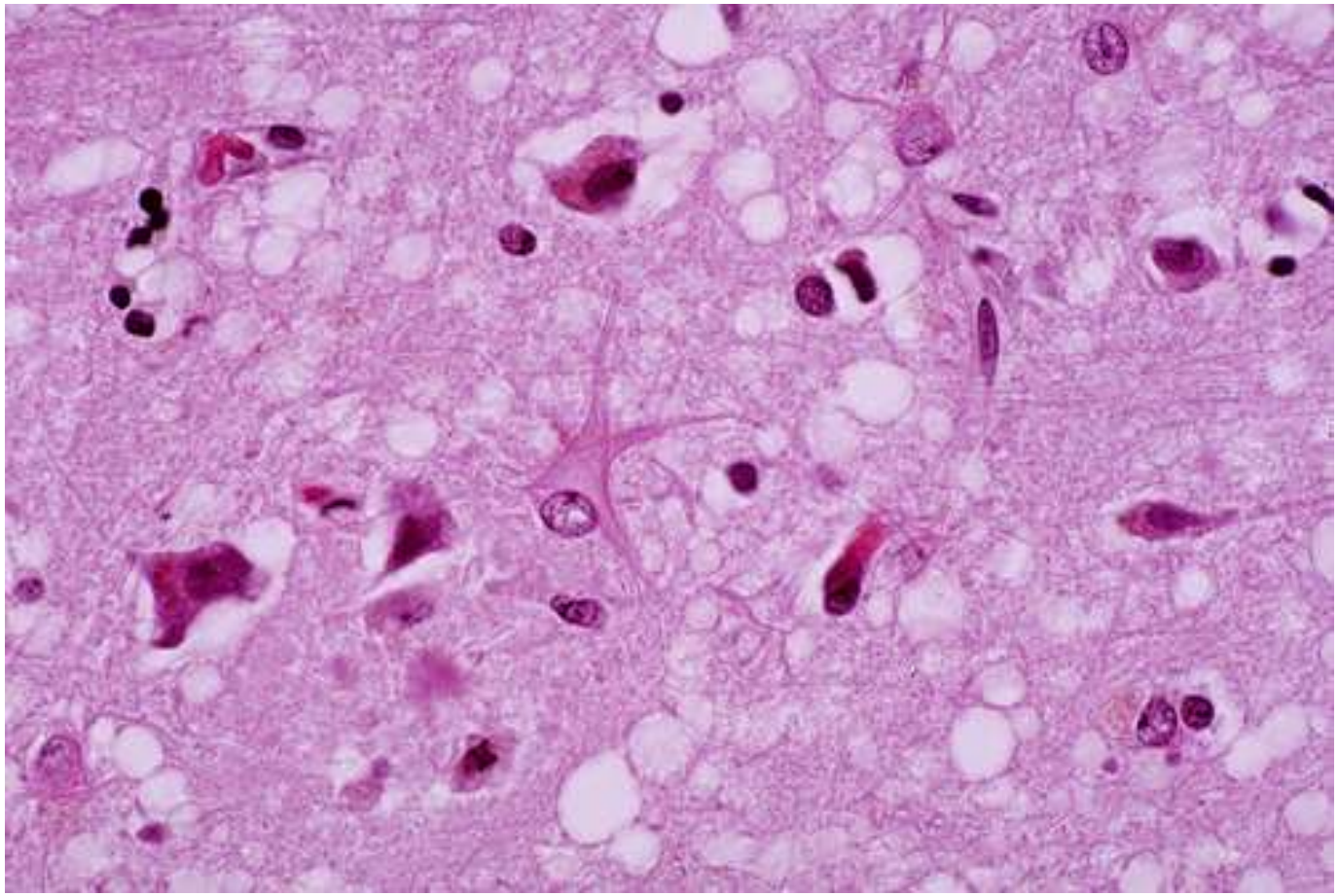
- This seed converts normal adjacent prions into abnormal ones by an unknown mechanism.
- As more PrPSc polymers are produced, they, in turn, act as seeds, propagating the conversion of normal to abnormal prions.
- This novel mechanism of disease is also implicated in the propagation of protein misfolding that is involved in Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis.

Pathology Of Prion Diseases

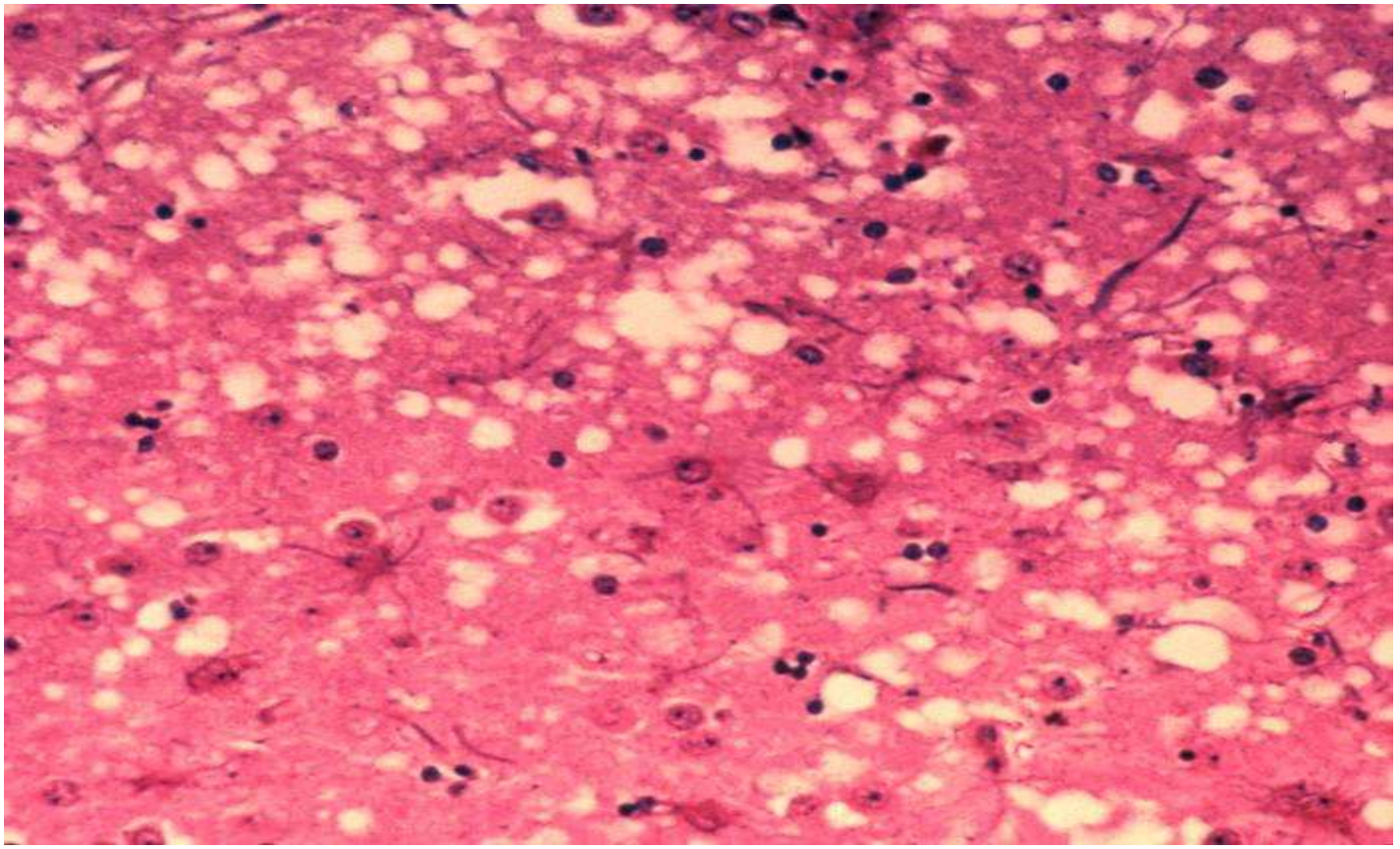
- During the incubation period and active clinical phase of prion diseases, PrPTSE is probably present (and can transmit the disease) in all tissues and body fluids, especially brain, nerves, and skeletal muscle.
- Pathology develops only in the brain.
- Initially, intracytoplasmic vacuoles appear in neurons.
- As the disease progresses, vacuolization becomes more pronounced and the cortical neuropil develops a spongy appearance, hence the term spongiform encephalopathy.

Pathology Of Prion Diseases

- Advanced cases show nerronal loss, gliosis, and brain atrophy.
- Cerebellar atrophy is usually severe.
- Unlike most cerebellar degenerations, there is more pronounced loss of granular neurons
- There is no inflammation. The CSF is normal.
- The changes are confined to the gray matter (the primary pathology involves the neuronal body).



Spongyform encephalophathy



TSE BY PRIONS

Scrapies

Ppt no. 169

Etiology

- Neurodegenerative disease caused by prions
- Replicate by converting normal proteins into copies of prions
- non-conventional disease agents

Host species

- Sheep
- Goat
- Mice
- Rat
- Monkey
- And other Lab animals

Geographical distribution

- World wide
- More in developing countries

Transmission

- Well understood in sheep
- To goats if kept with sheep
- Within herd
- During pregnancy
- Through urine

Symptoms

- Affected sheep will apart from flock
- Trembling
- Convulsions
- Scraping body with wall
- Drink small quantities of water often
- Blindness

END

Ppt no. 170

TSE BY PRIONS

Transmissible Mink Encephalopathy

Etiology

- Caused by prions
- Progressive and fatal neurodegenerative disease

Host species

- Minks
- Once signs are evident, death is inevitable
- Raccoons
- Skunks
- Ferrets
- Hamsters etc

Geographical distribution

- In USA from 1947 to 1985
- Canada
- Finland
- Germany

Transmission

- Orally
- Wounds on tongue
- Mucous membrane
- Skin

Symptoms

- Difficulty in eating and swallowing
- Hyperexcitable
- Incoordination
- Clenching of jaw
- Covulsions
- Death

END

Ppt no. 171

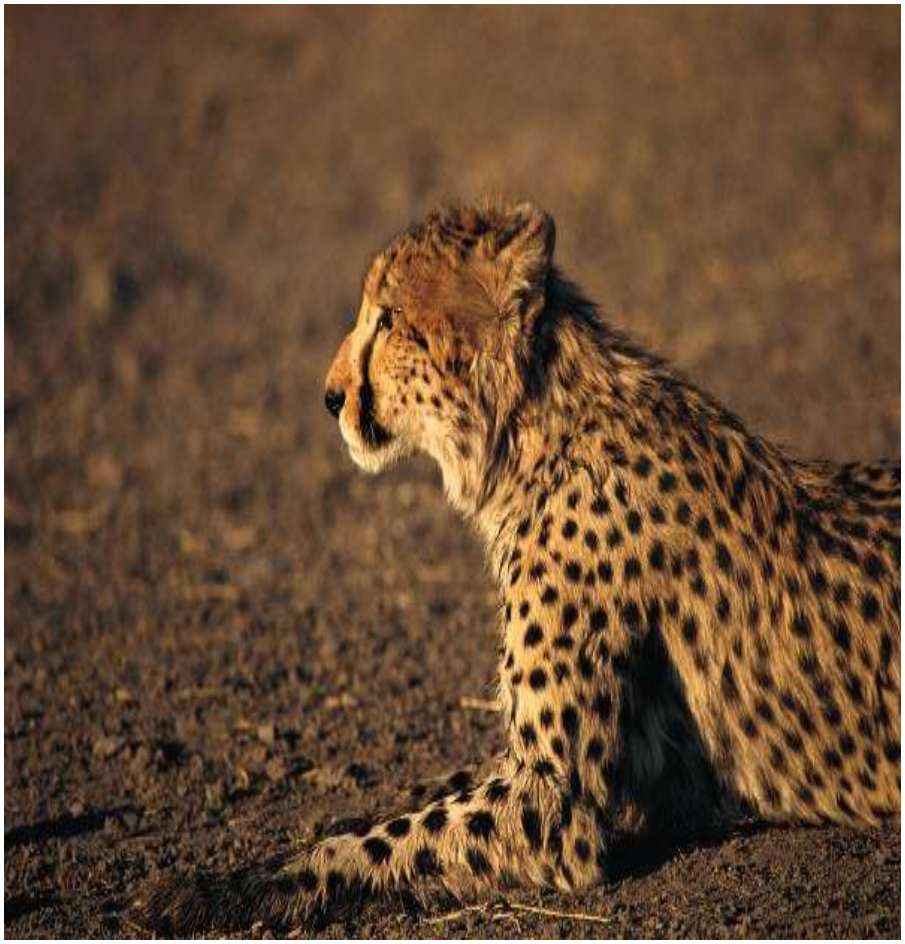
TSE BY PRIONS

Feline Spongiform Encephalopathy

Etiology

- Caused by prions
- Neurodegenerative disease
- Same agent responsible for Bovine Spongiform encephalopathy

Host species



- Domesticated cats
- Cheetahs
- Puma
- Tiger
- Golden cat
- Lion
- Leopard

Geographical distribution

- Most cases in UK
- Australia
- North America

Transmission

- Eating BSE contaminated tissue
- No horizontal transmission
- Possible maternal transmission
- Iatrogenic transmission in cheetah

Symptoms

- Aggression
- Hiding
- Gait abnormalities
- Hyperesthesia
- Excessive salivation
- Tremors
- Death

END

Ppt no. 172

TSE BY PRIONS

Chronic Wasting Disease

Etiology

- Caused by prions
- Contagious neurological disease
- Causes a characteristic spongy degeneration of the brains of infected animals resulting in emaciation, abnormal behavior, loss of bodily functions and death.

Host

- Mule deer
- white-tailed deer
- elk
- moose
- reindeer

Signs & symptoms

- Difficulties in movement
- Weight loss over time
- Listlessness
- Lowering of head
- Excessive salivation
- Nervousness
- tremors

Transmission

- Contact with infected animals
- Their body tissues
- Body fluids

END

Ppt no. 173

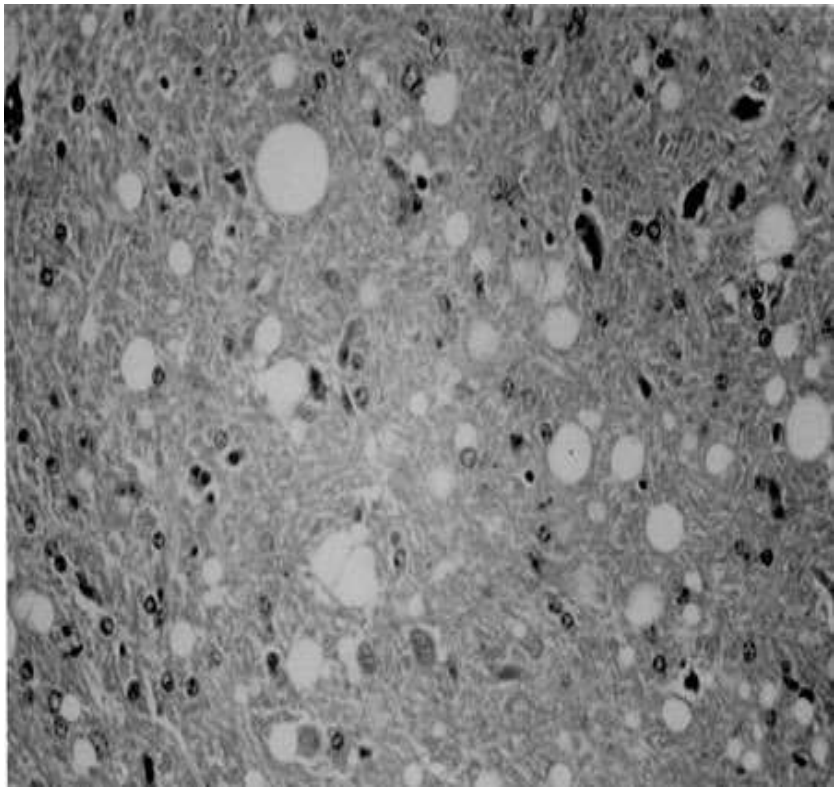
TSE BY PRIONS

Bovine Spongioform Encephalopathy

Etiology:

- Caused by prions
- Affects nervous system
- Also called **mad cow disease**

HOST



BSE shows spongy appearance of the brain tissue of infected cattle when sections are examined under a microscope.

Signs:

- Apprehension
- Hyperaesthesia
- Frequent licking with progressive paresis, & ataxia
- Circling

Transmission

- from the carcasses of scrapie-infected sheep.

- sheep by products infected with scrapie is used to feed cattle with the meat and bone mill (MBM)

END

Ppt no. 174

TSE BY PRIONS

In Human

Classic CJD (Creutzfeldt-Jakob disease)

- Fatal Familial Insomnia
- Kuru

Classic CJD (Creutzfeldt-Jakob disease):

- Most prevalent of spongiform diseases
- 1 out of million people
- 10% cases are inherited mutations in PRPN gene
- Strikes peopleage 50-75.

Symptoms:

dementia, muscle twitching, vision problems

Diagnosis:

- Post mortem definitive diagnosis
- Amyloid plaques surrounded by vacuoles
- Prion protein accumulation in cerebellum
- Spongiform appearance in gray matter

Fatal familial Insomnia:

- All case are inherited mutations in Prp gene
- Strikes people age 36-61
- Disruption of sleep/wake cycle leads to coma, then death.

Kuru:

Struck members of Fore tribe in 1950s and 1960s

Signs:

- Muscle weakness
- loss of coordination
- tremors
- inappropriate episodes of laughter or crying

Transmission:

Transmitted by ritual cannibalism as part of funeral ceremonies.

END

Ppt no. 175

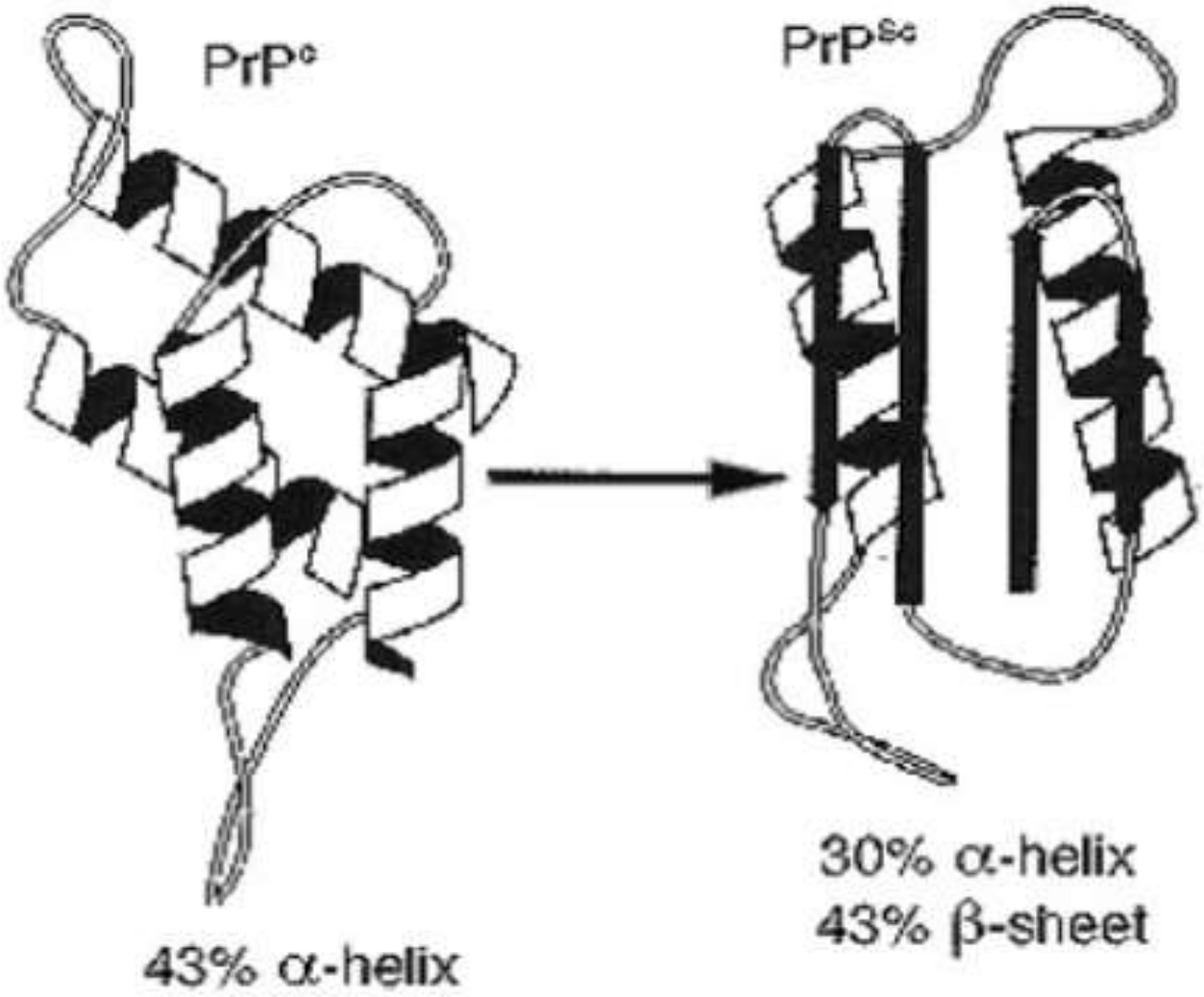
MOLECULAR BIOLOGY OF PRIONS

Prion protein:

- Abnormal isoform of normal protein
- Located in cell surface (brain, nervous system)
- Protects cell against oxidative stress (free radicals lacking electrons)

- PrP^{c} =prion protein cellular(normal form)
- PrP^{sc} =scrapie form (infectious form)

PrP^{c}	PrP^{Sc}
Rich in alpha helices	Rich in beta sheets
Soluble	Insoluble
PK (protease) sensitive • Definitive shelf life • Degrades when no longer needed	PK resistant
No aggregation • Cannot polymerize into aggregates	Aggregation • Allow polymerization of fibrils • Due to beta sheets


PrP^c
PrP^Sc
43% α-helix
30% α-helix
43% β-sheet

Ppt no. 176

DEVELOPMENT OF CULTURING METHOD FOR VIRUSES

Introduction

- First cell line developed in 1912 by Carrel and his coworkers
- From fragments of explanted chicken embryo heart
- Mantained by washing with Ringer solution

Introduction

- Developed in 1940s and 1950s
- By John Franklin Enders, Thomas Huckle Weller and Frederick Chapman Robbins

First culture

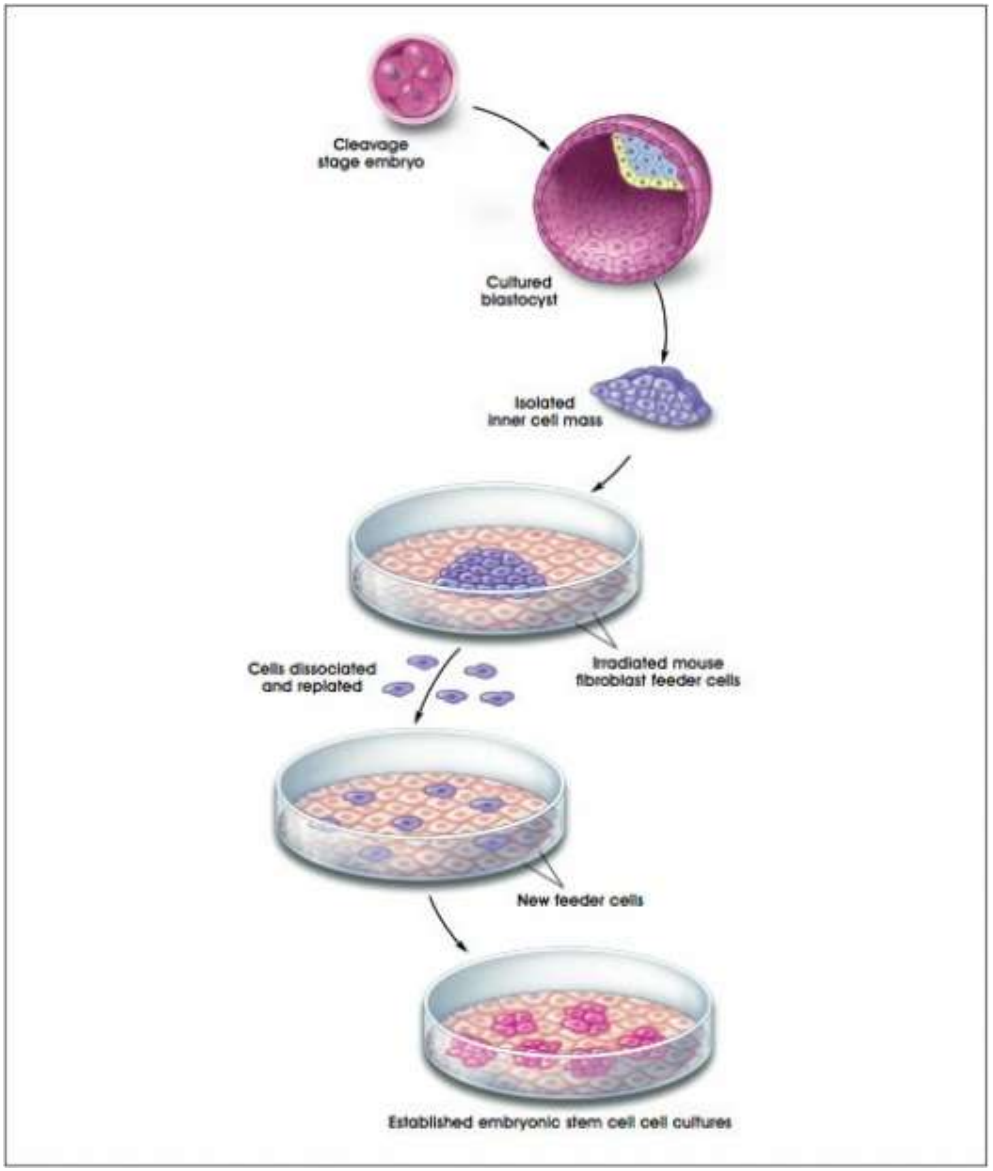
- Received Nobel Prize
- Culturing viruses in Monkey kidney cells

Media

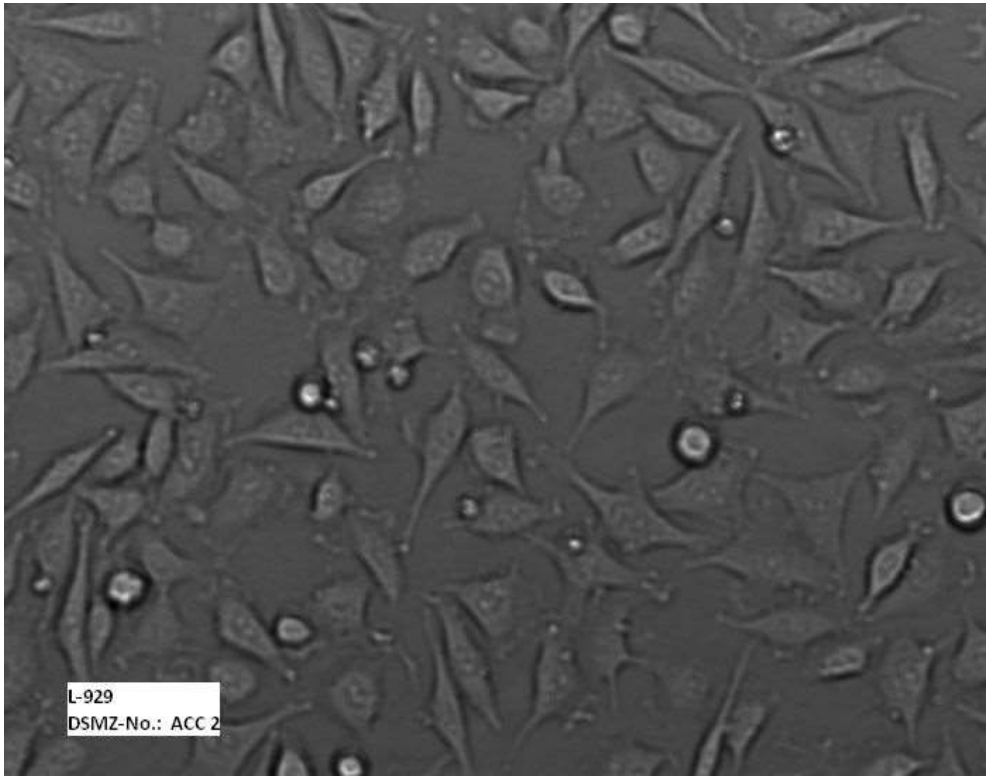
- Early use was naturally derived products from serum
- Later use of chemical-based synthetic media

Observation

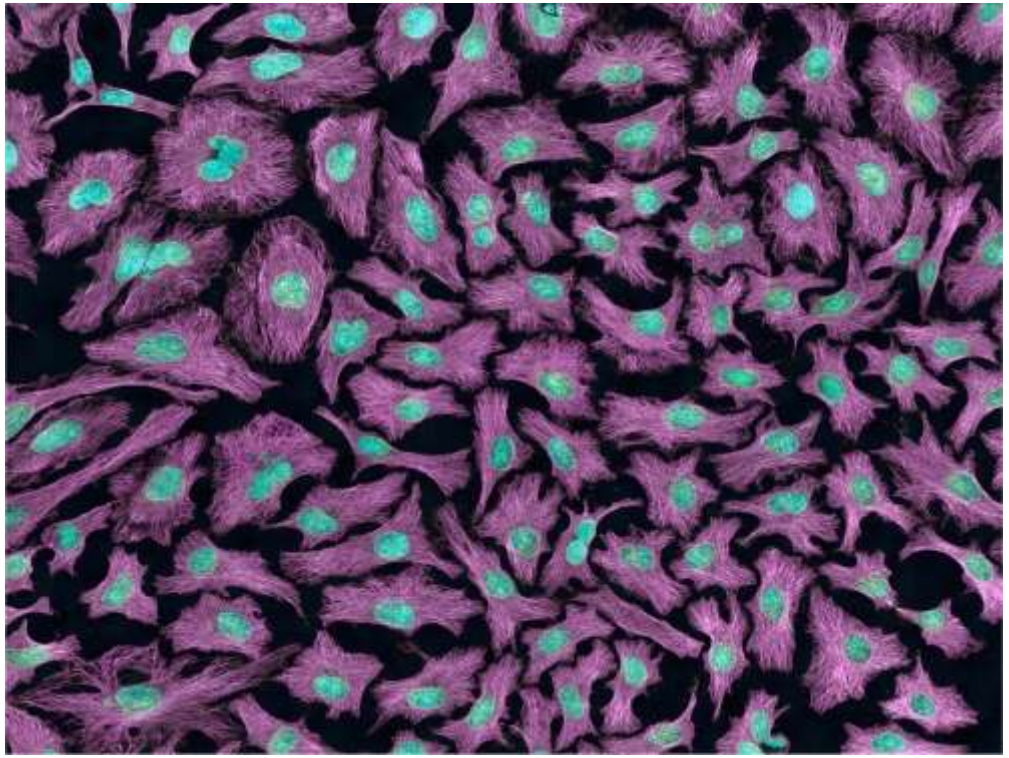
- Using modern microscopy techniques
- Possibility to observe inclusions in cells as a result of virus


Cleavage stage embryo
Cultured blastocyst
Isolated inner cell mass
Irradiated mouse fibroblast feeder cells
Cells dissociated and replated
New feeder cells
Established embryonic stem cell cell cultures

L- cell line



- Developed by Earle in 1948
- From subcutaneous mouse tissue



HeLa cell line

- In 50s and 60s
- Diploid cell lines
- From human tissue

Vero cell line

- In 50s and 60s
- Diploid cell lines
- From Simian tissue

Earliest derived cell lines

Name	Species and tissues	Morphology	Year of origin
L929	Mouse connective tissue	Fibroblast	1948
HeLa	Human cervix	Epithelial-like	1951
CHO	Chinese Hamster Ovary	Epithelial	1957
MDCK	Canine kidney	Epithelial	1958
WI-38	Human lung	Fibroblast	1961
BHK-21	Syrian Hamster Kidney	Fibroblast	1961
Vero	African Green monkey kidney	Epithelial	1962

Ppt no. 177

INTRODUCTON OF CELL CULTURE

introduction

- Cell culture is the process by which prokaryotic, eukaryotic or plant cells are grown under controlled conditions. But in practice it refers to the culturing of cells derived from animal cells.

- **Major development's in cell culture technology**
- First development was the use of antibiotics which inhibits the growth of contaminants.
- Second was the use of trypsin to remove adherent cells to subculture further from the culture vessel
- Third was the use of chemically defined culture medium.

importance

- Virology

  Cultivation of virus for vaccine production, also used to study there infectious cycle.

- Genetic Engineering

  Production of commercial proteins, large scale production of viruses for use in vaccine production e.g. polio, rabies, chicken pox, hepatitis B & measles

importance

- Gene therapy

  Cells having a functional gene can be replaced to cells which are having non-functional gene

END

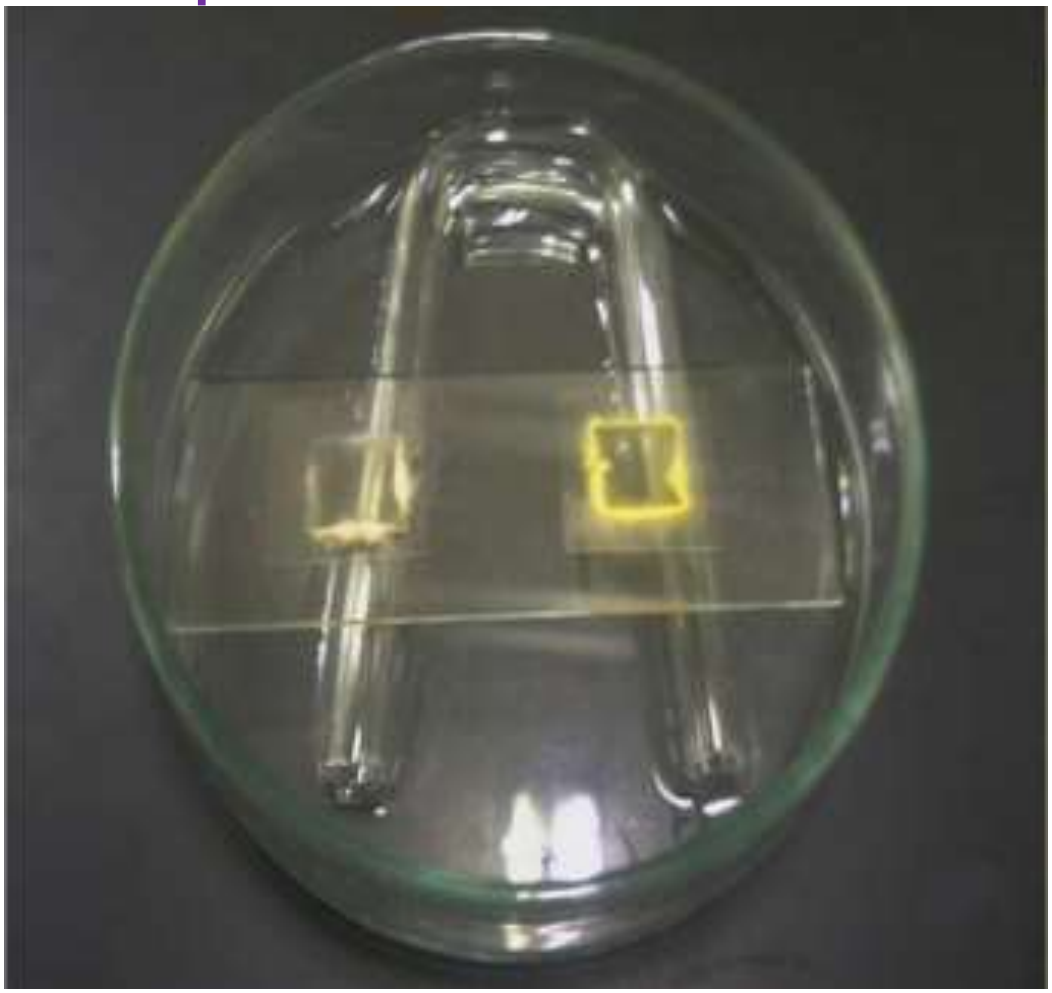
Ppt no. 178

SLIDE CULTURE

- Rapid **method** of preparing fungal colonies for examination and identification.
- Permits the fungi to be studied virtually insitu with as little disturbance as possible.

Procedure:

- A) Slide Culture Preparation



B) Application of Stain

- Place a drop of lactophenol cotton blue stain on a clean microscope slide.
- Remove the cover glass from the slide culture and discard the block of agar.
- Add a drop of 95% ethanol to the hyphae on the cover glass.

B) Application of Stain

- when alcohol has evaporated place the c over glass, mold sided own, on the drop of lactophenol cotton blue stain on the slide.
- Examine the slide under microscope

END

Ppt no. 179

Flask Culture

Laboratory flask

Laboratory flasks are vessels or containers that fall into the category of laboratory equipment known as glassware.

Erlenmeyer flask

- An Erlenmeyer flask, also known as a conical flask or titration flask, is a type of laboratory flask which features a flat bottom, a conical body

Round-bottom flask

Round-bottom flasks (also called round-bottomed flasks or RB flasks) are types of flasks having spherical bottoms

Fernbach flask

- A Fernbach flask is a type of flask suited for large volume cell culture where the culture requires a large surface area to volume ratio.

Ppt no. 180

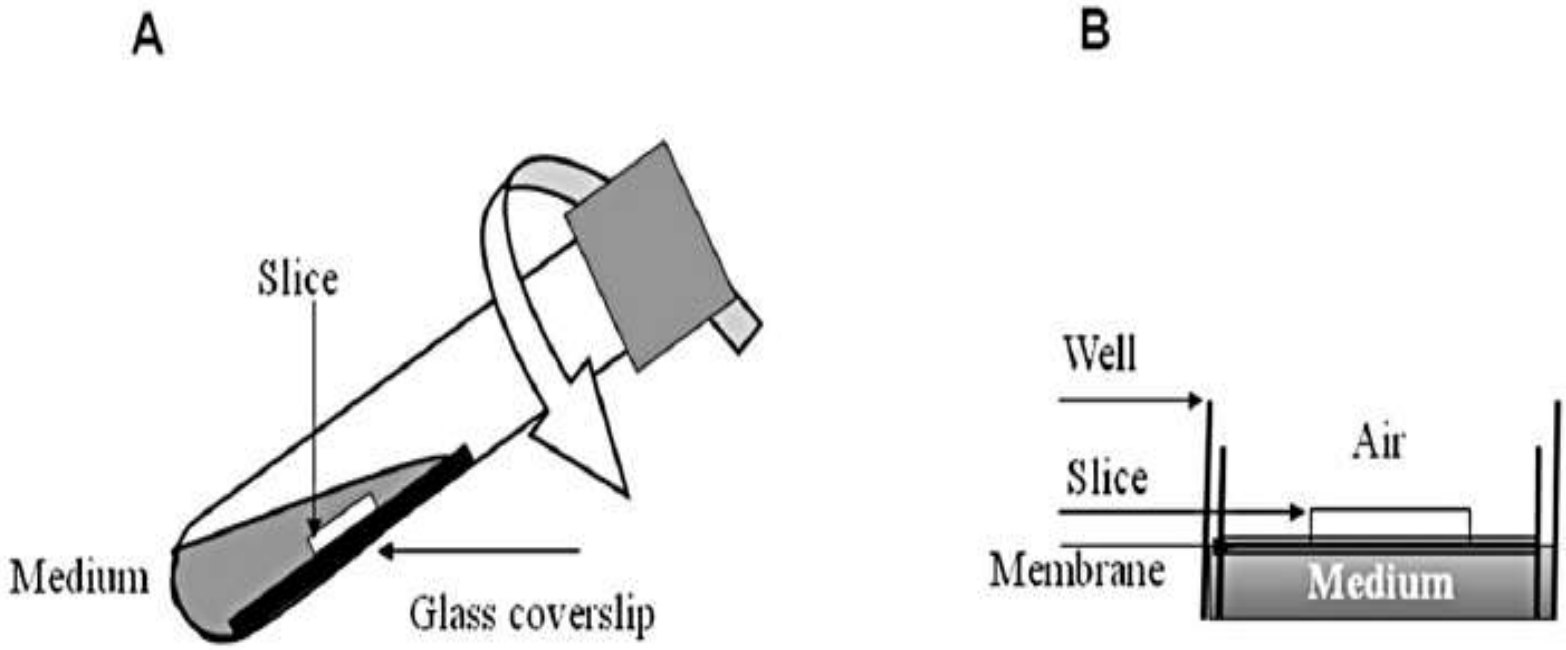
ROLLER TUBE CULTURES

INTRODUCTION

- Grown in test tubes
- Rotated during incubation

PROCEDURE

- Tissue explants placed in test tubes
- Small amount of plasma and agar distributed over the inside surface of tube, so fragments stay in place


A
Slice
Medium
Glass coverslip
B
Well
Air
Slice
Membrane
Medium

PROCEDURE

- As the tube is rotated 8 to 12 rpm
- Tissue explants submerged in nutrient media, only a portion of time

PROCEDURE

- Some cells don't grow satisfactorily if entirely submerged
- Other cells, especially those cultivated through many transfers will stick to glass surface and grown when completely submerged

END

Ppt no. 181

MONOLAYER TISSUE CULTURE

https://www.youtube.com/watch?v=CMRKKl9XSDU

Ppt no. 182

CYTOPATHIC EFFECTS ON TISSUE CULTURE

- **Definition:**

Cytopathic effect or **cytopathogenic effect CPE**) refers to structural changes in host cells that are caused by viral invasion.

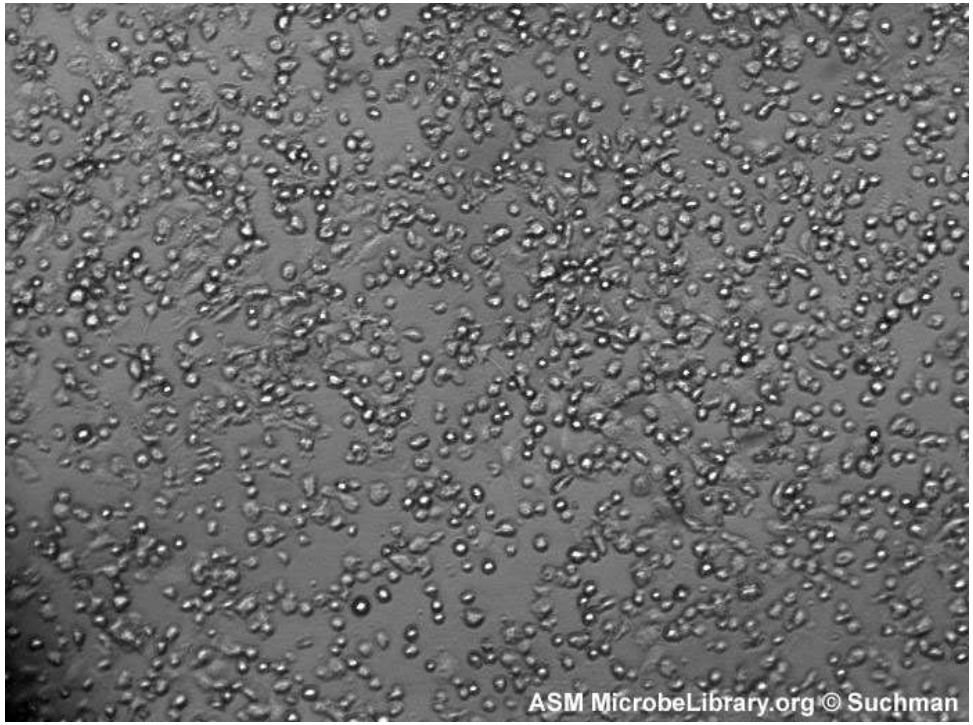
- **Total destruction of the cell monolayer**
- most severe form of CPE. All cells in the monolayer rapidly shrink, become dense (pyknosis), and detach from the glass within 72 hours.
- typical of most enteroviruses.

- **Subtotal destruction**
- detachment (death) of some but not all of the cells in the monolayer.
- togaviruses (alphaviruses), picornaviruses,
- can be observed using the 20X objective

- **Focal degeneration**

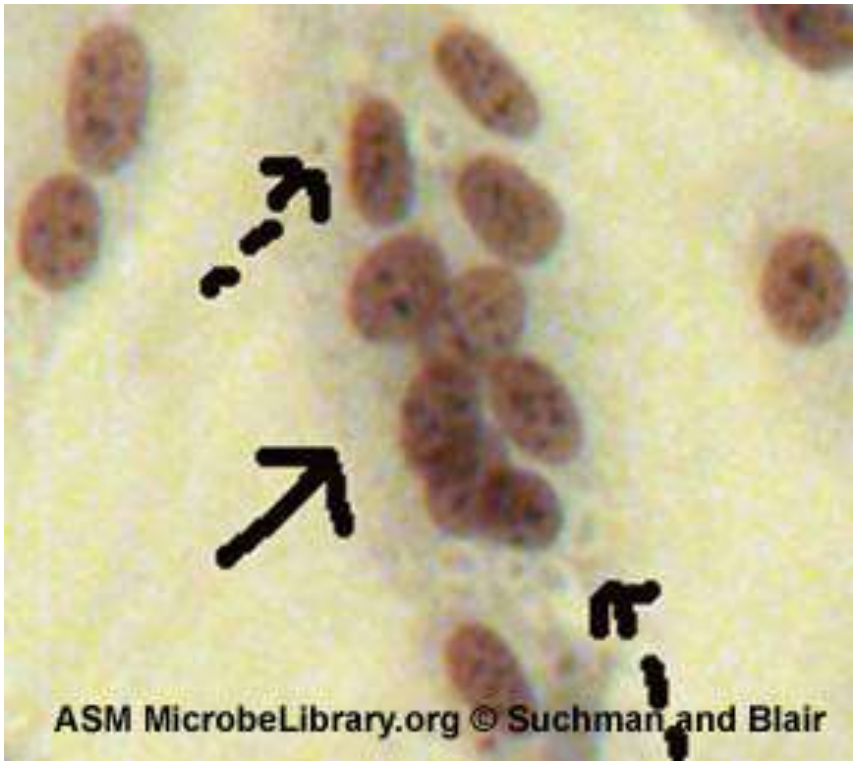
- Herpesviruses and poxviruses.
- viruses produce localized areas (foci) of infection.
- The focal nature of these lesions is due to direct cell-to-cell transfer of virus rather than diffusion through the extracellular medium.

- **Focal degeneration**
- Cells initially become enlarged, rounded, and refractile (more easily seen),then eventually detach from the growth surface leaving cleared areas surrounded by rounded up cells as the infection spreads concentrically



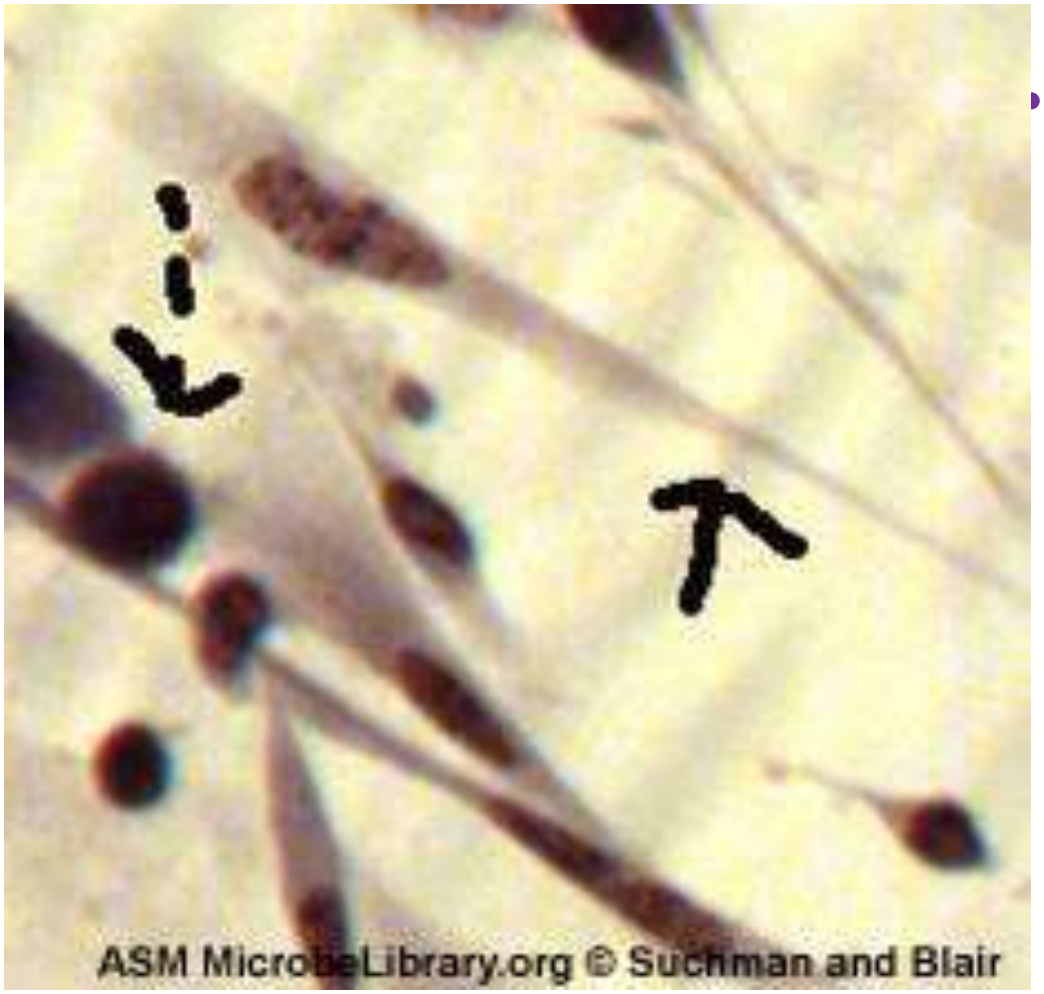
- **Swelling and clumping**
- typical of adenoviruses Infected cells greatly enlarge and clump together in "grape-like" clusters.

- **Foamy degeneration (vacuolization)**
- production of large and/or numerous cytoplasmic vacuoles.
- retroviruses, paramyxoviruses, and flaviviruses (pestiviruses) may cause vacuolization.
- Vacuolization is difficult to visualize without staining.



- **Cell fusion (syncytium or polykaryon formation)**
- fusion of the plasma membranes of four or more cells to produce an enlarged cell with four or more nuclei.
- paramyxoviruses, herpesviruses may produce syncytia as well as other forms of CPE

- **Inclusion bodies:**



cytoplasmic stranding (arrow) and nuclear inclusion bodies (dashed arrow)

Ppt no. 183

VIRUSES AND CANCER

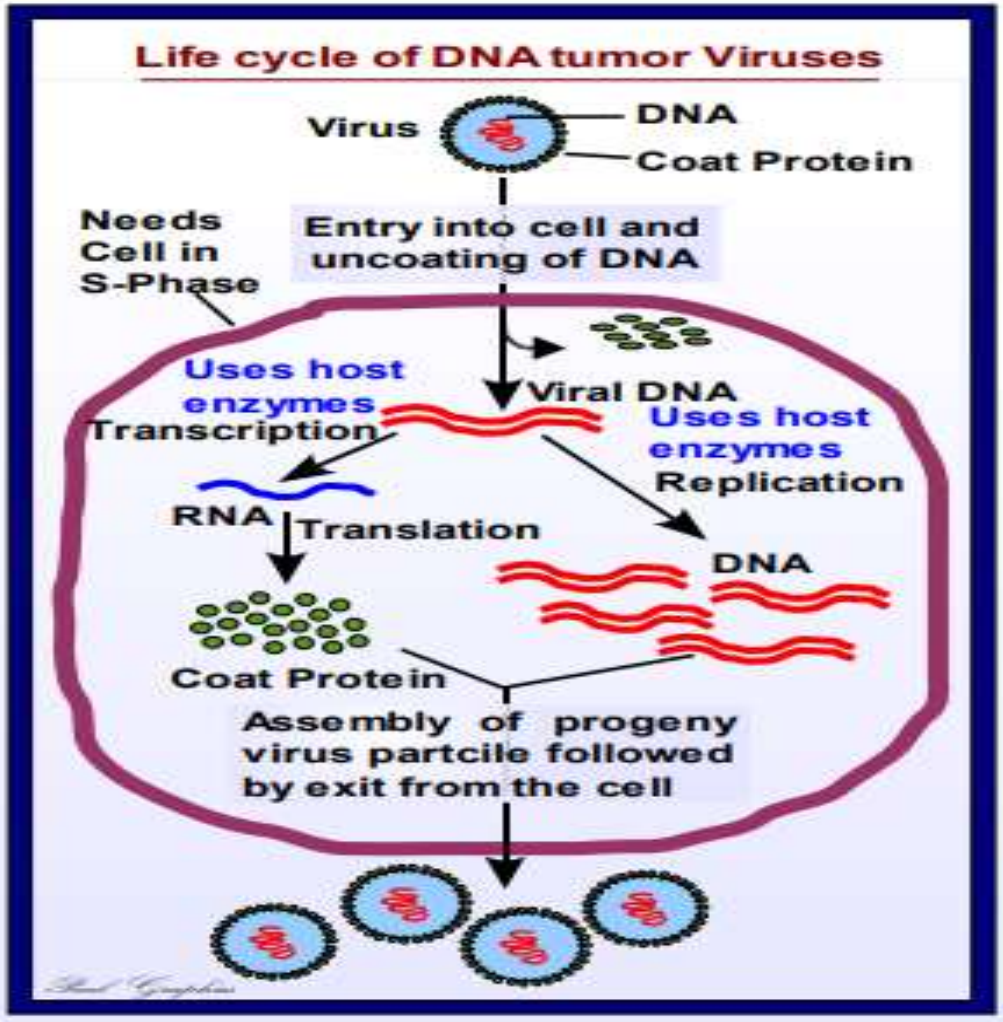
Etiology of cancer:

- Viruses (papilloma,Epstein-Barr,Hepatitis B, retrovirus)
- Radiation
- Environmental/industrial carcinogens
- Immunodeficiency syndromes
- Genetic susceptibility syndromes

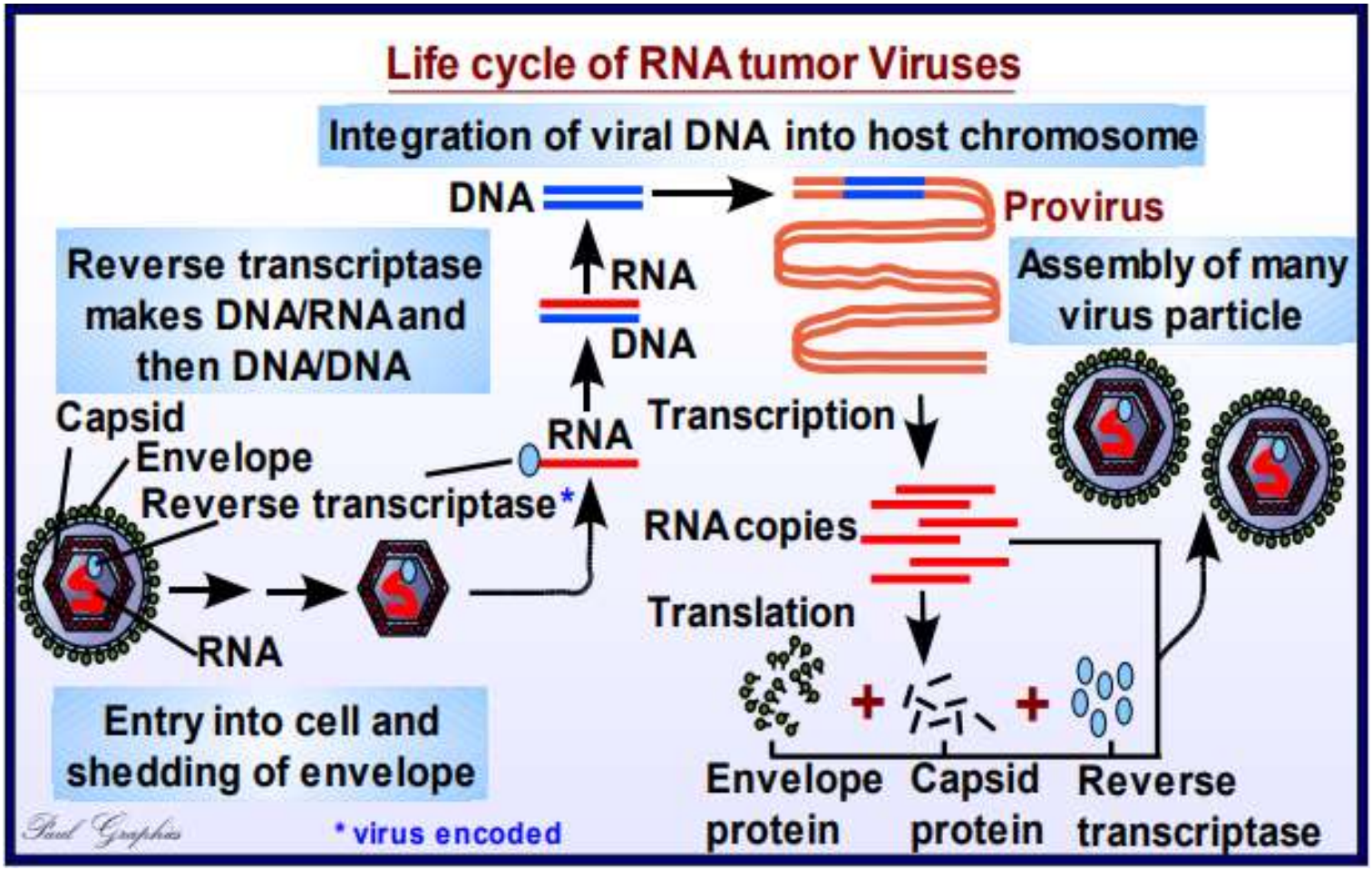
History:

- 1876: russian researcher reported transmission of a tumor from one dog to another
- 1908, **Wilhelm** and **Olaf Bang** in Copenhagen reported the extraact from chicken leukemia can cause tumor.
- In 1909 **Rous** found that extract from chicken sarcoma can cause tumor.

DNA tumor Viruses



RNA tumor Viruses



Ppt no. 184

TRANSFORMATION OF NORMAL CELL TO CANCEROUS CELL

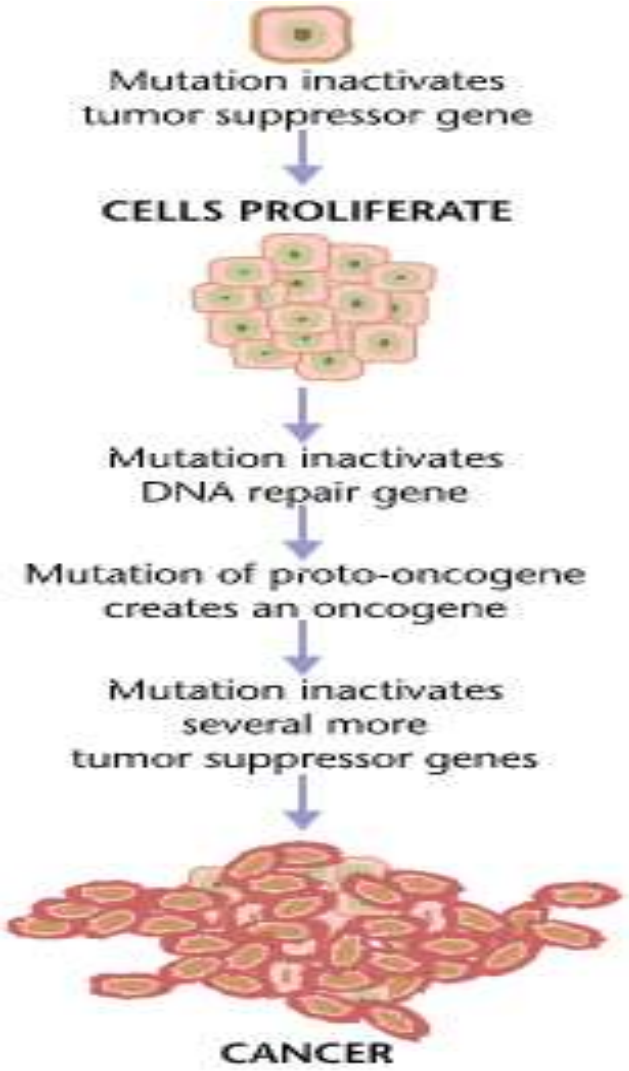
- Oncogene – a gene that when mutated or expressed at abnormally high levels contributes to converting a normal cell into a cancer cell
- Proto-oncogene – the "normal" cellular progenitors of oncogenes that function to promote the normal growth and division of cells

- **Proto-oncogene to oncogene:**
- An alteration occurs in a normal cellular gene (proto-oncogene) that makes the protein hyper-functional (oncogene)

- Proteins involved in the cell signaling pathways are products of proto-oncogenes
  - Proliferative
  - Anti-apoptotic (survival)
  - Angiogenic

- Tumor suppressors are recessive –
- require mutation of both alleles

- Oncogenes are dominant –
- mutation of 1 allele is sufficient

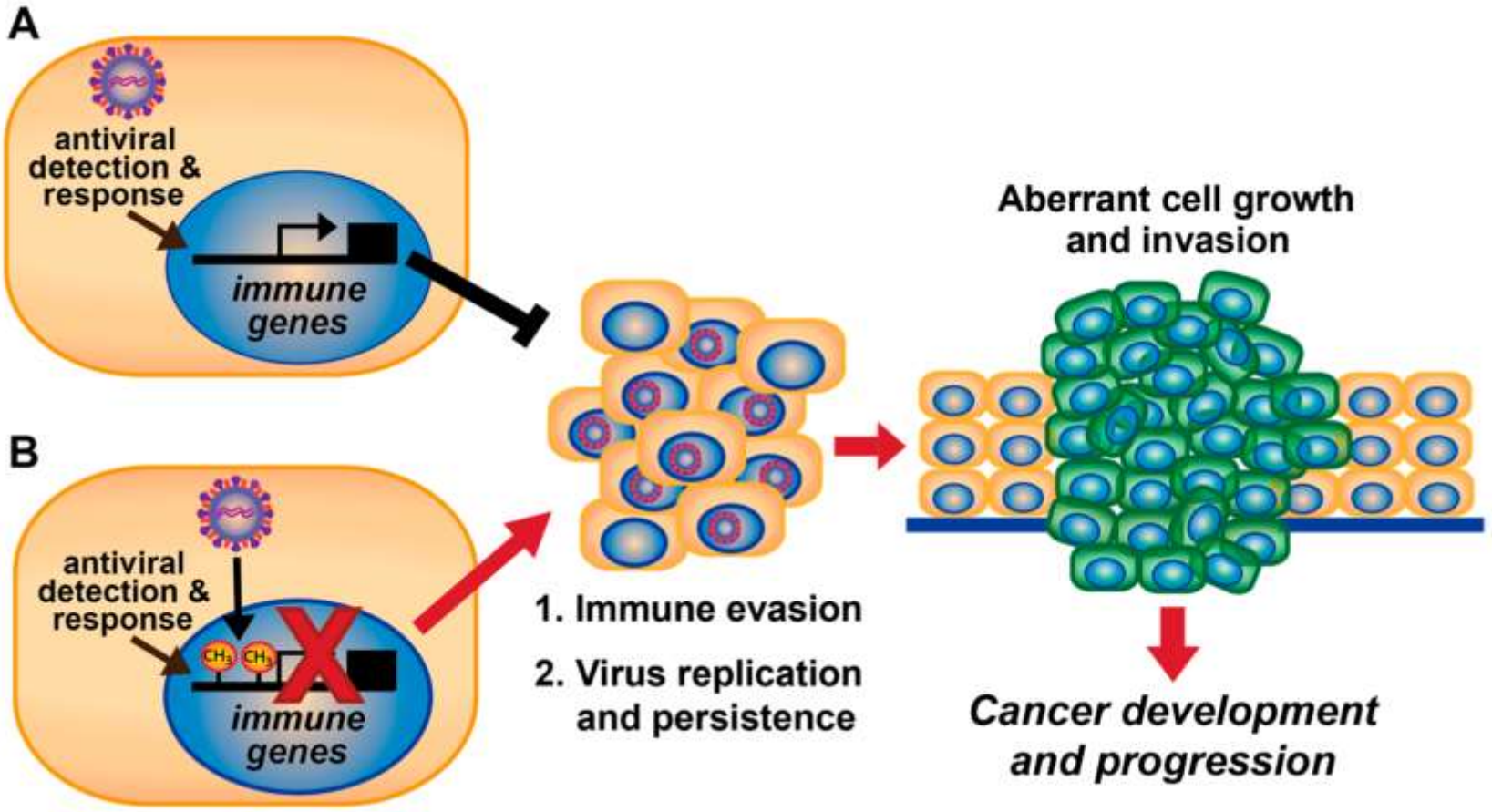

Mutation inactivates
tumor suppressor gene
CELLS PROLIFERATE
Mutation inactivates
DNA repair gene
Mutation of proto-oncogene
creates an oncogene
Mutation inactivates
several more
tumor suppressor genes
CANCER

Ppt no. 185

DNA TUMOR VIRUSES

DNA Oncogenic viruses

- Papilloma virus
- Polyoma virus
- Adenovirus
- Herpes virus
- Hepatitis B virus


A
antiviral
detection &
response
immune
genes
B
antiviral
detection &
response
CH3
CH3
immune
genes
1. Immune evasion
2. Virus replication
and persistence
Aberrant cell growth
and invasion
Cancer development
and progression

PAPILLOMA VIRUS

- DNA complexed with histone proteins
- Human penile, uterine, cervical and anal carcinomas
- Warts

PAPILLOMA VIRUS

POLYOMA VIRUS

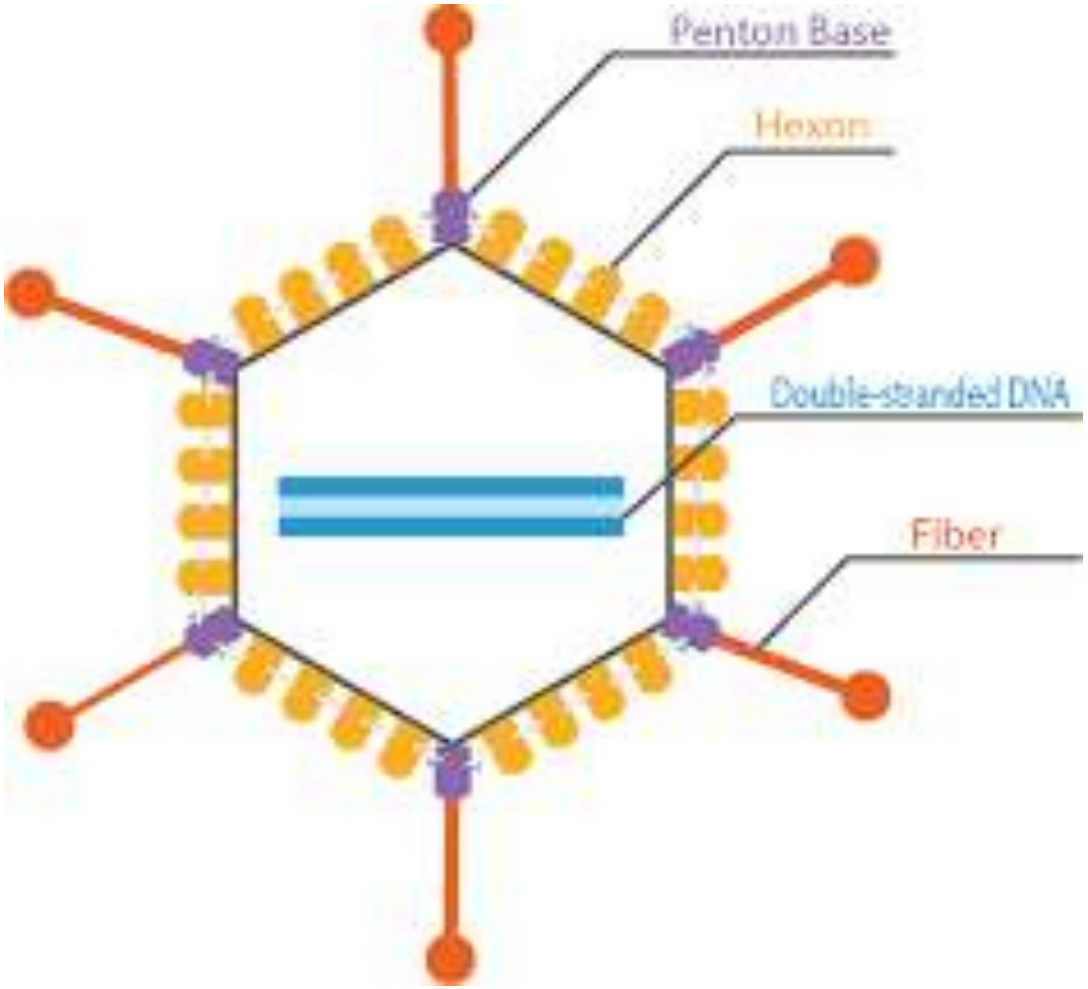
- Mouse Polyoma virus
- Simian Polyoma virus
- Human Polyoma Virus

POLYOMA VIRUS

ADENO VIRUS

- Highly oncogenic
- Host makes antibodies against own T-cells

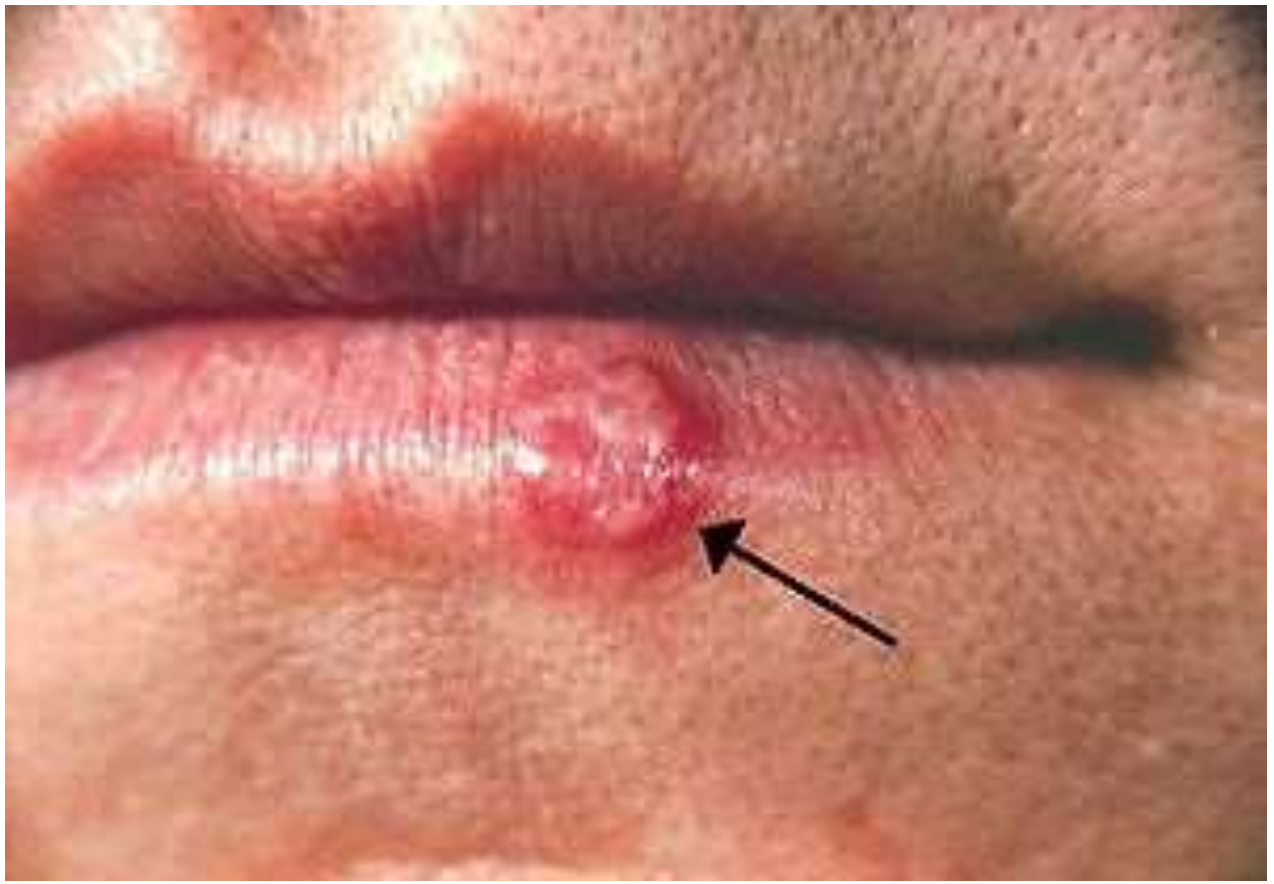
ADENO VIRUS



HERPES VIRUS

- Large size
- Epstien barr virus
- Kaposi Sarcoma virus
- Human cytomegalovirus

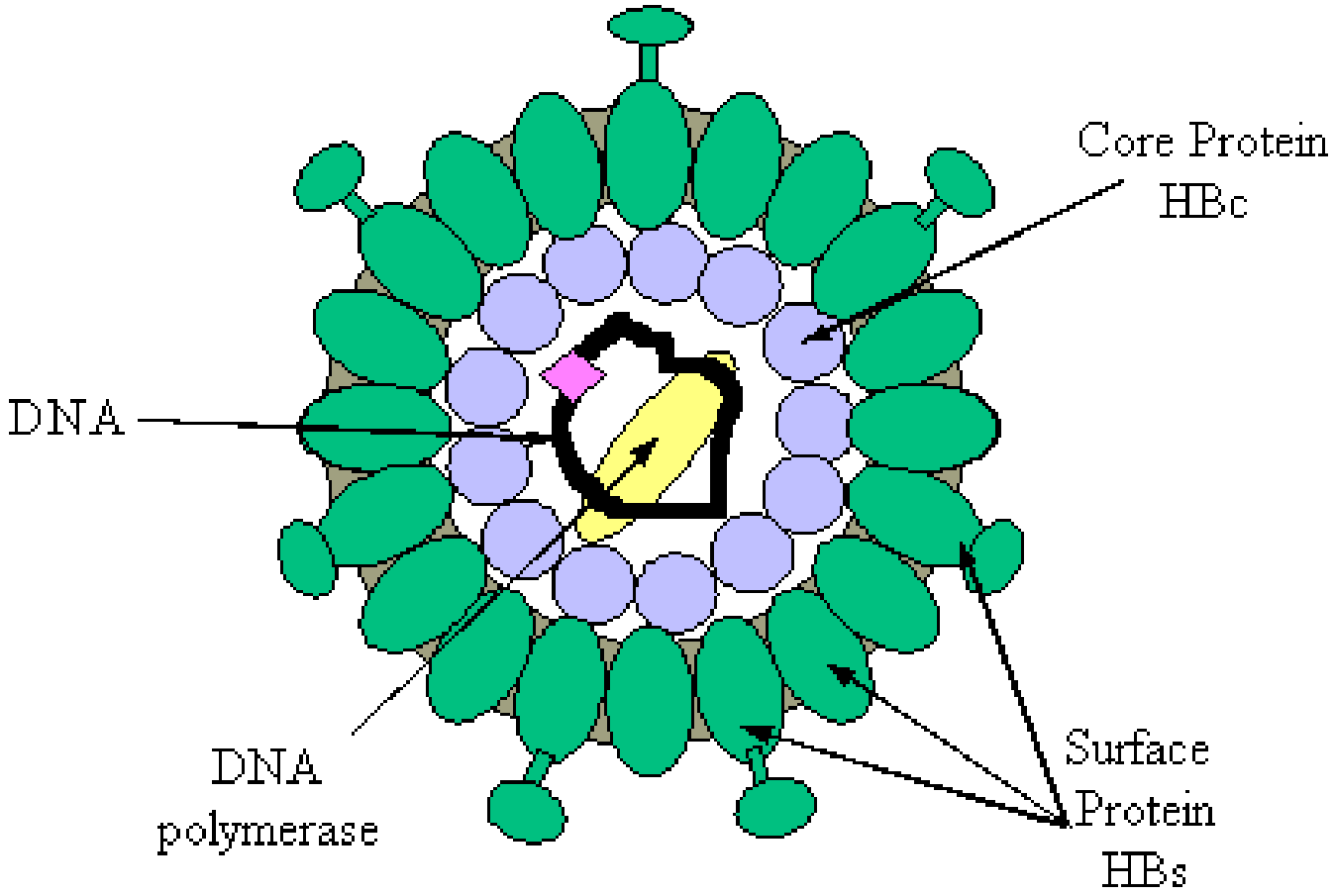
HERPES VIRUS



HEPADNA VIRUS

- Causes hepatitis B
- Hepatocellular carcinoma
- Reverse transcription like retroviruses

HEPADNA VIRUS



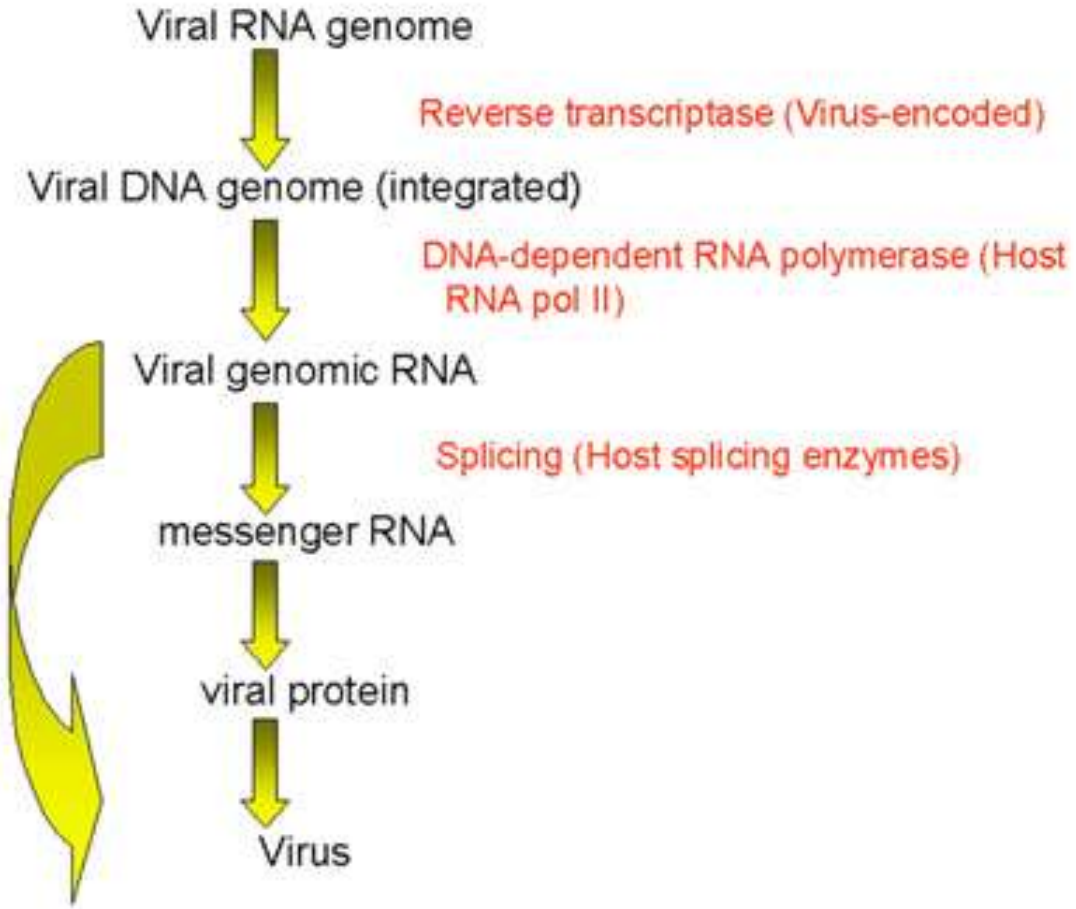
Ppt no. 186

RNA TUMOR VIRUSES

RNA Oncogenic viruses

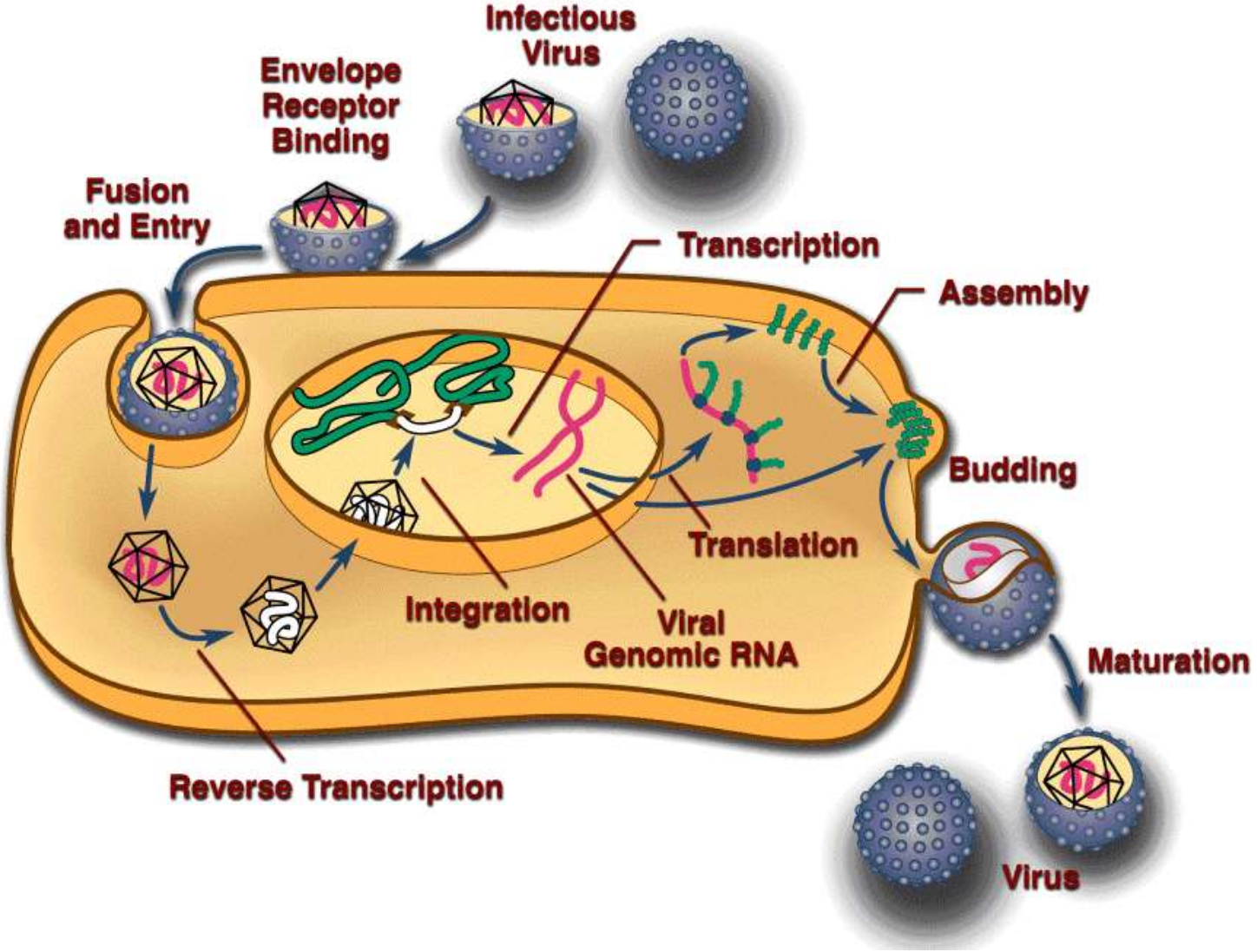
- Oncovirinae virus
- Lentivirinae virus
- Spumavirus

RNA Tumor Viruses



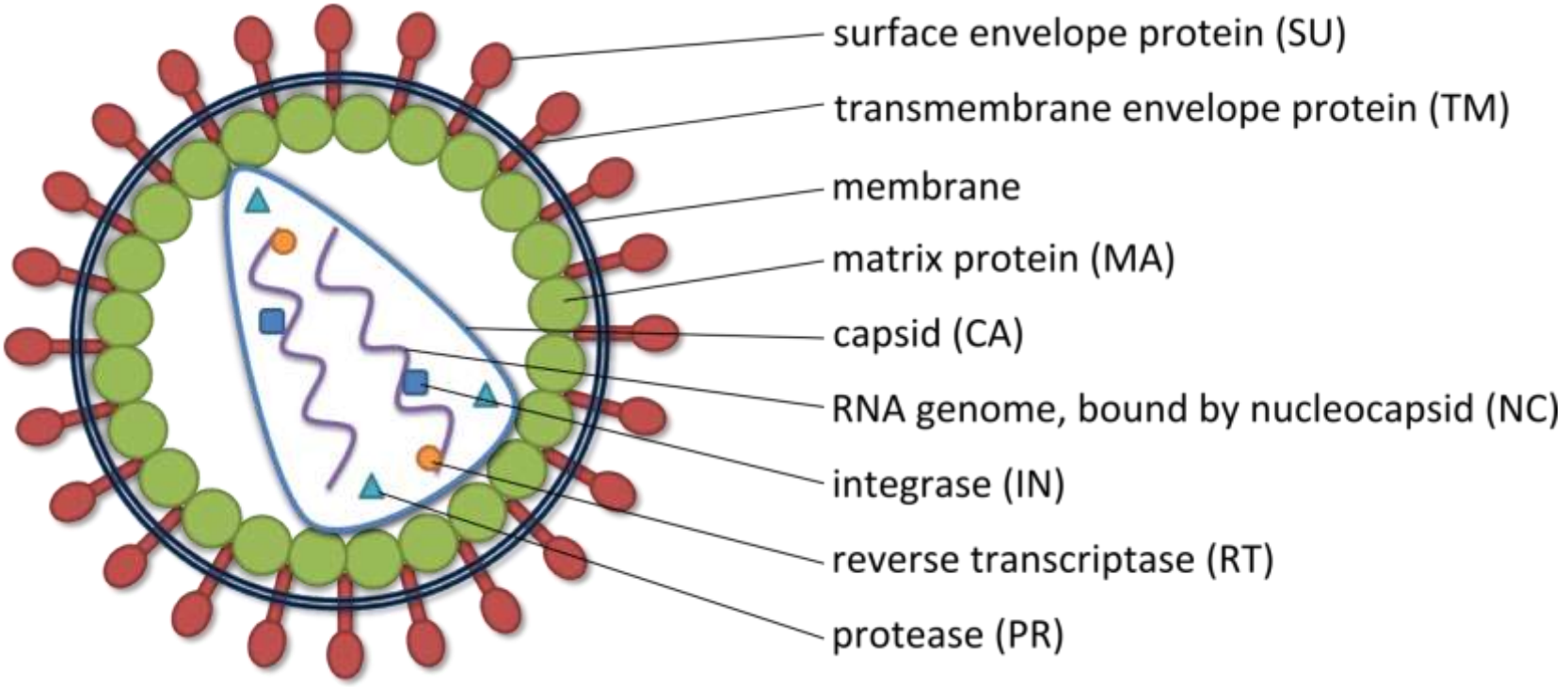
ONCOVIRINAE

- First discovered Rous- Sarcoma virus, showing slow neoplasms in chicken
- Human T-cell lymphotropic virus


Infectious Virus
Envelope Receptor Binding
Fusion and Entry
Transcription
Assembly
Budding
Translation
Integration
Viral Genomic RNA
Maturation
Reverse Transcription
Virus

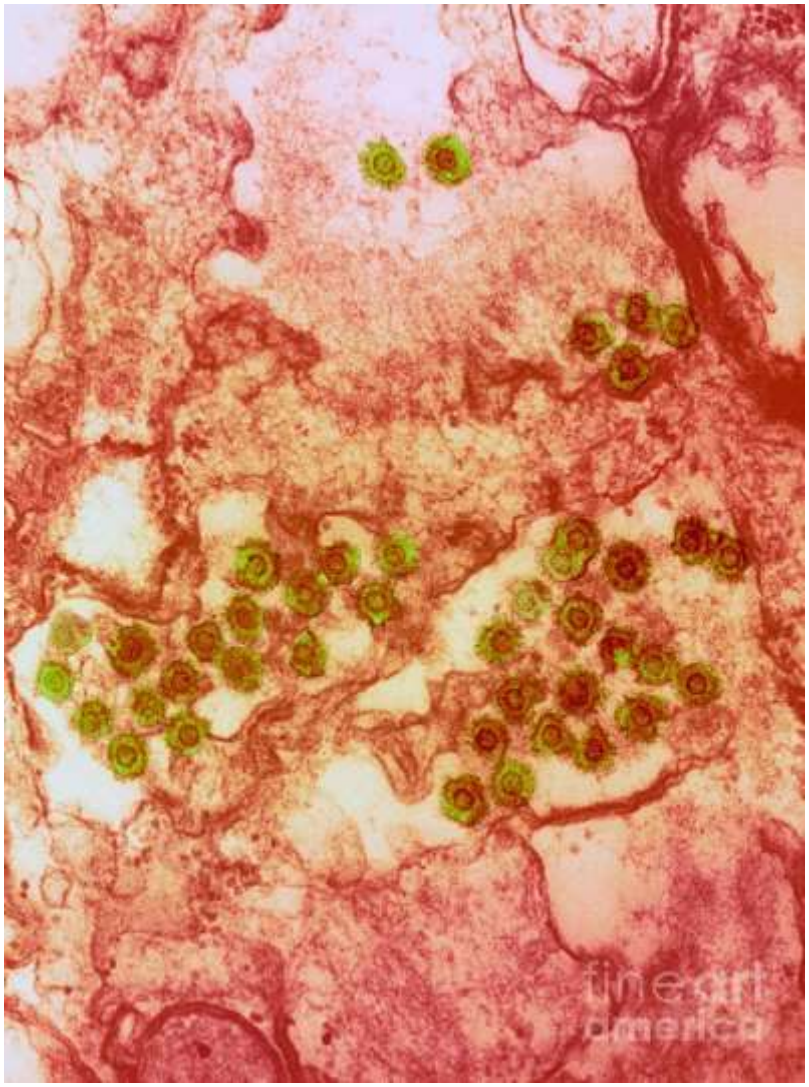
LENTIVIRINAE

- Long latent period of infections
- Disease of ungulates
- HIV that causes AIDS


surface envelope protein (SU)
transmembrane envelope protein (TM)
membrane
matrix protein (MA)
capsid (CA)
RNA genome, bound by nucleocapsid (NC)
integrase (IN)
reverse transcriptase (RT)
protease (PR)

SPUMAVIRINAE

- Foamy appearance of cells
- Syncytia formation of giant multinucleate cells
- Human foamy virus
- Simian foamy virus



Ppt no. 187

PERSISTENT VIRAL INFECTION

- Characterized as those in which the virus is not cleared but remains in specific cells of infected individuals.
- Involve stages of both silent and productive infection without rapidly killing or even producing excessive damage of the host cells.

- There are three types of overlapping persistent virus-host interaction
- latent,
- chronic
- slow infection

- PATHOGENESIS:
- modulation of virus
- cellular gene expression
- modification of the host immune response.

➢ Host immunosuppression → associated with reactivation of a number of persistent virus infections.

Bodily sites of persistence:

- Nervous system
- Glands and body surfaces
- Germ cells
- Blood

Importance:

- Reactivation
- Tumor causing
- Can cause progressive disease
- Persistence enables the parasite to be successful

Ppt no. 189

Plant Viruses and Viroids

- Plant viruses are viruses that affect plants.
- Like all other viruses, plant viruses are obligate intracellular parasites
- That do not have the molecular machinery to replicate without a host.
- Plant viruses can be pathogenic to higher plants.

Pepper mild mottle virus and leaf curl virus

General characteristics

- Most plant viruses are rod-shaped, with protein discs forming a tube surrounding the viral genome
- Isometric particles are another common structure
- They rarely have an envelope
- The great majority have an RNA genome, which is usually small and single stranded (ss), but some viruses have double-stranded (ds) RNA, ssDNA or dsDNA genomes

- Plant viruses are not as well understood as their animal counterparts
- One plant virus has become iconic: tobacco mosaic virus (TMV), the first virus to be discovered
- This and other viruses cause an estimated U.S$60 billion loss in crop yields worldwide each year
- Plant viruses are grouped into 73 genera and 49 families
- However, these figures relate only to cultivated plants, which represent only a tiny fraction of the total number of plant species.

Transmission

- To transmit from one plant to another and from one plant cell to another, plant viruses must use strategies that are usually different from animal viruses.
- Plants do not move, and so plant-to-plant transmission usually involves vectors (such as insects).

Transmission

- Plant cells are surrounded by solid cell walls, therefore transport through plasmodesmata
- Plants have specialized mechanisms for transporting mRNAs through plasmodesmata

Defense system

- Plant defenses against viral infection include, among other measures, the use of siRNA in response to dsRNA.
- Most plant viruses encode a protein to suppress this response.
- Plants also reduce transport through plasmodesmata in response to injury.

END

Ppt no. 190

Infection in Plants-I

Structure of plant Viruses

- These are extremely small and can only be observed under an electron microscope.
- The structure of a virus is given by its coat of proteins, which surround the viral genome.
- Assembly of viral particles takes place spontaneously.

Structure of plant Viruses

- Over 50% of known plant viruses are rod-shaped (flexuous or rigid).
- The length of the particle is normally dependent on the genome but it is usually between 300–500 nm with a diameter of 15–20 nm.
- Protein subunits can be placed around the circumference of a circle to form a disc.
- In the presence of the viral genome, the discs are stacked, then a tube is created with room for the nucleic acid genome in the middle

Structure of plant Viruses

- The second most common structure amongst plant viruses are isometric particles.
- They are 25-50 nm in diameter.
- In cases when there is only a single coat protein, the basic structure consists of 60 T subunits, where T is an integer.
- Some viruses may have 2 coat proteins that associate to form an icosahedral shaped particle.

Structure of plant Viruses

- There are three genera of Geminiviridae that consist of particles that are like two isometric particles stuck together.
- A very small number of plant viruses have, in addition to their coat proteins, a lipid envelope.
- This is derived from the plant cell membrane as the virus particle buds off from the cell

Ppt no. 191

Infection in Plants-II

Through sap

- Viruses can be spread by direct transfer of sap by contact of a wounded plant with a healthy one.
- During agricultural practices, as by damage caused by tools or hands, or naturally, as by an animal feeding.
- Examples, TMV, potato viruses and cucumber mosaic viruses

Insects

- Plant viruses need to be transmitted by a vector, most often insects such as leafhoppers.
- One class of viruses, the Rhabdoviridae, has been proposed to actually be insect viruses that have evolved to replicate in plants

Nematodes

- Soil-borne nematodes also have been shown to transmit viruses.
- They acquire and transmit them by feeding on infected roots.
- Viruses can be transmitted both non-persistently and persistently, but there is no evidence of viruses being able to replicate in nematodes.
- Examples; tobacco ringspot virus and tobacco rattle virus.

Plasmodiophorids

- A number of virus genera are transmitted, both persistently and non-persistently, by soil borne zoosporic protozoa.
- These protozoa are not phytopathogenic themselves, but parasitic.
- Transmission of the virus takes place when they become associated with the plant roots.
- Examples; Polymyxa graminis and Polymyxa betae.

- Plant virus transmission from generation to generation occurs in about 20% of plant viruses.
- When viruses are transmitted by seeds, the seed is infected in the generative cells and the virus is maintained in the germ cells and sometimes, but less often, in the seed coat.
- There does not seem to be a correlation between the location of the seed on the plant and its chances of being infected.
- Examples; families Leguminosae, Solanaceae, Compositae, Rosaceae, Cucurbitaceae, Gramineae.
- Bean common mosaic virus is transmitted through seeds.

Direct plant-to-human transmission

- The Pepper Mild Mottle Virus (PMMoV) may have moved on to infect humans.
- This is a very rare and highly unlikely event
- One possibility is that the virus does not infect human cells directly.
- Instead, the naked viral RNA may alter the function of the cells through a mechanism similar to RNA interference, in which the presence of certain RNA sequences can turn genes on and off

Ppt no. 192

Macroscopic Signs of Plant Viral Diseases

- Important for virus identification
- Used to name the disease
- Solely can not be used to characterize the virus as affected by many factors

local lesions

Necrotic

Chlorotic

Systemic lesions

- Appear in different patterns depending upon virus-host combinations
- Appear in sequential patterns and
- May comprise of different symptoms

Symptoms

- 1. Mosaic
- 2. Mottle
- 3. Yellowing
- 4. Chlorosis
- 5. Vein clearing

Symptoms

6. Vein banding
- Green vein banding
- Yellow vein banding

7. Leaf roll

8. Leaf curl

9. Streak

10. Blistering

END

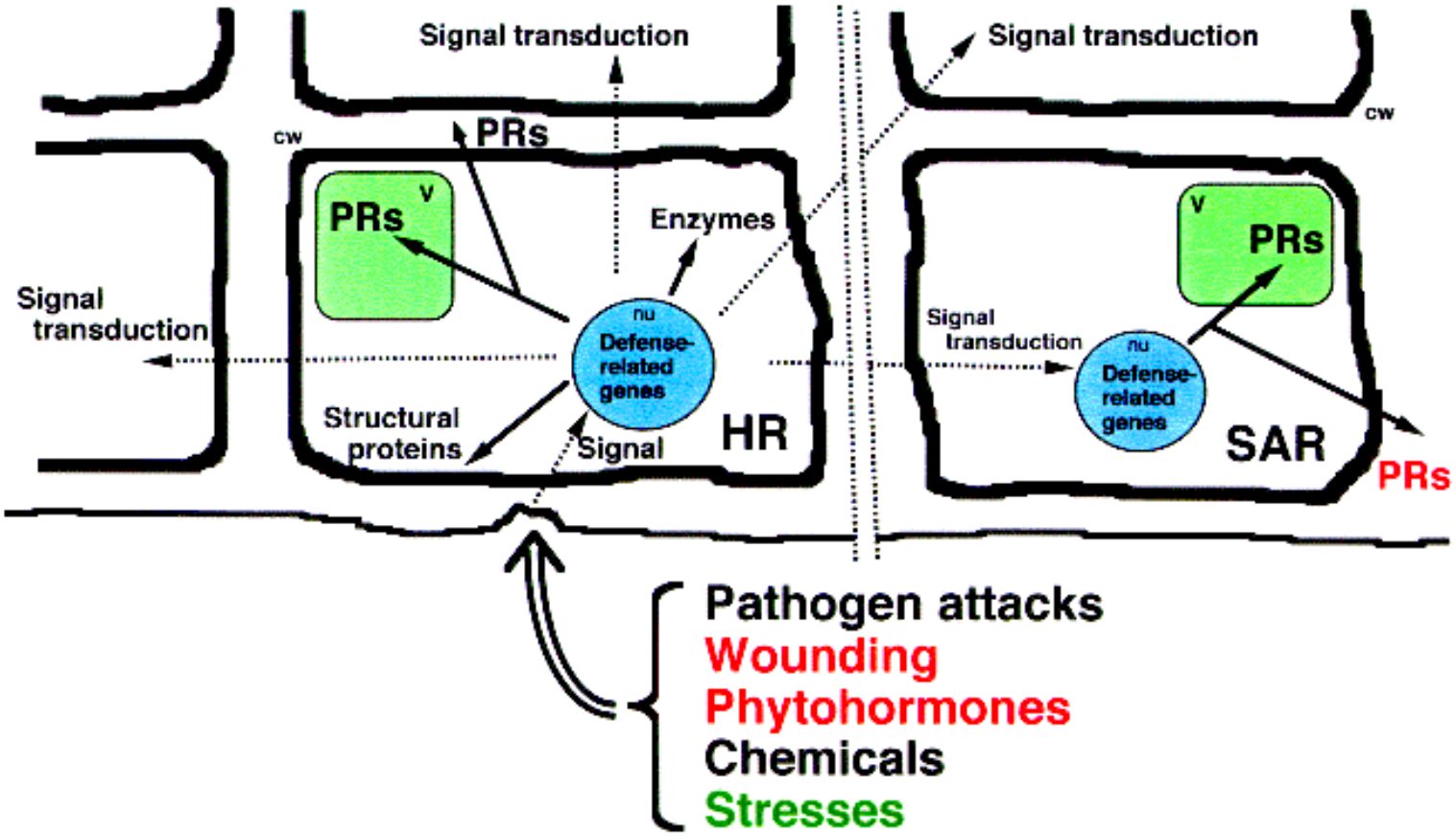
Ppt no. 193

Synthesis of Pathogen Related Proteins

Pathogen Related (PR) Proteins

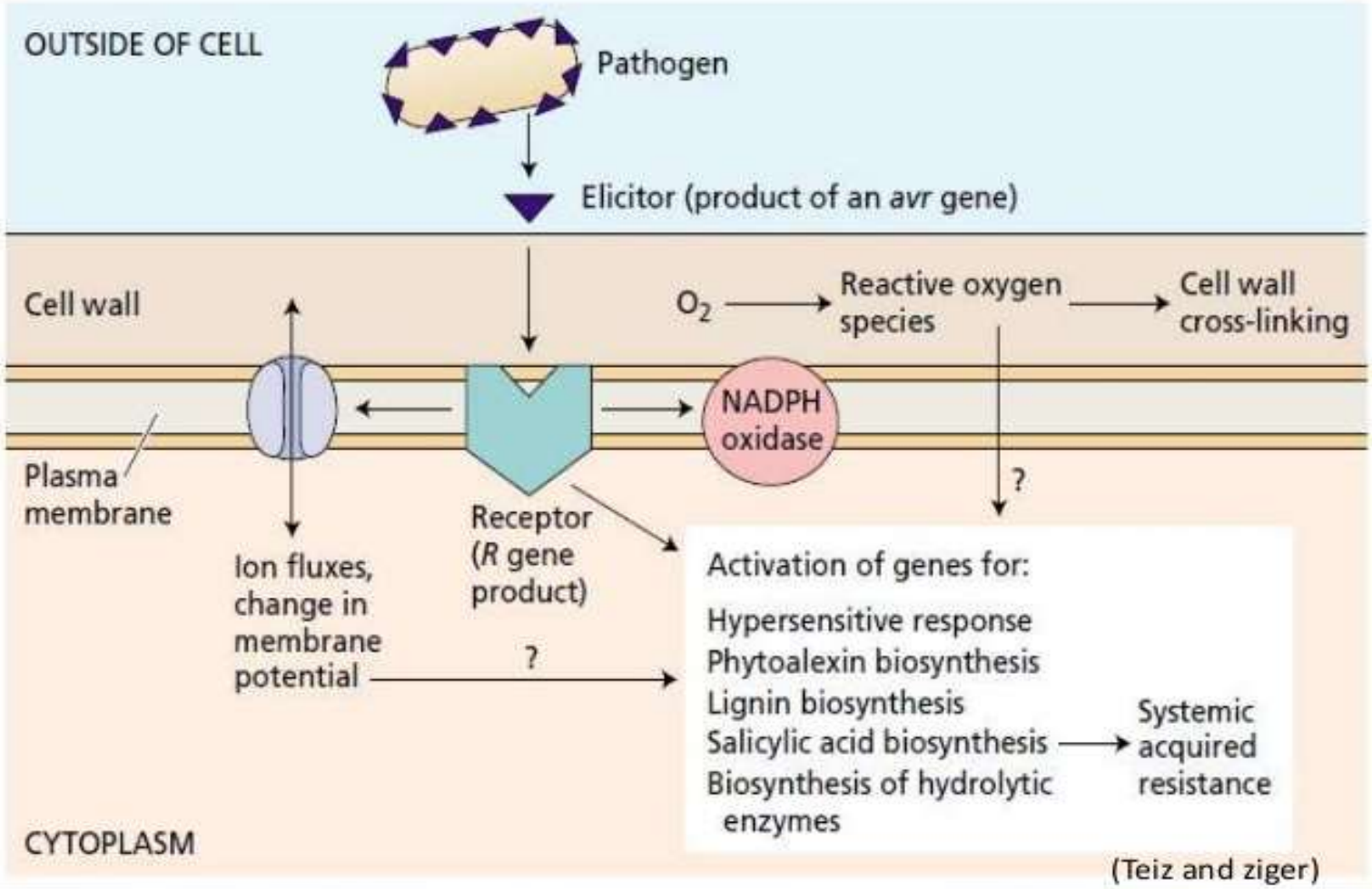
- Produced as a result of pathogen attack
- Infection activate genes that produce PR proteins

Plant Defense Responses



Nature

- Some are antimicrobial
- Some function as signal to spread news
- Salicyclic acid induces production of proteins



(Teiz and ziger)

Functions

- Antiviral action
- As well as antifungal and antibacterial
- Signals the message to nearby cells and tissues
- Stimulate cross-linkage of molecules

Functions

- Peroxidase
- Ribonuclease
- Lysozyme activity
- Hydrolytic and proteinase-inhibitory
- Inactivate viral proteins

END

Ppt no. 194

Virus Resistant Plants

Plant viruses

{Tulip breaking virus (TBV)}
(https://en.wikipedia.org/wiki/Tulip_breaking_virus)

{Tobacco mosaic virus (TMV) symptoms on tobacco leaf}

Modes of Transmission of plant viruses

- To transmit from one plant to another plant and from one plant cell to another, plant viruses must use strategies that are usually different from animal viruses.
- Plant do not move so plant plants transmission usually involves vectors (such as insect).

Modes of Transmission of plant viruses

- Insect – this forms the largest and most significant vector group and particularly include; Aphids, Whiteflies, Hoppers, Beetles etc.
- Through sap

– Eg TMV, Potato virus

- Nematodes

– Eg Tobacco ring spot virus

- Through seed

Mechanism of Plant resistance to viruses

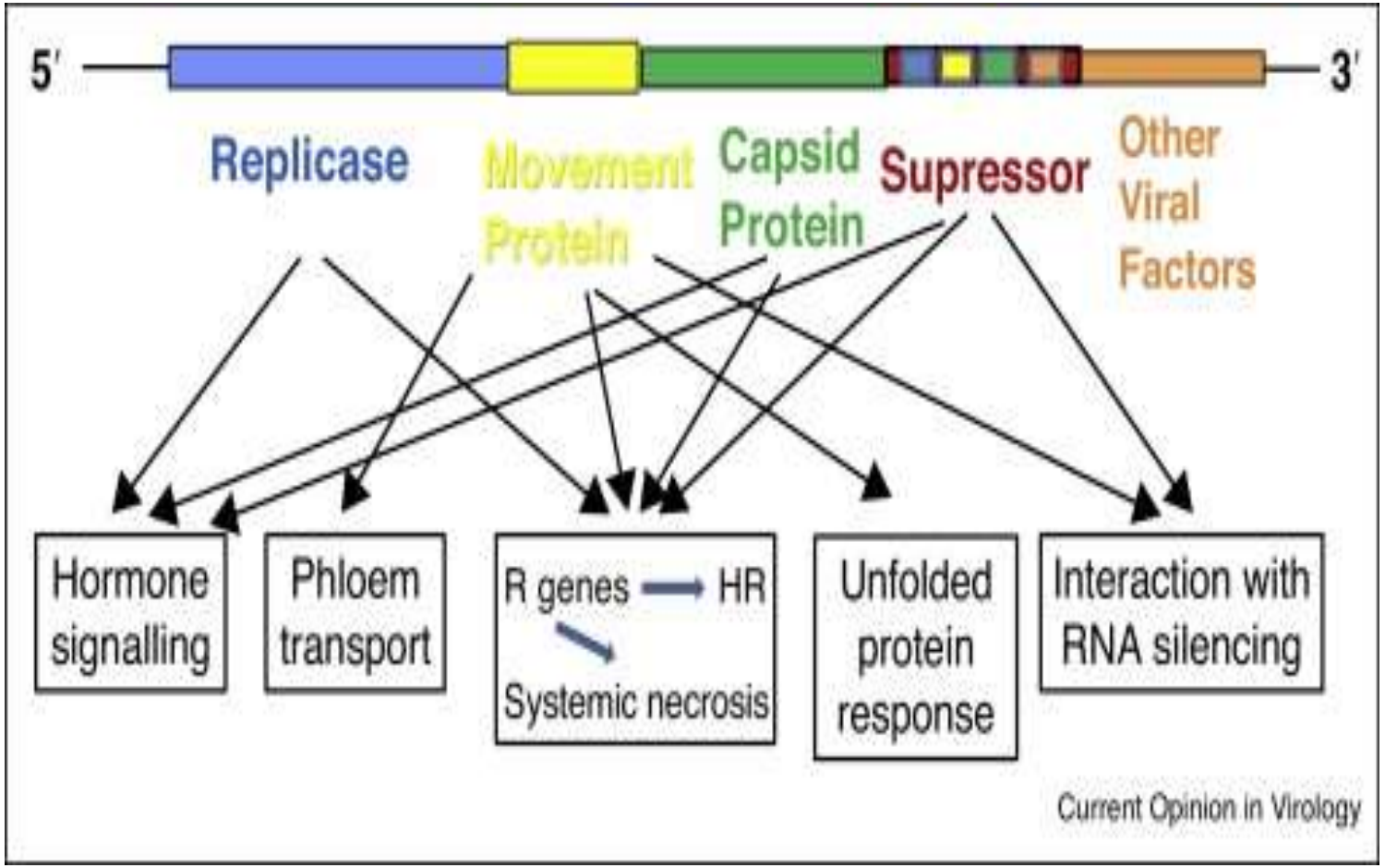
- There are approximately 450 species of plant-pathogenic viruses, which cause a range of diseases.
- The plant show two types of antiviral defense resistanse mechanism are
- 1) R gene mediated
- 2) RNA Silencing

Ppt no. 195

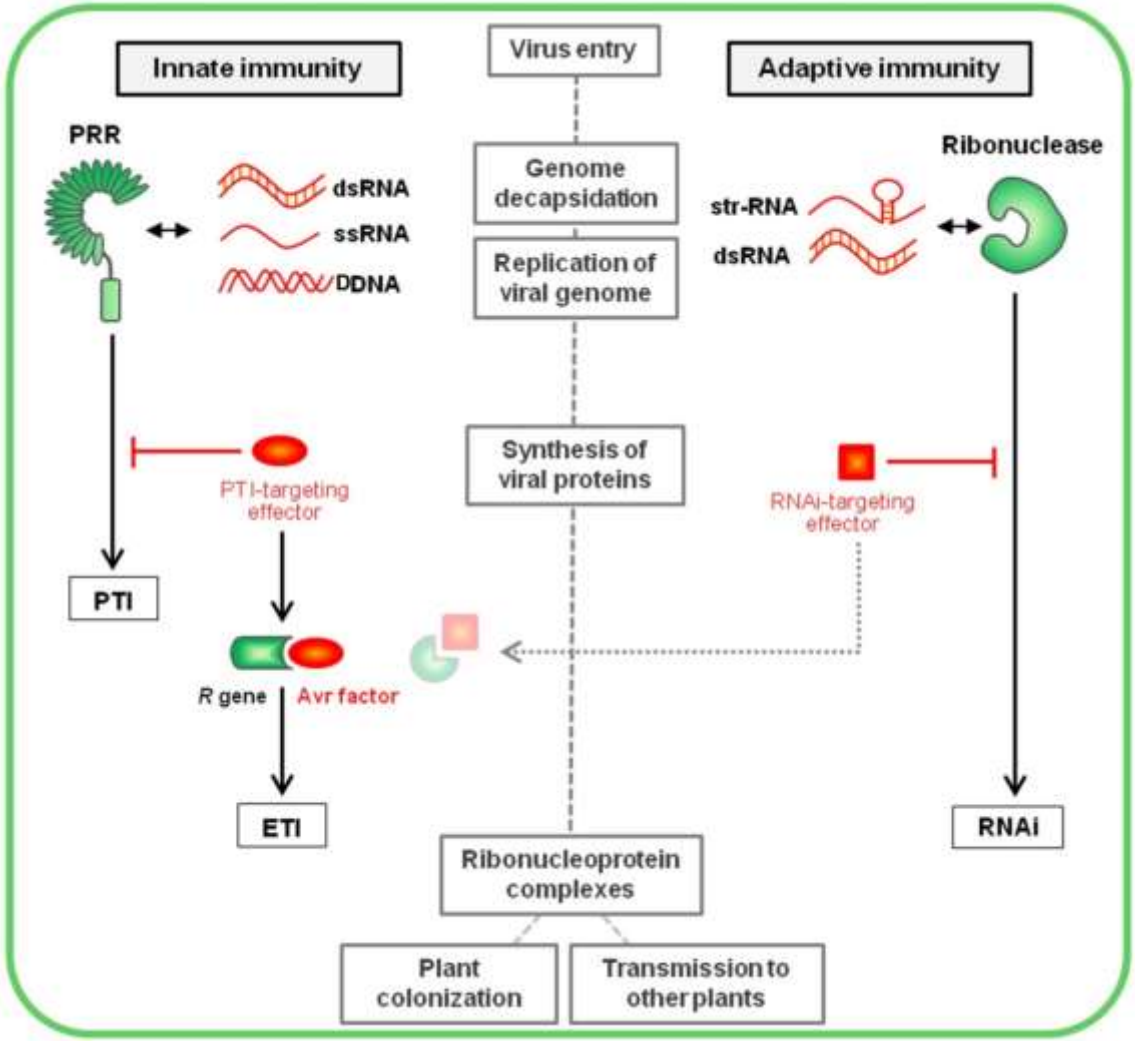
Pathogenesis of Plant Viral Diseases

Routes

- Seeds
- Grafting/ Vegetative propagation
- Vectors
- Mechanical

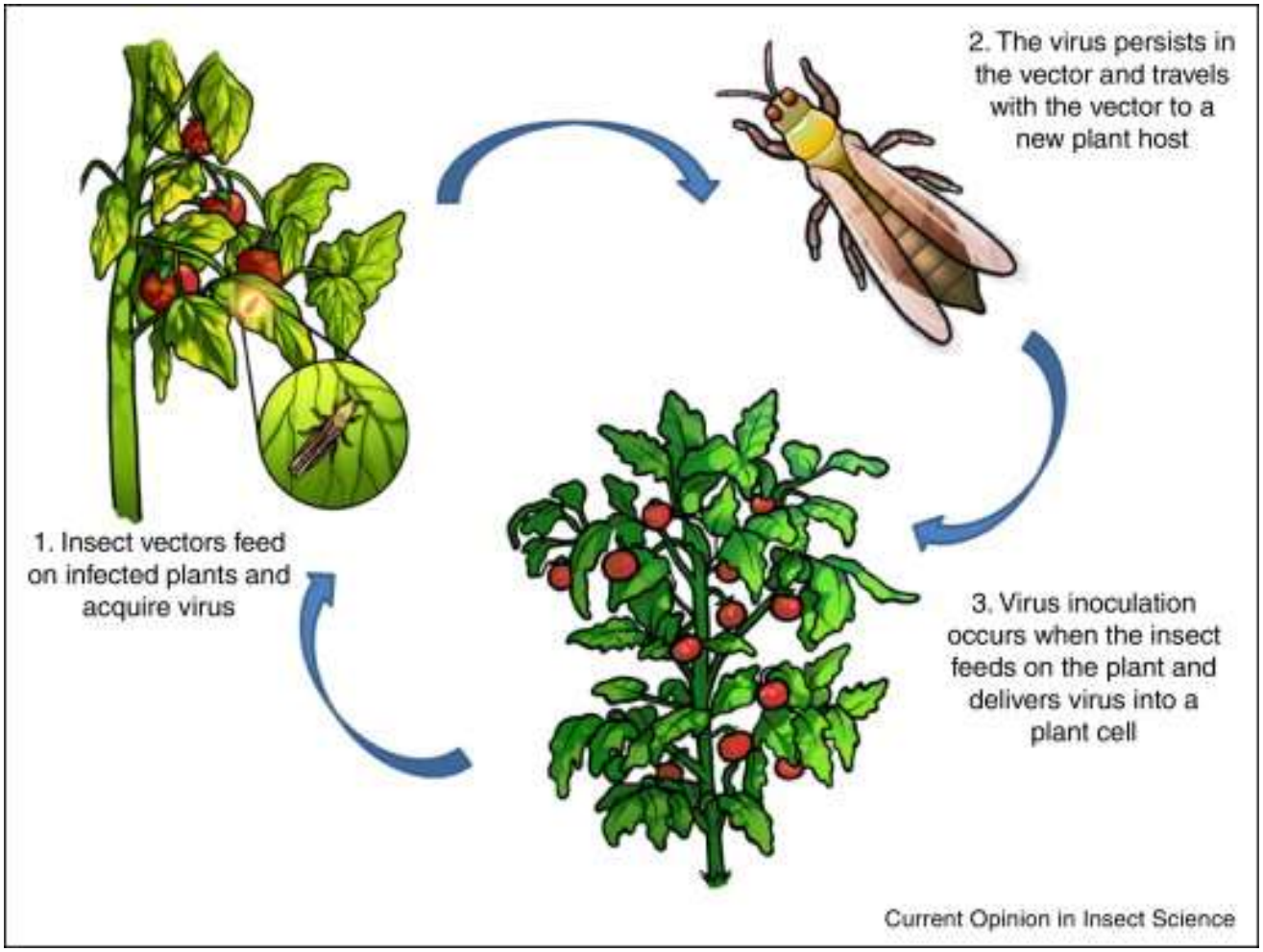

5′
3′
Replicase
Movement Protein
Capsid Protein
Supressor
Other Viral Factors
Hormone signalling
Phloem transport
R genes
HR
Systemic necrosis
Unfolded protein response
Interaction with RNA silencing
Current Opinion in Virology

- No specific receptors on cells
- Rely on mechanical breach for virus to enter through cell wall
- Usually with the help of insects


Innate immunity
Virus entry
Adaptive immunity
PRR
dsRNA
ssRNA
DDNA
Genome decapsidation
Replication of viral genome
Ribonuclease
str-RNA
dsRNA
PTI-targeting effector
Synthesis of viral proteins
RNAi-targeting effector
PTI
R gene
Avr factor
ETI
RNAi
Ribonucleoprotein complexes
Plant colonization
Transmission to other plants

Propagative Transmission

- Multiply in tissues of insects as well as host
- Most multiplication at site of infection
- Localized symptoms


2. The virus persists in the vector and travels with the vector to a new plant host
1. Insect vectors feed on infected plants and acquire virus
3. Virus inoculation occurs when the insect feeds on the plant and delivers virus into a plant cell
Current Opinion in Insect Science

Spread

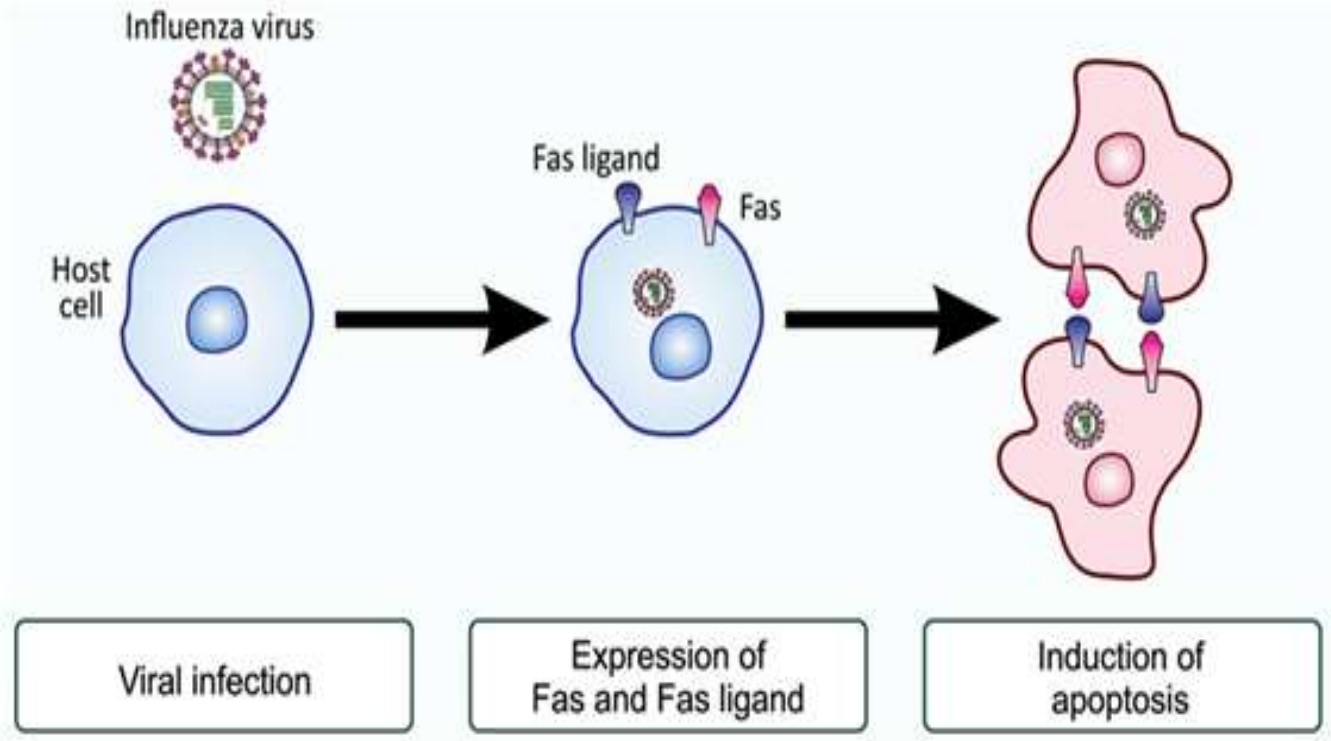
- Direct cell to cell
- Vascular system

Ppt no. 197

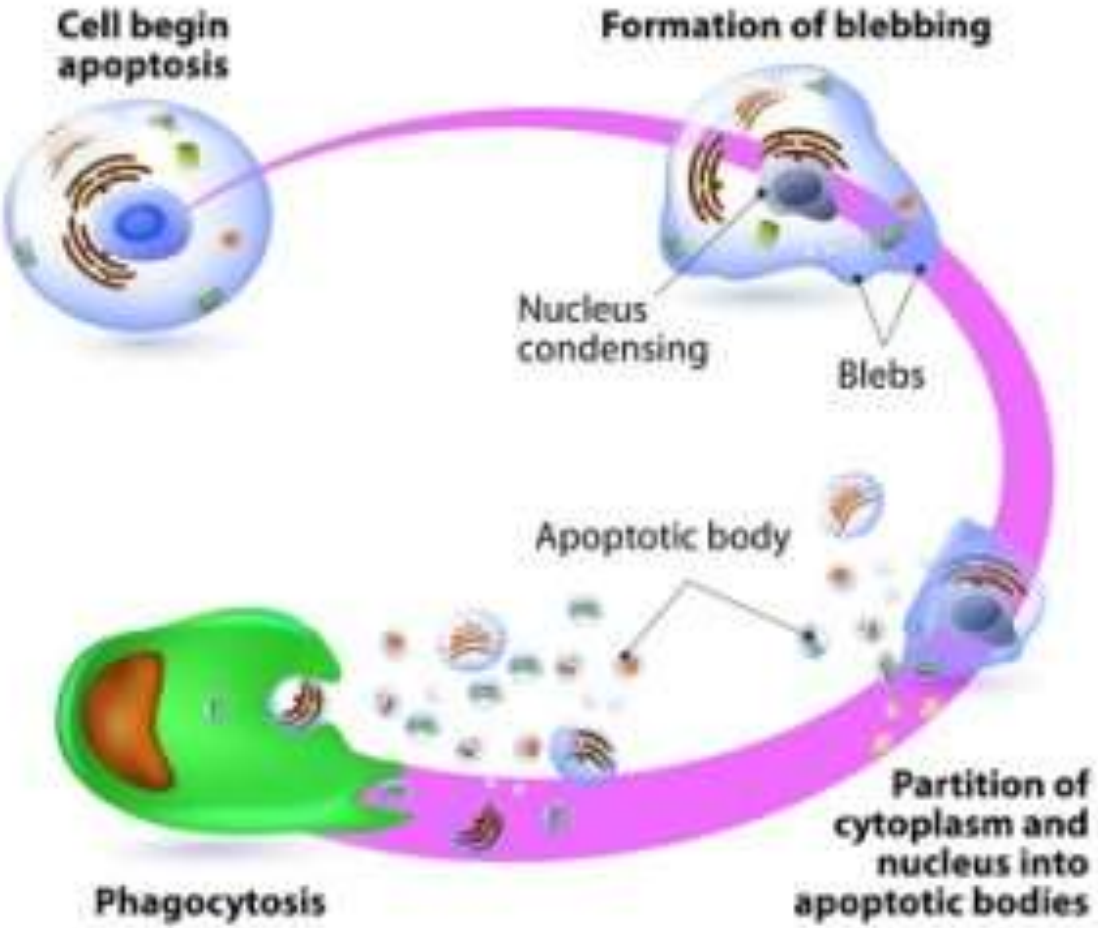
Apoptosis Caused by Viruses

Apoptosis

- Programmed cell death
- During cell development and cell killing by immune system


Influenza virus
Host cell
Fas ligand
Fas
Viral infection
Expression of Fas and Fas ligand
Induction of apoptosis

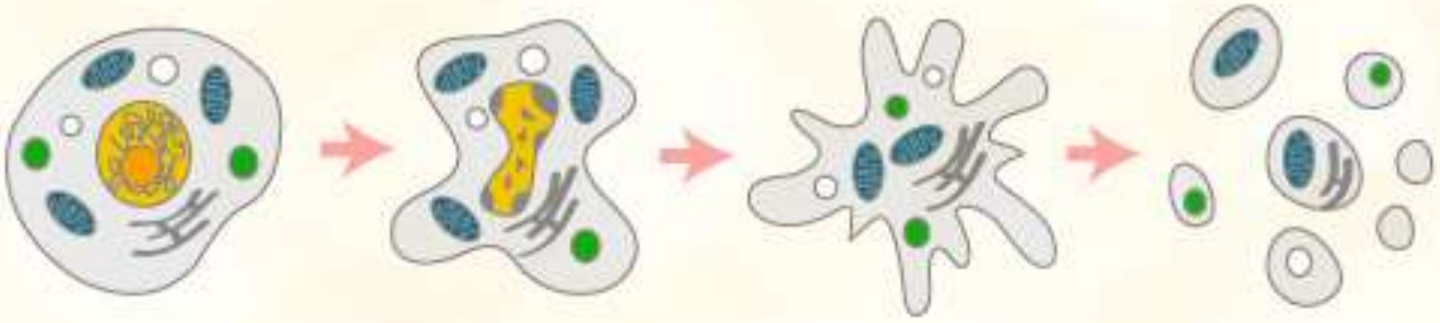
- Tightly regulated process
- Relies on complex molecule cascades for its control
- Shrinkage, condensation of chromatin, DNA fragmentation and blebbing


Cell begin apoptosis
Formation of blebbing
Nucleus condensing
Blebs
Apoptotic body
Partition of cytoplasm and nucleus into apoptotic bodies
Phagocytosis

Activation

- Immune effector cells like CTL's and NK cells release lytic granules
- Act on target cells by two mechanisms

Apoptosis



MECHANISM 1

- Release of cytokinins such as, perforin: peptide related to C9, forms a polyperforin and affects permeability of cell membrane
- Granzyme: serene proteases, affect cell pores

MECHANISM 2

- CTLs express a ligand, that binds on surface on target cell
- Triggers apoptosis
- Triggers cascade to initiate apoptosis

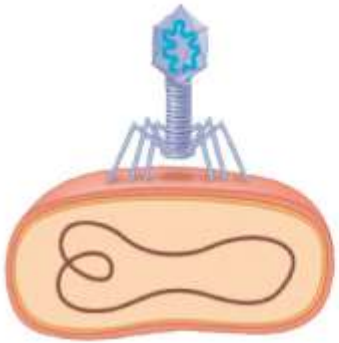
Ppt no. 198

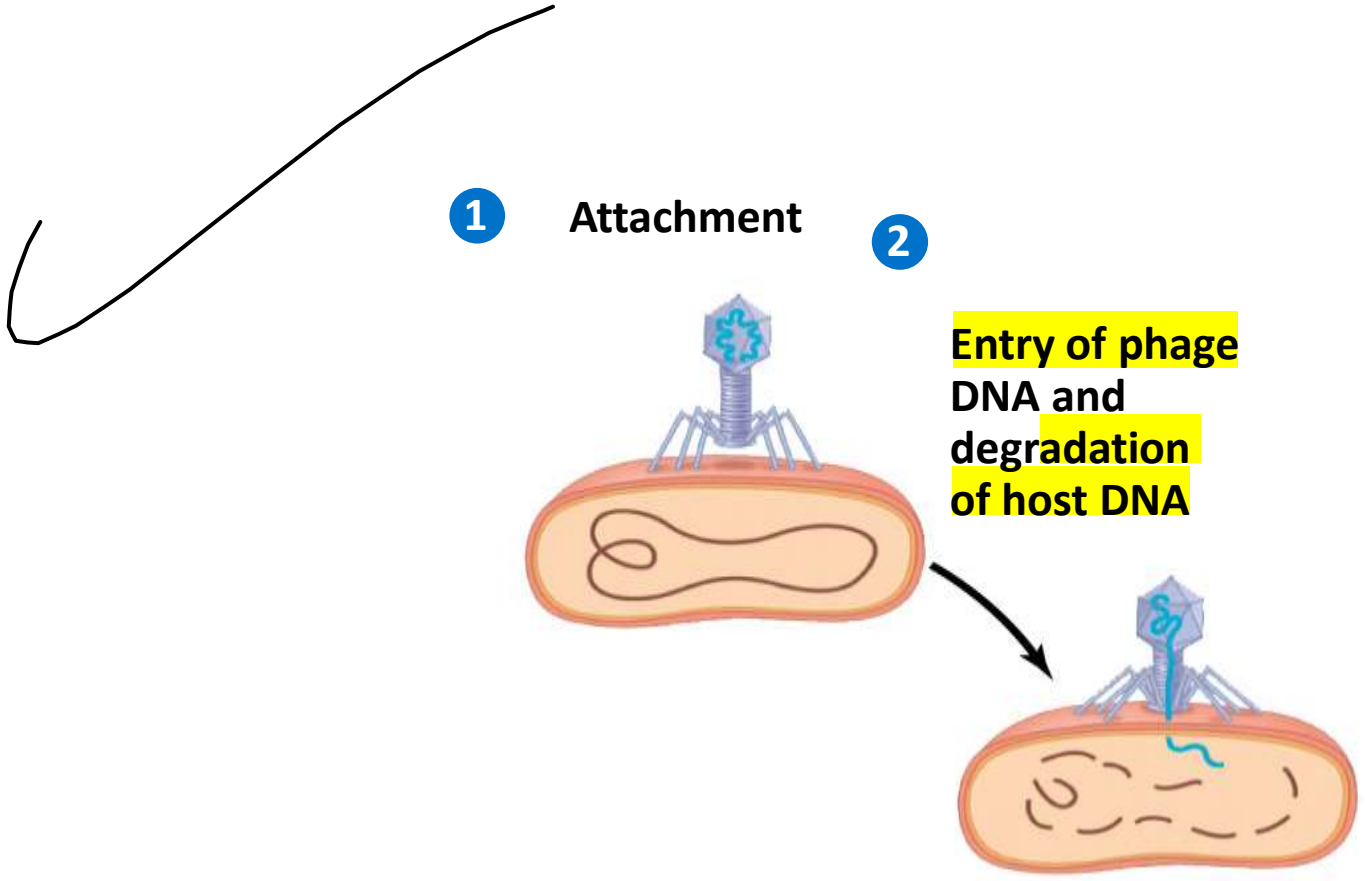
Lytic Cycle in Viruses

HOW DO PHAGES REPLICATE?

1. Lysogenic Cycle
2. Lytic Cycle

1 **Attachment**





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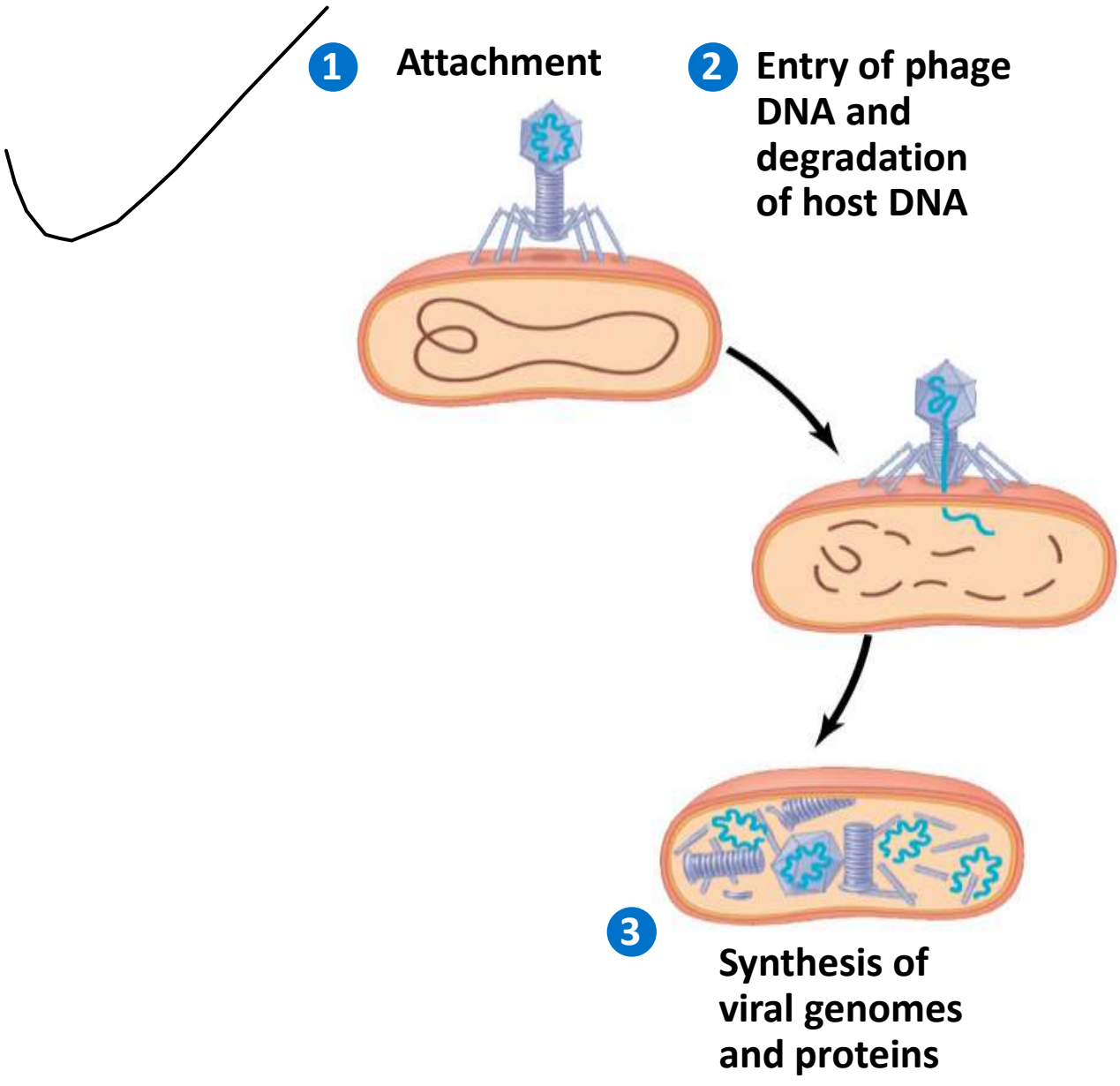

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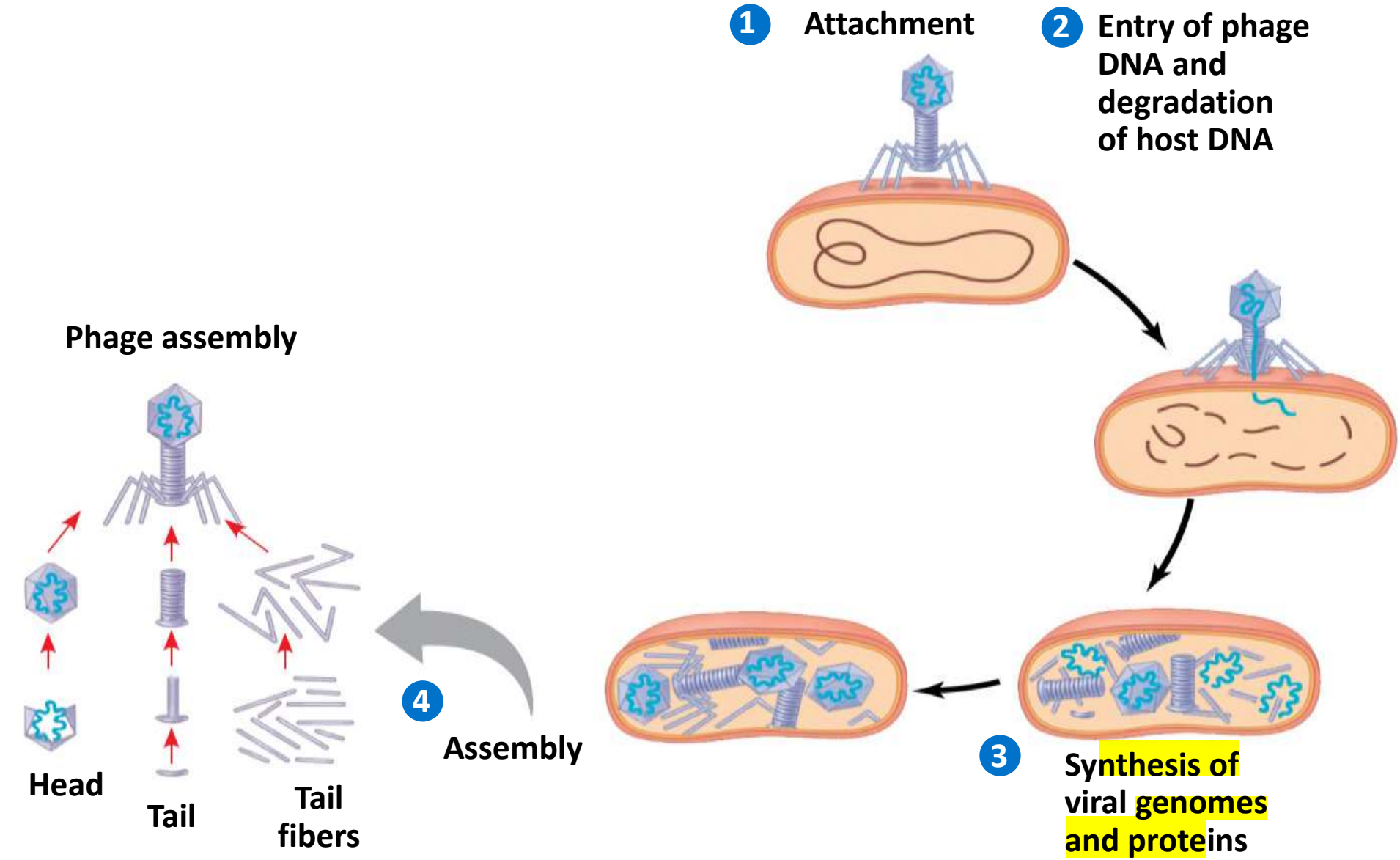

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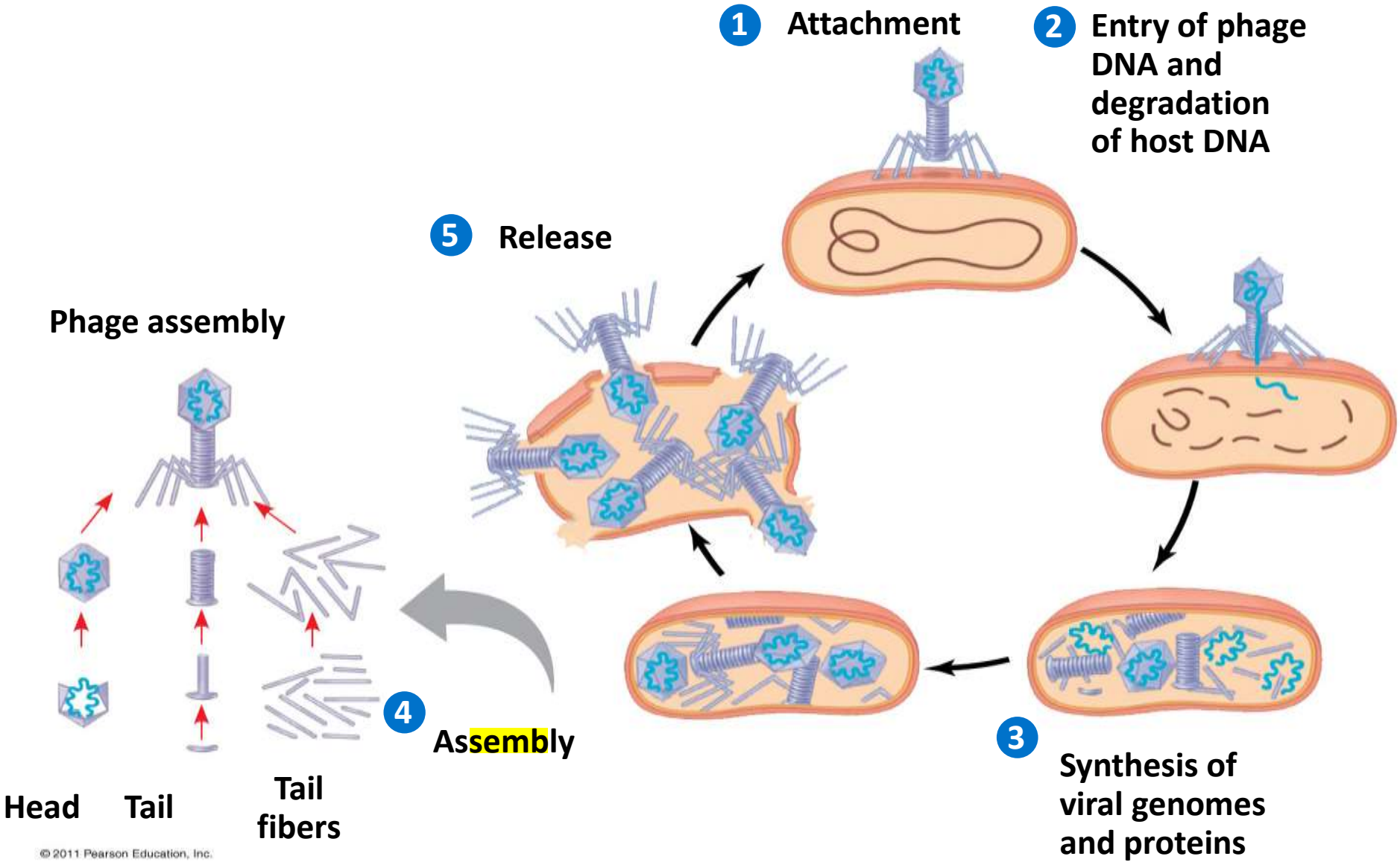
1 Attachment
2 Entry of phage DNA and degradation of host DNA
3 Synthesis of viral genomes and proteins
4 Assembly
5 Release
Phage assembly
Head
Tail
Tail fibers
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Ppt no. 199

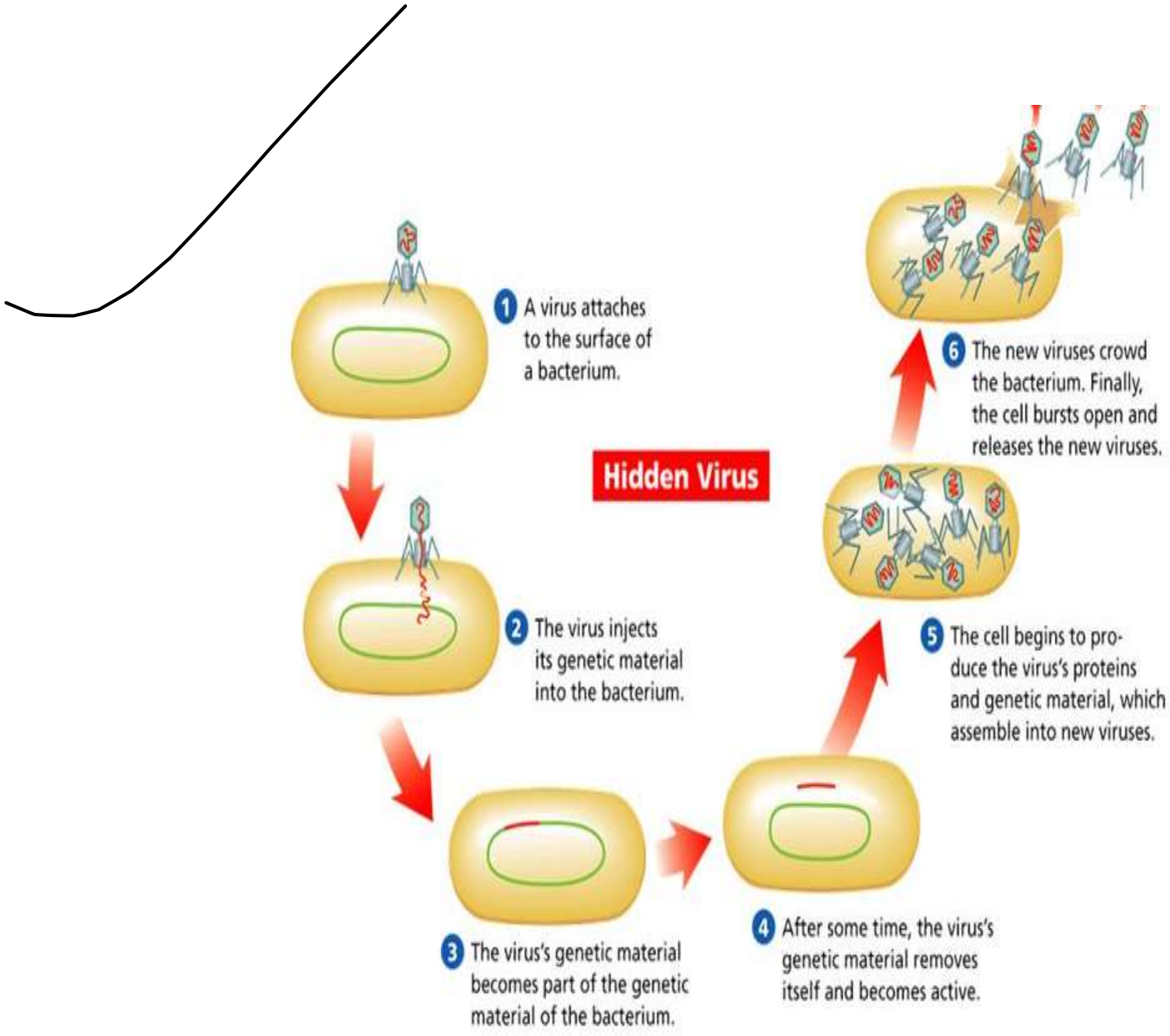
Lysogenic Cycle in Viruses

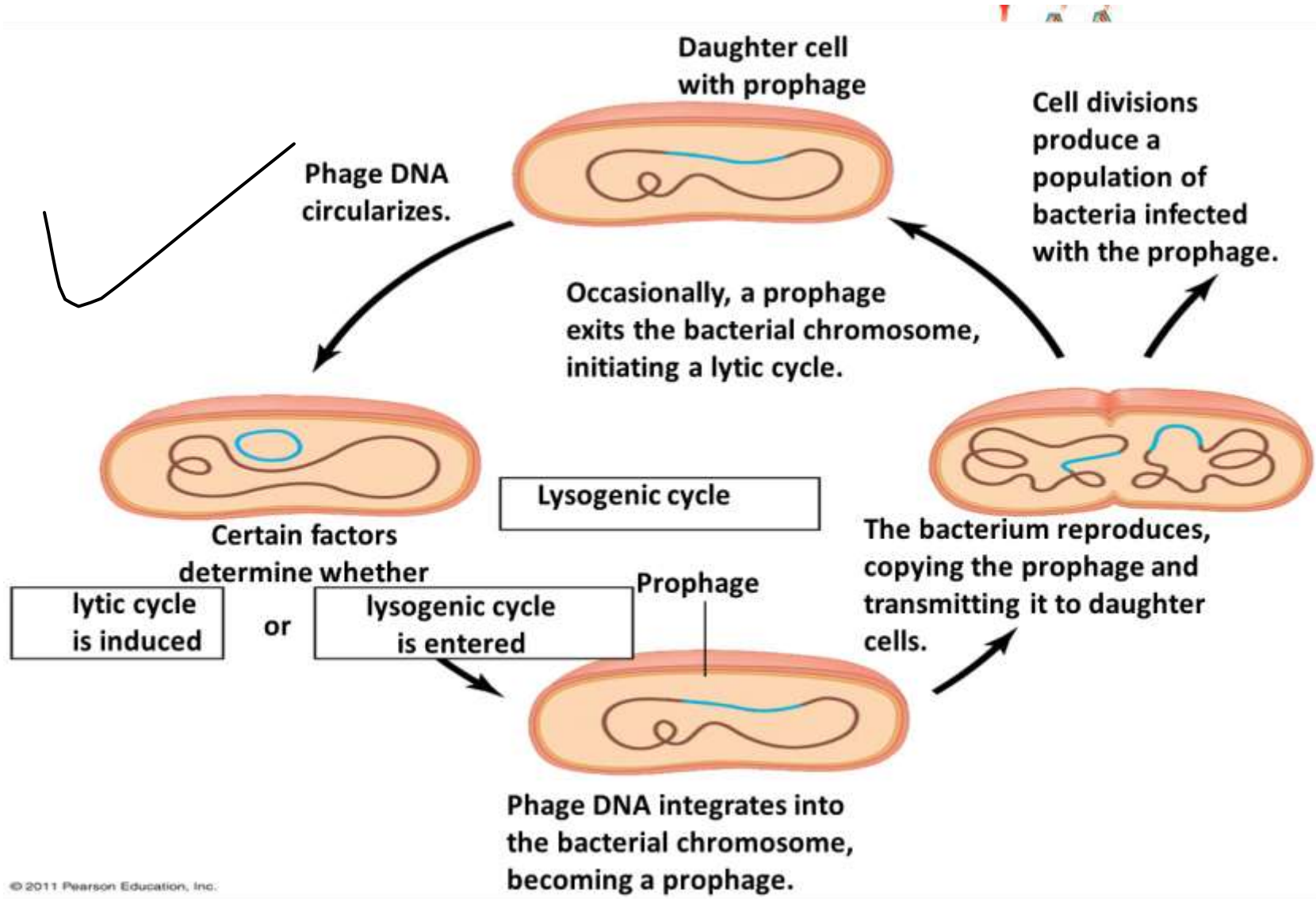
LYSOGEN OR LYSOGENIC

A **lysogen** or **lysogenic** bacterium is a bacterial cell in which a phage exists as DNA in its dormant state (prophage)

STEPS

1. Viral genome enter cells
2. Viral Genome integrates into host cell genome
3. Host DNA polymerase copies viral chromosomes
4. Host cell divides and viral chromosomes divide in to daughter cells
5. When virus is triggered, viral genome detaches and enters lytic cell


Hidden Virus
1 A virus attaches to the surface of a bacterium.
2 The virus injects its genetic material into the bacterium.
3 The virus's genetic material becomes part of the genetic material of the bacterium.
4 After some time, the virus's genetic material removes itself and becomes active.
5 The cell begins to produce the virus's proteins and genetic material, which assemble into new viruses.
6 The new viruses crowd the bacterium. Finally, the cell bursts open and releases the new viruses.



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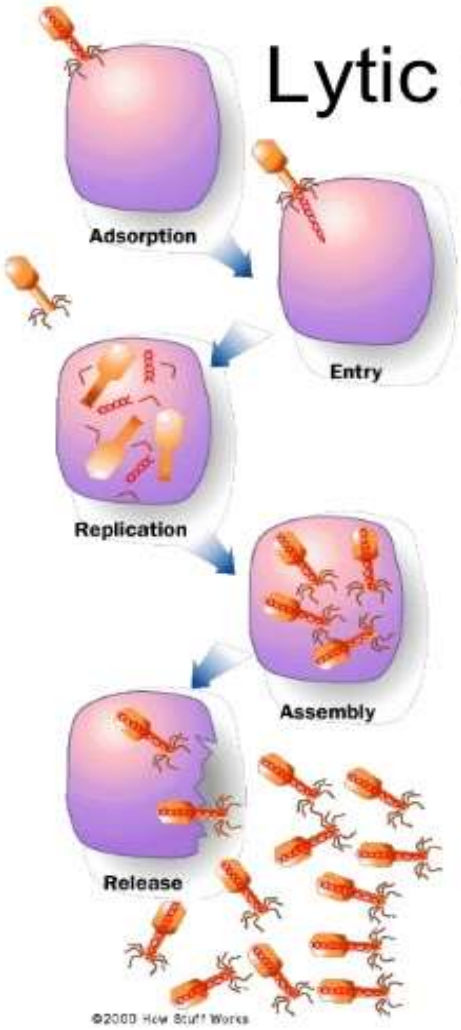
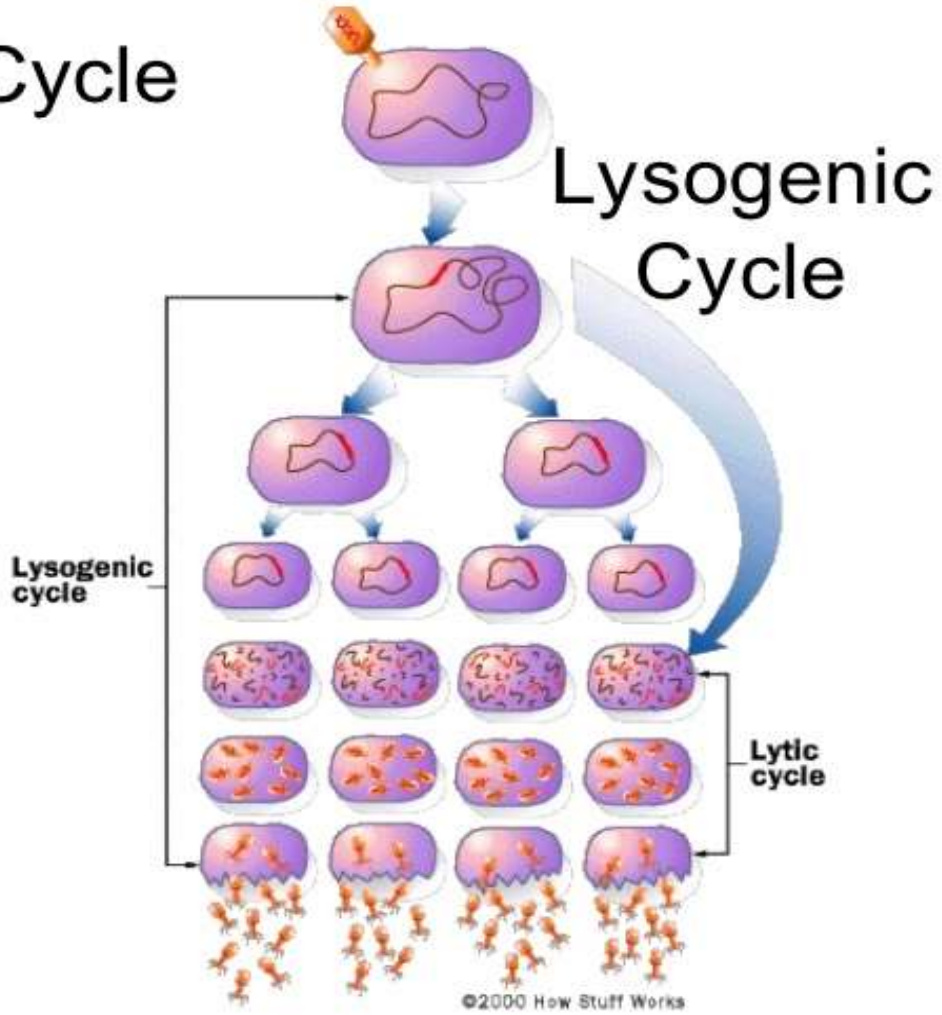
Ppt no. 200

Comparison of Lytic and Lysogenic

- Lytic cycle (virulent phage)
- Release of virus bursts and kills host cells (lysis)

- Lysogenic cycle (temperate phage)
- Viral DNA integrates into host genome (provirus)
- Can be transmitted to daughter cells.
- Can initiate lytic cycle in response to environmental signals (stress)


Lytic Cycle
Adsorption
Entry
Replication
Assembly
Release
©2000 How Stuff Works


Lysogenic Cycle
Lysogenic cycle
Lytic cycle
©2000 How Stuff Works

Advantages and Limits

Lytic cycle

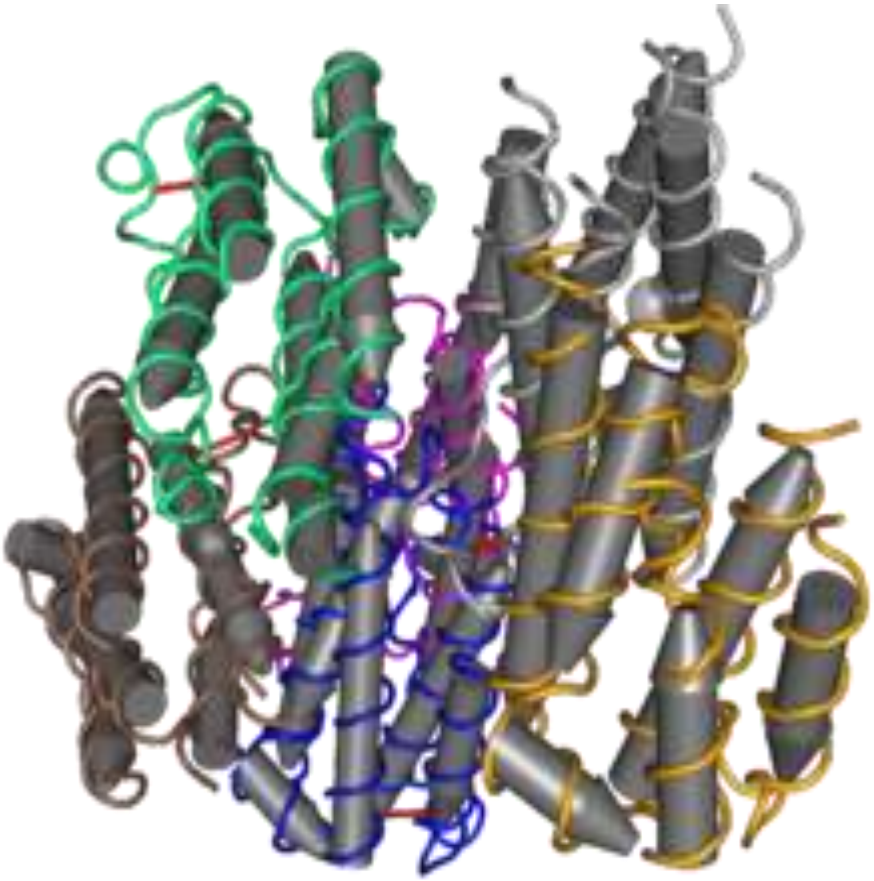
- Replication of new viruses is fast
- However, the host is also immediately killed preventing the viral genome from passing onto the next generation of host cells

Lysogenic cycle

- Many more viruses can be made because the viral genome is passed onto future generation of host cells
- However, replication is takes longer because it is dependent on the host cell's replication

Ppt no. 201

Interferons



- Natural antiviral compounds
- substances that have antiviral properties in adjacent, noninfected cells adjacent

Types of Interferons

Type I:

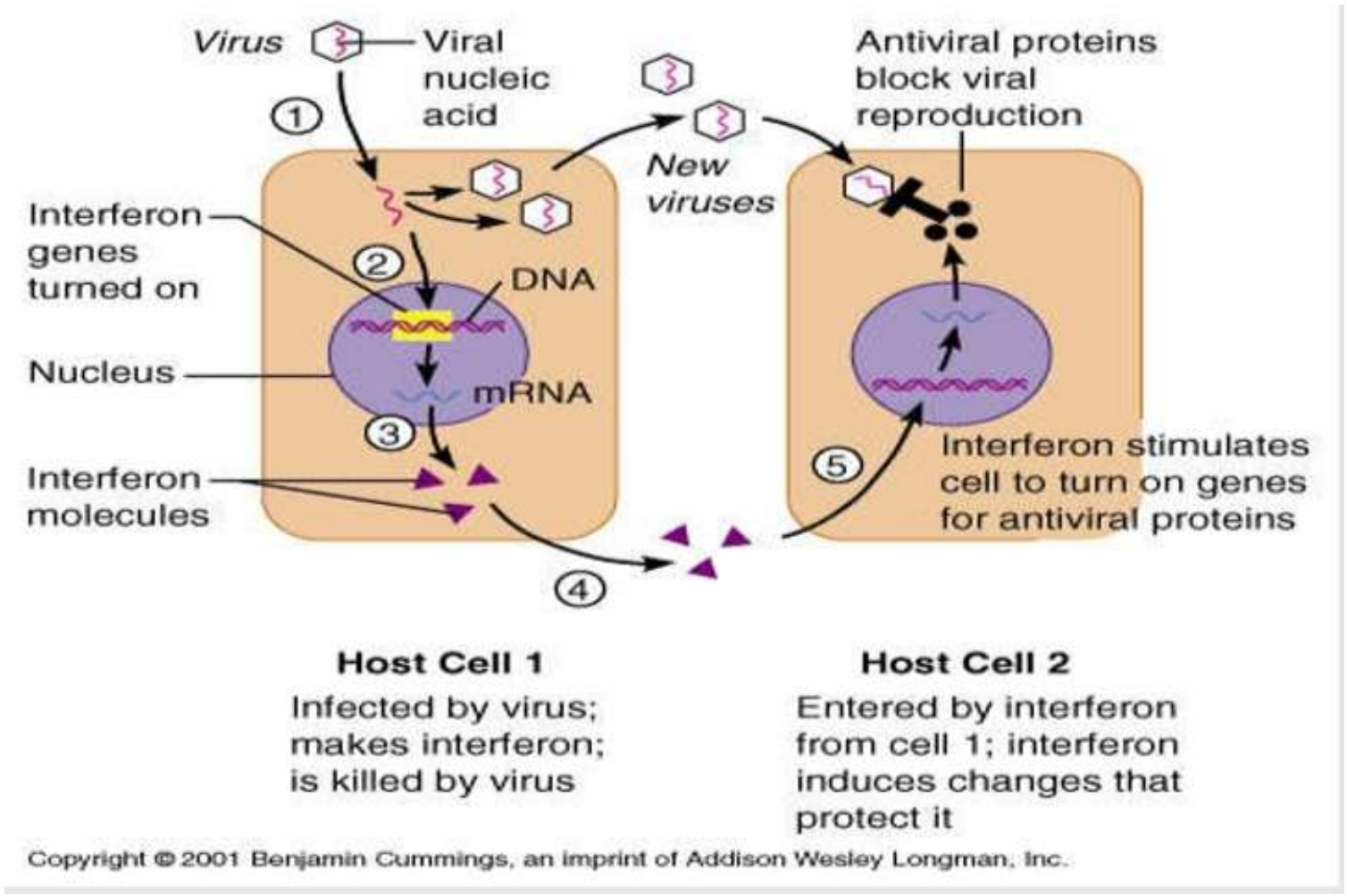
- (1) Interferon alpha = maximal antiviral activity
- (2) Interferon Beta = intermediate antiviral activity

Type II:

- Interferon Gamma = more lymphokine than antiviral

Regulation of Interferon Expression

- not expressed in a normal resting cell
- labile repressors bind to promoter elements, block transcription
- production of labile suppressors drop in viral infection and allows in viral infection and allows interferon synthesis to occur



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Clinical Uses:

IFN-A :

- treatment of viral infections: condylomata acuminata and chronic hepa B and C
- prophylactic or therapeutic agent in immunocomp. hosts (VZV, HSV 1 and 2)
- prophylaxis vs CMV in renal transplant
- treatment of AIDS-associated Kaposi's sarcoma and hairy cell leukemia

IFN-G: IFN-G:

- immunostimulant in oncologic and immunodeficiency disorders

Ppt no. 202

Evasion of Immune Response by Viruses

- When a virus infects a host, a complex immune response develops to eliminate the invading pathogen.
- Viruses, in turn, have evolved a profusion of strategies to escape from the immune system

Mechanisms of viral evasion occur at

- cellular level
- systemic level.

Viruses can avoid detection by both innate and adaptive immune responses

Targets of viral evasion proteins

- Pattern recognition receptors in infected cells
- Interferons
- dendritic cells
- T cell receptors
- Antibodies

- Viruses can express proteins that directly interfere with host processes or mimic host proteins and compete to bind specific receptors.
- Pathogens are able to directly deplete immune cells, and prevent their recruitment to the site of infection.
- In a state of latency, viruses remain dormant and undetectable in host cells

Ppt no. 203

Prevetion of Viral Infection

- Viruses cause a variety of diseases in animals, including humans, ranging from the common cold to potentially fatal illnesses like meningitis

- Vaccines may be prepared using live viruses, killed viruses, or molecular subunits of the virus.
- The killed viral vaccines and subunit viruses are both incapable of causing disease.
- Live viral vaccines are designed in the laboratory to cause few symptoms in recipients while giving them protective immunity against future infections.
- Polio vaccines.

- **Antiviral drugs.**
- have limited success in curing viral disease, but in many cases, they have been used to control and reduce symptoms for a wide variety of viral diseases.
- For most viruses, these drugs can inhibit the virus by blocking the actions of one or more of its proteins.

- Acyclovir-for genital herpes
- Tamiflu (oseltamivir) – for influenza

Ppt no. 204

Viral Vaccines

A vaccine is a biological preparation that provides active acquired immunity to a particular disease.

- A vaccination is the injection of a killed or weakened organism that produces immunity in the body against organism.
- An immunization is the process by which a person or animal becomes protected from a disease

History:

- 1796: Edward Jenner prevented smallpox by vaccination with cowpox

- 1885 Louis Pasteur successfully prevented rabies in Joseph Meister by post-exposure vaccination.

Why Vaccines?

- Vaccines are safe.
- Vaccines are effective.
- Vaccines work with immune system.
- Vaccines protect individuals and the community.

Ppt no. 205

Viral Vaccines

Types

When scientists create vaccines, they consider:

- How your immune system responds to the germ
- Who needs to be vaccinated against the germ
- The best technology or approach to create the vaccine

There are 4 main types of vaccines:

- Live-attenuated vaccines
- Inactivated vaccines
- Subunit, recombinant, polysaccharide, and conjugate vaccines
- Toxoid vaccines

Live-attenuated vaccines

- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease.
- Because these vaccines are so similar to the natural infection that they help prevent, they create a strong and long-lasting immune response.

Live vaccines are used to protect against:

- Measles, Mumps, Rubella (MMR combined vaccine)
- Rotavirus
- Smallpox
- Chickenpox
- Yellow fever

Inactivated vaccines

- Inactivated vaccines use the killed version of the germ that causes a disease.
- Inactivated vaccines usually don't provide immunity (protection) that's as strong as live vaccines.
- Inactivated vaccines are used to protect against:
  - Hepatitis A
  - Flu (shot only)
  - Polio (shot only)
  - Rabies

Subunit, recombinant, polysaccharide, and conjugate vaccines

- Use specific pieces of the virus – like its protein, sugar, or capsid
- They give a very strong immune response that's targeted to key parts of the virus.
- One limitation- need booster shots
  - Hepatitis B
  - HPV (Human Papillomaviurs)
  - Whooping cough (part of the DTaP combined vaccine)
  - Pneumococcal disease
  - Meniggococcal disease

Toxoid vaccines

- Use a toxin (harmful product) made by the virus that causes a disease.
- They create immunity to the parts of the virusthat cause a disease instead of the virus itself. That means the immune response is targeted to the toxin instead of the whole virus.
- You may need booster shots
- Toxoid vaccines are used to protect against:
- Diphtheria
- Tetanus

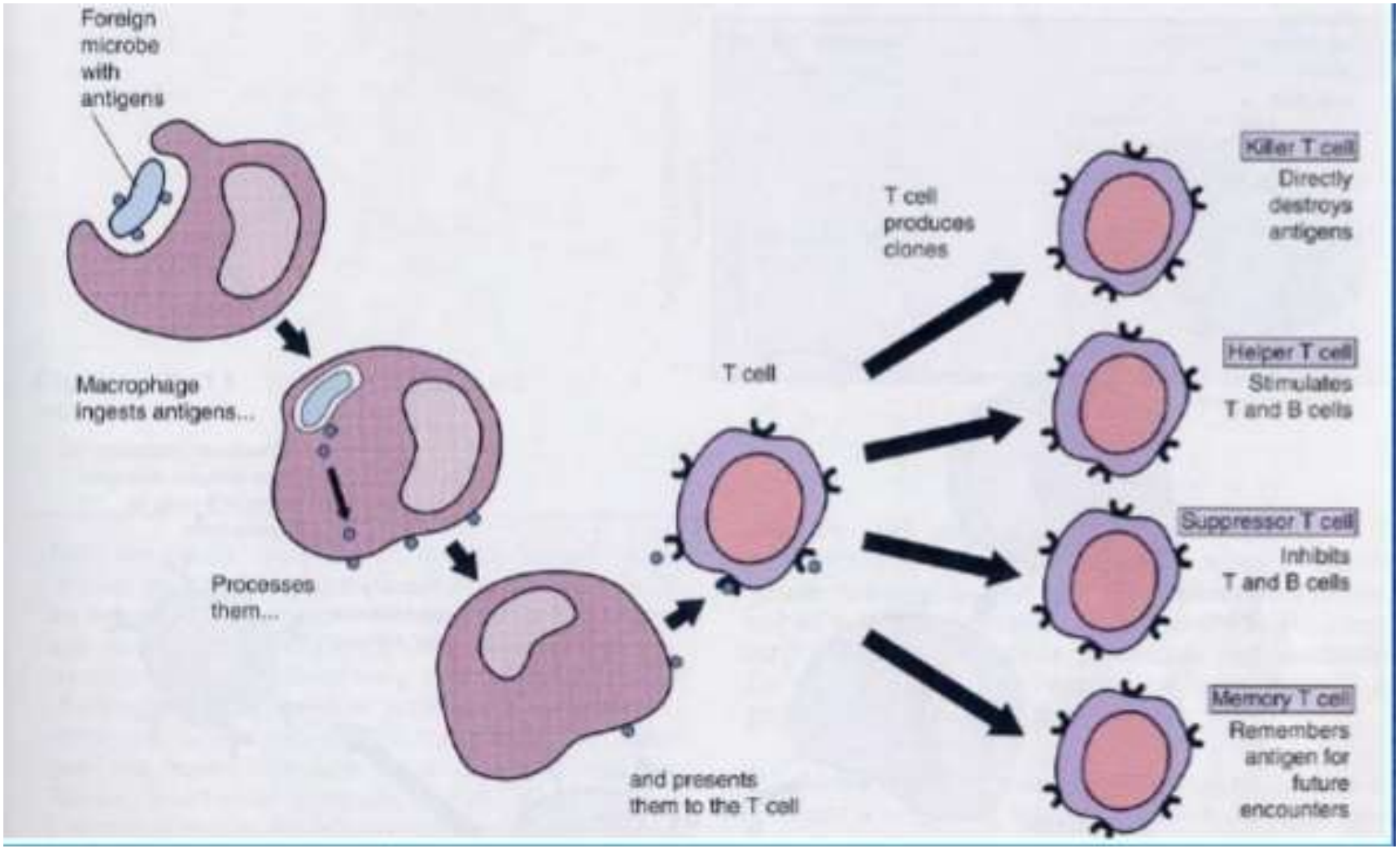
Ppt no. 206

Viral Vaccines

Mechanism

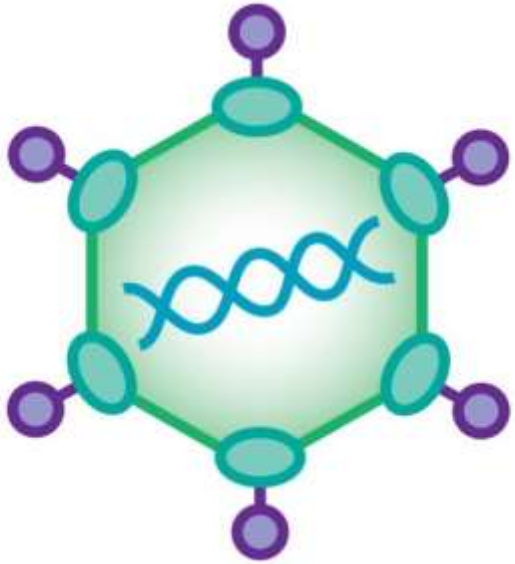
Mechanism of vaccines

- Vaccines are made from same organism or parts of them that causes disease.
- Introduced in body.
- Immune system reacts to the vaccine-make antibodies.
- Antibodies destroy vaccine germs
- Stay in body, giving specific immunity.


Foreign microbe with antigens
Macrophage ingests antigens...
Processes them...
and presents them to the T cell
T cell
T cell produces clones
Killer T cell
Directly destroys antigens
Helper T cell
Stimulates T and B cells
Suppressor T cell
Inhibits T and B cells
Memory T cell
Remembers antigen for future encounters

Ppt no. 207

Virus Vectors



SV 40 Based Vectors

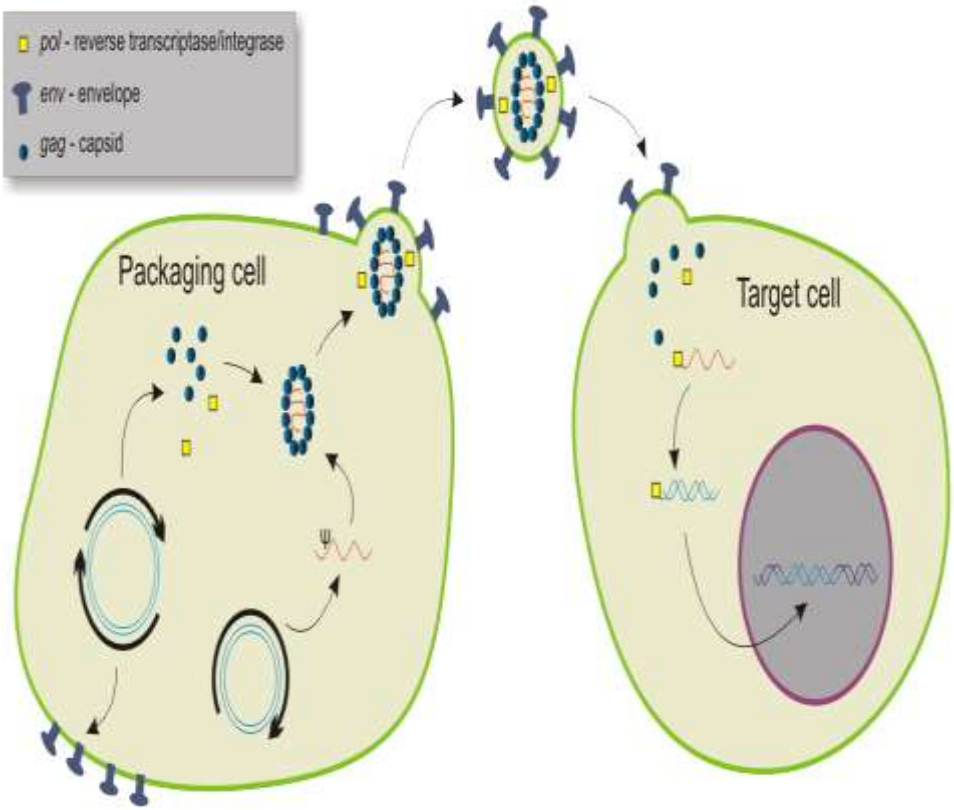
- First animal virus based vector
- Small virus with capsid and circular dsDNA
- Bidirection replication

Lytic Infection of Sv40

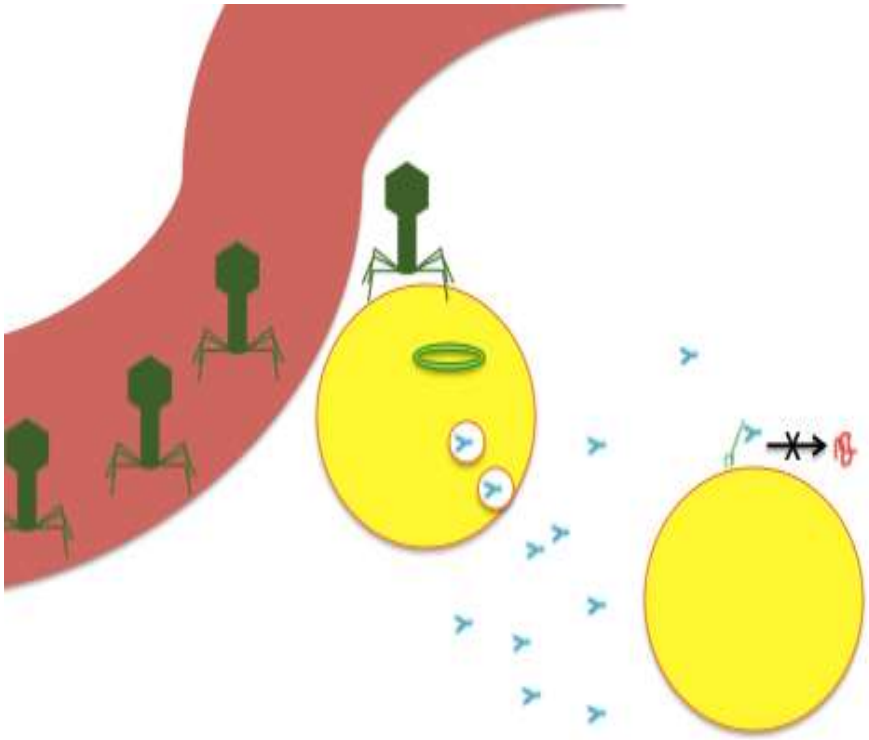
Three steps:

- Early mRNA synthesis
- Viral DNA replication
- Late mRNA synthesis

Vaccinia Virus Vector



- Large ds DNA
- Contains enzymes for replication
- Generated in vivo



Eppstein Barr Virus

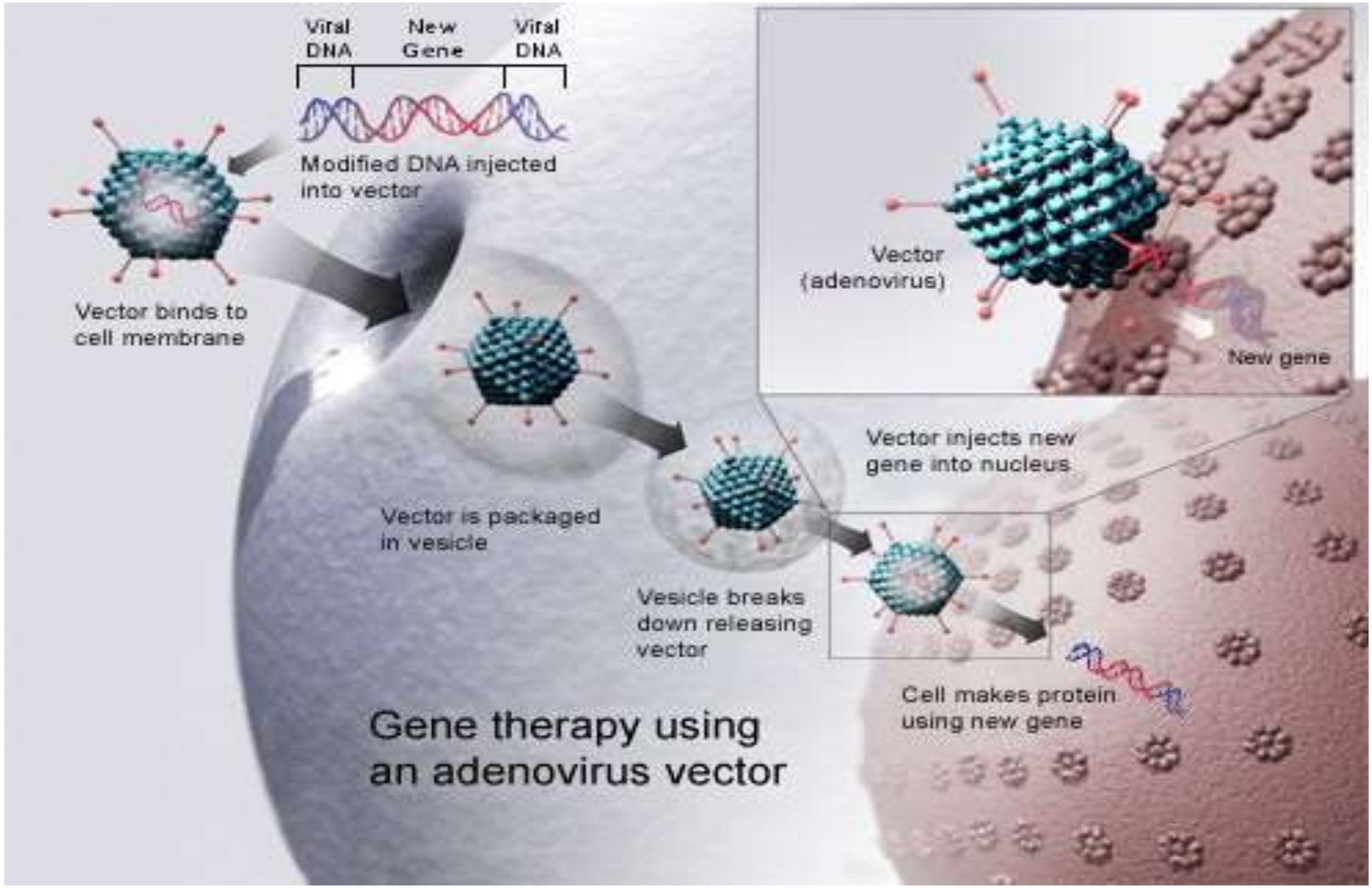
- Herpes virus with large genome
- Infects primates and canines
- Used for construction of episomal cDNA s

Ppt no. 208

Gene Therapy

- Experimental technique that uses genes to treat or prevent disease.
- Therapeutic delivery of nucleic acid into a patient's cells as a drug to treat disease

- First attempt at modifying human DNA was performed in 1980 by Martin Cline.


Viral DNA
New Gene
Viral DNA
Modified DNA injected into vector
Vector binds to cell membrane
Vector is packaged in vesicle
Vesicle breaks down releasing vector
Vector injects new gene into nucleus
Vector (adenovirus)
New gene
Cell makes protein using new gene
Gene therapy using an adenovirus vector

Vectors

- Viral
- Non viral
  - electroporation,
  - the gene gun,
  - sonoporation,
  - magnetofection,
  - oligonucleotides, lipoplexes, dendrimers, and inorganic nanoparticles

Hurdles:

- Short-lived nature
- Immune response
- Problems with viral vectors
- Multigene disorders
- Insertional mutagenesis
- Cost effective

END

Ppt no. 1209

Chemotherapy Against Viral Diseases

IDEAL DRUG

- Water soluble
- Chemically and metabolically stable
- Easily absorbed (apolar)

NOT

- Toxic
- Carcinogenic
- Allergenic
- Mutagenic
- Teratogenic

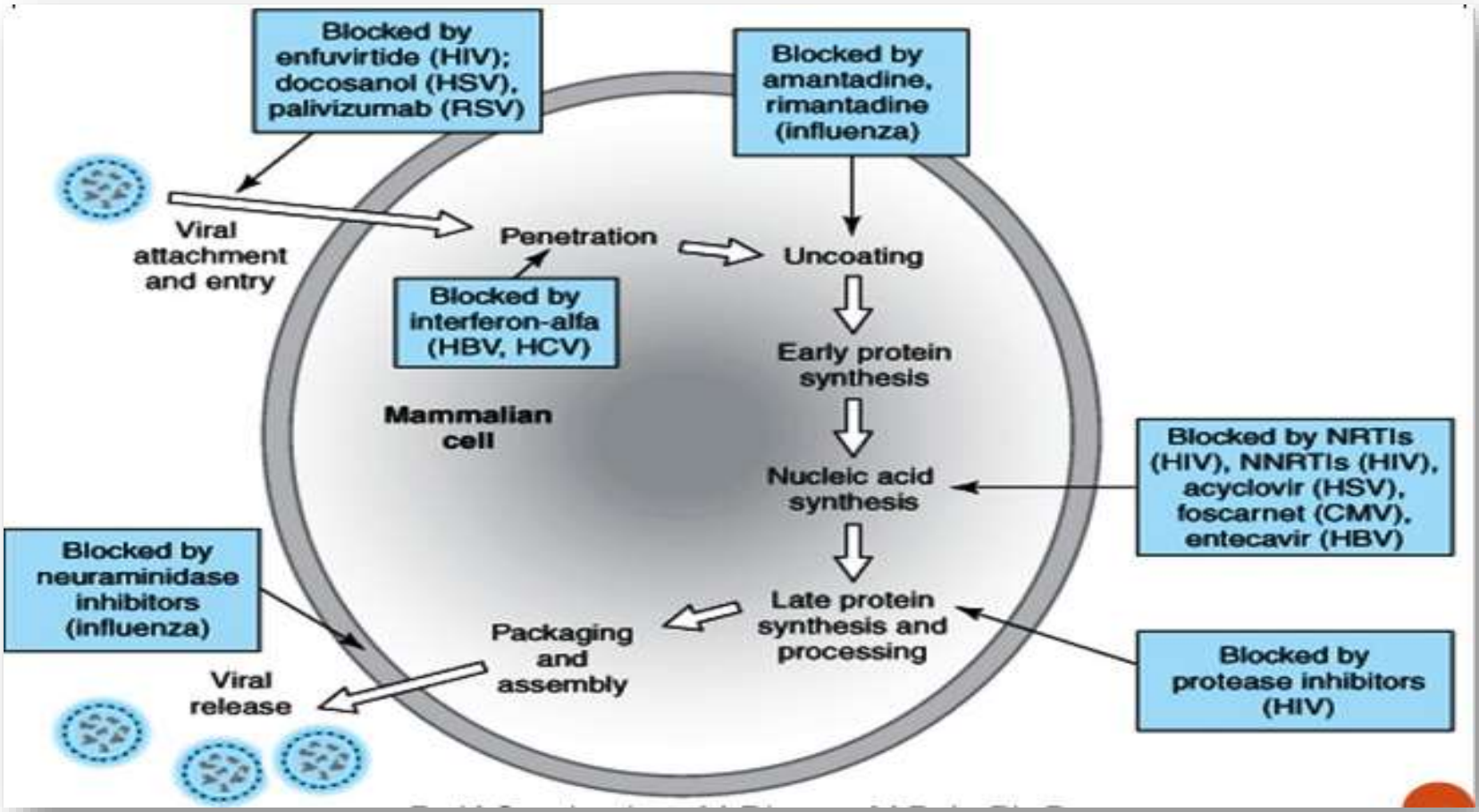
A successful drug must interfere with:

- A specific viral function e.g. enzyme necessary for viral life cycle Anti-Viral Chemotherapy
- A cellular function that the virus needs in order to replicate

If interfere with cellular function either:

- It must be crucial to virus but not the cell
- Only the virus-infected cell must be killed.

MECHANISM OF ANTIVIRAL AGENTS



Approved Antiviral Drugs

- Amantadine and rimantadine (influenza A virus)
- Ribavirin (respiratory syncytial virus)
- Idoxuridine and trifluridine (topical treatment of herpetic keratitis)
- Vidarabine and acyclovir (systemic treatment of herpes simplex virus and varicella-zoster virus infections)

- Famciclovir and valaciclovir (oral treatment of varicella-zoster virus infections)
- Ganciclovir and foscarnet (cytomegalovirus)
- Zidovudine, didanosine, zalcitabine and stavudine (human immunodeficiency virus)

Limitations of Antiviral Drugs

- Narrow antiviral spectrum
- Ineffectiveness against the latent virus
- Development of drug-resistant mutants
- Toxic side effects

Ppt no. 210

Virus Titration

Introduction

Introduction

- To understand the concentration of virus
- Infectivity titer
- Reed and Muench method or Karber formula

Standard

- LD50/mL
- Based on mortality of lab mice
- Infectivity titer
- Computation

Titer

- Frequency of production of specific response of host
- Necessary for viral reproduction
- Smallest amount of virus capable of producing manifestation in host

Reed and Muench method

Table 1 Calculation of virus titre in mice using the Reed and Muench method

Log_{10} virus dilution	Mice		Cumulative total			Percent mortality
	Died	Survived	Died	Survived	Total	
-1	10	0	57	0	57	57/57 × 100 = 100
-2	10	0	47	0	47	47/47 × 100 = 100
-3	10	0	37	0	37	37/37 × 100 = 100
-4	10	0	27	0	27	27/27 × 100 = 100
-5	10	0	17	0	17	17/17 × 100 = 100
-6	6	4	7	4	11	7/11 × 100 = 63
-7	1	9	1	13	14	1/14 × 100 = 7

Difference of logarithms = (63-50)/(63-7) = 0.23; $\log_{10}$ 50% end point dilution = -6 - (0.23 × 1) = -6.23; 50% end point dilution = $10^{-6.23}$; the titre of the virus = $10^{6.23}$ LD_{50}/mL.

Karber method

Midlength (mm)	Log Midlength (x_i)	Number of fish (n_i)	Mature (r_i)	Proportion of mature fish (p_i)	$X = X_{i+1} - X_i$	$q_i = 1 - p_i$	$(p_i x q_i)/(n_i - 1)$
85	1.9294	4	0	0	0.0483	1	0
95	1.9777	142	0	0	0.0435	1	0
105	2.0212	155	0	0	0.0395	1	0
115	2.0607	64	4	0.0625	0.0362	0.9375	0.0009
125	2.0969	51	25	0.4902	0.0334	0.5098	0.0050
135	2.1303	59	24	0.4068	0.0310	0.5932	0.0042
145	2.1614	29	11	0.3793	0.0290	0.6207	0.0084
155	2.1903	40	23	0.5750	0.0272	0.4250	0.0063
165	2.2175	27	15	0.5556	0.0256	0.4444	0.0095
175	2.2430	13	11	0.8462	0.0241	0.1538	0.0108
185	2.2672	4	2	0.5000	0.0229	0.5000	0.0833
195	2.2900	7	6	0.8571	0.0217	0.1429	0.0204
205	**2.3118**	4	4	**1**		0	0
Total		**599**	**125**	**5.6726**			**0.1488**

$m = 2.3118 + (0.0362/2) - (0.0362 \times 5.6726) = 2.1244$

Ppt no. 213

Factors Affecting Titration Procedures

Incubation Time

- Precise period
- Most viruses take precise time to reproduce
- Time between innoculum and appearance of symptom

Type of virus

- Plant virus or animal virus
- Conditions provided
- Optimum media

Procedure contamination

- Bacterial contamination
- Fungal contamination
- Careless lab work
- Improper cleanliness and sanitation
- Disruption of viral genome

Ppt no. 214

Bacteriophage Titration

Definition

- Determination of the number of phage particles in a stock

Definition

- Simplest type of titration
- Several dilution preparation
- Plating one or more measured samples of each dilution

Procedure

- Phage and bacteria introduced on agar surface
- Incubation
- Interruption of continuous bacterial layer by lysis of phage

Charging

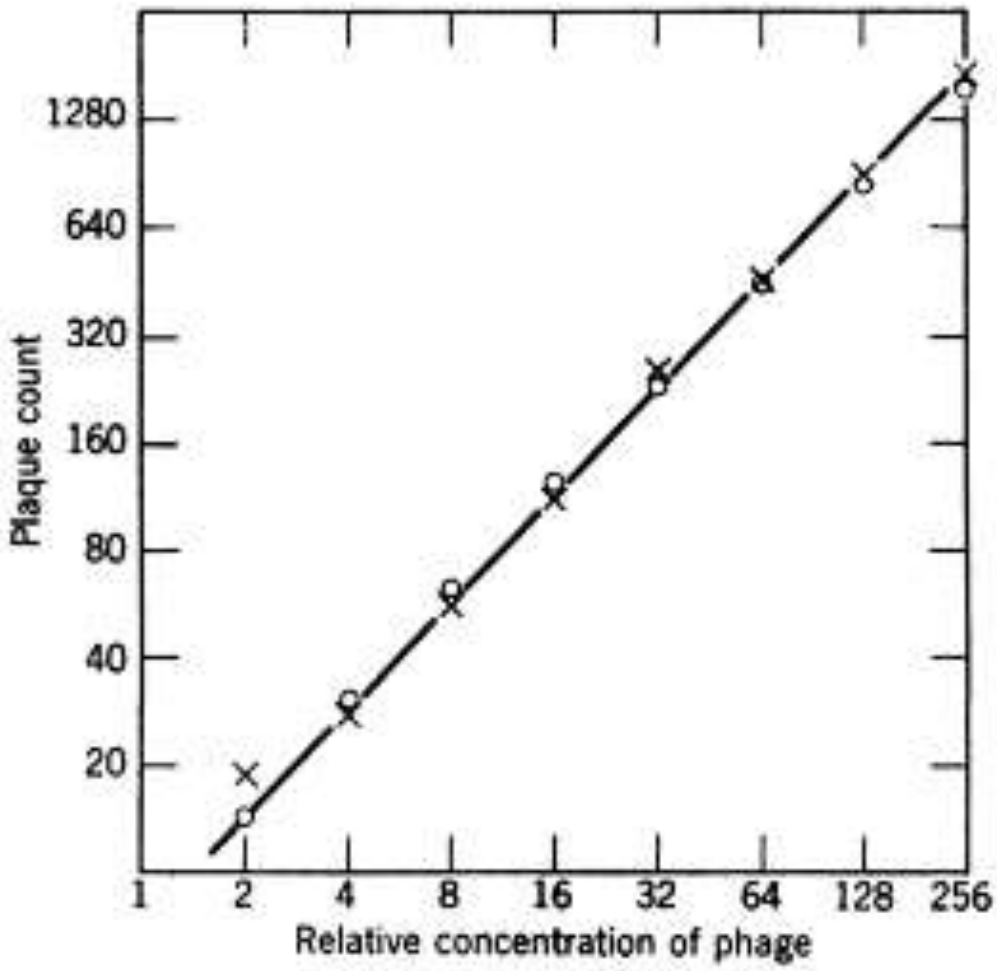
- Sometimes adhere to plastic or glass
- If particles adhere to pipettor tips during transfers, it may lower the amount of phage particles present
- One way to minimize this problem is to "charge" the pipettor by pipetting a sample of the solution up and down in the tip, and discharging the sample back into the original tube

Ppt no. 215

Plaque Count in Bacteriophage Titration

Plaque values

- Plaque counts analogous to colony counts for bacteria
- Values directly proportional to amount of phage plated


1280
640
320
160
80
40
20
Plaque count
1
2
4
8
16
32
64
128
256
Relative concentration of phage

- Plate cultures prepared by combining phage with host cells in **top agar**
- Top agar preparations contain lower conc. of agar (7 g/L) than normal solutions used to prepare agar plates (15 g/L)

- Low agar conc. allows progeny phage from lysed cells to diffuse through the media and infect neighbouring cells
- After lysis, **plaque** (zone of lysed cells) is produced on the plate
- The size of the plaques depend on how soon the bacteria in the agar reach stationary phase and stop reproducing
- Plaques will stop spreading at this point